CLASSIFICATION OF NUMERICAL GODEAUX SURFACES

JIHAO LIU, WENFEI LIU, AND LEI ZHANG

Abstract. We prove Reid's conjecture that the moduli space of numerical Godeaux surfaces has exactly five irreducible components, one for each cyclic fundamental group of order at most 5. In particular, there exist exactly five deformation classes of numerical Godeaux surfaces, and any numerical Godeaux surface with torsion-free Picard group is deformation equivalent to the Barlow surface.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

A numerical Godeaux surface is a smooth minimal projective surface S of general type with $K_S^2 = 1$ and $p_g(S) = q(S) = 0$. The surfaces are named after Godeaux [BCP11, Introduction], who considered the quotient of a quintic surface in $\mathbb{P}^3$ by a free action of $\mathbb{Z}/5$ [God31], see [CCML07, Section 1]. The canonical models of numerical Godeaux surfaces have a coarse moduli space [Gie77].

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The torsion subgroup of $\mathrm{Pic}(S)$ is cyclic of order at most 5 [Miy76, Lemma 11 and the Remark after Theorem 2'], see [SS20, Lemma 1.6]. See also [Rei78, Theorem 2.1].

[Bar84] constructed numerical Godeaux surfaces with $\pi_1(S) \cong \mathbb{Z}/2$, and [Bar85] constructed a simply connected numerical Godeaux surface, called the Barlow surface throughout the paper. [Wer94] constructed an example with $H_1(S, \mathbb{Z}) \cong \mathbb{Z}/2$. The Craighero–Gattazzo surface [CG94] is a smooth model of a singular quintic surface in $\mathbb{P}^3$, with trivial torsion subgroup of $\mathrm{Pic}(S)$ [DW99], and [RTU17] proved that it is simply connected. Further examples with trivial torsion subgroup of $\mathrm{Pic}(S)$ are in [LP07] and [SU16], see [FFP22, Section 1.1].

[BCP11, Conjecture 1] attributes the following conjecture to M. Reid.

Conjecture 1.1 ([BCP11, Conjecture 1])**.** *The moduli space of canonical models of numerical Godeaux surfaces has exactly five irreducible components, corresponding to each choice $\pi_1(S) \cong \mathbb{Z}/m$ for $1 \leq m \leq 5$.*

We refer the reader to Definitions 2.1 and 2.6 for related definitions.

The following results are known, where m denotes the order of the torsion subgroup of $\mathrm{Pic}(S)$.

(1) When $m = 5$, such surfaces exist and form an irreducible moduli space [Rei78, Introduction and Theorem 1.1] (the first such surfaces are due to [God31], and the irreducibility is due to [Miy76], see [Rei78, Introduction]). The quotient of the Fermat quintic by a free action of $\mathbb{Z}/5$ is one such surface [Ste18, Construction 9.1.1]. [CU18, Introduction] and [DR26, Introduction] say that $\pi_1(S) \cong \mathbb{Z}/5$ follows from [Rei78, Theorem 1.1].

(2) When $m = 4$, such surfaces exist and form an irreducible moduli space [Rei78, Introduction and Theorem 2.2]. A surface was also constructed independently in [Miy76], see [Rei78, Introduction]. [CU18, Introduction] and [DR26, Introduction] say that $\pi_1(S) \cong \mathbb{Z}/4$ follows from [Rei78, Theorem 2.2].

(3) When $m = 3$, such surfaces exist and form an irreducible moduli space [Rei78, Introduction and Theorem 3.9]. Such surfaces have $\pi_1(S) \cong \mathbb{Z}/3$ [CU18, Theorem 1.1]. See also [DR26, Introduction]. For each $m \in \{3, 4, 5\}$, the moduli space is of dimension 8 [DR26, Introduction].

(4) When $m = 2$, the moduli space is irreducible and unirational of dimension 8, and $\pi_1(S) \cong \mathbb{Z}/2$ [DR26, Corollary 7].

(5) When $m = 1$, that is, when the torsion subgroup of $\mathrm{Pic}(S)$ is trivial, part of the classification is known. [CL97, Theorem 7] states that the Barlow surface has a smooth versal deformation space of dimension 8, and [CP00, Theorem 0.1, last paragraph] states that the local moduli space of the Barlow surface and of the Craighero–Gattazzo surface is smooth of dimension 8. [CP00, Abstract and Introduction] provides a first step towards the classification for trivial torsion subgroup and for torsion subgroup $\mathbb{Z}/2$: the bicanonical system induces, on a blow-up of S, a fibration in curves of genus 2, 3, or 4, and the behavior of the bicanonical system divides the surfaces into four classes, of which the class of genus 2 is empty by [CP06, Theorem 5.1]. [Ste18, Chapter 1, p. 4] obtains a family of dimension 8 with trivial torsion subgroup, by a method in two steps explained below. [SS20, Corollary 5.2] constructs the family of Theorem 1.4. [SS24, Theorems 3.2 and 3.5] characterizes the reducible bicanonical fibers attached to torsion and the hyperelliptic bicanonical fibers, when the bicanonical system has no fixed part and four distinct base points. We refer the reader to [FFP22, Section 1.1] for a recap of known facts.

The proof in this paper depends on the following earlier results.

Theorem 1.2 ([Rei78, Introduction and Theorems 1.1, 2.2 and 3.9])**.** *For each $m \in \{3, 4, 5\}$, numerical Godeaux surfaces S whose torsion subgroup of $\mathrm{Pic}(S)$ is isomorphic to $\mathbb{Z}/m$ exist and form an irreducible moduli space.*

Theorem 1.3 ([DR26, Corollary 7])**.** *The moduli space of numerical Godeaux surfaces S whose torsion subgroup of $\mathrm{Pic}(S)$ is isomorphic to $\mathbb{Z}/2$ is irreducible and unirational of dimension 8, and any such S satisfies $\pi_1(S) \cong \mathbb{Z}/2$.*

Theorem 1.4 ([SS20, Propositions 3.4 and 5.6, Definition 3.5, Corollary 5.2, and Section 6])**.** *There exists an irreducible locally complete 8-dimensional family of numerical Godeaux surfaces with trivial fundamental group whose bicanonical system on the canonical model has four distinct base points. The Barlow surface is deformation equivalent to a general member of the family.*

Here locally complete means that any small deformation of a general member occurs in the family.

The proof in this paper also depends on the structure theorems for relatively minimal fibrations of genus 2 [CP06, Theorem 4.13] and for relatively minimal non-hyperelliptic fibrations of genus 3 with 2-connected fibers [CP06, Theorem 7.13].

We now explain why the case of trivial torsion subgroup is different. The known constructions for non-trivial torsion subgroup first construct the étale cover of S attached to the torsion subgroup and then take the quotient. See [Ste18, Chapter 1, p. 1] for [Rei78], and [DR26, Introduction] for the equations of the étale double covers when the torsion subgroup is $\mathbb{Z}/2$. When the torsion subgroup of $\mathrm{Pic}(S)$ is trivial, there exists no such cover. Thus [Ste18] and [SS20], following an idea of [Sch05], start directly from the canonical ring R of S, viewed as a module over the polynomial ring generated by a basis of $H^0(S, 2K_S)$ and a basis of $H^0(S, 3K_S)$, in degrees 2 and 3 [Ste18, Chapter 1, p. 2], [SS20, Introduction]. [SS20] recovers the resolution of R from the reduction of the resolution modulo the degree 2 generators. The twelve entries of the resolution that are linear forms in the degree 2 generators satisfy four quadratic equations, which define a complete intersection $\mathcal{Q} \subset \mathbb{P}^{11}$ whose coordinates are the twelve entries [SS20, Introduction].

The family of Theorem 1.4 is constructed in two steps: the first step is the choice of a line in $\mathcal{Q}$, and the second step is the solution of a linear system depending on the line. For a general line, the solution space is of dimension 4 [SS20, Theorem 5.1]. After the quotient by a three-dimensional torus, the dimension of the family of surfaces is at least the sum of the dimension of the family of lines and the projective dimension of the solution spaces minus 3, with equality if the stabilizer of a general point is finite. For general lines, the family of lines is of dimension 8 and the solution spaces are of projective dimension 3, which yields the family of dimension $8 + 3 - 3 = 8$ of Theorem 1.4 [SS20, Corollary 5.2].

The special loci in $\mathcal{Q}$ [SS20, Introduction] are supports of the homology of two complexes introduced in [SS20, Section 5]. Write the twelve coordinates of $\mathbb{P}^{11}$ as the non-zero entries of a 4×6 matrix, with three entries in each row and two in each column [SS24, Section 2, p. 4]. Then the special loci are the points where the matrix has rank at most 3, or a row vanishes, or a column vanishes [SS24, Section 2, pp. 5–6].

The points where a column vanishes form the union of the 24 components of [SS24, Section 2, pp. 5–6], since the representatives listed there have a vanishing column, the signed permutations of [SS20, Proposition 4.4] permute the columns, and, where a given column vanishes, two of the four quadratic equations of $\mathcal{Q}$ become products of two coordinates, whose branches give, by a direct calculation with the other two equations, two irreducible components with one vanishing column and four with two vanishing columns, all of dimension 5, so that the points with a vanishing column form $6 \cdot 2 + 6 \cdot 4/2 = 24$ irreducible components. A line that does not intersect the special loci leads only to surfaces in the family of Theorem 1.4, and the condition is only sufficient [SS20, Section 5, after Remark 5.3, p. 22].

For the second step, [SS20, Remark 2.8] introduces a condition on the first map of the resolution, called the ring condition, under which the cokernel of the first map carries a ring structure. The ring condition together with Du Val singularities of Proj of that ring is only sufficient for the cokernel to be a canonical ring [SS24, Section 2, p. 3]. Finally, [SS20] treats only surfaces whose bicanonical system has no fixed part and four distinct base points, which correspond to the dense orbit under PGL_4 of subschemes of length 4 of $\mathbb{P}^3$ [SS20, Remark 3.3 and Proposition 3.4], and it leaves the case of a non-reduced base scheme for later [SS20, Section 6].

A complete classification by the method of [SS20] depends on excluding lines that intersect the special loci [SS20, Section 6]. The following examples show that such lines cannot simply be excluded, and that the dimension of the solution space does not separate them. The Barlow surface has two hyperelliptic bicanonical fibers, counted with multiplicity [CP00, Theorem 0.1], see also [SS24, Remark 5.2], and the corresponding points of the line of the Barlow surface belong to the special loci, by [SS24, Corollary 3.6] and the description of the support of the zeroth homology of the second complex in [SS24, Section 2, p. 6]. See also [SS24, Theorem 3.5]. The solution space over the line of the Barlow surface is still of dimension 4 [SS20, Proof of Proposition 5.6].

The lines of surfaces with torsion subgroup $\mathbb{Z}/5$ and $\mathbb{Z}/3$ also intersect the special loci [SS20, Section 5, after Remark 5.3, p. 22]. Torsion subgroup $\mathbb{Z}/5$ corresponds to a 2-dimensional family of lines with solution spaces of projective dimension 9 [SS24, Theorem 4.4 and Table 1], and torsion subgroup $\mathbb{Z}/3$ corresponds to a 5-dimensional family of lines with solution spaces of projective dimension 6 [SS24, Theorem 4.7 and the table of Section 4.2]. In both cases, the family of surfaces is of dimension $2+9-3=5+6-3=8$ [SS24, Theorems 4.4 and 4.7].

An experiment over a finite field [SS24, Section 5.1, the table, row 1), p. 19] finds solutions that do not satisfy the ring condition, on a 7-dimensional family of lines with solution spaces of projective dimension 7, so that the family is of dimension at least $7+7-3=11$ after the quotient by the torus, and the scheme in $\mathbb{P}^1\times\mathbb{P}^3$ attached to such a solution decomposes into a $\mathbb{P}^1\times\mathbb{P}^1$ and a $(4,4)$-hypersurface in $\mathbb{P}^1\times\mathbb{P}^2$. Since $11\geq 8$, the solutions of row 1) fit the description of ghost components in [SS24, Section 2, p. 3]: components of the solution space that lead to surfaces which might not be numerical Godeaux surfaces but form families of dimension at least 8. [SS24, Introduction, p. 2] says that the authors could not determine the general stratification of the Fano scheme of lines with regard to the dimension of the solution space.

[DR26, Introduction] says that the case of trivial torsion subgroup is still open. [FR18, Introduction] points out that one cannot exclude a numerical Godeaux surface with trivial algebraic fundamental group and infinite topological fundamental group.

The goal of this paper is to prove Conjecture 1.1.

Main result. We prove the following classification.

Theorem 1.5. *Let $\mathcal{M}$ be the coarse moduli space of canonical models of numerical Godeaux surfaces. For $m\in\{1,\dots,5\}$, let $\mathcal{M}_m\subset\mathcal{M}$ be the locus of points whose minimal resolution S satisfies $\pi_1(S)\cong\mathbb{Z}/m$. Then*

$$\mathcal{M}=\bigsqcup_{m=1}^{5}\mathcal{M}_m,$$

and each $\mathcal{M}_m$ is non-empty, irreducible of dimension 8, and a connected component of $\mathcal{M}$. Moreover, $\mathcal{M}_m$ is the locus where the torsion subgroup of $\mathrm{Pic}(S)$ is isomorphic to $\mathbb{Z}/m$.

In particular, there exist exactly five deformation classes of numerical Godeaux surfaces, and two numerical Godeaux surfaces are deformation equivalent if and only if their fundamental groups have equal order.

Theorem 1.5 implies Conjecture 1.1.

The one new ingredient of Theorem 1.5 is the following. A numerical Godeaux surface S is torsion-free if the torsion subgroup of $\mathrm{Pic}(S)$ is trivial (see Definition 2.1). We prove that the locus of torsion-free surfaces in $\mathcal{M}$ is the closure of the Schreyer–Stenger family of Theorem 1.4 (Theorem 15.3). As a consequence, any torsion-free surface is deformation equivalent to the Barlow surface, and in particular simply connected. Thus there is no numerical Godeaux surface with trivial algebraic fundamental group and infinite topological fundamental group, since a surface with trivial algebraic fundamental group is torsion-free, as a non-trivial torsion line bundle defines a connected cyclic étale cover.

In the companion paper [LLZ26], we prove, independently of the classification here and using L^2 holomorphic forms on the universal cover, that the topological fundamental group of any numerical Godeaux surface S is finite.

Sketch of the proof. Theorem 1.5 follows from Definition-Proposition 15.4 and Lemma 15.1, where the cases $m \geq 2$ are cited (Theorems 1.3 and 1.2, with [CU18, Theorem 1.1], [CU18, Introduction], and [DR26, Introduction] for the fundamental group and the dimension when $m \in \{3, 4, 5\}$), and the case $m = 1$ is $\mathcal{M}_{\mathrm{tf}} = \mathcal{B}_{\mathrm{SS}}$ (Theorem 15.3, Definition 2.6), which we prove as follows.

(1) Since $\dim \mathcal{C} \geq 8$ for any irreducible component $\mathcal{C}$ of the analytic germ $(\mathcal{M}, x)$ at any point $x \in \mathcal{M}$ (Lemma 2.8), and $\mathcal{M}_{\mathrm{tf}} \backslash \mathcal{B}_{\mathrm{SS}}$ is analytically open (Lemma 15.1), it suffices to cover $\mathcal{M}_{\mathrm{tf}} \backslash \mathcal{B}_{\mathrm{SS}}$ by seven closed subsets with $\dim \leq 7$ (Construction-Proposition 15.2).
(2) They come from the genus of the bicanonical pencil, the partition of the base scheme, and events (Construction-Propositions 3.1 and 3.3, Definition 6.1). The bounds are Theorems 14.19, 6.9, 7.10, 9.1, 10.1, 11.1, and 12.1.
(3) In genus four, the lower blocks (Set-up 4.1, Theorem 2.10) are given by a matrix A and a solution in the space $\mathcal{U}_A$ of a linear system (Construction-Propositions 5.3 and 8.6), which, together with the fixed blocks and the quadrics in the first row of the presentation, determine at most one point of $\mathcal{M}$ (Lemma 4.5(2)), so a bound counts the parameters of A, plus $\dim \mathcal{U}_A$, minus the dimension of a group orbit.
(4) In genus three, [CP06, Theorem 7.13] and Lemma 14.17 yield a parametrization by a space of dimension at most 7.

The key new step is Construction-Proposition 8.4: a group preserving only the module structure and self-duality of $R/(s,t)$ (Construction-Theorem 3.2) acts on presentations and keeps the point of $\mathcal{M}$. The proof exhibits no deformation into the family of Theorem 1.4 and does not decide which subsets are empty.

(a) Upper bounds in place of deformations. [SS20, Proof of Proposition 5.6] moves the Barlow surface into the family of Theorem 1.4 by moving the line $\ell \subset \mathcal{Q}$ of the surface, since the solution space over ℓ is of dimension 4 and $h^1(\ell, N_{\ell|\mathcal{Q}}) = 0$. For another line, the solution space may be larger, and the deformations of the line may be obstructed, so we do not deform a given surface. Since $\chi(T_S) = -8$ and $h^0(S, T_S) = 0$, any component of the Kuranishi space of S is of dimension at least 8, whatever $h^2(S, T_S)$ is, and the bound passes to $\mathcal{M}$ by Theorem 2.7 (Lemma 2.8). We do not decide which solutions come from surfaces, since the known conditions are only sufficient [SS24, Section 2, p. 3]: the bounds count all solutions over the matrices A of torsion-free surfaces, apart from the exceptional presentations of type $(2, 2)$ in (c).

(b) Lines in $\mathcal{Q}$. For four distinct base points, A parametrizes a line in $\mathcal{Q}$ (Construction-Proposition 5.1). The lines of [SS24, Theorems 4.4 and 4.7] yield families of dimension 8 of surfaces with torsion, and those of row 1) of the table of [SS24, Section 5.1], with solutions that do not satisfy the ring condition, yield dimension at least 11, so we first use that S is torsion-free: no column of A vanishes identically (Lemma 5.7), and two columns of A, indexed by pairs in $\{0, 1, 2, 3\}$, with a common index have no common zero (Lemma 5.8). [SS20, Proof of Theorem 5.1] considers lines outside the special loci, and the 4×6 matrix in their description is A. On other lines, the generators V, W (Table 6) of the kernel on $\mathcal{Q}$ of the part of the linear system in the block C of Set-up 4.1 can become dependent at a point, which yields an extra solution after division by a linear form. Lemma 7.3 proves by explicit identities that V, W are independent at any point with no zero column.

Thus on a line of generic rank four without events, $\dim \mathcal{U}_A \leq 4 + r$, where r is the number of rows of A vanishing at a point of the line, and for $r \geq 1$ such lines form a set of dimension at most $7 - r$ (Construction-Proposition 7.6), so $(7 - r) + (4 + r) - 1 - 3 = 7$ with a torus orbit. For $r = 0$, the solutions form a bundle of rank 4 over an irreducible base of dimension 8 (Construction-Propositions 7.2 and 7.9), so the family of [SS20, Corollary 5.2] is dense in

the projectivization of the bundle (Proposition 7.8 and Construction-Proposition 7.9). The projectivized bundle is of dimension 11, the complement of the family is of dimension at most 10, and $10 - 3 = 7$ with a torus orbit.

If a column of A vanishes at a point p of the line, which is an event (Definition 6.1), then V, W are dependent at p. The largest family of lines with one event is of dimension 4, so the lines with one event form a set of dimension $4 + 3 = 7$ with a torus orbit. With one more solution, the count would be $7 + 5 - 1 - 3 = 8$, so the extra solution has to cancel outside a proper closed subset of the family. By Lemma 6.2, the event graph is one of five non-empty graphs, and by Construction-Propositions 6.6, 6.4, 6.5, and 6.3, lines with events form the eleven families of Table 5 up to symmetries. The pivot tables of Construction-Proposition 6.8 bound $\dim \mathcal{U}_A$ on them.

(c) The module group. For non-reduced base schemes (left for later in [SS20, Section 6]), the fixed blocks depend only on the algebra of the base scheme and a functional (Construction-Proposition 8.2). For collision type $(2, 1, 1)$ (Set-up 8.1), that is, partition $(2, 1, 1)$ of the base scheme, the count of A and $\mathcal{U}_A$ is $12 + 4 = 16$, where each point with a zero column yields one extra solution and one linear condition (Construction-Proposition 9.9 and Proposition 9.13). The graded algebra automorphisms of $R/(s, t)$, with GL_2 acting on s, t, yield orbits of dimension at most 5, and $16 - 5 = 11 > 7$, which is not enough.

Thus we use the group $\mathcal{G}$ of Construction-Proposition 8.4, with $\dim \mathcal{G} = 9 - n$ for n base points. By the generator degrees of the resolution, any element of $\mathcal{G}$ lifts uniquely to presentations of the ring R with other generators, transporting the multiplication of R. The group $\mathrm{GL}_2 \times \mathcal{G}^0$, of dimension $13 - n$, has orbits of dimension at least $12 - n$ (Lemma 8.7), so $16 - 9 = 7$ for type $(2, 1, 1)$, and $17 - 10 = 7$ for the non-exceptional presentations of type $(2, 2)$. For types $(3, 1)$ and (4), the group brings A to families of dimension at most 3, so $3 + 4 = 7$. Only on the exceptional presentations of type $(2, 2)$ is the linear system not enough, with $2 + 9 = 11$: there the multiplication of R yields four linear conditions, so $2 + 5 = 7$ (Construction-Proposition 10.10).

(d) Genus three. The pencil has a base point of multiplicity two (Profile I) or a fixed divisor (Profile II) (Construction-Proposition 14.1), so the base scheme is not curvilinear in the smooth locus of X, and instead of the blocks of (c) we use the relative canonical algebra of Construction-Proposition 14.2 [CP06, Lemma 7.2, Proposition 7.9]. [CP06, Theorem 7.13] considers nonhyperelliptic fibrations, and Corollary 13.6 provides this: a hyperelliptic involution yields a fibration of S of genus 2, which forces a bicanonical pencil of genus 4 (Theorem 13.5). By Construction-Proposition 14.2, the relation of degree 4 of the relative canonical algebra does not lift to the symmetric square of the generators of degree 2: the only sections of that square are the multiples of ζ^2, for the section ζ of $V_2(-5)$, which are non-zero in degree 4. Thus the image of the relation under the connecting homomorphism of the sequence in Construction-Proposition 14.2, which we call the *connecting class*, is non-zero.

The class excludes Profile I (Theorem 14.12), a separate argument excludes distinct centers of the blow-ups of Profile II (Proposition 14.7), and the class then forces the forms w, v of Construction-Proposition 14.13 to be proportional (Lemma 14.15) and leaves seven parameters (Lemma 14.17 and the proof of Theorem 14.19). By [CP06, Theorem 7.13], any such surface is given by a value of the seven parameters, and a value determines the surface by equations (Theorem 14.19).

The steps depending on the finite tables of exact eliminations in Appendix A are the families of pencil matrices with events, the non-vanishing of the 48×48 determinant of type $(2, 1, 1)$ (Table 33), the charts (Table 12) and the exceptional presentations of type $(2, 2)$, and the families of types $(3, 1)$ and (4).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we collect Miyaoka's results on the bicanonical system, the self-dual resolution of the canonical ring of Schreyer and Stenger, and the local structure of the moduli space. In Section 3 we show that the pencil obtained by resolving the bicanonical system has genus three

or four, we describe the quotient of the canonical ring by the bicanonical sections, and we list the five possible partitions of the base scheme of length four in genus four. In Section 4 we write the resolution of the canonical ring in blocks, and we show that the blocks below the first row, together with the six quadrics of the first row, determine the canonical model.

In Section 5 we consider the case of torsion-free surfaces with four distinct base points, where the first blocks are given by a line in $\mathcal{Q} \subset \mathbb{P}^{11}$ and the remaining blocks by the solutions of a linear system. In Section 6 we bound the moduli of such surfaces when two paired forms among the twelve binary linear forms of the line have a common zero. In Section 7 we bound the moduli of the other torsion-free surfaces with four distinct base points outside the closure of the Schreyer–Stenger family.

In Section 8 we construct a group acting on the resolutions when the base scheme is non-reduced. In Section 9 we bound the moduli of the torsion-free surfaces whose base scheme has partition $(2,1,1)$. In Section 10 we bound the moduli of the torsion-free surfaces whose base scheme has partition $(2,2)$. In Section 11 we bound the moduli of the torsion-free surfaces whose base scheme has partition $(3,1)$. In Section 12 we bound the moduli of the torsion-free surfaces whose base scheme has partition (4).

In Section 13 we show that a torsion-free surface with a genus-two fibration has a bicanonical pencil of genus four. In Section 14 we bound the moduli of the torsion-free surfaces whose bicanonical pencil has genus three. In Section 15 we prove Theorems 15.3 and 1.5. Appendix A collects the rank computations.

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2. Preliminaries

2.1. Numerical Godeaux surfaces.

Definition 2.1. A *numerical Godeaux surface* is a smooth minimal projective surface S of general type with $K_S^2 = 1$ and $p_g(S) = q(S) = 0$. We denote by $\pi : S \to X$ the minimal resolution of the canonical model X of S, and by

$$R := \bigoplus_{n \geq 0} R_n, \qquad R_n := H^0(S, nK_S),$$

the canonical ring of S. We say that S is *torsion-free* if the torsion subgroup of $\mathrm{Pic}(S)$ is trivial.

Lemma 2.2. *Let S be a numerical Godeaux surface. Then the following hold.*

(1) $\chi(\mathcal{O}_S) = 1$.
(2) For any integer $n \geq 2$, we have $H^i(S, nK_S) = 0$ for any $i > 0$, and we have $h^0(S, nK_S) = 1 + n(n-1)/2$. In particular, $h^0(S, 2K_S) = 2$ and $h^0(S, 3K_S) = 4$.
(3) For any integers m and i, we have $H^i(X, mK_X) = H^i(S, mK_S)$, and $H^1(S, mK_S) = 0$.
(4) For any divisor D on S, $D^2 + K_S \cdot D$ is even and $D^2 \leq (K_S \cdot D)^2$. If $D^2 = (K_S \cdot D)^2$, then $D - (K_S \cdot D)K_S$ is numerically trivial.

Proof. Since $p_g(S) = q(S) = 0$, we have $\chi(\mathcal{O}_S) = 1 - q(S) + p_g(S) = 1$, which is (1).

Let $n \geq 2$. Since $(n-1)K_S$ is nef and big (K_S is nef by [BHPV04, Chapter VII, Corollary 2.3], and big since $K_S^2 = 1 > 0$ [KM98, Proposition 2.61]), $H^i(S, -(n-1)K_S) = 0$ for any $i < 2$ [KM98,

Theorem 2.70]. Thus $H^i(S, nK_S) = 0$ for any $i > 0$ by Serre duality [Har77, Corollary III.7.7], and Riemann–Roch [Har77, Theorem V.1.6] implies that

$$h^0(S, nK_S) = \chi(\mathcal{O}_S) + \frac{1}{2}nK_S \cdot (n-1)K_S = 1 + \frac{n(n-1)}{2}.$$

This is (2).

Since X has canonical singularities (π contracts the (-2)-curves of S, so X has A-D-E singularities [BHPV04, Chapter VII, Section 5]), X is klt, and so X has rational singularities [KM98, Theorem 5.22]. Moreover, K_X is Cartier and $K_S = \pi^* K_X$ [KM98, Definition 4.4 and Theorem 4.20]. Thus the equality $H^i(X, mK_X) = H^i(S, mK_S)$ follows from [Har77, Exercises III.8.1 and III.8.3]. For $H^1(S, mK_S)$, the case $m \geq 2$ is (2), the cases $m = 0, 1$ follow from $q(S) = 0$ and Serre duality, and for $m < 0$, $H^1(S, mK_S)$ is Serre dual to $H^1(S, (1-m)K_S)$ with $1 - m \geq 2$.

For (4), let D be a divisor on S. By the Riemann–Roch theorem [Har77, Theorem V.1.6],

$$D^2 - K_S \cdot D = 2\big(\chi(\mathcal{O}_S(D)) - \chi(\mathcal{O}_S)\big),$$

so $D^2 - K_S \cdot D$ is even, and hence so is $D^2 + K_S \cdot D$. Let $D' := D - (K_S \cdot D)K_S$. Then $K_S \cdot D' = 0$ and $D'^2 = D^2 - (K_S \cdot D)^2$. Since $K_S^2 = 1$ and the intersection form on divisors modulo numerical equivalence has signature $(1, \rho - 1)$, where ρ is the Picard number of S [Har77, Theorem V.1.9, Remark V.1.9.1], the form is negative definite on the divisors orthogonal to K_S. Thus $D'^2 \leq 0$, with equality only if D' is numerically trivial. This is (4). □

Lemma 2.3. *Let S be a smooth projective surface with $p_g(S) = q(S) = 0$. Then the following hold:*

(1) $c_1 : \mathrm{Pic}(S) \to H^2(S, \mathbb{Z})$ is an isomorphism, and any numerically trivial line bundle on S is torsion.

(2) If S is a numerical Godeaux surface and $D \geq 0$ is a divisor with $2D \sim 2K_S$, then $\mathcal{O}_S(D - K_S)$ is a non-trivial 2-torsion line bundle.

Proof. The exponential sequence [Har77, Appendix B, Section 5] with GAGA [Har77, Appendix B, Theorem 2.1] and $H^1(S, \mathcal{O}_S) = H^2(S, \mathcal{O}_S) = 0$ implies that c_1 is an isomorphism. Let L be a numerically trivial line bundle on S. Since any class of $H^2(S, \mathbb{Z})$ is a Chern class, $c_1(L)$ pairs to zero with any class of $H^2(S, \mathbb{Z})$ (the intersection number of two line bundles is the cup product of their Chern classes [BHPV04, Chapter II, Section 10]). By Poincaré duality, $c_1(L)$ is torsion, and so L is torsion. This is (1).

For (2), $\mathcal{O}_S(D - K_S)^{\otimes 2}$ is trivial. If $\mathcal{O}_S(D - K_S)$ were trivial, then $D \in |K_S|$, which is not possible since $p_g(S) = 0$. □

Theorem 2.4 ([SS20, Proposition 1.1 and Remark 1.2])**.** *Let S be a numerical Godeaux surface and write $|2K_S| = |M| + Z$, where Z is the fixed divisor. Then a general member of $|M|$ is reduced and irreducible, and one of the following holds:*

(1) $Z = 0$ and $M^2 = 4$.

(2) $K_S \cdot Z = 0$, $Z^2 = -2$, $M^2 = 2$, and $M \cdot Z = 2$.

(3) $K_S \cdot Z = 0$, $Z^2 = -4$, $M^2 = 0$, and $M \cdot Z = 4$.

In any case, $K_S \cdot M = 2$. In particular, $|2K_X|$ has no fixed divisor and a general member of $|2K_X|$ is irreducible.

Proof of Theorem 2.4. We only need to show that $K_S \cdot M = 2$ and that $M^2 = 4$ in case (1), since the remaining part is [SS20, Proposition 1.1 and Remark 1.2], where Proposition 1.1 is quoted from [Miy76, Lemma 6]. We have $M + Z \sim 2K_S$ and $K_S^2 = 1$. Since $K_S \cdot Z = 0$ in cases (2) and (3) and $Z = 0$ in case (1), we have $K_S \cdot M = 2K_S^2 - K_S \cdot Z = 2$. In case (1), we have $M^2 = (2K_S)^2 = 4$. □

Theorem 2.5 ([SS24, Lemma 3.1])**.** *Let S be a numerical Godeaux surface, and let $C \in |2K_X|$. Then one of the following holds:*

(1) *C is embedded by ω_C, and the image of C under the tricanonical map is the complete intersection of a quadric and a cubic in $\mathbb{P}^3$.*
(2) *C is honestly hyperelliptic, that is, there exists a finite morphism $C \to \mathbb{P}^1$ of degree 2, and the image of C under the tricanonical map is a twisted cubic.*
(3) *$\pi^*C = D_1 + D_2$ with $D_i \in |K_S + \tau_i|$, where τ_1, τ_2 are non-trivial torsion classes and $\tau_1 + \tau_2 = 0$.*

The first case holds for a general member C. If S is torsion-free, then the third case does not occur.

The first two cases correspond to the dimensions 1 and 3 of the space of quadrics in Lemma 4.3.

2.2. **The moduli space.**

Definition 2.6. We denote by $\mathcal{M}$ the coarse moduli space of canonical models of numerical Godeaux surfaces, and by $[X] \in \mathcal{M}$ the point of the canonical model X of a numerical Godeaux surface S. For $m \in \{1, \dots, 5\}$, $\mathcal{M}_m$ is the locus of points $[X]$ with $\pi_1(S) \cong \mathbb{Z}/m$. We denote by $\mathcal{M}_{\mathrm{tf}}$ the locus of points $[X]$ with S torsion-free. Let $\mathcal{S}$ be the family of [SS20, Corollary 5.2], and let $\mathcal{B}_{\mathrm{SS}} \subset \mathcal{M}$ be the Zariski closure of the set of points $[X]$, where X is the canonical model of a fiber of $\mathcal{S}$. For a subset $\Sigma \subset \mathcal{M}$, we denote by $\overline{\Sigma}$ the Zariski closure of Σ in $\mathcal{M}$.

That $\mathcal{B}_{\mathrm{SS}}$ is irreducible and a connected component of $\mathcal{M}$ is part of Theorem 15.3.

Theorem 2.7 ([Cat13, Theorem 3.4, Remark 3.7, and Corollary 3.9])**.** *Let S be a numerical Godeaux surface with canonical model X, and let $\mathrm{Def}(S)$ and $\mathrm{Def}(X)$ be the Kuranishi spaces of S and X. Then the following hold.*

(1) *$\mathcal{M}$ is a quasi-projective scheme.*
(2) *$\mathrm{Aut}(X) = \mathrm{Aut}(S)$ is finite, and the germ $(\mathcal{M}, [X])$ is isomorphic to $\mathrm{Def}(X)/\mathrm{Aut}(X)$.*
(3) *The natural morphism $\mathrm{Def}(S) \to \mathrm{Def}(X)$ is finite and surjective.*

Proof. By [Cat13, Theorem 3.4], Gieseker's moduli space [Gie77] of canonical models with $K^2 = 1$ and $\chi = 1$ is the quotient by a special linear group of a locally closed subscheme of a Hilbert scheme of pluricanonically embedded canonical models, and it is a quasi-projective scheme. A canonical model X' in that space has rational singularities [KM98, Theorem 5.22], so $h^1(X', \mathcal{O}_{X'}) = q(S')$ for the minimal resolution S' of X', and $\chi(\mathcal{O}_{S'}) = 1$ implies that $p_g(S') = q(S')$. By semicontinuity [Har77, Theorem III.12.8], the condition $q = 0$ is open on the locally closed subscheme of the Hilbert scheme and invariant under the special linear group. Thus $\mathcal{M}$ is an open subscheme of Gieseker's moduli space (the Hilbert points are stable [Cat13, Section 3.2], so the quotient is geometric and an invariant open subset descends to an open subscheme), so $\mathcal{M}$ is quasi-projective, and the germ of $\mathcal{M}$ at any point $[X]$ is the germ of Gieseker's moduli space at $[X]$. This implies (1), and then (2) follows from [Cat13, Remark 3.7].

If K_S is not ample, then (3) is [BW74], see [Cat13, Corollary 3.9]. If K_S is ample, then $S = X$ and the morphism $\mathrm{Def}(S) \to \mathrm{Def}(X)$ is the identity. □

The following lemma is the lower bound against which the upper bounds on the dimensions of loci in $\mathcal{M}$ are compared.

Lemma 2.8. *Let $x \in \mathcal{M}$. Then $\dim \mathcal{C} \geq 8$ for any irreducible component $\mathcal{C}$ of the analytic germ $(\mathcal{M}, x)$. In particular, a non-empty analytically open subset of $\mathcal{M}$ is not contained in a Zariski closed subset of $\mathcal{M}$ of dimension at most 7.*

Proof. Write $x = [X]$, where X is the canonical model of a numerical Godeaux surface S. Riemann–Roch [Har77, Appendix A, Theorem 4.1 and Example 4.1.2] and Lemma 2.2(1) imply that

$$\chi(T_S) = 2K_S^2 - 10\chi(\mathcal{O}_S) = -8.$$

Since the tricanonical map of S is birational [Miy76, Theorem 3], see also [SS20, Theorem 1.3], $\mathrm{Bir}(S)$ is finite [Mat63, Section 2, Corollary 2]. Thus $H^0(S, T_S) = 0$, because $H^0(S, T_S)$ is the Lie algebra of $\mathrm{Aut}(S)$, and so

$$h^1(S, T_S) - h^2(S, T_S) = 8.$$

By [Kur62], see [Cat13, Theorem 1.8], $\mathrm{Def}(S)$ is the germ of the zero set of a holomorphic map $H^1(S, T_S) \to H^2(S, T_S)$. Thus any irreducible component A of the germ $\mathrm{Def}(S)$ corresponds to a minimal prime $\mathfrak{p}$ over an ideal generated by $h^2(S, T_S)$ elements in the local ring of convergent power series in $h^1(S, T_S)$ variables. By [Mat89, Theorems 13.5 and 17.4(i)] (the dimension of A is the Krull dimension of the local ring of A [GR84, Chapter 5, §1.2] and [Mat89, Theorem 13.4]),

$$\dim A = h^1(S, T_S) - \operatorname{ht} \mathfrak{p} \geq h^1(S, T_S) - h^2(S, T_S) = 8.$$

By Theorem 2.7, the composition

$$\mathrm{Def}(S) \to \mathrm{Def}(X) \to (\mathcal{M}, x)$$

is finite and surjective. Thus $(\mathcal{M}, x)$ is the union of the images of the finitely many irreducible components A_i of $\mathrm{Def}(S)$. The image of A_i is an analytic germ [GR84, Chapter 3, §1.3, Finite Mapping Theorem and Proposition], and it is irreducible, since a decomposition of the image of A_i pulls back to A_i. Thus any irreducible component $\mathcal{C}$ of $(\mathcal{M}, x)$ is the image of some A_i, and $\dim \mathcal{C} = \dim A_i \geq 8$ [GR84, Chapter 5, §4.1, Theorem (Invariance of Dimension)].

Finally, let $V \subset \mathcal{M}$ be a Zariski closed subset with $\dim V \leq 7$. If $x \in V$, then $\dim(V, x) \leq 7$ for the analytic germ (V, x), and so $(V, x) \neq (\mathcal{M}, x)$, since $\dim \mathcal{C} \geq 8$ for any irreducible component $\mathcal{C}$ of $(\mathcal{M}, x)$. Since $x \in \mathcal{M}$ is arbitrary, no non-empty analytically open subset of $\mathcal{M}$ is contained in V. □

The upper bounds on the dimensions of loci in $\mathcal{M}$ use the following count.

Lemma 2.9. *Let V and W be schemes of finite type, let $G \subset V \times W$ be a constructible subset, and let e be a non-negative integer. Assume that $\dim G_v = 0$ for any non-empty fiber G_v of $G \to V$ and that $\dim G_w \geq e$ for any non-empty fiber G_w of $G \to W$. Then*

$$\dim \overline{\mathrm{pr}_W(G)} \leq \dim \overline{\mathrm{pr}_V(G)} - e.$$

Proof. We may assume that $G \neq \emptyset$. Write G as a finite union of irreducible locally closed subsets G_α [Har77, Exercise II.3.18(a)] (decomposing each locally closed piece into irreducible components, with the reduced structure). Any non-empty fiber of $G_\alpha \to \overline{\mathrm{pr}_V(G_\alpha)}$ is zero-dimensional, so

$$\dim G_\alpha \leq \dim \overline{\mathrm{pr}_V(G_\alpha)} \leq \dim \overline{\mathrm{pr}_V(G)}$$

by [Har77, Exercise II.3.22(b)].

Let W' be an irreducible component of $\overline{\mathrm{pr}_W(G)}$ with $\dim W' = \dim \overline{\mathrm{pr}_W(G)}$. A general point $w \in W'$ belongs to $\mathrm{pr}_W(G)$ [Har77, Exercises II.3.18(b) and II.3.19], it is outside $\overline{\mathrm{pr}_W(G_\alpha)}$ for any α with $W' \not\subset \overline{\mathrm{pr}_W(G_\alpha)}$, and

$$\dim (G_\alpha)_w = \dim G_\alpha - \dim W'$$

for any α with $W' \subset \overline{\mathrm{pr}_W(G_\alpha)}$ [Har77, Exercise II.3.22(e)], where $W' = \overline{\mathrm{pr}_W(G_\alpha)}$ since W' is an irreducible component of $\overline{\mathrm{pr}_W(G)}$.

Since $G_w \neq \emptyset$ and $\dim G_w \geq e$, there exists an index α such that $(G_\alpha)_w \neq \emptyset$ and $\dim (G_\alpha)_w \geq e$. Then $W' \subset \overline{\mathrm{pr}_W(G_\alpha)}$, and so $\dim W' \leq \dim G_\alpha - e$. Thus

$$\dim \overline{\mathrm{pr}_W(G)} = \dim W' \leq \dim G_\alpha - e \leq \dim \overline{\mathrm{pr}_V(G)} - e.$$

□

2.3. **The canonical ring.**

Theorem 2.10 ([SS20, Lemmas 2.2–2.4 and 2.7, Proposition 2.5, and Theorem 2.6])**.** *Let S be a numerical Godeaux surface, s,t a basis of R_2, and $y_0,\dots,y_3$ a basis of R_3. Let S_0 be the polynomial ring in variables $s,t,y_0,\dots,y_3$ with weights $(2,2,3,3,3,3)$, each variable acting on R as multiplication by the section denoted by that letter. Then the following hold:*

(1) R is generated as an algebra in degrees at most 5 and is a finite S_0-module.
(2) X is the normalization of $\operatorname{Proj}(S_0/\operatorname{ann}_{S_0} R)$.
(3) R has a minimal free S_0-resolution

$$0 \leftarrow R \leftarrow F_0 \xleftarrow{d_1} F_1 \xleftarrow{d_2} F_1^*(-17) \xleftarrow{d_1^t} F_0^*(-17) \leftarrow 0$$

with d_2 alternating and

$$F_0 = S_0 \oplus S_0(-4)^{\oplus 4} \oplus S_0(-5)^{\oplus 3}, \qquad F_1 = S_0(-6)^{\oplus 6} \oplus S_0(-7)^{\oplus 12} \oplus S_0(-8)^{\oplus 8}.$$

(4) Let $z_0,\dots,z_3 \in R_4$ and $w_0,w_1,w_2 \in R_5$ be the images of the basis elements of positive degree of F_0. Then R is the quotient of the polynomial ring $S_0[z_0,\dots,z_3,w_0,w_1,w_2]$ by the ideal generated by the 26 relations given by the columns of d_1 and by 28 polynomials, one for each product of two of the elements $z_0,\dots,z_3,w_0,w_1,w_2$, each equal to the product minus an S_0-linear combination of $1,z_0,\dots,z_3,w_0,w_1,w_2$.

[SS20, Section 2] works with a numerical Godeaux surface with no further condition (see also [SS20, Remark 2.8] for the resolution), so Theorem 2.10 holds whether or not the base scheme of $|2K_X|$ is reduced, and it is the common starting point of the later sections. Moreover, [SS20, Theorem 2.6] is taken from [Ste22, Theorem 1.5 and Corollary 5.6].

3. The bicanonical pencil

In this section, we prove that the bicanonical pencil of S has genus 3 or 4, and we determine the quotient of the canonical ring by the bicanonical sections. The surfaces of genus four are divided according to the local lengths of the base scheme of $|2K_X|$, and the surfaces of genus three are treated in Section 14.

Construction-Proposition 3.1. Let S be a numerical Godeaux surface, let Z be the fixed divisor of $|2K_S|$, and write $|2K_S| = |M| + Z$. Let $\beta : Y \to S$ be obtained by successively blowing up the base points of $|M|$, stopping as soon as the transformed pencil is base point free (the process stops, since the transform M_k of M after k blow-ups has no fixed divisor, so $0 \le M_k^2 = M^2 - \sum_{i\le k} m_i^2$ [Har77, Propositions V.3.2 and V.3.6], where $m_i \ge 1$ is the multiplicity of the transform of a general member of $|M|$ at the i-th center), and let $f : Y \to \mathbb{P}^1$ be the morphism defined by the transformed pencil. Let $m_1,\dots,m_r$ be the multiplicities of the transforms of a general member of $|M|$ at the centers of the blow-ups of β, and let g be the genus of the general fiber of f. Then the following hold.

(1) β is not an isomorphism, and the general fiber of f is a connected smooth curve.
(2) $\sum_i m_i^2 = M^2$, and $2g-2 = K_S\cdot M + \sum_i m_i = 2 + \sum_i m_i$.
(3) $g \in \{3,4\}$.

We call g the *genus of the bicanonical pencil* of S. We denote by $\mathcal{L}_3 \subset \mathcal{M}_{\mathrm{tf}}$ the locus of points $[X]$ with S torsion-free whose bicanonical pencil has genus 3.

Proof. By Theorem 2.4, a general member of $|M|$ is reduced and irreducible, $K_S\cdot M = 2$, and $M^2 \in \{0,2,4\}$, with $M^2 = 4$ if $Z = 0$.

Let E_i be the total transform on Y of the i-th exceptional curve of β. At the i-th blow-up, the transform of a general member of $|M|$ is the pullback of the previous transform minus m_i

times the new exceptional curve [Har77, Proposition V.3.6]. Thus a general fiber F of f is the strict transform of a general member of $|M|$, and

$$F \sim \beta^* M - \sum_i m_i E_i.$$

Since a general member of $|M|$ is reduced and irreducible, F is connected. Since the transformed pencil is base point free, F is smooth [Har77, Corollary III.10.9].

We have $E_i \cdot E_j = -\delta_{ij}$, $\beta^* D \cdot E_i = 0$, and $\beta^* D \cdot \beta^* D' = D \cdot D'$ for any divisors D, D' on S [Har77, Proposition V.3.2]. Thus

$$0 = F^2 = M^2 - \sum_i m_i^2.$$

By $K_Y = \beta^* K_S + \sum_i E_i$ [Har77, Proposition V.3.3] and adjunction [Har77, Proposition V.1.5], we have

$$2g - 2 = F \cdot (F + K_Y) = F \cdot K_Y = K_S \cdot M + \sum_i m_i = 2 + \sum_i m_i.$$

This proves (2).

If $M^2 = 0$, then (2) implies that $r = 0$ and $g = 2$, which contradicts [CP06, Theorem 5.1], by which the fibration induced by the bicanonical pencil of a numerical Godeaux surface has fibers of genus 3 or 4. Thus $M^2 \in \{2, 4\}$. By (2), $\sum_i m_i^2 > 0$, so $r \geq 1$ and β is not an isomorphism. This proves (1).

Since any m_i is a positive integer, (2) leaves the following possibilities. If $M^2 = 2$, then $(m_i) = (1, 1)$ and $g = 3$. If $M^2 = 4$, then either $(m_i) = (1, 1, 1, 1)$ and $g = 4$, or $(m_i) = (2)$ and $g = 3$. This proves (3). □

Modulo (s, t), any presentation of R by generators and relations reduces to $R/(s, t)$. The next result describes $R/(s, t)$ by the coordinate algebra of the base scheme of $|2K_X|$ and a non-degenerate functional on it. Thus the case of a reduced base scheme and the case of a non-reduced base scheme can be treated together.

Construction-Theorem 3.2. Let S be a numerical Godeaux surface, s, t a basis of R_2, and B the scheme-theoretic base locus of $|2K_X|$. Then the following hold.

(1) B is a zero-dimensional complete intersection of length 4 in X.
(2) For $\mathcal{A} := H^0(B, \mathcal{O}_B)$, a formal generator τ of degree one, and a fixed trivialization of $\mathcal{O}_B(K_X)$, under which $H^0(B, \mathcal{O}_B(nK_X)) = \mathcal{A}\tau^n$ for any $n \geq 0$, there exists a linear functional $\lambda : \mathcal{A} \to \mathbb{C}$, unique up to a non-zero scalar, whose pairing $(a, b) \to \lambda(ab)$ is non-degenerate, such that the restriction map is an isomorphism from the graded algebra $R/(s, t)$ onto the subalgebra
$$Q(\mathcal{A}, \lambda) := \mathbb{C} \oplus \mathcal{A}\tau^3 \oplus \mathcal{A}\tau^4 \oplus \operatorname{Ker}(\lambda)\tau^5 \oplus \bigoplus_{n \geq 6} \mathcal{A}\tau^n$$
of $\mathcal{A}[\tau]$.
(3) The restriction map $R_3 \to \mathcal{A}\tau^3$ is an isomorphism.
(4) For a unit $g \in \mathcal{A}$, replacing the generator τ by $g\tau$ replaces λ by the functional $a \to \lambda(g^5 a)$, and any non-degenerate functional on $\mathcal{A}$ has the form $a \to \lambda(g^5 a)$ for a unit $g \in \mathcal{A}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1).

The singularities of X are Du Val [KM98, Theorem 4.20], so K_X is Cartier. We let $R_n := 0$ for $n < 0$. By Lemma 2.2(3) and the ampleness of K_X, we have $R_n = H^0(X, nK_X)$ for any integer n. By Theorem 2.4, s and t have no common curve on X. Since X is a normal surface, X is Cohen–Macaulay [Har77, Theorem II.8.22A]. Thus the local equations of s and t form a regular

sequence at any point of B. In particular, B is a zero-dimensional complete intersection in X, and for any integer n we have the Koszul sequence

$$0 \to \mathcal{O}_X((n-4)K_X) \to \mathcal{O}_X((n-2)K_X)^{\oplus 2} \to \mathcal{I}_B(nK_X) \to 0, \tag{3.1}$$

where the maps are $c \to (tc, -sc)$ and $(a, b) \to sa + tb$. By Lemma 2.2(3) and Serre duality on S [Har77, Corollary III.7.7], we have $\chi(\mathcal{O}_X(mK_X)) = \chi(\mathcal{O}_X((1-m)K_X))$ for any integer m. Thus (3.1) for $n = 0$ and Lemma 2.2 imply that

$$\chi(\mathcal{O}_B) = \chi(\mathcal{O}_X) - 2\chi(\mathcal{O}_X(3K_X)) + \chi(\mathcal{O}_X(5K_X)) = 1 - 8 + 11 = 4.$$

This proves (1).

Step 2. In this step we prove (3) and construct λ such that the restriction map is an isomorphism from $R/(s,t)$ onto $Q(\mathcal{A}, \lambda)$.

Since $H^1(X, (n-4)K_X) = 0$ by Lemma 2.2(3), the sequence (3.1) implies that

$$0 \to R_{n-4} \to R_{n-2}^{\oplus 2} \to H^0(X, \mathcal{I}_B(nK_X)) \to 0$$

is exact for any integer n. We obtain three consequences. First, the kernel of the restriction map $R_n \to H^0(B, \mathcal{O}_B(nK_X))$ is $sR_{n-2} + tR_{n-2} = (s,t)_n$. Second, s, t is a regular sequence on R: the element s is a non-zero-divisor since S is integral, and if $ta = sb$ for some $a, b \in R$, then $(-b, a) = (tc, -sc)$ for some $c \in R$, so $a \in (s)$. Third, for any integer n,

$$\dim(R/(s,t))_n = \dim R_n - 2\dim R_{n-2} + \dim R_{n-4}.$$

We have $\dim R_0 = 1$, $\dim R_1 = p_g(S) = 0$, and $\dim R_n = 1 + n(n-1)/2$ for $n \geq 2$ by Lemma 2.2(2). Thus $\dim(R/(s,t))_n$ equals $1, 0, 0$ for $n = 0, 1, 2$, equals $4, 4, 3$ for $n = 3, 4, 5$, and equals 4 for any $n \geq 6$.

The restriction maps commute with multiplication, so they embed $R/(s,t)$ into $\mathcal{A}[\tau]$ as a graded subalgebra. By (1), $\dim \mathcal{A}\tau^n = 4$. Comparing dimensions, $(R/(s,t))_n \to \mathcal{A}\tau^n$ is an isomorphism for $n = 3, 4$ and for any $n \geq 6$, and the image of $(R/(s,t))_5$ is $H\tau^5$, where H is a subspace of $\mathcal{A}$ with $\dim H = 3$. Since $R_1 = 0$, we have $(s,t)_3 = 0$. This proves (3).

Let $\lambda : \mathcal{A} \to \mathbb{C}$ be a linear functional with $\mathrm{Ker}(\lambda) = H$. Then the restriction map is an isomorphism from $R/(s,t)$ onto $Q(\mathcal{A}, \lambda)$. Any linear functional λ' such that the image of the restriction map is $Q(\mathcal{A}, \lambda')$ satisfies $\mathrm{Ker}(\lambda') = H$, so λ' is a non-zero scalar multiple of λ.

Step 3. In this step we prove that the pairing $(a, b) \to \lambda(ab)$ is non-degenerate, which is the rest of (2).

By (3), there exists $y \in R_3$ whose restriction is $1 \cdot \tau^3$. Since τ^3 is a non-zero-divisor in $\mathcal{A}[\tau]$, y is a non-zero-divisor on $R/(s,t)$. Thus s, t, y is a regular sequence on R. By **Step 2**, $R/(s,t,y)$ is isomorphic to the quotient of $Q(\mathcal{A}, \lambda)$ by

$$\tau^3 Q(\mathcal{A}, \lambda) = \mathbb{C}\tau^3 \oplus \mathcal{A}\tau^6 \oplus \mathcal{A}\tau^7 \oplus \mathrm{Ker}(\lambda)\tau^8 \oplus \bigoplus_{n \geq 9} \mathcal{A}\tau^n.$$

Thus

$$\big(R/(s,t,y)\big)_n \cong \begin{cases} \mathbb{C} & \text{if } n = 0, \\ \mathcal{A}/\mathbb{C} \cdot 1 & \text{if } n = 3, \\ \mathcal{A} & \text{if } n = 4, \\ \mathrm{Ker}(\lambda) & \text{if } n = 5, \\ \mathcal{A}/\mathrm{Ker}(\lambda) & \text{if } n = 8, \\ 0 & \text{otherwise.} \end{cases}$$

In particular, $\dim_{\mathbb{C}} R/(s,t,y) = 12$, and $R/(s,t,y)$ is an Artin local ring whose maximal ideal consists of the elements without a term of degree 0.

The ring R is Gorenstein [SS20, Section 2, p. 7]. Localizing at the ideal of elements of positive degree and taking the quotient by the regular sequence s, t, y, we obtain that $R/(s,t,y)$ is Gorenstein [BH98, Proposition 3.1.19]. Thus the socle of $R/(s,t,y)$ is one-dimensional [BH98,

Theorem 3.2.10]. The piece of degree 8 is one-dimensional, and it is annihilated by the maximal ideal since all pieces of degree at least 9 are zero. Thus the socle of $R/(s,t,y)$ is the piece of degree 8, which is isomorphic to $\mathcal{A}/\operatorname{Ker}(\lambda)$.

Let $a \in \mathcal{A}$ be a non-zero element. Then $a\tau^4$ has non-zero class in $R/(s,t,y)$, and the ideal generated by the class of $a\tau^4$ is non-zero, so the ideal contains the socle. Taking the term of degree 4 of a multiplier, we obtain $b \in \mathcal{A}$ with $ab\tau^8 \notin \operatorname{Ker}(\lambda)\tau^8$, that is, $\lambda(ab) \neq 0$. Thus the pairing

$$\mathcal{A} \times \mathcal{A} \to \mathcal{A}/\operatorname{Ker}(\lambda)$$

is perfect.

Step 4. In this step we prove (4).

Let $g \in \mathcal{A}$ be a unit. For any $a \in \mathcal{A}$, we have

$$a\tau^5 = ag^{-5}(g\tau)^5.$$

Thus an element $b(g\tau)^5$ with $b \in \mathcal{A}$ belongs to $\operatorname{Ker}(\lambda)\tau^5$ if and only if $\lambda(g^5 b) = 0$. Moreover, $\mathcal{A}(g\tau)^n = \mathcal{A}\tau^n$ for any $n \geq 0$. This proves the first claim of (4).

Let λ_1 be a non-degenerate functional on $\mathcal{A}$. Since $\dim \mathcal{A} = 4$, the non-degeneracy of λ implies that there exists a unique $w \in \mathcal{A}$ with $\lambda_1(a) = \lambda(wa)$ for any $a \in \mathcal{A}$. If w is not a unit, then $wc = 0$ for some non-zero $c \in \mathcal{A}$, so $\lambda_1(cb) = \lambda(wcb) = 0$ for any $b \in \mathcal{A}$, which contradicts the non-degeneracy of λ_1. Thus w is a unit. On any local factor of $\mathcal{A}$, we may write $w = c_0(1+e)$, where $c_0 \in \mathbb{C}$ is non-zero and e is nilpotent. A fifth root of c_0 and the finite binomial expansion of $(1+e)^{1/5}$ imply that there exists a unit $g \in \mathcal{A}$ with $g^5 = w$. Thus $\lambda_1(a) = \lambda(g^5 a)$ for any $a \in \mathcal{A}$. This proves the second claim of (4). □

Construction-Proposition 3.3. Notation and conditions as in Construction-Proposition 3.1. Let B be the scheme-theoretic base locus of $|2K_X|$. Assume that the bicanonical pencil of S has genus 4. Then the following hold.

(1) $Z = 0$, the morphism β consists of exactly four blow-ups, with $m_i = 1$ for any i, and the general member of $|2K_S|$ is a smooth nonhyperelliptic curve.
(2) B is supported in the smooth locus of X, it is of length 4, and it is *curvilinear*: for any point $x \in \operatorname{Supp} B$, the local algebra of B at x is isomorphic to $\mathbb{C}[z]/(z^n)$ for some integer $n \geq 1$.
(3) The integers n of (2) form one of the partitions

$$(1,1,1,1), \quad (2,1,1), \quad (2,2), \quad (3,1), \quad (4).$$

For such a partition $\mathbf{p}$, we denote by $\mathcal{M}_{\mathbf{p}} \subset \mathcal{M}_{\mathrm{tf}}$ the locus of points $[X]$ with S torsion-free whose bicanonical pencil has genus 4 and whose scheme B has partition $\mathbf{p}$.

Proof. By Construction-Proposition 3.1(2) and $g = 4$, we have $\sum_i m_i = 4$ and $\sum_i m_i^2 = M^2$. By Theorem 2.4, $M^2 \leq 4$. Since any m_i is a positive integer, we have

$$4 = \sum_i m_i \leq \sum_i m_i^2 = M^2 \leq 4.$$

Thus $M^2 = 4$, $r = 4$, and $m_i = 1$ for any i. By Theorem 2.4, $M^2 = 4$ implies that $Z = 0$.

Let C be a general member of $|2K_S| = |M|$. Away from the base points of $|M|$, the morphism β is an isomorphism and C is a general fiber of f, which is smooth by [Har77, Corollary III.10.9]. At a base point of $|M|$ on S, the multiplicity of C is one of the m_i, which equals 1, so C is smooth there. Hence the strict transform of C is isomorphic to C and is a general fiber of f. In particular, C is a connected smooth curve of genus 4.

By adjunction, $\omega_C = \mathcal{O}_C(3K_S)$, and we have the exact sequence

$$0 \to \mathcal{O}_S(K_S) \to \mathcal{O}_S(3K_S) \to \omega_C \to 0.$$

Since $p_g(S) = 0$ and $H^1(S, K_S) = 0$ by Lemma 2.2(3), the restriction map

$$H^0(S, 3K_S) \to H^0(C, \omega_C)$$

is an isomorphism. Thus the canonical map of C is the restriction to C of the tricanonical map of S. The tricanonical map of S is birational onto the image [SS20, Theorem 1.3], so it is an isomorphism onto the image on a dense open subset V of S. Since C is general, $C \cap V \neq \emptyset$. Thus the canonical map of C is birational onto the image. Since the canonical morphism of a hyperelliptic curve has degree 2 [Har77, Proposition IV.5.3], C is nonhyperelliptic. This proves (1).

By Construction-Theorem 3.2(1), B is of length 4. Let s, t be a basis of R_2, and let $x \in \operatorname{Supp} B$. If $x \in \operatorname{Sing} X$, then the pullbacks to S of s and t vanish on any exceptional curve over x, so $|2K_S|$ has a fixed divisor, which contradicts $Z = 0$. Thus $x \notin \operatorname{Sing} X$. Therefore $S \to X$ is an isomorphism over a neighborhood of x, and x is a base point of $|M|$. By (1), a general member of $|M|$ has multiplicity 1 at x. Possibly replacing s, t with another basis of R_2, we may assume that a local equation of s at x has non-zero differential. We choose analytic local coordinates u, v at x such that u is a local equation of s. Since B is zero-dimensional, the restriction to $\{u = 0\}$ of a local equation of t is a unit times v^n for some integer $n \geq 1$. Thus the local algebra of B at x is

$$\mathbb{C}\{u, v\}/(u, v^n) \cong \mathbb{C}[z]/(z^n).$$

This proves (2).

By (2), the integers n are positive and have sum 4, so they form a partition of 4. The five partitions in (3) are all the partitions of 4. This proves (3). □

Thus a surface whose scheme B is not curvilinear or satisfies $\operatorname{Supp} B \cap \operatorname{Sing} X \neq \emptyset$ has bicanonical pencil of genus 3.

4. Presentations of the canonical ring

In this section, we fix the language of presentations. The upper bounds for the dimensions of loci of $\mathcal{M}$ in later sections count lower blocks, and this section explains why counting lower blocks is enough.

The set-up below is [SS20, Notation 3.1].

Set-up 4.1. Let S, S_0, s, t, $y_0, \dots, y_3$, $z_0, \dots, z_3$, w_0, w_1, w_2, d_1, and d_2 be as follows.

(1) S is a numerical Godeaux surface.
(2) $S_0 := \mathbb{C}[s, t, y_0, \dots, y_3]$ is the polynomial ring with weights $(2, 2, 3, 3, 3, 3)$, and R is a graded S_0-module through a graded homomorphism $S_0 \to R$ sending s, t to a basis of R_2 and $y_0, \dots, y_3$ to a basis of R_3.
(3) $1, z_0, \dots, z_3, w_0, w_1, w_2$, of degrees $0, 4, 4, 4, 4, 5, 5, 5$, are the generators of R as an S_0-module.
(4) d_1, d_2 are the differentials of a resolution of R of the form in Theorem 2.10, in the generators above.
(5) The lower seven rows d_1' of d_1 and the matrix d_2 are

$$d_1' = \begin{pmatrix} A & B_1 & C \\ 0 & U & B_2 \end{pmatrix}, \qquad d_2 = \begin{pmatrix} O & N & B_3 \\ -N^t & B_4 & P \\ -B_3^t & -P^t & 0 \end{pmatrix},$$

with row blocks of d_1' of sizes $4, 3$ and column blocks of sizes $6, 12, 8$.
(6) The blocks $B_1, \dots, B_4$ are linear in $y_0, \dots, y_3$, the blocks A, U, P are linear in s, t, the blocks C, N are quadratic in s, t, and the entries of O are bilinear in (s, t) and $(y_0, \dots, y_3)$.
(7) The first six columns of d_1 are the relations

$$r_k(y) + \sum_{j=0}^{3} A_{jk}(s, t) z_j + v_k(s, t) = 0 \qquad (1 \leq k \leq 6),$$

where the r_k are quadrics in $y_0, \dots, y_3$ and the v_k are binary cubics.

We call (d_1, d_2) a *presentation* of S, and we call (d_1', d_2) the *lower blocks* of the presentation. We call any pair of matrices (d_1', d_2) with the blocks of (5) and the degrees of (6), with d_2 alternating and not necessarily coming from a surface, *lower blocks of the shape of* Set-up 4.1. For $u = [u_0 : u_1] \in \mathbb{P}^1$, we denote by $C_u \in |2K_X|$ the member defined by $u_1 s - u_0 t = 0$, by $A(u)$ and $v(u)$ the values of A and $v = (v_1, \dots, v_6)$ at $(s, t) = (u_0, u_1)$, and by $H_u \subset \mathbb{C}^6$ the kernel of the 5×6 matrix with rows $A(u)$ and $v(u)$. We say that the k-th column of A is a *zero column at* u if the k-th column of $A(u)$ is zero. For a 6×8 matrix Γ of binary linear forms, we call the base change $(d_1, d_2) \to (d_1 g^{-1}, g d_2 g^t)$, where

$$g = \begin{pmatrix} I_6 & 0 & \Gamma \\ 0 & I_{12} & 0 \\ 0 & 0 & I_8 \end{pmatrix},$$

the *gauge* of Γ.

Lemma 4.2. *Notation and conditions as in Set-up 4.1. Let Γ be a 6×8 matrix of binary linear forms, and let g be the matrix of the gauge of Γ. Then $(d_1 g^{-1}, g d_2 g^t)$ is a presentation of S whose first differential has the first six columns of d_1, with the blocks*

$$(O + B_3 \Gamma^t - \Gamma B_3^t,\ N - \Gamma P^t,\ C - A\Gamma)$$

in the places of (O, N, C), and with the other blocks equal to the blocks of (d_1, d_2). In particular, A, r_k, v_k, and the canonical model do not change.

Proof. The gauge is the base change of [SS20, Section 3, p. 17]. Since g^{-1} is g with Γ replaced by $-\Gamma$, matrix multiplication shows that $d_1 g^{-1}$ and $g d_2 g^t$ have the blocks above, and that the rows $(0, U, B_2)$ and the first six columns of d_1 do not change. Since g is a graded invertible matrix, the new matrices are the differentials of a resolution of the S_0-module R. □

The next lemma translates the geometry of C_u into rank conditions on $A(u)$ and $v(u)$. Sections 9–12 use the lemma to restrict the zero columns and the rank of A. The torsion-free hypothesis enters through [SS20, Theorem 1.7] and Theorem 2.5.

Lemma 4.3. *Notation and conditions as in Set-up 4.1. Then for any $u \in \mathbb{P}^1$, the map $c \to \sum_k c_k r_k$ is an isomorphism from H_u onto the kernel of*

$$\operatorname{Sym}^2 R_3 \to H^0(C_u, \omega_{C_u}^{\otimes 2}).$$

If moreover S is torsion-free, then for any $u \in \mathbb{P}^1$, the kernel is the space of quadrics vanishing on the scheme-theoretic image of C_u under the tricanonical map, $\dim H_u \in \{1, 3\}$, and the following hold:

1. *For a general u, $\dim H_u = 1$. In particular, A has generic rank four.*
2. *If $\operatorname{rank} A(u) \le 3$, then $\dim H_u = 3$, the image of C_u is a twisted cubic, and there exists no two-dimensional subspace $W \subset \operatorname{Ker} A(u)$ such that the quadrics $\sum_k c_k r_k$ with $c \in W$ have a common linear factor.*
3. *If $c \in H_u$ is non-zero and $\sum_k c_k r_k$ is a product of two linear forms, then $\dim H_u = 1$ and C_u is singular at any point whose image is a singular point of the quadric $\sum_k c_k r_k$.*
4. *For any $u \in \mathbb{P}^1$, the restriction $R_3 \to H^0(C_u, \omega_{C_u})$ is an isomorphism, and $\operatorname{rank} A(u) \ge 2$.*

Proof. Possibly changing the basis s, t, we may assume that $u = [0 : 1]$, that is, $C_u = \{s = 0\}$. Then $A(u)$ and $v(u)$ are the coefficients of t and t^3 in A and v. A change of the basis multiplies $A(u)$ and $v(u)$ by non-zero scalars, so H_u does not depend on the basis.

By adjunction [Har77, Theorem III.7.11] (Du Val singularities are hypersurface singularities [KM98, Theorem 4.20], so X and the Cartier divisor C_u are local complete intersections in projective space), $\omega_{C_u} \cong \mathcal{O}_{C_u}(3K_X)$. For any integer n, the exact sequence

$$0 \to \mathcal{O}_X((n-2)K_X) \xrightarrow{s} \mathcal{O}_X(nK_X) \to \mathcal{O}_{C_u}(nK_X) \to 0$$

and $H^1(X,(n-2)K_X)=0$ (Lemma 2.2) imply that

$$H^0(C_u,\mathcal{O}_{C_u}(nK_X))=R_n/sR_{n-2}.$$

For $n=3$, we have $\omega_{C_u}\cong\mathcal{O}_{C_u}(3K_X)$, and $H^0(X,K_X)=H^1(X,K_X)=0$ (Lemma 2.2), so $R_3\to H^0(C_u,\omega_{C_u})$ is an isomorphism, which is the first part of (4). For $n=6$, we have $H^0(C_u,\omega_{C_u}^{\otimes 2})=R_6/sR_4$, and by Lemma 2.2,

$$\dim R_6/sR_4=16-7=9.$$

Let $Q:=(F_0)_6/s(F_0)_4$, where $(F_0)_n$ is the degree n piece of F_0. Then Q has the basis consisting of the ten quadratic monomials in $y_0,\dots,y_3$, the elements $tz_0,\dots,tz_3$, and t^3. Thus $\dim Q=15$, and the surjection $Q\to R_6/sR_4$ has a kernel of dimension 6. The kernel of $(F_0)_6\to R_6$ is spanned by the first six columns of d_1, so the kernel of $Q\to R_6/sR_4$ is spanned by the images

$$\rho_k=r_k(y)+t\sum_{j=0}^{3}A_{jk}(u)z_j+t^3v_k(u)\qquad(1\le k\le 6).$$

Hence $\rho_1,\dots,\rho_6$ are linearly independent. For $c\in\mathbb{C}^6$, we have

$$\sum_k c_k\rho_k=\sum_k c_kr_k(y)+t\sum_{j=0}^{3}\Big(\sum_k A_{jk}(u)c_k\Big)z_j+t^3\sum_k v_k(u)c_k.$$

Thus a quadric q in $y_0,\dots,y_3$ vanishes in R_6/sR_4 if and only if $q=\sum_k c_kr_k$ for some $c\in H_u$, and for $c\in H_u$, the equality $\sum_k c_kr_k=0$ implies that $\sum_k c_k\rho_k=0$, hence $c=0$. Therefore, the map $c\to\sum_k c_kr_k$ is an isomorphism from H_u onto the kernel of

$$\operatorname{Sym}^2 R_3\to H^0(C_u,\omega_{C_u}^{\otimes 2}).$$

Now assume that S is torsion-free, so that the third case of Theorem 2.5 does not occur. Let $Y\subset\mathbb{P}^3$ be the scheme-theoretic image of C_u under the tricanonical map (the restriction to C_u is a morphism, since $|3K_X|$ has no base point by [SS20, Theorem 1.7]). By the definition of Y, the kernel above is the space of quadrics vanishing on Y.

If C_u is in the first case of Theorem 2.5, then C_u is isomorphic to $Y=\{q_0=h=0\}$ for a quadric q_0 and a cubic h. Thus the quadrics vanishing on Y are the multiples of q_0, and $\dim H_u=1$. If C_u is in the second case, let $f:C_u\to\mathbb{P}^1$ be the finite morphism of degree 2. Since $\deg\omega_{C_u}=3K_X\cdot 2K_X=6$, we have $p_a(C_u)=4$, so by [CFHR99, Lemma 3.5], C_u is a divisor of degree 10 in $\mathbb{P}(1,1,5)$ not passing through the vertex, f is the projection, and

$$f_*\mathcal{O}_{C_u}\cong\mathcal{O}_{\mathbb{P}^1}\oplus\mathcal{O}_{\mathbb{P}^1}(-5).$$

By adjunction, $\omega_{C_u}\cong f^*\mathcal{O}_{\mathbb{P}^1}(3)$, so

$$H^0(C_u,\omega_{C_u})=f^*H^0(\mathbb{P}^1,\mathcal{O}_{\mathbb{P}^1}(3)).$$

Thus the tricanonical map is the composition of f and the embedding of $\mathbb{P}^1$ as a twisted cubic. Since $\mathcal{O}_{\mathbb{P}^1}\to f_*\mathcal{O}_{C_u}$ is injective, Y is the twisted cubic, and, since quadrics restrict onto $H^0(\mathbb{P}^1,\mathcal{O}(6))$ (the multiplication map from Sym^2 of the binary cubics to the binary sextics is surjective),

$$\dim H_u=h^0(\mathbb{P}^3,\mathcal{O}(2))-h^0(\mathbb{P}^1,\mathcal{O}(6))=10-7=3.$$

Thus $\dim H_u\in\{1,3\}$. Moreover, if $\dim H_u=3$, then Y is integral and spans $\mathbb{P}^3$, so no non-zero product of two linear forms vanishes on Y.

(1) By Theorem 2.5, the first case holds for a general u, so $\dim H_u=1$. Then the 5×6 matrix with rows $A(u)$ and $v(u)$ has rank 5, so $\operatorname{rank}A(u)=4$.

(2) If $\operatorname{rank}A(u)\le 3$, then the 5×6 matrix has rank at most 4, so $\dim H_u\ge 2$, hence $\dim H_u=3$ and Y is a twisted cubic. Let $W\subset\operatorname{Ker}A(u)$ be a two-dimensional subspace, and suppose that the quadrics $\sum_k c_kr_k$ with $c\in W$ have a common linear factor. Since H_u is the kernel of $v(u)$ on $\operatorname{Ker}A(u)$, W contains a non-zero element $c\in H_u$. Then $\sum_k c_kr_k$ is non-zero, is a product of two linear forms, and vanishes on Y, which is not possible.

(3) Let $c \in H_u$ be a non-zero element such that $q_0 := \sum_k c_k r_k$ is a product of two linear forms. Since $c \to \sum_k c_k r_k$ is an isomorphism from H_u onto the space of quadrics vanishing on Y, $q_0 \neq 0$ and q_0 vanishes on Y, so $\dim H_u \neq 3$. Thus $\dim H_u = 1$, and C_u is isomorphic to $Y = \{q_0 = h = 0\}$ for a cubic h. Let p be a singular point of $\{q_0 = 0\}$ which is the image of a point of C_u. The Zariski tangent space of Y at p is the Zariski tangent space of $\{h = 0\}$ at p, of dimension at least 2. Hence C_u is singular at the point over p.

(4) The first part is proved above. Since $\dim H_u \leq 3$, the 5×6 matrix has rank at least 3, so $\operatorname{rank} A(u) \geq 2$. □

Lemma 4.4. *Let S and S' be numerical Godeaux surfaces with presentations in the sense of Set-up 4.1, both over the ring S_0, whose lower blocks coincide and whose quadrics $r_1, \dots, r_6$ coincide. Then the first differentials of the presentations differ by an invertible triangular change of generators, and the canonical models of S and S' are isomorphic.*

Proof. Let (d_1, d_2) be the presentation of S, and let $(\tilde{d}_1, d_2)$ be the presentation of S'. Let R' be the canonical ring of S', and let X' be the canonical model of S'. Let ρ_0 be the first row of d_1, let $\rho_1, \dots, \rho_7$ be the lower seven rows of d_1, and let $\tilde{\rho}_0$ be the first row of $\tilde{d}_1$.

We first show that some r_k is non-zero. By Lemma 2.2,

$$\dim \operatorname{Sym}^2 R_3 = 10 > 9 = \dim R_6/sR_4,$$

so there exist a non-zero quadric q in $y_0, \dots, y_3$ and an element $f \in R_4$ with $q(y) = sf$ in R. Since R_4 is spanned by $s^2, st, t^2, z_0, \dots, z_3$, the element $q - sf$ is a relation of degree 6 in F_0. Thus $q - sf$ is a constant linear combination of the first six columns of d_1 with coefficients c_k. Comparing the parts which are quadratic in $y_0, \dots, y_3$, we obtain $\sum_k c_k r_k = q \neq 0$.

Since $\tilde{d}_1 d_2 = 0$ and d_2 is alternating, $d_2 \tilde{\rho}_0^t = 0$. By the exactness of the resolution of R at $F_1^*(-17)$, we have

$$\tilde{\rho}_0 = \sum_{i=0}^{7} a_i \rho_i$$

for homogeneous elements $a_i \in S_0$, where $\deg a_0 = 0$, $\deg a_i = 4$ for $1 \leq i \leq 4$, and $\deg a_i = 5$ for $5 \leq i \leq 7$. Any homogeneous element of S_0 of degree 4 or 5 is in the ideal (s, t). The first six entries of ρ_0 and of $\tilde{\rho}_0$ are congruent to $r_1, \dots, r_6$ modulo (s, t), so $r_k \equiv a_0 r_k$ modulo (s, t) for any k. Since some r_k is non-zero, $a_0 = 1$. Thus

$$\tilde{d}_1 = \begin{pmatrix} 1 & a_1 & \cdots & a_7 \\ 0 & & I_7 & \end{pmatrix} d_1,$$

which is the first claim.

Hence R and R' are isomorphic as graded S_0-modules, and $\operatorname{ann}_{S_0} R = \operatorname{ann}_{S_0} R'$. By Theorem 2.10, both X and X' are the normalization of $\operatorname{Proj}(S_0/\operatorname{ann}_{S_0} R)$, so $X \cong X'$. □

Thus, with the quadrics fixed, the dimension of the set of points of $\mathcal{M}$ of numerical Godeaux surfaces with such presentations is at most the dimension of the set of their lower blocks, which is contained in the set of solutions of $d_1' d_2 = 0$. For fixed A, U, P and $B_1, \dots, B_4$, the equations $d_1' d_2 = 0$ are linear in C, N, O. The next lemma reduces upper bounds on $\mathcal{M}$ to the dimension of sets of lower blocks with fixed quadrics, and so to linear algebra.

Lemma 4.5. *The following hold.*

1. *Let T be a reduced scheme of finite type. A flat family over T of canonical models of numerical Godeaux surfaces induces a morphism $T \to \mathcal{M}$.*
2. *Let S_0 be the polynomial ring of Set-up 4.1, fix quadrics $r_1, \dots, r_6$ in $y_0, \dots, y_3$, and let $\mathcal{L}$ be an affine subspace of the space of lower blocks of the shape of Set-up 4.1. Let $\Xi_{\mathcal{L}} \subset \mathcal{L} \times \mathcal{M}$ be the set of pairs $(\ell, [X])$ for which ℓ is the lower blocks of a presentation with the quadrics $r_1, \dots, r_6$ of a numerical Godeaux surface with canonical model X. Then $\Xi_{\mathcal{L}}$ is constructible, and for any $\ell \in \mathcal{L}$ there exists at most one point $[X]$ with*

$(\ell, [X]) \in \Xi_{\mathcal{L}}$. *In particular, if* $V \subset \mathcal{L}$ *is a constructible subset and* $\mathcal{M}(V) \subset \mathcal{M}$ *is the image of* $\Xi_{\mathcal{L}} \cap (V \times \mathcal{M})$, *then* $\dim \overline{\mathcal{M}(V)} \le \dim V$.

Proof. (1) By [Cat13, Theorem 3.11], Gieseker's moduli space of canonical models with $K^2 = 1$ and $\chi = 1$ is the coarse moduli space for the functor of families of canonical models, so a flat family over T of canonical models of numerical Godeaux surfaces induces a morphism from T to Gieseker's moduli space. Since $\mathcal{M}$ is the open subscheme of Gieseker's moduli space defined by $q = 0$, we obtain a morphism $T \to \mathcal{M}$.

(2) The proof of (2) is long, so we divide it into several steps.

Step 1. In this step we define the coefficient space $\mathcal{P}$.

Let $\mathcal{P}$ be the affine scheme of finite type whose coordinates are the coordinates of $\mathcal{L}$ and the following coefficients: the coefficients of the first row of d_1, whose first six entries are $r_k + v_k$ for binary cubics v_k; the coefficients of the 28 products of two of the elements $z_0, \dots, z_3, w_0, w_1, w_2$, each written as an S_0-linear combination of $1, z_0, \dots, z_3, w_0, w_1, w_2$ [SS20, Lemma 2.7]; and the coefficients of finitely many elements of F_1 whose images under d_1 are the products of the columns of d_1 with the generators and the associators of the generators, for the multiplication on F_0 defined by the 28 products. The equations of $\mathcal{P}$ are $d_1 d_2 = 0$ and the polynomial equations which say that the images under d_1 are as described.

A point $x \in \mathcal{P}$ determines matrices $d_{1,x}, d_{2,x}$, the complex

$$\mathbb{F}_x : \ 0 \leftarrow F_0 \xleftarrow{d_{1,x}} F_1 \xleftarrow{d_{2,x}} F_1^*(-17) \xleftarrow{d_{1,x}^t} F_0^*(-17) \leftarrow 0,$$

and a graded commutative associative S_0-algebra $R'_x := \operatorname{coker} d_{1,x}$ with unit 1. By Theorem 2.10(4), a presentation with the quadrics $r_1, \dots, r_6$ and lower blocks in $\mathcal{L}$ of a numerical Godeaux surface S is the complex $\mathbb{F}_x$ of a point $x \in \mathcal{P}$ with $R'_x = R$ as S_0-algebras.

Step 2. In this step we stratify $\mathcal{P}_{\mathrm{red}}$ so that the complex $\mathbb{F}_x$ behaves well in all degrees at once.

By generic freeness [Mat89, Theorem 24.1] and noetherian induction, $\mathcal{P}_{\mathrm{red}}$ is a finite disjoint union of locally closed integral affine subschemes W_i such that, over $\mathcal{O}(W_i)[s, t, y_0, \dots, y_3]$, the kernels, images, and homology modules of the universal complex are free $\mathcal{O}(W_i)$-modules. Since the modules are graded, their graded pieces are finitely generated projective $\mathcal{O}(W_i)$-modules. Thus the formation of the kernels, images, and homology modules commutes with base change to any point of W_i (kernels, images, and homology are flat over $\mathcal{O}(W_i)$, so the short exact sequences relating them stay exact after tensoring with $k(x)$). In particular, the set of points x such that $\mathbb{F}_x$ is exact is the union of some of the W_i. We consider only those W_i from now on.

Let R' be the cokernel of the universal first differential over W_i. Then the graded pieces of R' are locally free over W_i, and the fiber of R' at x is R'_x. By Theorem 2.10(3), the Hilbert series of R'_x is the Hilbert series of the canonical ring of a numerical Godeaux surface. Since F_1 has no generators of degree less than 6, we have $R'_{x,0} = \mathbb{C}$ and $R'_{x,1} = 0$, and s, t and $y_0, \dots, y_3$ are bases of $R'_{x,2}$ and $R'_{x,3}$. Moreover, $\operatorname{Proj} R' \to W_i$ is a proper flat morphism with fibers $X_x := \operatorname{Proj} R'_x$ (the homogeneous localizations of R' are direct summands of localizations of the flat module R', and Proj commutes with base change).

Step 3. In this step we show that X_x is the canonical model of a numerical Godeaux surface for x in an open subset of W_i, and that $\mathbb{F}_x$ is a presentation of the surface.

By the Hilbert series, $\dim R'_x = 3$. Since R'_x has a free resolution of length 3 over S_0, R'_x is Cohen–Macaulay (by the Auslander–Buchsbaum formula [BH98, Theorem 1.3.3], since $\dim R'_x = 3$). Since $d_{2,x}$ is alternating, the complex $\mathbb{F}_x$ is self-dual, so $\operatorname{Ext}^3_{S_0}(R'_x, S_0(-17)) \cong R'_x$. The sum of the weights of S_0 is 16, so the canonical module of S_0 is $S_0(-16)$ [BH98, Example 3.6.10], and so the canonical module $\operatorname{Ext}^3_{S_0}(R'_x, S_0(-16))$ of R'_x [BH98, Proposition 3.6.12(b)] is isomorphic to $R'_x(1)$ as a graded S_0-module.

Let $W_i^\circ \subset W_i$ be the set of the points x such that X_x is normal with Du Val singularities, and such that at any point of X_x, some homogeneous elements of R'_x of coprime degrees do not vanish. The first condition is open by [Gro66, Théorème 12.2.4(iv)], [Elk78, Théorème 4],

and [Cat13, p. 184] (a relative Veronese embedding of $\operatorname{Proj} R'$ over an affine cover of W_i pulls the open set back). Since the generators of R'_x as an algebra have degrees $2,3,4,5$, the second condition fails at x if and only if x is in the image of one of the closed subsets of $\operatorname{Proj} R'$ defined by the vanishing of the elements

$$\{y_0,\dots,y_3,w_0,w_1,w_2\},\quad \{s,t,z_0,\dots,z_3,w_0,w_1,w_2\},\quad \{s,t,y_0,\dots,y_3,z_0,\dots,z_3\}.$$

Since $\operatorname{Proj} R' \to W_i$ is proper, W_i° is open in W_i.

Let $x \in W_i^\circ$. By the second condition, the homogeneous localization of R'_x at any point of X_x contains a unit of degree 1. Thus $\mathcal{O}_{X_x}(1) := (R'_x(1))^\sim$ is invertible and ample (a Veronese subring of R'_x is generated in degree one, so a power of $\mathcal{O}_{X_x}(1)$ is very ample), and $(R'_x(n))^\sim \cong \mathcal{O}_{X_x}(1)^{\otimes n}$ for any integer n. Let $\mathfrak{m}$ be the homogeneous maximal ideal of R'_x, and let $D_x := \operatorname{Spec} R'_x \backslash \{\mathfrak{m}\}$. The morphism $D_x \to X_x$ is affine (an affine morphism preserves cohomology [Har77, Exercise III.8.2]), and the direct image of $\mathcal{O}_{D_x}$ is $\bigoplus_{n\in\mathbb{Z}} \mathcal{O}_{X_x}(1)^{\otimes n}$, so

$$H^j(D_x,\mathcal{O}_{D_x}) = \bigoplus_{n\in\mathbb{Z}} H^j(X_x,\mathcal{O}_{X_x}(1)^{\otimes n}).$$

Since $\operatorname{depth} R'_x = 3$, [Har77, Exercise III.2.3(e), Exercise III.3.3(b), and Theorem III.3.5] and [Har77, Exercise III.3.4(b)] imply that for any integer n,

$$R'_{x,n} = H^0(X_x,\mathcal{O}_{X_x}(1)^{\otimes n}),\quad H^1(X_x,\mathcal{O}_{X_x}(1)^{\otimes n}) = 0,\quad H^2(X_x,\mathcal{O}_{X_x}(1)^{\otimes n}) = H^3_{\mathfrak{m}}(R'_x)_n.$$

Since $R'_{x,0} = \mathbb{C}$, X_x is connected. Since X_x is normal, X_x is integral, so $R'_x = \bigoplus_n H^0(X_x,\mathcal{O}_{X_x}(1)^{\otimes n})$ is a domain. For any integer m, let

$$\Omega_m := H^0(X_x,\omega_{X_x}\otimes\mathcal{O}_{X_x}(1)^{\otimes m}).$$

By Serre duality [Har77, Corollary III.7.7], [BH98, Theorem 3.6.19(a)], and the isomorphism of graded S_0-modules above, for any integer m, we have

$$\dim \Omega_m = h^2(X_x,\mathcal{O}_{X_x}(1)^{\otimes(-m)}) = \dim R'_{x,m+1}.$$

Let $\eta \in \Omega_{-1}$ be a non-zero element. Since ω_{X_x} is invertible (Du Val singularities are hypersurface singularities [KM98, Theorem 4.20]) and X_x is integral, $f \to f\eta$ is an injective graded R'_x-linear map $R'_x(1) \to \bigoplus_m \Omega_m$, and it is an isomorphism by the equality of dimensions. Let $\mathcal{K}$ be the cokernel of the injective morphism $\mathcal{O}_{X_x}(1) \to \omega_{X_x}$ defined by η. Since $H^1(X_x,\mathcal{O}_{X_x}(1)^{\otimes n}) = 0$ for any integer n, we have $H^0(X_x,\mathcal{K}\otimes\mathcal{O}_{X_x}(1)^{\otimes m}) = 0$ for any integer m, so $\mathcal{K} = 0$ since $\mathcal{O}_{X_x}(1)$ is ample. Thus $\omega_{X_x} \cong \mathcal{O}_{X_x}(1)$. Therefore, $R'_{x,n} = H^0(X_x,nK_{X_x})$ and $H^1(X_x,nK_{X_x}) = 0$ for any integer n. By Serre duality [Har77, Corollary III.7.7], $h^2(X_x,nK_{X_x}) = \dim R'_{x,1-n}$.

Let $\pi : S_x \to X_x$ be the minimal resolution. Since X_x is projective, so is S_x. Since X_x has Du Val singularities, X_x has canonical singularities [KM98, Theorem 4.20]. Since moreover K_{X_x} is Cartier ($\omega_{X_x} \cong \mathcal{O}_{X_x}(1)$), X_x has rational singularities [KM98, Corollary 5.24], so by [KM98, Definition 5.8],

$$\pi_*\mathcal{O}_{S_x} = \mathcal{O}_{X_x},\qquad R^j\pi_*\mathcal{O}_{S_x} = 0\quad (j>0).$$

Since π is an isomorphism over the smooth locus of X_x, we have $K_{S_x} \sim \pi^*K_{X_x} + E$ for a divisor E supported on the exceptional curves of π. For any exceptional curve C of π, we have $K_{S_x}\cdot C = 0$ [KM98, Definition 4.4] and $\pi^*K_{X_x}\cdot C = 0$, so $E\cdot C = 0$. Thus $E^2 = 0$, and $E = 0$ since the intersection matrix of the exceptional curves of π is negative definite [KM98, Lemma 3.40], that is, $K_{S_x} \sim \pi^*K_{X_x}$. By the projection formula [Har77, Exercise III.8.3], for any integer n,

$$\pi_*\mathcal{O}_{S_x}(nK_{S_x}) \cong \mathcal{O}_{X_x}(nK_{X_x}),\qquad R^j\pi_*\mathcal{O}_{S_x}(nK_{S_x}) = 0\quad (j>0),$$

so [Har77, Exercise III.8.1] implies that $H^j(S_x,nK_{S_x}) \cong H^j(X_x,nK_{X_x})$ for any integer $j \geq 0$, where the isomorphism for $j = 0$ is π^*. Thus, for any integer n,

$$h^0(S_x,nK_{S_x}) = \dim R'_{x,n},\qquad h^1(S_x,nK_{S_x}) = 0,\qquad h^2(S_x,nK_{S_x}) = \dim R'_{x,1-n}.$$

By **Step 2**, $R'_{x,n} = 0$ for $n < 0$, $R'_{x,0} = \mathbb{C}$, $R'_{x,1} = 0$, and $\dim R'_{x,2} = 2$. Hence

$$p_g(S_x) = q(S_x) = 0, \qquad \chi(\mathcal{O}_{S_x}) = 1, \qquad \chi(\mathcal{O}_{S_x}(2K_{S_x})) = 2,$$

and Riemann–Roch [Har77, Theorem V.1.6] implies that

$$K_{S_x}^2 = \chi(\mathcal{O}_{S_x}(2K_{S_x})) - \chi(\mathcal{O}_{S_x}) = 2 - 1 = 1.$$

Since K_{X_x} is ample, $|nK_{X_x}|$ has no base point for some $n > 0$, so $|nK_{S_x}|$ has no base point, and K_{S_x} is nef. Since moreover $K_{S_x}^2 = 1 > 0$, K_{S_x} is big [KM98, Proposition 2.61], and S_x is of general type. Since K_{S_x} is nef and $K_{S_x} \cdot C = -2 - C^2 = -1$ for any (-1)-curve C [Har77, Proposition V.1.5], S_x has no (-1)-curve, that is, S_x is minimal. Thus S_x is a numerical Godeaux surface (Definition 2.1). Moreover, π^* is an isomorphism of graded rings from $R'_x = \bigoplus_{n\geq 0} H^0(X_x, nK_{X_x})$ to the canonical ring of S_x, so the canonical model of S_x is $\operatorname{Proj} R'_x = X_x$. Finally, $(d_{1,x}, d_{2,x})$ is a presentation of S_x with the quadrics $r_1, \dots, r_6$ (the items of Set-up 4.1 hold by construction and by degrees).

Conversely, let S be a numerical Godeaux surface with a presentation with the quadrics $r_1, \dots, r_6$ and lower blocks in $\mathcal{L}$, and let $x \in \mathcal{P}$ be the point of **Step 1**. Then $\mathbb{F}_x$ is exact, and $X_x = X$ has Du Val singularities. Since K_X is ample and Cartier, for $n \gg 0$ and any point of X, some elements of R_n and of R_{n+1} do not vanish at the point. Thus $x \in W_i^\circ$ for some i.

Step 4. In this step we conclude the proof.

By (1), the flat family over the reduced scheme W_i° induces a morphism $\psi_i : W_i^\circ \to \mathcal{M}$ with $\psi_i(x) = [X_x]$. Let $p_i : W_i^\circ \to \mathcal{L}$ be the projection. By **Step 3**, $\Xi_{\mathcal{L}}$ is the union of the images of the morphisms (p_i, ψ_i). Thus $\Xi_{\mathcal{L}}$ is constructible by Chevalley's theorem [Har77, Exercise II.3.19]. By Lemma 4.4, there exists at most one point $[X]$ over any $\ell \in \mathcal{L}$.

Let $V \subset \mathcal{L}$ be a constructible subset. Then $\Xi_{\mathcal{L}} \cap (V \times \mathcal{M})$ is a constructible subset of $\mathcal{L} \times \mathcal{M}$, whose non-empty fibers over $\mathcal{L}$ are points and whose image in $\mathcal{L}$ is contained in V. Thus $\dim \overline{\mathcal{M}(V)} \leq \dim V$ by Lemma 2.9 with $e = 0$. □

5. Marked surfaces

In this section and the next two sections, we consider torsion-free numerical Godeaux surfaces whose bicanonical system on the canonical model has four distinct base points. In this section, we write down the standard presentations of [SS20, Definition 3.9], we define pencil matrices, we prove the bound of Proposition 5.5, which the dimension counts of Section 6 use, and we use that S is torsion-free to exclude lines in $\mathcal{Q}$ by Lemmas 5.7 and 5.8.

Construction-Proposition 5.1. Let S_0 be the polynomial ring of Set-up 4.1, let $B_1, \dots, B_4$ be the fixed blocks of Table 1, let a_{ij} with $i \neq j$ in $\{0,1,2,3\}$ be binary linear forms in s, t, and let

$$A = \begin{pmatrix} a_{01} & a_{02} & 0 & a_{03} & 0 & 0 \\ a_{10} & 0 & a_{12} & 0 & a_{13} & 0 \\ 0 & a_{20} & a_{21} & 0 & 0 & a_{23} \\ 0 & 0 & 0 & a_{30} & a_{31} & a_{32} \end{pmatrix},$$

$$q_0 := a_{12}a_{13} - a_{21}a_{23} + a_{31}a_{32}, \quad q_1 := a_{02}a_{03} - a_{30}a_{32} + a_{20}a_{23},$$
$$q_2 := a_{10}a_{13} - a_{01}a_{03} + a_{30}a_{31}, \quad q_3 := a_{01}a_{02} - a_{10}a_{12} + a_{20}a_{21}.$$

Then there exist unique matrices P, U of binary linear forms with $AB_3 + B_1P = 0$ and $UB_4 = B_2P^t$, and $UP = 0$ if and only if $q_0 = \cdots = q_3 = 0$. We call P, U the *blocks* of A, and we call A a *pencil matrix* if $q_0 = \cdots = q_3 = 0$. We index the columns of A, from left to right, by the pairs $\{0,1\}, \{0,2\}, \{1,2\}, \{0,3\}, \{1,3\}, \{2,3\}$, so that the column $\{i,j\}$ has the entries a_{ij}, a_{ji}. We number the rows and columns of C, O, P, U, and $B_1, \dots, B_4$ from 0. The k-th relation of Set-up 4.1 corresponds to column $k-1$ of A and of O. A linear change of the basis s, t is called a *binary reparametrization*, and it replaces the a_{ij} by their linear substitutions. We denote by $\mathcal{Q} \subset \mathbb{P}^{11}$ the subscheme with coordinates a_{ij} defined by $q_0, \dots, q_3$, and by $\mathcal{F}$ the scheme of lines in $\mathcal{Q}$ [Kol96, Theorem I.1.4]. A pencil matrix whose coefficient vectors of s and t in $\mathbb{C}^{12}$ are

linearly independent parametrizes a line in $\mathcal{Q}$, and any point of $\mathcal{F}$ arises in this way from a matrix unique up to binary reparametrization.

Now let S be a numerical Godeaux surface with $[X] \in \mathcal{M}_{(1,1,1,1)}$, so that S is torsion-free, $|2K_S|$ has no fixed divisor, and $|2K_S|$ has four distinct reduced base points $p_0, \dots, p_3$ (Construction-Proposition 3.3). We call the enumeration $p_0, \dots, p_3$ a *marking* of S, and we call S with a marking a *marked* surface. Choose a basis $y_0, \dots, y_3$ of R_3 with $y_i(p_j) \neq 0$ if and only if $i = j$ (Construction-Theorem 3.2(3)). Then S has a presentation as in Set-up 4.1, called *standard*, with

$$(r_1, \dots, r_6) = (y_0y_1, y_0y_2, y_1y_2, y_0y_3, y_1y_3, y_2y_3),$$

with the blocks $B_1, \dots, B_4$, with a pencil matrix A, and with the blocks P, U of A, so that the relation with $r_k = y_iy_j$ is

$$y_iy_j + a_{ij}z_i + a_{ji}z_j + v_k(s,t) = 0.$$

We call A a *standard pencil matrix* of S. The image of A under a binary reparametrization is a standard pencil matrix of S.

TABLE 1. The fixed blocks $B_1, \dots, B_4$ of a standard presentation.

$$B_1 = \begin{pmatrix} y_1 & y_2 & y_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & y_0 & y_2 & y_3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & y_0 & y_1 & y_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_0 & y_1 & y_2 \end{pmatrix}$$

$$B_2 = \begin{pmatrix} -y_3 & 0 & y_2 & 0 & -y_1 & 0 & 0 & 0 \\ -y_3 & -y_3 & y_2 & y_2 & 0 & 0 & -y_0 & 0 \\ 0 & 0 & 0 & -y_2 & 0 & y_1 & 0 & -y_0 \end{pmatrix}$$

$$B_3 = \begin{pmatrix} -y_2 & 0 & -y_3 & 0 & 0 & 0 & 0 & 0 \\ y_1 & -y_1 & 0 & 0 & -y_3 & 0 & 0 & 0 \\ 0 & y_0 & 0 & 0 & 0 & 0 & -y_3 & 0 \\ 0 & 0 & y_1 & -y_1 & y_2 & -y_2 & 0 & 0 \\ 0 & 0 & 0 & y_0 & 0 & 0 & y_2 & -y_2 \\ 0 & 0 & 0 & 0 & 0 & y_0 & 0 & y_1 \end{pmatrix}$$

B_4 is block diagonal: for $i \in \{0,1,2,3\}$, with $j_0 < j_1 < j_2$ the indices other than i, the block in rows and columns $3i, 3i+1, 3i+2$ is $\begin{pmatrix} 0 & y_{j_2} & -y_{j_1} \\ -y_{j_2} & 0 & y_{j_0} \\ y_{j_1} & -y_{j_0} & 0 \end{pmatrix}$

Proof of Construction-Proposition 5.1. Let $[y_j]$ denote the coefficient of y_j, and let $j_0 < j_1 < j_2$ be as in Table 1. For any binary linear forms a_{ij}, define

$$P_{3i+m,n} := -[y_{j_m}](AB_3)_{i,n},$$

$$U_{l,3i} := [y_{j_2}](B_2P^t)_{l,3i+1}, \quad U_{l,3i+1} := [y_{j_0}](B_2P^t)_{l,3i+2}, \quad U_{l,3i+2} := [y_{j_1}](B_2P^t)_{l,3i}.$$

A direct calculation shows that $AB_3 + B_1P = 0$, $UB_4 = B_2P^t$, $B_1B_4 = 0$, and $B_2B_3^t = 0$, and that the non-zero entries of UP are $\pm q_k$, with all four of $q_0, \dots, q_3$ occurring. Thus $UP = 0$ if and only if $q_0 = \cdots = q_3 = 0$. If x is a vector with entries in $\mathbb{C}[s,t]$ and $B_1x = 0$ or $xB_4 = 0$, then $x = 0$, so P and U are unique.

The presentation and the form of A are [SS20, Propositions 3.6 and 3.11, Definition 3.9, and Remark 3.10]. A binary reparametrization does not change R, and $B_1, \dots, B_4$ do not involve s, t, so the linear substitution in a standard presentation is a standard presentation of S. The equations $q_0 = \cdots = q_3 = 0$ say that the map $[s:t] \to [a_{ij}(s,t)]$ takes values in $\mathcal{Q}$, and two bases of a line differ by a binary reparametrization. □

Theorem 5.2 ([SS20, Proposition 4.4] and [Ste18, Section 8.3]). *The torus $(\mathbb{C}^*)^4$ acts on $\mathcal{Q}$ by $a_{ij} \to \lambda_i\lambda_j^2 a_{ij}$, and the symmetric group $\mathfrak{S}_4$ acts on $\mathcal{Q}$ by signed permutations, where the transposition $\sigma = (k, k+1)$ acts by $a_{ij} \to -\epsilon_{ij} a_{\sigma(i)\sigma(j)}$, with $\epsilon_{ij} = -1$ if exactly one of i, j belongs to $\{k, k+1\}$ and $\epsilon_{ij} = 1$ otherwise. Let A be a standard pencil matrix of a surface S as in Construction-Proposition 5.1. Then the image of A under either action is a standard pencil matrix of S for a marking of S. More precisely, the element $(\lambda_0, \dots, \lambda_3)$ of the torus corresponds to the change $y_i \to \lambda_i^2 y_i$, $z_i \to \lambda_i z_i$ with the marking $p_0, \dots, p_3$, and the transposition $\sigma = (k, k+1)$ corresponds to the change $y_k \to -y_{k+1}$, $y_{k+1} \to -y_k$, $y_m \to y_m$ for $m \neq k, k+1$, $z_m \to -z_{\sigma(m)}$ with the marking $p_{\sigma(0)}, \dots, p_{\sigma(3)}$.*

Proof. The actions on $\mathcal{Q}$ are [SS20, Proposition 4.4], and the diagonal changes for the torus are [Ste18, Section 8.3]. Since the transpositions $(k, k+1)$ generate $\mathfrak{S}_4$, it is enough to prove, for any element of the torus and for a transposition $(k, k+1)$, that the image of A is a standard pencil matrix of S with the change of y_i, z_i of the theorem. Let (d_1, d_2) be a standard presentation of S with standard pencil matrix A, and let $j_0 < j_1 < j_2$ be as in Table 1.

Let φ be the automorphism of S_0 which fixes s, t and sends y_i to $\lambda_i^{-2} y_i$. Let D_0 be the diagonal matrix with entries $1, \lambda_0^{-1}, \dots, \lambda_3^{-1}, 1, 1, 1$. Let D_1 be the diagonal matrix with entries $\lambda_i^2\lambda_j^2$ at the column $\{i, j\}$ of the first block, $\lambda_i\lambda_{j_m}^2$ at the column $3i + m$ of the second block, and

$$\lambda_3^2, \lambda_3^2, \lambda_2^2, \lambda_2^2, \lambda_1^2, \lambda_1^2, \lambda_0^2, \lambda_0^2$$

on the third block. A direct calculation shows that

$$\big(D_0\varphi(d_1)D_1,\ (\lambda_0\lambda_1\lambda_2\lambda_3)^2 D_1^{-1}\varphi(d_2)D_1^{-1}\big)$$

has the blocks $B_1, \dots, B_4$, the quadrics y_iy_j, and the pencil matrix with entries $\lambda_i\lambda_j^2 a_{ij}$. The pair is a presentation of S for the sections s, t, $\lambda_i^2 y_i$, $\lambda_i z_i$, and w_0, w_1, w_2. Since $(\lambda_i^2 y_i)(p_j) \neq 0$ if and only if $i = j$, the new matrix is a standard pencil matrix of S.

Let $\sigma = (k, k+1)$, and let φ be the automorphism of S_0 which fixes s, t and y_m for $m \neq k, k+1$ and which sends y_k to $-y_{k+1}$ and y_{k+1} to $-y_k$. Write y'_m for the section $\varphi(y_m)$ of R_3, and let $z'_m := -z_{\sigma(m)}$. Then $y'_iy'_j = \epsilon_{ij} y_{\sigma(i)}y_{\sigma(j)}$ with the sign ϵ_{ij} of the theorem. Multiplying the relation with quadric $y_{\sigma(i)}y_{\sigma(j)}$ by ϵ_{ij}, we obtain

$$y'_iy'_j - \epsilon_{ij} a_{\sigma(i)\sigma(j)} z'_i - \epsilon_{ij} a_{\sigma(j)\sigma(i)} z'_j + \epsilon_{ij} v = 0,$$

where v is the cubic of that relation. Let $\delta_m := -1$ for $m \in \{k, k+1\}$ and $\delta_m := 1$ otherwise, so that $\varphi(y_m) = \delta_m y_{\sigma(m)}$ and $\epsilon_{ij} = \delta_i\delta_j$. Let

$$E_1 := \begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix}, \quad E_2 := \begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix}, \quad E_3 := \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}.$$

Let D_0 be the block diagonal matrix with the entry 1 on the first row, with the entry -1 at row i and column $\sigma(i)$ on the rows of $z_0, \dots, z_3$, and with

$$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & -1 \end{pmatrix}$$

on the rows of w_0, w_1, w_2 for $k = 0, 1, 2$, respectively. Let D_1 be the block diagonal matrix defined as follows. On the first block, the column $\{i, j\}$ has the entry ϵ_{ij} at the row $\{\sigma(i), \sigma(j)\}$. On the second block, the column $3i + m$, with $l = j_m$ the index with y_l in row i of B_1 at that column, has the entry $-\delta_l$ at the row of the column of $y_{\sigma(l)}$ in row $\sigma(i)$ of B_1. On the third block, D_1 is $\mathrm{diag}(E_1, E_1, E_3)$, $\mathrm{diag}(E_2, E_3, E_1)$, $\mathrm{diag}(E_3, E_2, E_2)$ for $k = 0, 1, 2$, respectively. A direct calculation shows that

$$\big(D_0\varphi(d_1)D_1,\ -D_1^{-1}\varphi(d_2)D_1^{-t}\big)$$

has the blocks $B_1, \dots, B_4$, the quadrics y_iy_j, and the pencil matrix with entries $-\epsilon_{ij} a_{\sigma(i)\sigma(j)}$. The factor -1 restores the sign of B_4. On the three rows and columns of index i, if $i \notin \{k, k+1\}$,

then σ exchanges two neighbours among j_0, j_1, j_2 and $\delta_{j_0}\delta_{j_1}\delta_{j_2} = 1$. If $i \in \{k, k+1\}$, then σ keeps the order of j_0, j_1, j_2 and $\delta_{j_0}\delta_{j_1}\delta_{j_2} = -1$. In both cases the change without the factor -1 sends the block to its negative. The factor -1 keeps $d_1 d_2 = 0$ and the alternation of d_2. The pair is a presentation of S for the sections s, t, y'_m, z'_m, and for three linear combinations of w_0, w_1, w_2. Since $y'_i(p_{\sigma(j)}) \neq 0$ if and only if $i = j$, the new matrix is a standard pencil matrix of S for the marking $p_{\sigma(0)}, \dots, p_{\sigma(3)}$. □

We define the space $\mathcal{U}_A$ and bound $\dim \mathcal{U}_A$ for any pencil matrix, so we will not need to decide whether a pencil matrix is a standard pencil matrix of a surface.

Construction-Proposition 5.3. Let A be a pencil matrix, with the fixed blocks $B_1, \dots, B_4$ and the blocks P, U of Construction-Proposition 5.1. For a standard presentation, the base changes of Lemma 4.2 normalize the block O to the shape of Table 2, with twelve binary linear coefficients $o_0, \dots, o_{11}$. The equation $AO - B_1 N^t - C B_3^t = 0$ then determines twelve entries of C and determines N by $N_{n,3i+m} = [y_{j_m}](AO - CB_3^t)_{i,n}$ for $0 \leq m \leq 2$, where $[y_j]$ denotes the coefficient of y_j and $j_0 < j_1 < j_2$ are as in Table 1, and the other entries of C are the twenty binary quadratic forms $c_0, \dots, c_{19}$ of Table 2, so that (C, O) has $3 \cdot 20 + 2 \cdot 12 = 84$ coefficients. We denote by $\mathcal{U}_A$ the vector space of all tuples $\mathbf{u} = (C, O)$ of the normalized shape of Table 2 satisfying

$$AO - B_1 N^t - C B_3^t = 0, \quad AN - CP^t = 0, \quad UN^t = 0.$$

We say that $\mathbf{u} \in \mathcal{U}_A$ is *geometric* if $\mathbf{u}$ is the tuple of a normalized standard presentation with standard pencil matrix A of a surface as in Construction-Proposition 5.1. By Lemma 4.4, a geometric $\mathbf{u}$ determines the canonical model X of the surface up to isomorphism, and we say that $\mathbf{u}$ *defines* the point $[X]$ of $\mathcal{M}$.

TABLE 2. The normalized shape of the blocks C and O.

	slots (i,j) with $C_{i,j} = c_r$, in the order of r				
$c_0, \dots, c_4$	$(0,0)$	$(0,2)$	$(0,4)$	$(0,6)$	$(0,7)$
$c_5, \dots, c_9$	$(1,1)$	$(1,3)$	$(1,4)$	$(1,5)$	$(1,6)$
$c_{10}, \dots, c_{14}$	$(2,1)$	$(2,2)$	$(2,3)$	$(2,5)$	$(2,7)$
$c_{15}, \dots, c_{19}$	$(3,0)$	$(3,1)$	$(3,3)$	$(3,5)$	$(3,7)$

$C_{1,0} = C_{1,1}$, $C_{1,2} = C_{1,3}$, $C_{2,4} = C_{2,5}$, $C_{2,6} = C_{2,7}$, and any other entry of C is zero.

	slots (i,j,k) of O, in the order of r			
$o_0, \dots, o_3$	$(0,1,0)$	$(0,2,1)$	$(0,3,0)$	$(0,4,1)$
$o_4, \dots, o_7$	$(1,2,2)$	$(1,3,0)$	$(1,5,2)$	$(2,4,1)$
$o_8, \dots, o_{11}$	$(2,5,2)$	$(3,4,3)$	$(3,5,3)$	$(4,5,3)$

$O_{i,j} = o_r y_k$ and $O_{j,i} = -O_{i,j}$ for the slot (i,j,k) of number r,
and any other entry of O is zero.

Proof of Construction-Proposition 5.3. By Lemma 4.2, the gauge of Γ replaces O by

$$O + B_3 \Gamma^t - \Gamma B_3^t.$$

The map $\Gamma \to B_3\Gamma^t - \Gamma B_3^t$ separates the coefficients of s, t. On constant 6×8 matrices, a space of dimension 48, the map takes values in the alternating 6×6 matrices of linear forms in $y_0, \dots, y_3$, a space of dimension 60. It is injective with image complementary to the twelve slots of O in Table 2: with the matrix units of Γ in lexicographic order followed by the twelve slots, and with the coordinates (i,j,k), $i < j$, in lexicographic order, the 60×60 integer matrix has determinant 16. Thus the normal form of [SS20, Section 3, p. 17] exists and Γ is unique.

Row i of B_1N^t involves only y_j with $j \neq i$, and the coefficient of y_{j_m} in the entry of column n is $N_{n,3i+m}$. Thus

$$N_{n,3i+m} = [y_{j_m}](AO - CB_3^t)_{i,n}.$$

In row i of $AO - B_1N^t - CB_3^t = 0$, take the coefficient of y_i at the twelve positions whose column has an index pair not containing i. At any such position AO has no y_i, so the coefficient involves only C and does not involve A. We obtain

$$C_{0,1} = C_{0,3} = C_{0,5} = C_{1,7} = C_{2,0} = C_{3,2} = C_{3,4} = C_{3,6} = 0$$

and

$$C_{1,0} = C_{1,1}, \quad C_{1,2} = C_{1,3}, \quad C_{2,4} = C_{2,5}, \quad C_{2,6} = C_{2,7}.$$

The other twelve coefficients of y_i in row i, at positions whose column has an index pair containing i, do not involve C and depend on A. For instance, row 1 and column 0 yield

$$a_{12}o_1 + a_{13}o_3 = 0.$$

The twelve equations at such positions stay among the defining equations of $\mathcal{U}_A$.

The block equations of $d_1'd_2 = 0$ are $AO - B_1N^t - CB_3^t = 0$, $AN + B_1B_4 - CP^t = 0$, $AB_3 + B_1P = 0$, $UN^t + B_2B_3^t = 0$, $UB_4 = B_2P^t$, and $UP = 0$. Since $B_1B_4 = 0$ and $B_2B_3^t = 0$ (a direct calculation with Table 1), the tuple of a normalized standard presentation belongs to $\mathcal{U}_A$. □

Lemma 5.4. *Let (d_1, d_2) be a presentation of a numerical Godeaux surface S with the blocks $B_1, \dots, B_4$ of Table 1 and the quadrics y_iy_j of Construction-Proposition 5.1, and let $\lambda \in \mathbb{C}^*$. Then S has a presentation for the sections s, t, $\lambda^{-1}y_i$, $\lambda^{-2}z_i$, and $\lambda^{-3}w_i$, with the blocks $A, U, P, B_1, \dots, B_4$ and the quadrics of (d_1, d_2), and with $\lambda^{-2}(C, O, N)$ in the place of (C, O, N).*

Proof. We do not assume that S is torsion-free or that $y_0, \dots, y_3$ are adapted to a marking. Let φ_λ be the automorphism of S_0 which fixes s, t and sends y_i to λy_i, and let

$$D_0 := \operatorname{diag}(\lambda^{-2}, I_4, \lambda I_3), \quad D_1 := \operatorname{diag}(I_6, \lambda^{-1}I_{12}, \lambda^{-2}I_8).$$

Let

$$\tilde{d}_1 := D_0\varphi_\lambda(d_1)D_1, \quad \tilde{d}_2 := \lambda^{-3}D_1^{-1}\varphi_\lambda(d_2)D_1^{-1}.$$

Then $\tilde{d}_1\tilde{d}_2 = 0$, $\tilde{d}_2\tilde{d}_1^t = 0$, $\tilde{d}_2$ is alternating, and the complex with the differentials $\tilde{d}_1, \tilde{d}_2, \tilde{d}_1^t$ is exact since φ_λ is an automorphism and D_0, D_1 are invertible. The lower blocks acquire the factors

	A	B_1	C	U	B_2	O	N	B_3	B_4	P
from φ_λ	1	λ	1	1	λ	λ	1	λ	λ	1
from D_0, D_1, λ^{-3}	1	λ^{-1}	λ^{-2}	1	λ^{-1}	λ^{-3}	λ^{-2}	λ^{-1}	λ^{-1}	1
total	1	1	λ^{-2}	1	1	λ^{-2}	λ^{-2}	1	1	1

In the first row of $\tilde{d}_1$, the first six entries are $y_iy_j + \lambda^{-2}v_k$, since the factor λ^2 of φ_λ on y_iy_j cancels the factor λ^{-2} of D_0. Thus $(\tilde{d}_1, \tilde{d}_2)$ is a presentation of S for the sections s, t, $\lambda^{-1}y_i$, $\lambda^{-2}z_i$, and $\lambda^{-3}w_i$, with the blocks $A, U, P, B_1, \dots, B_4$ and the quadrics of (d_1, d_2), and with $\lambda^{-2}(C, O, N)$ in the place of (C, O, N). □

By Lemma 5.4, a non-zero geometric tuple $\mathbf{u}$ and the tuples $\rho\mathbf{u}$ with $\rho \in \mathbb{C}^*$ define one point of $\mathcal{M}$ (with $\lambda^{-2} = \rho$ in the lemma, the sections $\lambda^{-1}y_i$ satisfy $(\lambda^{-1}y_i)(p_j) \neq 0$ if and only if $i = j$, and scaling keeps the normalized shape of Table 2). The term -1 in Proposition 5.5 is $-\dim \mathbb{C}^*\mathbf{u}$.

Proposition 5.5. *Notation and conditions as in Construction-Proposition 5.3. Let V be a variety carrying an algebraic family of pencil matrices A_x with $x \in V$, not necessarily coming from surfaces, and let $\mathcal{M}_V \subset \mathcal{M}$ be the set of points defined by geometric tuples over V. Assume that $\dim \mathcal{U}_{A_x} \leq d$ for any $x \in V$, where $d \geq 1$. Then $\dim \overline{\mathcal{M}_V} \leq \dim V + d - 1$.*

Proof. Let $I_V \subset V \times \mathbb{C}^{84}$ be the set of pairs $(x, \mathbf{u})$ with $\mathbf{u} \in \mathcal{U}_{A_x}$. The coefficients of A_x are regular functions on V, and the equations of $\mathcal{U}_{A_x}$ are linear in $\mathbf{u}$, so I_V is a closed cone. Let $\mathcal{L}$ be the affine space of all lower blocks of the shape of Set-up 4.1 with the fixed blocks of Table 1. Let $I_V \to \mathcal{L}$ be the morphism which sends $(x, \mathbf{u})$ to the lower blocks with A_x, C, O, the blocks P, U of A_x, and the matrix N of Construction-Proposition 5.3 for A_x.

Let $G \subset I_V \times \mathcal{M}$ be the preimage of $\Xi_{\mathcal{L}}$, for the quadrics $y_i y_j$. By Lemma 4.5(2), G is constructible, and for any $(x, \mathbf{u}) \in I_V$ there exists at most one point $[X]$ with $(x, \mathbf{u}, [X]) \in G$. Let $(x, \mathbf{u}, [X]) \in G$ and let $\rho \in \mathbb{C}^*$. By Lemma 5.4 with $\lambda^{-2} = \rho$, the lower blocks determined by $(A_x, \rho\mathbf{u})$ are the lower blocks of a presentation of a surface with canonical model X, so $(x, \rho\mathbf{u}, [X]) \in G$. Thus G is invariant under $\mathbf{u} \to \rho\mathbf{u}$.

Let $G^*, G^0 \subset G$ be the parts with $\mathbf{u} \neq 0$ and with $\mathbf{u} = 0$. Both are constructible, and any non-empty fiber of $G^* \to I_V$ or of $G^0 \to I_V$ is a point. Any non-empty fiber of $G^* \to \mathcal{M}$ contains a set $\{x\} \times \mathbb{C}^*\mathbf{u}$, of dimension 1, by the invariance of G. Since $\dim I_V \leq \dim V + d$ [Har77, Exercise II.3.22(c)], Lemma 2.9 for $G^* \subset I_V \times \mathcal{M}$ with $e = 1$ implies that the closure of the image of G^* in $\mathcal{M}$ is of dimension at most $\dim V + d - 1$. The image of G^0 in I_V is contained in $V \times \{0\}$, so Lemma 2.9 for $G^0 \subset I_V \times \mathcal{M}$ with $e = 0$ implies that the closure of the image of G^0 in $\mathcal{M}$ is of dimension at most $\dim V$, and $\dim V \leq \dim V + d - 1$ since $d \geq 1$.

A surface as in Construction-Proposition 5.1 is a numerical Godeaux surface with the blocks of Table 1 and the quadrics $y_i y_j$, so $\mathcal{M}_V$ is contained in the image of G in $\mathcal{M}$. Thus $\dim \overline{\mathcal{M}_V} \leq \dim V + d - 1$. □

The equations of Construction-Proposition 5.3 do not by themselves yield the bounds ≤ 7 of Theorems 6.9 and 7.10, since the eight-dimensional families of surfaces with torsion $\mathbb{Z}/5$ and $\mathbb{Z}/3$ of [SS24, Theorems 4.4 and 4.7] are also given by lines in $\mathcal{Q}$. Thus we first use that S is torsion-free to exclude lines in $\mathcal{Q}$, by Lemmas 5.7 and 5.8.

Lemma 5.6. *Notation and conditions as in Construction-Proposition 5.1. Let $l \in \{0, 1, 2, 3\}$. Then any member of $|2K_S|$ is smooth at p_l, $p_l \notin \Gamma$ for any irreducible curve Γ on S with $K_S \cdot \Gamma = 0$, and π is an isomorphism near p_l.*

Proof. Since the base scheme of $|2K_S|$ is reduced at p_l, the local equations of s, t at p_l generate the maximal ideal of $\mathcal{O}_{S,p_l}$. Thus any member of $|2K_S|$ is smooth at p_l. Let Γ be an irreducible curve on S with $K_S \cdot \Gamma = 0$, and suppose that $p_l \in \Gamma$. Let G be a member of $|2K_S|$ not containing Γ. Since $p_l \in G \cap \Gamma$, we have $G \cdot \Gamma > 0$ [Har77, Proposition V.1.4], which contradicts $G \cdot \Gamma = 2K_S \cdot \Gamma = 0$. Thus $p_l \notin \Gamma$. Since π contracts exactly the irreducible curves Γ with $K_S \cdot \Gamma = 0$ (since $K_S = \pi^* K_X$ and K_X is ample), π is an isomorphism near p_l. □

Lemma 5.7 excludes columns of A that vanish identically. The line of [SS24, Example 4.3] for surfaces with torsion $\mathbb{Z}/5$ has the columns $\{0, 3\}$ and $\{1, 2\}$ vanishing identically in the coordinates of [SS20, Proposition 4.4], so Lemma 5.7 depends on the assumption that S is torsion-free.

Lemma 5.7. *Notation and conditions as in Construction-Proposition 5.1. Then for any distinct $i, j \in \{0, 1, 2, 3\}$, there exists no binary cubic h such that $y_i y_j = h(s, t)$ in R_6. In particular, a_{ij} and a_{ji} are not both identically zero.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we describe the members of $|2K_S|$ at the base points, and we bound the self-intersection numbers of some divisors $D \geq 0$.

Since K_S is nef, $K_S \cdot \Gamma \geq 0$ for any irreducible curve Γ on S. The intersection number of two curves without common irreducible component is the sum of the local intersection numbers, and the local intersection number at a common point p is the length of $\mathcal{O}_{S,p}/(f, g)$, where f, g are the local equations at p [Har77, Proposition V.1.4].

Let $l \in \{0, 1, 2, 3\}$. Since the base scheme of $|2K_S|$ is reduced at p_l, two distinct members of $|2K_S|$ have local intersection number 1 at p_l. Since $|2K_S|$ is a pencil without fixed divisor, two

distinct members of $|2K_S|$ have no common irreducible component, and the set of their common points is $\{p_0, p_1, p_2, p_3\}$.

By Lemma 5.6, $K_S \cdot C \geq 1$ for any irreducible curve C on S containing some p_l.

Let D be a divisor on S. By Lemma 2.2(4), $D^2 \leq (K_S \cdot D)^2$, $D^2 + K_S \cdot D$ is even, and if $D^2 = (K_S \cdot D)^2$, then $D - (K_S \cdot D)K_S$ is numerically trivial. Since $[X] \in \mathcal{M}_{(1,1,1,1)}$, S is torsion-free, so by Lemma 2.3(1), if $D^2 = (K_S \cdot D)^2$, then $D \sim (K_S \cdot D)K_S$.

Now assume that $D \geq 0$. If $K_S \cdot D = 1$, then D^2 is odd and $D^2 \leq 1$, and $D^2 = 1$ would imply that $D \sim K_S$, which contradicts $p_g(S) = 0$. Thus

$$K_S \cdot D = 1 \quad \text{implies} \quad D^2 \leq -1.$$

If $K_S \cdot D = 2$ and $p_l \notin \operatorname{Supp} D$ for some l, then D^2 is even and $D^2 \leq 4$, and $D^2 = 4$ would imply that $D \sim 2K_S$, so that D is a member of $|2K_S|$ and $p_l \in \operatorname{Supp} D$. Thus

$$K_S \cdot D = 2 \text{ and } p_l \notin \operatorname{Supp} D \text{ for some } l \quad \text{imply} \quad D^2 \leq 2.$$

Step 2. In this step we assume that a relation $y_i y_j = h(s,t)$ exists, and we split $\operatorname{div}(y_i)$ along the factors of h.

Suppose that there exist distinct $i, j \in \{0,1,2,3\}$ and a binary cubic h such that $y_i y_j = h(s,t)$ in R_6. Since S is integral, R is a domain. Since y_i, y_j are non-zero, $h \neq 0$. We write

$$h = c \prod_\alpha L_\alpha^{m_\alpha},$$

where $c \in \mathbb{C}^*$, the L_α are pairwise non-proportional binary linear forms, the m_α are positive integers, and $\sum_\alpha m_\alpha = 3$. Let $G_\alpha := \operatorname{div}(L_\alpha(s,t))$, $Y_i := \operatorname{div}(y_i)$, and $Y_j := \operatorname{div}(y_j)$. Then $Y_i + Y_j = \sum_\alpha m_\alpha G_\alpha$. Since distinct G_α have no common irreducible component by **Step 1**, we may write

$$Y_i = \sum_\alpha D_\alpha, \quad Y_j = \sum_\alpha E_\alpha, \quad D_\alpha + E_\alpha = m_\alpha G_\alpha,$$

where $D_\alpha, E_\alpha \geq 0$ and $\operatorname{Supp} D_\alpha, \operatorname{Supp} E_\alpha \subset \operatorname{Supp} G_\alpha$.

Fix α. By Lemma 5.6, G_α is smooth at p_i and at p_j. Let C'_α and C''_α be the irreducible components of G_α containing p_i and p_j respectively, so that $\operatorname{mult}_{C'_\alpha} G_\alpha = \operatorname{mult}_{C''_\alpha} G_\alpha = 1$. Since $y_i(p_i) \neq 0$, we have $p_i \notin \operatorname{Supp} Y_i$, so

$$\operatorname{mult}_{C'_\alpha} D_\alpha = 0 \quad \text{and} \quad \operatorname{mult}_{C'_\alpha} E_\alpha = m_\alpha.$$

Since $y_j(p_j) \neq 0$, we have $p_j \notin \operatorname{Supp} Y_j$, so

$$\operatorname{mult}_{C''_\alpha} E_\alpha = 0 \quad \text{and} \quad \operatorname{mult}_{C''_\alpha} D_\alpha = m_\alpha.$$

In particular, $C'_\alpha \neq C''_\alpha$. By **Step 1**, $K_S \cdot C'_\alpha \geq 1$ and $K_S \cdot C''_\alpha \geq 1$. Since K_S is nef and $K_S \cdot G_\alpha = 2$, we have

$$K_S \cdot C'_\alpha = K_S \cdot C''_\alpha = 1,$$

and $K_S \cdot \Gamma = 0$ for any other irreducible component Γ of G_α. Thus

$$K_S \cdot D_\alpha = K_S \cdot E_\alpha = m_\alpha.$$

Let $l \in \{0,1,2,3\}$. By Lemma 5.6, p_l is not contained in any irreducible component Γ of G_α with $K_S \cdot \Gamma = 0$. Since G_α is smooth at p_l, exactly one of C'_α, C''_α contains p_l. Thus near p_l, either $D_\alpha = 0$ or $D_\alpha = m_\alpha G_\alpha$. Let

$$J_\alpha := \{l \mid D_\alpha = m_\alpha G_\alpha \text{ near } p_l\}.$$

Then $j \in J_\alpha$ and $i \notin J_\alpha$. Let G be a member of $|2K_S|$ different from G_α. By **Step 1**, D_α and G have common points only among p_0, p_1, p_2, p_3, and any $l \in J_\alpha$ contributes m_α to $G \cdot D_\alpha$. Thus

$$2m_\alpha = 2(K_S \cdot D_\alpha) = G \cdot D_\alpha = m_\alpha |J_\alpha|,$$

so $|J_\alpha| = 2$. Similarly, since G_α and G_β have local intersection number 1 at any p_l, for any $\alpha \neq \beta$, we have

$$D_\alpha \cdot D_\beta = m_\alpha m_\beta |J_\alpha \cap J_\beta|.$$

Let k, k' be the indices different from i, j. Then $J_\alpha \in \{\{j,k\},\{j,k'\}\}$ for any α. Since $y_i(p_k) = y_i(p_{k'}) = 0$, we have $p_k, p_{k'} \in \operatorname{Supp} Y_i$. Since $Y_i = \sum_\alpha D_\alpha$, both k and k' belong to $\bigcup_\alpha J_\alpha$. Thus there exist α, β with $J_\alpha = \{j,k\}$ and $J_\beta = \{j,k'\}$. In particular, h has at least two non-proportional linear factors.

Step 3. In this step we run through the multiplicities m_α and conclude the proof.

By **Step 2**, either h has two non-proportional linear factors with multiplicities 2 and 1, or h has three pairwise non-proportional linear factors with multiplicity 1. We also have

$$Y_i^2 = (3K_S)^2 = 9.$$

First suppose that $h = cL_1^2 L_2$. By **Step 2**, $J_1 \neq J_2$, so $|J_1 \cap J_2| = 1$ and $D_1 \cdot D_2 = 2$. Moreover, $K_S \cdot D_1 = 2$, and $p_i \notin \operatorname{Supp} D_1$ since $i \notin J_1$. Since $K_S \cdot D_2 = 1$, **Step 1** implies that $D_1^2 \leq 2$ and $D_2^2 \leq -1$. Thus

$$Y_i^2 = D_1^2 + D_2^2 + 2(D_1 \cdot D_2) \leq 2 - 1 + 4 = 5,$$

which contradicts $Y_i^2 = 9$.

Now suppose that $h = cL_1 L_2 L_3$. By **Step 2**, possibly renumbering the L_α, we may assume that $J_1 = J_2 \neq J_3$. Then

$$D_1 \cdot D_2 + D_1 \cdot D_3 + D_2 \cdot D_3 = 2 + 1 + 1 = 4.$$

Since $K_S \cdot D_\alpha = 1$, **Step 1** implies that $D_\alpha^2 \leq -1$ for $\alpha = 1, 2, 3$. Thus

$$Y_i^2 = \sum_{\alpha=1}^{3} D_\alpha^2 + 2(D_1 \cdot D_2 + D_1 \cdot D_3 + D_2 \cdot D_3) \leq -3 + 8 = 5,$$

which contradicts $Y_i^2 = 9$. Thus there exists no binary cubic h such that $y_i y_j = h(s,t)$ in R_6.

Finally, let m be the index with $r_m = y_i y_j$, and suppose that a_{ij} and a_{ji} are both identically zero. By Construction-Proposition 5.1, the relation

$$y_i y_j + a_{ij} z_i + a_{ji} z_j + v_m(s,t) = 0$$

becomes $y_i y_j = -v_m(s,t)$. This contradicts the first conclusion with $h = -v_m$. □

Lemma 5.8 excludes columns of A with a common index that vanish at one point of $\mathbb{P}^1$, and thus the lines of rows 1) and 2.2) of the table of [SS24, Section 5.1, p. 19], which [SS24] obtains by an experiment over a finite field. Up to the action of $\mathfrak{S}_4$, any line of row 1) passes through a general point of a locus of matrices of the second type of [SS24, Section 2, p. 5], on which $a_{0j} = a_{j0} = 0$ for $j = 1, 2, 3$, so the columns $\{0,1\},\{0,2\},\{0,3\}$ of A vanish at the corresponding point of $\mathbb{P}^1$. Up to the action of $\mathfrak{S}_4$, any line of row 2.2) passes through a general point of a locus of matrices of [SS24, Section 2, p. 6] on which $a_{03} = a_{13} = a_{23} = a_{31} = a_{32} = 0$, so the columns $\{1,3\},\{2,3\}$ of A vanish at the corresponding point of $\mathbb{P}^1$.

Lemma 5.8. *Notation and conditions as in Construction-Proposition 5.1. Let u be a point of $\mathbb{P}^1$. Then the set of pairs $\{i,j\}$ with $a_{ij}(u) = a_{ji}(u) = 0$ is a matching on $\{0,1,2,3\}$: two distinct such pairs have no common index. In particular, the coefficient vectors of s and t in A are linearly independent.*

Proof. Suppose that there exist pairwise distinct indices i, j, k such that

$$a_{ij}(u) = a_{ji}(u) = a_{ik}(u) = a_{ki}(u) = 0,$$

and let l be the fourth index.

By Lemma 5.6, π is an isomorphism near p_l. Since $\pi^* C_u$ is a member of $|2K_S|$, C_u is smooth at $\pi(p_l)$.

Let v_{ij} be the v_m with $r_m = y_iy_j$, and let $e_{ij} \in \mathbb{C}^6$ be the standard basis vector of the column with $r_m = y_iy_j$. Let v_{ik} be the v_m with $r_m = y_iy_k$, and let $e_{ik} \in \mathbb{C}^6$ be the standard basis vector of the column with $r_m = y_iy_k$. Let

$$c := v_{ik}(u)e_{ij} - v_{ij}(u)e_{ik}$$

if $v_{ij}(u)$ and $v_{ik}(u)$ are not both zero, and let $c := e_{ij}$ otherwise. Then $c \neq 0$. Since the columns $\{i,j\}$ and $\{i,k\}$ of $A(u)$ are zero, $A(u)c = 0$. By the choice of c, $v(u)c = 0$. Thus $c \in H_u$, and the quadric $\sum_m c_m r_m$ is

$$y_i\big(v_{ik}(u)y_j - v_{ij}(u)y_k\big)$$

in the first case and y_iy_j in the second case. In both cases, $\sum_m c_m r_m$ is a product of two linear forms in y_i, y_j, y_k.

Since $y_m(p_l) \neq 0$ if and only if $m = l$, the image of $\pi(p_l)$ under the tricanonical map is the point of $\mathbb{P}^3$ at which $y_m = 0$ for $m \neq l$. Since $l \notin \{i,j,k\}$, both linear forms vanish at the image of $\pi(p_l)$, so the image of $\pi(p_l)$ is a singular point of the quadric $\sum_m c_m r_m$. By Lemma 4.3(3), C_u is singular at $\pi(p_l)$, which contradicts the smoothness of C_u at $\pi(p_l)$. This proves the first conclusion.

Finally, suppose that the coefficient vectors of s and t in A are linearly dependent. Then there exists a non-zero binary linear form L such that any a_{ij} is a constant multiple of L. At the zero u of L, we have $a_{ij}(u) = 0$ for any $i \neq j$, so $\{0,1\}$ and $\{0,2\}$ are two distinct pairs with the common index 0 in the set of the first conclusion. This is a contradiction. □

6. Pencil matrices with events

In this section, we prove Theorem 6.9. The argument of Section 7 works on pencil matrices with no zero column at any point of $\mathbb{P}^1$, so the pencil matrices with an event are treated here by explicit families, listed according to the shape of the event graph.

6.1. **The event graph and three events.**

In this subsection, we define the event graph of a pencil matrix, determine the shapes of the event graph of a standard pencil matrix, and write down normal forms for the standard pencil matrices whose event graph has three edges.

Definition 6.1. Let A be a pencil matrix. The *event graph* J_A of A is the graph on $\{0,1,2,3\}$ with an edge $\{i,j\}$ if and only if a_{ij} and a_{ji} have a common zero on $\mathbb{P}^1$. An edge of J_A is called an *event*. We say that A is *event-free* if J_A has no edge. We denote by $\mathcal{M}_{\mathrm{ev}} \subset \mathcal{M}$ the set of points having a standard pencil matrix with at least one event, and by $\mathcal{M}_{\mathrm{ne}} \subset \mathcal{M}$ the set of points having an event-free standard pencil matrix. We call the action of the torus of Theorem 5.2 the *diagonal action*. By a *family* of dimension b we mean a non-empty open subset $\mathcal{V} \subset \mathbb{C}^b$ together with, for any point of $\mathcal{V}$, a matrix of the form of A of Construction-Proposition 5.1, whose coefficients are regular functions on $\mathcal{V}$.

An event means that the column $\{i,j\}$ of A vanishes at a point of $\mathbb{P}^1$. Since two binary linear forms have a common zero on $\mathbb{P}^1$ if and only if they are linearly dependent, $\{i,j\}$ is not an edge of J_A if and only if a_{ij} and a_{ji} are linearly independent. By Construction-Proposition 5.1 and Theorem 5.2, signed permutations, binary reparametrization, and the diagonal action send a standard pencil matrix to a standard pencil matrix. Since $\mathbb{C}^*$ is divisible, an integer $k \times k$ matrix with non-zero determinant defines a surjective endomorphism of $(\mathbb{C}^*)^k$.

Lemma 6.2. *Let A be a standard pencil matrix. Then the following hold.*

1. *J_A has at most three edges, and up to relabeling it is the empty graph, one edge, a path with two edges, two disjoint edges, a triangle, or a path with three edges.*
2. *For any edge $\{i,j\}$ of J_A, the forms a_{ij} and a_{ji} have exactly one common zero on $\mathbb{P}^1$, and two adjacent edges of J_A have distinct common zeros.*

Proof. We first prove (2). Let $\{i,j\}$ be an edge of J_A. By Lemma 5.7, at least one of a_{ij}, a_{ji} is a non-zero binary linear form, and a non-zero binary linear form has exactly one zero on $\mathbb{P}^1$. Thus a_{ij} and a_{ji} have exactly one common zero. Two adjacent edges of J_A have distinct common zeros by Lemma 5.8.

We now prove (1). A signed permutation relabels the vertices of J_A. Thus we may relabel the indices $0,1,2,3$ in the proof below.

Recall from Construction-Proposition 5.1 that a pencil matrix satisfies $q_0 = q_1 = q_2 = q_3 = 0$, where

$$q_0 = a_{12}a_{13} - a_{21}a_{23} + a_{31}a_{32}, \quad q_1 = a_{02}a_{03} - a_{30}a_{32} + a_{20}a_{23},$$

$$q_2 = a_{10}a_{13} - a_{01}a_{03} + a_{30}a_{31}, \quad q_3 = a_{01}a_{02} - a_{10}a_{12} + a_{20}a_{21}.$$

Step 1. In this step we prove that J_A has no vertex of degree three.

Suppose that J_A has a vertex of degree three. Possibly relabeling the indices, we may assume that $\{0,1\},\{0,2\},\{0,3\}$ are edges of J_A. For $k=1,2,3$, let L_k be a non-zero binary linear form whose zero u_k is the common zero of a_{0k} and a_{k0}. By (2), u_1,u_2,u_3 are distinct. Write

$$a_{0k} = c_{0k}L_k, \quad a_{k0} = c_{k0}L_k \quad (k=1,2,3),$$

where $(c_{0k}, c_{k0}) \neq (0,0)$ by Lemma 5.7. The three identities $q_3 = q_2 = q_1 = 0$ read

$$\begin{aligned} c_{01}c_{02}L_1L_2 - c_{10}L_1a_{12} + c_{20}L_2a_{21} &= 0,\\ c_{10}L_1a_{13} - c_{01}c_{03}L_1L_3 + c_{30}L_3a_{31} &= 0,\\ c_{02}c_{03}L_2L_3 - c_{30}L_3a_{32} + c_{20}L_2a_{23} &= 0. \end{aligned}$$

We prove that c_{10}, c_{20}, c_{30} are non-zero. Suppose that one of them is zero. Possibly relabeling $1,2,3$, we may assume that $c_{10} = 0$, hence $c_{01} \neq 0$. If $c_{20} = 0$, then $c_{02} \neq 0$ and $q_3 = c_{01}c_{02}L_1L_2 \neq 0$, a contradiction. If $c_{30} = 0$, then $c_{03} \neq 0$ and $q_2 = -c_{01}c_{03}L_1L_3 \neq 0$, a contradiction. Thus $c_{20}c_{30} \neq 0$, and $q_3 = q_2 = 0$ imply

$$a_{21} = -\frac{c_{01}c_{02}}{c_{20}}L_1, \quad a_{31} = \frac{c_{01}c_{03}}{c_{30}}L_1.$$

Hence $a_{21}(u_1) = a_{31}(u_1) = 0$, and $q_0(u_1) = 0$ implies $a_{12}(u_1)a_{13}(u_1) = 0$. Thus the column $\{1,2\}$ or the column $\{1,3\}$ vanishes at u_1. Since the column $\{0,1\}$ vanishes at u_1, we obtain a contradiction to Lemma 5.8.

Since c_{10}, c_{20}, c_{30} are non-zero and u_1,u_2,u_3 are distinct, evaluating $q_3 = 0$ at u_1,u_2, $q_2 = 0$ at u_1,u_3, and $q_1 = 0$ at u_2,u_3 implies

$$a_{21}(u_1) = a_{12}(u_2) = a_{31}(u_1) = a_{13}(u_3) = a_{32}(u_2) = a_{23}(u_3) = 0.$$

Thus $a_{ij} = c_{ij}L_j$ for any distinct $i,j \in \{1,2,3\}$, where $c_{ij} \in \mathbb{C}$. For any distinct $i,j \in \{1,2,3\}$, the columns $\{0,i\}$ and $\{i,j\}$ do not both vanish at u_i by Lemma 5.8. Since $a_{ji}(u_i) = 0$ and the column $\{0,i\}$ vanishes at u_i, we have $a_{ij}(u_i) \neq 0$, hence $c_{ij} \neq 0$. Finally,

$$q_0 = c_{12}c_{13}L_2L_3 - c_{21}c_{23}L_1L_3 + c_{31}c_{32}L_1L_2,$$

so $q_0(u_1) = c_{12}c_{13}L_2(u_1)L_3(u_1) \neq 0$, a contradiction.

Step 2. In this step we prove that J_A is not a four-cycle.

Suppose that J_A is a four-cycle. Possibly relabeling the indices, we may assume that the edges of J_A are $\{0,1\},\{1,2\},\{2,3\},\{0,3\}$. Let L_1,L_2,L_3,L_4 be non-zero binary linear forms whose zeros u_1,u_2,u_3,u_4 are the respective common zeros of the columns $\{0,1\},\{1,2\},\{2,3\},\{0,3\}$. Write

$$\begin{aligned} a_{01} &= c_{01}L_1, \quad a_{12} = c_{12}L_2, \quad a_{23} = c_{23}L_3, \quad a_{30} = c_{30}L_4,\\ a_{10} &= c_{10}L_1, \quad a_{21} = c_{21}L_2, \quad a_{32} = c_{32}L_3, \quad a_{03} = c_{03}L_4. \end{aligned}$$

By Lemma 5.7, the constants attached to one edge are not both zero. By (2), u_1 and u_3 are both different from u_2 and u_4. Since $\{0,2\}$ and $\{1,3\}$ are not edges of J_A, the forms $a_{02}, a_{20}, a_{13}, a_{31}$ are non-zero. The identities $q_0 = q_1 = q_2 = q_3 = 0$ read

$$\begin{aligned} c_{12}L_2a_{13} - c_{21}c_{23}L_2L_3 + c_{32}L_3a_{31} &= 0,\\ c_{03}L_4a_{02} - c_{30}c_{32}L_3L_4 + c_{23}L_3a_{20} &= 0,\\ c_{10}L_1a_{13} - c_{01}c_{03}L_1L_4 + c_{30}L_4a_{31} &= 0,\\ c_{01}L_1a_{02} - c_{10}c_{12}L_1L_2 + c_{21}L_2a_{20} &= 0. \end{aligned}$$

We prove that

$$c_{01}c_{03} = c_{10}c_{12} = c_{21}c_{23} = c_{30}c_{32} = 0.$$

If $c_{01}c_{03} \neq 0$, then $q_3(u_2) = 0$ and $q_1(u_3) = 0$ imply $a_{02}(u_2) = a_{02}(u_3) = 0$, which is not possible since $a_{02} \neq 0$ and $u_2 \neq u_3$. If $c_{10}c_{12} \neq 0$, then $q_2(u_4) = 0$ and $q_0(u_3) = 0$ imply $a_{13}(u_4) = a_{13}(u_3) = 0$, which is not possible since $a_{13} \neq 0$ and $u_3 \neq u_4$. If $c_{21}c_{23} \neq 0$, then $q_3(u_1) = 0$ and $q_1(u_4) = 0$ imply $a_{20}(u_1) = a_{20}(u_4) = 0$, which is not possible since $a_{20} \neq 0$ and $u_1 \neq u_4$. If $c_{30}c_{32} \neq 0$, then $q_2(u_1) = 0$ and $q_0(u_2) = 0$ imply $a_{31}(u_1) = a_{31}(u_2) = 0$, which is not possible since $a_{31} \neq 0$ and $u_1 \neq u_2$.

Each of the pairs (c_{01}, c_{10}), (c_{12}, c_{21}), (c_{23}, c_{32}), (c_{30}, c_{03}) has a non-zero entry, and each of the pairs (c_{01}, c_{03}), (c_{10}, c_{12}), (c_{21}, c_{23}), (c_{30}, c_{32}) has a zero entry. Thus either

$$c_{01}, c_{12}, c_{23}, c_{30} \neq 0 \quad \text{and} \quad c_{10} = c_{21} = c_{32} = c_{03} = 0,$$

or

$$c_{10}, c_{21}, c_{32}, c_{03} \neq 0 \quad \text{and} \quad c_{01} = c_{12} = c_{23} = c_{30} = 0.$$

In the first case $q_3 = c_{01}L_1a_{02} \neq 0$, and in the second case $q_3 = c_{21}L_2a_{20} \neq 0$, a contradiction.

Step 3. In this step we conclude the proof.

By **Step 1**, J_A is a disjoint union of paths, cycles, and isolated vertices. If J_A contains a cycle, then by **Step 2** the cycle is a triangle, and the remaining vertex is isolated. If J_A contains no cycle, then J_A is the empty graph, one edge, a path with two edges, two disjoint edges, or a path with three edges. In any case, J_A has at most three edges. □

Construction-Proposition 6.3. Let A be a standard pencil matrix whose event graph has three edges. Then, up to signed permutations, binary reparametrization, and the diagonal action, A belongs to one of the following two families of dimension 2, with $(g,h) \in (\mathbb{C}^*)^2$:

(1) (triangle)

$$\begin{aligned} a_{01} &= s, & a_{02} &= 0, & a_{03} &= gt,\\ a_{10} &= 0, & a_{12} &= t, & a_{13} &= -gh(s+t),\\ a_{20} &= s+t, & a_{21} &= 0, & a_{23} &= hs,\\ a_{30} &= s, & a_{31} &= gt, & a_{32} &= h(s+t); \end{aligned}$$

(2) (path)

$$\begin{aligned} a_{01} &= 0, & a_{02} &= g(s+t), & a_{03} &= -s/g,\\ a_{10} &= s, & a_{12} &= t, & a_{13} &= s+t,\\ a_{20} &= s, & a_{21} &= t, & a_{23} &= s+t,\\ a_{30} &= h(s+t), & a_{31} &= -s/h, & a_{32} &= 0. \end{aligned}$$

Moreover, any matrix of families (1) and (2) is a pencil matrix.

Proof. By Lemma 6.2(1), J_A is a triangle or a path with three edges. A signed permutation relabels the vertices of J_A.

Case 1. J_A is a triangle.

Possibly relabeling the indices, we may assume that the edges of J_A are $\{0,1\},\{1,2\},\{0,2\}$. Let L_1, L_2, L_3 be non-zero binary linear forms whose zeros u_1, u_2, u_3 are the respective common zeros of the columns $\{0,1\},\{1,2\},\{0,2\}$. By Lemma 6.2(2), u_1, u_2, u_3 are distinct. Write

$$a_{01} = c_{01}L_1, \quad a_{10} = c_{10}L_1, \quad a_{12} = c_{12}L_2, \quad a_{21} = c_{21}L_2, \quad a_{02} = c_{02}L_3, \quad a_{20} = c_{20}L_3,$$

where the constants attached to one edge are not both zero by Lemma 5.7.

The quadratic forms L_1L_2, L_1L_3, L_2L_3 are linearly independent, as evaluating a linear relation at u_1, u_2, u_3 shows. Since

$$q_3 = c_{01}c_{02}L_1L_3 - c_{10}c_{12}L_1L_2 + c_{20}c_{21}L_2L_3 = 0,$$

we have $c_{01}c_{02} = c_{10}c_{12} = c_{20}c_{21} = 0$. Each of the pairs (c_{01}, c_{10}), (c_{12}, c_{21}), (c_{02}, c_{20}) has a non-zero entry. Thus either $c_{01}, c_{12}, c_{20} \neq 0$ and $c_{10} = c_{21} = c_{02} = 0$, or $c_{10}, c_{21}, c_{02} \neq 0$ and $c_{01} = c_{12} = c_{20} = 0$. Possibly exchanging the indices $1, 2$ and renaming L_1, L_2, L_3 accordingly, we may assume that

$$a_{01} = c_{01}L_1, \quad a_{12} = c_{12}L_2, \quad a_{20} = c_{20}L_3, \quad a_{10} = a_{21} = a_{02} = 0,$$

where $c_{01}c_{12}c_{20} \neq 0$.

The identities $q_2 = q_1 = q_0 = 0$ now read

$$a_{30}a_{31} = a_{01}a_{03}, \quad a_{30}a_{32} = a_{20}a_{23}, \quad a_{31}a_{32} = -a_{12}a_{13}.$$

Since a_{01}, a_{12}, a_{20} are non-zero, if $a_{3j} = 0$ for some $j \in \{0,1,2\}$, then $q_2 = q_1 = q_0 = 0$ imply $a_{j3} = 0$, which contradicts Lemma 5.7. Thus a_{30}, a_{31}, a_{32} are non-zero. Moreover, u_1 is a zero of $a_{30}a_{31}$, u_3 is a zero of $a_{30}a_{32}$, and u_2 is a zero of $a_{31}a_{32}$. Since u_1, u_2, u_3 are distinct and each of a_{30}, a_{31}, a_{32} has exactly one zero, the zeros of (a_{30}, a_{31}, a_{32}) are either (u_1, u_2, u_3) or (u_3, u_1, u_2).

Suppose that the zeros of (a_{30}, a_{31}, a_{32}) are (u_3, u_1, u_2). Then $a_{01}a_{03} = a_{30}a_{31}$ is a non-zero multiple of L_1L_3, hence $a_{03}(u_3) = 0$. Thus the column $\{0,3\}$ vanishes at u_3. Since the column $\{0,2\}$ also vanishes at u_3, we obtain a contradiction to Lemma 5.8.

Thus there exist non-zero constants c_{30}, c_{31}, c_{32} with

$$a_{30} = c_{30}L_1, \quad a_{31} = c_{31}L_2, \quad a_{32} = c_{32}L_3,$$

and the three identities above imply

$$a_{03} = \frac{c_{30}c_{31}}{c_{01}}L_2, \quad a_{13} = -\frac{c_{31}c_{32}}{c_{12}}L_3, \quad a_{23} = \frac{c_{30}c_{32}}{c_{20}}L_1.$$

A binary reparametrization sends u_1, u_2, u_3 to the zeros of $s, t, s+t$. Possibly rescaling L_1, L_2, L_3 and changing the constants c_{ij} accordingly, we may assume that $L_1 = s$, $L_2 = t$, and $L_3 = s + t$. Under the diagonal action, the constants c_{01}, c_{12}, c_{20} are multiplied by

$$\lambda_0\lambda_1^2, \quad \lambda_1\lambda_2^2, \quad \lambda_2\lambda_0^2.$$

The exponent matrix has determinant 9. Thus we may choose $\lambda_0, \lambda_1, \lambda_2$ with $c_{01} = c_{12} = c_{20} = 1$. Under the diagonal action, the constant c_{30} is multiplied by $\lambda_3\lambda_0^2$, so we may choose λ_3 with $c_{30} = 1$. Let $g := c_{31}$ and $h := c_{32}$. Then A belongs to family (1).

Case 2. J_A is a path with three edges.

Possibly relabeling the indices, we may assume that the edges of J_A are $\{0,1\},\{1,2\},\{2,3\}$. Let L_1, L_2, L_3 be non-zero binary linear forms whose zeros u_1, u_2, u_3 are the respective common zeros of the columns $\{0,1\},\{1,2\},\{2,3\}$. Write

$$a_{01} = c_{01}L_1, \quad a_{10} = c_{10}L_1, \quad a_{12} = c_{12}L_2, \quad a_{21} = c_{21}L_2, \quad a_{23} = c_{23}L_3, \quad a_{32} = c_{32}L_3,$$

where the constants attached to one edge are not both zero by Lemma 5.7. By Lemma 6.2(2), $u_2 \neq u_1$ and $u_2 \neq u_3$. Since $\{0,2\},\{0,3\},\{1,3\}$ are not edges of J_A, the forms $a_{02}, a_{20}, a_{03}, a_{30}, a_{13}, a_{31}$

are non-zero. The identities $q_0 = q_1 = q_2 = q_3 = 0$ read

$$\begin{aligned} c_{12}L_2a_{13} - c_{21}c_{23}L_2L_3 + c_{32}L_3a_{31} &= 0,\\ a_{02}a_{03} - c_{32}L_3a_{30} + c_{23}L_3a_{20} &= 0,\\ c_{10}L_1a_{13} - c_{01}L_1a_{03} + a_{30}a_{31} &= 0,\\ c_{01}L_1a_{02} - c_{10}c_{12}L_1L_2 + c_{21}L_2a_{20} &= 0. \end{aligned}$$

We first prove that $c_{01}c_{32} = 0$. Suppose that $c_{01}c_{32} \neq 0$. Then $q_3(u_2) = 0$ and $q_0(u_2) = 0$ imply $a_{02}(u_2) = a_{31}(u_2) = 0$. Since $u_3 \neq u_2$ and $u_1 \neq u_2$, we have $a_{02}(u_3) \neq 0$ and $a_{31}(u_1) \neq 0$. Thus $q_1(u_3) = 0$ and $q_2(u_1) = 0$ imply $a_{03}(u_3) = a_{30}(u_1) = 0$. Hence there exist non-zero constants $c_{02}, c_{31}, c_{03}, c_{30}$ with

$$a_{02} = c_{02}L_2, \quad a_{31} = c_{31}L_2, \quad a_{03} = c_{03}L_3, \quad a_{30} = c_{30}L_1.$$

If $c_{21} \neq 0$, then $q_3(u_1) = 0$ implies $a_{20} = c_{20}L_1$ for some $c_{20} \in \mathbb{C}$, and dividing $q_1 = 0$ by L_3 implies

$$c_{02}c_{03}L_2 + (c_{20}c_{23} - c_{30}c_{32})L_1 = 0,$$

which is not possible since $c_{02}c_{03} \neq 0$ and $u_1 \neq u_2$. Thus $c_{21} = 0$, hence $c_{12} \neq 0$. Now $q_0 = 0$ implies

$$a_{13} = -\frac{c_{31}c_{32}}{c_{12}}L_3,$$

and dividing $q_2 = 0$ by L_1 implies

$$-\left(\frac{c_{10}c_{31}c_{32}}{c_{12}} + c_{01}c_{03}\right)L_3 + c_{30}c_{31}L_2 = 0,$$

which is not possible since $c_{30}c_{31} \neq 0$ and $u_2 \neq u_3$. Thus $c_{01}c_{32} = 0$.

The relabeling $i \to 3 - i$ preserves the edges of J_A and exchanges c_{01} and c_{32} up to sign. Possibly relabeling the indices, we may assume that $c_{01} = 0$, hence $c_{10} \neq 0$. Now $q_3 = 0$ reads

$$c_{21}a_{20} = c_{10}c_{12}L_1.$$

If $c_{21} = 0$, then $c_{12} \neq 0$, and the identity is not possible. Thus $c_{21} \neq 0$. Since $a_{20} \neq 0$, we have $c_{12} \neq 0$ and

$$a_{20} = \frac{c_{10}c_{12}}{c_{21}}L_1.$$

Suppose that $c_{32} \neq 0$. Then $q_0(u_2) = 0$ implies $a_{31}(u_2) = 0$. Since $c_{01} = 0$, the identity $q_2 = 0$ reads $c_{10}L_1a_{13} = -a_{30}a_{31}$, hence $a_{13}(u_2) = 0$. Thus the column $\{1, 3\}$ vanishes at u_2, which is not possible since $\{1, 3\}$ is not an edge of J_A. Thus $c_{32} = 0$, hence $c_{23} \neq 0$. Now $q_0 = 0$ implies

$$a_{13} = \frac{c_{21}c_{23}}{c_{12}}L_3,$$

and $q_1 = q_2 = 0$ read

$$a_{02}a_{03} = -\frac{c_{10}c_{12}c_{23}}{c_{21}}L_1L_3, \quad a_{30}a_{31} = -\frac{c_{10}c_{21}c_{23}}{c_{12}}L_1L_3.$$

Since $a_{20}(u_1) = 0$ and $\{0, 2\}$ is not an edge of J_A, we have $a_{02}(u_1) \neq 0$. Thus $a_{02} = c_{02}L_3$ and $a_{03} = c_{03}L_1$ with $c_{02}c_{03} \neq 0$. In particular, $L_3(u_1) \neq 0$, hence $u_1 \neq u_3$. Since $a_{13}(u_3) = 0$ and $\{1, 3\}$ is not an edge of J_A, we have $a_{31}(u_3) \neq 0$. Thus $a_{30} = c_{30}L_3$ and $a_{31} = c_{31}L_1$ with $c_{30}c_{31} \neq 0$. Moreover,

$$c_{02}c_{03} = -\frac{c_{10}c_{12}c_{23}}{c_{21}}, \quad c_{30}c_{31} = -\frac{c_{10}c_{21}c_{23}}{c_{12}}.$$

A binary reparametrization sends u_1, u_2, u_3 to the zeros of $s, t, s + t$. Possibly rescaling L_1, L_2, L_3 and changing the constants c_{ij} accordingly, we may assume that $L_1 = s$, $L_2 = t$, and $L_3 = s + t$. Under the diagonal action, the constants $c_{10}, c_{12}, c_{21}, c_{23}$ are multiplied by

$$\lambda_1\lambda_0^2, \quad \lambda_1\lambda_2^2, \quad \lambda_2\lambda_1^2, \quad \lambda_2\lambda_3^2.$$

The exponent matrix has determinant -12. Thus we may choose $\lambda_0, \lambda_1, \lambda_2, \lambda_3$ with $c_{10} = c_{12} = c_{21} = c_{23} = 1$. Then $a_{20} = s$, $a_{13} = s + t$, and $c_{02}c_{03} = c_{30}c_{31} = -1$. Let $g := c_{02}$ and $h := c_{30}$. Then A belongs to family (2).

Conversely, direct substitution implies that $q_0 = q_1 = q_2 = q_3 = 0$ on families (1) and (2). □

6.2. **Two events.** In this subsection, we determine the normal forms of the standard pencil matrices whose event graph has exactly two edges, which are used in Construction-Proposition 6.8.

Construction-Proposition 6.4. Let A be a standard pencil matrix whose event graph has exactly two edges, and assume that the edges are adjacent. Then, up to signed permutations, binary reparametrization, and the diagonal action, A belongs to the following family of dimension 3, with $(p, q, w) \in \mathbb{C}^3$, $pqw(p-1)(q-1) \neq 0$, and $D := (1-p)(1-q)$:

$$a_{01} = 0,\ a_{02} = w(s+pt),\ a_{03} = -(s+qt)/(Dw),\ a_{10} = s,\ a_{12} = t,$$

$a_{13} = ((p+q-pq)s+pqt)/D$, $a_{20} = s$, $a_{21} = t$, $a_{23} = (s+pqt)/D$, $a_{30} = -a_{13}$, $a_{31} = s$, $a_{32} = t$. Moreover, any such matrix is a pencil matrix.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we normalize the edges of J_A and show that $a_{01} = 0$ after possibly reversing the path formed by the edges.

After a signed permutation, the edges of J_A are $\{0,1\}$ and $\{1,2\}$. By Lemma 6.2(2), the forms a_{01}, a_{10} have exactly one common zero u_1, the forms a_{12}, a_{21} have exactly one common zero u_2, and $u_1 \neq u_2$. After a binary reparametrization, u_1 is the point $s = 0$ and u_2 is the point $t = 0$. Thus

$$a_{01} = c_{01}s, \quad a_{10} = c_{10}s, \quad a_{12} = c_{12}t, \quad a_{21} = c_{21}t,$$

where $c_{01}, c_{10}, c_{12}, c_{21} \in \mathbb{C}$, and $(c_{01}, c_{10}) \neq (0,0)$ and $(c_{12}, c_{21}) \neq (0,0)$ by Lemma 5.7.

Let $\{i,j\}$ be one of $\{0,2\}, \{0,3\}, \{1,3\}, \{2,3\}$. Since $\{i,j\}$ is not an edge of J_A and linearly dependent binary linear forms have a common zero on $\mathbb{P}^1$, the forms a_{ij} and a_{ji} are non-zero and linearly independent. The identities $q_0 = q_1 = q_2 = q_3 = 0$ read

$$\begin{aligned} &c_{12}ta_{13} - c_{21}ta_{23} + a_{31}a_{32} = 0,\\ &a_{02}a_{03} - a_{30}a_{32} + a_{20}a_{23} = 0,\\ &c_{10}sa_{13} - c_{01}sa_{03} + a_{30}a_{31} = 0,\\ &c_{01}sa_{02} - c_{10}c_{12}st + c_{21}ta_{20} = 0. \end{aligned} \tag{6.1}$$

Suppose that $c_{01}c_{21} \neq 0$. Setting $t = 0$ and $s = 0$ in the fourth identity of (6.1), we obtain $a_{02} = c_{02}t$ and $a_{20} = c_{20}s$ with $c_{02}c_{20} \neq 0$. If a_{32} were a multiple of t, then setting $t = 0$ in the second identity would imply that a_{23} is a multiple of t, so $\{2,3\}$ would be an edge of J_A. Thus a_{32} is not a multiple of t, and setting $t = 0$ in the first identity implies that a_{31} is a non-zero multiple of t. Setting $s = 0$ in the third identity, we see that a_{30} is a multiple of s. Setting $s = 0$ in the second identity, we see that a_{03} is a multiple of s, so $\{0,3\}$ is an edge of J_A, a contradiction. Thus $c_{01}c_{21} = 0$.

The signed permutation attached to $\sigma = (0,2)$ followed by the exchange of s and t preserves the set of edges and exchanges c_{01} and c_{21} up to sign. Since $c_{01}c_{21} = 0$, possibly replacing A with the image of A under the signed permutation and the exchange, we may assume that $c_{01} = 0$. Then $a_{01} = 0$ and $c_{10} \neq 0$. The fourth identity of (6.1) now reads $c_{21}a_{20} = c_{10}c_{12}s$. If $c_{21} = 0$, then $c_{12} \neq 0$ and $c_{10}c_{12}s = 0$, which is impossible. Thus $c_{21} \neq 0$. Since $a_{20} \neq 0$, we have $c_{12} \neq 0$ and $a_{20} = (c_{10}c_{12}/c_{21})s$.

Step 2. In this step we determine $a_{30}, a_{31}, a_{32}, a_{23}$ in terms of a_{13} and normalize the scalars.

The third identity of (6.1) now reads $a_{30}a_{31} = -c_{10}sa_{13}$. Since a_{13} and a_{31} are non-zero and not proportional, unique factorization in $\mathbb{C}[s,t]$ implies that

$$a_{31} = c_{31}s, \quad a_{30} = c_{30}a_{13}, \quad c_{31}c_{30} = -c_{10},$$

with $c_{31}c_{30} \neq 0$. Setting $t = 0$ in the first identity, we see that $a_{32} = c_{32}t$ with $c_{32} \neq 0$. Dividing the first identity by t, we obtain

$$a_{23} = \frac{c_{12}a_{13} + c_{31}c_{32}s}{c_{21}}.$$

The diagonal action multiplies $c_{10}, c_{12}, c_{21}, c_{31}$ by

$$\lambda_1\lambda_0^2, \quad \lambda_1\lambda_2^2, \quad \lambda_2\lambda_1^2, \quad \lambda_3\lambda_1^2$$

respectively. The exponent matrix of the characters on $c_{10}, c_{12}, c_{21}, c_{31}$ has determinant -6. Thus we may assume that $c_{10} = c_{12} = c_{21} = c_{31} = 1$, that is,

$$a_{20} = s, \quad a_{31} = s, \quad a_{30} = -a_{13}, \quad a_{32} = c_{32}t, \quad a_{23} = a_{13} + c_{32}s.$$

Let $\kappa := c_{32}^{-1}$, and let $\eta \in \mathbb{C}^*$ be a number with $\eta^6 = \kappa$. Apply the substitution $t \to \kappa t$ and then the diagonal action of

$$(\lambda_0, \lambda_1, \lambda_2, \lambda_3) = (\eta, \eta^{-2}, \eta^{-2}, \eta^4).$$

The characters on $a_{10}, a_{20}, a_{31}, a_{32}$ equal 1, and the characters on a_{12}, a_{21} equal $\eta^{-6} = \kappa^{-1}$. Thus $a_{10}, a_{12}, a_{20}, a_{21}, a_{31}$ do not change and a_{32} becomes t. Therefore, we may assume that $c_{32} = 1$, so that

$$a_{32} = t, \quad a_{23} = a_{13} + s.$$

Step 3. In this step we factor $a_{02}a_{03}$ and conclude the proof.

Write $a_{13} = \alpha s + \beta t$. Since a_{13}, a_{31} are linearly independent and $a_{31} = s$, we have $\beta \neq 0$. Since $a_{23} = (\alpha+1)s + \beta t$ and $a_{32} = t$ are linearly independent, we have $\alpha + 1 \neq 0$. The second identity of (6.1) reads

$$a_{02}a_{03} = -ta_{13} - s(a_{13} + s) = -(\alpha+1)s^2 - (\alpha+\beta)st - \beta t^2.$$

Since the coefficients of s^2 and t^2 are non-zero, both a_{02} and a_{03} have non-zero coefficients of s and of t. Thus

$$a_{02} = w(s + pt), \quad a_{03} = -\frac{(\alpha+1)(s+qt)}{w},$$

with $pqw \neq 0$. Comparing the coefficients, we obtain

$$p + q = \frac{\alpha+\beta}{\alpha+1}, \quad pq = \frac{\beta}{\alpha+1}.$$

It follows that

$$D = (1-p)(1-q) = \frac{1}{\alpha+1} \neq 0, \quad \alpha = \frac{p+q-pq}{D}, \quad \beta = \frac{pq}{D}.$$

Thus $a_{13} = ((p+q-pq)s + pqt)/D$, $a_{30} = -a_{13}$, $a_{03} = -(s+qt)/(Dw)$, and $a_{23} = a_{13} + s = (s + pqt)/D$, which are the forms of the statement.

Conversely, for any $(p, q, w) \in \mathbb{C}^3$ with $pqw(p-1)(q-1) \neq 0$, direct substitution implies that $q_0 = q_1 = q_2 = q_3 = 0$. □

Construction-Proposition 6.5. Let A be a standard pencil matrix whose event graph consists of two disjoint edges $\{i, j\}$ and $\{k, l\}$, and let $u, u' \in \mathbb{P}^1$ be the common zeros of a_{ij}, a_{ji} and of a_{kl}, a_{lk}. Then, up to signed permutations, binary reparametrization, and the diagonal action, A belongs to one of the five families of Table 3, which lists their dimensions: to (I), (II), or (III) if $u \neq u'$, and to (IV) or (V) if $u = u'$. Moreover, any matrix of Table 3 is a pencil matrix.

Proof of Construction-Proposition 6.5. The proof is long, so we divide it into several steps.

After a signed permutation, the edges of J_A are $\{0, 1\}$ and $\{2, 3\}$. The common zero of each edge is unique by Lemma 6.2(2). For any $i \in \{0, 1\}$ and $j \in \{2, 3\}$, the pair $\{i, j\}$ is not an edge of J_A, and linearly dependent binary linear forms have a common zero on $\mathbb{P}^1$, so a_{ij} and a_{ji} are non-zero and linearly independent.

TABLE 3. The five families of Construction-Proposition 6.5.

family	dim	coordinates and entries
(I)	3	$a_{01}=a_{10}=s$, $a_{23}=a_{32}=t$, $a_{02}=t$, $a_{20}=ps$, $a_{13}=qt$, $a_{31}=ws$, $a_{12}=(pws+t)/(1-pq)$, $a_{21}=(ws+qt)/(1-pq)$, $a_{03}=(pws+qt)/(1-w)$, $a_{30}=(ps+qt)/(1-w)$, $pqw(1-pq)(1-w)\neq 0$.
(II)	3	$\Delta:=1-pqw$, $a_{01}=a_{10}=s$, $a_{23}=a_{32}=t$, $a_{02}=t$, $a_{13}=pt$, $a_{30}=qs$, $a_{21}=ws$, $a_{03}=(qw(1-pq)s+p(1-q)t)/\Delta$, $a_{12}=(qw(1-w)s+(1-pw)t)/\Delta$, $a_{20}=(q(1-w)s-p(1-q)t)/\Delta$, $a_{31}=(w(1-pq)s+p(pw-1)t)/\Delta$, $pqw\Delta(1-pq)(1-w)(1-q)(1-pw)\neq 0$.
(III)	2	$a_{01}=a_{10}=s$, $a_{23}=a_{32}=t$, $a_{02}=t$, $a_{13}=t$, $a_{30}=s$, $a_{21}=s$, $a_{03}=ps+(q+1)t$, $a_{12}=(1-p)s-qt$, $a_{20}=(1-p)s-(q+1)t$, $a_{31}=ps+qt$, $pq(p-1)(q+1)\neq 0$.
(IV)	2	$\gamma:=(1+p^2q)/(p(1+pq^2))$, $\delta:=(1+pq^2)/(q(1+p^2q))$, $a_{01}=a_{10}=a_{23}=a_{32}=s$, $a_{02}=t$, $a_{20}=ps$, $a_{03}=\gamma s$, $a_{30}=\gamma t+ps$, $a_{12}=\delta s$, $a_{21}=(\delta s-t)/p$, $a_{13}=(\gamma-pq)s-\gamma qt$, $a_{31}=qs$, $pq(1+p^2q)(1+pq^2)\neq 0$.
(V)	1	$a_{01}=a_{10}=a_{23}=a_{32}=s$, $a_{02}=t$, $a_{20}=-s$, $a_{03}=ws$, $a_{30}=wt-s$, $a_{12}=s/w$, $a_{21}=t-s/w$, $a_{13}=(w-1)s+wt$, $a_{31}=-s$, $w\neq 0$.

Step 1. In this step we suppose that $u\neq u'$ and determine which forms vanish at the points u,u'.

After a binary reparametrization, u is the point $s=0$ and u' is the point $t=0$. Thus $a_{01}=c_{01}s$, $a_{10}=c_{10}s$, $a_{23}=c_{23}t$, and $a_{32}=c_{32}t$, where $c_{01},c_{10},c_{23},c_{32}\in\mathbb{C}$, and the identities $q_0=q_1=q_2=q_3=0$ read

$$\begin{aligned} a_{12}a_{13}&=t(c_{23}a_{21}-c_{32}a_{31}),\\ a_{02}a_{03}&=t(c_{32}a_{30}-c_{23}a_{20}),\\ a_{30}a_{31}&=s(c_{01}a_{03}-c_{10}a_{13}),\\ a_{20}a_{21}&=s(c_{10}a_{12}-c_{01}a_{02}). \end{aligned} \tag{6.2}$$

Setting $t=0$ in the first and the second identity, we see that one of a_{12},a_{13} and one of a_{02},a_{03} are multiples of t. Suppose that a_{02} and a_{12} are both multiples of t. Then the fourth identity implies that $a_{20}a_{21}$ is a multiple of st. Since $a_{20}a_{21}\neq 0$, one of a_{20},a_{21} is a multiple of t, so $\{0,2\}$ or $\{1,2\}$ is an edge of J_A, a contradiction. Suppose that a_{03} and a_{13} are both multiples of t. Then the third identity implies that $a_{30}a_{31}$ is a multiple of st, so one of a_{30},a_{31} is a multiple of t, and $\{0,3\}$ or $\{1,3\}$ is an edge of J_A, a contradiction. Thus the forms among $a_{02},a_{03},a_{12},a_{13}$ that are multiples of t are either a_{02},a_{13} or a_{03},a_{12}. Possibly replacing A with the image of A under the signed permutation attached to $\sigma=(0,1)$, we may assume that

$$a_{02}=c_{02}t,\quad a_{13}=c_{13}t,\quad c_{02}c_{13}\neq 0.$$

Setting $s=0$ in the third and the fourth identity of (6.2), we see that one of a_{30},a_{31} and one of a_{20},a_{21} are multiples of s. Suppose that a_{20} and a_{30} are both multiples of s. Then the second identity implies that $a_{02}a_{03}$ is a multiple of st. Since $a_{02}=c_{02}t$, a_{03} is a multiple of s, so $\{0,3\}$ is an edge of J_A, a contradiction. Suppose that a_{21} and a_{31} are both multiples of s. Then the first identity implies that $a_{12}a_{13}$ is a multiple of st. Since $a_{13}=c_{13}t$, a_{12} is a multiple of s, so $\{1,2\}$ is an edge of J_A, a contradiction. Thus precisely one of the following cases holds, where $c_{20},c_{31},c_{30},c_{21}$ are non-zero numbers:

- Case 1: $a_{20}=c_{20}s$ and $a_{31}=c_{31}s$.
- Case 2: $a_{30}=c_{30}s$ and $a_{21}=c_{21}s$.

Step 2. In this step we still suppose that $u \neq u'$, we show that $c_{01}, c_{10}, c_{23}, c_{32}$ are non-zero, and we normalize $c_{01}, c_{10}, c_{23}, c_{32}, c_{02}$ to 1.

In Case 1, the identities (6.2) read

$$\begin{aligned} c_{13}a_{12} &= c_{23}a_{21} - c_{32}c_{31}s, \quad & c_{02}a_{03} &= c_{32}a_{30} - c_{23}c_{20}s, \\ c_{31}a_{30} &= c_{01}a_{03} - c_{10}c_{13}t, \quad & c_{20}a_{21} &= c_{10}a_{12} - c_{01}c_{02}t. \end{aligned} \tag{6.3}$$

If $c_{01} = 0$ or $c_{32} = 0$, then the fourth or the first identity of (6.3) implies that a_{12}, a_{21} are proportional. If $c_{10} = 0$ or $c_{23} = 0$, then the third or the second identity of (6.3) implies that a_{03}, a_{30} are proportional. Since a_{12}, a_{21} and a_{03}, a_{30} are linearly independent, we have $c_{01}c_{10}c_{23}c_{32} \neq 0$ in Case 1.

In Case 2, the identities (6.2) read

$$\begin{aligned} c_{13}a_{12} &= c_{23}c_{21}s - c_{32}a_{31}, \quad & c_{02}a_{03} &= c_{32}c_{30}s - c_{23}a_{20}, \\ c_{30}a_{31} &= c_{01}a_{03} - c_{10}c_{13}t, \quad & c_{21}a_{20} &= c_{10}a_{12} - c_{01}c_{02}t. \end{aligned} \tag{6.4}$$

If $c_{10} = 0$, then the fourth identity of (6.4) implies that a_{20} is a multiple of t, so $\{0, 2\}$ is an edge of J_A. If $c_{01} = 0$, then the third identity implies that a_{31} is a multiple of t, so $\{1, 3\}$ is an edge of J_A. If $c_{32} = 0$, then the first identity implies that a_{12} is a multiple of s, so $\{1, 2\}$ is an edge of J_A. If $c_{23} = 0$, then the second identity implies that a_{03} is a multiple of s, so $\{0, 3\}$ is an edge of J_A. Since J_A has no edge other than $\{0, 1\}$ and $\{2, 3\}$, we have $c_{01}c_{10}c_{23}c_{32} \neq 0$ in Case 2.

The diagonal action multiplies $c_{01}, c_{10}, c_{23}, c_{32}$ by

$$\lambda_0\lambda_1^2, \quad \lambda_1\lambda_0^2, \quad \lambda_2\lambda_3^2, \quad \lambda_3\lambda_2^2$$

respectively. The exponent matrix of the characters on $c_{01}, c_{10}, c_{23}, c_{32}$ has determinant 9. Thus we may assume that $c_{01} = c_{10} = c_{23} = c_{32} = 1$.

Let $\mu := c_{02}^{-1}$. Apply the substitution $s \to \mu^{-3}s$ and then the diagonal action of $(\mu, \mu, 1, 1)$. The characters on a_{01}, a_{10} equal μ^3 and the characters on a_{23}, a_{32} equal 1, so $c_{01} = c_{10} = c_{23} = c_{32} = 1$ still holds. The forms a_{02}, a_{13} are multiplied by μ, and the multiples of s in Case 1 and in Case 2 stay multiples of s. Thus we may assume that $c_{02} = 1$.

Step 3. In this step we suppose that $u \neq u'$ and that Case 1 holds, and we obtain family (I).

The identities (6.3) read

$$c_{13}a_{12} = a_{21} - c_{31}s, \quad a_{03} = a_{30} - c_{20}s, \quad c_{31}a_{30} = a_{03} - c_{13}t, \quad c_{20}a_{21} = a_{12} - t.$$

Eliminating a_{21} and a_{03}, we obtain

$$(1 - c_{20}c_{13})a_{12} = c_{20}c_{31}s + t, \quad (1 - c_{31})a_{30} = c_{20}s + c_{13}t.$$

If $c_{20}c_{13} = 1$, then $c_{20}c_{31}s + t = 0$, which is impossible. If $c_{31} = 1$, then $c_{20}s + c_{13}t = 0$, which is impossible. Thus $(1 - c_{20}c_{13})(1 - c_{31}) \neq 0$ and

$$a_{12} = \frac{c_{20}c_{31}s + t}{1 - c_{20}c_{13}}, \quad a_{21} = \frac{c_{31}s + c_{13}t}{1 - c_{20}c_{13}}, \quad a_{03} = \frac{c_{31}c_{20}s + c_{13}t}{1 - c_{31}}, \quad a_{30} = \frac{c_{20}s + c_{13}t}{1 - c_{31}}.$$

Setting $(c_{20}, c_{13}, c_{31}) = (p, q, w)$, we see that A belongs to family (I).

Step 4. In this step we suppose that $u \neq u'$ and that Case 2 holds, and we obtain family (II) or family (III).

The identities (6.4) read

$$c_{13}a_{12} = c_{21}s - a_{31}, \quad a_{03} = c_{30}s - a_{20}, \quad c_{30}a_{31} = a_{03} - c_{13}t, \quad c_{21}a_{20} = a_{12} - t. \tag{6.5}$$

Eliminating a_{12} and a_{03}, we obtain

$$\begin{aligned} (1 - c_{13}c_{30}c_{21})a_{20} &= c_{30}(1 - c_{21})s - c_{13}(1 - c_{30})t, \\ (1 - c_{13}c_{30}c_{21})a_{31} &= c_{21}(1 - c_{13}c_{30})s + c_{13}(c_{13}c_{21} - 1)t. \end{aligned} \tag{6.6}$$

Suppose that $1 - c_{13}c_{30}c_{21} \neq 0$. Set $(c_{13}, c_{30}, c_{21}) = (p, q, w)$. Then (6.6) and (6.5) imply the forms of family (II), with $\Delta = 1 - pqw \neq 0$. Since $a_{02} = t$ and a_{20} are linearly independent, the coefficient $q(1-w)/\Delta$ of s in a_{20} is non-zero. Since $a_{13} = pt$ and a_{31} are linearly independent, the

coefficient $w(1-pq)/\Delta$ of s in a_{31} is non-zero. Since $a_{30} = qs$ and a_{03} are linearly independent, the coefficient $p(1-q)/\Delta$ of t in a_{03} is non-zero. Since $a_{21} = ws$ and a_{12} are linearly independent, the coefficient $(1-pw)/\Delta$ of t in a_{12} is non-zero. Thus A belongs to family (II).

Suppose that $1 - c_{13}c_{30}c_{21} = 0$. Then the second identity of (6.6) implies that $c_{13}c_{30} = 1$ and $c_{13}c_{21} = 1$. Since $c_{13}c_{30}c_{21} = 1$, we obtain $c_{13} = c_{30} = c_{21} = 1$. Thus $a_{13} = t$, $a_{30} = s$, and $a_{21} = s$. Write $a_{31} = ps + qt$. Then (6.5) implies that

$$a_{03} = ps + (q+1)t, \quad a_{12} = (1-p)s - qt, \quad a_{20} = (1-p)s - (q+1)t.$$

The linear independence of the pairs a_{02}, a_{20}, and a_{13}, a_{31}, and a_{30}, a_{03}, and a_{21}, a_{12} is equivalent to $1 - p \neq 0$, $p \neq 0$, $q + 1 \neq 0$, and $q \neq 0$ respectively. Thus A belongs to family (III).

Step 5. In this step we suppose that $u = u'$ and reduce A to a form with parameters $\alpha, \beta, \gamma, \delta$.

Recall that the edges of J_A are $\{0,1\}$ and $\{2,3\}$. After a binary reparametrization, $u = u'$ is the point $s = 0$. Thus $a_{01} = c_{01}s$, $a_{10} = c_{10}s$, $a_{23} = c_{23}s$, and $a_{32} = c_{32}s$, where $c_{01}, c_{10}, c_{23}, c_{32} \in \mathbb{C}$. Setting $s = 0$ in $q_0 = q_1 = q_2 = q_3 = 0$, we see that the products

$$a_{12}a_{13}, \quad a_{02}a_{03}, \quad a_{30}a_{31}, \quad a_{20}a_{21}$$

vanish at $s = 0$. Let $i \in \{0,1\}$ and $j \in \{2,3\}$. Since $\{i,j\}$ has a common index with $\{0,1\}$, Lemma 5.8 implies that a_{ij} and a_{ji} do not both vanish at $s = 0$. Each of the four pairs $\{i,j\}$ contributes a form which does not vanish at $s = 0$. Each of the four products has at most one factor which does not vanish at $s = 0$, and any form a_{ij} or a_{ji} is a factor of exactly one of the four products. Thus equality holds in both counts: each product has exactly one factor which does not vanish at $s = 0$, and for each pair $\{i,j\}$, exactly one of a_{ij}, a_{ji} does not vanish at $s = 0$.

Suppose that a_{02} does not vanish at $s = 0$. Then a_{20} vanishes at $s = 0$, so a_{21} does not, so a_{12} vanishes at $s = 0$, so a_{13} does not, so a_{31} vanishes at $s = 0$, so a_{30} does not, and a_{03} vanishes at $s = 0$. Suppose that a_{02} vanishes at $s = 0$. Then a_{03} does not, so a_{30} vanishes at $s = 0$, so a_{31} does not, so a_{13} vanishes at $s = 0$, so a_{12} does not, so a_{21} vanishes at $s = 0$, and a_{20} does not. Thus the forms which do not vanish at $s = 0$ are either $a_{02}, a_{21}, a_{13}, a_{30}$ or $a_{03}, a_{31}, a_{12}, a_{20}$. Possibly replacing A with the image of A under the signed permutation attached to $\sigma = (0,1)$, we may assume that $a_{02}, a_{21}, a_{13}, a_{30}$ have non-zero coefficients of t and that

$$a_{20} = \alpha s, \quad a_{31} = \beta s, \quad a_{03} = \gamma s, \quad a_{12} = \delta s.$$

By the paragraph before **Step 1**, the forms $a_{20}, a_{31}, a_{03}, a_{12}$ are non-zero, that is, $\alpha\beta\gamma\delta \neq 0$.

Dividing the identities $q_0 = q_1 = q_2 = q_3 = 0$ by s, we obtain

$$\delta a_{13} - c_{23}a_{21} + c_{32}\beta s = 0, \quad \gamma a_{02} - c_{32}a_{30} + c_{23}\alpha s = 0,$$
$$c_{10}a_{13} - c_{01}\gamma s + \beta a_{30} = 0, \quad c_{01}a_{02} - c_{10}\delta s + \alpha a_{21} = 0.$$

Comparing the coefficients of t in the four identities, we obtain $c_{23}, c_{32}, c_{10}, c_{01} \neq 0$. The characters on $c_{01}, c_{10}, c_{23}, c_{32}$ are those of **Step 2**, with determinant 9, so we may assume that $c_{01} = c_{10} = c_{23} = c_{32} = 1$. Then

$$a_{21} = \delta a_{13} + \beta s, \quad a_{30} = \gamma a_{02} + \alpha s, \quad a_{13} = \gamma s - \beta a_{30}, \quad a_{02} = \delta s - \alpha a_{21}. \tag{6.7}$$

Eliminating a_{21}, a_{30}, a_{13}, we obtain

$$(1 - \alpha\beta\gamma\delta)a_{02} = (\delta - \alpha\gamma\delta + \alpha^2\beta\delta - \alpha\beta)s.$$

Since a_{02} has a non-zero coefficient of t, we have $1 - \alpha\beta\gamma\delta = 0$ and $\delta - \alpha\gamma\delta + \alpha^2\beta\delta - \alpha\beta = 0$. Dividing the second equation by δ and using $1/\delta = \alpha\beta\gamma$, we obtain

$$\alpha\beta\gamma\delta = 1, \quad 1 + \alpha^2\beta - \gamma(\alpha + \alpha^2\beta^2) = 0. \tag{6.8}$$

A binary reparametrization fixing s sends a_{02} to t, and it does not change the forms that are multiples of s. Then (6.7) implies that

$$\begin{aligned} &a_{01} = a_{10} = a_{23} = a_{32} = s, \quad a_{02} = t, \quad a_{20} = \alpha s, \quad a_{03} = \gamma s, \quad a_{30} = \gamma t + \alpha s, \\ &a_{12} = \delta s, \quad a_{21} = (\delta s - t)/\alpha, \quad a_{13} = (\gamma - \alpha\beta)s - \beta\gamma t, \quad a_{31} = \beta s. \end{aligned} \tag{6.9}$$

Step 6. In this step we obtain family (IV) and family (V) and conclude the proof.

Suppose that $1+\alpha\beta^2 \neq 0$. Set $\alpha = p$ and $\beta = q$. Then (6.8) implies that

$$\gamma = \frac{1+p^2q}{p(1+pq^2)}, \quad \delta = \frac{1+pq^2}{q(1+p^2q)},$$

and $1+p^2q \neq 0$ since $\gamma \neq 0$. By (6.9), A belongs to family (IV).

Suppose that $1+\alpha\beta^2 = 0$. Since $\alpha+\alpha^2\beta^2 = \alpha(1+\alpha\beta^2) = 0$, the second equation of (6.8) implies that $1+\alpha^2\beta = 0$. Thus $\alpha^2\beta = \alpha\beta^2 = -1$, so $\alpha = \beta = \zeta$ with $\zeta^3 = -1$, and $\delta = 1/(\zeta^2\gamma)$. Let $\theta := \zeta^2$, so that $\theta^3 = 1$. The diagonal action of

$$(\lambda_0, \lambda_1, \lambda_2, \lambda_3) = (1, 1, \theta, \theta)$$

followed by the substitution $t \to \theta^{-2}t$ preserves $a_{01}, a_{10}, a_{23}, a_{32}, a_{02}$. It sends γ to $\theta^2\gamma$, it sends the coefficient of a_{12} to $\theta^2\delta = 1/(\theta^2\gamma)$, and it sends α, β to $\theta\zeta = -1$. Set $w := \theta^2\gamma$. Then (6.9) with $\alpha = \beta = -1$, $\gamma = w$, and $\delta = 1/w$ implies that A belongs to family (V).

The domains of the families are non-empty open subsets of $\mathbb{C}^3, \mathbb{C}^3, \mathbb{C}^2, \mathbb{C}^2, \mathbb{C}$ respectively. Conversely, direct substitution implies that $q_0 = q_1 = q_2 = q_3 = 0$ on each of the five families. □

6.3. **One event and the moduli bound.** In this subsection, we classify the standard pencil matrices with exactly one event, we define the matrix T of the lower system, and we prove Theorem 6.9.

Construction-Proposition 6.6. Let A be a standard pencil matrix whose event graph has exactly one edge. Then, up to signed permutations, binary reparametrization, and the diagonal action, A belongs to one of the three families of Table 4, of dimensions $4, 3, 3$. Moreover, any matrix of Table 4 is a pencil matrix.

TABLE 4. The three families of pencil matrices with exactly one event.

family	dim	coordinates and entries
I	4	$(p,q,w,v) \in \mathbb{C}^4$ with $pqwv(1-q)(p-q)(w+v)(1+v(p-q))$ $(1+p(p-q)(w+v))(p(p-q)v+q) \neq 0$, and, with $\alpha := wp+v(p-q)$, $l := (1-q)(w+v)$, and $m := p(1+v(p-q))/(p-q)$, $a_{13} = \alpha s + wt$, $\quad a_{23} = a_{13} - l(t+ps)$, $\quad a_{32} = -mqs + (1-m)t$, $a_{12} = mqs/(lp) + (l-w)t/l$, $\quad a_{03} = ((m-1)\alpha + (1-q)v)s/m + (m-1)wt/m$.
II	3	$(p,q,w) \in \mathbb{C}^3$ with $pqw(1-q)(p-q)(p(p-q)w-q)(1+p(p-q)w-q) \neq 0$, and, with $l := (1-q)(p(p-q)w-q)/(p(p-q))$, $a_{13} = (wp - q/p)s + wt$, $\quad a_{23} = a_{13} - l(t+ps)$, $\quad a_{32} = -qs$, $a_{12} = qs/(lp) + (l-w)t/l$, $\quad a_{03} = -q(1-q)s/(p(p-q))$.
III	3	$(p,q,w) \in \mathbb{C}^3$ with $pqw(w-p+q-1) \neq 0$, and $a_{13} = (w-1)s + wt$, $\quad a_{23} = a_{13} - p(t+s)$, $\quad a_{32} = -qs + (1-q)t$, $a_{12} = qs/p + (p-w)t/p$, $\quad a_{03} = ((q-1)(w-1)+p)s/q + (q-1)wt/q$.
all		$a_{01} = a_{10} = a_{20} = a_{31} = s$, $\quad a_{02} = t$, $\quad a_{21} = a_{12} - t$, $\quad a_{30} = a_{03} - a_{13}$.

Proof of Construction-Proposition 6.6. The proof is long, so we divide it into several steps.

Step 1. In this step we normalize the entries $a_{01}, a_{10}, a_{20}, a_{31}, a_{02}$.

Possibly applying a signed permutation, we may assume that the edge of J_A is $\{0,1\}$. By Lemma 6.2(2), the forms a_{01} and a_{10} have exactly one common zero on $\mathbb{P}^1$. Possibly applying a binary reparametrization, we may assume that a_{01} and a_{10} vanish at $s = 0$, and we write $a_{01} = c_{01}s$ and $a_{10} = c_{10}s$ with $c_{01}, c_{10} \in \mathbb{C}$. For any $\{i,j\} \neq \{0,1\}$, the forms a_{ij} and a_{ji} have no common zero on $\mathbb{P}^1$, thus a_{ij} and a_{ji} are linearly independent. In particular, neither of a_{ij} and a_{ji} vanishes identically.

The identities $q_3 = 0$ and $q_2 = 0$ read

$$a_{20}a_{21} = s(c_{10}a_{12} - c_{01}a_{02}), \qquad a_{30}a_{31} = s(c_{01}a_{03} - c_{10}a_{13}). \tag{6.10}$$

Thus s divides a_{20} or a_{21}, and s divides a_{30} or a_{31}. Suppose that s divides a_{20} and a_{30}. By $q_1 = 0$, we have $a_{02}a_{03} = a_{30}a_{32} - a_{20}a_{23}$, thus s divides $a_{02}a_{03}$. This is not possible: s divides a_{20} and a_{30}, and $\{0,2\}, \{0,3\}$ are not edges of J_A, so neither of a_{02} and a_{03} vanishes at $s = 0$. Suppose that s divides a_{21} and a_{31}. By $q_0 = 0$, we have $a_{12}a_{13} = a_{21}a_{23} - a_{31}a_{32}$, thus s divides $a_{12}a_{13}$. This is not possible: s divides a_{21} and a_{31}, and $\{1,2\}, \{1,3\}$ are not edges of J_A, so neither of a_{12} and a_{13} vanishes at $s = 0$.

Thus either s divides a_{20} and a_{31}, or s divides a_{21} and a_{30}. Possibly applying the signed permutation attached to $\sigma = (0,1)$, we may assume that $a_{20} = c_{20}s$ and $a_{31} = c_{31}s$ with $c_{20}c_{31} \neq 0$. Then (6.10) implies

$$c_{20}a_{21} = c_{10}a_{12} - c_{01}a_{02}, \qquad c_{31}a_{30} = c_{01}a_{03} - c_{10}a_{13}.$$

If $c_{01} = 0$, then a_{12} and a_{21} are proportional, and $\{1,2\}$ is an edge of J_A. If $c_{10} = 0$, then a_{03} and a_{30} are proportional, and $\{0,3\}$ is an edge of J_A. Thus $c_{01}c_{10}c_{20}c_{31} \neq 0$.

The diagonal action multiplies $(c_{01}, c_{10}, c_{20}, c_{31})$ by

$$(\lambda_0\lambda_1^2,\ \lambda_0^2\lambda_1,\ \lambda_0^2\lambda_2,\ \lambda_1^2\lambda_3).$$

The exponent matrix has determinant -3. Thus we may assume that $c_{01} = c_{10} = c_{20} = c_{31} = 1$. Since a_{02} and $a_{20} = s$ are linearly independent, possibly applying a binary reparametrization fixing s, we may assume that $a_{02} = t$. We obtain

$$a_{01} = a_{10} = a_{20} = a_{31} = s, \qquad a_{02} = t, \qquad a_{21} = a_{12} - t, \qquad a_{30} = a_{03} - a_{13}.$$

Now $q_2 = q_3 = 0$ hold, and $q_0 = 0$ and $q_1 = 0$ are equivalent to

$$a_{12}(a_{13} - a_{23}) = -ta_{23} - sa_{32}, \qquad a_{03}(t - a_{32}) = -a_{13}a_{32} - sa_{23}. \tag{6.11}$$

Step 2. In this step we introduce the roots p, q and show that either $p \neq q$ and $q \neq 1$, or $p = q = 1$.

Suppose that s divides $a_{13} - a_{23}$, possibly $a_{13} - a_{23} = 0$. By the first equation of (6.11) at $s = 0$, s divides a_{23}, thus s divides a_{13}. Since $a_{31} = s$, $\{1,3\}$ is an edge of J_A, which is not possible. Suppose that s divides $t - a_{32}$, possibly $t - a_{32} = 0$. Then a_{32} does not vanish at $s = 0$, and the second equation of (6.11) at $s = 0$ implies that s divides a_{13}, which is again not possible. Thus we may write

$$a_{13} - a_{23} = l(t + ps), \qquad t - a_{32} = m(t + qs), \qquad a_{13} = \alpha s + wt,$$

with $lmw \neq 0$. The right sides of (6.11) vanish at $(s,t) = (1,-p)$ and at $(s,t) = (1,-q)$ respectively, that is,

$$p(\alpha - wp) = (m-1)p - mq, \qquad (1-q)(\alpha - wq) = l(p-q). \tag{6.12}$$

Suppose that $q = 0$. By (6.12), $\alpha = lp$, thus $a_{23} = (w-l)t$ and $a_{32} = (1-m)t$. Then $\{2,3\}$ is an edge of J_A, which is not possible. Suppose that $p = 0$. By (6.12), $mq = 0$, thus $q = 0$, which is not possible. Therefore $pq \neq 0$. If $p = q$, then (6.12) implies $\alpha - wp = -1$ and $1 - q = 0$, thus $p = q = 1$. If $q = 1$, then (6.12) implies $l(p-1) = 0$, thus $p = 1$.

Step 3. In this step we treat the case $p \neq q$, which leads to families I and II.

For binary linear forms $f = f_1 s + f_2 t$ and $g = g_1 s + g_2 t$, we let $\det(f,g) := f_1 g_2 - f_2 g_1$. Then $\det(f,g) \neq 0$ if and only if f and g have no common zero on $\mathbb{P}^1$. Let $v := (\alpha - wp)/(p-q)$. Then $\alpha = wp + v(p-q)$ and $\alpha - wq = (w+v)(p-q)$, thus (6.12) is equivalent to

$$l = (1-q)(w+v), \qquad m = \frac{p(1+v(p-q))}{p-q}.$$

The definitions of l and m imply the forms a_{23} and a_{32} of family I. By (6.12), the right sides of (6.11) are divisible by $l(t+ps)$ and by $m(t+qs)$ respectively. A direct calculation, comparing

the coefficients of s^2 and t^2 and using $lp = (1-q)(\alpha+qv)$, shows that the quotients are the forms a_{12} and a_{03} of family I. A direct calculation shows

$$\det(a_{02}, a_{20}) = -1, \qquad \det(a_{13}, a_{31}) = -w, \qquad \det(a_{12}, a_{21}) = -\frac{mq}{lp},$$

$$\det(a_{03}, a_{30}) = -\frac{w(1-q)v}{m}, \qquad \det(a_{23}, a_{32}) = -qv(1+p(p-q)(w+v)).$$

Since J_A has no edge other than $\{0,1\}$, the determinants above are non-zero. Together with $lm \neq 0$, we obtain

$$pqwv(1-q)(p-q)(w+v)(1+v(p-q))(1+p(p-q)(w+v)) \neq 0. \tag{6.13}$$

We have $m-1 = (p(p-q)v+q)/(p-q)$. If $p(p-q)v+q \neq 0$, then A belongs to family I. If $p(p-q)v+q = 0$, then $m = 1$ and $v = -q/(p(p-q))$. In this case, $\alpha = wp - q/p$, $l = (1-q)(p(p-q)w-q)/(p(p-q))$, and the forms $a_{13}, a_{23}, a_{32}, a_{12}, a_{03}$ of family I specialize to those of family II. Since $1+v(p-q) = (p-q)/p$, $w+v = (p(p-q)w-q)/(p(p-q))$, and $1+p(p-q)(w+v) = 1+p(p-q)w-q$, the inequality (6.13) is equivalent to the inequality of family II. Thus A belongs to family II.

Step 4. In this step we treat the case $p = q = 1$, which leads to family III, and we conclude the proof.

By (6.12), $\alpha = w-1$, and l, m, w are free. As in **Step 3**, comparing the coefficients of s^2 and t^2 in (6.11) implies

$$a_{12} = \frac{ms + (l-w)t}{l}, \qquad a_{03} = \frac{((m-1)(w-1)+l)s + (m-1)wt}{m}.$$

Since $p = q = 1$ in this case, we use the letters (p, q) for (l, m) in the rest of this step, and we obtain the forms of family III. A direct calculation shows

$$\det(a_{02}, a_{20}) = -1, \quad \det(a_{13}, a_{31}) = -w, \quad \det(a_{12}, a_{21}) = -\frac{q}{p}, \quad \det(a_{03}, a_{30}) = -\frac{wp}{q},$$

$$\det(a_{23}, a_{32}) = w - p + q - 1.$$

Since J_A has no edge other than $\{0,1\}$, we have $pqw(w-p+q-1) \neq 0$. Thus A belongs to family III.

The polynomials defining the families are non-zero, thus the domains of the families are non-empty open subsets of $\mathbb{C}^4$, $\mathbb{C}^3$, and $\mathbb{C}^3$. Conversely, a direct calculation shows that q_0, q_1, q_2, q_3 vanish identically on any matrix of Table 4, so that any such matrix is a pencil matrix. □

A pencil matrix with an event has a column that vanishes at a point of $\mathbb{P}^1$, so the argument of Section 7 is not available, and we compute the rank of the matrix T below on each family.

Construction-Proposition 6.7. Notation and conditions as in Construction-Proposition 5.3. For (C, O) of the normalized shape, let N be the matrix given by the formula of Construction-Proposition 5.3. We denote by T the 552×84 coefficient matrix of the three equations of Construction-Proposition 5.3 in the 84 coefficients of (C, O), so that $\mathcal{U}_A = \operatorname{Ker} T$, with rows and columns numbered from 0 as follows.

(1) Column $3r+e$, where $0 \leq r \leq 19$ and $0 \leq e \leq 2$, is the coefficient of $s^{2-e}t^e$ in c_r.
(2) Column $60+2r+e$, where $0 \leq r \leq 11$ and $0 \leq e \leq 1$, is the coefficient of $s^{1-e}t^e$ in o_r.
(3) Row $12(6i+j)+m$, where $0 \leq i \leq 3$, $0 \leq j \leq 5$, and $0 \leq m \leq 11$, is the coefficient of the m-th term of the list $s^2y_0, \ldots, s^2y_3, sty_0, \ldots, sty_3, t^2y_0, \ldots, t^2y_3$ in the entry (i, j) of $AO - B_1N^t - CB_3^t$.
(4) Row $288+4(12i+j)+m$, where $0 \leq i \leq 3$, $0 \leq j \leq 11$, and $0 \leq m \leq 3$, is the coefficient of the m-th term of the list s^3, s^2t, st^2, t^3 in the entry (i, j) of $AN - CP^t$.
(5) Row $480+4(6i+j)+m$, where $0 \leq i \leq 2$, $0 \leq j \leq 5$, and $0 \leq m \leq 3$, is the coefficient of the m-th term of the list s^3, s^2t, st^2, t^3 in the entry (i, j) of UN^t.

Let M be a matrix with entries in the coordinate ring of an affine variety. A *pivot table* of M is a list of triples (r, c, β), with β a unit of the ring, such that, starting from M and taking the triples in order, the entry of the current matrix in row r and column c equals β, and the next matrix is obtained by the replacement

$$M_{ij} \to M_{ij} - M_{ic}M_{rj}/\beta$$

followed by the removal of row r and column c, where rows and columns keep their numbers. If M has a pivot table with k triples, and M_k is the matrix left after the k triples, then $\operatorname{rank} M = k + \operatorname{rank} M_k$ at any point of the variety. In particular, $\operatorname{rank} M \geq k$ at any point of the variety.

Proof. By Construction-Proposition 5.3, the entries of N are linear in the 84 coefficients of (C, O), so the rows of T are $288 + 192 + 72 = 552$ linear equations in the $60 + 24 = 84$ coefficients, and $\mathcal{U}_A = \operatorname{Ker} T$.

Since β is a unit, a step of a pivot table consists of row and column operations that clear row r and column c except for the entry β, followed by the removal of row r and column c, so the step lowers the rank by exactly one at any point. □

By Proposition 5.5, on a family of dimension b, the goal is $b + (84 - \operatorname{rank} T) - 1 \leq 7$, that is, $\operatorname{rank} T \geq 76 + b$, where T is the matrix of Construction-Proposition 6.7. Family I of Table 4 has $b = 4$, which is one less than the dimension $8 - 3 = 5$ of the event-free lines modulo the torus (Section 7), so we need $\operatorname{rank} T \geq 80$ at a general point of family I, and 80 is the rank of T at a general event-free line, where $\dim \mathcal{U}_A = 4$.

Construction-Proposition 6.8. On each family of Construction-Propositions 6.3, 6.4, 6.5, and 6.6, let T be the matrix of Construction-Proposition 6.7, and let b be the dimension of the family. Then $\operatorname{rank} T \geq 76 + b$ at any point of the family, except on families I and III of Table 4, where $\operatorname{rank} T \geq 75 + b$ at any point and $\operatorname{rank} T \geq 76 + b$ off a proper closed subset of the family.

TABLE 5. The dimension b of each family, the lower bound for $\operatorname{rank} T$, and the pivot table of Appendix A that implies the bound.

family	b	rank proved	pivot table
triangle	2	78	Table 22
path	2	78	Table 23
adjacent	3	79	Table 24
disjoint (I)	3	79	Table 25
disjoint (II)	3	79	Table 26
disjoint (III)	2	78	Table 27
disjoint (IV)	2	78	Table 28
disjoint (V)	1	77	Table 29
one event I	4	79 at any point, 80 off $H_0 = 0$, see Lemma A.1	Table 30
one event II	3	80	Table 31
one event III	3	78 at any point, 79 off $G = 0$, see Lemma A.1	Table 32

Proof of Construction-Proposition 6.8. The ranks come from the replacement of Construction-Proposition 6.7, carried out exactly over the localized polynomial coordinate ring of the family, each equality of an entry with β being checked by substitution in the definition of T. By Construction-Proposition 6.7, the pivot tables of Table 5 imply $\operatorname{rank} T \geq 76 + b$ on the triangle, path, adjacent, and disjoint families, and $\operatorname{rank} T \geq 80$ on family II of Table 4, where $76 + b = 79$.

We consider family I of Table 4, where $b = 4$. By Lemma A.1, after the 78 triples of Table 30, the residual entries $(328, 49)$ and $(374, 49)$ equal a unit times $p(w + v) - v$ and a unit times

$q(w+v)-v$. Since

$$(p(w+v)-v)-(q(w+v)-v)=(p-q)(w+v)$$

is a unit, the residual entries $(328,49)$ and $(374,49)$ do not both vanish at any point. Thus, by Construction-Proposition 6.7, $\operatorname{rank} T \geq 79 = 75+b$ at any point. The 2×2 minor on the rows $328, 374$ and the columns $49, 9$ of the matrix left after the 78 triples equals a unit times the polynomial H_0 of Lemma A.1. The coefficient of p^3w^2 in H_0 equals 1, thus H_0 is a non-zero polynomial, and $H_0=0$ defines a proper closed subset of the family. Off $H_0=0$, we have $\operatorname{rank} T \geq 80 = 76+b$.

We consider family III of Table 4, where $b=3$. By Lemma A.1, after the 78 triples of Table 32, the residual entry $(405,74)$ equals $w(p-w)(w-1)/p^2$. Since w and p are units, $\operatorname{rank} T \geq 79 = 76+b$ off the proper closed subset $G=0$, where $G=(p-w)(w-1)$, and $\operatorname{rank} T \geq 78 = 75+b$ at any point. □

Theorem 6.9. *We have* $\dim \overline{\mathcal{M}_{\mathrm{ev}}} \leq 7$.

Proof. Let $[X] \in \mathcal{M}_{\mathrm{ev}}$, and let A be a standard pencil matrix with at least one event of a marked surface S with canonical model X. By Lemma 5.7, A has no identically zero column. By Lemma 6.2(1), up to relabeling, J_A is one edge, a path with two edges, two disjoint edges, a triangle, or a path with three edges. By Construction-Propositions 6.6, 6.4, 6.5, and 6.3, after signed permutations and the diagonal action (Theorem 5.2) and a binary reparametrization (Construction-Proposition 5.1), which all preserve the property of being a standard pencil matrix of S for a marking of S, A belongs to a family $\mathcal{V}$ of Table 5. By Construction-Proposition 5.3, a standard presentation of S with pencil matrix A has a normalized tuple, which is geometric, so $[X] \in \mathcal{M}_{\mathcal{V}}$, with $\mathcal{M}_{\mathcal{V}}$ the set $\mathcal{M}_V$ of Proposition 5.5 for $V=\mathcal{V}$. Thus $\mathcal{M}_{\mathrm{ev}}$ is contained in the union of the $\mathcal{M}_{\mathcal{V}}$ over the finitely many families $\mathcal{V}$ of Table 5, and it suffices to prove that $\dim \overline{\mathcal{M}_{\mathcal{V}}} \leq 7$ for any such $\mathcal{V}$.

Let $\mathcal{V}$ be a family of dimension b. We have $\dim \mathcal{U}_{A_x} = 84 - \operatorname{rank} T$ at any $x \in \mathcal{V}$ (Construction-Proposition 6.7). If $\operatorname{rank} T \geq 76+b$ at any point of $\mathcal{V}$, Construction-Proposition 6.8 and Proposition 5.5 with $d=8-b\geq 1$, since $b\leq 4$, imply

$$\dim \overline{\mathcal{M}_{\mathcal{V}}} \leq b+(8-b)-1=7.$$

Otherwise, $\mathcal{V}$ is family I or family III of Table 4, and we let $Z \subset \mathcal{V}$ be the proper closed subset of Construction-Proposition 6.8. Since $\mathcal{V}$ is integral, $\dim Z < b$ [Har77, Exercise I.1.10(d)], and Z may be empty. Proposition 5.5 with $d=8-b\geq 1$ implies

$$\dim \overline{\mathcal{M}_{\mathcal{V}\setminus Z}} \leq b+(8-b)-1=7.$$

For any irreducible component Z' of Z, we have $\dim Z' \leq b-1$, and Proposition 5.5 with $d=9-b\geq 1$ implies

$$\dim \overline{\mathcal{M}_{Z'}} \leq (b-1)+(9-b)-1=7.$$

Since $\mathcal{M}_{\mathcal{V}}$ is contained in the union of $\mathcal{M}_{\mathcal{V}\setminus Z}$ and the finitely many $\mathcal{M}_{Z'}$, we have $\dim \overline{\mathcal{M}_{\mathcal{V}}} \leq 7$. Since the number of families is finite, $\dim \overline{\mathcal{M}_{\mathrm{ev}}} \leq 7$. □

7. Event-free pencil matrices

In this section, we prove Theorem 7.10, that is,

$$\dim \overline{\mathcal{M}_{\mathrm{ne}} \setminus \mathcal{B}_{\mathrm{SS}}} \leq 7.$$

We remove $\mathcal{B}_{\mathrm{SS}}$, since a general member of the family $\mathcal{S}$ of Definition 2.6 has an event-free standard pencil matrix (Proposition 7.8) and $\mathcal{S}$ is an 8-dimensional family (Theorem 1.4). Compared with [SS20, Theorem 5.1], whose proof considers lines that do not intersect the special loci, the new points of this section are that the lines with an event-free pencil matrix may pass through points at which the rank of the pencil matrix drops (Lemma 7.3), that the pencil matrix may have vanishing rows (Construction-Proposition 7.6), and that such lines form an irreducible variety (Construction-Proposition 7.2).

7.1. **Lines on intersections of two quadrics.** In this subsection, we show that the lines in $\mathcal{F}$ with an event-free pencil matrix form an irreducible variety of dimension eight.

A singular intersection of two quadrics may contain more lines than a smooth one, so the relative scheme of lines of Construction-Proposition 7.2 could have a component over the discriminant. The following lemma bounds the lines uniformly, also on singular intersections.

Lemma 7.1. *Let P_0, P_1 be independent quadratic forms on a six-dimensional complex vector space, let $W := V(P_0, P_1) \subset \mathbb{P}^5$, and let $F_1(W)$ be the scheme of lines in W. Assume that some member of their pencil is non-singular and any non-zero member of their pencil has rank at least four. Then $\dim F_1(W) \leq 2$, and for any point $x \in W$ the lines in W through x form a family of dimension at most one. If W is smooth, then only finitely many lines in W pass through any given point.*

Proof. Let V_6 be the six-dimensional vector space, so that $\mathbb{P}^5 = \mathbb{P}(V_6)$. The scheme W and the hypotheses depend only on the pencil of quadrics spanned by P_0 and P_1. Possibly replacing P_0, P_1 with another basis of the pencil of quadrics, we may assume that P_0 is non-singular. Since P_0 is irreducible and P_1 is not proportional to P_0, W is a complete intersection and $\dim W = 3$.

Step 1. In this step we bound the lines in W through a point.

Let $x \in W$, and let $x \in V_6$ also denote a non-zero vector representing the point x. Let $H_x \subset V_6$ be the hyperplane orthogonal to x with respect to P_0. Since P_0 is non-singular and $P_0(x) = 0$, the restriction of P_0 to H_x has radical $\langle x \rangle$, so P_0 induces a non-singular quadratic form P_0' on the four-dimensional vector space $H_x/\langle x \rangle$. A line in $V(P_0)$ through x corresponds to a point of $V(P_0') \subset \mathbb{P}(H_x/\langle x \rangle) = \mathbb{P}^3$.

Suppose first that $x \in \mathrm{Sing}(W)$. Since W is defined by two equations in $\mathbb{P}^5$ and $\dim W = 3$, we have $x \in \mathrm{Sing}(W)$ if and only if the differentials of P_0, P_1 at x are dependent (cf. [Har77, Exercise I.5.8]). The direction that independent differentials at x imply that W is smooth at x holds for any closed subscheme of $\mathbb{P}^5$ defined by two equations. Thus some non-zero member P_x of the pencil of quadrics has x in the radical of P_x. Since P_0 is non-singular, P_x is not proportional to P_0, so $W = V(P_0, P_x)$. The form P_x induces a quadratic form P_x' on $H_x/\langle x \rangle$, and the lines in W through x correspond to the points of $V(P_0', P_x') \subset \mathbb{P}^3$. Suppose that $P_x' = cP_0'$ for some $c \in \mathbb{C}$. Then the non-zero member $P_x - cP_0$ vanishes on H_x. Thus $P_x - cP_0$ is the product of two linear forms and

$$\mathrm{rank}(P_x - cP_0) \leq 2,$$

a contradiction. Since P_0' is irreducible, $\dim V(P_0', P_x') \leq 1$.

Suppose now that $x \notin \mathrm{Sing}(W)$. Let $H_x' \subset V_6$ be the hyperplane orthogonal to x with respect to P_1. Then $H_x \neq H_x'$, and the lines in W through x correspond to points of the plane $\mathbb{P}((H_x \cap H_x')/\langle x \rangle) = \mathbb{P}^2$ on which both induced forms vanish. Since $\mathrm{rank}\, P_0' = 4$, the restriction of P_0' to $(H_x \cap H_x')/\langle x \rangle$ has rank at least two. Let $C_x \subset \mathbb{P}^2$ be the conic defined by the restriction of P_0'. The lines in W through x correspond to points of a closed subset of C_x, and $\dim C_x = 1$.

Step 2. In this step we prove that $\dim F_1(W) \leq 2$.

By **Step 1**, any point of $\mathrm{Sing}(W)$ is in the radical of a singular member of the pencil of quadrics. Since P_0 is non-singular, the determinant of the pencil of quadrics is a non-zero binary form, so the pencil of quadrics has only finitely many singular members up to scalar. The radical of a singular member is a vector space of dimension at most two, since the member has rank at least four. Thus $\dim \mathrm{Sing}(W) \leq 1$. By **Step 1**, the closed subset of $F_1(W)$ of lines intersecting $\mathrm{Sing}(W)$ is of dimension at most $1 + 1 = 2$.

Let $L \subset W$ be a line with $L \cap \mathrm{Sing}(W) = \emptyset$, and let $N_{L/W}$ be the normal bundle of L in W. Since W is a complete intersection of two quadrics and smooth along L, there exists an exact sequence

$$0 \to N_{L/W} \to \mathcal{O}_L(1)^{\oplus 4} \to \mathcal{O}_L(2)^{\oplus 2} \to 0.$$

Thus $N_{L/W}$ is of rank two and $\deg N_{L/W} = 4 - 4 = 0$. By [Har77, Exercise V.2.6], $N_{L/W} \cong \mathcal{O}_L(a) \oplus \mathcal{O}_L(-a)$ for some integer $a \geq 0$. Since $N_{L/W} \subset \mathcal{O}_L(1)^{\oplus 4}$, we have $a \leq 1$. By [Har77,

Theorem III.5.1], we have

$$H^1(L, N_{L/W}) = 0 \text{ and } h^0(L, N_{L/W}) = 2.$$

By [Kol96, Theorem I.2.8, Lemma I.2.12, and Proposition I.2.14], the vanishing of $H^1(L, N_{L/W})$ implies that $F_1(W)$ is smooth at the point of $F_1(W)$ representing L, with local dimension two. Thus $\dim F_1(W) \leq 2$.

Step 3. In this step we consider the case when W is smooth.

By [Rei72], the variety of lines on a smooth complete intersection of two quadrics in $\mathbb{P}^5$ is isomorphic to the Jacobian of a hyperelliptic curve of genus 2. See also [SS20, Corollary 4.2]. Thus $F_1(W)$ is an abelian surface.

Let $x \in W$, and suppose that infinitely many lines in W pass through x. Since W is smooth, $x \notin \mathrm{Sing}(W)$. Recall the conic C_x of **Step 1**, defined by a form of rank at least two. By **Step 1**, the closed subset of C_x corresponding to the lines in W through x contains an irreducible curve C. Since C is an irreducible component of the plane conic C_x, we have $C \cong \mathbb{P}^1$. Sending a point of C to the corresponding line in W defines a non-constant morphism $\mathbb{P}^1 \to F_1(W)$ (by the universal property of the Fano scheme [Kol96, Theorem I.1.4]). This contradicts [BL04, Proposition 4.9.5], since a morphism from $\mathbb{P}^1$ to an abelian surface is constant. □

[Ste18, Lemma 7.3.2] states only the dimension 8 of $\mathcal{F}$, and [SS20, Section 5, after Corollary 5.2] refers only to the numerical proof of irreducibility in [Ste18]. Here we need an irreducible variety of lines, since otherwise the density in Construction-Proposition 7.9 would fail. Any line in $\mathcal{F}$ with an event-free pencil matrix is contained in a $\mathbb{P}^5$ on which q_0, q_1 vanish, so it is a line on an intersection of two quadrics.

Construction-Proposition 7.2. Notation as in Construction-Proposition 5.1. Let $\mathcal{F}^{\mathrm{ne}} \subset \mathcal{F}$ be the open subset of lines on which a_{ij} and a_{ji} are linearly independent for any $i \neq j$. Let

$$T := \{(b, c, d, e, f, g) \in \mathbb{A}^6 \mid be(cf - dg) \neq 0\}.$$

For $\tau = (b, c, d, e, f, g) \in \mathbb{A}^6$, on $\mathbb{P}^5$ with coordinates (u, v, x, y, z, w), let

$$P_\tau := vz - uw + ebx^2 + ecxz - gbxw - gczw,$$

$$R_\tau := u(fx + gy) - v(dx + cy) + eby^2 - edyz + fbyw - fdzw.$$

Let

$$\mathcal{J} := \{(\tau, L) \mid \tau \in T,\ L \in F_1(V(P_\tau, R_\tau))\}.$$

Then $\mathcal{J}$ is irreducible, $\dim \mathcal{J} = 8$, and $\dim \mathcal{J}_\tau \leq 2$ for any fiber $\mathcal{J}_\tau$ of $\mathcal{J} \to T$. For any $\tau \in T$, the forms P_τ, R_τ satisfy the hypotheses of Lemma 7.1. There exists a non-empty open subset $T_{\mathrm{sm}} \subset T$ such that $V(P_\tau, R_\tau)$ is smooth and $F_1(V(P_\tau, R_\tau))$ is an abelian surface for any $\tau \in T_{\mathrm{sm}}$. The substitution

$$a_{01} = u,\ a_{10} = v,\ a_{23} = x,\ a_{32} = y,\ a_{13} = z,\ a_{03} = w,\ a_{12} = cy + dx,\ a_{21} = -by + dz,$$

$$a_{31} = -bx - cz,\ a_{02} = fx + gy,\ a_{30} = -ex + gw,\ a_{20} = -ey - fw$$

defines a morphism $\mathcal{J} \to \mathcal{F}$ which is injective over $\mathcal{F}^{\mathrm{ne}}$ and whose image contains $\mathcal{F}^{\mathrm{ne}}$. If a point of $\mathcal{J}$ over (b, c, d, e, f, g) has image in $\mathcal{F}^{\mathrm{ne}}$, then $df \neq 0$. In particular, $\mathcal{F}^{\mathrm{ne}}$ is non-empty, irreducible, and of dimension 8.

Proof. The proof is long, so we divide it into several steps. We write $\delta := cf - dg$.

Step 1. In this step we check the hypotheses of Lemma 7.1 for any $\tau \in T$, and we construct T_{sm}.

Let $\tau = (b, c, d, e, f, g) \in T$, and let Hess denote the Hessian matrix in the coordinates (u, v, x, y, z, w). The forms P_τ and R_τ are independent, since the coefficients of uw and vz in P_τ are -1 and 1, the form R_τ has no term uw or vz, and the coefficient of y^2 in R_τ is $eb \neq 0$.

For any $h \in \mathbb{C}$, the terms of $P_\tau + hR_\tau$ involving u or v are

$$u(-w + hfx + hgy) + v(z - hdx - hcy).$$

Thus the block of $\operatorname{Hess}(P_\tau + hR_\tau)$ with rows u, v and columns z, w has determinant one. Since $P_\tau + hR_\tau$ has no term u^2, uv, v^2, the block with rows and columns u, v vanishes, so the four-by-four Hessian minor on u, v, z, w is equal to $1^2 = 1$, and $\operatorname{rank}(P_\tau + hR_\tau) \geq 4$. The block of $\operatorname{Hess}(R_\tau)$ with rows u, v and columns x, y is

$$\begin{pmatrix} f & g \\ -d & -c \end{pmatrix},$$

with determinant $-\delta$. Since R_τ has no term u^2, uv, v^2, the four-by-four Hessian minor of R_τ on u, v, x, y is equal to $(-\delta)^2 = (cf - dg)^2 \neq 0$, and $\operatorname{rank} R_\tau \geq 4$.

The form P_τ does not involve y, and

$$P_\tau = (v + ecx - gcw)z - (u + gbx)w + ebx^2.$$

Thus $\det \operatorname{Hess}(P_\tau) = 0$, and the five-by-five Hessian minor of P_τ on u, v, x, z, w is equal to $2eb$. Since the row y of $\operatorname{Hess}(P_\tau)$ vanishes, the coefficient of h in $\det \operatorname{Hess}(P_\tau + hR_\tau)$ is the product of the entry of $\operatorname{Hess}(R_\tau)$ in row y and column y, which is $2eb$, and the five-by-five Hessian minor of P_τ on u, v, x, z, w. Thus

$$\det \operatorname{Hess}(P_\tau + hR_\tau) = 4b^2e^2h + O(h^2). \tag{7.1}$$

Since $4b^2e^2 \neq 0$, some member of the pencil of quadrics spanned by P_τ, R_τ is non-singular. Thus P_τ, R_τ satisfy the hypotheses of Lemma 7.1, and $\dim \mathcal{J}_\tau \leq 2$.

By (7.1), $\det \operatorname{Hess}(\lambda P_\tau + \mu R_\tau)$ is a non-zero binary form of degree six in λ, μ for any $\tau \in T$. Let $T_{\mathrm{sm}} \subset T$ be the open subset of points τ at which the discriminant of $\det \operatorname{Hess}(\lambda P_\tau + \mu R_\tau)$ is non-zero. A direct calculation shows that at the point $(b, c, d, e, f, g) = (1, 0, 1, 1, 1, 1) \in T$, we have

$$\det \operatorname{Hess}(P_\tau + hR_\tau) = -h\big(h(h+1)^4 - 4\big). \tag{7.2}$$

The right-hand side of (7.2) is of degree six in h, so $(\lambda, \mu) = (0, 1)$ is not a zero of the binary form. Let $p(h) := h(h+1)^4 - 4$. Then

$$p(0) = -4, \quad p'(h) = (h+1)^3(5h+1), \quad p(-1) = -4, \quad p\left(-\frac{1}{5}\right) = -\frac{12756}{3125}.$$

Thus the binary form has six distinct zeros at the point $(1, 0, 1, 1, 1, 1)$, and T_{sm} is non-empty.

Let $\tau \in T_{\mathrm{sm}}$. By [Rei72, Proposition 2.1], $V(P_\tau, R_\tau)$ is smooth. By [Rei72] (see also [SS20, Corollary 4.2]), $F_1(V(P_\tau, R_\tau))$ is an abelian surface.

Step 2. In this step we prove that $\mathcal{J}$ is irreducible and $\dim \mathcal{J} = 8$.

Let Z be an irreducible component of $\mathcal{J}$. A coordinate chart of $T \times \operatorname{Gr}(2, 6)$ is an open subset of $\mathbb{A}^{14}$. In such a coordinate chart, $\mathcal{J}$ is cut out by six polynomials, namely the three coefficients of each of P_τ, R_τ on the tautological two-plane. Let Z° be the intersection of Z with a coordinate chart that intersects Z. The closure $\overline{Z^\circ}$ of Z° in $\mathbb{A}^{14}$ is an irreducible component of the zero set of the six polynomials, so $\dim Z \geq \dim Z^\circ = \dim \overline{Z^\circ} \geq 14 - 6 = 8$ by [Har77, Exercise I.1.10(a), Proposition I.1.10, and Exercise I.1.9]. By **Step 1**, any fiber of $Z \to T$ is of dimension at most two. By [Har77, Exercise II.3.22(c)], we have

$$8 \leq \dim Z \leq \dim \overline{\operatorname{Im}(Z \to T)} + 2 \leq 6 + 2 = 8.$$

Thus $\dim Z = 8$ and Z dominates T. Since T_{sm} is non-empty, $\mathcal{J}$ is non-empty.

Let Z_1, Z_2 be irreducible components of $\mathcal{J}$. By [Har77, Exercise II.3.22(e)], for a general closed point $\tau \in T_{\mathrm{sm}}$, we have

$$\dim (Z_1)_\tau = \dim (Z_2)_\tau = 2.$$

Since $\mathcal{J}_\tau$ is an irreducible surface by **Step 1**, we have $(Z_1)_\tau = (Z_2)_\tau = \mathcal{J}_\tau$. Thus $\dim(Z_1 \cap Z_2) \geq 6 + 2 = 8$, and $Z_1 = Z_2$.

Step 3. In this step we construct the morphism and prove that the morphism is injective over $\mathcal{F}^{\mathrm{ne}}$ and that the image contains $\mathcal{F}^{\mathrm{ne}}$.

With rows and columns indexed by $0,1,2,3$, let

$$M_0 := \begin{pmatrix} 0 & a_{12} & a_{21} & a_{31} \\ -a_{12} & 0 & a_{32} & a_{23} \\ -a_{21} & -a_{32} & 0 & a_{13} \\ -a_{31} & -a_{23} & -a_{13} & 0 \end{pmatrix}, \quad M_1 := \begin{pmatrix} 0 & a_{02} & a_{30} & a_{20} \\ -a_{02} & 0 & a_{23} & a_{32} \\ -a_{30} & -a_{23} & 0 & a_{03} \\ -a_{20} & -a_{32} & -a_{03} & 0 \end{pmatrix}.$$

The Pfaffians of M_0 and M_1 are q_0 and q_1 respectively.

For $\tau = (b,c,d,e,f,g) \in \mathbb{A}^6$, let $\varphi_\tau : \mathbb{P}^5 \to \mathbb{P}^{11}$ be the linear embedding defined by the substitution in the statement. The rows with index $1,2,3$ of the equations

$$M_0 \begin{pmatrix} 1 \\ b \\ c \\ d \end{pmatrix} = 0, \quad M_1 \begin{pmatrix} 1 \\ e \\ f \\ g \end{pmatrix} = 0 \tag{7.3}$$

are the expressions of a_{12}, a_{21}, a_{31} and of a_{02}, a_{30}, a_{20} in the substitution, and the rows with index 0 follow from the other rows. Thus $\varphi_\tau(\mathbb{P}^5)$ is the linear subspace of $\mathbb{P}^{11}$ defined by (7.3). A direct calculation shows that

$$\varphi_\tau^* q_0 = \varphi_\tau^* q_1 = 0, \quad \varphi_\tau^* q_2 = P_\tau, \quad \varphi_\tau^* q_3 = R_\tau. \tag{7.4}$$

Thus φ_τ maps $V(P_\tau, R_\tau)$ into $\mathcal{Q}$ for any $\tau \in T$, and the maps φ_τ define a morphism (by the universal property of the Fano scheme [Kol96, Theorem I.1.4], since the matrix of φ_τ is polynomial in τ)

$$\varphi : \mathcal{J} \to \mathcal{F}, \quad (\tau, L) \to \varphi_\tau(L).$$

Let $\ell \in \mathcal{F}^{\mathrm{ne}}$, and regard the a_{ij} as linear forms in s,t on ℓ. Let M be either M_0 or M_1 on ℓ, and write $M = s\omega + t\omega'$ with constant alternating matrices ω, ω'. Since a_{23} and a_{32} are entries of M and are linearly independent, ω and ω' are linearly independent. Since q_0 and q_1 vanish on ℓ, $\mathrm{Pf}(\lambda\omega + \mu\omega') = 0$ for any $\lambda, \mu \in \mathbb{C}$. Thus we may write $\omega = \alpha_1 \wedge \alpha_2$ and $\omega' = \alpha_3 \wedge \alpha_4$ for covectors $\alpha_1, \dots, \alpha_4$, and

$$\omega \wedge \omega' = 0.$$

Thus $\mathrm{Span}(\alpha_1, \alpha_2) \cap \mathrm{Span}(\alpha_3, \alpha_4)$ is of dimension one, and $\mathrm{Span}(\alpha_1, \dots, \alpha_4)$ is of dimension three. The common kernel of ω and ω' is the annihilator of $\mathrm{Span}(\alpha_1, \dots, \alpha_4)$. Thus there exist vectors $\beta = (\beta_0, \beta_1, \beta_2, \beta_3)$ and $\gamma = (\gamma_0, \gamma_1, \gamma_2, \gamma_3)$, unique up to scalar, such that $M_0\beta = 0$ and $M_1\gamma = 0$ on ℓ.

Suppose that $\beta_0 = 0$. Then the row of $M_0\beta = 0$ with index 1 is $a_{32}\beta_2 + a_{23}\beta_3 = 0$, so $\beta_2 = \beta_3 = 0$. The row with index 2 is then $-a_{32}\beta_1 = 0$, so $\beta_1 = 0$, a contradiction. Thus $\beta_0 \neq 0$. Suppose that $\gamma_0 = 0$. Then the row of $M_1\gamma = 0$ with index 1 is $a_{23}\gamma_2 + a_{32}\gamma_3 = 0$, so $\gamma_2 = \gamma_3 = 0$. The row with index 2 is then $-a_{23}\gamma_1 = 0$, so $\gamma_1 = 0$, a contradiction. Thus $\gamma_0 \neq 0$. We normalize

$$\beta = (1, b, c, d), \quad \gamma = (1, e, f, g).$$

By (7.3), $\ell \subset \varphi_\tau(\mathbb{P}^5)$, where $\tau := (b,c,d,e,f,g) \in \mathbb{A}^6$.

Now let $\tau = (b,c,d,e,f,g) \in \mathbb{A}^6$ be any point with $\ell \subset \varphi_\tau(\mathbb{P}^5)$, and regard x, y, z, w as the forms $a_{23}, a_{32}, a_{13}, a_{03}$ on ℓ. We have the following.

- If $b = 0$, then $a_{31} = -cz$ and $a_{13} = z$ are linearly dependent.
- If $e = 0$, then $a_{30} = gw$ and $a_{03} = w$ are linearly dependent.
- If $d = 0$, then $a_{12} = cy$ and $a_{21} = -by$ are linearly dependent.
- If $f = 0$, then $a_{02} = gy$ and $a_{20} = -ey$ are linearly dependent.

Since $\ell \in \mathcal{F}^{\mathrm{ne}}$, we have $bdef \neq 0$. In particular, $df \neq 0$ if a point of $\mathcal{J}$ over τ has image in $\mathcal{F}^{\mathrm{ne}}$. Suppose that $\delta = 0$. Then

$$a_{12} = dx + cy = \frac{d}{f}(fx + gy) = \frac{d}{f} a_{02}.$$

Let $\xi \in \ell$ be a zero of a_{02}. Then $a_{12}(\xi) = 0$, and $q_3 = a_{01}a_{02} - a_{10}a_{12} + a_{20}a_{21} = 0$ implies that $a_{20}(\xi)a_{21}(\xi) = 0$. If $a_{20}(\xi) = 0$, then a_{02} and a_{20} both vanish at ξ and are linearly dependent. If

$a_{21}(\xi) = 0$, then a_{12} and a_{21} both vanish at ξ and are linearly dependent. This is not possible since $\ell \in \mathcal{F}^{\mathrm{ne}}$. Thus $\delta \neq 0$ and $\tau \in T$.

Let $L := \varphi_\tau^{-1}(\ell)$. By (7.4), P_τ and R_τ vanish on L. Thus $(\tau, L) \in \mathcal{J}$ and $\varphi(\tau, L) = \ell$, so the image of φ contains $\mathcal{F}^{\mathrm{ne}}$.

Let $(\tau, L) \in \mathcal{J}$ be a point with $\ell := \varphi(\tau, L) \in \mathcal{F}^{\mathrm{ne}}$. By (7.3), $(1, b, c, d)$ and $(1, e, f, g)$ are vectors in the common kernels of M_0 and M_1 on ℓ. Since β and γ are unique up to scalar, ℓ determines τ. Since φ_τ is an embedding, $L = \varphi_\tau^{-1}(\ell)$. Thus φ is injective over $\mathcal{F}^{\mathrm{ne}}$.

Step 4. In this step we conclude the proof.

Consider the following binary linear forms in independent variables s, t:

(7.5)

$$\begin{aligned}
&a_{01} = s, && a_{02} = -\tfrac{1}{2}s - \sqrt{-1}t, && a_{03} = -s,\\
&a_{10} = t, && a_{12} = -t, && a_{13} = (1+\sqrt{-1})s + \sqrt{-1}t,\\
&a_{20} = \tfrac{1}{2}\big(-\sqrt{-1}s + (1+\sqrt{-1})t\big), && a_{21} = \sqrt{-1}s + (\sqrt{-1}-1)t, && a_{23} = \sqrt{-1}s + (\sqrt{-1}-1)t,\\
&a_{30} = s + \sqrt{-1}t, && a_{31} = -s - t, && a_{32} = s + \sqrt{-1}t.
\end{aligned}$$

A direct calculation shows that (7.5) satisfies $q_0 = q_1 = q_2 = q_3 = 0$. Since $a_{01} = s$ and $a_{10} = t$, the coefficient vectors of s and t are linearly independent, so (7.5) parametrizes a line in $\mathcal{Q}$. The determinants of the coefficients of a_{ij}, a_{ji} with respect to s, t, for $\{i, j\} = \{0,1\}, \{0,2\}, \{0,3\}, \{1,2\}, \{1,3\}, \{2,3\}$ in order, are

$$1, \quad \frac{1-\sqrt{-1}}{4}, \quad -\sqrt{-1}, \quad \sqrt{-1}, \quad -1, \quad -\sqrt{-1}. \tag{7.6}$$

By (7.6), the line of (7.5) is a point of $\mathcal{F}^{\mathrm{ne}}$, so $\mathcal{F}^{\mathrm{ne}}$ is non-empty.

The subset $\mathcal{F}^{\mathrm{ne}}$ is open in $\mathcal{F}$, since the linear independence of a_{ij} and a_{ji} is the non-vanishing of a two-by-two determinant on a coordinate chart of $\mathrm{Gr}(2, 12)$. Thus $\varphi^{-1}(\mathcal{F}^{\mathrm{ne}})$ is a non-empty open subset of $\mathcal{J}$. By **Step 2**, $\varphi^{-1}(\mathcal{F}^{\mathrm{ne}})$ is irreducible and $\dim \varphi^{-1}(\mathcal{F}^{\mathrm{ne}}) = 8$. By **Step 3**, the morphism

$$\varphi^{-1}(\mathcal{F}^{\mathrm{ne}}) \to \mathcal{F}^{\mathrm{ne}}$$

is bijective. Since a bijective morphism of schemes of finite type over $\mathbb{C}$ preserves irreducibility and dimension [Har77, Exercise II.3.22], $\mathcal{F}^{\mathrm{ne}}$ is irreducible and $\dim \mathcal{F}^{\mathrm{ne}} = 8$. □

7.2. **Saturation of the lower system.** In this subsection, we prove Lemma 7.3.

On a line in $\mathcal{Q}$, the sheaf of solutions with $O = 0$ can be larger than the restriction of the sheaf of solutions on $\mathcal{Q}$, since two generators that become dependent at a point yield an extra global solution after division by a linear form. The assumption of generic rank four in Lemma 7.3 holds for a standard pencil matrix of any torsion-free surface by Lemma 4.3(1), so a line on which it fails is not parametrized by a standard pencil matrix of a torsion-free surface.

Lemma 7.3. *Let A be an event-free pencil matrix of generic rank four, and let r be the number of rows of A whose three entries have a common zero on $\mathbb{P}^1$. Let $\mathcal{I}_A \subset \mathcal{O}_{\mathbb{P}^1}(2)^{\oplus 20}$ be the sheaf of local solutions with $O = 0$ of the equations of Construction-Proposition 5.3, in the twenty free entries of C. Then $\mathcal{I}_A \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. In particular, the projection from $\mathcal{U}_A$ to the coefficients of O is injective. Moreover,*

$$4 \leq \dim \mathcal{U}_A \leq 4 + r.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we define the map Λ and the columns V, W, and we prove that the rank of $[V\ W]$ is invariant under signed permutations.

Column $\{i, j\}$ of A has the entries a_{ij} and a_{ji}. Since A is event-free, a_{ij} and a_{ji} have no common zero, so no column vanishes at a point of $\mathbb{P}^1$, and a_{ij}, a_{ji} are linearly independent. Thus the forms a_{ij} define a linear embedding $\mathbb{P}^1 \to \mathcal{Q} \subset \mathbb{P}^{11}$, and $\mathcal{O}_{\mathcal{Q}}(d)$ pulls back to $\mathcal{O}_{\mathbb{P}^1}(d)$.

The twenty free entries of C are $c_0, \dots, c_{19}$ of Table 2, and that table gives the other entries of C. With $O = 0$, the formula of Construction-Proposition 5.3 determines N from C. We denote by

$$\Lambda_{\mathcal{Q}} : \mathcal{O}_{\mathcal{Q}}(2)^{\oplus 20} \to \mathcal{O}_{\mathcal{Q}}(3)^{\oplus 66}, \quad C \to (AN - CP^t, UN^t)$$

the map on $\mathcal{Q}$, with N given by the formula of Construction-Proposition 5.3, and by

$$\Lambda : \mathcal{O}_{\mathbb{P}^1}(2)^{\oplus 20} \to \mathcal{O}_{\mathbb{P}^1}(3)^{\oplus 66}$$

the map on $\mathbb{P}^1$ given by the formula defining $\Lambda_{\mathcal{Q}}$. Then $\mathcal{I}_A = \operatorname{Ker} \Lambda$.

By [SS20, Section 5, p. 22], the equations of the standard resolution in normal form consist of $q_0, \dots, q_3$ and 42 relations linear in the 20 variables of C and the 12 variables of O. The matrix m_a of the 42 relations has twelve rows that involve only O, which form a matrix l_1, and thirty further rows whose part in the variables of C is a matrix l_2.

By Set-up 4.1, the product $d_1' d_2$ has the blocks $AO - B_1 N^t - C B_3^t$, $AN + B_1 B_4 - CP^t$, $AB_3 + B_1 P$, $-UN^t - B_2 B_3^t$, $UB_4 - B_2 P^t$, and UP. Since $B_1 B_4 = B_2 B_3^t = 0$ and the non-zero entries of UP are $\pm q_k$, with all four of $q_0, \dots, q_3$ occurring, by a direct calculation with Table 1, and since $AB_3 + B_1 P = 0$ and $UB_4 = B_2 P^t$ by Construction-Proposition 5.1, the equation $d_1' d_2 = 0$ in normal form is the system of $q_0, \dots, q_3$ and the three equations of Construction-Proposition 5.3. After we eliminate N, the coefficients of y_k in the first equation yield the rows of l_1, and for $O = 0$ the first equation holds identically, and the other two equations are the 66 entries of $\Lambda_{\mathcal{Q}} c$, where $c = (c_0, \dots, c_{19})^t$. Thus the ideals $(q_0, \dots, q_3, \Lambda_{\mathcal{Q}} c)$ and $(q_0, \dots, q_3, l_2 c)$ of $\mathbb{C}[a_{ij}, c_0, \dots, c_{19}]$ are equal, since both are the ideal of the relations of $d_1' d_2 = 0$ with $O = 0$. The entries of $\Lambda_{\mathcal{Q}} c$ and of $l_2 c$ have bidegree $(1,1)$ in (a_{ij}, c_m), and q_k has bidegree $(2,0)$. Comparing the parts of bidegree $(1,1)$, we see that the span over $\mathbb{C}$ of the 66 rows of $\Lambda_{\mathcal{Q}}$ equals the span of the 30 rows of l_2. Thus $\operatorname{Ker} \Lambda_{\mathcal{Q}} = \operatorname{Ker} l_2$ over $\mathcal{Q}$. For arbitrary O, as in the case $O = 0$, the ideal of the relations of $d_1' d_2 = 0$ is generated by $q_0, \dots, q_3$ and the 42 relations of m_a, and also by $q_0, \dots, q_3$ and the equations of Construction-Proposition 5.3 after we eliminate N. Comparing the elements of the ideal that are linear in $c_0, \dots, c_{19}, o_0, \dots, o_{11}$ modulo $q_0, \dots, q_3$, we see that the module over $\mathbb{C}[a_{ij}]/(q_0, \dots, q_3)$ generated by the rows of m_a equals the module generated by the rows of the equations of Construction-Proposition 5.3. Thus $\operatorname{Ker} m_a$ is the sheaf of solutions of the equations of Construction-Proposition 5.3. By [SS20, Section 5, p. 22], $\operatorname{Ker} l_2$ is free, so

$$\operatorname{Ker} \Lambda_{\mathcal{Q}} \cong \mathcal{O}_{\mathcal{Q}}(-1)^{\oplus 2},$$

with two generators whose entries are cubic in the a_{ij}.

A direct calculation shows that either column of Table 6, substituted for C, satisfies $AN - CP^t = UN^t = 0$ modulo $q_0, \dots, q_3$. We expand the products and use

$$a_{12}a_{13} = a_{21}a_{23} - a_{31}a_{32}, \quad a_{02}a_{03} = a_{30}a_{32} - a_{20}a_{23},$$

$$a_{10}a_{13} = a_{01}a_{03} - a_{30}a_{31}, \quad a_{10}a_{12} = a_{01}a_{02} + a_{20}a_{21}.$$

At the point of $\mathcal{Q}$ with

$$(a_{01}, a_{02}, a_{03}, a_{10}, a_{12}, a_{13}, a_{20}, a_{21}, a_{23}, a_{30}, a_{31}, a_{32}) = \left(1, 1, 1, 1, 1, -\frac{1}{3}, 0, 0, 1, 2, \frac{2}{3}, \frac{1}{2}\right),$$

we have $V_0 W_5 - W_0 V_5 = -\frac{1}{9}$. Since $\operatorname{Ker} \Lambda_{\mathcal{Q}}$ is free on two generators of degree three, V and W are constant linear combinations of generators of $\operatorname{Ker} \Lambda_{\mathcal{Q}}$, with an invertible matrix of constants. Thus

$$[V \ W] : \mathcal{O}_{\mathcal{Q}}(-1)^{\oplus 2} \to \mathcal{O}_{\mathcal{Q}}(2)^{\oplus 20}$$

is an isomorphism onto $\operatorname{Ker} \Lambda_{\mathcal{Q}}$.

We turn to the invariance of the rank. Fix a transposition $\sigma = (k, k+1)$ of Theorem 5.2, acting on a_{ij}, y_i, and z_i as in the theorem, so that $\sigma(z) = S_z z$ with a signed permutation matrix S_z. Let $\mathbf{r} = (y_0 y_1, y_0 y_2, y_1 y_2, y_0 y_3, y_1 y_3, y_2 y_3)$, and let Π be the signed permutation matrix with $\mathbf{r}(\sigma(y)) = \mathbf{r}(y)\Pi$. Since $\mathbf{r}, B_3, B_2^t$ are the differentials of the minimal free resolution of the

TABLE 6. The cubic columns V and W, where V_m and W_m are the values of c_m of Table 2.

m	c_m	V_m	W_m
0	$C_{0,0}$	$a_{03}a_{21}a_{23}$	$a_{03}(a_{21}a_{23}-a_{31}a_{32})$
1	$C_{0,2}$	$-a_{02}a_{21}a_{23}$	$-a_{02}(a_{21}a_{23}-a_{31}a_{32})$
2	$C_{0,4}$	$a_{01}a_{21}a_{23}$	$a_{01}(a_{21}a_{23}-a_{31}a_{32})$
3	$C_{0,6}$	$-a_{01}a_{20}a_{23}$	$-a_{01}a_{20}a_{23}+a_{01}a_{30}a_{32}-a_{02}a_{30}a_{31}$
4	$C_{0,7}$	$-a_{03}a_{20}a_{21}$	$-a_{02}a_{30}a_{31}-a_{03}a_{20}a_{21}$
5	$C_{1,1}$	$a_{13}a_{30}a_{32}$	$-a_{13}(a_{20}a_{23}-a_{30}a_{32})$
6	$C_{1,3}$	$-a_{12}a_{30}a_{32}$	$a_{12}(a_{20}a_{23}-a_{30}a_{32})$
7	$C_{1,4}$	$-a_{31}(a_{10}a_{32}-a_{12}a_{30})$	$a_{10}a_{21}a_{23}-a_{10}a_{31}a_{32}+a_{12}a_{30}a_{31}$
8	$C_{1,5}$	$a_{12}a_{30}a_{31}$	$a_{12}a_{30}a_{31}+a_{13}a_{20}a_{21}$
9	$C_{1,6}$	$a_{10}a_{30}a_{32}$	$-a_{10}(a_{20}a_{23}-a_{30}a_{32})$
10	$C_{2,1}$	$a_{23}(a_{10}a_{13}+a_{30}a_{31})$	$a_{23}a_{30}a_{31}$
11	$C_{2,2}$	$-a_{21}(a_{20}a_{23}-a_{30}a_{32})$	$-a_{32}(a_{20}a_{31}-a_{21}a_{30})$
12	$C_{2,3}$	$-a_{10}a_{12}a_{23}+a_{21}a_{30}a_{32}$	$a_{21}a_{30}a_{32}$
13	$C_{2,5}$	$-a_{21}(a_{10}a_{13}+a_{30}a_{31})$	$-a_{21}a_{30}a_{31}$
14	$C_{2,7}$	$a_{20}(a_{10}a_{13}+a_{30}a_{31})$	$a_{20}a_{30}a_{31}$
15	$C_{3,0}$	$-a_{30}(a_{21}a_{23}-a_{31}a_{32})$	$a_{23}(a_{20}a_{31}-a_{21}a_{30})$
16	$C_{3,1}$	$-a_{10}a_{13}a_{32}$	$a_{20}a_{23}a_{31}$
17	$C_{3,3}$	$a_{10}a_{12}a_{32}$	$a_{20}a_{21}a_{32}$
18	$C_{3,5}$	$-a_{10}a_{12}a_{31}$	$-a_{20}a_{21}a_{31}$
19	$C_{3,7}$	$a_{10}a_{12}a_{30}$	$a_{20}a_{21}a_{30}$

coordinate ring of the four coordinate points [SS20, Remark 3.8, (4)–(6)], uniqueness of minimal graded resolutions implies that there exist $E_8 \in \mathrm{GL}_8(\mathbb{C})$ and $D_3 \in \mathrm{GL}_3(\mathbb{C})$ with

$$\Pi B_3(\sigma(y)) = B_3(y)E_8, \quad E_8 B_2(\sigma(y))^t = B_2(y)^t D_3.$$

The Koszul blocks yield a signed permutation matrix E_{12} with

$$S_z^t B_1(\sigma(y)) = B_1(y)E_{12}, \quad E_{12}B_4(\sigma(y))E_{12}^t = -B_4(y).$$

The sign is -1 on all four 3×3 blocks, since σ acts on the three variables of any block by a signed permutation of determinant -1. By the invariance of the matrix A in [SS20, Proposition 4.4], we have

$$A(\sigma(a)) = S_z^t A(a)\Pi^{-1}.$$

Since P, U are determined by A, we obtain

$$P(\sigma(a)) = E_{12}P(a)E_8^{-1}, \quad U(\sigma(a)) = -D_3^{-t}U(a)E_{12}^{-1}.$$

We substitute $\sigma(y)$ for y in $B_1N^t + CB_3^t = 0$. It follows that $(C,N) \to (S_z^t C E_8^t, \Pi N E_{12}^t)$ preserves the solutions of $B_1N^t + CB_3^t = 0$. Thus the map preserves the shape of C of Table 2, which is the condition that, for any i, the coefficient of y_i in row i of CB_3^t vanishes at the three columns whose index pair does not contain i, and it defines $G_\sigma \in \mathrm{GL}_{20}(\mathbb{C})$. The map carries $AN - CP^t$ and UN^t at a to $S_z^t(AN-CP^t)E_{12}^t$ and $-D_3^{-t}(UN^t)\Pi^t$ at $\sigma(a)$. So G_σ carries $\operatorname{Ker}\Lambda_{\mathcal{Q}}$ at a to $\operatorname{Ker}\Lambda_{\mathcal{Q}}$ at $\sigma(a)$. Since $\operatorname{Ker}\Lambda_{\mathcal{Q}}$ is free on generators of degree three,

$$[V\ W](\sigma(a)) = G_\sigma[V\ W](a)g_\sigma \quad \text{modulo } q_0,\dots,q_3$$

with $g_\sigma \in \mathrm{GL}_2(\mathbb{C})$. A direct calculation shows that $g_\sigma = \begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix}$ for $\sigma = (0,1),(2,3)$ and that g_σ is the interchange of V, W for $\sigma = (1,2)$. Hence the rank of $[V\ W]$ at $\sigma(a)$ equals the rank at a, for any point a of $\mathcal{Q}$. By Theorem 5.2, the transpositions σ act by $a_{ij} \to \pm a_{\sigma(i)\sigma(j)}$. So the group of signed permutations permutes the six columns of A, and it preserves the condition that no column vanishes.

Step 2. In this step we prove that $[V\ W]$ has rank two at any point of $\mathcal{Q}$ at which no column of A vanishes.

Let a be such a point. For any pair $\{i,j\}$, fix a non-zero coordinate among a_{ij}, a_{ji} of a, say a_{ij}, and orient $i \to j$. We obtain a tournament on $\{0,1,2,3\}$. A vertex with no incoming edge is a source, and a vertex with no outgoing edge is a sink. If no vertex is a source but some vertex is a sink, then the other three vertices form a directed triangle, since a vertex with two out-neighbors among them would also beat the sink, and hence would be a source. If no vertex is a source or a sink, then all out-degrees are in $\{1,2\}$ with sum 6, so two vertices i,j have out-degree 2 and two vertices k,l have out-degree 1. Say $i \to j$ and $k \to l$. Then $j \to k$ and $j \to l$, next $l \to i$ and $i \to k$, so the tournament is unique up to relabeling. We call it the *strong tournament*.

By **Step 1**, we may relabel with a signed permutation. Thus a is in one of the following three charts:

(1) a tournament with a source, where $a_{01}, a_{02}, a_{03}, a_{12}$ are non-zero;
(2) the tournament with a sink and no source, where $a_{10}, a_{20}, a_{30}, a_{12}, a_{23}, a_{31}$ are non-zero;
(3) the strong tournament, where $a_{01}, a_{12}, a_{23}, a_{30}, a_{02}, a_{13}$ are non-zero.

We use the diagonal action $a_{ij} \to \lambda_i \lambda_j^2 a_{ij}$ of Theorem 5.2. In any line of Table 6, all monomials of V_m and of W_m have one common character of the torus. Hence $[V\ W](\lambda \cdot a) = D_\lambda [V\ W](a)$ with an invertible diagonal matrix D_λ, and the torus preserves the rank of $[V\ W]$. The exponent matrices of the characters of

$$(a_{01}, a_{02}, a_{03}, a_{12}), \quad (a_{10}, a_{20}, a_{30}, a_{12}), \quad (a_{01}, a_{12}, a_{23}, a_{30})$$

have determinants 12, 6, and -15. Since $\mathbb{C}^*$ is divisible, an integer exponent matrix with non-zero determinant defines a surjective homomorphism of tori. Thus the four coordinates can be set equal to one in the three charts, respectively.

Suppose that $[V\ W]$ has rank less than two at a. Then $W = 0$ at a, or $V + \mu W = 0$ at a for some $\mu \in \mathbb{C}$. The identity tables, Tables 7, 8, and 9, exclude both. A line of an identity table with term q_j and coefficient h contributes hq_j, and a line with term c_m and coefficient h contributes $h(V_m + \mu W_m)$ in the second column and hW_m in the third column. An empty entry is zero. After the substitution of the four normalized coordinates, the lines of any column sum to 1. Table 7 holds in the polynomial ring in the remaining coordinates and μ, Table 8 holds in that ring localized at $a_{23}a_{31}$, and Table 9 holds in that ring localized at $a_{02}a_{13}$. Each identity is checked by multiplying the polynomials of the table and collecting terms. At a, all terms of the relevant column vanish, and $a_{23}a_{31}$, respectively $a_{02}a_{13}$, is non-zero, which contradicts the sum 1. Thus $[V\ W]$ has rank two at a.

TABLE 7. The identity table of a tournament with a source, with $a_{01} = a_{02} = a_{03} = a_{12} = 1$.

term	coefficient for $V + \mu W$	coefficient for W
q_1	$-\frac{1}{2}(a_{20}a_{23}a_{30}a_{31}\mu + a_{20}a_{23} - a_{30}a_{31}\mu - 2)$	1
q_2	$\frac{1}{2}(2a_{20}a_{23}a_{30}a_{32} + a_{20}a_{23} - 2a_{30}a_{32})$	
q_3	$\frac{1}{2}a_{30}a_{31}\mu(a_{20}a_{23} - 1)$	
c_0	$-a_{30}^2(\mu+1)(a_{20}a_{23} - 1)$	
c_3	$-\frac{1}{2}(a_{20}a_{23}a_{30}a_{31}\mu + a_{20}a_{23} - a_{30}a_{31}\mu)$	
c_4	$\frac{1}{2}a_{30}a_{31}\mu(a_{20}a_{23} - 1)$	
c_6	$\frac{1}{2}(a_{20}a_{23}a_{30}a_{31}\mu - a_{20}a_{23} - a_{30}a_{31}\mu)$	-1
c_{10}	$-\frac{1}{2}a_{20}$	
c_{15}	$-a_{30}(\mu+1)(a_{20}a_{23} - 1)$	
c_{16}	$a_{30}(a_{20}a_{23} - 1)$	
c_{18}	$-\frac{1}{2}a_{30}\mu(a_{20}a_{23} - 1)$	

Step 3. In this step we prove that $\mathcal{I}_A \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$.

TABLE 8. The identity table of the tournament with a sink and no source, with $a_{10} = a_{20} = a_{30} = a_{12} = 1$.

term	coefficient for $V + \mu W$	coefficient for W
q_0	$\frac{a_{32}-1}{2a_{23}}$	
c_6	$\frac{a_{31}(a_{32}-1)}{2a_{23}}$	
c_7	$-\frac{a_{32}+1}{2a_{23}}$	
c_{10}		$\frac{1}{a_{23}a_{31}}$
c_{11}	$\frac{1}{a_{23}}$	
c_{12}	$-\frac{1}{a_{23}}$	
c_{14}	$\frac{a_{32}+1}{2a_{23}}$	
c_{15}	$-\frac{a_{32}+1}{2a_{23}}$	
c_{16}	$\frac{1}{a_{23}}$	

TABLE 9. The identity table of the strong tournament, with $a_{01} = a_{12} = a_{23} = a_{30} = 1$.

term	coefficient for $V + \mu W$	coefficient for W
q_0	$\frac{a_{02}\mu+a_{02}-a_{10}\mu}{a_{02}a_{13}}$	$\frac{1}{a_{13}}$
q_3	$-\frac{\mu}{a_{02}}$	
c_1	$-\frac{1}{a_{02}a_{13}}$	$-\frac{1}{a_{02}a_{13}}$
c_2	$-\frac{a_{02}+a_{10}}{2a_{02}a_{13}}$	
c_6	$\frac{a_{31}(a_{02}+a_{10})}{2a_{02}a_{13}}$	
c_7	$-\frac{1}{a_{02}a_{13}}$	
c_8	$\frac{1}{a_{02}a_{13}}$	
c_{15}	$-\frac{a_{02}+a_{10}}{2a_{02}a_{13}}$	

By **Step 2**, the restriction of $[V\ W]$ to $\mathbb{P}^1$ has rank two at any point of $\mathbb{P}^1$. Thus the image

$$\mathcal{I}' \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$$

of $[V\ W]$ is a subbundle of $\mathcal{O}_{\mathbb{P}^1}(2)^{\oplus 20}$, the quotient $\mathcal{O}_{\mathbb{P}^1}(2)^{\oplus 20}/\mathcal{I}'$ is locally free, and $\mathcal{I}' \subset \mathcal{I}_A$.

We use the complex $\mathcal{C}_2$ of [SS20, Section 5, pp. 22–23], with the differentials l_2 and g_2, which is a complex of free modules of ranks $2, 20, 30, 12$. By [SS20, Section 5, p. 24], $H_2(\mathcal{C}_2) = H_3(\mathcal{C}_2) = 0$, and $\mathcal{C}_2$ has homology only in $H_0(\mathcal{C}_2) = \operatorname{coker} g_2$ and $H_1(\mathcal{C}_2) = \operatorname{Ker} g_2/\operatorname{Im} l_2$. The supports of $H_0(\mathcal{C}_2)$ and $H_1(\mathcal{C}_2)$ are computed using Macaulay2 in [SS20, Section 5, p. 24] and [SS24, Section 2, pp. 5–6].

By [SS24, Section 2, pp. 5–6], $\operatorname{Supp} H_1(\mathcal{C}_2)$ has 24 irreducible components, which are the translates under the group of signed permutations of the components with the ideals

$$(a_{03}, a_{13}, a_{23}, a_{31}, a_{32}, a_{21}a_{20} - a_{12}a_{10} + a_{02}a_{01}),$$

$$(a_{03}, a_{12}, a_{23}, a_{32}, a_{21}a_{20} + a_{02}a_{01}, a_{31}a_{30} + a_{13}a_{10}).$$

Both ideals contain a_{23} and a_{32}, so the column $\{2, 3\}$ vanishes on both components. By **Step 1**, a column vanishes at any point of $\operatorname{Supp} H_1(\mathcal{C}_2)$. Since no column vanishes at a point of $\mathbb{P}^1$, the line in $\mathcal{Q}$ is disjoint from $\operatorname{Supp} H_1(\mathcal{C}_2)$.

By [SS20, Section 5, p. 24] and [SS24, Section 2, p. 6], the minimal primes of $\operatorname{Supp} H_0(\mathcal{C}_2)$ are exactly the minimal primes of the ideal of 4×4 minors of A in $\mathcal{Q}$. Since A has generic rank

four, a general point x of $\mathbb{P}^1$ is not contained in $\operatorname{Supp} H_0(\mathcal{C}_2)$. Thus $\mathcal{C}_2$ is exact at x. Since the modules of $\mathcal{C}_2$ are free, $l_2(x)$ has rank $20-2=18$, and hence so does $\Lambda(x)$. Thus $\mathcal{I}_A = \operatorname{Ker}\Lambda$ has rank two.

The quotient $\mathcal{I}_A/\mathcal{I}'$ embeds in the locally free sheaf $\mathcal{O}_{\mathbb{P}^1}(2)^{\oplus 20}/\mathcal{I}'$, and $\mathcal{I}_A/\mathcal{I}'$ is a torsion sheaf since both $\mathcal{I}_A$ and $\mathcal{I}'$ have rank two. A locally free sheaf on $\mathbb{P}^1$ has no non-zero torsion subsheaf, so

$$\mathcal{I}_A = \mathcal{I}' \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}.$$

A tuple of $\mathcal{U}_A$ with $O=0$ is a global section of $\mathcal{I}_A$, and $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-1)) = 0$. Thus the projection from $\mathcal{U}_A$ to the coefficients of O is injective.

Step 4. In this step we prove that $4 \le \dim \mathcal{U}_A \le 4+r$.

We first prove the lower bound. Let $\mathcal{E} = \operatorname{Ker} m_a$ be the kernel sheaf on $\mathcal{Q}$ of [SS20, Section 5, p. 23], with domain

$$\mathcal{O}_{\mathcal{Q}}(2)^{\oplus 20} \oplus \mathcal{O}_{\mathcal{Q}}(1)^{\oplus 12}.$$

By [SS20, Section 5, (8), p. 23], where the vanishing of the boundary map is verified computationally, there exists a complex

$$0 \to \mathcal{O}_{\mathcal{Q}}(-1)^{\oplus 2} \to \mathcal{E} \to \mathcal{O}_{\mathcal{Q}}^{\oplus 4} \to 0$$

which is exact except at the last position, because of $H_1(\mathcal{C}_2)$. Since the line is disjoint from $\operatorname{Supp} H_1(\mathcal{C}_2)$, the complex is a short exact sequence of vector bundles on a neighborhood of the line, since an extension of locally free sheaves is locally free. We restrict to $\mathbb{P}^1$, which keeps the sequence exact since the last term is locally free, and obtain

$$h^0(\mathbb{P}^1, \mathcal{E}|_{\mathbb{P}^1}) = 4,$$

since

$$h^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-1)) = h^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-1)) = 0.$$

We prove that the natural map

$$\mathcal{E}|_{\mathbb{P}^1} \to \mathcal{O}_{\mathbb{P}^1}(2)^{\oplus 20} \oplus \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus 12}$$

is injective. Since A has generic rank four, no row of A vanishes identically, so at a general point x of $\mathbb{P}^1$, no row of A vanishes. The complex $\mathcal{C}_1$ of [SS20, Section 5, p. 22] is a direct sum of four Koszul complexes on the rows of A. Hence $\mathcal{C}_1$ is split exact near x, since an entry of any row is a unit near x. By **Step 3**, $\mathcal{C}_2$ is exact at x, and hence split exact near x. So near x, $\operatorname{Ker} l_1$ and $\operatorname{Ker} l_2$ are direct summands of free modules. On the fiber at x, the subbundle $\mathcal{O}(-1)^{\oplus 2}$ of $\mathcal{E}$ maps injectively to the C-summand since $\operatorname{Ker} l_2$ is a direct summand, and the quotient $\mathcal{O}^{\oplus 4}$ of $\mathcal{E}$ maps injectively to the O-summand since $\operatorname{Ker} l_1$ is a direct summand. Thus the natural map is injective on the fiber at x. Since $\mathcal{E}|_{\mathbb{P}^1}$ is locally free on $\mathbb{P}^1$, a generically zero kernel is torsion, and hence zero. So the four independent global sections of $\mathcal{E}|_{\mathbb{P}^1}$ are elements of $\mathcal{U}_A$, and $\dim \mathcal{U}_A \ge 4$.

We now prove the upper bound. Let $(C,O) \in \mathcal{U}_A$. By Construction-Proposition 5.3, $N_{b,3i+e} = [y_{j_e}](AO - CB_3^t)_{i,b}$ for $0 \le e \le 2$, with $j_0 < j_1 < j_2$ as in Table 1, and the twelve determined entries of C are the four equalities and eight zeros of Table 2, which do not involve O. By Table 2, O is alternating, and $O_{b,b'}$ is a binary linear coefficient times y_k for any slot (b,b',k) of the table. For $k \in \{0,1,2,3\}$, let I_k be the set of three columns whose index pair contains k, so that

$$I_0 = \{0,1,3\}, \quad I_1 = \{0,2,4\}, \quad I_2 = \{1,2,5\}, \quad I_3 = \{3,4,5\}.$$

Any slot with y_k has $b, b' \in I_k$, row k of B_1 has no y_k, and the rows of B_3 indexed by I_k have no y_k. Let f_k be the row vector of the three entries of row k of A in the columns I_k, and let ω_k be the alternating 3×3 matrix of the coefficients of y_k in O on I_k. We take the coefficient of y_k in entry (k,b), $b \in I_k$, of $AO - B_1N^t - CB_3^t = 0$, and we obtain

$$f_k \omega_k = 0.$$

The four systems $f_k\omega_k = 0$ are the equations $l_1 O = 0$, one for each row of A, where l_1 is the direct sum of the four skew 3×3 blocks of [SS20, Section 5, p. 22].

Fix k. Since A is event-free, $f_k \neq 0$. We write $f_k = gf'$, where g is the greatest common divisor of the entries of f_k, so that the entries of f' have no common zero. Let w be the row vector $(\omega_{k,23}, -\omega_{k,13}, \omega_{k,12})$, with the entries of ω_k indexed by $1, 2, 3$ in the order of I_k. The equation $f_k\omega_k = 0$ says that the 2×2 minors of the matrix with rows f' and w vanish. Since the entries of f' are coprime, $w = hf'$ with a binary form h of degree $\deg g$. If the entries of f_k have no common zero, then g is constant, and the solutions ω_k form a vector space of dimension 1. If the entries of f_k have a common zero, then g is linear, f' is constant, and h is any binary linear form, so the solutions ω_k form a vector space of dimension 2. We sum over k and obtain that the coefficients of O allowed by $l_1O = 0$ form a vector space Ω with

$$\dim \Omega = 4 + r.$$

By **Step 3**, the projection $\mathcal{U}_A \to \Omega$ is injective. Thus $\dim \mathcal{U}_A \leq 4 + r$. □

7.3. **Vanishing rows.** In this subsection, we bound the dimension of the set of points of $\mathcal{M}$ having an event-free standard pencil matrix with a row whose three entries have a common zero on $\mathbb{P}^1$.

To bound the dimension of such a set, we subtract the dimension of a torus orbit from the dimension of an incidence. We need the dimension of the orbit at any event-free line, not only at a general one.

Lemma 7.4. *Let $\mathbb{T} := (\mathbb{C}^*)^4/\mathbb{C}^*$ act on $\mathcal{F}$ by $a_{ij} \to \lambda_i\lambda_j^2 a_{ij}$. Then any line in $\mathcal{F}^{\mathrm{ne}}$ has finite stabilizer in $\mathbb{T}$. In particular, $\dim \mathbb{T} \cdot L = 3$ for any line L in $\mathcal{F}^{\mathrm{ne}}$.*

Proof. Let $L = \mathbb{P}(\Pi)$ be a line in $\mathcal{F}^{\mathrm{ne}}$, where $\Pi \subset \mathbb{C}^{12}$ is a two-dimensional subspace. Then the restrictions $a_{ij}|_\Pi$ are non-zero, and $a_{ij}|_\Pi$ and $a_{ji}|_\Pi$ are linearly independent for any $i \neq j$.

Suppose that the stabilizer of L in $\mathbb{T}$ is of positive dimension. The identity component of a closed subgroup of a torus is a torus [Bor91, Sections 8.4–8.5], so the stabilizer contains a non-trivial one-parameter subgroup of $\mathbb{T}$. Since the exact sequence of the quotient map $(\mathbb{C}^*)^4 \to \mathbb{T}$ splits, we lift the one-parameter subgroup to $c \to (c^{m_0}, \dots, c^{m_3})$ in $(\mathbb{C}^*)^4$, where $m_0, \dots, m_3$ are integers which are not all equal. It acts on the coordinate a_{ij} of $\mathbb{C}^{12}$ with the weight

$$\omega_{ij} := m_i + 2m_j, \qquad \text{and} \qquad \omega_{ij} - \omega_{ji} = m_j - m_i.$$

Since Π is invariant, and an invariant subspace W of a diagonalizable action is the direct sum of the intersections of W with the weight spaces, Π is the direct sum of the intersections of Π with the weight spaces of $\mathbb{C}^{12}$. The form a_{ij} vanishes on any weight space of weight different from ω_{ij} and $a_{ij}|_\Pi \neq 0$, so ω_{ij} is a weight of Π for any $i \neq j$.

Case 1. Π has exactly one weight. Then $\omega_{ij} = \omega_{ji}$ for any $i \neq j$, so $m_0 = m_1 = m_2 = m_3$, a contradiction.

Case 2. Π has two weights. Then both weight spaces of Π are lines. If $\omega_{ij} = \omega_{kl}$, then $a_{ij}|_\Pi$ and $a_{kl}|_\Pi$ vanish on the weight space of Π of weight different from ω_{ij}, so they are proportional. Since $a_{ij}|_\Pi$ and $a_{ji}|_\Pi$ are linearly independent, $\omega_{ij} \neq \omega_{ji}$ for any $i \neq j$, that is, $m_0, \dots, m_3$ are pairwise distinct. Thus $m_0 + 2m_1$, $m_0 + 2m_2$, and $m_0 + 2m_3$ are three distinct weights of Π, a contradiction.

Hence the stabilizer of L is of dimension zero, so it is finite, and $\dim \mathbb{T} \cdot L = \dim \mathbb{T} = 3$. □

The diagonal action extends to the lower spaces of all pencil matrices, not only of the standard pencil matrices.

Construction-Proposition 7.5. Notation as in Construction-Proposition 5.3. Let A be a pencil matrix, let $\lambda \in (\mathbb{C}^*)^4$, and let $\lambda \cdot A$ be the pencil matrix with entries $\lambda_i\lambda_j^2 a_{ij}$, that is,

$$\lambda \cdot A = \operatorname{diag}(\lambda_0^{-1}, \dots, \lambda_3^{-1}) A \operatorname{diag}(\lambda_a^2\lambda_b^2)_{(a,b)},$$

where the columns of A are indexed by the pairs (a, b) of the relations with quadric $y_a y_b$. Then there exists a linear isomorphism

$$\Phi_\lambda : \mathcal{U}_A \to \mathcal{U}_{\lambda \cdot A}$$

which multiplies each of the 84 coefficients by a Laurent monomial in $\lambda_0, \dots, \lambda_3$. Moreover, let $\mathbf{u} \in \mathcal{U}_A$ and $\rho \in \mathbb{C}^*$. If the lower blocks determined by $(A, \mathbf{u})$ are the lower blocks of a presentation with the quadrics $y_i y_j$ of a numerical Godeaux surface S', then the lower blocks determined by $(\lambda \cdot A, \rho\Phi_\lambda(\mathbf{u}))$ are the lower blocks of a presentation with the quadrics $y_i y_j$ of S'.

Proof. Let $\pi := \lambda_0\lambda_1\lambda_2\lambda_3$ and $\mu_i := \lambda_i^2$. Let α_0 be the diagonal matrix with the entry π on the generators $1, w_0, w_1, w_2$ and the entry π/λ_i on the generator z_i, and let

$$\alpha_z := \operatorname{diag}(\pi/\lambda_0, \dots, \pi/\lambda_3), \qquad \alpha_1 := \operatorname{diag}(\alpha^A, \alpha^{B_1}, \alpha^{B_3}),$$

where $\alpha^A, \alpha^{B_1}, \alpha^{B_3}$ are the diagonal matrices with

$$\alpha^A_{(a,b)} = \frac{\pi}{\mu_a\mu_b}, \qquad \alpha^{B_1}_{(i,j)} = \frac{\pi}{\lambda_i\mu_j}, \qquad \alpha^{B_3}_{\theta} = \frac{\pi}{\mu_{i_\theta}}$$

on the six columns (a, b) of A, the twelve columns (i, j) of B_1, and the eight columns θ of B_3, respectively, where $(i_0, \dots, i_7) := (3, 3, 2, 2, 1, 1, 0, 0)$, so that $\{a, b, k, i_\theta\} = \{0, 1, 2, 3\}$ for any non-zero entry $\pm y_k$ of column θ of B_3 in the row (a, b). A superscript σ on a matrix with entries in $\mathbb{C}[s, t, y_0, \dots, y_3]$ denotes the substitution $y_m \to y_m/\mu_m$, for any m, in the entries.

Let (d_1, d_2) be a pair of matrices of the shape of Set-up 4.1 with the quadrics $y_i y_j$ and the fixed blocks $B_1, \dots, B_4$. We consider the pair

$$\big(\alpha_0 d_1^\sigma \alpha_1^{-1},\ \alpha_1 d_2^\sigma \alpha_1\big). \tag{7.7}$$

A direct calculation with Table 1 shows that the blocks $r_k, B_1, \dots, B_4$ of (7.7) are equal to the blocks $r_k, B_1, \dots, B_4$ of (d_1, d_2) and the other blocks become

$$\widehat{A} = \lambda \cdot A, \quad \widehat{P} = \alpha^{B_1} P \alpha^{B_3}, \quad \widehat{U} = \pi U (\alpha^{B_1})^{-1}, \quad \widehat{C}_{i,\theta} = \frac{\mu_{i_\theta}}{\lambda_i} C_{i,\theta},$$

$$\widehat{N}_{(a,b),(i,j)} = \alpha^A_{(a,b)} \alpha^{B_1}_{(i,j)} N_{(a,b),(i,j)},$$

$$\widehat{O}_{(a,b),(a',b')} = \alpha^A_{(a,b)} \alpha^A_{(a',b')} O^\sigma_{(a,b),(a',b')}.$$

Here the rows of N and the rows and columns of O are indexed by the six pairs (a, b), and the columns of N are indexed by the twelve pairs (i, j). Thus any monomial slot of C and O is multiplied by a non-zero constant, which is a Laurent monomial in $\lambda_0, \dots, \lambda_3$, and $(\widehat{C}, \widehat{O})$ is of the normalized shape of Table 2.

Now let A be a pencil matrix. By Theorem 5.2, $\lambda \cdot A$ is a pencil matrix. Since $\alpha_z B_1^\sigma (\alpha^{B_1})^{-1} = B_1$, $\pi B_2^\sigma (\alpha^{B_3})^{-1} = B_2$, $\alpha^A B_3^\sigma \alpha^{B_3} = B_3$, and $\alpha^{B_1} B_4^\sigma \alpha^{B_1} = B_4$, we have

$$(\lambda \cdot A) B_3 + B_1 \widehat{P} = \alpha_z (A B_3 + B_1 P)^\sigma \alpha^{B_3}, \qquad \widehat{U} B_4 - B_2 \widehat{P}^t = \pi (U B_4 - B_2 P^t)^\sigma \alpha^{B_1},$$

so $\widehat{P}$ and $\widehat{U}$ are the blocks P, U of $\lambda \cdot A$. For $\mathbf{u} = (C, O) \in \mathcal{U}_A$ with the block N determined by $\mathbf{u}$, we let $\Phi_\lambda(\mathbf{u}) := (\widehat{C}, \widehat{O})$. Then

$$\widehat{A}\widehat{O} - B_1 \widehat{N}^t - \widehat{C} B_3^t = \alpha_z (AO - B_1 N^t - C B_3^t)^\sigma \alpha^A,$$

$$\widehat{A}\widehat{N} - \widehat{C}\widehat{P}^t = \alpha_z (AN - CP^t) \alpha^{B_1}, \qquad \widehat{U}\widehat{N}^t = \pi U N^t \alpha^A,$$

so $\Phi_\lambda(\mathbf{u}) \in \mathcal{U}_{\lambda \cdot A}$. The map Φ_λ is linear, it multiplies each of the 84 coefficients by a Laurent monomial in $\lambda_0, \dots, \lambda_3$, and the inverse of Φ_λ is $\Phi_{\lambda^{-1}}$.

Assume that the lower blocks determined by $(A, \mathbf{u})$ are the lower blocks of a presentation (d_1, d_2) with the quadrics $y_i y_j$ of S'. We use the generators

$$y'_i := \mu_i y_i, \qquad z'_i := \lambda_i z_i, \qquad w'_k := w_k$$

of the canonical ring of S', with s, t fixed, which is the change of [Ste18, Section 8.3, pp. 100–101]. Since $y_m = y'_m/\mu_m$ and $z_i = z'_i/\lambda_i$, the columns of $\alpha_0 d_1^\sigma$, read as relations among the new generators, are π times the relations given by the columns of d_1. The matrices α_0 and α_1 are invertible, and the substitution of the superscript σ is induced by an automorphism of the polynomial ring. Thus (7.7) is a presentation of S' with the quadrics $y'_i y'_j$, and the lower blocks of (7.7) are determined by $(\lambda \cdot A, \Phi_\lambda(\mathbf{u}))$.

It remains to treat the scalar ρ. Possibly replacing $(A, \mathbf{u})$ with $(\lambda \cdot A, \Phi_\lambda(\mathbf{u}))$ and y_i, z_i, w_k with y_i', z_i', w_k', we may assume that $\lambda = (1,1,1,1)$. Let $\kappa \in \mathbb{C}^*$ with $\kappa^{-2} = \rho$. By Lemma 5.4, applied with the scalar κ, S' has a presentation with the quadrics $y_i y_j$, with the blocks $A, U, P, B_1, \dots, B_4$ of (d_1, d_2), and with $\rho(C, O, N)$ in the place of (C, O, N). Thus the lower blocks determined by $(A, \rho\mathbf{u})$ are the lower blocks of a presentation with the quadrics $y_i y_j$ of S'. □

The rows whose entries have a common zero are the only possible excess in the bound of Lemma 7.3, each such row adds at most one, and the codimension of the locus of lines with such a row offsets the excess.

Construction-Proposition 7.6. Notation as in Construction-Proposition 7.2. For a line in $\mathcal{F}^{\mathrm{ne}}$, we call row i of A a *vanishing row* if the three forms a_{ij} with $j \neq i$ have a common zero on $\mathbb{P}^1$. Let r be the number of vanishing rows of A, let $Z_r \subset \mathcal{F}^{\mathrm{ne}}$ be the locus of lines with exactly r vanishing rows, and let $\mathcal{M}_{\mathrm{row}}$ be the set of points of $\mathcal{M}_{\mathrm{ne}}$ having an event-free standard pencil matrix with $r \geq 1$. Then $r \leq 2$ for any line in $\mathcal{F}^{\mathrm{ne}}$, $\dim Z_1 \leq 6$, $\dim Z_2 \leq 5$, and $\dim \overline{\mathcal{M}_{\mathrm{row}}} \leq 7$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove that $r \leq 2$ for any line in $\mathcal{F}^{\mathrm{ne}}$.

Let $L \in \mathcal{F}^{\mathrm{ne}}$, and identify $\mathbb{P}^1$ with L. The twelve forms a_{ij} are non-zero on L. If row i is a vanishing row, then there exist a linear form θ_i on L with zero $\xi_i \in L$ and non-zero constants γ_{ij} with $a_{ij} = \gamma_{ij}\theta_i$ for any $j \neq i$. If rows $i \neq j$ are vanishing rows, then $\xi_i \neq \xi_j$, since otherwise a_{ij} and a_{ji} have a common zero.

By Theorem 5.2, the signed permutations act on $\mathcal{F}$, preserve $\mathcal{F}^{\mathrm{ne}}$, and send vanishing rows to vanishing rows. Suppose that $r \geq 3$. Possibly replacing L with the image of L under a signed permutation, we may assume that rows $0, 1, 2$ are vanishing rows. Then $q_3 = 0$ implies

$$\gamma_{01}\gamma_{02}\theta_0^2 - \gamma_{10}\gamma_{12}\theta_1^2 + \gamma_{20}\gamma_{21}\theta_2^2 = 0.$$

Choose coordinates on L with $\theta_0 = s$ and $\theta_1 = t$. Then $\theta_2 = \alpha s + \beta t$ with $\alpha\beta \neq 0$, since ξ_0, ξ_1, ξ_2 are pairwise distinct. The coefficient of st implies $\gamma_{20}\gamma_{21} = 0$, a contradiction.

Step 2. In this step we describe the common zeros of rows 0 and 1 by finite covers of open subsets of T.

Recall from Construction-Proposition 7.2 that $T = \{(b,c,d,e,f,g) \in \mathbb{A}^6 \mid be(cf - dg) \neq 0\}$, that for $\tau = (b,c,d,e,f,g) \in T$ we have

$$P_\tau = vz - uw + ebx^2 + ecxz - gbxw - gczw,$$
$$R_\tau = u(fx + gy) - v(dx + cy) + eby^2 - edyz + fbyw - fdzw,$$

and that the substitution is

$$a_{01} = u,\ a_{10} = v,\ a_{23} = x,\ a_{32} = y,\ a_{13} = z,\ a_{03} = w,\ a_{12} = cy + dx,\ a_{21} = -by + dz,$$
$$a_{31} = -bx - cz,\ a_{02} = fx + gy,\ a_{30} = -ex + gw,\ a_{20} = -ey - fw.$$

Let $\delta := cf - dg$, so that $be\delta \neq 0$ on T. We let

$$T_0 := \{\tau \in T \mid df \neq 0\},\quad T_0^g := T_0 \cap \{g \neq 0\},\quad T_0^c := T_0 \cap \{c \neq 0\},\quad T_{01} := T_0^g \cap T_0^c.$$

The sets T_0, T_0^g, T_0^c, and T_{01} are open subsets of the irreducible variety T of dimension 6, and they are non-empty, since they contain $(1,1,1,1,1,2)$. Let $L \in \mathcal{F}^{\mathrm{ne}}$. By Construction-Proposition 7.2, L is the image of a unique point of $\mathcal{J}$ over some $\tau \in T$, and $\tau \in T_0$. We identify L with the corresponding line in $V(P_\tau, R_\tau)$, and we regard the a_{ij} as the linear forms in (u, v, x, y, z, w) of the substitution above.

Assume that row 0 of L is a vanishing row. Then $u = w = fx + gy = 0$ at ξ_0. Since $a_{02}(\xi_0) = 0$, we have $a_{20}(\xi_0) \neq 0$. Since $a_{20} = -ey - fw$, the coordinate y is non-zero at ξ_0, and we normalize $y = 1$. Thus $x = -g/f$ at ξ_0. Since $a_{03}(\xi_0) = 0$, we have $a_{30}(\xi_0) = eg/f \neq 0$, so $g \neq 0$ and $\tau \in T_0^g$.

The equation $R_\tau(\xi_0)=0$ implies $v=ef(b-dz)/\delta$, and the equation $P_\tau(\xi_0)=0$, multiplied by $f^2\delta/e$, is

$$\phi_0(z) := f^3z(b-dz) - cgf\delta z + bg^2\delta = 0. \tag{7.8}$$

Let $\mathcal{Y}_0 \subset T_0^g \times \mathbb{A}^1$ be the closed subscheme defined by (7.8). Since ϕ_0 is of degree two with the unit leading coefficient $-f^3d$, the coordinate ring of $\mathcal{Y}_0$ is free of rank two over the coordinate ring of T_0^g, so $\mathcal{Y}_0$ is finite over T_0^g. For $(\tau,z)\in\mathcal{Y}_0$, we let

$$\xi_0(\tau,z) := \left(0, \frac{ef(b-dz)}{\delta}, -\frac{g}{f}, 1, z, 0\right).$$

Then $\xi_0(\tau,z)\in V(P_\tau,R_\tau)$, and the zero of row 0 of L is $\xi_0(\tau,z)$ for some $(\tau,z)\in\mathcal{Y}_0$.

Assume that row 1 of L is a vanishing row. Then $v=z=dx+cy=0$ at ξ_1. Since $a_{12}(\xi_1)=0$ and $a_{21}=-by+dz$, we have $y\neq 0$ at ξ_1, and we normalize $y=1$. Thus $x=-c/d$ at ξ_1. Since $a_{13}(\xi_1)=0$, we have $a_{31}(\xi_1)=bc/d\neq 0$, so $c\neq 0$ and $\tau\in T_0^c$. The equation $R_\tau(\xi_1)=0$ implies $u=bd(e+fw)/\delta$, and the equation $P_\tau(\xi_1)=0$, multiplied by $d^2\delta/b$, is

$$\phi_1(w) := -d^3w(e+fw) + gcd\delta w + ec^2\delta = 0. \tag{7.9}$$

Let $\mathcal{Y}_1\subset T_0^c\times\mathbb{A}^1$ be the closed subscheme defined by (7.9). Since ϕ_1 is of degree two with the unit leading coefficient $-d^3f$, $\mathcal{Y}_1$ is finite over T_0^c. For $(\tau,w)\in\mathcal{Y}_1$, we let

$$\xi_1(\tau,w) := \left(\frac{bd(e+fw)}{\delta}, 0, -\frac{c}{d}, 1, 0, w\right).$$

Then $\xi_1(\tau,w)\in V(P_\tau,R_\tau)$, and the zero of row 1 of L is $\xi_1(\tau,w)$ for some $(\tau,w)\in\mathcal{Y}_1$.

Step 3. In this step we prove that $\dim Z_1\le 6$.

Let $\mathcal{J}_0\subset\mathcal{Y}_0\times_T\mathcal{J}$ be the closed subset of triples (τ,z,L), where $(\tau,z)\in\mathcal{Y}_0$ and $L\in F_1(V(P_\tau,R_\tau))$ is a line through $\xi_0(\tau,z)$. By Construction-Proposition 7.2, P_τ,R_τ satisfy the hypotheses of Lemma 7.1 for any $\tau\in T$. Since $\mathcal{Y}_0$ is finite over T_0^g, Lemma 7.1 implies that the fibers of $\mathcal{J}_0\to T_0^g$ are of dimension at most one, and they are finite over $T_{\mathrm{sm}}\cap T_0^g$, since $V(P_\tau,R_\tau)$ is smooth for $\tau\in T_{\mathrm{sm}}$. The set $T_{\mathrm{sm}}\cap T_0^g$ is non-empty, since T_{sm} and T_0^g are non-empty open subsets of the irreducible variety T. Thus the complement of $T_{\mathrm{sm}}\cap T_0^g$ in T_0^g is of dimension at most 5, and [Har77, Exercise II.3.22] implies

$$\dim\mathcal{J}_0\le\max(6+0,5+1)=6.$$

By **Step 2**, the image of $\mathcal{J}_0$ in $\mathcal{F}$ contains any line in $\mathcal{F}^{\mathrm{ne}}$ whose row 0 is a vanishing row. The signed permutations of Theorem 5.2 cover the other three rows. Hence the locus of lines in $\mathcal{F}^{\mathrm{ne}}$ with at least one vanishing row is of dimension at most 6. In particular, $\dim Z_1\le 6$.

Step 4. In this step we prove that $\dim Z_2\le 5$.

Let $\mathcal{Y}_{01}$ be the closed subscheme of $\mathcal{Y}_0\times_{T_{01}}\mathcal{Y}_1$ defined by

$$P_\tau(\xi_0+\xi_1)=R_\tau(\xi_0+\xi_1)=0, \tag{7.10}$$

where ξ_0,ξ_1 are the vectors above. Since $\xi_0,\xi_1\in V(P_\tau,R_\tau)$, the equations (7.10) hold if and only if the line through ξ_0 and ξ_1 is contained in $V(P_\tau,R_\tau)$. We have $\xi_0\neq\xi_1$: on T_{01},

$$\phi_0\left(\frac{b}{d}\right) = -\frac{bg\delta^2}{d}\neq 0,$$

so the coordinate v of ξ_0 is non-zero, whereas the coordinate v of ξ_1 is zero.

The scheme $\mathcal{Y}_{01}$ is finite over T_{01}, so the image D of $\mathcal{Y}_{01}$ in T_{01} is closed [Har77, Exercise II.3.5(b)]. We show that $D\neq T_{01}$. Let $(b,c,d,e,f,g)=(1,1,1,1,1,2)$, so that $\delta=-1$. Then (7.8) and (7.9) become

$$z^2-3z+4=0, \qquad w^2+3w+1=0,$$

and

$$\xi_0=(0,z-1,-2,1,z,0), \qquad \xi_1=(-1-w,0,-1,1,0,w).$$

At a point of the fiber of $\mathcal{Y}_0 \times_{T_{01}} \mathcal{Y}_1$, we have

$$P_\tau(\xi_0+\xi_1) = 4 - z + 4w - 2zw, \qquad R_\tau(\xi_0+\xi_1) = 2 - z + w - zw.$$

If $P_\tau(\xi_0+\xi_1) = R_\tau(\xi_0+\xi_1) = 0$, then $P_\tau(\xi_0+\xi_1) - 2R_\tau(\xi_0+\xi_1) = z + 2w = 0$, and substituting $z = -2w$ in $R_\tau(\xi_0+\xi_1) = 0$ implies $2w^2 + 3w + 2 = 0$. Since $w^2 + 3w + 1 = 0$, we obtain $-3w = (2w^2+3w+2) - 2(w^2+3w+1) = 0$, so $w = 0$, which contradicts $w^2+3w+1 = 0$. Thus $D \neq T_{01}$. Since T_{01} is irreducible of dimension 6, we have $\dim D \leq 5$ and $\dim \mathcal{Y}_{01} \leq 5$.

Let $L \in \mathcal{F}^{\mathrm{ne}}$ be a line whose rows 0 and 1 are vanishing rows. By **Step 2**, $\tau \in T_{01}$, and L is the line through $\xi_0(\tau, z)$ and $\xi_1(\tau, w)$ for a point $(\tau, z, w) \in \mathcal{Y}_{01}$. The line depends algebraically on (τ, z, w), by the non-zero Plücker vector $\xi_0 \wedge \xi_1$. Thus the locus of L in $\mathcal{F}^{\mathrm{ne}}$ is of dimension at most 5 [Har77, Exercise II.3.22]. Any line in Z_2 has exactly two vanishing rows, and the signed permutations of Theorem 5.2 send any of the six pairs of rows to the rows $0, 1$. Hence $\dim Z_2 \leq 5$.

Step 5. In this step we prove that $\dim \overline{\mathcal{M}_{\mathrm{row}}} \leq 7$.

Fix the quadrics $y_i y_j$ of Construction-Proposition 5.1, and let $\mathcal{L}$ be the affine space of all lower blocks of the shape of Set-up 4.1. Let $G := \Xi_{\mathcal{L}} \subset \mathcal{L} \times \mathcal{M}$ be the constructible set of Lemma 4.5(2) for the quadrics $r_1, \ldots, r_6$.

Fix $r \in \{1, 2\}$. We cover $\mathrm{Gr}(2, 12)$ by the finitely many standard affine charts. On a chart, a two-dimensional subspace of $\mathbb{C}^{12}$ has a canonical basis, namely the basis in reduced echelon form with respect to the pivot coordinates of the chart, and the coefficients of the basis are regular on the chart. Fix a chart. For a line L of $\mathcal{F}$ in the chart, the canonical basis defines a pencil matrix A_L of L. Let Z'_r be the set of lines $L \in Z_r$ in the chart with A_L of generic rank four. We send a pair $(L, \mathbf{u})$ with $L \in Z'_r$ and $\mathbf{u} \in \mathcal{U}_{A_L}$ to the lower blocks determined by $(A_L, \mathbf{u})$: the blocks P, U are given by A_L, and N and the twelve remaining entries of C are given by the first block equation of Construction-Proposition 5.3. The map is regular and injective. Let $V \subset \mathcal{L}$ be the image, $V^* \subset V$ the image of the pairs with $\mathbf{u} \neq 0$, and $V^0 \subset V$ the image of the pairs with $\mathbf{u} = 0$. Then V, V^*, and V^0 are constructible by Chevalley's theorem [Har77, Exercise II.3.19], since in a chart a vanishing row is a closed condition, exactly r vanishing rows a locally closed one, and generic rank four an open one. By Lemma 7.3, $\dim \mathcal{U}_{A_L} \leq 4 + r$ for any $L \in Z'_r$, so [Har77, Exercise II.3.22] implies

$$\dim V^* \leq \dim Z_r + 4 + r, \qquad \dim V^0 \leq \dim Z_r.$$

Let $G^* := G \cap (V^* \times \mathcal{M})$ and $G^0 := G \cap (V^0 \times \mathcal{M})$. By Lemma 4.5(2), any non-empty fiber of $G^* \to V^*$ or $G^0 \to V^0$ is a point.

We bound the fibers over $\mathcal{M}$ from below. Let $(\ell, [X]) \in G$, where ℓ is determined by $(A_L, \mathbf{u})$ with $L \in Z'_r$. The surface in the definition of G need not be marked, so $\mathbf{u}$ need not be geometric, and we use Construction-Proposition 7.5 instead of Theorem 5.2. Let $\lambda \in (\mathbb{C}^*)^4$ and $\rho \in \mathbb{C}^*$. By Construction-Proposition 7.5, the lower blocks determined by $(\lambda \cdot A_L, \rho\Phi_\lambda(\mathbf{u}))$ are the lower blocks of a presentation with the quadrics $y_i y_j$ of a surface with canonical model X. Assume that $\lambda \cdot L$ is contained in the chart. Then $\lambda \cdot L \in Z'_r$, since the entries of A_L are multiplied by non-zero scalars, which preserves $\mathcal{F}^{\mathrm{ne}}$ and the vanishing rows, and since $\lambda \cdot A_L$ is the product of A_L with invertible diagonal matrices (Construction-Proposition 7.5), which preserves the generic rank, as does the binary reparametrization from $\lambda \cdot A_L$ to $A_{\lambda \cdot L}$ (Construction-Proposition 5.1). A linear change of s, t sends a presentation of a surface S' to a presentation of S' and preserves the slots of C and O. Let h_λ be the linear change of s, t which sends $\lambda \cdot A_L$ to $A_{\lambda \cdot L}$, and let $\mathbf{u}_{\lambda,\rho} \in \mathcal{U}_{A_{\lambda \cdot L}}$ be the image of $\rho\Phi_\lambda(\mathbf{u})$ under h_λ, so that $\mathbf{u}_{\lambda,\rho} = \rho\mathbf{u}_{\lambda,1}$. Let $\mathcal{K} \subset \mathcal{L}$ be the set of lower blocks determined by $(A_{\lambda \cdot L}, \mathbf{u}_{\lambda,\rho})$ for any λ with $\lambda \cdot L$ in the chart and any ρ. Then $\mathcal{K}$ is contained in the fiber of G over $[X]$.

Assume that $\mathbf{u} \neq 0$. By Lemma 7.4, $\dim \mathbb{T} \cdot L = 3$, and the intersection of $\mathbb{T} \cdot L$ with the chart is a non-empty open subset of $\mathbb{T} \cdot L$. Thus the projection from $\mathcal{K}$ to the lines has image of dimension 3. Any fiber of the projection contains the ray of scalars $\rho\mathbf{u}_{\lambda,1}$, of dimension 1, since $\mathbf{u}_{\lambda,1} \neq 0$. Thus $\mathcal{K} \subset V^*$. Since h_λ depends regularly on λ, $\mathcal{K}$ is the image of an irreducible variety

under a regular map, and $\dim \mathcal{K} \geq 4$, since the image of the projection is three-dimensional and the fibers are of dimension at least one. By Lemma 2.9, applied to $G^* \subset \mathcal{L} \times \mathcal{M}$, we have

$$\dim \overline{\mathrm{pr}_{\mathcal{M}}(G^*)} \leq \dim V^* - 4 \leq \dim Z_r + r.$$

By **Steps 3 and 4**, the right-hand side is at most $6 + 1 = 7$ for $r = 1$ and at most $5 + 2 = 7$ for $r = 2$.

Assume that $\mathbf{u} = 0$. Then $\mathbf{u}_{\lambda,\rho} = 0$, so $\mathcal{K} \subset V^0$, and $\mathcal{K}$ is identified with the intersection of $\mathbb{T} \cdot L$ with the chart, so $\dim \mathcal{K} = 3$ by Lemma 7.4. By Lemma 2.9, applied to $G^0 \subset \mathcal{L} \times \mathcal{M}$, we have

$$\dim \overline{\mathrm{pr}_{\mathcal{M}}(G^0)} \leq \dim V^0 - 3 \leq \dim Z_r - 3 \leq 3.$$

It remains to prove that the sets $\mathrm{pr}_{\mathcal{M}}(G^*)$ and $\mathrm{pr}_{\mathcal{M}}(G^0)$ cover $\mathcal{M}_{\mathrm{row}}$, where r and the chart vary. Let $[X] \in \mathcal{M}_{\mathrm{row}}$, and let A be an event-free standard pencil matrix of a surface S with canonical model X and $r \geq 1$. Since a_{ij} and a_{ji} have no common zero, they are linearly independent, so A parametrizes a line $L \in \mathcal{F}^{\mathrm{ne}}$. By **Step 1**, $L \in Z_1 \cup Z_2$. Since S is torsion-free, A is of generic rank four by Lemma 4.3(1). By Construction-Proposition 5.1, a binary reparametrization sends A to a standard pencil matrix of S, and it preserves the slots of C and O, so we may assume that $A = A_L$ for a chart containing L. By Construction-Proposition 5.3, S has a standard presentation with a geometric tuple $\mathbf{u} \in \mathcal{U}_{A_L}$. Thus $L \in Z_r'$ and $[X]$ is contained in $\mathrm{pr}_{\mathcal{M}}(G^*)$ or $\mathrm{pr}_{\mathcal{M}}(G^0)$. There exist finitely many values of r and finitely many charts, so $\dim \overline{\mathcal{M}_{\mathrm{row}}} \leq 7$. □

7.4. **The rank-four bundle and the bound.** In this subsection, we prove Theorem 7.10.

To compare the family $\mathcal{S}$ with $\mathcal{M}$, we first construct a morphism from the parameter space of the family to $\mathcal{M}$.

Construction-Proposition 7.7. Let $\mathcal{D}$ be the parameter space of [SS20, Corollary 5.2], with the reduced structure, so that $\mathcal{D}$ is irreducible of dimension $8+3$ and the family $\mathcal{S}$ of Definition 2.6 is the family over $\mathcal{D}$ modulo the action of $(\mathbb{C}^*)^3$. Then the map sending a point of $\mathcal{D}$ to the point $[X]$ of the canonical model X of the surface of the point is a morphism $\mathcal{D} \to \mathcal{M}$, and $\mathcal{B}_{\mathrm{SS}}$ is the Zariski closure of the image of the morphism.

Proof. By [SS20, Introduction, p. 2], the resolution of the canonical ring of a surface of the family is given by a solution of the equations in the unfolding parameters, and by [SS20, Corollary 5.2], any point of $\mathcal{D}$ yields a numerical Godeaux surface, with the resolution given by the point. By [SS20, Introduction, p. 2], a point of $\mathcal{D}$ is a line in $\mathcal{Q}$ together with the class, up to a common scalar, of a solution over the line. Thus any closed point of $\mathcal{D}$ has an open neighborhood $\mathcal{D}''$ on which, after we take the basis of the line given by a standard affine chart of $\mathrm{Gr}(2, 12)$ and a representative of the class with regular coefficients, the blocks of d_1' and d_2 are matrices over $\mathcal{O}_{\mathcal{D}''}[s, t, y_0, \ldots, y_3]$, and at any closed point x of $\mathcal{D}''$, they are the lower blocks of a presentation with the quadrics $y_i y_j$ of the surface of x. Another basis or another representative changes the presentation by a binary reparametrization of Construction-Proposition 5.1 or by Lemma 5.4, and not the surface. We denote by R_x the canonical ring of the surface of x.

A first row of d_1 is a row ρ with $\rho d_2 = 0$, whose first six entries are the quadrics $y_i y_j$ plus binary cubics. At a closed point x, Theorem 2.10(3) and the alternation of d_2 imply that the rows ρ with $\rho d_2 = 0$ are the linear combinations of the rows of d_1 with coefficients in S_0, so the dimension of the solutions in any degree does not depend on x. Since $\mathcal{D}''$ is reduced, the solutions form a vector bundle on $\mathcal{D}''$ [Har77, Exercise II.5.8(c)]. At a closed point, the quadratic part in $y_0, \ldots, y_3$ of the first six entries of such a row ρ is a constant multiple of $(y_i y_j)$, since the other rows of d_1 contribute only binary cubics there, and the constant is 1 for the first row of the presentation. Thus the coefficient of $y_0 y_1$ in the first entry is a linear form on the vector bundle which is non-zero at any closed point, and any closed point of $\mathcal{D}''$ has an open neighborhood $\mathcal{D}' \subset \mathcal{D}''$ over which there exists a first row of d_1 with regular coefficients. Let R' be the cokernel of the first differential over $\mathcal{D}'$. At a closed point x, the first row differs from the first row of the presentation by a linear combination of the other rows of d_1, with coefficients in S_0, so the

fiber of R' at x is R_x as a graded S_0-module, and the graded pieces of R' have fibers of constant dimension, and they are locally free [Har77, Exercise II.5.8(c)].

Possibly shrinking $\mathcal{D}'$, the products of the generators $1, z_0, \dots, z_3, w_0, w_1, w_2$ are the unknowns of a linear system over $\mathcal{O}_{\mathcal{D}'}$, namely the S_0-bilinearity on the columns of d_1 together with the unit 1. At a closed point x, the multiplication of R_x is a solution. Let μ be a solution at x, and let $m \in R_x$ be homogeneous. By Theorem 2.10(2), there exists a non-zero homogeneous $\sigma \in S_0/\operatorname{ann}_{S_0} R_x$ with $\sigma m \in S_0/\operatorname{ann}_{S_0} R_x$. Then $\sigma\mu(m, m') = \mu(\sigma m, m') = \sigma m m'$, and R_x is a domain, so $\mu(m, m') = mm'$. Thus the multiplication of R_x is the only solution at x, and the matrix of the system has zero kernel at any closed point. Hence, near any closed point, some maximal minor of the matrix is invertible, and Cramer's rule yields a solution of the equations of the minor with regular coefficients. The value of the solution at any closed point x is the multiplication of R_x, so the solution satisfies all equations at any closed point, and hence over the reduced scheme $\mathcal{D}'$. Commutativity and associativity hold at any closed point, and hence over $\mathcal{D}'$. Therefore, R' is a graded $\mathcal{O}_{\mathcal{D}'}$-algebra with fiber R_x at any closed point x, and $\operatorname{Proj} R' \to \mathcal{D}'$ is a flat family of canonical models (the graded pieces of R' are locally free, the homogeneous localizations of R' are direct summands of localizations of R', and Proj commutes with base change).

By Lemma 4.5(1), the flat family induces a morphism $\mathcal{D}' \to \mathcal{M}$, which sends a closed point to the point of the statement. Since $\mathcal{D}$ is reduced and $\mathcal{M}$ is a separated scheme by Theorem 2.7(1), the morphisms on such open subsets $\mathcal{D}'$ glue to a morphism $\mathcal{D} \to \mathcal{M}$. The fibers of $\mathcal{S}$ are the surfaces of the points of $\mathcal{D}$ by [SS20, Corollary 5.2], so Definition 2.6 implies that $\mathcal{B}_{\mathrm{SS}}$ is the Zariski closure of the image of the morphism. □

The family $\mathcal{S}$ is constructed over a general line, and we now locate that line in $\mathcal{F}^{\mathrm{ne}}$.

Proposition 7.8. *Notation as in Construction-Proposition 7.7. Then a general member of the family $\mathcal{S}$ of Definition 2.6 has an event-free standard pencil matrix, and the lines in $\mathcal{F}^{\mathrm{ne}}$ of the points of $\mathcal{D}$ are dense in $\mathcal{F}^{\mathrm{ne}}$.*

Proof. By Theorem 1.4 and [SS20, Proposition 3.4], any member of $\mathcal{S}$ is a torsion-free surface as in Construction-Proposition 5.1, with bicanonical pencil of genus 4 by Construction-Proposition 3.1(2), so any member of $\mathcal{S}$ has a standard pencil matrix.

Let S be a general member of $\mathcal{S}$, with canonical model X. Any small deformation of S occurs in $\mathcal{S}$, so the germ of the base of $\mathcal{S}$ at the point of S surjects onto $\operatorname{Def}(S)$. By Theorem 2.7, $\operatorname{Def}(S) \to \operatorname{Def}(X)$ is finite and surjective, and $(\mathcal{M}, [X])$ is isomorphic to $\operatorname{Def}(X)/\operatorname{Aut}(X)$. Thus the set of points $[X']$, where X' is the canonical model of a fiber of $\mathcal{S}$, contains an analytically open neighborhood of $[X]$. Lemma 2.8 implies that $\mathcal{B}_{\mathrm{SS}}$ is not contained in a Zariski closed subset of $\mathcal{M}$ of dimension at most 7. Therefore, $\dim \mathcal{B}_{\mathrm{SS}} \geq 8$.

By Construction-Proposition 7.7, we have a morphism $\mathcal{D} \to \mathcal{M}$, and $\mathcal{B}_{\mathrm{SS}}$ is the Zariski closure of the image of the morphism. Since $\dim \mathcal{B}_{\mathrm{SS}} \geq 8$ and $\dim \overline{\mathcal{M}_{\mathrm{ev}}} \leq 7$ by Theorem 6.9, the image of the morphism is not contained in $\overline{\mathcal{M}_{\mathrm{ev}}}$. Since $\mathcal{D}$ is irreducible, we may assume that $[X] \notin \overline{\mathcal{M}_{\mathrm{ev}}}$. Thus no standard pencil matrix of S has an event. Therefore a_{ij} and a_{ji} have no common zero, so they are linearly independent, for any $i \neq j$ and any standard pencil matrix of S. In particular, S has an event-free standard pencil matrix, and the line of any standard pencil matrix of S is a point of $\mathcal{F}^{\mathrm{ne}}$.

By [SS20, Introduction, Theorem 5.1, and Corollary 5.2], $\mathcal{D}$ is irreducible of dimension $8 + 3$, and the solution space over a general line of the family is of dimension 4. Let $\mathcal{E} \subset \mathcal{F}$ be the set of lines of the points of $\mathcal{D}$. Thus $\mathcal{E}$ is an irreducible constructible set [Har77, Exercise II.3.19]. The points of $\mathcal{D}$ over a general element of $\mathcal{E}$ are classes, up to a common scalar, of elements of the solution space over that line, so the fiber of $\mathcal{D} \to \mathcal{E}$ over a general element of $\mathcal{E}$ is contained in the projectivization $\mathbb{P}^3$ of the solution space over that line, and hence the fiber is of dimension at most 3. Therefore, $\dim \mathcal{E} \geq 11 - 3 = 8$ [Har77, Exercise II.3.22]. A general element of $\mathcal{E}$ is a point of $\mathcal{F}^{\mathrm{ne}}$ by the paragraph above, and $\mathcal{F}^{\mathrm{ne}}$ is irreducible with $\dim \mathcal{F}^{\mathrm{ne}} = 8$ by Construction-Proposition 7.2. Hence $\mathcal{E} \cap \mathcal{F}^{\mathrm{ne}}$ is dense in $\mathcal{F}^{\mathrm{ne}}$. □

On the lines of $\mathcal{F}^{\mathrm{ne}}$ with no vanishing row and with A of generic rank four, the solution spaces satisfy $\dim \mathcal{U}_A = 4$, so all lower solutions form one vector bundle, and the family of [SS20, Corollary 5.2] is dense in the projectivization of the vector bundle.

Construction-Proposition 7.9. Notation as in Construction-Proposition 7.6. Let $\mathcal{F}^\circ \subset \mathcal{F}^{\mathrm{ne}}$ be the open subset, with the reduced structure, of lines with no vanishing row and with A of generic rank four, and let $\mathcal{H}$ be the tautological bundle of rank 2 on $\mathcal{F}^\circ$. Then the spaces $\mathcal{U}_A$, with the twenty binary quadratic entries of C read in $\operatorname{Sym}^2 \mathcal{H}^*$ and the twelve binary linear coefficients of O read in $\mathcal{H}^*$, form a subbundle $\mathcal{W}$ of rank 4 of

$$(\mathbb{C}^{20} \otimes \operatorname{Sym}^2 \mathcal{H}^*) \oplus (\mathbb{C}^{12} \otimes \mathcal{H}^*),$$

and the projective bundle $\mathbb{P}(\mathcal{W})$ of one-dimensional subspaces of $\mathcal{W}$ is irreducible of dimension 11.

Let $\mathcal{R} \subset \mathbb{P}(\mathcal{W}) \times \mathcal{M}$ be the set of pairs $(p, [X])$ such that p is the class of a non-zero tuple whose lower blocks are the lower blocks of a presentation with the quadrics $y_i y_j$ of a numerical Godeaux surface with canonical model X, and let $\mathcal{R}_0 \subset \mathcal{F}^\circ \times \mathcal{M}$ be the set of pairs $(L, [X])$ such that the lower blocks of the zero tuple over L are the lower blocks of a presentation with the quadrics $y_i y_j$ of a numerical Godeaux surface with canonical model X. Then $\mathcal{R}$ and $\mathcal{R}_0$ are constructible, and any fiber of $\mathcal{R} \to \mathbb{P}(\mathcal{W})$ or of $\mathcal{R}_0 \to \mathcal{F}^\circ$ contains at most one point.

Moreover, there exists a dense open subset $\mathcal{P} \subset \mathbb{P}(\mathcal{W})$ such that $[X] \in \mathcal{B}_{\mathrm{SS}}$ for any $(p, [X]) \in \mathcal{R}$ with $p \in \mathcal{P}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that $\mathcal{F}^\circ$ is irreducible with $\dim \mathcal{F}^\circ = 8$.

A row of A is not a vanishing row if and only if the three entries of the row span the space of binary linear forms, and A has generic rank four if and only if some coefficient of some 4×4 minor of A is non-zero. Both conditions are open on $\mathcal{F}^{\mathrm{ne}}$ and are preserved by binary reparametrizations.

By Construction-Proposition 7.6, $\dim Z_1 \leq 6$ and $\dim Z_2 \leq 5$, so the lines of $\mathcal{F}^{\mathrm{ne}}$ with no vanishing row form a dense open subset of the irreducible $\mathcal{F}^{\mathrm{ne}}$. By Proposition 7.8, the subset contains the line of a point of the space $\mathcal{D}$ of Construction-Proposition 7.7, which is the line of a standard pencil matrix of a member of $\mathcal{S}$, and by Lemma 4.3(1) that pencil matrix has generic rank four. Hence $\mathcal{F}^\circ$ is non-empty, and $\mathcal{F}^\circ$ is an open subset of $\mathcal{F}^{\mathrm{ne}}$, so $\mathcal{F}^\circ$ is irreducible with $\dim \mathcal{F}^\circ = 8$ by Construction-Proposition 7.2.

Step 2. In this step we construct $\mathcal{W}$.

Let

$$\mathcal{V} := (\mathbb{C}^{20} \otimes \operatorname{Sym}^2 \mathcal{H}^*) \oplus (\mathbb{C}^{12} \otimes \mathcal{H}^*),$$

so that $\operatorname{rank} \mathcal{V} = 20 \cdot 3 + 12 \cdot 2 = 84$. On a standard affine chart of $\mathrm{Gr}(2, 12)$, the basis of $\mathcal{H}$ in reduced echelon form with respect to the pivot coordinates of the chart, the *canonical basis* of $\mathcal{H}$, expresses the a_{ij} as binary linear forms whose coefficients are regular functions, and it expresses a point of $\mathcal{V}$ as an 84-coefficient tuple of the normalized shape of Table 2.

Consider the coefficient matrix of Construction-Proposition 6.7, whose kernel is $\mathcal{U}_A$. The entries of the matrix are regular functions of the coefficients of A on the chart. By Lemma 7.3, $\dim \mathcal{U}_A = 4$ on $\mathcal{F}^\circ$, so the rank of the matrix equals 80 at any closed point. Near a point of the chart, some 80×80 minor is invertible, and row and column operations bring the matrix to a form with an identity block of size 80 and a remaining block. The entries of the remaining block vanish at any closed point, and they are zero regular functions since $\mathcal{F}^\circ$ is reduced and of finite type. Thus the kernel is a free direct summand of rank 4 (see [Har77, Exercise II.5.8(c)]).

A change of basis of $\mathcal{H}$ acts on $\mathcal{V}$ through $\operatorname{Sym}^2 \mathcal{H}^*$ and $\mathcal{H}^*$, it preserves the normalized slots, and the equations of Construction-Proposition 5.3 transform tensorially. Thus, on the principal $\mathrm{GL}(2)$-bundle of bases of $\mathcal{H}$, the kernel construction is $\mathrm{GL}(2)$-equivariant, and it

descends to a subbundle $\mathcal{W} \subset \mathcal{V}$ of rank 4 with fibers $\mathcal{U}_A$. By **Step 1**, $\mathbb{P}(\mathcal{W})$ is irreducible with $\dim \mathbb{P}(\mathcal{W}) = 8 + 3 = 11$.

Step 3. In this step we show that $\mathcal{R}$ and $\mathcal{R}_0$ are constructible with at most one point in any fiber.

A change of basis of $\mathcal{H}$ is a binary reparametrization of Construction-Proposition 5.1. A binary reparametrization replaces a presentation of a surface S with a presentation of S, and preserves the canonical ring, hence the canonical model. Thus the definitions of $\mathcal{R}$ and $\mathcal{R}_0$ do not depend on the basis of $\mathcal{H}$.

Let $\mathcal{L}$ be the affine space of lower blocks of the shape of Set-up 4.1 with the fixed blocks of Table 1, and consider the quadrics $r_k = y_i y_j$. On a chart of $\mathrm{Gr}(2, 12)$ with the canonical basis of $\mathcal{H}$, the formula for N and the equations for P, U of Construction-Proposition 5.1 define a morphism from $\mathcal{W}$ over the chart to $\mathcal{L}$, which sends $(A, \mathbf{u})$ to the lower blocks determined by $(A, \mathbf{u})$. By Lemma 4.5(2), the preimage of $\Xi_{\mathcal{L}}$ in the product of $\mathcal{W}$ over the chart with $\mathcal{M}$ is constructible. Removing the zero section from the preimage and taking the image in $\mathbb{P}(\mathcal{W}) \times \mathcal{M}$, we obtain a constructible set by [Har77, Exercise II.3.19]. The restriction of the preimage to the zero section is constructible. Since finitely many charts cover $\mathcal{F}^\circ$, the sets $\mathcal{R}$ and $\mathcal{R}_0$ are constructible.

Let $(p, [X])$ and $(p, [X'])$ be points of $\mathcal{R}$, let L be the line of p, and fix a basis of $\mathcal{H}$ at L, with pencil matrix A. Let $\mathbf{u} \in \mathcal{U}_A$ be a representative of p. Since the definition of $\mathcal{R}$ does not depend on the basis of $\mathcal{H}$, there exist $\rho, \rho' \in \mathbb{C}^*$ such that the lower blocks determined by $(A, \rho\mathbf{u})$ are the lower blocks of a presentation with the quadrics $y_i y_j$ of a surface with canonical model X, and the lower blocks determined by $(A, \rho'\mathbf{u})$ are the lower blocks of a presentation with the quadrics $y_i y_j$ of a surface with canonical model X'. By Construction-Proposition 7.5 with $\lambda = (1, 1, 1, 1)$, for which $\lambda \cdot A = A$ and Φ_λ is the identity, the lower blocks determined by $(A, \mathbf{u})$ are the lower blocks of such presentations of both surfaces. Thus $[X] = [X']$ by Lemma 4.5(2).

Let $(L, [X])$ and $(L, [X'])$ be points of $\mathcal{R}_0$, and fix a basis of $\mathcal{H}$ at L, with pencil matrix A. Since the definition of $\mathcal{R}_0$ does not depend on the basis of $\mathcal{H}$, the lower blocks determined by $(A, 0)$ are the lower blocks of a presentation with the quadrics $y_i y_j$ of a surface with canonical model X and of a presentation with the quadrics $y_i y_j$ of a surface with canonical model X'. Thus $[X] = [X']$ by Lemma 4.5(2).

Step 4. In this step we construct $\mathcal{P}$.

By [SS20, Section 3, pp. 17–18, Section 5, p. 22, Introduction, p. 2, and Corollary 5.2], a point of $\mathcal{D}$ is a line together with the class, up to a common scalar, of a tuple of twenty binary quadratic c-entries and twelve binary linear o-entries, with N eliminated. By Proposition 7.8 and **Step 1**, some point of $\mathcal{D}$ has line in $\mathcal{F}^\circ$. Since $\mathcal{D}$ is irreducible and $\mathcal{F}^\circ$ is open in $\mathcal{F}$, the points of $\mathcal{D}$ with line in $\mathcal{F}^\circ$ form a non-empty open subset of $\mathcal{D}$, of dimension 11. Sending a point $(L, [\mathbf{u}])$ of the open subset to the class of $\mathbf{u}$ over L defines an injective morphism from the open subset to $\mathbb{P}(\mathcal{W})$. The members of the family of [SS20, Corollary 5.2] are geometric, so the image of the open subset is contained in the projection of $\mathcal{R}$, and the image is constructible [Har77, Exercise II.3.19], of dimension 11 since the fibers of an injective morphism are points [Har77, Exercise II.3.22(b)]. Since $\mathbb{P}(\mathcal{W})$ is irreducible with $\dim \mathbb{P}(\mathcal{W}) = 11$, the image is dense in $\mathbb{P}(\mathcal{W})$, and hence the image contains a dense open subset $\mathcal{P} \subset \mathbb{P}(\mathcal{W})$.

Let $(p, [X]) \in \mathcal{R}$ with $p \in \mathcal{P}$. Then p is the class of the tuple of a member of $\mathcal{S}$ with canonical model X', so $(p, [X']) \in \mathcal{R}$ and $[X'] \in \mathcal{B}_{\mathrm{SS}}$. By **Step 3**, $[X] = [X']$, and hence $[X] \in \mathcal{B}_{\mathrm{SS}}$. □

Theorem 7.10. *We have*

$$\dim \overline{\mathcal{M}_{\mathrm{ne}} \backslash \mathcal{B}_{\mathrm{SS}}} \leq 7.$$

Proof. The subset $\mathcal{F}^\circ$ is invariant under the torus $\mathbb{T}$, because the entries of A are multiplied by non-zero scalars, which preserves events and vanishing rows, and because $\lambda \cdot A$ is the product of A with invertible diagonal matrices (Construction-Proposition 7.5), which preserves the generic rank.

Let

$$\mathcal{R}' := \mathcal{R} \cap \big(\mathbb{P}(\mathcal{W}) \times (\mathcal{M}\backslash\mathcal{B}_{\mathrm{SS}})\big), \qquad \mathcal{R}'_0 := \mathcal{R}_0 \cap \big(\mathcal{F}^\circ \times (\mathcal{M}\backslash\mathcal{B}_{\mathrm{SS}})\big).$$

By Construction-Proposition 7.9, $\mathcal{R}'$ and $\mathcal{R}'_0$ are constructible, any non-empty fiber of $\mathcal{R}' \to \mathbb{P}(\mathcal{W})$ or of $\mathcal{R}'_0 \to \mathcal{F}^\circ$ consists of one point, and the projection of $\mathcal{R}'$ to $\mathbb{P}(\mathcal{W})$ is contained in the closed complement of $\mathcal{P}$, which is of dimension at most 10.

Let $(p, [X]) \in \mathcal{R}'$, where p is the class of a non-zero tuple $\mathbf{u} \in \mathcal{U}_A$ over a line L. For any $\lambda \in (\mathbb{C}^*)^4$, Construction-Proposition 7.5 implies that the lower blocks determined by $(\lambda \cdot A, \Phi_\lambda(\mathbf{u}))$ are the lower blocks of a presentation with the quadrics $y_i y_j$ of a numerical Godeaux surface with canonical model X, and $\Phi_\lambda(\mathbf{u}) \neq 0$. Thus the fiber of $\mathcal{R}'$ over $[X]$ contains the classes of $\Phi_\lambda(\mathbf{u})$ over $\lambda \cdot L$. The image of the fiber in $\mathcal{F}^\circ$ contains $\mathbb{T} \cdot L$, and $\dim \mathbb{T} \cdot L = 3$ by Lemma 7.4, so the fiber is of dimension at least 3. By Lemma 2.9 with $e = 3$, we have

$$\dim \overline{\mathrm{pr}_{\mathcal{M}}(\mathcal{R}')} \leq \dim \overline{\mathrm{pr}_{\mathbb{P}(\mathcal{W})}(\mathcal{R}')} - 3 \leq 10 - 3 = 7.$$

Let $(L, [X]) \in \mathcal{R}'_0$. Since $\Phi_\lambda(0) = 0$, Construction-Proposition 7.5 implies that $(\lambda \cdot L, [X]) \in \mathcal{R}'_0$ for any $\lambda \in (\mathbb{C}^*)^4$. Thus the fiber of $\mathcal{R}'_0$ over $[X]$ contains $\mathbb{T} \cdot L$, and Lemmas 7.4 and 2.9 with $e = 3$ imply that

$$\dim \overline{\mathrm{pr}_{\mathcal{M}}(\mathcal{R}'_0)} \leq \dim \mathcal{F}^\circ - 3 = 8 - 3 = 5.$$

Let $[X] \in \mathcal{M}_{\mathrm{ne}}\backslash\mathcal{B}_{\mathrm{SS}}$, with an event-free standard pencil matrix A with line $L \in \mathcal{F}^{\mathrm{ne}}$. If A has a vanishing row, then $[X] \in \mathcal{M}_{\mathrm{row}}$. Assume that A has no vanishing row. By Lemma 4.3(1), L is a point of $\mathcal{F}^\circ$. Let $\mathbf{u} \in \mathcal{U}_A$ be the geometric tuple of the normalized standard presentation of Construction-Proposition 5.3. If $\mathbf{u} \neq 0$, then the class of $\mathbf{u}$ and $[X]$ define a point of $\mathcal{R}'$. If $\mathbf{u} = 0$, then $(L, [X]) \in \mathcal{R}'_0$. Therefore

$$\overline{\mathcal{M}_{\mathrm{ne}}\backslash\mathcal{B}_{\mathrm{SS}}} \subset \overline{\mathcal{M}_{\mathrm{row}}} \cup \overline{\mathrm{pr}_{\mathcal{M}}(\mathcal{R}')} \cup \overline{\mathrm{pr}_{\mathcal{M}}(\mathcal{R}'_0)}.$$

By Construction-Proposition 7.6, $\dim \overline{\mathcal{M}_{\mathrm{row}}} \leq 7$, and Theorem 7.10 follows. □

8. Non-reduced base schemes

In this section, we construct the fixed blocks, the normalized lower space, and the module group, a group acting on presentations (Construction-Proposition 8.4), for the partitions $(2,1,1)$, $(2,2)$, $(3,1)$, (4) of Construction-Proposition 3.3, for which the scheme B is not reduced. We use them in Sections 9, 10, 11, and 12 to bound the moduli of each type.

Set-up 8.1. Let $\mathbf{p}$, n, $\mathcal{A}$, λ, S_1, $\mathcal{N}$, and S be as follows.

(1) $\mathbf{p}$ is one of the partitions $(2,1,1)$, $(2,2)$, $(3,1)$, (4), and n is the number of parts of $\mathbf{p}$.
(2) $\mathcal{A}$ is the product of the algebras $\mathbb{C}[z]/(z^p)$ over the parts p of $\mathbf{p}$, and λ is a linear functional on $\mathcal{A}$ whose pairing $(a, a') \to \lambda(aa')$ is non-degenerate.
(3) $S_1 := \mathbb{C}[y_0, y_1, y_2, y_3]$ and $\mathcal{N} := Q(\mathcal{A}, \lambda)$.
(4) S is a torsion-free numerical Godeaux surface with $[X] \in \mathcal{M}_{\mathbf{p}}$, and for a trivialization of $\mathcal{O}_B(K_X)$ the algebra and the functional of Construction-Theorem 3.2 are $\mathcal{A}$ and λ.

We call $\mathbf{p}$ the *collision type* of S, and we call the blocks A, U, P of a presentation of S the *first stage* of the presentation.

By Construction-Theorem 3.2(4), replacing the trivialization of $\mathcal{O}_B(K_X)$ by a unit multiple turns any non-degenerate functional on $\mathcal{A}$ into the functional fixed in Section 9, Section 10, Section 11, or Section 12. Thus (4) holds for any S with $[X] \in \mathcal{M}_{\mathbf{p}}$ and any such λ, and all results of this section hold for any λ with non-degenerate pairing.

The first stage and the blocks O, N, C of a presentation vary, and the fixed blocks do not: the minimal resolution of $R/(s,t)$ splits by the residue of the degree modulo 3, and the pieces depend only on $(\mathcal{A}, \lambda)$.

Construction-Proposition 8.2. Notation and conditions as in Set-up 8.1(1)–(3). Fix a basis b_0, b_1, b_2, b_3 of $\mathcal{A}$ such that $1, b_1, b_2, b_3$ is also a basis of $\mathcal{A}$. Let S_1 act on $\mathcal{N}$ through $y_i \to b_i\tau^3$, write $y(a) := \sum_i a_i y_i$ for $a = \sum_i a_i b_i$, and let $y_* := y(1)$. We denote by

$$\mathcal{N}_0 := \mathbb{C} \oplus \bigoplus_{m\geq 1} \mathcal{A}\tau^{3m}, \quad \mathcal{N}_1 := \bigoplus_{m\geq 0} \mathcal{A}\tau^{3m+4}, \quad \mathcal{N}_2 := \operatorname{Ker}(\lambda)\tau^5 \oplus \bigoplus_{m\geq 1} \mathcal{A}\tau^{3m+5}$$

the three S_1-submodules of $\mathcal{N}$ given by the residue of the degree. The quotient $\mathcal{N}/\tau^3\mathcal{N}$ is non-zero exactly in degrees $0, 3, 4, 5, 8$, of dimensions $1, 3, 4, 3, 1$, with degree 8 piece $\mathcal{A}/\operatorname{Ker}\lambda$. The *self-duality* of $\mathcal{N}$ is the isomorphism $\operatorname{Ext}^3_{S_1}(\mathcal{N}, S_1(-17)) \cong \mathcal{N}$ inducing the pairing

$$(\mathcal{N}/\tau^3\mathcal{N})_i \times (\mathcal{N}/\tau^3\mathcal{N})_{8-i} \to \mathcal{A}/\operatorname{Ker}\lambda \to \mathbb{C},$$

with the last map induced by λ. A minimal resolution $F_\bullet$ of $\mathcal{N}$ is *self-dual* if there is an isomorphism

$$\operatorname{Hom}(F_\bullet, S_1(-17)) \cong F_\bullet$$

lifting the self-duality of $\mathcal{N}$, and the middle differential of $F_\bullet$ is alternating. Let M_j be the matrix of multiplication by b_j on $\mathcal{A}$, let $\Lambda := (\lambda(b_ib_k))_{i,k}$, and let $D_j := y_jI_4 - y_*M_j$ for $j \in \{1, 2, 3\}$. We define

$$B_1 := \begin{pmatrix} D_1 & D_2 & D_3 \end{pmatrix}, \qquad B_4 := \begin{pmatrix} 0 & D_3\Lambda^{-1} & -D_2\Lambda^{-1} \\ -D_3\Lambda^{-1} & 0 & D_1\Lambda^{-1} \\ D_2\Lambda^{-1} & -D_1\Lambda^{-1} & 0 \end{pmatrix}.$$

Let $r_1, \ldots, r_6$ be a basis of the quadrics in $y_0, \ldots, y_3$ vanishing on the image of $\operatorname{Spec}\mathcal{A}$ in $\mathbb{P}^3$ under $y_i \to b_i$, that is, $r_k(b_0, \ldots, b_3) = 0$ in $\mathcal{A}$. Let B_3, B_2 be matrices of linear forms in $y_0, \ldots, y_3$ of sizes 6×8 and 3×8 with $(r_1, \ldots, r_6)B_3 = 0$ and $B_3B_2^t = 0$. Assume that the reductions modulo y_* of $r_1, \ldots, r_6$, of the columns of B_3, and of the columns of B_2^t are a basis of $(y_1, y_2, y_3)^2$, a basis of the linear first syzygies of $(y_1, y_2, y_3)^2$, and a basis of the linear second syzygies of $(y_1, y_2, y_3)^2$. For matrices A, P, U of binary linear forms, the equations

$$AB_3 + B_1P = 0, \quad UB_4 - B_2P^t = 0, \quad UP = 0$$

are the *first-stage identities* of type **p**, and a tuple satisfying them is a *first-stage tuple*. We call $B_1, \ldots, B_4$ the *fixed blocks*. Then $B_1, \ldots, B_4$ are blocks of a self-dual minimal S_1-resolution of $\mathcal{N}$. Moreover, for any 4×6 matrix A of binary linear forms, there exists at most one pair (P, U) of matrices of binary linear forms with $AB_3 + B_1P = 0$ and $UB_4 - B_2P^t = 0$.

Proof. The proof is long, so we divide it into several steps. Since $1, b_1, b_2, b_3$ is a basis of $\mathcal{A}$, the forms y_*, y_1, y_2, y_3 are a basis of the linear forms of S_1. By the formulas for $\mathcal{N}_0, \mathcal{N}_1, \mathcal{N}_2$, the pieces of $\mathcal{N}/\tau^3\mathcal{N}$ in degrees $0, 3, 4, 5, 8$ are $\mathbb{C}$, $\mathcal{A}/\mathbb{C}$, $\mathcal{A}$, $\operatorname{Ker}\lambda$, and $\mathcal{A}/\operatorname{Ker}\lambda$, and the other pieces are zero.

Step 1. In this step we resolve $\mathcal{N}_0$.

The homomorphism $S_1 \to \mathcal{N}_0$ sending y_i to $b_i\tau^3$ kills $r_1, \ldots, r_6$ because $r_k(b_0, \ldots, b_3) = 0$, and it is onto because $y(x)y_*^{m-1}$ maps to $x\tau^{3m}$ for any $x \in \mathcal{A}$ and $m \geq 1$. We have $\dim(\mathcal{N}_0)_{3m} = 4$ for any $m \geq 1$. Since the reductions modulo y_* of $r_1, \ldots, r_6$ are a basis of $(y_1, y_2, y_3)^2$, the multiplication by y_* on $S_1/(r_1, \ldots, r_6)$ is onto from degree $3m$ to degree $3m + 3$ for any $m \geq 1$, and $\dim(S_1/(r_1, \ldots, r_6))_3 = 4$. Thus $\dim(S_1/(r_1, \ldots, r_6))_{3m} \leq 4$ for any $m \geq 1$, and $S_1/(r_1, \ldots, r_6) \cong \mathcal{N}_0$.

Since y_* acts on $\mathcal{N}_0$ as multiplication by the non-zero-divisor τ^3 of $\mathcal{A}[\tau]$, the element y_* is regular on $\mathcal{N}_0$, and

$$\mathcal{N}_0/y_*\mathcal{N}_0 \cong \mathbb{C}[y_1, y_2, y_3]/(y_1, y_2, y_3)^2.$$

The matrix

$$\begin{pmatrix} y_1 & y_2 & y_3 & 0 \\ 0 & y_1 & y_2 & y_3 \end{pmatrix}$$

has maximal minors $y_1^2, y_1y_2, y_1y_3, y_2^2 - y_1y_3, y_2y_3, y_3^2$, which generate $(y_1, y_2, y_3)^2$ of height $3 = 4-2+1$. Thus the Eagon–Northcott complex is a minimal resolution of $\mathbb{C}[y_1, y_2, y_3]/(y_1, y_2, y_3)^2$, linear after the first step, with Betti numbers $(1, 6, 8, 3)$ [Eis95, Theorem A2.10, Corollary A2.12] (after localizing at the graded maximal ideal). By the assumption on B_3 and B_2 (such B_3, B_2 exist, since y_* is regular on $\mathcal{N}_0$), the reduction modulo y_* of

$$0 \leftarrow \mathcal{N}_0 \leftarrow S_1 \xleftarrow{(r_1,\dots,r_6)} S_1(-6)^{\oplus 6} \xleftarrow{B_3} S_1(-9)^{\oplus 8} \xleftarrow{B_2^t} S_1(-12)^{\oplus 3} \leftarrow 0 \tag{8.1}$$

is exact, and exactness lifts to (8.1) by the graded Nakayama lemma [Eis95, Exercise 4.6(a)].

Step 2. In this step we resolve $\mathcal{N}_1$.

The module $\mathcal{N}_1$ is free over $\mathbb{C}[y_*]$ with basis $b_0\tau^4, \dots, b_3\tau^4$, and y_j acts on $\mathcal{N}_1$ as y_*M_j. The matrices D_1, D_2, D_3 commute because M_1, M_2, M_3 commute. Adjoining y_3, y_2, y_1 one at a time, we see that D_1, D_2, D_3 is a regular sequence of operators on $S_1^{\oplus 4}$ with quotient $\mathcal{N}_1$, so the Koszul complex of D_1, D_2, D_3 is a resolution of $\mathcal{N}_1$. Both $M_j^t\Lambda$ and ΛM_j have (i,k) entry $\lambda(b_ib_jb_k)$, so $M_j^t\Lambda = \Lambda M_j$. Thus $D_j\Lambda^{-1}$ is symmetric, $D_j^t\Lambda = \Lambda D_j$, and B_4 is alternating. After base changes by Λ of the last two free modules, the Koszul complex is

$$0 \leftarrow \mathcal{N}_1 \leftarrow S_1(-4)^{\oplus 4} \xleftarrow{B_1} S_1(-7)^{\oplus 12} \xleftarrow{B_4} S_1(-10)^{\oplus 12} \xleftarrow{B_1^t} S_1(-13)^{\oplus 4} \leftarrow 0. \tag{8.2}$$

Since $B_4^t = -B_4$, the dual of (8.2) with values in $S_1(-17)$ is (8.2) up to signs. Modulo y_*, the complex (8.2) is the Koszul resolution of $\mathcal{A}\tau^4$ over $\mathbb{C}[y_1, y_2, y_3]$ with middle blocks $y_j\Lambda^{-1}$, which differs from the standard one only by the base changes by Λ, and the dual of the standard resolution has cokernel $(\mathcal{A}\tau^4)^\vee$. Thus the isomorphism of complexes induces on $\mathcal{A}\tau^4$ the pairing with Gram matrix $\pm\Lambda$ (we normalize the duality by the Koszul complex). Choosing the sign of the isomorphism on (8.2), we obtain the pairing $(a, a') \to \lambda(aa')$.

Step 3. In this step we resolve $\mathcal{N}_2$.

By **Step 1**, $\mathcal{N}_0$ is the homogeneous coordinate ring of the image of $\operatorname{Spec}\mathcal{A}$ in $\mathbb{P}^3$. Let $\mathfrak{m}$ be the graded maximal ideal of S_1. Since y_* is regular on $\mathcal{N}_0$ and τ^3 becomes a unit in $\mathcal{A}[\tau, \tau^{-1}]$, the local cohomology module $H^1_{\mathfrak{m}}(\mathcal{N}_0)$ is the cokernel of the inclusion of $\mathcal{N}_0$ into $\bigoplus_{m\in\mathbb{Z}} \mathcal{A}\tau^{3m}$, with the piece $\mathcal{A}/\mathbb{C}$ in degree 0 and the piece $\mathcal{A}$ in degree $-3m$ for $m \geq 1$. Since $\omega_{S_1} = S_1(-12)$ [BH98, Example 3.6.10], the graded local duality [BH98, Theorem 3.6.19(b)] implies that $\operatorname{Ext}^3_{S_1}(\mathcal{N}_0, S_1(-12))$ has the piece $(\mathcal{A}/\mathbb{C})^\vee$ in degree 0 and the piece $\mathcal{A}^\vee$ in degree $3m$ for $m \geq 1$. The map $a \to \lambda(a\cdot)$ sends $\mathcal{A}$ onto $\mathcal{A}^\vee$ and $\operatorname{Ker}\lambda$ onto $(\mathcal{A}/\mathbb{C})^\vee$, and it commutes with the action of S_1. Thus

$$\operatorname{Ext}^3_{S_1}(\mathcal{N}_0, S_1(-17)) \cong \operatorname{Ker}(\lambda)\tau^5 \oplus \bigoplus_{m\geq 1} \mathcal{A}\tau^{3m+5} = \mathcal{N}_2.$$

Since $\mathcal{N}_0$ is Cohen–Macaulay of codimension 3, the dual of (8.1) is a resolution of $\mathcal{N}_2$ [BH98, Theorem 3.3.10(c)] (after localizing at the graded maximal ideal), that is,

$$0 \leftarrow \mathcal{N}_2 \leftarrow S_1(-5)^{\oplus 3} \xleftarrow{B_2} S_1(-8)^{\oplus 8} \xleftarrow{-B_3^t} S_1(-11)^{\oplus 6} \xleftarrow{(r_1,\dots,r_6)^t} S_1(-17) \leftarrow 0. \tag{8.3}$$

By construction, the duality between (8.1) and (8.3) induces the pairing $(a, a') \to \lambda(aa')$ in degrees 3 and 5, and in degrees 0 and 8.

Step 4. In this step we conclude that $B_1, \dots, B_4$ are blocks of a self-dual resolution.

The direct sum of (8.1), (8.2), and (8.3) is a minimal S_1-resolution of $\mathcal{N}$, with free modules ordered by the degrees of the generators. The first differential of the direct sum has the diagonal blocks $(r_1, \dots, r_6)$, B_1, and B_2, and the second differential is the alternating matrix

$$\begin{pmatrix} 0 & 0 & B_3 \\ 0 & B_4 & 0 \\ -B_3^t & 0 & 0 \end{pmatrix}.$$

The third differential is the transpose of the first one. By **Steps 2 and 3**, the duality of the resolution lifts the self-duality of $\mathcal{N}$.

Step 5. In this step we prove the uniqueness of (P, U).

Let $c = (c_1, c_2, c_3)$ with $c_j \in \mathbb{C}^4$ be a constant vector with $B_1 c = 0$. The coefficient of y_j in the basis y_*, y_1, y_2, y_3 of $B_1 c = \sum_j D_j c_j$ is c_j, so $c = 0$. If P, P' both satisfy $AB_3 + B_1 P = 0$, then the coefficients of s and of t in $P - P'$ are killed by B_1, so $P = P'$. Let $e = (e_1, e_2, e_3)$ be a constant row vector with $eB_4 = 0$. The coefficients of y_3 and y_2 in the first column block of eB_4 are $-e_2\Lambda^{-1}$ and $e_3\Lambda^{-1}$, and the coefficient of y_3 in the second column block is $e_1\Lambda^{-1}$, so $e = 0$. Thus U is determined by P through $UB_4 - B_2 P^t = 0$. □

Lemma 8.3. *Notation and conditions as in Set-up 8.1 and Construction-Proposition 8.2, with the trivialization of Set-up 8.1(4). Let $y_i \in R_3$ and $z_i \in R_4$ be elements restricting to $b_i\tau^3$ and $b_i\tau^4$ for $i \in \{0, 1, 2, 3\}$. Then S has a presentation with the generators y_i, z_i, with the quadrics $r_1, \dots, r_6$, and with the fixed blocks $B_1, \dots, B_4$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce a resolution of R modulo (s, t).

By Construction-Theorem 3.2(3), $y_0, \dots, y_3$ is a basis of R_3. By Theorem 2.10, R has a resolution of length 3 over S_0, so R is Cohen–Macaulay of dimension 3 [BH98, Theorem 1.3.3] (after localizing at the graded maximal ideal), and $\dim R/(s, t) = \dim \mathcal{N} = 1$ by Construction-Theorem 3.2. Hence s, t is a regular sequence on R, and the resolution reduced modulo (s, t) is a minimal S_1-resolution of $\mathcal{N}$ [BH98, Proposition 1.1.5]. Since $z_0, \dots, z_3$ restrict to a basis of $\mathcal{A}\tau^4$, a change of basis of F_0 brings them to the generators of degree 4. By degrees, O, N, C, A, U, P and the binary cubics vanish modulo (s, t), and the reduction of d_1 consists of the quadrics, a block linear in $y_0, \dots, y_3$ from degree 7 to degree 4, and a block linear in $y_0, \dots, y_3$ from degree 8 to degree 5.

Step 2. In this step we normalize the quadrics, B_1, B_3, and B_2.

Choose the basis of the degree 6 part of F_1 so that the quadrics are $r_1, \dots, r_6$: they are a basis of the quadrics vanishing on the image of $\operatorname{Spec} \mathcal{A}$, since y_i restricts to $b_i\tau^3$. Choose the basis of the degree 7 part so that the block is B_1: the columns of B_1 are the twelve linear relations

$$y_j z_i - y_* \sum_k (M_j)_{ki} z_k$$

of $\mathcal{N}$, since z_i restricts to $b_i\tau^4$. The columns of the block of d_2 in the position of B_3 are a basis of the linear syzygies of $r_1, \dots, r_6$, so the basis of the degree 8 part moves the block to B_3. Then the basis of w_0, w_1, w_2 moves the next block to B_2. The base changes above are congruences of d_2, which stays alternating.

Step 3. In this step we determine the block of d_2 in the position of B_4 up to a unit $h \in \mathcal{A}$.

Let $\tilde{B}_4$ be the block of d_2 in the position of B_4. Then $B_1\tilde{B}_4 = 0$, so by the Koszul resolution (8.2),

$$\tilde{B}_4 = B_4 \operatorname{diag}(\Lambda, \Lambda, \Lambda) E$$

for some $E \in \mathrm{GL}_{12}(\mathbb{C})$. Comparing the coefficients of y_1, y_2, y_3 in $\tilde{B}_4^t = -\tilde{B}_4$, we have $E = \operatorname{diag}(E_0, E_0, E_0)$ with E_0 symmetric. The coefficient of y_* implies that $M_j E_0$ is symmetric for $j \in \{1, 2, 3\}$. Since $M_j^t \Lambda = \Lambda M_j$, the matrix $E_0\Lambda$ commutes with M_1, M_2, M_3. Hence $E_0 = M_h\Lambda^{-1}$ (the commutant of $\mathcal{A}$ in $\operatorname{End}(\mathcal{A})$ is $\mathcal{A}$), where M_h is the multiplication by a unit $h \in \mathcal{A}$, and the blocks of $\tilde{B}_4$ are $\pm D_j M_h \Lambda^{-1}$.

Step 4. In this step we show that h is a scalar.

The alternating d_2 induces an S_0-linear isomorphism $\psi : R \to \operatorname{Ext}^3_{S_0}(R, S_0(-17))$ of degree 0. Since $\omega_{S_0} = S_0(-16)$ and $\omega_R \cong R(1)$ [SS20, Section 2, p. 8], the module $\operatorname{Ext}^3_{S_0}(R, S_0(-17))$ is isomorphic to R as an R-module, and the composite of ψ with such an isomorphism is an S_0-linear automorphism f of R of degree 0. By Theorem 2.10(2), for homogeneous $x \in R$ there exists a

non-zero homogeneous $\sigma \in S_0/\operatorname{ann}_{S_0} R$ with $\sigma x \in S_0/\operatorname{ann}_{S_0} R$. Then $\sigma f(x) = f(\sigma x) = \sigma x f(1)$, and R is a domain, so $f(x) = xf(1)$ with $f(1) \in R_0 = \mathbb{C}$. Thus ψ is R-linear.

Since y_* acts on $R/(s,t) = \mathcal{N}$ as multiplication by the non-zero-divisor τ^3 of $\mathcal{A}[\tau]$, the sequence s, t, y_* is regular on R. Let $C_0 := R/(s,t,y_*)$. Since s, t, y_* have degrees $2, 2, 3$, we have $\omega_{C_0} \cong (\omega_R/(s,t,y_*)\omega_R)(7)$ [BH98, Corollary 3.6.14], and $\omega_{C_0} \cong \operatorname{Hom}_{\mathbb{C}}(C_0, \mathbb{C})$ since $\dim C_0 = 0$ [BH98, Theorem 3.6.19(a)]. Thus, reducing ψ modulo (s,t,y_*), we obtain a C_0-linear isomorphism $C_0 \to \operatorname{Hom}_{\mathbb{C}}(C_0, \mathbb{C})(-8)$, that is, the pairing $(x, x') \to \lambda'(xx')$ for a linear functional λ' on

$$(C_0)_8 = \mathcal{A}\tau^8/\operatorname{Ker}(\lambda)\tau^8.$$

So $\lambda' = \kappa\lambda$ with $\kappa \in \mathbb{C}^*$, and on $(C_0)_4 = \mathcal{A}\tau^4$ the pairing is $(a, a') \to \kappa\lambda(aa')$. Modulo y_*, the class of degree one modulo 3 of the reduced complex is the Koszul resolution of $\mathcal{A}\tau^4$ with middle blocks $y_j M_h \Lambda^{-1}$, and the self-duality of the complex induces on $\mathcal{A}\tau^4$ the pairing with Gram matrix

$$\pm(M_h\Lambda^{-1})^{-1} = \pm\Lambda M_h^{-1},$$

that is, $(a, a') \to \pm\lambda(h^{-1}aa')$ (we normalize the duality by the Koszul complex). Comparing with the pairing $(a, a') \to \kappa\lambda(aa')$, we have $\kappa\lambda(aa') = \pm\lambda(h^{-1}aa')$ for any $a, a' \in \mathcal{A}$. Thus $\kappa = \pm h^{-1}$ by the non-degeneracy of λ, and $h \in \mathbb{C}^*$.

Step 5. In this step we restore B_4.

Replace d_2 by $h^{-1}d_2$, which keeps $d_1 d_2 = 0$ and the alternation. Multiply the basis of the degree 8 part of F_1 by h to restore B_3 and P, and rescale w_0, w_1, w_2 by one common scalar to restore B_2. The generators y_i, z_i do not change. □

To bound the moduli, we have to reduce the parameters of a presentation by a large group, so we keep only the S_1-module $\mathcal{N}$ and the self-duality up to a scalar, and not the algebra structure.

Construction-Proposition 8.4. Notation and conditions as in Set-up 8.1(1)–(3) and Construction-Proposition 8.2. For $\phi \in \operatorname{Aut}(\mathcal{A})$, let b_ϕ be the unique unit of $\mathcal{A}$ with $\lambda(b_\phi\phi(a)) = \lambda(a)$ for any $a \in \mathcal{A}$. Let $\mathcal{G}$ be the algebraic group of quadruples (ϕ, v, χ, w), where v, w are units of $\mathcal{A}$, $\chi \in \mathbb{C}^*$, and $w^2 = \chi v b_\phi$. We call $\mathcal{G}$ the *module group*. Then $\dim \mathcal{G} = 9 - n$, and the formulas, for $k \geq 0$,

$$a\tau^{3k} \to v^k\phi(a)\tau^{3k}, \quad a\tau^{3k+4} \to wv^k\phi(a)\tau^{3k+4}, \quad a\tau^{3k+5} \to \chi b_\phi v^k \phi(a)\tau^{3k+5}$$

with $a \in \mathcal{A}$, except that $a \in \mathbb{C}$ in the first formula for $k = 0$ and $a \in \operatorname{Ker}\lambda$ in the third formula for $k = 0$, define an action of $\mathcal{G}$ on $\mathcal{N}$, semilinear for $y(a) \to y(v\phi(a))$, which preserves the self-duality of $\mathcal{N}$ up to a scalar. Any element of $\mathcal{G}$ lifts uniquely to a chain map of the minimal S_1-resolution of $\mathcal{N}$, given by matrices with constant entries. We say that a presentation of a numerical Godeaux surface has the *fixed blocks* if the blocks in the places of $B_1, \dots, B_4$ in Set-up 4.1 are the fixed blocks and the quadrics are $r_1, \dots, r_6$. Let $\mathcal{G}^0$ be the identity component of $\mathcal{G}$, and let $J := \operatorname{GL}_2 \times \mathcal{G}^0$. Then $\dim J = 13 - n$, the group $\operatorname{GL}_2 \times \mathcal{G}$ acts on the presentations with the fixed blocks and on the gauge classes of their lower blocks, the action preserves the isomorphism class of the canonical model, and an element of $\mathcal{G}$ sends the block A to $G_z A G_r^{-1}$, where G_z and G_r are the blocks in degrees 4 and 6 of the constant matrices of the chain lift of the element.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct the group and the action on $\mathcal{N}$, and we compute $\dim \mathcal{G}$.

The functional $\lambda \circ \phi^{-1}$ has non-degenerate pairing, so it equals $\lambda(b_\phi \cdot)$ for a unique unit b_ϕ, and

$$\lambda(b_\phi\phi(b_{\phi'})\phi\phi'(a)) = \lambda(b_{\phi'}\phi'(a)) = \lambda(a)$$

implies that $b_{\phi\phi'} = b_\phi\phi(b_{\phi'})$. Thus

$$(\phi, v, \chi, w)(\phi', v', \chi', w') = (\phi\phi', v\phi(v'), \chi\chi', w\phi(w'))$$

preserves the equation $w^2 = \chi v b_\phi$, and it is the group law of $\mathcal{G}$ (associativity and inverses, with $b_{\phi^{-1}} = \phi^{-1}(b_\phi^{-1})$, follow directly from the cocycle identity). The third formula of the statement preserves $\mathrm{Ker}(\lambda)\tau^5$ because $\lambda(b_\phi\phi(a)) = \lambda(a)$, and the three formulas compose by the group law. Since $y(x)$ acts on $\mathcal{N}$ as the multiplication by $x\tau^3$, a direct calculation shows that

$$g(y(x)m) = y(v\phi(x))g(m)$$

for any $g = (\phi, v, \chi, w) \in \mathcal{G}$, $x \in \mathcal{A}$, and $m \in \mathcal{N}$.

The square map on the group of units of $\mathcal{A}$ is finite and onto: on a factor of $\mathcal{A}$, a unit is $\alpha(1+\nu)$ with $\alpha \in \mathbb{C}^*$ and ν nilpotent, and the square roots are $\pm\sqrt{\alpha}$ times the finite binomial series of $(1+\nu)^{1/2}$. Thus $(\phi, v, \chi, w) \to (\phi, v, \chi)$ is a finite surjective morphism onto $\mathrm{Aut}(\mathcal{A}) \times \mathcal{A}^* \times \mathbb{C}^*$. An automorphism of $\mathbb{C}[z]/(z^p)$ sends z to $\sum_{i=1}^{p-1} \alpha_i z^i$ with $\alpha_1 \neq 0$, so $\dim \mathrm{Aut}(\mathcal{A}) = 4 - n$. Therefore $\dim \mathcal{G} = (4-n) + 4 + 1 = 9 - n$.

Step 2. In this step we show that $g = (\phi, v, \chi, w) \in \mathcal{G}$ preserves the self-duality up to a scalar.

The form $y(v)$ restricts to $v\tau^3$, which is regular on $\mathcal{N}$, and g carries $\tau^3\mathcal{N}$ onto $v\tau^3\mathcal{N}$. The self-duality of $\mathcal{N}$ induces on $\mathcal{N}/v\tau^3\mathcal{N}$ the pairing with values in $\mathcal{A}\tau^8/v\,\mathrm{Ker}(\lambda)\tau^8$ and with functional $\lambda_v(x) := \lambda(v^{-1}x)$, since division by $v\tau^3$ in place of τ^3 multiplies the value by v^{-1}. By $w^2 = \chi v b_\phi$, we have

$$\lambda_v(w\phi(a) \cdot w\phi(a')) = \lambda(\chi b_\phi \phi(aa')) = \chi\lambda(aa')$$

in degrees 4 and 4,

$$\lambda_v(v\phi(a) \cdot \chi b_\phi \phi(a')) = \chi\lambda(aa')$$

in degrees 3 and 5, and

$$\lambda_v(\chi b_\phi v\phi(a)) = \chi\lambda(a)$$

in degrees 0 and 8.

The self-duality composed with the inverse of the transport of the self-duality by g is an S_1-linear automorphism of $\mathcal{N}$ of degree 0. There exists no non-zero S_1-linear map of degree 0 between two different modules among $\mathcal{N}_0, \mathcal{N}_1, \mathcal{N}_2$, since their degrees differ modulo 3. Let e be an S_1-linear endomorphism of $\mathcal{N}_1$ of degree 0, and let e_4 be the map on $\mathcal{A}$ with $e(a\tau^4) = e_4(a)\tau^4$. Since $y(x)$ and y_* commute with e and τ^3 is regular, $e_4(xa) = xe_4(a)$, so e is the multiplication by $e_4(1)$. The module $\mathcal{N}_0$ is cyclic, so an S_1-linear endomorphism of $\mathcal{N}_0$ of degree 0 is a scalar, and an S_1-linear endomorphism of degree 0 of the dual $\mathcal{N}_2$ of $\mathcal{N}_0$ is a scalar. Thus the automorphism is a scalar on $\mathcal{N}_0$, a scalar on $\mathcal{N}_2$, and the multiplication by the unit $e_4(1)$ on $\mathcal{N}_1$. Let Σ_g be the automorphism of S_1 with $\Sigma_g(y(x)) = y(v\phi(x))$. The transport by Σ_g multiplies the generator $dy_0 \wedge dy_1 \wedge dy_2 \wedge dy_3$ of $\omega_{S_1} \cong S_1(-12)$ by the determinant of the substitution $y(x) \to y(v\phi(x))$, which is a non-zero scalar, so by the three equations above, there exists $\mu \in \mathbb{C}^*$ such that the transported pairings in degrees 4 and 4, in degrees 3 and 5, and in degrees 0 and 8 are μ times the pairings of the self-duality. By the non-degeneracy of λ, the unit $e_4(1)$ and the scalars on $\mathcal{N}_0$ and on $\mathcal{N}_2$ are all equal to μ, so the automorphism is a scalar.

Step 3. In this step we lift $g = (\phi, v, \chi, w) \in \mathcal{G}$ to the resolution.

Let $\bar{F}_0, \bar{F}_1, \bar{F}_2, \bar{F}_3$ be the free modules of the resolution of Construction-Proposition 8.2, with differentials $\bar{d}_1, \bar{d}_2, \bar{d}_3$. Thus $\bar{d}_1$ has the diagonal blocks $(r_1, \ldots, r_6)$, B_1, B_2, the matrix $\bar{d}_2$ has the blocks B_3, B_4, $-B_3^t$, and $\bar{d}_3 = \bar{d}_1^t$. The generators have degrees

$$\{0, 4, 5\}, \quad \{6, 7, 8\}, \quad \{9, 10, 11\}, \quad \{12, 13, 17\}.$$

Let S_1 act on a second copy of $\mathcal{N}$ and of $\bar{F}_\bullet$ through Σ_g. Since Σ_g is a graded automorphism of S_1, the second copy of $\bar{F}_\bullet$ is a graded free resolution of the second copy of $\mathcal{N}$, and g is a homomorphism of graded S_1-modules from $\mathcal{N}$ to the second copy by **Step 1**. By the comparison theorem [Wei94, Theorem 2.2.6] for graded S_1-modules, g lifts to a Σ_g-semilinear chain map of $\bar{F}_\bullet$. Two lifts differ by a homotopy of degree 0 from $\bar{F}_i$ to $\bar{F}_{i+1}$, which vanishes because all generator degrees of $\bar{F}_{i+1}$ are larger than those of $\bar{F}_i$. Thus the lift is unique. The differences between distinct generator degrees of one module $\bar{F}_i$ are not non-zero multiples of 3, so the

lift consists of matrices G_0, G_1, G_2, G_3 with constant entries, block diagonal with respect to the generator degrees, and

$$\bar{d}_i G_i = G_{i-1}\Sigma_g(\bar{d}_i) \quad \text{for } i \in \{1,2,3\}. \tag{8.4}$$

The equations (8.4) are linear in G_i with the fixed coefficient matrix $\bar{d}_i$ and a unique solution, so $G_0, \dots, G_3$ are regular functions on $\mathcal{G}$. By uniqueness, $G_i(gg') = G_i(g)G_i(g')$. The dual of the lift of g lifts the dual of g, so by **Step 2** and uniqueness there exists $\kappa_g \in \mathbb{C}^*$ with

$$G_2 = \kappa_g (G_1^t)^{-1}. \tag{8.5}$$

Step 4. In this step we let $\mathrm{GL}_2 \times \mathcal{G}$ act on presentations.

Let (d_1, d_2) be a presentation with the fixed blocks of a numerical Godeaux surface with canonical ring R, through the homomorphism $\rho : S_0 \to R$. The definition of the fixed blocks contains no condition on the restriction of y_i, z_j to B. Let $(\gamma, g) \in \mathrm{GL}_2 \times \mathcal{G}$, and let Σ be the automorphism of S_0 extending Σ_g with $\Sigma((s,t)) = (s,t)\gamma$. Let

$$(\gamma, g) \cdot (d_1, d_2) := (G_0 \Sigma(d_1) G_1^{-1}, G_1 \Sigma(d_2) G_2^{-1}).$$

By (8.4), the reductions modulo (s,t) of the new differentials are $\bar{d}_1, \bar{d}_2$. Since G_i is block diagonal with constant entries, the blocks keep their forms in Set-up 4.1. In particular, for $\gamma = 1$, the block A becomes $G_z A G_r^{-1}$, where G_z and G_r are the blocks of G_0 and G_1 in degrees 4 and 6, since Σ fixes s, t. The new second differential is $\kappa_g^{-1} G_1 \Sigma(d_2) G_1^t$ by (8.5), so it is alternating. The element g fixes the generator 1, so the substitution $y(x) \to y(v\phi(x))$ changes $r_1, \dots, r_6$ by a constant matrix, which is the degree 6 block of G_1, and the quadrics return to $r_1, \dots, r_6$. The cokernel of $G_0 \Sigma(d_1) G_1^{-1}$ is R, with S_0 acting through $\rho \circ \Sigma^{-1}$ and with generators moved by G_0^{-1}. Thus $(\gamma, g) \cdot (d_1, d_2)$ is a presentation with the fixed blocks of the ring R with other generators (the new complex is the Σ-twist of the old one followed by a constant change of bases, and its third differential is the transpose of the new first differential), and the canonical model does not change. By $G_i(gg') = G_i(g)G_i(g')$ and $\Sigma_{gg'} = \Sigma_g \Sigma_{g'}$, it is a group action.

The multiplication of R is transported through the module isomorphism, and the ring so obtained is isomorphic to R.

Let G_8 be the block of G_1 in degree 8. The constant matrices of the chain lift send the gauge of Γ to the gauge of $G_r \Sigma(\Gamma) G_8^{-1}$, so $\mathrm{GL}_2 \times \mathcal{G}$ acts on the gauge classes of lower blocks. We have $\dim J = 4 + (9 - n) = 13 - n$. □

Nilpotent elements of $\mathcal{A}$ and nilpotent derivations yield elements of $\mathcal{G}^0$ depending polynomially on a parameter, so a normalization by them holds at any point of the first stage, not only on an open subset.

Lemma 8.5. *Notation and conditions as in Set-up 8.1(1)–(3) and Construction-Proposition 8.4. Let δ be a nilpotent derivation of $\mathcal{A}$, let a be a nilpotent element of $\mathcal{A}$, and for $\epsilon \in \mathbb{C}$, let*

$$g_\epsilon := (\exp(\epsilon\delta), \exp(\epsilon a), 1, w_\epsilon) \in \mathcal{G}^0,$$

where w_ϵ is the square root of $\exp(\epsilon a) b_{\exp(\epsilon\delta)}$ equal to 1 at $\epsilon = 0$. We call (δ, a) a nilpotent field. *Let G_0, G_1, G_2 be the matrices of the chain lift of g_ϵ, let G_z, G_5 be the blocks of G_0 in degrees $4, 5$, let G_r, G_7 be the blocks of G_1 in degrees $6, 7$, and let G_9 be the block of G_2 in degree 9. Let G_y be the matrix of the action of g_ϵ on the span of $y_0, \dots, y_3$, with the convention $g_\epsilon(y_i) = \sum_j (G_y)_{ji} y_j$, and likewise for the free generators $z_0, \dots, z_3$ of degree 4 of the resolution of Construction-Proposition 8.2 and for $r_1, \dots, r_6$, with the rows of A indexed by $z_0, \dots, z_3$ and the columns of A indexed by $r_1, \dots, r_6$. Let L_y, L_z, L_r be the derivatives at $\epsilon = 0$ of G_y, G_z, G_r. Then the following hold.*

(1) Any first-stage tuple (A, P, U) transforms polynomially in ϵ under g_ϵ, by

$$(A, P, U) \to (G_z A G_r^{-1}, G_7 P G_9^{-1}, G_5 U G_7^{-1}),$$

the derivative of A at $\epsilon = 0$ is $L_z A - A L_r$, and the first-stage identities are preserved.

(2) *If* $\delta(a) = 0$*, then* $g_{\epsilon+\epsilon'} = g_\epsilon g_{\epsilon'}$*, and the derivative at any* ϵ *of* $A_\epsilon := G_z A G_r^{-1}$ *is* $L_z A_\epsilon - A_\epsilon L_r$*. We call* $\epsilon \to g_\epsilon$ *the* flow *of* (δ, a)*.*

Proof. The element g_ϵ acts by $y(x) \to y(\exp(\epsilon a)\exp(\epsilon\delta)(x))$ and $z(x) \to z(w_\epsilon \exp(\epsilon\delta)(x))$, where $z(x) := \sum_i x_i z_i$ for $x = \sum_i x_i b_i$. The element $b_{\exp(\epsilon\delta)}$ is the image of $\lambda \circ \exp(-\epsilon\delta)$ under the fixed isomorphism $\mathcal{A}^\vee \cong \mathcal{A}$, so it is polynomial in ϵ because δ is nilpotent. On each residue field of $\mathcal{A}$, it is a nowhere-zero polynomial with value 1 at $\epsilon = 0$, so it is 1 plus a nilpotent element. Thus w_ϵ is polynomial in ϵ by the finite binomial series.

Let $\bar{d}_i$ be the reductions modulo (s, t) of the differentials with the fixed blocks, so that the matrices of the chain lift of g_ϵ satisfy $\bar{d}_i G_i = G_{i-1}\Sigma(\bar{d}_i)$ with $\Sigma(y(x)) = y(\exp(\epsilon a)\exp(\epsilon\delta)(x))$. The matrix G_0 is polynomial in ϵ. The lifting equations are linear in G_i with a fixed coefficient matrix and a unique solution, so G_1, G_2 are polynomial in ϵ. Their determinants are nowhere-zero polynomials, so constant, and the inverse matrices are polynomial in ϵ.

Let G_8 be the block of G_1 in degree 8, and let G_{10} be the block of G_2 in degree 10. Since A, P, U do not depend on $y_0, \dots, y_3$, the formula of (1) is the action of g_ϵ on the first stage of a presentation with the fixed blocks. The matrices of the formula are polynomial in ϵ, and the derivative of A at $\epsilon = 0$ is $L_z A - A L_r$.

We prove that the first-stage identities are preserved for any first-stage tuple (A, P, U). By the lifting equations, $B_1 G_7 = G_z \Sigma(B_1)$, $B_3 G_9 = G_r \Sigma(B_3)$, $B_2 G_8 = G_5 \Sigma(B_2)$, and $B_4 G_{10} = G_7 \Sigma(B_4)$. Since g_ϵ preserves the self-duality up to a scalar and the chain lift is unique, there exists $\kappa \in \mathbb{C}^*$ with $G_9 = \kappa (G_8^t)^{-1}$ and $G_{10} = \kappa (G_7^t)^{-1}$. Thus a direct calculation shows that

$$G_z A G_r^{-1} B_3 + B_1 G_7 P G_9^{-1} = G_z \Sigma(A B_3 + B_1 P) G_9^{-1} = 0,$$
$$G_5 U G_7^{-1} B_4 - B_2 (G_7 P G_9^{-1})^t = \kappa^{-1} G_5 \Sigma(U B_4 - B_2 P^t) G_7^t = 0,$$
$$G_5 U G_7^{-1} G_7 P G_9^{-1} = G_5 U P G_9^{-1} = 0.$$

We prove (2). Assume that $\delta(a) = 0$. Then $\exp(\epsilon\delta)$ fixes a, so the second coordinate of $g_\epsilon g_{\epsilon'}$ is $\exp((\epsilon + \epsilon')a)$. The fourth coordinates $w_{\epsilon+\epsilon'}$ and $w_\epsilon \exp(\epsilon\delta)(w_{\epsilon'})$ of $g_{\epsilon+\epsilon'}$ and $g_\epsilon g_{\epsilon'}$ have equal squares, by $b_{\phi\phi'} = b_\phi \phi(b_{\phi'})$, and both of them are 1 plus a nilpotent element, so they are equal. Thus $g_{\epsilon+\epsilon'} = g_\epsilon g_{\epsilon'}$. Since the chain lift is multiplicative by the uniqueness of the chain lift, the derivative of $A_{\epsilon+\epsilon'}$ at $\epsilon' = 0$ is $L_z A_\epsilon - A_\epsilon L_r$. □

Construction-Proposition 8.6. Let $\mathbf{p}$ be one of the partitions $(2,1,1)$, $(2,2)$, $(3,1)$, (4), with the fixed blocks $B_1, \dots, B_4$ of the corresponding section, and let A be a first-stage matrix of type $\mathbf{p}$, with binary linear entries satisfying the first-stage identities of type $\mathbf{p}$ and not necessarily coming from a surface, with P, U determined by A. An element g of $\mathrm{GL}_2 \times \mathcal{G}$ acts on the affine space of all lower blocks with the fixed blocks by the constant matrices of the chain lift and the linear substitutions in $s, t, y_0, \dots, y_3$, and we denote the action by $\ell \to g \cdot \ell$. Then the following hold.

(1) The map $\Gamma \to B_3 \Gamma^t - \Gamma B_3^t$ has a 48×48 minor of non-zero constant determinant, listed for each type in Appendix A, so any triple (O, N, C) is moved by a unique Γ, through the formula of Lemma 4.2, to a triple with O supported on twelve fixed slots, where a *slot* is the coefficient of y_k in an entry O_{ij}, with binary linear coefficients $o_0, \dots, o_{11}$.

(2) The equation $AO - B_1 N^t - C B_3^t = 0$ then determines N and twelve entries of C, and leaves twenty entries $c_0, \dots, c_{19}$ of C, binary quadratic and free.

(3) We denote by $\mathcal{U}_A$ the subspace of $\mathbb{C}^{84}$ of the coefficient vectors of $(o_0, \dots, o_{11}, c_0, \dots, c_{19})$ with $AO - B_1 N^t - C B_3^t = 0$, $AN - C P^t = 0$, $U N^t = 0$, and we call $\mathcal{U}_A$ the *normalized lower space* and a point of $\mathcal{U}_A$ *geometric* if the lower blocks determined by A and the point are the lower blocks of a presentation with the fixed blocks, in the sense of Construction-Proposition 8.4, of a surface as in Set-up 8.1.

(4) The map $(g, \ell) \to g * \ell$ is an algebraic action of J on the affine space $\mathcal{L}_\mathbf{p}$ of lower blocks with the fixed blocks and with O supported on the twelve slots, where for $g \in J$ and $\ell \in \mathcal{L}_\mathbf{p}$ we denote by $g * \ell \in \mathcal{L}_\mathbf{p}$ the point in the gauge class of $g \cdot \ell$.

Proof. The space of matrices Γ is the tensor product of the binary linear forms with $\mathbb{C}^{48}$, and the map $\Gamma \to B_3\Gamma^t - \Gamma B_3^t$ is the tensor product of the binary linear forms with a linear map from $\mathbb{C}^{48}$ to the 60 slots of an alternating 6×6 matrix O. The minor on the complement of the twelve slots has determinant 16 for $(2,1,1)$, Table 33, determinant 16 for $(2,2)$, Table 34, determinant -16 for $(3,1)$, Table 46, and determinant 32 for (4), Table 50. Thus for any O there exists a unique Γ such that $O + B_3\Gamma^t - \Gamma B_3^t$ is supported on the twelve slots, and Γ is a polynomial function of O. Thus the formula of Lemma 4.2 moves (O,N,C) to the triple $(O+B_3\Gamma^t-\Gamma B_3^t, N-\Gamma P^t, C-A\Gamma)$, whose block O is supported on the twelve slots.

Comparing the coefficients of $y_0,\dots,y_3$, the equation $AO - B_1N^t - CB_3^t = 0$ is a system of 96 linear equations in the 72 entries of N and the 32 entries of C, with a constant coefficient matrix. For $(2,2)$, $(3,1)$, and (4), the matrix has a constant 84×84 minor of determinant 1, -1, and -1 respectively, listed in Appendix A, which solves N and twelve entries of C. For $(2,1,1)$, comparing the coefficients solves N and twelve entries of C without division, with the formulas of Construction-Proposition 9.10(1), and the comparison of coefficients does not use Set-up 9.4. The other twenty entries of C are binary quadratic and free. The matrices P,U exist and are determined by A by Construction-Proposition 8.2, so the three equations are linear and homogeneous in the $12\cdot 2 + 20\cdot 3 = 84$ scalar coefficients of the o_i and the c_j. Thus $\mathcal{U}_A$ is a vector space for any first-stage tuple, and only the word geometric refers to surfaces.

By the uniqueness of the chain lift in Construction-Proposition 8.4, $\ell \to g\cdot\ell$ is a group action on the affine space of all lower blocks with the fixed blocks, and it sends the gauge of a matrix Γ to the gauge of a 6×8 matrix of binary linear forms. Thus $g\cdot(g'*\ell)$ and $(gg')\cdot\ell$ belong to one gauge class for any $g,g'\in J$ and $\ell\in\mathcal{L}_{\mathbf{p}}$. By the uniqueness of Γ, we have $g*(g'*\ell) = (gg')*\ell$. The matrices of the chain lift are regular functions on J and Γ is a polynomial function of O, so $(g,\ell)\to g*\ell$ is an algebraic action. □

The 48×48 minor of Construction-Proposition 8.6 implies that Γ is determined by $B_3\Gamma^t - \Gamma B_3^t$, so the gauge acts freely on the triples (O,N,C) and we work modulo the gauge.

Below, $J\cdot\ell$ is the orbit of the gauge class, equivalently the orbit of the representative in $\mathcal{L}_{\mathbf{p}}$ under $g*\ell$. To turn $\dim J$ into a dimension of a fiber over a moduli point, the stabilizer has to be small: finiteness of $\operatorname{Aut}(X)$ reduces the connected stabilizer to the grading torus, so $\dim J\cdot\ell \geq 12-n$.

Lemma 8.7. *Notation and conditions as in Set-up 8.1(1)–(3) and Construction-Proposition 8.4. We call*

$$\{(\alpha^2 I_2, (\mathrm{id}_{\mathcal{A}}, \alpha^3, \alpha^5, \alpha^4)) : \alpha\in\mathbb{C}^*\}\subset J$$

the grading torus. *Let ℓ be the class modulo the gauge of Lemma 4.2 of the lower blocks of a presentation with the fixed blocks of a numerical Godeaux surface, with $A\neq 0$, and let $L\subset J$ be the stabilizer of ℓ. Then the identity component of L is contained in the grading torus. In particular,* $\dim J\cdot\ell\geq 12-n$.

Proof. For $\alpha\in\mathbb{C}^*$, the quadruple $(\mathrm{id}_{\mathcal{A}},\alpha^3,\alpha^5,\alpha^4)$ belongs to $\mathcal{G}$ since $(\alpha^4)^2 = \alpha^5\alpha^3$ and $b_{\mathrm{id}_{\mathcal{A}}} = 1$, so the grading torus is contained in J. Let R be the canonical ring of the surface, X the canonical model, $\rho: S_0\to R$ the homomorphism of the presentation (d_1,d_2), and $I_\rho := \operatorname{Ker}\rho = \operatorname{ann}_{S_0} R$. Let $g=(\gamma,(\phi,v,\chi,w))\in L$, and let Σ be the automorphism of S_0 with $\Sigma((s,t)) = (s,t)\gamma$ and $\Sigma(y(x)) = y(v\phi(x))$. By Construction-Proposition 8.4, $g\cdot(d_1,d_2)$ is a presentation of R through $\rho\circ\Sigma^{-1}$, so the annihilator of the cokernel of the first differential is $\Sigma(I_\rho)$. Since $g\in L$, a gauge of Lemma 4.2 moves the lower blocks of $g\cdot(d_1,d_2)$ to those of (d_1,d_2), keeping the quadrics $r_1,\dots,r_6$ and the cokernel of the first differential. By Lemma 4.4, the first differentials differ by an invertible triangular change of generators, so $\Sigma(I_\rho) = I_\rho$. Thus Σ induces a graded automorphism of S_0/I_ρ, and an automorphism of X by Theorem 2.10(2). We obtain a homomorphism $L\to\operatorname{Aut}(X)$. Let S be the minimal resolution of X. Then $\operatorname{Aut}(X)$ embeds in $\operatorname{Bir}(S)$, and $\operatorname{Bir}(S)$ is finite by [SS20, Theorem 1.3] and [Mat63, Section 2, Corollary 2]. Thus the identity component L^0 of L, which is a connected algebraic group, acts trivially on X.

Let $g \in L^0$, and let Σ also denote the induced automorphism of the homogeneous fraction field of R, which is the homogeneous fraction field of S_0/I_ρ by Theorem 2.10(2). Let $y \in R_3$ be a non-zero element, and let $f := \Sigma(y/s)/(y/s)$, an element of degree 0. The elements of degree 0 are rational functions on X, and Σ fixes them, so we have $\Sigma(x) = f^m x$ for homogeneous x of degree m. Since X is normal, R is integrally closed in the homogeneous fraction field, and R is finite over S_0/I_ρ, so Σ, which preserves S_0/I_ρ, extends to R and sends $x \in R_m$ to $f^m x \in R_m$. Since K_X is ample, $|mK_X|$ is base point free for a sufficiently divisible m [Har77, Theorem II.7.6]. For any point of X, choose $x \in R_m$ not vanishing at the point. Then $f^m x \in R_m$ implies that f^m is regular at the point. Thus f^m is regular on the projective variety X, so f is a constant $\alpha \in \mathbb{C}^*$.

Since s, t and $y_0, \dots, y_3$ are bases of R_2 and R_3, we have $\gamma = \alpha^2 I_2$ and $v\phi(x) = \alpha^3 x$ for any $x \in \mathcal{A}$. Taking $x = 1$, we have $v = \alpha^3$, and then $\phi = \mathrm{id}_{\mathcal{A}}$. As $b_{\mathrm{id}_{\mathcal{A}}} = 1$, the equation of $\mathcal{G}$ is $w^2 = \chi\alpha^3$. Thus w has no nilpotent part on any factor of $\mathcal{A}$, and the values of w on two factors differ by a sign. Since L^0 is connected and contains the identity, w is a scalar $\beta \in \mathbb{C}^*$ with $\beta^2 = \chi\alpha^3$.

The element g acts by $(s,t) \to \alpha^2(s,t)$, $y_i \to \alpha^3 y_i$, $z_i \to \beta z_i$, and $r_k \to \alpha^6 r_k$, so it sends A to $\beta\alpha^2\alpha^{-6}A = (\beta/\alpha^4)A$. The gauge keeps A, and $A \neq 0$, so $\beta = \alpha^4$ and $\chi = \alpha^5$. Thus g belongs to the grading torus. In particular, $\dim L \leq 1$, and since $\dim J = 13 - n$ by Construction-Proposition 8.4, we have $\dim J \cdot \ell \geq 12 - n$. □

For a surface as in Set-up 8.1(4), we have $A \neq 0$ for any presentation by Lemma 4.3(1). The surface in Lemma 8.7 need not be torsion-free, because the orbit counts of the type sections use the set $\Xi_{\mathcal{L}}$ of Lemma 4.5(2), which is defined for all numerical Godeaux surfaces with the fixed blocks.

9. The collision type $(2,1,1)$

In this section, we prove the following theorem.

Theorem 9.1. *We have* $\dim \overline{\mathcal{M}_{(2,1,1)}} \leq 7$.

We bound the first-stage tuples and the normalized lower spaces, and we subtract the orbits of J. The key point of this section is Proposition 9.13: the space of solutions with vanishing O gains one dimension for each point of $\mathbb{P}^1$ at which A has a zero column, and the linear conditions in Subsection 9.4 cancel the gain.

9.1. The first stage.

In this subsection, we determine the first stage of a presentation of collision type $(2,1,1)$ and bound the dimension of the space of first-stage tuples.

Construction-Proposition 9.2. Notation and conditions as in Set-up 8.1 with $\mathbf{p} = (2,1,1)$. Choose the basis (e, z, p, q) of $\mathcal{A} = \mathbb{C}[z]/(z^2) \times \mathbb{C} \times \mathbb{C}$, where e, p, q are the idempotents, and normalize $\lambda(e) = 0$ and $\lambda(z) = \lambda(p) = \lambda(q) = 1$. Then S has a presentation with $y_0, \dots, y_3$ restricting to $(e, z, p, q)\tau^3$ and $z_0, \dots, z_3$ restricting to $(e, z, p, q)\tau^4$,

$$(r_1, \dots, r_6) = (y_0y_2, y_0y_3, y_1^2, y_1y_2, y_1y_3, y_2y_3), \quad A = \begin{pmatrix} -d & -E & l & 0 & 0 & 0 \\ -a & -b & 2c & -d & -E & 0 \\ G & 0 & 0 & F & 0 & -h \\ 0 & k & 0 & 0 & I & -j \end{pmatrix},$$

where $a, b, c, d, E, F, G, h, I, j, k, l$ are binary linear forms with $l \neq 0$ and

$$aE + bd - Gh - kj = al + 2cd - F^2 = bl + 2cE - I^2 = dE - hF - jI = 0.$$

The blocks $B_1, \dots, B_4$ are the fixed blocks of Table 10, and P, U are the matrices of Table 10, which are determined by A. Let $\mathcal{P}_{(2,1,1)}$ be the locally closed subset of the affine space of 12-tuples of binary linear forms defined by $l \neq 0$ and the four identities above. Then $\dim \mathcal{P}_{(2,1,1)} \leq 12$.

TABLE 10. The fixed blocks and the blocks P, U of collision type $(2,1,1)$, with $(b_1,b_2,b_3)=(z,p,q)$ in Construction-Proposition 8.2.

$$y_*=y_0+y_2+y_3,\qquad \Lambda=\Lambda^{-1}=\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix},$$

$$M_z=\begin{pmatrix}0&0&0&0\\1&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix},\quad M_p=\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&1&0\\0&0&0&0\end{pmatrix},\quad M_q=\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&1\end{pmatrix},$$

$$D_1=y_1I_4-y_*M_z,\quad D_2=y_2I_4-y_*M_p,\quad D_3=y_3I_4-y_*M_q,$$

$$B_1=\begin{pmatrix}D_1&D_2&D_3\end{pmatrix},\qquad B_4=\begin{pmatrix}0&D_3\Lambda^{-1}&-D_2\Lambda^{-1}\\-D_3\Lambda^{-1}&0&D_1\Lambda^{-1}\\D_2\Lambda^{-1}&-D_1\Lambda^{-1}&0\end{pmatrix},$$

$$B_2=\begin{pmatrix}y_1&0&-y_3&y_2&0&0&-y_0&0\\0&y_1&-y_3&0&0&0&0&-y_0\\0&0&0&0&y_3&-y_2&y_1&0\end{pmatrix},\qquad B_3=\begin{pmatrix}y_3&y_3&y_1&0&0&0&0&0\\-y_2&0&0&y_1&0&0&0&0\\0&0&0&0&y_2&y_3&0&0\\0&0&-y_0&0&-y_1&0&y_3&y_3\\0&0&0&-y_0&0&-y_1&-y_2&0\\0&-y_0&0&0&0&0&0&-y_1\end{pmatrix},$$

$$P=\begin{pmatrix}0&0&d&E&0&0&0&0\\0&0&a&b&-d&-E&0&0\\0&0&-G&0&F&0&0&-h\\0&0&0&-k&0&I&0&-j\\-E&0&0&0&-l&0&0&0\\-b&0&d&E&-2c&0&-E&0\\0&h&-F&0&0&0&0&0\\k&j&0&-I&0&0&I&0\\d&d&0&0&0&-l&0&0\\a&a&d&E&0&-2c&d&d\\-G&-G+h&-F&0&0&0&-F&-F\\0&j&0&-I&0&0&0&0\end{pmatrix},$$

$$U=\begin{pmatrix}-d-E&0&F&I&a&d&-G&0&b&E&0&-k\\-d&0&F&0&a&d&-G+h&j&0&0&-h&-j\\-2c&-l&0&0&d&0&-F&0&E&0&0&-I\end{pmatrix}.$$

Proof of Construction-Proposition 9.2. The proof is long, so we divide it into several steps.

Step 1. In this step we obtain a presentation with the blocks $B_1,\dots,B_4$ of Table 10.

Let λ_0 be the linear functional on $\mathcal{A}$ with $\lambda_0(e)=0$ and $\lambda_0(z)=\lambda_0(p)=\lambda_0(q)=1$. In the basis (e,z,p,q), the pairing of λ_0 has Gram matrix Λ of Table 10, so λ_0 is non-degenerate. Possibly replacing τ with $g\tau$ for a unit g of $\mathcal{A}$, we may assume that $\lambda=\lambda_0$ (Construction-Theorem 3.2(4)). By Construction-Theorem 3.2(2)(3), we may choose $y_0,\dots,y_3\in R_3$ and $z_0,\dots,z_3\in R_4$ restricting to $(e,z,p,q)\tau^3$ and $(e,z,p,q)\tau^4$.

We have $r_1(e,z,p,q)=\cdots=r_6(e,z,p,q)=0$ in $\mathcal{A}$, y_* restricts to $e+p+q=1$, direct multiplication shows that $(r_1,\dots,r_6)B_3=0$ and $B_3B_2^t=0$, and modulo y_* the quadrics $r_1,\dots,r_6$ are a basis of $(y_1,y_2,y_3)^2$ and the columns of B_3, B_2^t are the linear first and second syzygies, so the hypotheses of Construction-Proposition 8.2 hold.

By Lemma 8.3, S has a presentation with the generators y_i, z_i, with the quadrics $r_1,\dots,r_6$, and with the fixed blocks $B_1,\dots,B_4$ of Table 10.

Step 2. In this step we determine A, P, and U, and we prove the four identities.

By Set-up 4.1, the entries of A, P, and U are binary linear forms. Taking the first row and third column, the second row and second column, and the second row and third column of

$d_1' d_2 = 0$, we obtain

$$AB_3 + B_1 P = 0, \qquad UB_4 - B_2 P^t = 0, \qquad UP = 0. \tag{9.1}$$

For a column vector $\xi = (\xi_0, \xi_1, \xi_2, \xi_3)^t$ with entries in $S_1[s,t]$, let

$$\mathrm{ev}(\xi) := \xi_0(e,z,p,q)e + \xi_1(e,z,p,q)z + \xi_2(e,z,p,q)p + \xi_3(e,z,p,q)q \in \mathcal{A}[s,t],$$

where $\xi_i(e,z,p,q)$ is the substitution $(y_0,y_1,y_2,y_3) \to (e,z,p,q)$ in ξ_i. Since y_* restricts to 1 and M_z, M_p, M_q are the matrices of multiplication by z,p,q, we have

$$\mathrm{ev}(D_1\xi) = z\,\mathrm{ev}(\xi) - z\,\mathrm{ev}(\xi) = 0, \qquad \mathrm{ev}(D_2\xi) = p\,\mathrm{ev}(\xi) - p\,\mathrm{ev}(\xi) = 0,$$
$$\mathrm{ev}(D_3\xi) = q\,\mathrm{ev}(\xi) - q\,\mathrm{ev}(\xi) = 0.$$

Thus ev vanishes on the columns of B_1P.

Let $A^{(1)}, \dots, A^{(6)} \in \mathcal{A}[s,t]$ be the columns of A regarded as elements of $\mathcal{A}[s,t]$ through the basis (e,z,p,q). Applying ev to the eight columns of $AB_3 = -B_1P$, we obtain

$$qA^{(1)} - pA^{(2)} = 0, \quad qA^{(1)} - eA^{(6)} = 0, \quad zA^{(1)} - eA^{(4)} = 0, \quad zA^{(2)} - eA^{(5)} = 0,$$
$$pA^{(3)} - zA^{(4)} = 0, \quad qA^{(3)} - zA^{(5)} = 0, \quad qA^{(4)} - pA^{(5)} = 0, \quad qA^{(4)} - zA^{(6)} = 0.$$

We have $e^2 = e$, $ez = z$, $z^2 = 0$, $p^2 = p$, $q^2 = q$, and $ep = eq = zp = zq = pq = 0$. We compare coordinates in the basis (e,z,p,q):

(1) the first two equations show that the q-coordinate of $A^{(1)}$, the p-coordinate of $A^{(2)}$, and the e- and z-coordinates of $A^{(6)}$ vanish;
(2) the next two equations show that the e-coordinates of $A^{(4)}, A^{(5)}$ vanish, and that the z-coordinates of $A^{(4)}, A^{(5)}$ are the e-coordinates of $A^{(1)}, A^{(2)}$;
(3) the next two equations show that the p- and q-coordinates of $A^{(3)}$ vanish;
(4) the last two equations show that the q-coordinate of $A^{(4)}$ and the p-coordinate of $A^{(5)}$ vanish, and they imply no further condition.

Naming the twelve remaining coordinates, we obtain the matrix A of the statement.

A direct calculation shows that the matrices P, U of Table 10 satisfy the first two equations of (9.1). By Construction-Proposition 8.2, they are the blocks P, U of the presentation, and they are determined by A. Let

$$Q_1 := aE + bd - Gh - kj, \quad Q_2 := al + 2cd - F^2, \quad Q_3 := bl + 2cE - I^2, \quad Q_4 := dE - hF - jI.$$

A direct calculation shows that

$$UP = \begin{pmatrix} 0 & Q_1 & 0 & 0 & -Q_2 & -Q_3 & 0 & Q_4 \\ -Q_1 & 0 & 0 & 0 & -Q_2 & 0 & -Q_4 & 0 \\ 0 & Q_4 & -Q_2 & -Q_3 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus $Q_1 = Q_2 = Q_3 = Q_4 = 0$ by (9.1).

Step 3. In this step we show that $l \neq 0$.

Let x be the double point of B, $\mathfrak{m}$ the maximal ideal of $\mathcal{O}_{X,x}$, and τ a local generator of $\mathcal{O}_X(K_X)$ at x extending the trivialization of Construction-Theorem 3.2. We write

$$\bar{s} := s/\tau^2, \quad \bar{t} := t/\tau^2, \quad \bar{y}_1 := y_1/\tau^3, \quad \bar{z}_0 := z_0/\tau^4, \quad \bar{z}_1 := z_1/\tau^4$$

in $\mathcal{O}_{X,x}$. By Construction-Proposition 3.3(2), $\mathcal{O}_{X,x}$ is a regular local ring of dimension 2. The ideal $\mathfrak{a}$ of B at x is $(\bar{s},\bar{t})$, and $\mathcal{O}_{X,x}/\mathfrak{a} = \mathbb{C}[z]/(z^2)$ with $\bar{y}_1$ restricting to z and $\bar{z}_1 \in \mathfrak{m}$. Hence $\mathfrak{m}^2 \subset \mathfrak{a}$. Since $\mathcal{O}_{X,x}/\mathfrak{m}^2$ is of length 3 and $\mathcal{O}_{X,x}/\mathfrak{a}$ is of length 2, we may choose $\mu \in \mathfrak{a}\backslash\mathfrak{m}^2$. Then $\mu, \bar{y}_1$ are regular parameters of $\mathcal{O}_{X,x}$ and $\mathfrak{a} = (\mu, \bar{y}_1^2)$ by comparing lengths.

We claim that $\bar{y}_1^2 \notin \mathfrak{m}\mathfrak{a}$. Otherwise, modulo μ, $\bar{y}_1^2$ would belong to $(\bar{y}_1^3)$ in the discrete valuation ring $\mathcal{O}_{X,x}/(\mu)$, which is impossible.

The third relation of Set-up 4.1(7), divided by τ^6, is

$$\bar{y}_1^2 + l(\bar{s},\bar{t})\bar{z}_0 + 2c(\bar{s},\bar{t})\bar{z}_1 + v_3(\bar{s},\bar{t}) = 0.$$

If $l = 0$, then

$$2c(\bar{s}, \bar{t})\bar{z}_1 + v_3(\bar{s}, \bar{t}) \in \mathfrak{ma},$$

hence $\bar{y}_1^2 \in \mathfrak{ma}$, a contradiction. Thus $l \neq 0$.

Step 4. In this step we show that $\dim \mathcal{P}_{(2,1,1)} \leq 12$. The bound is for all tuples in $\mathcal{P}_{(2,1,1)}$, whether or not they come from a surface. In each case below, we add the dimension of the base and the bound on the dimension of the fibers (cf. [Har77, Exercise II.3.22]).

The group GL_2 acts on $\mathcal{P}_{(2,1,1)}$ by linear substitutions in s, t, and the map from $\mathcal{P}_{(2,1,1)}$ to $\mathbb{C}[s,t]_1 \backslash \{0\}$ sending $(a, \ldots, l)$ to l is equivariant. Since the action on $\mathbb{C}[s,t]_1 \backslash \{0\}$ is transitive and $\dim \mathbb{C}[s,t]_1 = 2$, it suffices to show that the fiber $\mathcal{P}_s$ of $\mathcal{P}_{(2,1,1)}$ over $l = s$ satisfies

$$\dim \mathcal{P}_s \leq 10.$$

We write the coefficients of a binary linear form with one and two primes, $a = a's + a''t$, and similarly for the other forms. On $\mathcal{P}_s$, the equations $Q_2 = Q_3 = 0$ read $as = F^2 - 2cd$ and $bs = I^2 - 2cE$. They have a solution (a, b) if and only if

$$(F'')^2 = 2c''d'', \qquad (I'')^2 = 2c''E'', \tag{9.2}$$

and then (a, b) is unique. Substituting a, b in $Q_4 = Q_1 = 0$, we obtain

$$dE = hF + jI, \tag{9.3}$$

$$Gh + kj = \Theta, \qquad \text{where } \Theta := \frac{EF^2 + dI^2 - 4cdE}{s}. \tag{9.4}$$

By (9.2), s divides $EF^2 + dI^2 - 4cdE$, so Θ is a binary quadric. Thus $\mathcal{P}_s$ is the set of tuples $(c, d, E, F, G, h, I, j, k)$ satisfying (9.2), (9.3), and (9.4). Interchanging (a, d, F, G, h) and (b, E, I, k, j) preserves l, Q_1, Q_4 and interchanges Q_2, Q_3. We consider the following cases according to the span of h, j.

Case 1. h, j are linearly independent. We have $\dim\{(h, j)\} = 4$. Fix (h, j).

If $c'' \neq 0$, then (9.2) implies that $d'' = (F'')^2/(2c'')$ and $E'' = (I'')^2/(2c'')$, and the coefficient of t^2 in (9.3) is the equation

$$(F'')^2(I'')^2 = 4(c'')^2(h''F'' + j''I''),$$

which defines a proper closed subset of the space of (c'', F'', I'') with $c'' \neq 0$. If $c'' = 0$, then (9.2) implies that $F'' = I'' = 0$, and the coefficient of t^2 in (9.3) is $d''E'' = 0$. Thus

$$\dim\{(c'', d'', E'', F'', I'')\} \leq 2.$$

Fix $(c'', d'', E'', F'', I'')$. Then c', d', E' are arbitrary, and the coefficients of s^2 and st in (9.3) determine F', I', because they are linear in (F', I') with coefficient matrix

$$\begin{pmatrix} h' & j' \\ h'' & j'' \end{pmatrix},$$

whose determinant is non-zero. Thus

$$\dim\{(c, d, E, F, I, h, j)\} \leq 4 + 2 + 3 = 9.$$

The map $(G, k) \to Gh + kj$ from $\mathbb{C}[s,t]_1^{\oplus 2}$ onto $\mathbb{C}[s,t]_2$ has one-dimensional kernel, so (9.4) adds exactly 1 to the dimension. Thus the locus of Case 1 in $\mathcal{P}_s$ is of dimension at most 10.

Case 2. $h = \alpha\kappa$ and $j = \beta\kappa$ with $\kappa = \kappa's + t$ and $(\alpha, \beta) \neq (0, 0)$. We have $\dim\{(\kappa', \alpha, \beta)\} = 3$. The map $(G, k) \to \kappa(\alpha G + \beta k)$ has image $\kappa\mathbb{C}[s,t]_1$ and two-dimensional kernel. Hence the solutions (G, k) of (9.4) for a given (c, d, E, F, I, h, j) are empty or form an affine space of dimension 2. So it suffices to show that

$$\dim\{(c, d, E, F, I, h, j)\} \leq 8.$$

By (9.3), κ divides d or E. Possibly interchanging (a, d, F, G, h) and (b, E, I, k, j), we may assume that $d = \delta\kappa$ with $\delta \in \mathbb{C}$. Then $d'' = \delta$.

Case 2.1. $E = \epsilon\kappa$ with $\epsilon \in \mathbb{C}$. Then (9.3) reads

$$\alpha F + \beta I = \delta\epsilon\kappa.$$

Fix (κ', α, β). The tuples $(\delta, \epsilon, c, F, I)$ satisfying $\alpha F + \beta I = \delta\epsilon\kappa$ form an irreducible variety W with $\dim W = 2 + 2 + 2 = 6$: two scalars, a free form c, and a two-dimensional affine space of pairs (F, I). The first equation of (9.2) is $(F'')^2 = 2c''\delta$, which defines a proper closed subset of W, because c'' is a free coordinate on W. Thus the locus of Case 2.1 in W is of dimension at most 5, and $\dim\{(c, d, E, F, I, h, j)\} \le 3 + 5 = 8$.

Case 2.2. κ does not divide E. By (9.4), κ divides Θ. We evaluate $s\Theta = EF^2 + dI^2 - 4cdE$ at the root of κ. Since d vanishes there, EF^2 vanishes there. Since E does not vanish there, κ divides F. We write $F = \phi\kappa$ with $\phi \in \mathbb{C}$. Then (9.3) reads

$$\delta E = \alpha\phi\kappa + \beta I.$$

If $\beta \neq 0$, then $\delta E = \alpha\phi\kappa + \beta I$ determines I. We have $\dim\{(\delta, \phi, E, c)\} = 6$, and the first equation of (9.2) is $\phi^2 = 2c''\delta$, which defines a proper closed subset of the space of (δ, ϕ, E, c). Thus $\dim\{(c, d, E, F, I, h, j)\} \le 3 + 5 = 8$.

If $\beta = 0$, then $\alpha \neq 0$. Since κ does not divide E, the equation $\delta E = \alpha\phi\kappa$ implies that $\delta = \phi = 0$, so $d = F = 0$. We have $\dim\{(h, j)\} = 2$ and $\dim\{(c, E, I)\} \le 6$. Thus

$$\dim\{(c, d, E, F, I, h, j)\} \le 8.$$

Case 3. $h = \alpha s$ and $j = \beta s$ with $(\alpha, \beta) \neq (0, 0)$. The map $(G, k) \to s(\alpha G + \beta k)$ has image $s\mathbb{C}[s, t]_1$ and two-dimensional kernel. Hence the solutions (G, k) of (9.4) for a given (c, d, E, F, I, h, j) are empty or form an affine space of dimension 2. So it suffices to show that

$$\dim\{(c, d, E, F, I, h, j)\} \le 7.$$

By (9.3), s divides d or E. Possibly interchanging (a, d, F, G, h) and (b, E, I, k, j), we may assume that $d = \delta s$ with $\delta \in \mathbb{C}$. Then $d'' = 0$, so (9.2) implies that $F'' = 0$. We write $F = \phi s$ with $\phi \in \mathbb{C}$. Then (9.3) reads

$$\delta E = \alpha\phi s + \beta I.$$

Case 3.1. $\beta \neq 0$. Then $\delta E = \alpha\phi s + \beta I$ determines I, and $I'' = \delta E''/\beta$. The tuples $(\alpha, \beta, \delta, \phi, E, c)$ form an irreducible parameter space W' with $\dim W' = 8$. The second equation of (9.2) reads

$$E''\big(\delta^2 E'' - 2\beta^2 c''\big) = 0,$$

which defines a proper closed subset of W'. Thus $\dim\{(c, d, E, F, I, h, j)\} \le 7$.

Case 3.2. $\beta = 0$. Then $\alpha \neq 0$ and $\delta E = \alpha\phi s$. If $\delta \neq 0$, then $E = E's$ and ϕ is determined, and we have $\dim\{(\alpha, \delta, E', I, c)\} = 7$. If $\delta = 0$, then $\phi = 0$, and we have $\dim\{(\alpha, E, I, c)\} = 7$. Thus $\dim\{(c, d, E, F, I, h, j)\} \le 7$.

Case 4. $h = j = 0$. By (9.3), $dE = 0$. Possibly interchanging (a, d, F, G, h) and (b, E, I, k, j), we may assume that $d = 0$. Then (9.2) implies that $F'' = 0$. We write $F = \phi s$ with $\phi \in \mathbb{C}$. Then $Q_2 = 0$ implies that $a = \phi^2 s$, and $Q_1 = 0$ reads $\phi^2 sE = 0$.

Case 4.1. $E \neq 0$. Then $\phi = 0$, so $F = 0$. The forms c, E, I satisfy the equation $(I'')^2 = 2c''E''$ of (9.2), which defines a proper closed subset of the space of (c, E, I), so $\dim\{(c, E, I)\} \le 5$. The forms G, k are arbitrary, so $\dim\{(G, k)\} = 4$. Thus the locus of Case 4.1 in $\mathcal{P}_s$ is of dimension at most 9.

Case 4.2. $E = 0$. Then (9.2) implies that $I'' = 0$. We write $I = \iota s$ with $\iota \in \mathbb{C}$. The forms c, G, k are arbitrary and ϕ, ι are scalars. Thus the locus of Case 4.2 in $\mathcal{P}_s$ is of dimension at most 8.

Combining Cases 1–4, we have $\dim \mathcal{P}_s \le 10$, hence $\dim \mathcal{P}_{(2,1,1)} \le 12$. □

9.2. **Column restrictions.** In this subsection, we prove Lemma 9.3, which restricts the zeros of the columns of the matrix A of a presentation of S. The elimination in Subsections 9.3 and 9.4 uses that at most one column of $A(u)$ is zero for any $u \in \mathbb{P}^1$, which holds for the matrix of a presentation of S by Lemma 9.3.

Lemma 9.3. *Notation and conditions as in Construction-Proposition 9.2, where A is the matrix of a presentation of S. Then:*

(1) Any column of A is non-zero as a vector of binary linear forms.
(2) For any $u \in \mathbb{P}^1$, columns $1,2,3$ of $A(u)$ are non-zero and at most one of columns $4,5,6$ of $A(u)$ is zero.
(3) The forms c, l are linearly independent, and l, d, E have no common zero.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the form of the relations, name the points of B, and define the exchange of y_2, y_3.

Recall that

$$A = \begin{pmatrix} -d & -E & l & 0 & 0 & 0 \\ -a & -b & 2c & -d & -E & 0 \\ G & 0 & 0 & F & 0 & -h \\ 0 & k & 0 & 0 & I & -j \end{pmatrix}, \qquad (r_1, \dots, r_6) = (y_0y_2, y_0y_3, y_1^2, y_1y_2, y_1y_3, y_2y_3),$$

that $l \neq 0$, and that

$$aE + bd = Gh + kj, \qquad al + 2cd = F^2, \qquad bl + 2cE = I^2, \qquad dE = hF + jI. \tag{9.5}$$

We index the columns of A by i, i' and the rows by $\nu \in \{0,1,2,3\}$, so that the relations of Set-up 4.1 read

$$r_i(y) + \sum_\nu A_{\nu i}(s,t) z_\nu + v_i(s,t) = 0.$$

Recall that for $u = [u_0 : u_1]$, the member C_u is defined by $u_1 s - u_0 t = 0$, that $A(u), v(u)$ are the values at $(s,t) = (u_0, u_1)$, and that H_u is the kernel of the matrix with rows $A(u), v(u)$ (Set-up 4.1). We write a vector of H_u as $\mathbf{w}$, with entries $\mathbf{w}_i$.

Let x_e, x_p, x_q be the points of $\operatorname{Supp} B$ at which e, p, q respectively do not vanish. Since $y_0, \dots, y_3$ restrict to $(e, z, p, q)\tau^3$, the images of x_e, x_p, x_q under the tricanonical map are $[1:0:0:0]$, $[0:0:1:0]$, and $[0:0:0:1]$ respectively. By Construction-Proposition 3.3(2), x_e, x_p, x_q are smooth points of X.

The *exchange* is the simultaneous interchange

$$a \leftrightarrow b, \quad d \leftrightarrow E, \quad F \leftrightarrow I, \quad G \leftrightarrow k, \quad h \leftrightarrow j, \quad y_2 \leftrightarrow y_3, \quad z_2 \leftrightarrow z_3.$$

It interchanges columns $1,2$, columns $4,5$, and the last two rows of A, it fixes r_3, r_6, and it preserves the four identities in (9.5), the rank of $A(u)$, the point x_e, and the form l. Below, a statement deduced by the exchange is proved by the interchanged argument, which uses only the shape of A, the identities in (9.5), and columns and pairs of columns already excluded. We do not apply the exchange to the blocks of Table 10, which are not used in the proof.

Step 2. In this step we prove three statements on two columns whose quadrics have a common linear factor.

Let i, i' be columns of A with $r_i = y_\rho y_\sigma$ and $r_{i'} = y_\rho y_\theta$, where $\sigma \neq \theta$. For instance, (i, i') can be any of $(1,4), (2,5), (3,4), (3,5), (4,5), (4,6), (5,6)$.

(a) Columns i, i' of A are not both identically zero.
(b) If columns i, i' of $A(u)$ are both zero at a point $u \in \mathbb{P}^1$, then $\operatorname{rank} A(u) \geq 4$.
(c) If in addition $x \in \operatorname{Supp} B$ is a point whose image is a common zero of $y_\rho, y_\sigma, y_\theta$, then C_u is singular at x.

(a) Suppose that columns i, i' of A are both identically zero. Then the relations read $y_\rho y_\sigma + v_i(s,t) = 0$ and $y_\rho y_\theta + v_{i'}(s,t) = 0$ in R. Since X is integral, R is a domain, and $y_0, \dots, y_3$ are a basis of R_3. Thus $v_i, v_{i'}$ are non-zero binary cubics, and

$$v_{i'}(s,t)y_\sigma - v_i(s,t)y_\theta = 0 \quad \text{in } R.$$

Let $u = [u_0 : u_1]$ be a point with $v_i(u)v_{i'}(u) \neq 0$. On C_u we have $s = u_0\psi$ and $t = u_1\psi$ for a section ψ of $2K_X|_{C_u}$. The zero locus of ψ is B, which is finite by Construction-Theorem 3.2(1), and the Cartier divisor C_u on the normal surface X has no embedded points, since X is Cohen–Macaulay [Har77, Theorem II.8.22A]. Thus ψ is a non-zero-divisor on $\mathcal{O}_{C_u}$, and therefore $v_{i'}(u)y_\sigma - v_i(u)y_\theta$ vanishes on C_u. The kernel of $R_3 \to H^0(C_u, 3K_X|_{C_u})$ is $H^0(X, K_X) = 0$ since $p_g = 0$ (see also Lemma 4.3(4)), a contradiction.

(b) Let $u \in \mathbb{P}^1$ be a point at which columns i, i' of $A(u)$ are both zero. The span of the i-th and i'-th basis vectors of $\mathbb{C}^6$ is contained in $\operatorname{Ker} A(u)$, and the corresponding quadrics have the common linear factor y_ρ. By Lemma 4.3(2), $\operatorname{rank} A(u) \geq 4$.

(c) Let u be as in (b). There exists a non-zero vector $\mathbf{w}$ with $\mathbf{w}_i v_i(u) + \mathbf{w}_{i'} v_{i'}(u) = 0$ and with all other entries zero. Then $\mathbf{w} \in H_u$ and

$$\sum_{i''} \mathbf{w}_{i''} r_{i''} = y_\rho(\mathbf{w}_i y_\sigma + \mathbf{w}_{i'} y_\theta).$$

If $x \in \operatorname{Supp} B$ is a point whose image is a common zero of $y_\rho, y_\sigma, y_\theta$, then the image of x is a singular point of $y_\rho(\mathbf{w}_i y_\sigma + \mathbf{w}_{i'} y_\theta)$, so C_u is singular at x by Lemma 4.3(3).

Step 3. In this step we prove (1).

Column 3 is non-zero since $l \neq 0$. Suppose that column 1 is identically zero, that is, $d = a = G = 0$. The identity $al + 2cd = F^2$ implies $F = 0$. Thus column 4 is also identically zero, which contradicts **Step 2**(a) for the pair $(1, 4)$. By the exchange, column 2 is non-zero.

Suppose that column 6 is identically zero, that is, $h = j = 0$. The identity $dE = hF + jI$ implies $dE = 0$. By the exchange, we may assume that $d = 0$. If $E \neq 0$, then $aE + bd = Gh + kj$ implies $a = 0$, and $al + 2cd = F^2$ implies $F = 0$. Thus column 4 is identically zero, which contradicts **Step 2**(a) for the pair $(4, 6)$.

If $E = 0$, then $al = F^2$ and $bl = I^2$. Since l is a non-zero linear form, there exist constants $\phi, \iota \in \mathbb{C}$ with

$$F = \phi l, \quad a = \phi^2 l, \quad I = \iota l, \quad b = \iota^2 l.$$

Let u be the zero of l. Then the first row of $A(u)$ is zero, so $\operatorname{rank} A(u) \leq 3$. Moreover, columns $4, 5$ of $A(u)$ are zero, which contradicts **Step 2**(b). Thus column 6 is non-zero.

Suppose that column 4 is identically zero, that is, $d = F = 0$. The identity $al + 2cd = F^2$ implies $a = 0$, and $dE = hF + jI$ implies $jI = 0$. If $j = 0$, then $aE + bd = Gh + kj$ implies $Gh = 0$. Thus $G = 0$, so column 1 is identically zero, or $h = 0$, so column 6 is identically zero. Both are excluded above, so $j \neq 0$ and $I = 0$.

If $E = 0$, then column 5 is also identically zero, which contradicts **Step 2**(a) for the pair $(4, 5)$. Thus $E \neq 0$. Let u be the zero of E. Then columns $4, 5$ of $A(u)$ are zero, and $bl + 2cE = I^2$ implies $b(u)l(u) = 0$. If $l(u) = 0$, then the first row of $A(u)$ is zero. If $l(u) \neq 0$, then $b(u) = 0$, and the first two rows of $A(u)$ are

$$(0, 0, l(u), 0, 0, 0), \qquad (0, 0, 2c(u), 0, 0, 0).$$

In both cases $\operatorname{rank} A(u) \leq 3$, which contradicts **Step 2**(b). Thus column 4 is non-zero. By the exchange, column 5 is non-zero.

Step 4. In this step we show that at any u, the columns in the following pairs are not both zero.

columns	$(1,4)$	$(2,5)$	$(3,4)$	$(3,5)$
quadrics	$y_0y_2,\ y_1y_2$	$y_0y_3,\ y_1y_3$	$y_1^2,\ y_1y_2$	$y_1^2,\ y_1y_3$
point	x_q	x_p	x_q	x_p

Since x_p is a smooth point of X and a reduced point of B, after trivializing K_X at x_p the sections s, t are a basis of $\mathfrak{m}/\mathfrak{m}^2$, where $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}_{X,x_p}$. Thus $u_1 s - u_0 t \notin \mathfrak{m}^2$, and C_u is smooth at x_p for any u. Replacing x_p by x_q, we see that C_u is smooth at x_q for any u. Suppose that the columns in a pair of the table are both zero at u. The linear factors of the quadrics vanish at the image of the indicated point, so **Step 2**(c) implies that C_u is singular at that point, a contradiction.

Step 5. In this step we show that C_u is smooth at x_e for any u with $l(u) \neq 0$, and we exclude the pair $(4, 5)$ at any u.

Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{X,x_e}$, let τ be a local generator of $\mathcal{O}_X(K_X)$ at x_e extending the trivialization of Construction-Theorem 3.2, and write $\bar{s} := s/\tau^2$, $\bar{t} := t/\tau^2$, $\bar{y}_1 := y_1/\tau^3$, $\bar{z}_0 := z_0/\tau^4$, $\bar{z}_1 := z_1/\tau^4$ in $\mathcal{O}_{X,x_e}$. Since x_e is a smooth point of X, $\mathcal{O}_{X,x_e}$ is a regular local ring of dimension 2. The ideal $\mathfrak{a}$ of B at x_e is $(\bar{s}, \bar{t})$, and $\mathcal{O}_{X,x_e}/\mathfrak{a} = \mathbb{C}[z]/(z^2)$ with $\bar{y}_1$ restricting to z. Hence $\mathfrak{m}^2 \subset \mathfrak{a}$. Since $\mathcal{O}_{X,x_e}/\mathfrak{m}^2$ is of length 3 and $\mathcal{O}_{X,x_e}/\mathfrak{a}$ is of length 2, we may choose $\mu \in \mathfrak{a} \backslash \mathfrak{m}^2$. Since $\mathfrak{m}^2 \subset \mathfrak{a}$ and $\bar{y}_1 \notin \mathfrak{a}$, the elements $\mu, \bar{y}_1$ are regular parameters of $\mathcal{O}_{X,x_e}$, and $\mathfrak{a} = (\mu, \bar{y}_1^2)$ since $(\mu, \bar{y}_1^2) \subset \mathfrak{a}$ and both quotients are of length 2.

The relation for $r_3 = y_1^2$, divided by τ^6, is

$$\bar{y}_1^2 + l(\bar{s}, \bar{t})\bar{z}_0 + 2c(\bar{s}, \bar{t})\bar{z}_1 + v_3(\bar{s}, \bar{t}) = 0.$$

By Construction-Proposition 9.2, $\bar{z}_0$ is a unit and $\bar{z}_1 \in \mathfrak{m}$. Since $c(\bar{s}, \bar{t}) \in \mathfrak{a}$, $v_3(\bar{s}, \bar{t}) \in \mathfrak{a}^3$, and $\bar{z}_0 - \bar{z}_0(x_e) \in \mathfrak{m}$, we have

$$\bar{y}_1^2 + \bar{z}_0(x_e) l(\bar{s}, \bar{t}) \in \mathfrak{m}\mathfrak{a}.$$

Since $\mathfrak{m}\mathfrak{a} \subset \mathfrak{m}^2$ and $\bar{y}_1^2 \in \mathfrak{m}^2$, we obtain $l(\bar{s}, \bar{t}) \in \mathfrak{m}^2$. The elements $\bar{s}, \bar{t}$ are a basis of $\mathfrak{a}/\mathfrak{m}\mathfrak{a}$. Since $\mu, \bar{y}_1^2$ is also a basis of $\mathfrak{a}/\mathfrak{m}\mathfrak{a}$, the kernel of

$$\mathfrak{a}/\mathfrak{m}\mathfrak{a} \to \mathfrak{m}/\mathfrak{m}^2$$

is spanned by the class of $\bar{y}_1^2$. Thus any member of the pencil other than the divisor of $l(s, t)$ is smooth at x_e. The divisor of $l(s, t)$ is C_u for the zero u of l.

Suppose that columns $4, 5$ of $A(u)$ are zero. The forms y_1, y_2, y_3 vanish at the image of x_e, so C_u is singular at x_e by **Step 2**(c). Thus $l(u) = 0$. Since $d(u) = E(u) = l(u) = 0$, the first row of $A(u)$ is zero, so $\operatorname{rank} A(u) \leq 3$, which contradicts **Step 2**(b).

Step 6. In this step we show that columns $1, 2, 3$ of $A(u)$ are non-zero for any u, and we prove (3).

Suppose that column 1 of $A(u)$ is zero, that is, $d(u) = a(u) = G(u) = 0$. The identity $al + 2cd = F^2$ at u implies $F(u) = 0$. Thus column 4 of $A(u)$ is zero, which contradicts **Step 4**. By the exchange, column 2 of $A(u)$ is non-zero.

Suppose that column 3 of $A(u)$ is zero, that is, $l(u) = c(u) = 0$. The identities $al + 2cd = F^2$ and $bl + 2cE = I^2$ at u imply $F(u) = I(u) = 0$, and $dE = hF + jI$ at u implies $d(u)E(u) = 0$. If $d(u) = 0$, then column 4 of $A(u)$ is zero, and if $E(u) = 0$, then column 5 of $A(u)$ is zero. In both cases we obtain a contradiction to **Step 4**. In particular, c, l have no common zero. Since l is a non-zero binary linear form, c, l are linearly independent.

Suppose that l, d, E vanish at u. The identities $al + 2cd = F^2$ and $bl + 2cE = I^2$ at u imply $F(u) = I(u) = 0$. Thus columns $4, 5$ of $A(u)$ are zero, which contradicts **Step 5**.

Step 7. In this step we exclude the pairs $(4, 6)$ and $(5, 6)$ at any u. Together with **Step 5**, this shows that at most one of columns $4, 5, 6$ of $A(u)$ is zero.

Suppose that columns $4, 6$ or columns $5, 6$ of $A(u)$ are zero. The forms y_1, y_2, y_3 vanish at the image of x_e, so C_u is singular at x_e by **Step 2**(c). Thus $l(u) = 0$ by **Step 5**. By the exchange, we may assume that columns $4, 6$ of $A(u)$ are zero, that is, $d(u) = F(u) = h(u) = j(u) = 0$.

By **Step 6**, $c(u) \neq 0$. By **Step 5**, column 5 of $A(u)$ is non-zero, that is, $(E(u), I(u)) \neq (0, 0)$. The identity $bl + 2cE = I^2$ at u reads $I(u)^2 = 2c(u)E(u)$, so $E(u) \neq 0$ and $I(u) \neq 0$. The identity $aE + bd = Gh + kj$ at u reads $a(u)E(u) = 0$, so $a(u) = 0$.

Thus a, d, F, h, j vanish at the unique zero of l, and there exist constants $\alpha, \delta, \phi, \eta, \kappa \in \mathbb{C}$ with

$$a = \alpha l, \quad d = \delta l, \quad F = \phi l, \quad h = \eta l, \quad j = \kappa l.$$

The identity $al + 2cd = F^2$ reads $(\alpha - \phi^2)l + 2\delta c = 0$. By **Step 6**, c, l are linearly independent, so $\delta = 0$ and $\alpha = \phi^2$. The identity $dE = hF + jI$ now reads

$$\eta\phi l + \kappa I = 0.$$

Since $I(u) \neq 0$, the forms I, l are linearly independent, so $\kappa = 0$ and $\eta\phi = 0$. The identity $aE + bd = Gh + kj$ now reads $\phi^2 E = \eta G$. If $\phi \neq 0$, then $\eta = 0$ and $E = 0$, which contradicts $E(u) \neq 0$. Thus $\phi = 0$, so $a = d = F = j = 0$ and $\eta G = 0$. By (1), column 1 of A is non-zero, so $G \neq 0$ and $\eta = 0$. Then $h = j = 0$, which contradicts (1). □

9.3. **The solutions with vanishing O block.** In this subsection, we determine the blocks (C, N) with $d_1' d_2 = 0$ at $O = 0$ for a first stage of type $(2, 1, 1)$.

Set-up 9.4. Let $(a, b, c, d, E, F, G, h, I, j, k, l) \in \mathcal{P}_{(2,1,1)}$ and A be as follows.

(1) A is the matrix of Construction-Proposition 9.2 with the entries $a, \dots, l$, and A has generic rank four.
(2) The forms c, l are linearly independent, and l, d, E have no common zero.
(3) Any column of A is non-zero as a vector of binary linear forms.
(4) For any $u \in \mathbb{P}^1$, columns $1, 2, 3$ of $A(u)$ are non-zero and at most one of columns $4, 5, 6$ of $A(u)$ is zero.

We denote by $\mathcal{P}^{\mathrm{good}} \subset \mathcal{P}_{(2,1,1)}$ the set of such tuples, and we fix the blocks $B_1, \dots, B_4, P, U$ of Construction-Proposition 9.2. Let

$$Q_1 := aE + bd - Gh - kj, \quad Q_2 := al + 2cd - F^2, \quad Q_3 := bl + 2cE - I^2, \quad Q_4 := dE - hF - jI,$$

so that $Q_1 = Q_2 = Q_3 = Q_4 = 0$. When binary linear forms f, g are proportional and $g \neq 0$, we write $f = f_g g$ with $f_g \in \mathbb{C}$. The rows and columns of C, N, O and of the matrices X, Y below are counted from 0, and the columns of A are counted from 1.

Condition (2) says that for any $u \in \mathbb{P}^1$, column 3 of $A(u)$, which is the column of y_1^2, and the first row of $A(u)$, which is the row of z_0, are non-zero, and the elimination at the zeros of l uses both facts.

Lemma 9.3 and Lemma 4.3(1) show that the tuple of any presentation of a surface as in Construction-Proposition 9.2 belongs to $\mathcal{P}^{\mathrm{good}}$. The elimination below uses only the conditions of Set-up 9.4.

Construction-Proposition 9.5. Notation and conditions as in Set-up 9.4. For a vector $\mathbf{c} = (c_0, \dots, c_{19})$ with coordinates in $\mathbb{C}[s, t]$, let

$$C(\mathbf{c}) := \begin{pmatrix} c_0 & 0 & 0 & 0 & -c_4 & -c_5 & c_1 & -c_3 \\ c_2 & c_3 & c_4 & c_5 & c_6 & c_7 & c_8 & c_9 \\ 0 & c_{10} & c_{11} & c_{12} & 0 & c_{13} & 0 & c_{14} \\ -c_{15} & c_{15} & c_{16} & c_{17} & c_{18} & 0 & -c_{19} & c_{19} \end{pmatrix},$$

and let $N(\mathbf{c})$ be the 6×12 matrix obtained by placing the second of the following matrices to the right of the first:

$$\begin{pmatrix} 0 & -c_4 & -c_{11} & -c_{16} & 0 & 0 \\ 0 & -c_5 & -c_{12} & -c_{17} & c_0 & c_2 \\ 0 & 0 & 0 & 0 & c_4 & -c_6 \\ -c_4 & c_6 & 0 & c_{18} & 0 & -c_4 \\ -c_5 & c_7 & c_{13} & 0 & c_1 & -c_5 + c_8 \\ -c_3 & c_9 & c_{14} & c_{19} & 0 & -c_3 \end{pmatrix},$$

$$\begin{pmatrix} 0 & 0 & -c_0 & -c_2-c_3 & -c_{10} & 0 \\ 0 & -c_{15} & 0 & 0 & 0 & 0 \\ 0 & -c_{18} & c_5 & -c_7 & -c_{13} & 0 \\ -c_{11} & -c_{16} & -c_1+c_3 & -c_4-c_8-c_9 & -c_{11}-c_{14} & -c_{16} \\ -c_{12} & -c_{17}-c_{19} & 0 & -c_5 & -c_{12} & -c_{17} \\ -c_{10} & -c_{15} & 0 & -c_3 & -c_{10} & -c_{15} \end{pmatrix}.$$

Let $X := AN(\mathbf{c}) - C(\mathbf{c})P^t$ and $Y := UN(\mathbf{c})^t$. Let L be the 66×20 matrix of binary linear forms such that $L\mathbf{c}$ is the vector of the entries of X and Y. Let $\xi = (\xi_0, \dots, \xi_{19})$ and $\eta = (\eta_0, \dots, \eta_{19})$ be the vectors of binary cubic forms

$$\begin{aligned} &\xi_0 = -ldE, && \xi_1 = d(bl-2cE), && \xi_2 = aEl, && \xi_3 = l(bd-aE), \\ &\xi_4 = -dE^2, && \xi_5 = d^2E, && \xi_6 = -aE^2, && \xi_7 = bd^2, \\ &\xi_8 = a(2cE-bl), && \xi_9 = 2c(bd-aE), && \xi_{10} = FI^2, && \xi_{11} = hI^2, \\ &\xi_{12} = FIj, && \xi_{13} = j(Fk-GI), && \xi_{14} = -GI^2, && \xi_{15} = -F^2I, \\ &\xi_{16} = -FIh, && \xi_{17} = -F^2j, && \xi_{18} = h(Fk-GI), && \xi_{19} = F^2k, \end{aligned}$$

and

$$\begin{aligned} &\eta_0 = \eta_1 = \eta_3 = \eta_4 = \eta_5 = 0, && \eta_2 = l(Fh-Ij), && \eta_6 = -E(Fh-Ij), \\ &\eta_7 = d(Fh-Ij), && \eta_8 = -2Ghl + I^2d + 2aEl, && \eta_9 = -l(Gh+kj) - 4cdE, \\ &\eta_{10} = -FEl, && \eta_{11} = -Ehl, && \eta_{12} = dhl, \\ &\eta_{13} = -FdE, && \eta_{14} = GEl, && \eta_{15} = -Idl, \\ &\eta_{16} = Ejl, && \eta_{17} = -djl, && \eta_{18} = -IdE, \\ &\eta_{19} = kdl. \end{aligned}$$

Then:

(1) A pair (C, N) of matrices of binary quadratic forms of the sizes in Set-up 4.1 satisfies $B_1N^t + CB_3^t = 0$ if and only if $(C, N) = (C(\mathbf{c}), N(\mathbf{c}))$ for a vector $\mathbf{c}$ of binary quadratic forms, which is unique since the coordinates of $\mathbf{c}$ are entries of C.
(2) The coordinates $c_0, \dots, c_{19}$ of the vector $\mathbf{c}$ of (1) are the twenty free entries of Construction-Proposition 8.6 in the case $O = 0$.
(3) Eighteen entries of X are zero, namely, $X(0, i') = 0$ for $i' \in \{0, 1, 4, 6, 8, 11\}$, $X(1, i') = 0$ for $i' \in \{0, 1, 5, 9\}$, $X(2, i') = 0$ for $i' \in \{2, 4, 6, 10\}$, and $X(3, i') = 0$ for $i' \in \{3, 7, 8, 11\}$.
(4) $L\mathbf{c} = 0$ if and only if the following equations hold, where we write next to each equation the entries of X and Y, up to the sign written, equal to the left side:

$$Fc_4 + dc_{11} + Ec_{12} - hc_3 = 0 \quad (X(0,2), X(1,6), -X(2,0), -X(2,5)) \tag{9.6}$$

$$Ic_5 + dc_{16} + Ec_{17} - jc_3 = 0 \quad (X(0,3), X(1,11), -X(3,0), -X(3,9)) \tag{9.7}$$

$$bc_0 + Ec_1 - Ec_2 - 2cc_4 - lc_6 = 0 \quad (X(0,5), -X(1,4)) \tag{9.8}$$

$$-kc_0 - Ic_1 + Ec_{15} - lc_{18} = 0 \quad (X(0,7), -X(3,4)) \tag{9.9}$$

$$-ac_0 - dc_1 + dc_2 + 2dc_3 - 2cc_5 - lc_7 = 0 \quad (X(0,9), -X(1,8)) \tag{9.10}$$

$$Gc_0 + Fc_1 - Fc_3 + dc_{10} - lc_{13} = 0 \quad (X(0,10), -X(2,8)) \tag{9.11}$$

$$Gc_4 - Fc_6 + hc_9 + ac_{11} + bc_{12} - Ec_{13} = 0 \quad (X(1,2), -X(2,1)) \tag{9.12}$$

$$kc_5 - Ic_7 + jc_9 + ac_{16} + bc_{17} - dc_{18} = 0 \quad (X(1,3), -X(3,1)) \tag{9.13}$$

$$-kc_2 - jc_3 + Ic_5 - Ic_8 + bc_{15}$$

$$+dc_{16} + Ec_{17} - 2cc_{18} + Ec_{19} = 0 \quad (X(1,7), -X(3,5)) \tag{9.14}$$

$$
\begin{aligned}
&Gc_2 + (G-h)c_3 + Fc_4 + Fc_8 + Fc_9 \\
&+ac_{10} + dc_{11} + Ec_{12} - 2cc_{13} + dc_{14} = 0 \quad (X(1,10), -X(2,9)) \qquad (9.15)\\
&kc_{12} - Ic_{13} + jc_{14} - Gc_{16} + Fc_{18} - hc_{19} = 0 \quad (X(2,3), -X(3,2)) \qquad (9.16)\\
&-jc_{10} + Ic_{12} + hc_{15} - Fc_{16} = 0 \quad (X(2,7), X(2,11), -X(3,6), -X(3,10)) \qquad (9.17)\\
&-bc_0 - Ec_2 - Ec_3 - Fc_{11} - Ic_{16} = 0 \quad (Y(0,0)) \qquad (9.18)\\
&ac_0 + dc_2 - Fc_{12} - Ic_{17} = 0 \quad (Y(0,1)) \qquad (9.19)\\
&ac_4 + bc_5 - dc_6 - Ec_7 = 0 \quad (Y(0,2)) \qquad (9.20)\\
&-bc_1 + bc_3 - Ec_8 - Ec_9 \\
&+Gc_{11} + kc_{16} + Ic_{18} = 0 \quad (Y(0,3)) \qquad (9.21)\\
&ac_1 + dc_8 + Gc_{12} + Fc_{13} + kc_{17} = 0 \quad (Y(0,4)) \qquad (9.22)\\
&Gc_{10} + Fc_{14} + kc_{15} + Ic_{19} = 0 \quad (Y(0,5)) \qquad (9.23)\\
&hc_{10} - Fc_{11} = 0 \quad (Y(1,0)) \qquad (9.24)\\
&ac_0 + dc_2 - Fc_{12} - jc_{15} = 0 \quad (Y(1,1)) \qquad (9.25)\\
&ac_4 - dc_6 + hc_{13} - jc_{18} = 0 \quad (Y(1,2)) \qquad (9.26)\\
&Gc_{11} + hc_{14} = 0 \quad (Y(1,3)) \qquad (9.27)\\
&ac_1 + dc_8 + Gc_{12} + Fc_{13} - jc_{19} = 0 \quad (Y(1,4)) \qquad (9.28)\\
&Gc_{10} + Fc_{14} = 0 \quad (Y(1,5)) \qquad (9.29)\\
&-Ec_0 + lc_4 = 0 \quad (Y(2,0)) \qquad (9.30)\\
&dc_0 + lc_5 = 0 \quad (Y(2,1)) \qquad (9.31)\\
&dc_4 + Ec_5 = 0 \quad (Y(2,2)) \qquad (9.32)\\
&-Ec_1 + Ec_3 + 2cc_4 - lc_6 + Fc_{11} + Ic_{16} = 0 \quad (Y(2,3)) \qquad (9.33)\\
&dc_1 + 2cc_5 - lc_7 + Fc_{12} + Ic_{17} = 0 \quad (Y(2,4)) \qquad (9.34)\\
&2cc_3 - lc_9 + Fc_{10} + Ic_{15} = 0 \quad (Y(2,5)) \qquad (9.35)
\end{aligned}
$$

(5) For $u \in \mathbb{P}^1$, a vector $\mathbf{c} \in \mathbb{C}^{20}$ belongs to $\operatorname{Ker} L(u)$ if and only if (9.6)–(9.35) hold with $a, \dots, l$ replaced by the values $a(u), \dots, l(u)$.

(6) For any vector $\mathbf{c}$,

$$X(0,9) - Y(2,4) - Y(0,1) = -2\big(ac_0 + 2cc_5 + d(c_1 - c_3)\big). \tag{9.36}$$

(7) $L\xi = L\eta = 0$.

Proof. The equations $B_1 N^t + C B_3^t = 0$, $AN - CP^t = 0$, and $UN^t = 0$ are identities of matrices over S_0, so we may compare the coefficients of $y_0, \dots, y_3$.

We prove (1) and (2). By Construction-Proposition 8.2, with $(b_1, b_2, b_3) = (z, p, q)$ and $y_* = y_0 + y_2 + y_3$, the three blocks of B_1 are $y_i I_4 - y_* M_i$ for $i = 1, 2, 3$. Let N_1, N_2, N_3 be the 4×6 blocks of N^t, consisting of rows 0 to 3, rows 4 to 7, and rows 8 to 11 of N^t, and let $B_3^{(i)}$ be the coefficient of y_i in B_3 (Table 10). Comparing the coefficients of y_0, y_1, y_2, y_3 in $B_1 N^t + C B_3^t = 0$, we obtain

$$N_1 = -C\big(B_3^{(1)}\big)^t, \quad N_2 = C\big(B_3^{(0)} - B_3^{(2)}\big)^t, \quad N_3 = C\big(B_3^{(0)} - B_3^{(3)}\big)^t,$$

and

$$M_1 N_1 + M_2 N_2 + M_3 N_3 = C\big(B_3^{(0)}\big)^t.$$

Substituting the first three equations into the last one, a direct calculation shows that the last equation is equivalent to the twelve equations

$$C(0,1) = C(0,2) = C(0,3) = C(2,0) = C(2,4) = C(2,6) = C(3,5) = 0,$$
$$C(0,4) + C(1,2) = C(0,5) + C(1,3) = C(0,7) + C(1,1) = 0,$$

$$C(3,0)+C(3,1)=C(3,6)+C(3,7)=0.$$

All coefficients in the twelve equations are constants, so no division occurs, and the solutions are the matrices $C=C(\mathbf{c})$, where

$$(c_0,c_1)=(C(0,0),C(0,6)),\quad (c_2,\dots,c_9)=(C(1,0),\dots,C(1,7)),$$
$$(c_{10},\dots,c_{14})=(C(2,1),C(2,2),C(2,3),C(2,5),C(2,7)),$$
$$(c_{15},\dots,c_{19})=(C(3,1),C(3,2),C(3,3),C(3,4),C(3,7)).$$

Then the formulas for N_1,N_2,N_3 imply $N=N(\mathbf{c})$. At $O=0$, the equation $AO-B_1N^t-CB_3^t=0$ of Construction-Proposition 8.6 is $B_1N^t+CB_3^t=0$, so $c_0,\dots,c_{19}$ are the twenty free entries there.

Parts (3) and (4) are direct calculations from $C(\mathbf{c})$, $N(\mathbf{c})$, and the matrices P,U of Table 10. Part (5) follows from (4), since $L(u)$ is the value of L at u. Part (6) follows from (9.10), (9.34), and (9.19), whose left sides are $X(0,9)$, $Y(2,4)$, and $Y(0,1)$.

We prove (7). Substituting ξ and η into the left sides of (9.6)–(9.35), a direct calculation shows that any of the 60 polynomials is zero modulo $Q_1,\dots,Q_4$, and the reduction uses only

$$Gh+kj=aE+bd,\quad F^2=al+2cd,\quad I^2=bl+2cE,\quad Fh+Ij=dE.$$

For example, (9.30) and (9.31) are immediate for ξ, since $-E\xi_0+l\xi_4=ldE^2-ldE^2$ and $d\xi_0+l\xi_5=-ld^2E+ld^2E$, and for η, since $\eta_0=\eta_4=\eta_5=0$. For (9.35), the vector ξ satisfies

$$2c\xi_3-l\xi_9+F\xi_{10}+I\xi_{15}=2cl(bd-aE)-2cl(bd-aE)+F^2I^2-F^2I^2=0,$$

and the vector η satisfies

$$\begin{aligned}2c\eta_3-l\eta_9+F\eta_{10}+I\eta_{15}&=l^2(Gh+kj)+4cldE-F^2El-I^2dl\\&=l\big(l(aE+bd)+4cdE-(al+2cd)E-(bl+2cE)d\big)=0.\end{aligned}$$

□

Lemma 9.6. *Notation and conditions as in Set-up 9.4. Then d,E,h,j are non-zero forms. If column 6 of $A(u)$ is zero for some $u\in\mathbb{P}^1$, then $d(u)=E(u)=0$, $l(u)F(u)I(u)\neq 0$, and columns $4,5$ of A are nowhere zero. Moreover, at most two points of $\mathbb{P}^1$ have a zero column of A, and if two points of $\mathbb{P}^1$ have a zero column of A, then one point is a zero of column 4 and the other point is a zero of column 5.*

Proof. The columns of A are

$$\begin{pmatrix}-d\\-a\\G\\0\end{pmatrix},\ \begin{pmatrix}-E\\-b\\0\\k\end{pmatrix},\ \begin{pmatrix}l\\2c\\0\\0\end{pmatrix},\ \begin{pmatrix}0\\-d\\F\\0\end{pmatrix},\ \begin{pmatrix}0\\-E\\0\\I\end{pmatrix},\ \begin{pmatrix}0\\0\\-h\\-j\end{pmatrix}.$$

Suppose that column 6 of $A(u)$ is zero for some $u\in\mathbb{P}^1$. Then $h(u)=j(u)=0$, and $Q_4=0$ implies $d(u)E(u)=0$. If $d(u)=0$ and $E(u)\neq 0$, then $Q_1=0$ implies $a(u)=0$, and $Q_2=0$ implies $F(u)=0$, so column 4 of $A(u)$ is zero, which contradicts Set-up 9.4(4). If $E(u)=0$ and $d(u)\neq 0$, then $Q_1=0$ implies $b(u)=0$, and $Q_3=0$ implies $I(u)=0$, so column 5 of $A(u)$ is zero, which contradicts Set-up 9.4(4). Thus $d(u)=E(u)=0$. By Set-up 9.4(2), $l(u)\neq 0$. Since columns $4,5$ of $A(u)$ are non-zero, $F(u)\neq 0$ and $I(u)\neq 0$.

We prove that $h\neq 0$ and $j\neq 0$. If $h=j=0$, then column 6 of A is zero, which contradicts Set-up 9.4(3). Suppose that $h=0$ and $j\neq 0$, and let u be the zero of j. Column 6 of $A(u)$ is zero, so $d(u)=E(u)=0$ and $I(u)\neq 0$. Thus $d=d_jj$ and $E=E_jj$, and $Q_4=0$ reads $d_jE_jj^2=jI$. Hence $I=d_jE_jj$, which contradicts $I(u)\neq 0$. Suppose that $j=0$ and $h\neq 0$, and let u be the zero of h. Column 6 of $A(u)$ is zero, so $d=d_hh$, $E=E_hh$, and $F(u)\neq 0$. Now $Q_4=0$ implies $F=d_hE_hh$, which contradicts $F(u)\neq 0$.

We prove that $d\neq 0$ and $E\neq 0$. Suppose that $d=0$, and let u be the zero of l. Now $Q_2=0$ reads $al=F^2$, so $F(u)=0$, and l divides F and a. Thus $a(u)=0$. Set-up 9.4(2) implies $c(u)\neq 0$

and $E(u) \neq 0$, and $Q_3 = 0$ implies $I(u)^2 = 2c(u)E(u) \neq 0$. Then $Q_4 = 0$ implies $j(u) = 0$. Since column 1 of $A(u)$ is non-zero, $G(u) \neq 0$, and $Q_1 = 0$ implies $h(u) = 0$. Hence columns $4, 6$ of $A(u)$ are zero, which contradicts Set-up 9.4(4). Suppose that $E = 0$, and let u be the zero of l. Now $Q_3 = 0$ reads $bl = I^2$, so $I(u) = 0$, and l divides I and b. Thus $b(u) = 0$. Set-up 9.4(2) implies $c(u) \neq 0$ and $d(u) \neq 0$, and $Q_2 = 0$ implies $F(u)^2 = 2c(u)d(u) \neq 0$. Then $Q_4 = 0$ implies $h(u) = 0$. Since column 2 of $A(u)$ is non-zero, $k(u) \neq 0$, and $Q_1 = 0$ implies $j(u) = 0$. Hence columns $5, 6$ of $A(u)$ are zero, which contradicts Set-up 9.4(4).

Suppose again that column 6 of $A(u)$ is zero. Since d, E are non-zero forms with $d(u) = E(u) = 0$, the point u is the unique zero of d and the unique zero of E. Since $F(u)I(u) \neq 0$, columns $4, 5$ of A are nowhere zero. Thus, if column 6 has a zero, then exactly one point of $\mathbb{P}^1$ has a zero column of A.

By Set-up 9.4(3), any column of A has at most one zero, and by Set-up 9.4(4), columns $1, 2, 3$ have no zero. If column 6 has no zero, then only columns $4, 5$ have zeros. Hence at most two points of $\mathbb{P}^1$ have a zero column of A, and if two such points exist, then one point is a zero of column 4 and the other point is a zero of column 5. □

Lemma 9.7. *Notation and conditions as in Set-up 9.4 and Construction-Proposition 9.5. Then for a general $u \in \mathbb{P}^1$, the vectors $\xi(u), \eta(u)$ are linearly independent and span* $\operatorname{Ker} L(u)$.

Proof. By Lemma 9.6 and $l \neq 0$, the forms l, d, E, h, j are non-zero. Suppose that $F = I = 0$. Then $Q_2 = Q_3 = 0$ imply that l divides cd and cE. Since c, l are linearly independent, l divides d and E, which contradicts Set-up 9.4(2). Thus the forms F, I are not both zero.

Let

$$\Phi_1 := (2cd - al)F + Gdl, \quad \Phi_2 := (2cE - bl)I + kEl,$$

$$\Psi_1 := FEj + Ihd, \quad \Psi_2 := l(IG - Fk)j + FI^2d, \quad \Psi_3 := lGEj + dFIE + dhkl.$$

Suppose that $F \neq 0$ and $I \neq 0$. We claim that Ψ_1, Φ_1, Φ_2 are not all zero. Suppose that $\Psi_1 = \Phi_1 = \Phi_2 = 0$. Then for a general point $v \in \mathbb{P}^1$, the row vector

$$\left(-\frac{2c(v)}{l(v)},\ 1,\ \frac{d(v)}{F(v)},\ \frac{E(v)}{I(v)}\right)$$

is non-zero, and the product of the row vector with $A(v)$ is

$$\left(\frac{\Phi_1(v)}{l(v)F(v)},\ \frac{\Phi_2(v)}{l(v)I(v)},\ 0,\ 0,\ 0,\ -\frac{\Psi_1(v)}{F(v)I(v)}\right) = 0,$$

which contradicts the generic rank four of A.

Let u be a general point of $\mathbb{P}^1$. Then $l(u), d(u), E(u), h(u), j(u)$ are non-zero, $F(u) \neq 0$ if $F \neq 0$, $I(u) \neq 0$ if $I \neq 0$, and, if $F \neq 0$ and $I \neq 0$, one of $\Psi_1(u), \Phi_1(u), \Phi_2(u)$ is non-zero. Let $\mathbf{c} \in \operatorname{Ker} L(u)$. In this proof the coordinates $c_0, \dots, c_{19}$ are scalars, and in any equation that contains a coordinate c_i, the letters $a, \dots, l$ stand for the values $a(u), \dots, l(u)$. By Construction-Proposition 9.5(5), the coordinates of $\mathbf{c}$ satisfy (9.6)–(9.35), and by Construction-Proposition 9.5(7), $\xi(u), \eta(u) \in \operatorname{Ker} L(u)$. We prove that $\mathbf{c}$ is a linear combination of $\xi(u), \eta(u)$. Since F, I are not both zero, the following three cases are exhaustive.

Case 1. Suppose that $F \neq 0$ and $I \neq 0$. We have $\xi_0(u) = -l(u)d(u)E(u) \neq 0$, $\eta_0 = 0$, and $\eta_{10}(u) = -F(u)E(u)l(u) \neq 0$, so $\xi(u), \eta(u)$ are linearly independent. Possibly replacing $\mathbf{c}$ with a vector $\mathbf{c} - \alpha\xi(u) - \beta\eta(u)$ for some $\alpha, \beta \in \mathbb{C}$, we may assume that $c_0 = c_{10} = 0$. We prove that $\mathbf{c} = 0$.

By (9.30), (9.31), (9.24), and (9.29), we have $c_4 = c_5 = c_{11} = c_{14} = 0$. Then (9.6), (9.7), (9.18), and (9.19) read

$$Ec_{12} = hc_3, \quad dc_{16} + Ec_{17} = jc_3, \quad Ic_{16} = -E(c_2 + c_3), \quad Fc_{12} + Ic_{17} = dc_2.$$

Multiplying the second equation by I and substituting the other three equations, we obtain

$$Ijc_3 = d(Ic_{16}) + E(Ic_{17}) = -dE(c_2 + c_3) + E(dc_2 - Fc_{12}) = -(dE + Fh)c_3.$$

Thus $(dE+Fh+Ij)c_3=0$, and $Q_4=0$ implies $2dEc_3=0$. Hence $c_3=0$, so

$$c_{12}=0,\quad c_{16}=-\frac{Ec_2}{I},\quad c_{17}=\frac{dc_2}{I}.$$

Since $c_0=c_5=0$, (9.36) and $L(u)\mathbf{c}=0$ imply $d(c_1-c_3)=0$, so $c_1=0$. Then (9.11) implies $c_{13}=0$. The equations (9.25), (9.8), (9.34), (9.35), (9.9), (9.23), and (9.28), namely the entries $Y(1,1)$, $X(0,5)$, $Y(2,4)$, $Y(2,5)$, $X(0,7)$, $Y(0,5)$, and $Y(1,4)$, imply respectively

$$c_2=\frac{jc_{15}}{d},\quad c_6=-\frac{Ec_2}{l},\quad c_7=\frac{dc_2}{l},\quad c_9=\frac{Ic_{15}}{l},$$
$$c_{18}=\frac{Ec_{15}}{l},\quad c_{19}=-\frac{kc_{15}}{I},\quad c_8=\frac{jc_{19}}{d}=-\frac{kc_2}{I}.$$

So any coordinate of $\mathbf{c}$ is a multiple of c_{15}, and it remains to prove that $c_{15}=0$.

After substitution of the zeros, (9.17) reads $-Fc_{16}+hc_{15}=0$, (9.15), which is the entry $X(1,10)$, reads $F(c_8+c_9)+Gc_2=0$, and (9.16), which is the entry $X(2,3)$, reads $Fc_{18}-Gc_{16}-hc_{19}=0$. Substituting $c_2=jc_{15}/d$, $c_8=-kc_2/I$, $c_9=Ic_{15}/l$, $c_{16}=-Ec_2/I$, $c_{18}=Ec_{15}/l$, and $c_{19}=-kc_{15}/I$, and multiplying respectively by dI, by dIl, and by dIl, we obtain

$$\Psi_1(u)c_{15}=\Psi_2(u)c_{15}=\Psi_3(u)c_{15}=0.$$

A direct calculation shows the identities

$$\Psi_2-E\Phi_1=lFQ_1-FdQ_3-lGQ_4,\quad F\Psi_3-lG\Psi_1-d^2\Phi_2=dlIQ_1-dIEQ_2-dlkQ_4,$$

so $\Psi_2=E\Phi_1$ and $F\Psi_3=lG\Psi_1+d^2\Phi_2$.

If $c_{15}\neq 0$, then $\Psi_1(u)=\Psi_2(u)=\Psi_3(u)=0$, and the identities $\Psi_2=E\Phi_1$ and $F\Psi_3=lG\Psi_1+d^2\Phi_2$, with $E(u)F(u)\neq 0$ and $d(u)\neq 0$, imply $\Phi_1(u)=\Phi_2(u)=0$, which contradicts the choice of u. Hence $c_{15}=0$ and $\mathbf{c}=0$.

Case 2. Suppose that $F\neq 0$ and $I=0$. As in Case 1, $\xi_0(u)\neq 0$, $\eta_0=0$, and $\eta_{10}(u)\neq 0$, so $\xi(u),\eta(u)$ are linearly independent, and we may assume that $c_0=c_{10}=0$ and prove that $\mathbf{c}=0$. As in Case 1, $c_4=c_5=c_{11}=c_{14}=0$. Since $I=0$, (9.18) implies $c_2=-c_3$, and (9.19) and (9.6) read

$$Fc_{12}=-dc_3,\quad Ec_{12}=hc_3.$$

Thus $(Fh+dE)c_3=0$. Since $Fh=dE$ by $Q_4=0$, we obtain $2dEc_3=0$. Hence $c_2=c_3=c_{12}=0$. By (9.25), $jc_{15}=0$, so $c_{15}=0$. By (9.17), $c_{16}=0$. By (9.7), $c_{17}=0$. By (9.35), $c_9=0$. Since $c_0=c_5=0$, (9.36) and $L(u)\mathbf{c}=0$ imply $d(c_1-c_3)=0$, so $c_1=0$. Then (9.8), (9.11), (9.34), (9.15), (9.9), and (9.28) imply respectively

$$c_6=c_{13}=c_7=c_8=c_{18}=c_{19}=0.$$

Case 3. Suppose that $F=0$ and $I\neq 0$. We have $\xi_0(u)\neq 0$, $\eta_0=0$, $\xi_{15}=0$, and $\eta_{15}(u)=-I(u)d(u)l(u)\neq 0$, so $\xi(u),\eta(u)$ are linearly independent, and we may assume that $c_0=c_{15}=0$. We prove that $\mathbf{c}=0$. By (9.30) and (9.31), $c_4=c_5=0$. By (9.24), $hc_{10}=0$, so $c_{10}=0$. By (9.23), (9.25), and (9.19), we obtain $c_{19}=c_2=c_{17}=0$. Then (9.7) and (9.18) read

$$dc_{16}=jc_3,\quad Ic_{16}=-Ec_3.$$

Thus $(dE+Ij)c_3=0$. Since $Ij=dE$ by $Q_4=0$, we obtain $2dEc_3=0$. Hence $c_3=c_{16}=0$. By (9.35), (9.17), and (9.6), we obtain $c_9=c_{12}=c_{11}=0$. Since $c_0=c_5=0$, (9.36) and $L(u)\mathbf{c}=0$ imply $d(c_1-c_3)=0$, so $c_1=0$. Then (9.11), (9.9), (9.8), (9.10), (9.15), and (9.28) imply respectively

$$c_{13}=c_{18}=c_6=c_7=c_{14}=c_8=0.$$

□

Lemma 9.8. *Notation and conditions as in Set-up 9.4 and Construction-Proposition 9.5. Let $u\in\mathbb{P}^1$. Then:*

(1) If no column of $A(u)$ is zero, then $\xi(u),\eta(u)$ are linearly independent.

(2) *If some column of $A(u)$ is zero, and ϱ is a non-zero linear form vanishing at u, then one of the following holds.*
   (a) *ϱ divides any coordinate of ξ, and $(\xi/\varrho)(u), \eta(u)$ are linearly independent.*
   (b) *ϱ divides any coordinate of η, and $\xi(u), (\eta/\varrho)(u)$ are linearly independent.*

Proof. We prove (1). Assume that no column of $A(u)$ is zero.

Suppose that $l(u)d(u)E(u) \neq 0$. Then $\xi_0(u) \neq 0$ and $\eta_0 = 0$, so it suffices to prove that $\eta(u) \neq 0$. If $I(u) \neq 0$, then $\eta_{15}(u) = -I(u)d(u)l(u) \neq 0$. If $I(u) = 0$, then $Q_4 = 0$ implies $F(u)h(u) = d(u)E(u) \neq 0$, so $\eta_2(u) = l(u)d(u)E(u) \neq 0$.

Suppose that $d(u)E(u) \neq 0$ and $l(u) = 0$. By Set-up 9.4(2), $c(u) \neq 0$. By $Q_2 = Q_3 = 0$, $F(u)^2 = 2c(u)d(u) \neq 0$ and $I(u)^2 = 2c(u)E(u) \neq 0$. Then $\xi_{10}(u) = F(u)I(u)^2 \neq 0$, $\eta_{10}(u) = 0$, and $\eta_{13}(u) = -F(u)d(u)E(u) \neq 0$, so $\xi(u), \eta(u)$ are linearly independent.

Suppose that $d(u) = 0$ and $E(u) \neq 0$. Since column 4 of $A(u)$ is non-zero, $F(u) \neq 0$, and $Q_2 = 0$ implies $a(u)l(u) = F(u)^2 \neq 0$. Thus $\eta_{10}(u) = -F(u)E(u)l(u) \neq 0$. If $I(u) \neq 0$, then $\xi_{15}(u) = -F(u)^2I(u) \neq 0$ and $\eta_{15}(u) = 0$, so $\xi(u), \eta(u)$ are linearly independent. If $I(u) = 0$, then $Q_4 = 0$ implies $h(u) = 0$, and $Q_1 = 0$ implies $k(u)j(u) = a(u)E(u) \neq 0$. Thus $\xi_{17}(u) = -F(u)^2j(u) \neq 0$, and $\xi_{10}(u) = 0$ with $\eta_{10}(u) \neq 0$, so $\xi(u), \eta(u)$ are linearly independent.

Suppose that $E(u) = 0$ and $d(u) \neq 0$. Since column 5 of $A(u)$ is non-zero, $I(u) \neq 0$, and $Q_3 = 0$ implies $b(u)l(u) = I(u)^2 \neq 0$. One of $F(u), h(u), G(u)$ is non-zero. Otherwise $Q_4 = 0$ implies $j(u) = 0$, and $Q_1 = 0$ implies $b(u)d(u) = 0$, a contradiction. Now

$$(\xi_{10}(u), \xi_{11}(u), \xi_{14}(u)) = I(u)^2(F(u), h(u), -G(u)) \neq 0, \quad \eta_{10}(u) = \eta_{11}(u) = \eta_{14}(u) = 0,$$

and $\eta_{15}(u) = -I(u)d(u)l(u) \neq 0$, so $\xi(u), \eta(u)$ are linearly independent.

Suppose that $d(u) = E(u) = 0$. By Set-up 9.4(2), $l(u) \neq 0$. Since columns $4, 5$ of $A(u)$ are non-zero, $F(u)I(u) \neq 0$. Since column 6 of $A(u)$ is non-zero and $h(u)F(u) + j(u)I(u) = 0$ by $Q_4 = 0$, we have $h(u)j(u) \neq 0$. Then $\xi_2(u) = 0$, $\eta_2(u) = 2l(u)F(u)h(u) \neq 0$, $\xi_{10}(u) = F(u)I(u)^2 \neq 0$, and $\eta_{10}(u) = 0$, so $\xi(u), \eta(u)$ are linearly independent. This proves (1).

We prove (2). By Set-up 9.4(4), exactly one of columns $4, 5, 6$ of $A(u)$ is zero. By Lemma 9.6, $l(u) \neq 0$ if column 6 of $A(u)$ is zero, so the following five cases are exhaustive. The linear independence in (a) and (b) does not change if ϱ is multiplied by a non-zero scalar.

Case 1. Suppose that column 4 of $A(u)$ is zero and $l(u) \neq 0$. Then $d(u) = F(u) = 0$, $Q_2 = 0$ implies $a(u) = 0$, and since column 1 of $A(u)$ is non-zero, $G(u) \neq 0$. If $I(u) \neq 0$, then $Q_4 = 0$ implies $j(u) = 0$, and $Q_1 = 0$ implies $h(u) = 0$, so column 6 of $A(u)$ is also zero, a contradiction. Hence $I(u) = 0$. Since column 5 of $A(u)$ is non-zero, $E(u) \neq 0$. The forms a, d, F, I vanish at u, so they are multiples of ϱ, and any coordinate of ξ belongs to the ideal generated by a, d, F, I. Thus ϱ divides any coordinate of ξ. Since $d \neq 0$ by Lemma 9.6, $d_\varrho \neq 0$, so $(\xi/\varrho)_0(u) = -l(u)d_\varrho E(u) \neq 0$. Moreover, $\eta_0 = 0$ and $\eta_{14}(u) = G(u)E(u)l(u) \neq 0$. Thus $(\xi/\varrho)(u), \eta(u)$ are linearly independent.

Case 2. Suppose that column 5 of $A(u)$ is zero and $l(u) \neq 0$. Then $E(u) = I(u) = 0$, $Q_3 = 0$ implies $b(u) = 0$, and since column 2 of $A(u)$ is non-zero, $k(u) \neq 0$. If $F(u) \neq 0$, then $Q_4 = 0$ implies $h(u) = 0$, and $Q_1 = 0$ implies $j(u) = 0$, so column 6 of $A(u)$ is also zero, a contradiction. Hence $F(u) = 0$. Since column 4 of $A(u)$ is non-zero, $d(u) \neq 0$. The forms b, E, F, I vanish at u, so they are multiples of ϱ, and any coordinate of ξ belongs to the ideal generated by b, E, F, I. Thus ϱ divides any coordinate of ξ. Since $E \neq 0$, $E_\varrho \neq 0$, so $(\xi/\varrho)_0(u) = -l(u)d(u)E_\varrho \neq 0$. If $h(u) = 0$, then $Q_1 = 0$ implies $j(u) = 0$, so column 6 of $A(u)$ is also zero, a contradiction. Hence $h(u) \neq 0$, so $\eta_{12}(u) = d(u)h(u)l(u) \neq 0$, while $\eta_0 = 0$. Thus $(\xi/\varrho)(u), \eta(u)$ are linearly independent.

Case 3. Suppose that column 4 of $A(u)$ is zero and $l(u) = 0$. We may take $\varrho = l$. Since $d(u) = F(u) = 0$, we have $d = d_l l$ and $F = F_l l$. By Set-up 9.4(2), $E(u)c(u) \neq 0$, and $Q_3 = 0$ implies $I(u)^2 = 2c(u)E(u) \neq 0$. Then $Q_4 = 0$ implies $j(u) = 0$, so $j = j_l l$, and $j_l \neq 0$ since $j \neq 0$. Since column 6 of $A(u)$ is non-zero, $h(u) \neq 0$. Dividing $Q_2 = 0$ and $Q_4 = 0$ by l, we obtain

$$a = F_l^2 l - 2d_l c, \quad d_l E = F_l h + j_l I.$$

The forms d, F, j, l are multiples of l, and any coordinate of η belongs to the ideal generated by d, F, j, l, so l divides any coordinate of η. We have $\xi_{11}(u) = h(u)I(u)^2 \neq 0$ and $(\eta/l)_{11}(u) = -E(u)h(u) \neq 0$. Suppose that $(\eta/l)(u) = \mu\xi(u)$ with $\mu \in \mathbb{C}$. Then

$$\mu = -\frac{E(u)}{I(u)^2} = -\frac{1}{2c(u)}.$$

Now

$$(\eta/l)_6 = -E(F_l h - j_l I) = -d_l E^2 + 2j_l EI,$$

and $\xi_6(u) = -a(u)E(u)^2 = 2d_l c(u)E(u)^2$, so $\mu\xi_6(u) = -d_l E(u)^2$. Comparing with the value of $(\eta/l)_6$ at u, we obtain $2j_l E(u)I(u) = 0$, a contradiction. Thus $\xi(u), (\eta/l)(u)$ are linearly independent.

Case 4. Suppose that column 5 of $A(u)$ is zero and $l(u) = 0$. We may take $\varrho = l$. Since $E(u) = I(u) = 0$, we have $E = E_l l$ and $I = I_l l$. By Set-up 9.4(2), $d(u)c(u) \neq 0$, and $Q_2 = 0$ implies $F(u)^2 = 2c(u)d(u) \neq 0$. Then $Q_4 = 0$ implies $h(u) = 0$, so $h = h_l l$, and $h_l \neq 0$ since $h \neq 0$. Since column 6 of $A(u)$ is non-zero, $j(u) \neq 0$. Dividing $Q_3 = 0$ and $Q_4 = 0$ by l, we obtain

$$b = I_l^2 l - 2E_l c, \quad dE_l = h_l F + I_l j.$$

The forms E, I, h, l are multiples of l, and any coordinate of η belongs to the ideal generated by E, I, h, l, so l divides any coordinate of η. We have $\xi_{17}(u) = -F(u)^2 j(u) \neq 0$ and $(\eta/l)_{17}(u) = -d(u)j(u) \neq 0$. Suppose that $(\eta/l)(u) = \mu\xi(u)$ with $\mu \in \mathbb{C}$. Then

$$\mu = \frac{d(u)}{F(u)^2} = \frac{1}{2c(u)}.$$

Now

$$(\eta/l)_7 = d(Fh_l - I_l j) = -E_l d^2 + 2h_l dF,$$

and $\xi_7(u) = b(u)d(u)^2 = -2E_l c(u)d(u)^2$, so $\mu\xi_7(u) = -E_l d(u)^2$. Comparing with the value of $(\eta/l)_7$ at u, we obtain $2h_l d(u)F(u) = 0$, a contradiction. Thus $\xi(u), (\eta/l)(u)$ are linearly independent.

Case 5. Suppose that column 6 of $A(u)$ is zero. By Lemma 9.6, $d(u) = E(u) = 0$ and $l(u)F(u)I(u) \neq 0$, and d, E, h, j are non-zero forms. Thus $d = d_\varrho \varrho$, $E = E_\varrho \varrho$, $h = h_\varrho \varrho$, and $j = j_\varrho \varrho$ with $h_\varrho j_\varrho \neq 0$. Dividing $Q_4 = 0$ by ϱ and taking the value at u, we obtain $h_\varrho F(u) + j_\varrho I(u) = 0$. Any coordinate of η belongs to the ideal generated by d, E, h, j, so ϱ divides any coordinate of η, and

$$(\eta/\varrho)_2(u) = l(u)\big(F(u)h_\varrho - I(u)j_\varrho\big) = 2l(u)F(u)h_\varrho \neq 0.$$

Moreover, $\xi_2(u) = a(u)E(u)l(u) = 0$ and $\xi_{10}(u) = F(u)I(u)^2 \neq 0$. Thus $\xi(u), (\eta/\varrho)(u)$ are linearly independent. □

Any point with a zero column contributes exactly one dimension to the solutions with $O = 0$, and at most two such points exist. Since $B_1 B_4 = 0$ and $B_3 B_2^t = 0$ by Construction-Proposition 8.2, the relation $d_1' d_2 = 0$ of Set-up 4.1 at $O = 0$ implies the three equations of the next result.

Construction-Proposition 9.9. Notation and conditions as in Set-up 9.4. Let $\mathcal{V}_A$ be the vector space of pairs (C, N) of matrices of binary quadratic forms of the sizes in Set-up 4.1 with

$$B_1 N^t + C B_3^t = 0, \quad AN - CP^t = 0, \quad UN^t = 0,$$

and let $e(A)$ be the number of points of $\mathbb{P}^1$ at which some column of A is zero. Then $e(A) \leq 2$ and $\dim \mathcal{V}_A = e(A)$.

Proof. By Lemma 9.6, $e(A) \leq 2$. From now on, the coordinates $c_0, \ldots, c_{19}$ are binary quadratic forms. By Construction-Proposition 9.5(1), the map $(C, N) \to \mathbf{c}$ is an isomorphism from $\mathcal{V}_A$ onto the vector space of vectors $\mathbf{c}$ of binary quadratic forms with $L\mathbf{c} = 0$.

Let $\mathcal{K}$ be the kernel of the map of sheaves

$$\mathcal{O}(2)^{\oplus 20} \to \mathcal{O}(3)^{\oplus 66}$$

on $\mathbb{P}^1$ defined by L. Since the binary forms of degree n are the sections of $\mathcal{O}(n)$ [Har77, Theorem III.5.1(a)], we have $\mathcal{V}_A \cong H^0(\mathbb{P}^1, \mathcal{K})$. Let

$$\psi : \mathcal{O}(-1)^{\oplus 2} \to \mathcal{O}(2)^{\oplus 20}$$

be the map sending (f, g) to $f\xi + g\eta$. By Construction-Proposition 9.5(7), the image $\mathcal{K}_0$ of ψ is contained in $\mathcal{K}$. By Lemma 9.7, ψ is injective, so $\mathcal{K}_0 \cong \mathcal{O}(-1)^{\oplus 2}$. Moreover, by Lemma 9.7, on a dense open subset of $\mathbb{P}^1$, the matrix $L(u)$ has constant rank 18, and $\operatorname{Ker} L(u)$ is spanned by $\xi(u), \eta(u)$. Thus $\mathcal{K}_0 = \mathcal{K}$ on a dense open subset of $\mathbb{P}^1$.

Let $u \in \mathbb{P}^1$. We trivialize all sheaves at u by powers of a linear form not vanishing at u, so that ξ, η are vectors over the local ring $\mathcal{O}_{\mathbb{P}^1,u}$, the stalk of $\mathcal{O}(2)^{\oplus 20}$ at u is the free module $\mathcal{O}_{\mathbb{P}^1,u}^{\oplus 20}$, and $(\mathcal{K}_0)_u$ is generated by ξ, η. Let ζ_1, ζ_2 be the following elements of $\mathcal{O}_{\mathbb{P}^1,u}^{\oplus 20}$. If no column of $A(u)$ is zero, then $(\zeta_1, \zeta_2) := (\xi, \eta)$. If some column of $A(u)$ is zero, let ϱ be a non-zero linear form vanishing at u, and let (ζ_1, ζ_2) be $(\xi/\varrho, \eta)$ or $(\xi, \eta/\varrho)$, according to Lemma 9.8(2).

By Lemma 9.8, the values of ζ_1, ζ_2 at u are linearly independent, so, by Nakayama's lemma, the submodule $\mathcal{T}_u$ generated by ζ_1, ζ_2 is a direct summand of $\mathcal{O}_{\mathbb{P}^1,u}^{\oplus 20}$. Since $L\xi = L\eta = 0$ and ϱ is a non-zero-divisor, $\mathcal{T}_u \subset \mathcal{K}_u$. The quotient $\mathcal{K}_u/\mathcal{T}_u$ is a submodule of the free module $\mathcal{O}_{\mathbb{P}^1,u}^{\oplus 20}/\mathcal{T}_u$. Since $(\mathcal{K}_0)_u \subset \mathcal{T}_u$ and $\mathcal{K}_0 = \mathcal{K}$ on a dense open subset of $\mathbb{P}^1$, the quotient $\mathcal{K}_u/\mathcal{T}_u$ is a torsion module. Thus $\mathcal{K}_u = \mathcal{T}_u$.

If no column of $A(u)$ is zero, then $\mathcal{K}_u = (\mathcal{K}_0)_u$. If some column of $A(u)$ is zero, then

$$\mathcal{K}_u/(\mathcal{K}_0)_u \cong \mathcal{O}_{\mathbb{P}^1,u}/\mathfrak{m}_u,$$

since ϱ generates $\mathfrak{m}_u$. Hence we have an exact sequence

$$0 \to \mathcal{O}(-1)^{\oplus 2} \to \mathcal{K} \to \bigoplus_u \mathcal{O}_{\mathbb{P}^1,u}/\mathfrak{m}_u \to 0,$$

where u runs over the $e(A)$ points of $\mathbb{P}^1$ at which some column of A is zero, and the last term is the direct sum of the skyscraper sheaves at u. Since $H^i(\mathbb{P}^1, \mathcal{O}(-1)) = 0$ for $i = 0, 1$ [Har77, Theorem III.5.1(a), (d)], the long exact sequence of cohomology implies that $\dim \mathcal{V}_A = h^0(\mathbb{P}^1, \mathcal{K}) = e(A)$. □

9.4. **The lower system modulo gauge.** In this subsection, we prove that $\dim \mathcal{U}_A \le 4$ for the matrix A of any tuple in $\mathcal{P}^{\mathrm{good}}$.

Construction-Proposition 9.10. Notation and conditions as in Set-up 9.4. Let (O, N, C) be a triple of the block shapes in Set-up 4.1 with

$$AO - B_1 N^t - C B_3^t = 0, \quad AN - CP^t = 0, \quad UN^t = 0.$$

Let $X := AN - CP^t$ and $Y := UN^t$, with entries $X(i, i')$ and $Y(i, i')$. Assume that O is supported on the twelve slots

$$(0,1,0), (0,2,0), (0,3,2), (0,5,2), (1,2,0), (1,3,0),$$
$$(1,4,3), (1,5,3), (2,3,0), (2,4,0), (3,5,2), (4,5,3),$$

where the slot (i, i', m) is the coefficient of y_m in $O_{ii'}$, that is,

$$O = \begin{pmatrix} 0 & o_0 y_0 & o_1 y_0 & o_2 y_2 & 0 & o_3 y_2 \\ -o_0 y_0 & 0 & o_4 y_0 & o_5 y_0 & o_6 y_3 & o_7 y_3 \\ -o_1 y_0 & -o_4 y_0 & 0 & o_8 y_0 & o_9 y_0 & 0 \\ -o_2 y_2 & -o_5 y_0 & -o_8 y_0 & 0 & 0 & o_{10} y_2 \\ 0 & -o_6 y_3 & -o_9 y_0 & 0 & 0 & o_{11} y_3 \\ -o_3 y_2 & -o_7 y_3 & 0 & -o_{10} y_2 & -o_{11} y_3 & 0 \end{pmatrix}$$

with binary linear forms $o_0, \dots, o_{11}$. Then:

(1) There exist binary quadratic forms $c_0, \dots, c_{19}$ with

$$C = \begin{pmatrix} c_0 & 0 & Eo_5 - lo_8 & -lo_9 & bo_5 - 2co_8 - c_4 & -2co_9 - c_5 & c_1 & -c_3 \\ c_2 & c_3 & c_4 & c_5 & c_6 & c_7 & c_8 & c_9 \\ 0 & c_{10} & c_{11} & c_{12} & 0 & c_{13} & 0 & c_{14} \\ -c_{15} & c_{15} & c_{16} & c_{17} & c_{18} & 0 & -c_{19} & c_{19} \end{pmatrix},$$

and the rows of N, from top to bottom, are

$$\begin{aligned}
&(-Eo_5 + lo_8, -c_4, -c_{11}, -c_{16}, 0, do_2 - Eo_5 + lo_8, 0, ko_0 + jo_3,\\
&\quad -c_0, -Eo_5 + lo_8 - c_2 - c_3, -c_{10}, ko_0),\\
&(lo_9, -c_5, -c_{12}, -c_{17}, c_0, lo_9 + c_2, Fo_5 - Go_0, -c_{15},\\
&\quad 0, Eo_6 + lo_9, Fo_5 - Go_0 + ho_7, 0),\\
&(0, 0, 0, 0, -bo_5 + 2co_8 + c_4, -c_6, Fo_8 - Go_1, Io_9 - ko_4 - c_{18},\\
&\quad 2co_9 + c_5, -c_7, Fo_8 - Go_1 - c_{13}, Io_9 - ko_4),\\
&(bo_5 - 2co_8 - c_4, c_6, 0, c_{18}, -do_2, -ao_2 + bo_5 - 2co_8 - c_4, -c_{11},\\
&\quad -ko_5 + jo_{10} - c_{16}, -c_1 + c_3, bo_5 - 2co_8 - c_4 - c_8 - c_9, -c_{11} - c_{14}, -ko_5 - c_{16}),\\
&(-2co_9 - c_5, c_7, c_{13}, 0, c_1, -2co_9 - c_5 + c_8, -c_{12}, -c_{17} - c_{19},\\
&\quad -Eo_6, -bo_6 - 2co_9 - c_5, ho_{11} - c_{12}, -c_{17}),\\
&(-c_3, c_9, c_{14}, c_{19}, -do_3, -ao_3 - do_{10} - c_3, -c_{10}, -c_{15},\\
&\quad -Eo_7, -bo_7 - Eo_{11} - c_3, -c_{10}, -c_{15}).
\end{aligned}$$

(2) The forms $o_0, \dots, o_{11}$ satisfy

$$\begin{gathered}
-Eo_0 + lo_1 = do_0 + lo_4 = do_1 + Eo_4 = 0,\\
-bo_0 + 2co_1 + Eo_5 - lo_8 = ao_0 + 2co_4 - do_5 - lo_9 = ao_1 + bo_4 - do_8 - Eo_9 = 0,\\
Fo_2 - ho_3 = Go_2 + ho_{10} = Go_3 + Fo_{10} = 0,\\
Io_6 - jo_7 = ko_6 + jo_{11} = ko_7 + Io_{11} = 0.
\end{gathered} \tag{9.37}$$

(3) The following entries of X and Y, and the difference $Y(0,4) - Y(1,4)$, are used below:

$$\begin{aligned}
0 = X(0,2) &= Fc_4 + dc_{11} + Ec_{12} - hc_3 - Fbo_5 + 2Fco_8 + GEo_5 - Glo_8,\\
0 = X(0,3) &= Ic_5 + dc_{16} + Ec_{17} - jc_3 + 2Ico_9 - klo_9,\\
0 = X(0,7) &= -Ic_1 - kdo_0 - klo_4 - kc_0 - djo_3 + Ec_{15} - lc_{18},\\
0 = X(0,10) &= Fc_1 - Fc_3 + GEo_0 - Glo_1 + Gc_0 + dc_{10} - Eho_7 - lc_{13},\\
0 = X(1,2) &= -Fc_6 + Gc_4 + ac_{11} + bc_{12} - Ec_{13} + hc_9,\\
0 = X(1,3) &= -Ic_7 + kc_5 + ac_{16} + bc_{17} - dc_{18} + jc_9,\\
0 = Y(0,0) &= -Fc_{11} - Ic_{16} - k^2o_0 - bc_0 + d^2o_2 - Ec_2 - Ec_3,\\
0 = Y(0,1) &= -FGo_5 - Fc_{12} + G^2o_0 - Ic_{17} + ac_0 + dc_2 + E^2o_6,\\
0 = Y(1,0) &= -Fc_{11} + d^2o_2 + hc_{10} + j^2o_3,\\
0 = Y(1,1) &= -FGo_5 - Fc_{12} + G^2o_0 + ac_0 + dc_2 - h^2o_7 - jc_{15},\\
0 = Y(1,3) &= Gc_{11} - 2ado_2 + hc_{14} + j^2o_{10},\\
0 = Y(2,0) &= -Iko_0 + 2cEo_5 - 2clo_8 - Ec_0 + lc_4,\\
0 = Y(2,1) &= -F^2o_5 + FGo_0 - 2clo_9 + dc_0 + lc_5,\\
0 = Y(0,4) - Y(1,4) &= kc_{17} + jc_{19} - 2bEo_6 + h^2o_{11}.
\end{aligned}$$

(4) There exist unique constants κ, ν and unique rational functions ρ, σ of degree zero in s, t with

$$o_0 = \kappa l, \quad o_1 = \kappa E, \quad o_4 = -\kappa d, \quad o_5 = \nu l - 2\kappa c, \quad o_8 = \nu E - \kappa b, \quad o_9 = \kappa a - \nu d, \\ (o_2, o_3, o_{10}) = \rho(h, F, -G), \quad (o_6, o_7, o_{11}) = \sigma(j, I, -k). \tag{9.38}$$

Proof. Comparing the coefficients of $y_0, \dots, y_3$ in $AO = B_1 N^t + C B_3^t$ with the matrices of Table 10 implies (1) and (2). A direct calculation with (1) and the matrices A, P, U implies (3).

We prove (4). Since l, d, E have no common zero, one of d, E is non-zero at the zero of l, so the first two equations of (9.37) imply that o_0 vanishes at the zero of l. Thus $o_0 = \kappa l$ for a constant κ, and $o_1 = \kappa E$, $o_4 = -\kappa d$. The fourth and the fifth equations of (9.37) then read

$$E(o_5 + 2\kappa c) = l(o_8 + \kappa b), \qquad d(o_5 + 2\kappa c) = l(\kappa a - o_9).$$

Since l, d, E have no common zero, $o_5 + 2\kappa c = \nu l$ for a constant ν, and hence $o_8 = \nu E - \kappa b$ and $o_9 = \kappa a - \nu d$. By Lemma 9.6, $h, j \neq 0$. Let $\rho := o_2/h$ and $\sigma := o_6/j$. The last two lines of (9.37) imply $o_3 = \rho F$, $o_{10} = -\rho G$, $o_7 = \sigma I$, and $o_{11} = -\sigma k$. The uniqueness follows from $l, h, j \neq 0$. □

Lemma 9.11. *Notation and conditions as in Construction-Proposition 9.10. Then ρ and σ are constants.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we find the possible poles of ρ and σ.

Write $\rho = f_1/f_2$ with coprime binary forms f_1, f_2 and $\deg f_1 = \deg f_2$. Since $\rho h, \rho F, \rho G$ are the forms $o_2, o_3, -o_{10}$, f_2 divides h, F, G. Since $h \neq 0$, $\deg f_2 \leq 1$. Thus, if ρ is not a constant, then F, G, h have a common zero u and ρ has a simple pole at u and no other pole. Since $\sigma j, \sigma I, \sigma k$ are the forms $o_6, o_7, -o_{11}$ and $j \neq 0$, the argument for ρ implies that, if σ is not a constant, then k, I, j have a common zero u and σ has a simple pole at u and no other pole.

Step 2. Let $u \in \mathbb{P}^1$ be a point with $F(u) = G(u) = h(u) = 0$. In this step we prove that $l(u), d(u), E(u), I(u), j(u)$ are non-zero.

Suppose that $d(u) = 0$. Then column 4 of $A(u)$ is zero, so columns $1, 2, 3, 5, 6$ of $A(u)$ are non-zero by Set-up 9.4(4). Since column 6 of $A(u)$ is non-zero and $h(u) = 0$, $j(u) \neq 0$. By $Q_4(u) = 0$, $I(u) = 0$. Since column 5 of $A(u)$ is non-zero, $E(u) \neq 0$. Since column 1 of $A(u)$ is non-zero, $a(u) \neq 0$. By $Q_2(u) = 0$, $l(u) = 0$, and by $Q_3(u) = 0$, $c(u) = 0$. This contradicts the independence of c, l. Thus $d(u) \neq 0$.

Suppose that $j(u) = 0$. Then column 6 of $A(u)$ is zero, so columns $1, \dots, 5$ of $A(u)$ are non-zero by Set-up 9.4(4). By $Q_4(u) = 0$, $E(u) = 0$. Since column 5 of $A(u)$ is non-zero, $I(u) \neq 0$. By $Q_3(u) = 0$, $b(u)l(u) \neq 0$, and by $Q_1(u) = 0$, $b(u)d(u) = 0$, a contradiction. Thus $j(u) \neq 0$.

If $l(u) = 0$, then $Q_2(u) = 0$ implies $c(u)d(u) = 0$, while $c(u) \neq 0$ by the independence of c, l. Thus $l(u) \neq 0$.

Finally, $Q_4(u) = 0$ reads $d(u)E(u) = j(u)I(u)$, so $E(u), I(u)$ are both non-zero or both zero. If both are zero, then $Q_3(u) = 0$ implies $b(u) = 0$, and $Q_1(u) = 0$ implies $k(u)j(u) = 0$, so $k(u) = 0$ and column 2 of $A(u)$ is zero, which contradicts Set-up 9.4(4).

Step 3. Let u be as in **Step 2**, and suppose that ρ has a pole at u. In this step we determine the forms $a, \dots, l$.

All the equations of Construction-Proposition 9.10 are identities between binary forms, so after a linear change of s, t we may assume that $l = s$ and that t vanishes at u. We work on $s = 1$, and we write any first-stage form x as $x = \bar{x} + x't$ with constants $\bar{x}, x'$, and we denote by $x(0)$ the value of a form x at $t = 0$. Thus $l = 1$, $F = F't$, $G = G't$, $h = h't$, $c' \neq 0$ by the independence of c, l, and $\bar{d}, \bar{E}, \bar{I}, \bar{j}$ are non-zero by **Step 2**. By **Step 1**, $\rho = \rho_0 + \varpi/t$ with constants ρ_0 and $\varpi \neq 0$, so

$$o_2 = h'(\varpi + \rho_0 t), \quad o_3 = F'(\varpi + \rho_0 t), \quad o_{10} = -G'(\varpi + \rho_0 t).$$

The constant terms of Q_2, Q_3, Q_4, Q_1 imply

$$\bar{a} = -2\bar{c}\bar{d}, \quad \bar{b} = \bar{I}^2 - 2\bar{c}\bar{E}, \quad \bar{j} = \frac{\bar{d}\bar{E}}{\bar{I}}, \quad \bar{k} = \frac{\bar{I}(\bar{I}^2 - 4\bar{c}\bar{E})}{\bar{E}}.$$

At $t = 0$, the equation $Y(1,0) = 0$ reads $\bar{d}^2 o_2(0) + \bar{j}^2 o_3(0) = 0$, and the equation $Y(1,3) - 4\bar{c}Y(1,0) = 0$ reads $\bar{j}^2(o_{10}(0) - 4\bar{c}o_3(0)) = 0$. Since $\varpi \neq 0$,

$$h' = -\frac{\bar{E}^2}{\bar{I}^2}F', \qquad G' = -4\bar{c}F'. \tag{9.39}$$

Since $h \neq 0$, $F' \neq 0$.

Write $d = \bar{d}(1 + \delta_d t)$, $E = \bar{E}(1 + \delta_E t)$, and $I = \bar{I}(1 + \delta_I t)$, and let $\alpha := 2c'\bar{E}/\bar{I}^2 \neq 0$. The coefficients of t^2 in Q_2 and Q_3 imply

$$F'^2 = 2c'\bar{d}\delta_d, \qquad \delta_E = \frac{\delta_I^2}{\alpha}.$$

Let $\tilde{a} := a + 2\bar{c}d$, $\tilde{b} := b + 2\bar{c}E$, $\tilde{G} := G + 4\bar{c}F$, $\tilde{k} := k + 4\bar{c}I$, and $\tilde{c} := c - \bar{c}l$. A direct calculation shows

$$\tilde{a}E + \tilde{b}d - \tilde{G}h - \tilde{k}j = Q_1 + 4\bar{c}Q_4, \quad \tilde{a}l + 2\tilde{c}d - F^2 = Q_2, \quad \tilde{b}l + 2\tilde{c}E - I^2 = Q_3.$$

By the definition of $\tilde{c}$ and (9.39), $\tilde{c} = c't$ and $\tilde{G} = 0$. By the last two identities above and the coefficients of t^2 in Q_2 and Q_3, $\tilde{a} = -2c'\bar{d}t$ and $\tilde{b} = \bar{I}^2(1 + (2\delta_I - \alpha)t)$. The constant and linear coefficients of Q_4 and $Q_1 + 4\bar{c}Q_4$ imply

$$j = \bar{j}\big(1 + (\delta_d + \delta_E - \delta_I)t\big), \qquad \tilde{k} = \frac{\bar{I}^3}{\bar{E}}\big(1 + (3\delta_I - 2\alpha - \delta_E)t\big).$$

The coefficients of t^2 of Q_4 and $Q_1 + 4\bar{c}Q_4$, divided by $\bar{d}\bar{E}$ and $\bar{d}\bar{I}^2$ respectively, imply

$$\begin{aligned} \delta_d(\delta_E - \delta_I + \alpha) - \delta_I\delta_E + \delta_I^2 &= 0, \\ \delta_d(\delta_E - \delta_I + \alpha) - \alpha\delta_E - (3\delta_I - 2\alpha - \delta_E)(\delta_E - \delta_I) &= 0. \end{aligned}$$

Subtracting and substituting $\delta_E = \delta_I^2/\alpha$, we obtain

$$\delta_I(\delta_I - 2\alpha)(\delta_I^2 - \alpha\delta_I + \alpha^2) = 0.$$

If $\delta_I^2 - \alpha\delta_I + \alpha^2 = 0$, then $\delta_E - \delta_I + \alpha = 0$, and the first equation above multiplied by α reads $\delta_I^2(\alpha - \delta_I) = 0$. Then $\delta_I = 0$ or $\delta_I = \alpha$, and either value contradicts $\delta_I^2 - \alpha\delta_I + \alpha^2 = 0$ since $\alpha \neq 0$. If $\delta_I = 0$, then $\delta_E = 0$, the first equation above implies $\delta_d = 0$, and then $F' = 0$, a contradiction. Therefore $\delta_I = 2\alpha$, $\delta_d = 4\alpha/3$, and $\delta_E = 4\alpha$. Thus

$$\begin{gathered} l = 1, \quad c = \bar{c} + c't, \quad d = \bar{d}\Big(1 + \frac{4\alpha}{3}t\Big), \quad E = \bar{E}(1 + 4\alpha t), \quad F = F't, \\ h = -\frac{\bar{E}^2}{\bar{I}^2}F't, \quad G = -4\bar{c}F't, \quad I = \bar{I}(1 + 2\alpha t), \quad j = \frac{\bar{d}\bar{E}}{\bar{I}}\Big(1 + \frac{10\alpha}{3}t\Big), \\ a = -2c'\bar{d}t - 2\bar{c}d, \quad b = \bar{I}^2(1 + 3\alpha t) - 2\bar{c}E, \quad k = \frac{\bar{I}^3}{\bar{E}} - 4\bar{c}I, \end{gathered} \tag{9.40}$$

where $\bar{d} = 3\bar{I}^2F'^2/(16c'^2\bar{E})$ by $F'^2 = 2c'\bar{d}\delta_d$, $\bar{c}$ is arbitrary, and $c', \bar{E}, \bar{I}, F'$ are non-zero.

Step 4. In this step we prove that ρ is a constant.

We denote by $[t]x$ the coefficient of t in x, in the coordinates of **Step 3** on $s = 1$. With the notation $x(0)$ of **Step 3**, $X(0,2)(0)$ is the value at $t = 0$ of the entry $X(0,2)$. We have the identity

$$(\bar{I}^2 - 2\bar{c}\bar{E})X(0,2)(0) + \bar{E}^2X(0,10)(0) - \bar{E}X(1,2)(0) + \frac{\bar{d}\bar{I}^2}{F'}[t]Y(1,0) = \frac{27F'^6\bar{I}^4}{512c'^5}\varpi. \tag{9.41}$$

To prove (9.41), note that F, G, h vanish at $t = 0$, so the terms in o_i of $X(0,2)$ and $X(0,10)$ vanish at $t = 0$, and $X(1,2)$ has no such term. Thus

$$\begin{aligned}&(\bar{I}^2 - 2\bar{c}\bar{E})X(0,2)(0) + \bar{E}^2 X(0,10)(0) - \bar{E}X(1,2)(0)\\ &= (\bar{I}^2 - 2\bar{c}\bar{E})(\bar{d}c_{11}(0) + \bar{E}c_{12}(0)) + \bar{E}^2(\bar{d}c_{10}(0) - c_{13}(0)) - \bar{E}(\bar{a}c_{11}(0) + \bar{b}c_{12}(0) - \bar{E}c_{13}(0))\\ &= \bar{d}\bar{I}^2 c_{11}(0) + \bar{d}\bar{E}^2 c_{10}(0).\end{aligned}$$

By (9.39),

$$\frac{\bar{d}\bar{I}^2}{F'}[t](-Fc_{11} + hc_{10}) = -\bar{d}\bar{I}^2 c_{11}(0) - \bar{d}\bar{E}^2 c_{10}(0),$$

so the terms in c_i of the left-hand side of (9.41) cancel. Moreover,

$$d^2 o_2 + j^2 o_3 = (d^2 h' + j^2 F')(\varpi + \rho_0 t),$$

and $\bar{d}^2 h' + \bar{j}^2 F' = 0$ by (9.39). By (9.40), $d' = 4\alpha\bar{d}/3$ and $j' = 10\alpha\bar{j}/3$, so

$$\frac{\bar{d}\bar{I}^2}{F'}[t](d^2 o_2 + j^2 o_3) = 2\varpi\frac{\bar{d}\bar{I}^2}{F'}(\bar{d}d'h' + \bar{j}j'F') = 4\alpha\bar{d}^3\bar{E}^2\varpi = \frac{27F'^6\bar{I}^4}{512c'^5}\varpi.$$

This proves (9.41). Since the left-hand side of (9.41) is zero and $F', \bar{I}, c'$ are non-zero, $\varpi = 0$, a contradiction. By **Step 1**, ρ is a constant.

Step 5. In this step we prove that σ is a constant.

Suppose that σ is not a constant, and let u be the common zero of k, I, j of **Step 1**. The exchange of the five pairs $(a,b),(d,E),(F,I),(G,k),(h,j)$ interchanges columns $1,2$ and columns $4,5$ of A up to the order of the rows, fixes Q_1, Q_4, interchanges Q_2, Q_3, and preserves the conditions of Set-up 9.4. Possibly replacing the first-stage forms with the exchanged forms, the argument of **Step 2** implies that $l(u), d(u), E(u), F(u), h(u)$ are non-zero.

In the coordinates of **Step 3**, we have $\sigma = \sigma_0 + \varpi/t$ with constants σ_0 and $\varpi \neq 0$, so $o_6 = j'(\varpi + \sigma_0 t)$, $o_7 = I'(\varpi + \sigma_0 t)$, and $o_{11} = -k'(\varpi + \sigma_0 t)$. The constant terms of Q_3 and Q_4 imply $\bar{b} = -2\bar{c}\bar{E}$ and $\bar{h} = \bar{d}\bar{E}/\bar{F}$. By Construction-Proposition 9.10(3),

$$Y(0,1) - Y(1,1) = -Ic_{17} + jc_{15} + E^2 o_6 + h^2 o_7,$$

so at $t = 0$ we have $\bar{E}^2 o_6(0) + \bar{h}^2 o_7(0) = 0$. At $t = 0$, the equation $Y(0,4) - Y(1,4) = 0$ reads $4\bar{c}\bar{E}^2 o_6(0) + \bar{h}^2 o_{11}(0) = 0$. Since $\varpi \neq 0$,

$$j' = -\frac{\bar{d}^2}{\bar{F}^2}I', \qquad k' = -4\bar{c}I'.$$

Since $j \neq 0$, $I' \neq 0$. The first-stage part of **Step 3** depends only on $Q_1, \dots, Q_4$, on $l = 1$, on (9.39), and on the non-vanishing of the constants there. Possibly replacing the first-stage forms with the exchanged forms, the argument of **Step 3** implies that, with $\alpha := 2c'\bar{d}/\bar{F}^2$,

$$\begin{gathered}l = 1, \quad c = \bar{c} + c't, \quad E = \bar{E}\Big(1 + \frac{4\alpha}{3}t\Big), \quad d = \bar{d}(1 + 4\alpha t), \quad I = I't,\\ j = -\frac{\bar{d}^2}{\bar{F}^2}I't, \quad k = -4\bar{c}I't, \quad F = \bar{F}(1 + 2\alpha t), \quad h = \frac{\bar{E}\bar{d}}{\bar{F}}\Big(1 + \frac{10\alpha}{3}t\Big),\\ b = -2c'\bar{E}t - 2\bar{c}E, \quad a = \bar{F}^2(1 + 3\alpha t) - 2\bar{c}d, \quad G = \frac{\bar{F}^3}{\bar{d}} - 4\bar{c}F,\end{gathered}$$

where $\bar{E} = 3\bar{F}^2 I'^2/(16c'^2\bar{d})$, $\bar{c}$ is arbitrary, and $c', \bar{d}, \bar{F}, I'$ are non-zero.

We have the identity

$$\begin{aligned}(\bar{F}^2 - 2\bar{c}\bar{d})X(0,3)(0) + \bar{d}^2 X(0,7)(0) - \bar{d}X(1,3)(0)&\\ + \frac{\bar{E}\bar{F}^2}{I'}[t]\big(Y(0,1) - Y(1,1)\big) = \frac{27I'^6\bar{F}^4}{512c'^5}\varpi.&\end{aligned} \tag{9.42}$$

The equations of Construction-Proposition 9.10(3) are not preserved by the exchange of the five pairs, so we prove (9.42) directly. Since k, I, j vanish at $t = 0$, the terms in o_i of $X(0,3)$ and $X(0,7)$ vanish at $t = 0$, and $X(1,3)$ has no such term. Thus

$$\begin{aligned}&(\bar{F}^2 - 2\bar{c}\bar{d})X(0,3)(0) + \bar{d}^2X(0,7)(0) - \bar{d}X(1,3)(0)\\ &= (\bar{F}^2 - 2\bar{c}\bar{d})(\bar{d}c_{16}(0) + \bar{E}c_{17}(0)) + \bar{d}^2(\bar{E}c_{15}(0) - c_{18}(0)) - \bar{d}(\bar{a}c_{16}(0) + \bar{b}c_{17}(0) - \bar{d}c_{18}(0))\\ &= \bar{E}\bar{F}^2c_{17}(0) + \bar{d}^2\bar{E}c_{15}(0).\end{aligned}$$

Since

$$[t](-Ic_{17} + jc_{15}) = -I'c_{17}(0) - \frac{\bar{d}^2}{\bar{F}^2}I'c_{15}(0),$$

the terms in c_i of the left-hand side of (9.42) cancel. Moreover,

$$E^2o_6 + h^2o_7 = (E^2j' + h^2I')(\varpi + \sigma_0 t),$$

and $\bar{E}^2j' + \bar{h}^2I' = 0$. Since $E' = 4\alpha\bar{E}/3$ and $h' = 10\alpha\bar{h}/3$,

$$\frac{\bar{E}\bar{F}^2}{I'}[t](E^2o_6 + h^2o_7) = 2\varpi\frac{\bar{E}\bar{F}^2}{I'}(\bar{E}E'j' + \bar{h}h'I') = 4\alpha\bar{E}^3\bar{d}^2\varpi = \frac{27I'^6\bar{F}^4}{512c'^5}\varpi.$$

This proves (9.42). Since the left-hand side of (9.42) is zero and $I', \bar{F}, c'$ are non-zero, $\varpi = 0$, a contradiction. □

Lemma 9.12. *Notation and conditions as in Construction-Proposition 9.10, and let $u \in \mathbb{P}^1$. Then, with all forms evaluated at u and all denominators non-zero:*

(1) If column 4 of $A(u)$ is zero and $l(u) \neq 0$, then

$$\sigma + \frac{G^2l}{E^2j}\kappa = 0. \tag{9.43}$$

(2) If column 4 of $A(u)$ is zero and $l(u) = 0$, then

$$\sigma + \frac{GI^3}{E^3h}\kappa = 0. \tag{9.44}$$

(3) If column 5 of $A(u)$ is zero and $l(u) \neq 0$, then

$$\rho - \frac{k^2l}{d^2h}\kappa = 0. \tag{9.45}$$

(4) If column 5 of $A(u)$ is zero and $l(u) = 0$, then

$$\rho - \frac{F^3k}{d^3j}\kappa = 0. \tag{9.46}$$

(5) If column 6 of $A(u)$ is zero, then

$$\nu + \Big(\frac{2c}{l} - \frac{k}{I} - \frac{G}{F}\Big)\kappa = 0. \tag{9.47}$$

Moreover, let $\mathcal{U}_A$ be the normalized lower space of Construction-Proposition 8.6, a point of which we identify with the triple (O, N, C) it determines, and let $e(A)$ be as in Construction-Proposition 9.9. Then the image of the linear map $\mathcal{U}_A \to \mathbb{C}^4$ sending a triple to its constants $(\kappa, \nu, \rho, \sigma)$ of Lemma 9.11 is of dimension at most $4 - e(A)$.

Proof. By Lemma 9.11, $\kappa, \nu, \rho, \sigma$ are constants. We evaluate all forms at u. The fractions below are homogeneous of degree zero in s, t. By Set-up 9.4(4), columns $1, 2, 3$ of $A(u)$ are non-zero, and at most one of columns $4, 5, 6$ of $A(u)$ is zero.

(1) We have $d = F = 0$, so $Q_2 = 0$ implies $a = 0$, and since column 1 of $A(u)$ is non-zero, $G \neq 0$. By $Q_4 = 0$, $jI = 0$. If $j = 0$, then $Q_1 = 0$ implies $h = 0$, so columns 4 and 6 of $A(u)$ are zero, a contradiction. Thus $j \neq 0$ and $I = 0$, and since column 5 of $A(u)$ is non-zero, $E \neq 0$. By (9.38), the equation $Y(0,1) = 0$ reads $G^2\kappa l + E^2\sigma j = 0$. Dividing by E^2j, we obtain (9.43).

(2) We have $d = F = l = 0$, so $E \neq 0$ by Set-up 9.4(2), and $c \neq 0$ by the independence of c, l. By $Q_3 = 0$, $I^2 = 2cE \neq 0$. By $Q_4 = 0$, $j = 0$. Since column 6 of $A(u)$ is non-zero, $h \neq 0$. By $Q_1 = 0$, $a = Gh/E$. Since column 1 of $A(u)$ is non-zero, $G \neq 0$. By (9.38), $o_0 = 0$, $o_5 = -2\kappa c$, and $o_7 = \sigma I$. Thus

$$X(0,10) = Gc_0 - EhI\sigma, \qquad Y(2,0) = -4\kappa c^2 E - Ec_0.$$

Since $4c^2 = I^4/E^2$, we obtain

$$0 = -\frac{X(0,10)}{IEh} - \frac{GY(2,0)}{IE^2h} = \sigma + \frac{GI^3}{E^3h}\kappa.$$

(3) We have $E = I = 0$, so $Q_3 = 0$ implies $b = 0$, and since column 2 of $A(u)$ is non-zero, $k \neq 0$. By $Q_4 = 0$, $hF = 0$. If $h = 0$, then $Q_1 = 0$ implies $j = 0$, so columns 5 and 6 of $A(u)$ are zero, a contradiction. Thus $h \neq 0$ and $F = 0$, and since column 4 of $A(u)$ is non-zero, $d \neq 0$. By (9.38), the equation $Y(0,0) = 0$ reads $-k^2\kappa l + d^2\rho h = 0$. Dividing by d^2h, we obtain (9.45).

(4) We have $E = I = l = 0$, so $d \neq 0$ by Set-up 9.4(2), and $c \neq 0$ by the independence of c, l. By $Q_2 = 0$, $F^2 = 2cd \neq 0$. By $Q_4 = 0$, $h = 0$. Since column 6 of $A(u)$ is non-zero, $j \neq 0$. By $Q_1 = 0$, $b = kj/d$. Since column 2 of $A(u)$ is non-zero, $k \neq 0$. By (9.38), $o_0 = 0$, $o_5 = -2\kappa c$, and $o_3 = \rho F$. Thus

$$X(0,7) = -kc_0 - djF\rho, \qquad Y(2,1) = 2\kappa cF^2 + dc_0.$$

Since $2c = F^2/d$, we obtain

$$0 = -\frac{X(0,7)}{Fdj} - \frac{kY(2,1)}{Fd^2j} = \rho - \frac{F^3k}{d^3j}\kappa.$$

(5) By Lemma 9.6, $d = E = h = j = 0$ and $lFI \neq 0$. By $Q_3 = 0$, $b = I^2/l$. By (9.38), $o_0 = \kappa l$, $o_5 = \nu l - 2\kappa c$, and $o_8 = -\kappa b$. Thus

$$X(0,2) = Fc_4 - Fbo_5 - 2Fc\kappa b + Gl\kappa b, \qquad Y(2,0) = -Ik\kappa l + 2cl\kappa b + lc_4.$$

We obtain

$$0 = -\frac{X(0,2)}{FI^2} + \frac{Y(2,0)}{I^2l} = \nu + \Big(\frac{2c}{l} - \frac{k}{I} - \frac{G}{F}\Big)\kappa.$$

Since u is arbitrary, any point with a zero column is as in one of (1)–(5). If $e(A) = 1$, then the linear condition of (1)–(5) is non-zero, since the coefficient of σ, ρ, or ν is 1. If $e(A) = 2$, then by Lemma 9.6 one point is a zero of column 4 and the other point is a zero of column 5. Thus the first condition has coefficient 1 on σ and 0 on ρ, and the second condition has coefficient 1 on ρ and 0 on σ, so the conditions are linearly independent. Since $e(A) \leq 2$ (Construction-Proposition 9.9), the image is of dimension at most $4 - e(A)$. □

Proposition 9.13. *Notation and conditions as in Set-up 9.4. Then the normalized lower space $\mathcal{U}_A$ of Construction-Proposition 8.6 satisfies* $\dim \mathcal{U}_A \leq 4$.

Proof. By Lemma 9.11, we have the linear map $\mathcal{U}_A \to \mathbb{C}^4$ sending a point to $(\kappa, \nu, \rho, \sigma)$. By (9.38) and $l, h, j \neq 0$, a point is in the kernel of the map if and only if $O = 0$. For $O = 0$, the three equations of Construction-Proposition 8.6 are the equations of $\mathcal{V}_A$, so the kernel is isomorphic to $\mathcal{V}_A$, and $\dim \mathcal{V}_A = e(A)$ by Construction-Proposition 9.9. By Lemma 9.12, the image of the map is of dimension at most $4 - e(A)$. Therefore

$$\dim \mathcal{U}_A \leq e(A) + (4 - e(A)) = 4.$$

□

9.5. **The moduli bound.** In this subsection, we prove Theorem 9.1.

Lemma 9.14. *Notation and conditions as in Set-up 9.4. Let ℓ be the lower blocks of a presentation with the fixed blocks of a numerical Godeaux surface, with O supported on the twelve slots of Construction-Proposition 8.6 and with block A, and let $g \in J$. Then the block A of $g * \ell$ is the matrix of a tuple in $\mathcal{P}^{\mathrm{good}}$.*

Proof. By Construction-Proposition 8.4 and Lemma 4.2, $g * \ell$ is again the lower blocks of a presentation with the fixed blocks of a numerical Godeaux surface. By Lemma 4.2, the block A of $g * \ell$ is the block A of $g \cdot \ell$. Since $(g, \ell) \to g * \ell$ is an action of J (Construction-Proposition 8.6), it suffices to consider g in a set of generators of $J = \mathrm{GL}_2 \times \mathcal{G}^0$.

An element of GL_2 only substitutes s, t linearly, so the four identities of Construction-Proposition 9.2, the generic rank of A, and the conditions (2)–(4) of Set-up 9.4 are preserved, with u moved.

The map $(\phi, v, \chi, w) \to (\phi, v, \chi)$ has finite fibers, any unit of $\mathcal{A}$ is the product of a unit with vanishing coefficient of z and a unit $1 + \theta z$ with $\theta \in \mathbb{C}$, and the automorphisms in the identity component of $\mathrm{Aut}(\mathcal{A})$ are $z \to \alpha z$ with $\alpha \in \mathbb{C}^*$. Thus $\mathcal{G}^0$ is generated by the elements of $\mathcal{G}^0$ with $\phi(z) = \alpha z$, $v = (\beta, \beta_p, \beta_q)$, and $w = (\gamma, \gamma_p, \gamma_q)$ on the three factors, together with the elements $(\mathrm{id}_{\mathcal{A}}, 1 + \theta z, 1, 1 + (\theta/2)z)$ for $\theta \in \mathbb{C}$ (the z-coefficient of w is determined by $w^2 = \chi v b_\phi$, since $b_\phi = (\alpha^{-1}, 1, 1)$ has no z-term).

First consider an element with $\phi(z) = \alpha z$, $v = (\beta, \beta_p, \beta_q)$, and $w = (\gamma, \gamma_p, \gamma_q)$. Since $b_\phi = (\alpha^{-1}, 1, 1)$, we have

$$\gamma^2 = \chi\beta/\alpha, \quad \gamma_p^2 = \chi\beta_p, \quad \gamma_q^2 = \chi\beta_q.$$

In the basis (e, z, p, q) of Construction-Proposition 9.2, Construction-Proposition 8.4 implies that the element multiplies $y_0, \dots, y_3$ by $\beta, \beta\alpha, \beta_p, \beta_q$ and $z_0, \dots, z_3$ by $\gamma, \gamma\alpha, \gamma_p, \gamma_q$, hence the monomial r_m by the product of the scalars of both factors of r_m. We divide the m-th relation of Set-up 4.1(7) by the scalar of r_m. Then the entry A_{im} is multiplied by the scalar of z_i over the scalar of r_m, so the forms are multiplied by the scalars

$$\begin{aligned}
&d : \frac{\gamma}{\beta\beta_p}, \quad E : \frac{\gamma}{\beta\beta_q}, \quad l : \frac{\gamma}{\beta^2\alpha^2}, \quad c : \frac{\gamma}{\beta^2\alpha},\\
&a : \frac{\gamma\alpha}{\beta\beta_p}, \quad b : \frac{\gamma\alpha}{\beta\beta_q}, \quad G : \frac{\gamma_p}{\beta\beta_p}, \quad F : \frac{\gamma_p}{\alpha\beta\beta_p},\\
&h : \frac{\gamma_p}{\beta_p\beta_q}, \quad k : \frac{\gamma_q}{\beta\beta_q}, \quad I : \frac{\gamma_q}{\alpha\beta\beta_q}, \quad j : \frac{\gamma_q}{\beta_p\beta_q},
\end{aligned}$$

and both occurrences of d in A are multiplied by $\gamma/(\beta\beta_p)$, and both occurrences of E in A are multiplied by $\gamma/(\beta\beta_q)$. Thus A is multiplied on the left and on the right by invertible diagonal matrices, so, for any $u \in \mathbb{P}^1$, the new matrix at u has rank rank $A(u)$, and the zero columns of the new matrix at u are the zero columns of $A(u)$. Hence the conditions (1), (3), and (4) of Set-up 9.4 are preserved. Since any of the twelve forms is multiplied by a non-zero scalar, the condition (2) and $l \neq 0$ are preserved. The four identities, in the order of Construction-Proposition 9.2, are multiplied by

$$\frac{\chi}{\beta\beta_p\beta_q}, \quad \frac{\chi}{\beta^2\alpha^2\beta_p}, \quad \frac{\chi}{\beta^2\alpha^2\beta_q}, \quad \frac{\chi}{\alpha\beta\beta_p\beta_q}.$$

Next consider the element with $\phi = \mathrm{id}_{\mathcal{A}}$, $v = 1 + \theta z$, $\chi = 1$, and $w = 1 + (\theta/2)z$. We substitute $y_0 \to y_0 + \theta y_1$ and $z_0 \to z_0 + (\theta/2)z_1$, then replace the first relation by itself minus θ times the fourth, and the second by itself minus θ times the fifth. The quadrics $r_1, \dots, r_6$ are restored, and we obtain

$$a \to a - \theta d/2, \quad b \to b - \theta E/2, \quad c \to c + \theta l/4, \quad G \to G - \theta F, \quad k \to k - \theta I,$$

and the forms d, E, F, h, I, j, l do not change. A direct calculation shows that $al + 2cd - F^2$, $bl + 2cE - I^2$, and $dE - hF - jI$ are preserved, and that

$$aE + bd - Gh - kj \to aE + bd - Gh - kj - \theta(dE - hF - jI).$$

Thus the four identities are preserved. The new matrix is A with $\theta/2$ times the row of z_0 added to the row of z_1, with θ times column 4 subtracted from column 1, and with θ times column 5 subtracted from column 2. Hence, for any $u \in \mathbb{P}^1$, the new matrix at u has rank $\operatorname{rank} A(u)$, so the condition (1) is preserved. The forms $c + \theta l/4, l$ are linearly independent if and only if c, l are, and the forms l, d, E do not change, so the condition (2) is preserved.

The entries of columns $4, 5, 6$ do not change, and the entries l and $2c + \theta l/2$ of column 3 vanish at u if and only if l and c vanish at u. Suppose that the new column 1 is zero at $u \in \mathbb{P}^1$. Then $d(u) = a(u) = 0$ and $G(u) = \theta F(u)$. The identity $al + 2cd - F^2 = 0$ at u implies that $F(u) = 0$, so $G(u) = 0$. Thus column 1 of $A(u)$ is zero, which contradicts the condition (4) for A. Suppose that the new column 2 is zero at $u \in \mathbb{P}^1$. Then $E(u) = b(u) = 0$ and $k(u) = \theta I(u)$. The identity $bl + 2cE - I^2 = 0$ at u implies that $I(u) = 0$, so $k(u) = 0$. Thus column 2 of $A(u)$ is zero, which contradicts the condition (4) for A. Hence the conditions (3) and (4) are preserved. □

Proof of Theorem 9.1. The proof is long, so we divide it into several steps.

Step 1. In this step we introduce an incidence $\mathcal{I}$ and show that $\dim \mathcal{I} \leq 16$.

Fix the blocks $B_1, \dots, B_4$ and the quadrics $r_1, \dots, r_6$ of Construction-Proposition 9.2. Let $\mathcal{L}$ be the affine space of all lower blocks of the shape of Set-up 4.1 with the fixed blocks $B_1, \dots, B_4$ of Construction-Proposition 9.2. Let $\mathcal{I} \subset \mathcal{L}$ be the set of points with A the matrix of a tuple in $\mathcal{P}^{\mathrm{good}}$, with (P, U) the pair determined by A, with O supported on the twelve slots of Construction-Proposition 8.6, and with

$$AO - B_1 N^t - C B_3^t = 0, \quad AN - CP^t = 0, \quad UN^t = 0.$$

Since the points of $\mathcal{I}$ have the fixed blocks and O supported on the twelve slots, $\mathcal{I}$ is contained in the affine space $\mathcal{L}_{(2,1,1)} \subset \mathcal{L}$ of Construction-Proposition 8.6(4), so $g * \ell$ is defined for $\ell \in \mathcal{I}$ and $g \in J$. The conditions (1)–(4) of Set-up 9.4 fail on the images in $\mathcal{P}_{(2,1,1)}$ of closed subsets of $\mathbb{P}^1 \times \mathcal{P}_{(2,1,1)}$, or on closed subsets of $\mathcal{P}_{(2,1,1)}$, and $\mathbb{P}^1$ is proper, so $\mathcal{P}^{\mathrm{good}}$ is open in $\mathcal{P}_{(2,1,1)}$ and $\mathcal{I}$ is constructible.

The fiber of $\mathcal{I} \to \mathcal{P}^{\mathrm{good}}$ over a tuple with matrix A is a linear copy of $\mathcal{U}_A$, since N and twelve entries of C are determined by Construction-Proposition 8.6. By Proposition 9.13, $\dim \mathcal{U}_A \leq 4$. We stratify $\mathcal{I}$ into irreducible locally closed pieces. By [Har77, Exercise II.3.22(e)], the relative dimension of any piece over the closure of the image of the piece equals the dimension of a general fiber, so it is at most 4. Since $\dim \mathcal{P}_{(2,1,1)} \leq 12$ (Construction-Proposition 9.2), we have

$$\dim \mathcal{I} \leq 12 + 4 = 16.$$

Step 2. In this step we introduce a constructible set $\Theta \subset \mathcal{L} \times \mathcal{M}$ and show that $\mathcal{M}_{(2,1,1)}$ is contained in the image of Θ in $\mathcal{M}$.

Let $\Theta := \Xi_{\mathcal{L}} \cap (\mathcal{I} \times \mathcal{M})$, for the set $\Xi_{\mathcal{L}}$ of Lemma 4.5(2) with the quadrics $r_1, \dots, r_6$. By Lemma 4.5(2) and **Step 1**, Θ is constructible.

Let $[X] \in \mathcal{M}_{(2,1,1)}$, where X is the canonical model of S. By the definition of $\mathcal{M}_{(2,1,1)}$ in Construction-Proposition 3.3, S satisfies Set-up 8.1 with $\mathbf{p} = (2, 1, 1)$. By Construction-Proposition 9.2, S has a presentation with the fixed blocks, and with A the matrix of a tuple in $\mathcal{P}_{(2,1,1)}$. Lemma 9.3 implies the conditions (2)–(4) of Set-up 9.4, and Lemma 4.3(1) implies the condition (1), so the tuple is in $\mathcal{P}^{\mathrm{good}}$. By Construction-Proposition 8.6 and Lemma 4.2, the gauge of a unique Γ moves the presentation to a presentation of S that keeps A, U, P, and r_k and has O supported on the twelve slots. Let ℓ be the lower blocks of the new presentation. Since $B_1, \dots, B_4$ only involve $y_0, \dots, y_3$, the three equations of **Step 1** are among the terms of $d_1' d_2 = 0$ of positive degree in s, t. Thus $\ell \in \mathcal{I}$, and $(\ell, [X]) \in \Theta$.

Step 3. In this step we conclude the proof by Lemma 2.9.

In Lemma 2.9, we take $V = \mathcal{L}$, $W = \mathcal{M}$, $G = \Theta$, and $e = 9$. A non-empty fiber of Θ over $\ell \in \mathcal{L}$ is one point, by Lemma 4.5(2).

Let $(\ell, [X]) \in \Theta$ and $g \in J$. By Construction-Proposition 8.4 and Lemma 4.2, $g * \ell$ is the lower blocks of a presentation with the fixed blocks of a numerical Godeaux surface with canonical model isomorphic to X. We show that $g * \ell \in \mathcal{I}$. By Lemma 9.14, the block A of $g * \ell$ is the matrix of a tuple in $\mathcal{P}^{\mathrm{good}}$. Since $g * \ell$ is the lower blocks of a presentation, the equation $d_1' d_2 = 0$ implies that its blocks satisfy $AB_3 + B_1 P = 0$ and $UB_4 - B_2 P^t = 0$. Thus its blocks P, U are the pair determined by its block A (Construction-Proposition 8.2). The block O of $g * \ell$ is supported on the twelve slots, by the definition of $g * \ell$. The chain lift of g preserves $d_1' d_2 = 0$, so $g * \ell$ satisfies the three equations of **Step 1**, as in **Step 2**. Thus $(g * \ell, [X]) \in \Theta$, so the fiber of Θ over $[X]$ contains $J * \ell$.

By Construction-Proposition 8.6, the gauge class of $g \cdot \ell$ contains exactly one point of $\mathcal{L}_{(2,1,1)}$, which is $g * \ell$, so the stabilizer of ℓ for the action $(g, \ell) \to g * \ell$ equals the stabilizer of the class of ℓ modulo the gauge. The block A of ℓ is the matrix of a tuple in $\mathcal{P}^{\mathrm{good}} \subset \mathcal{P}_{(2,1,1)}$, so its entry l is non-zero by the definition of $\mathcal{P}_{(2,1,1)}$. Thus $A \neq 0$, and Lemma 8.7 with $n = 3$ implies that

$$\dim J * \ell \geq 12 - 3 = 9.$$

Thus any non-empty fiber of Θ over $\mathcal{M}$ is of dimension at least 9.

By Lemma 2.9, **Step 1**, and **Step 2**, we have

$$\dim \overline{\mathcal{M}_{(2,1,1)}} \leq \dim \overline{\mathrm{pr}_{\mathcal{M}}(\Theta)} \leq \dim \overline{\mathrm{pr}_{\mathcal{L}}(\Theta)} - 9 \leq \dim \mathcal{I} - 9 \leq 16 - 9 = 7.$$

□

10. The collision type $(2, 2)$

In this section, we prove the following theorem.

Theorem 10.1. *We have* $\dim \overline{\mathcal{M}_{(2,2)}} \leq 7$.

We divide the presentations of type $(2, 2)$ into non-exceptional and exceptional presentations (Definition 10.3). For the non-exceptional presentations, we count orbits as for the type $(2, 1, 1)$ in Section 9, and for the exceptional presentations, we use the multiplication of the ring R.

10.1. The first stage.

In this subsection, we compute the first stage of a presentation of type $(2, 2)$.

Construction-Proposition 10.2. Notation and conditions as in Set-up 8.1 with $\mathbf{p} = (2, 2)$. Choose the basis $(1, z, \mathbf{e}, w)$ of $\mathcal{A} = \mathbb{C}[z]/(z^2) \times \mathbb{C}[w]/(w^2)$, where $\mathbf{e}$ is the identity of the second factor, and normalize $\lambda(1) = \lambda(\mathbf{e}) = 0$ and $\lambda(z) = \lambda(w) = 1$. Then S has a presentation with $y_0, \dots, y_3$ restricting to $(\tau^3, z\tau^3, \mathbf{e}\tau^3, w\tau^3)$ and $z_0, \dots, z_3$ restricting to $(\tau^4, z\tau^4, \mathbf{e}\tau^4, w\tau^4)$,

$$(r_1, \dots, r_6) = (y_1^2, y_1 y_2, y_1 y_3, -y_0 y_2 + y_2^2, -y_0 y_3 + y_2 y_3, y_3^2),$$

$$A = \begin{pmatrix} -a_1 & 0 & 0 & -a_5 + a_9 & -a_8 & 0 \\ a_0 & a_5 - a_9 & a_8 & a_4 & a_7 & 0 \\ a_1 & a_3 & 0 & a_5 & a_8 & a_{10} \\ 0 & a_2 & a_3 & a_6 & a_9 & a_{11} \end{pmatrix},$$

where $a_0, \dots, a_{11}$ are binary linear forms with $a_1 \neq 0$, $a_{10} \neq 0$, and

$$a_0(a_5 - a_9) + a_1 a_4 + 2a_2 a_3 = a_0 a_8 + a_1 a_7 + a_3^2 = a_{10} a_2 + a_{11} a_3 + a_8^2 = a_{10} a_6 + a_{11} a_9 + 2a_7 a_8 = 0.$$

The blocks $B_1, \dots, B_4$ are the fixed blocks of Table 11, and P, U are determined by A. The form a_1 (respectively a_{10}) defines the unique member of $|2K_X|$ that is singular at the point of the support of B with image $[1 : 0 : 0 : 0]$ (respectively $[1 : 0 : 1 : 0]$) under $[y_0 : \cdots : y_3]$.

TABLE 11. The fixed blocks $B_1, \dots, B_4$ and the matrices P, U determined by A, for the collision type $(2,2)$, with $(b_1, b_2, b_3) = (z, \mathbf{e}, w)$ in Construction-Proposition 8.2.

$$B_1 = \begin{pmatrix} y_1 & 0 & 0 & 0 & y_2 & 0 & 0 & 0 & y_3 & 0 & 0 & 0 \\ -y_0 & y_1 & 0 & 0 & 0 & y_2 & 0 & 0 & 0 & y_3 & 0 & 0 \\ 0 & 0 & y_1 & 0 & -y_0 & 0 & -y_0+y_2 & 0 & 0 & 0 & y_3 & 0 \\ 0 & 0 & 0 & y_1 & 0 & 0 & 0 & -y_0+y_2 & -y_0 & 0 & -y_0 & y_3 \end{pmatrix}$$

$$B_2 = \begin{pmatrix} -y_3 & y_2 & 0 & -y_1 & y_1 & 0 & 0 & 0 \\ 0 & 0 & y_3 & y_0-y_2 & -y_0 & 0 & y_1 & 0 \\ 0 & 0 & 0 & 0 & y_3 & y_0-y_2 & 0 & y_1 \end{pmatrix}$$

$$B_3 = \begin{pmatrix} y_2 & y_3 & 0 & 0 & 0 & 0 & 0 & 0 \\ -y_1 & 0 & -y_0+y_2 & y_3 & 0 & 0 & 0 & 0 \\ 0 & -y_1 & 0 & -y_0 & -y_0+y_2 & y_3 & 0 & 0 \\ 0 & 0 & -y_1 & 0 & 0 & 0 & y_3 & 0 \\ 0 & 0 & 0 & -y_1 & -y_1 & 0 & -y_2 & y_3 \\ 0 & 0 & 0 & 0 & 0 & -y_1 & 0 & y_0-y_2 \end{pmatrix}$$

$$B_4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & y_3 & 0 & 0 & 0 & -y_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & y_3 & 0 & -y_3 & 0 & -y_2 & 0 & y_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & -y_3 & 0 & y_3 & 0 & y_2 & 0 & y_0-y_2 \\ 0 & 0 & 0 & 0 & 0 & 0 & y_3 & -y_0 & 0 & 0 & y_0-y_2 & 0 \\ 0 & -y_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1 & 0 & 0 \\ -y_3 & 0 & y_3 & 0 & 0 & 0 & 0 & 0 & y_1 & -y_0 & -y_1 & 0 \\ 0 & y_3 & 0 & -y_3 & 0 & 0 & 0 & 0 & 0 & -y_1 & 0 & y_1 \\ 0 & 0 & -y_3 & y_0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1 & 0 \\ 0 & y_2 & 0 & 0 & 0 & -y_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ y_2 & 0 & -y_2 & 0 & -y_1 & y_0 & y_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -y_2 & 0 & -y_0+y_2 & 0 & y_1 & 0 & -y_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -y_0+y_2 & 0 & 0 & 0 & -y_1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$P = \begin{pmatrix} 0 & 0 & -a_5+a_9 & -a_8 & -a_8 & 0 & 0 & 0 \\ a_5-a_9 & a_8 & a_4 & a_7 & a_7 & 0 & 0 & 0 \\ a_3 & 0 & a_5 & a_8 & a_8 & a_{10} & 0 & 0 \\ a_2 & a_3 & a_6 & a_9 & a_9 & a_{11} & 0 & 0 \\ a_1 & 0 & 0 & 0 & 0 & 0 & -a_8 & 0 \\ -a_0 & 0 & -a_5+a_9 & 0 & -a_8 & 0 & a_7 & 0 \\ -a_1 & 0 & -a_3 & 0 & 0 & 0 & a_8 & a_{10} \\ 0 & 0 & -a_2 & 0 & -a_3 & 0 & a_9 & a_{11} \\ 0 & a_1 & 0 & 0 & 0 & 0 & a_5-a_9 & a_8 \\ 0 & -a_0 & 0 & -a_5+a_9 & 0 & -a_8 & -a_4 & -a_7 \\ 0 & -a_1 & 0 & -a_3 & 0 & 0 & -a_5 & -a_8 \\ 0 & 0 & 0 & -a_2 & 0 & -a_3 & -a_6 & -a_9 \end{pmatrix}$$

$$U = \begin{pmatrix} a_0 & -a_1 & 0 & 0 & a_2+a_5-a_9 & 0 & a_2 & a_3 & a_3+a_8 & 0 & a_3 & 0 \\ -a_2-a_5+a_9 & 0 & -a_2 & -a_3 & -a_4-a_6 & a_5-a_9 & -a_6 & -a_9 & -a_7-a_9 & a_8 & -a_9 & 0 \\ -a_3-a_8 & 0 & -a_3 & 0 & -a_7-a_9 & a_8 & -a_9 & 0 & -a_{11} & 0 & -a_{11} & -a_{10} \end{pmatrix}$$

Proof of Construction-Proposition 10.2. We take $(b_0, b_1, b_2, b_3) = (1, z, \mathbf{e}, w)$ in Construction-Proposition 8.2, so that $y_* = y_0$, the k-th column of M_j consists of the coordinates of $b_j b_k$ in the basis (b_0, b_1, b_2, b_3), and the matrix Λ of the functional of the statement satisfies

$$\Lambda^{-1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

In particular, the functional is non-degenerate. The formulas of Construction-Proposition 8.2 for B_1, B_4, with $y_* = y_0$, the matrices M_j, and Λ above, yield the matrices of Table 11. By Construction-Theorem 3.2(4), possibly replacing the trivialization of $\mathcal{O}_B(K_X)$, we may assume that λ is the functional of the statement. Let $\mathbf{b} := (b_0, b_1, b_2, b_3)$. We write $B_j(\mathbf{b})$ for the value of B_j at $y_i = b_i$.

A direct calculation shows that $r_k(b_0, \dots, b_3) = 0$ for $1 \le k \le 6$, that $(r_1, \dots, r_6)B_3 = 0$, and that $B_3 B_2^t = 0$. Modulo y_0, the quadrics $r_1, \dots, r_6$ are a basis of $(y_1, y_2, y_3)^2$, and the columns of B_3 and B_2^t are the linear first and second syzygies of $(y_1, y_2, y_3)^2$. Let $y_i \in R_3$ and $z_i \in R_4$ be elements restricting to $b_i\tau^3$ and $b_i\tau^4$ (Construction-Theorem 3.2(2)). By Lemma 8.3, S has a presentation with the generators y_i, z_i, with the quadrics $r_1, \dots, r_6$, and with the blocks $B_1, \dots, B_4$ of Table 11.

Next we bound the coefficient matrices of A. Let I_B be the kernel of $\mathbb{C}[y_1, y_2, y_3] \to \mathcal{A}$, $y_i \to b_i$. By Construction-Proposition 8.2, the part in degrees divisible by 3 of the resolution of $\mathcal{N}$ is a resolution of $\mathcal{N}_0$ with the differentials $(r_1, \dots, r_6)$, B_3, B_2^t. Setting $y_0 = 1$ preserves exactness, since $y_0 - 1$ is a non-zero-divisor on $\mathcal{N}_0$ and on the free modules. Moreover, $\mathcal{N}_0/(y_0 - 1)\mathcal{N}_0 \cong \mathcal{A}$ with $y_i \to b_i$, since y_0 restricts to the unit τ^3. Thus the specializations of $(r_1, \dots, r_6)$ and B_3 at $y_0 = 1$ present I_B. Thus I_B/I_B^2 is the cokernel of $B_3(\mathbf{b}) : \mathcal{A}^8 \to \mathcal{A}^6$, and

$$\operatorname{Hom}_{\mathcal{A}}(I_B/I_B^2, \mathcal{A}) = \{\nu \in \mathcal{A}^6 \mid \nu B_3(\mathbf{b}) = 0\}.$$

Near the zeros $(0,0,0)$ and $(0,1,0)$ of I_B, the ideal I_B is generated by the regular sequences y_2, y_3, y_1^2 and $y_1, y_2 - 1, y_3^2$ respectively. Thus I_B/I_B^2 is locally free of rank 3 over $\mathcal{A}$ [Mat89, Theorem 16.2(i)], and

$$\dim_{\mathbb{C}} \operatorname{Hom}_{\mathcal{A}}(I_B/I_B^2, \mathcal{A}) = 12.$$

For a constant 4×6 matrix A_0, we set

$$\nu_k := \sum_{i=0}^{3} A_{0,ik} b_i \qquad (1 \le k \le 6).$$

Since $b_0, \dots, b_3$ is a basis of $\mathcal{A}$, the map $A_0 \to \nu = (\nu_1, \dots, \nu_6)$ is a bijection from constant 4×6 matrices onto $\mathcal{A}^6$. If $A_0 B_3 + B_1 P_0 = 0$ for a constant matrix P_0, then $A_0 B_3(\mathbf{b}) + B_1(\mathbf{b}) P_0 = 0$, and multiplying on the left by $(b_0, \dots, b_3)$ implies that $\nu B_3(\mathbf{b}) = 0$, because $(b_0, \dots, b_3)B_1(\mathbf{b}) = 0$. Thus the constant matrices A_0 for which such a matrix P_0 exists form a vector space of dimension at most 12.

A direct calculation shows that, for the matrix A of the statement and the matrices P, U of Table 11, with $a_0, \dots, a_{11}$ indeterminates, we have

$$AB_3 + B_1 P = 0, \qquad UB_4 - B_2 P^t = 0,$$

and the non-zero entries of UP are, up to sign, the four expressions in the statement. The twelve coefficient matrices of $a_0, \dots, a_{11}$ in A are linearly independent, since the entries of A recover $a_0, \dots, a_{11}$. Moreover, they belong to the vector space above: specialize $a_0, \dots, a_{11}$ to constants in the first equation. Thus they are a basis of the vector space above. By Set-up 4.1(5), the blocks of the presentation satisfy $AB_3 + B_1 P = 0$, $UP = 0$, and $UB_4 - B_2 P^t = 0$. Thus the coefficients of s and t in A belong to the vector space above, and A has the form in the statement, with binary linear forms $a_0, \dots, a_{11}$. By Construction-Proposition 8.2, the pair (P, U) is the pair of Table 11. Thus $UP = 0$ is equivalent to the four quadratic identities of the statement.

It remains to consider a_1 and a_{10}. Let x be the point of the support of B with image $[1 : 0 : 0 : 0]$, a smooth point of X by Construction-Proposition 3.3. Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{X,x}$, let τ be a local generator of $\mathcal{O}_X(K_X)$ at x extending the trivialization, and let $\bar{s} := s/\tau^2$, $\bar{t} := t/\tau^2$, $\bar{y}_i := y_i/\tau^3$, $\bar{z}_j := z_j/\tau^4$. The ideal $\mathfrak{a}$ of B at x is $(\bar{s}, \bar{t})$, and $\bar{y}_1$ restricts to z in $\mathcal{O}_{X,x}/\mathfrak{a} = \mathbb{C}[z]/(z^2)$. Thus $\mathfrak{a} \not\subset \mathfrak{m}^2$, since $\mathbb{C}[z]/(z^2)$ has embedding dimension 1. For $\mu \in \mathfrak{a} \setminus \mathfrak{m}^2$, the elements $\mu, \bar{y}_1$ are regular parameters of $\mathcal{O}_{X,x}$, $\mathfrak{a} = (\mu, \bar{y}_1^2)$, and $\bar{y}_1^2 \notin \mathfrak{m}\mathfrak{a}$, since $\mathcal{O}_{X,x}/(\mu)$ is

a discrete valuation ring with uniformizer $\bar{y}_1$, and $\mathfrak{a}$ is not principal. The relation of $r_1 = y_1^2$ divided by τ^6 is

$$\bar{y}_1^2 + a_1(\bar{s},\bar{t})(\bar{z}_2 - \bar{z}_0) + a_0(\bar{s},\bar{t})\bar{z}_1 + v_1(\bar{s},\bar{t}) = 0,$$

where $\bar{z}_0$ is a unit, $\bar{z}_1, \bar{z}_2 \in \mathfrak{m}$, and $v_1(\bar{s},\bar{t}) \in \mathfrak{a}^3$. Thus

$$a_1(\bar{s},\bar{t}) \equiv \bar{z}_0^{-1}\bar{y}_1^2 \mod \mathfrak{m}\mathfrak{a}.$$

In particular, $a_1 \neq 0$ and $a_1(\bar{s},\bar{t}) \in \mathfrak{m}^2$. Any binary linear form independent of a_1 takes a value outside $\mathfrak{m}^2$ at $(\bar{s},\bar{t})$, since $\mathfrak{a} \not\subset \mathfrak{m}^2$. Thus a_1 defines the unique member of $|2K_X|$ singular at x.

At the point of the support of B with image $[1:0:1:0]$, we replace $\bar{y}_1$ with $\bar{y}_3$ and the relation of r_1 with the relation of $r_6 = y_3^2$, which is

$$\bar{y}_3^2 + a_{10}(\bar{s},\bar{t})\bar{z}_2 + a_{11}(\bar{s},\bar{t})\bar{z}_3 + v_6(\bar{s},\bar{t}) = 0,$$

where $\bar{z}_2$ is a unit and $\bar{z}_3$ belongs to the maximal ideal. The argument above implies that $a_{10} \neq 0$ and that a_{10} defines the unique member of $|2K_X|$ singular at that point. □

Definition 10.3. A presentation as in Construction-Proposition 10.2 is *exceptional* if $\operatorname{rank} A(u) = 2$ for some $u \in \mathbb{P}^1$, and *non-exceptional* otherwise. We denote by $\mathcal{M}^{\mathrm{exc}}_{(2,2)}$ the set of points of $\mathcal{M}_{(2,2)}$ having an exceptional presentation, and by $\mathcal{M}^{\circ}_{(2,2)}$ the set of points of $\mathcal{M}_{(2,2)}$ having a non-exceptional presentation.

We treat exceptional presentations separately, because they are outside the charts of Table 12, and $\dim \mathcal{U}_A = 9$ over them (Construction-Proposition 10.10).

10.2. **Non-exceptional presentations.** In this subsection, we prove Proposition 10.9.

We start with the values of the first stage at the zeros of a_1 and a_{10}.

Lemma 10.4. *Notation and conditions as in Construction-Proposition 10.2.*

(1) Assume that a_1, a_{10} are linearly independent, and choose $s = a_1$ and $t = a_{10}$. Then $a_8(0,1) \neq 0$ and $a_3(1,0) \neq 0$.

(2) Assume that a_{10} is a non-zero multiple of a_1, and evaluate at the zero of a_1. Then a_3, a_8 are either both zero or both non-zero there. If they are both zero, then $a_0 a_{11} \neq 0$, $a_5 = a_9 = 0$, and $\operatorname{rank} A = 2$ there.

(3) A presentation is exceptional if and only if a_{10} is a non-zero multiple of a_1 and a_3, a_8 vanish at the zero of a_1.

Proof. Let $b := a_5 - a_9$ and $d := a_9$. Then

$$A = \begin{pmatrix} -a_1 & 0 & 0 & -b & -a_8 & 0 \\ a_0 & b & a_8 & a_4 & a_7 & 0 \\ a_1 & a_3 & 0 & a_5 & a_8 & a_{10} \\ 0 & a_2 & a_3 & a_6 & d & a_{11} \end{pmatrix},$$

and the first-stage identities are

$$\begin{aligned} a_0 b + a_1 a_4 + 2a_2 a_3 = 0, \quad a_0 a_8 + a_1 a_7 + a_3^2 = 0, \\ a_{10} a_2 + a_{11} a_3 + a_8^2 = 0, \quad a_{10} a_6 + a_{11} d + 2a_7 a_8 = 0. \end{aligned} \tag{10.1}$$

We use Lemma 4.3(3) in the following form. Let $u \in \mathbb{P}^1$.

(a) If the first and third columns of A are zero columns at u, then $a_{10}(u) = 0$.

(b) If the third and sixth columns of A are zero columns at u, then $a_1(u) = 0$.

We prove (a). The space H_u contains a non-zero vector c with $c_k = 0$ for $k \neq 1, 3$, because the only condition on c_1, c_3 is $c_1 v_1(u) + c_3 v_3(u) = 0$. The quadric

$$c_1 r_1 + c_3 r_3 = y_1(c_1 y_1 + c_3 y_3)$$

is a product of two linear forms, and it is singular at $[1:0:1:0]$. By Lemma 4.3(3), C_u is singular at the point of the support of B with image $[1:0:1:0]$. By Construction-Proposition 10.2, C_u is the member defined by a_{10}, that is, $a_{10}(u) = 0$. For (b), we use the

quadric $c_3r_3 + c_6r_6 = y_3(c_3y_1 + c_6y_3)$, which is singular at $[1 : 0 : 0 : 0]$, and the member defined by a_1.

Proof of (1). Let $u = (0, 1)$, so that $a_1(u) = 0$ and $a_{10}(u) = 1$, and assume that $a_8(u) = 0$. By the second identity of (10.1), $a_3(u) = 0$. By the third identity, $a_2(u) = 0$. If $a_0(u) = 0$, then the first and third columns of A are zero columns at u, which contradicts (a). Thus $a_0(u) \neq 0$, and the first identity implies that $b(u) = 0$. Therefore, the second and third columns of A are zero columns at u and the first row of $A(u)$ is zero. Thus $\operatorname{rank} A(u) \leq 3$, the span of the second and third basis vectors of $\mathbb{C}^6$ is contained in $\operatorname{Ker} A(u)$, and the corresponding quadrics $r_2 = y_1y_2$ and $r_3 = y_1y_3$ have the common linear factor y_1, which contradicts Lemma 4.3(2). Hence $a_8(0, 1) \neq 0$.

Let $u = (1, 0)$, so that $a_1(u) = 1$ and $a_{10}(u) = 0$, and assume that $a_3(u) = 0$. By the third identity of (10.1), $a_8(u) = 0$. By the second identity, $a_7(u) = 0$. If $a_{11}(u) = 0$, then the third and sixth columns of A are zero columns at u, which contradicts (b). Thus $a_{11}(u) \neq 0$, and the fourth identity implies that $d(u) = 0$. Therefore, the third and fifth columns of A are zero columns at u. The sum of the first and third rows of A is

$$(0, a_3, 0, d, 0, a_{10}),$$

which vanishes at u. Thus $\operatorname{rank} A(u) \leq 3$, the span of the third and fifth basis vectors of $\mathbb{C}^6$ is contained in $\operatorname{Ker} A(u)$, and the corresponding quadrics $r_3 = y_1y_3$ and $r_5 = y_3(y_2 - y_0)$ have the common linear factor y_3, which contradicts Lemma 4.3(2). Hence $a_3(1, 0) \neq 0$.

Proof of (2). Let u be the zero of a_1, so that $a_{10}(u) = 0$. The second and third identities of (10.1) at u are

$$a_0(u)a_8(u) + a_3(u)^2 = 0, \qquad a_{11}(u)a_3(u) + a_8(u)^2 = 0.$$

Thus $a_3(u), a_8(u)$ are either both zero or both non-zero.

Assume that $a_3(u) = a_8(u) = 0$. Then the third column of A is a zero column at u, and the first and third rows of $A(u)$ are $(0, 0, 0, -b(u), 0, 0)$ and $(0, 0, 0, a_5(u), 0, 0)$. Thus $\operatorname{rank} A(u) \leq 3$. If $a_0(u) = 0$, then the first and third columns of A are zero columns at u, so the span of the first and third basis vectors of $\mathbb{C}^6$ is contained in $\operatorname{Ker} A(u)$, and the corresponding quadrics $r_1 = y_1^2$ and $r_3 = y_1y_3$ have the common linear factor y_1. If $a_{11}(u) = 0$, then the third and sixth columns of A are zero columns at u, so the span of the third and sixth basis vectors of $\mathbb{C}^6$ is contained in $\operatorname{Ker} A(u)$, and the corresponding quadrics $r_3 = y_1y_3$ and $r_6 = y_3^2$ have the common linear factor y_3. Both possibilities contradict Lemma 4.3(2). Thus $a_0(u)a_{11}(u) \neq 0$. The first and fourth identities at u are $a_0(u)b(u) = 0$ and $a_{11}(u)d(u) = 0$, hence $b(u) = d(u) = 0$, and $a_5(u) = a_9(u) = 0$. Therefore, the first and third rows of $A(u)$ vanish. The minor of $A(u)$ on rows $2, 4$ and columns $1, 6$ is $a_0(u)a_{11}(u) \neq 0$, so $\operatorname{rank} A(u) = 2$.

Proof of (3). If a_{10} is a non-zero multiple of a_1 and a_3, a_8 vanish at the zero of a_1, then the presentation is exceptional by (2).

Conversely, assume that a_1, a_{10} are linearly independent, or that a_{10} is a non-zero multiple of a_1 and a_3, a_8 do not both vanish at the zero of a_1. In the second situation, a_3, a_8 are both non-zero at the zero of a_1 by (2). We show that $\operatorname{rank} A(u) \neq 2$ for any $u \in \mathbb{P}^1$.

Let $u \in \mathbb{P}^1$ with $a_1(u)a_{10}(u) \neq 0$, and assume that $\operatorname{rank} A(u) = 2$. The first and sixth columns of $A(u)$ are linearly independent, so they span the columns of $A(u)$. The first row of $A(u)$ has a non-zero entry in column 1 and zero entries in columns $2, 3, 6$. Thus the second and third columns of $A(u)$ are multiples of the sixth column. The third row of $A(u)$ has the non-zero entry $a_{10}(u)$ in column 6 and a zero entry in column 3, so the third column of $A(u)$ is zero. In particular, $a_3(u) = 0$. Then the third row of $A(u)$ has a zero entry in column 2, so the second column of $A(u)$ is zero. Thus the span of the second and third basis vectors of $\mathbb{C}^6$ is contained in $\operatorname{Ker} A(u)$, and the corresponding quadrics $r_2 = y_1y_2$ and $r_3 = y_1y_3$ have the common linear factor y_1, which contradicts Lemma 4.3(2).

Assume that a_1, a_{10} are linearly independent, and choose $s = a_1$ and $t = a_{10}$. By (1), at the zero $(1,0)$ of a_{10} and at the zero $(0,1)$ of a_1, we have respectively

$$\det\begin{pmatrix} -a_1 & 0 & 0 \\ a_1 & a_3 & 0 \\ 0 & a_2 & a_3 \end{pmatrix} = -a_1 a_3^2 \neq 0, \qquad \det\begin{pmatrix} 0 & -a_8 & 0 \\ a_8 & a_7 & 0 \\ 0 & a_8 & a_{10} \end{pmatrix} = a_8^2 a_{10} \neq 0,$$

where the first matrix is the submatrix of A on rows $1,3,4$ and columns $1,2,3$, and the second matrix is the submatrix of A on rows $1,2,3$ and columns $3,5,6$. Thus $\operatorname{rank} A(u) \geq 3$ at both zeros.

Assume that a_{10} is a non-zero multiple of a_1 and a_3, a_8 are non-zero at the zero of a_1. At the zero of a_1, the submatrix of A on rows $1,2,3$ and columns $2,3,5$ satisfies

$$\det\begin{pmatrix} 0 & 0 & -a_8 \\ b & a_8 & a_7 \\ a_3 & 0 & a_8 \end{pmatrix} = a_3 a_8^2 \neq 0.$$

Thus $\operatorname{rank} A(u) \geq 3$ at the zero of a_1. Since a_1, a_{10} are non-zero binary linear forms, they are either linearly independent or proportional, so the cases above cover any $u \in \mathbb{P}^1$. □

The next result describes the elements of the module group that we use to normalize the first stage.

Construction-Proposition 10.5. Notation and conditions as in Construction-Proposition 10.2, and let $b := a_5 - a_9$ and $d := a_9$. We write an element of $\mathcal{G}$ as $(\phi, (\rho_1, \rho_2), \chi, (\omega_1, \omega_2))$, where (ρ_1, ρ_2) is the unit v and (ω_1, ω_2) is the unit w of Construction-Proposition 8.4, since w is a generator of $\mathcal{A}$ in this section. The following hold for the action of Construction-Proposition 8.4 on presentations.

(1) The elements

$$\left(\mathrm{id}_{\mathcal{A}}, (1+\delta_1 z, 1+\delta_2 w), 1, \left(1+\frac{\delta_1}{2}z, 1+\frac{\delta_2}{2}w\right)\right)$$

with $\delta_1, \delta_2 \in \mathbb{C}$ form a subgroup $\mathcal{G}_{\mathrm{sh}} \subset \mathcal{G}^0$, and such an element fixes a_1, a_{10}, a_3, a_8 and sends a_0, a_{11}, b, d to $a_0 - \frac{\delta_1}{2}a_1$, $a_{11} + \frac{\delta_2}{2}a_{10}$, $b - \delta_2 a_8$, $d + \delta_1 a_3$, respectively.

(2) The elements $(\phi, (\rho_1, \rho_2), \chi, (\omega_1, \omega_2))$ with $\phi(z) = \alpha_1 z$, $\phi(w) = \alpha_2 w$, and $\omega_i^2 = \chi\rho_i/\alpha_i$ for $i = 1, 2$ form a torus $\mathcal{G}_{\mathrm{tor}} \subset \mathcal{G}^0$, and such an element multiplies $a_1, a_{10}, a_3, a_8, b, d, a_0, a_{11}$ by

$$\frac{\omega_1}{\rho_1^2\alpha_1^2}, \quad \frac{\omega_2}{\rho_2^2\alpha_2^2}, \quad \frac{\omega_2}{\rho_1\rho_2\alpha_1}, \quad \frac{\omega_1}{\rho_1\rho_2\alpha_2}, \quad \frac{\omega_1}{\rho_1\rho_2}, \quad \frac{\omega_2}{\rho_1\rho_2}, \quad \frac{\omega_1}{\rho_1^2\alpha_1}, \quad \frac{\omega_2}{\rho_2^2\alpha_2},$$

respectively.

(3) Let $\iota := (\phi_0, 1, 1, 1) \in \mathcal{G}$, where $\phi_0 \in \mathrm{Aut}(\mathcal{A})$ interchanges the factors $\mathbb{C}[z]/(z^2)$ and $\mathbb{C}[w]/(w^2)$ of $\mathcal{A}$ with $\phi_0(z) = w$. Then ι carries a presentation of S to another presentation of S, and it sends the tuple $(a_1, a_{10}, a_3, a_8, b, d, a_0, a_{11}, a_2, a_7, a_4, a_6)$ to

$$(-a_{10}, -a_1, a_8, a_3, -d, -b, a_{11}, a_0, -a_7, -a_2, a_6, a_4).$$

Proof. Let $y_0' := y_0 - y_2$ and $z_0' := z_0 - z_2$. Then y_0', y_1, y_2, y_3 restrict to $(1-\mathbf{e})\tau^3, z\tau^3, \mathbf{e}\tau^3, w\tau^3$, and z_0', z_1, z_2, z_3 restrict to $(1-\mathbf{e})\tau^4, z\tau^4, \mathbf{e}\tau^4, w\tau^4$. We have $r_4 = -y_0' y_2$ and $r_5 = -y_0' y_3$. We change the signs of the fourth and fifth relations of Set-up 4.1, so that the quadrics of the relations are the monomials

$$y_1^2, \quad y_1 y_2, \quad y_1 y_3, \quad y_0' y_2, \quad y_0' y_3, \quad y_3^2.$$

Since $z_0 = z_0' + z_2$, the matrix of the relations in the basis z_0', z_1, z_2, z_3 has the sum of the first and third rows of A as third row. Thus the matrix of the relations is

$$A' := \begin{pmatrix} -a_1 & 0 & 0 & b & a_8 & 0 \\ a_0 & b & a_8 & -a_4 & -a_7 & 0 \\ 0 & a_3 & 0 & -d & 0 & a_{10} \\ 0 & a_2 & a_3 & -a_6 & -d & a_{11} \end{pmatrix}.$$

In the convention of Lemma 8.5, if an element acts on the spans of y_0', y_1, y_2, y_3 and of z_0', z_1, z_2, z_3 by matrices G_y and G_z, and G_r is the matrix induced by G_y on the span of the six monomials, then the element sends A' to $G_z A' G_r^{-1}$. Since $a_1, a_{10} \neq 0$, the forms a_2, a_7, a_4, a_6 are determined by the identities of Construction-Proposition 10.2 and the other eight forms, so they do not appear in (1) and (2).

Proof of (1). Since $z^2 = w^2 = 0$ and $b_{\mathrm{id}_{\mathcal{A}}} = 1$, we have

$$\left(1 + \frac{\delta_1}{2} z, 1 + \frac{\delta_2}{2} w\right)^2 = (1 + \delta_1 z, 1 + \delta_2 w) = 1 \cdot (1 + \delta_1 z, 1 + \delta_2 w) \cdot b_{\mathrm{id}_{\mathcal{A}}},$$

so the element belongs to $\mathcal{G}$. Moreover, $\mathcal{G}_{\mathrm{sh}} \cong \mathbb{C}^2$ is connected, so $\mathcal{G}_{\mathrm{sh}} \subset \mathcal{G}^0$. By Construction-Proposition 8.4, the element acts by

$$y_0' \to y_0' + \delta_1 y_1, \quad y_2 \to y_2 + \delta_2 y_3, \quad z_0' \to z_0' + \frac{\delta_1}{2} z_1, \quad z_2 \to z_2 + \frac{\delta_2}{2} z_3,$$

and it fixes y_1, y_3, z_1, z_3. Thus G_z adds $\frac{\delta_1}{2}$ times the first row to the second row and $\frac{\delta_2}{2}$ times the third row to the fourth row. Moreover, G_r^{-1} is given by

$$y_1 y_2 \to y_1 y_2 - \delta_2 y_1 y_3, \quad y_0' y_3 \to y_0' y_3 - \delta_1 y_1 y_3, \quad y_0' y_2 \to y_0' y_2 - \delta_2 y_0' y_3 - \delta_1 y_1 y_2 + \delta_1 \delta_2 y_1 y_3,$$

and it fixes $y_1^2, y_1 y_3, y_3^2$. A direct calculation shows that

$$G_z A' G_r^{-1} = \begin{pmatrix} -a_1 & 0 & 0 & b - \delta_2 a_8 & a_8 & 0 \\ a_0 - \frac{\delta_1}{2} a_1 & b - \delta_2 a_8 & a_8 & -a_4^\sharp & -a_7 - \frac{\delta_1}{2} a_8 & 0 \\ 0 & a_3 & 0 & -d - \delta_1 a_3 & 0 & a_{10} \\ 0 & a_2 - \frac{\delta_2}{2} a_3 & a_3 & -a_6^\sharp & -d - \delta_1 a_3 & a_{11} + \frac{\delta_2}{2} a_{10} \end{pmatrix},$$

where

$$a_4^\sharp := a_4 + \frac{\delta_1}{2} b - \delta_2 a_7 - \frac{\delta_1 \delta_2}{2} a_8, \qquad a_6^\sharp := a_6 - \frac{\delta_2}{2} d + \delta_1 a_2 - \frac{\delta_1 \delta_2}{2} a_3.$$

Comparing with A' implies (1).

Proof of (2). Since $\lambda(1) = \lambda(\mathbf{e}) = 0$ and $\lambda(z) = \lambda(w) = 1$, we have $b_\phi = (\alpha_1^{-1}, \alpha_2^{-1})$. Thus the condition of Construction-Proposition 8.4 is $\omega_i^2 = \chi \rho_i / \alpha_i$ for $i = 1, 2$. The parameters $(\alpha_1, \alpha_2, \chi, \omega_1, \omega_2)$ in $(\mathbb{C}^*)^5$ define an isomorphism of groups onto $\mathcal{G}_{\mathrm{tor}}$, with $\rho_i = \omega_i^2 \alpha_i / \chi$. Thus $\mathcal{G}_{\mathrm{tor}}$ is a connected torus, and $\mathcal{G}_{\mathrm{tor}} \subset \mathcal{G}^0$.

The element multiplies y_0', y_1, y_2, y_3 by $\rho_1, \rho_1\alpha_1, \rho_2, \rho_2\alpha_2$ and multiplies z_0', z_1, z_2, z_3 by $\omega_1, \omega_1\alpha_1, \omega_2, \omega_2\alpha_2$. Thus the six monomials are multiplied by

$$\rho_1^2 \alpha_1^2, \quad \rho_1 \rho_2 \alpha_1, \quad \rho_1 \rho_2 \alpha_1 \alpha_2, \quad \rho_1 \rho_2, \quad \rho_1 \rho_2 \alpha_2, \quad \rho_2^2 \alpha_2^2,$$

respectively. An entry of A' in the row of a z and the column of a monomial is multiplied by the factor of the z divided by the factor of the monomial. We read off (2) from the entries of A' at the positions (row, column) $(1,1)$, $(3,6)$, $(3,2)$, $(1,5)$, $(1,4)$, $(3,4)$, $(2,1)$, $(4,6)$.

Proof of (3). Since $\lambda \circ \phi_0 = \lambda$, we have $b_{\phi_0} = 1$, so $\iota \in \mathcal{G}$. By Construction-Proposition 8.4, ι lifts to a chain map of the minimal S_1-resolution of $\mathcal{N}$, given by matrices with constant entries. The element ι is a change of generators of R:

$$y_0 \to y_0, \quad y_1 \to y_3, \quad y_2 \to y_0 - y_2, \quad y_3 \to y_1,$$

and $z_0 \to z_0$, $z_1 \to z_3$, $z_2 \to z_0 - z_2$, $z_3 \to z_1$. Thus the ring R does not change, and we obtain a presentation of S with canonical model $\operatorname{Proj} R$. Since $\lambda \circ \phi_0 = \lambda$, the composition of ϕ_0 with the

identification of the coordinate algebra of B with $\mathcal{A}$ is again an identification with $(\mathcal{A}, \lambda)$, and the new generators restrict as in Construction-Proposition 10.2. The quadrics transform by

$$(r_1, \dots, r_6) \to (r_6, -r_5, r_3, r_4, -r_2, r_1).$$

Thus ι interchanges y_0' with y_2, y_1 with y_3, z_0' with z_2, and z_1 with z_3, and A' becomes

$$\begin{pmatrix} a_{10} & 0 & 0 & -d & a_3 & 0 \\ a_{11} & -d & a_3 & -a_6 & a_2 & 0 \\ 0 & a_8 & 0 & b & 0 & -a_1 \\ 0 & -a_7 & a_8 & -a_4 & b & a_0 \end{pmatrix}.$$

Comparing with A' implies (3). □

The next result is the list of normal forms for the first stage of a non-exceptional presentation.

Construction-Proposition 10.6. Notation and conditions as in Construction-Proposition 10.5. Let $\mathcal{G}_3 \subset \mathcal{G}_{\mathrm{tor}}$ be the subtorus defined by $\chi = \omega_2 = 1$, and let $\mathcal{G}_2 \subset \mathcal{G}_3$ be the subtorus defined by $\alpha_2 = 1$. For $i \in \{2, 3\}$, let $J_i \subset J$ be the subgroup $\mathrm{GL}_2 \times_{\mathcal{G}_{\mathrm{sh}}} \mathcal{G}_i$, so that $\dim J_i = 6 + i$. Let (d_1, d_2) be the image under an element of J of a non-exceptional presentation. Then there exist a chart of Table 12 among NN, ZN, ZZ, PN, of parameter dimensions $3, 3, 4, 3$, and an element g of J_3 for NN and ZN, and of J_2 for ZZ and PN, such that g sends the first stage of (d_1, d_2) to a point of the chart. For the chart ZN only, we may have to replace (d_1, d_2) with the image of (d_1, d_2) under ι.

TABLE 12. The charts of non-exceptional presentations of type $(2, 2)$, where a_2, a_7, a_4, a_6 are uniquely determined by the first-stage identities of Construction-Proposition 10.2, since $a_1, a_{10} \neq 0$.

chart	a_1	a_{10}	a_3	a_8	b	d	a_0	a_{11}	parameters
NN	s	t	$s+t$	$ps+t$	$es-2t$	$-(2/p)s+ft$	$-t$	$-p^2 s$	$p \neq 0,\ e, f \in \mathbb{C}$
ZN	s	t	s	$s+t$	$es+kt$	$-2s+ft$	0	$-s$	$e, f, k \in \mathbb{C}$
ZZ	s	t	s	t	$es+kt$	$hs+ft$	0	0	$e, f, h, k \in \mathbb{C}$
PN	s	s	t	$ps+t$	$es-4pt$	$fs+2pt$	$-t$	$-t$	$e, f, p \in \mathbb{C}$

Proof of Construction-Proposition 10.6. Let $\nu := \omega_1/\omega_2$. Since $\rho_i = \omega_i^2 \alpha_i/\chi$, we have

$$\frac{\rho_1}{\rho_2} = \frac{\nu^2 \alpha_1}{\alpha_2}.$$

Thus $\mathcal{G}_3$ has coordinates $(\nu, \alpha_1, \alpha_2)$, and $\mathcal{G}_2$ has coordinates (ν, α_1). By Construction-Proposition 8.4, the composition of $(\phi, (\rho_1, \rho_2), \chi, (\omega_1, \omega_2))$ and $(\phi', (\rho_1', \rho_2'), \chi', (\omega_1', \omega_2'))$ has second coordinate $(\rho_1, \rho_2)\phi(\rho_1', \rho_2')$. Thus conjugation by an element of $\mathcal{G}_{\mathrm{tor}}$ sends the element of $\mathcal{G}_{\mathrm{sh}}$ with parameters (δ_1, δ_2) to the element with parameters $(\alpha_1\delta_1, \alpha_2\delta_2)$. Therefore, $\mathcal{G}_{\mathrm{sh}}\mathcal{G}_2$ and $\mathcal{G}_{\mathrm{sh}}\mathcal{G}_3$ are subgroups of $\mathcal{G}^0$ of dimensions 4 and 5, and $\dim J_i = 4 + 2 + i = 6 + i$. Likewise, $\mathcal{G}_{\mathrm{sh}}\mathcal{G}_{\mathrm{tor}}$ is a connected subgroup of $\mathcal{G}^0$ of dimension $7 = \dim \mathcal{G}$, so $\mathcal{G}^0 = \mathcal{G}_{\mathrm{sh}}\mathcal{G}_{\mathrm{tor}}$.

By Construction-Proposition 8.4, (d_1, d_2) is a presentation with the fixed blocks, so the first-stage identities hold for (d_1, d_2), and the blocks P, U are determined by A (Construction-Proposition 10.2). By Construction-Proposition 10.5(1)(2), an element of $\mathcal{G}^0$ keeps the shape of A and multiplies a_1, a_{10}, a_3, a_8 by non-zero scalars, and an element of GL_2 substitutes s, t linearly. Thus the conclusions of Lemma 10.4 on a non-exceptional presentation hold for the forms a_1, a_{10}, a_3, a_8 of (d_1, d_2). Since $a_5 = b + d$ and $a_9 = d$, it suffices to bring $a_1, a_{10}, a_3, a_8, b, d, a_0, a_{11}$ to a row of Table 12.

Case 1. Assume that a_1, a_{10} are linearly independent. An element of GL_2 brings them to $a_1 = s$ and $a_{10} = t$. Write

$$a_3 = \beta s + \eta t, \qquad a_8 = \xi s + \gamma t.$$

Then $\beta\gamma \neq 0$ by Lemma 10.4(1), since $\gamma = a_8(0,1)$ and $\beta = a_3(1,0)$.

Assume that $\eta \neq 0$ and $\xi = 0$. By Construction-Proposition 10.5(3), the image under ι of (d_1, d_2) has $(a_1, a_{10}, a_3, a_8) = (-t, -s, \gamma t, \beta s + \eta t)$. Here (3) holds for (d_1, d_2), since by Construction-Proposition 8.4, ι sends the block A of any presentation with the fixed blocks to $G_z A G_r^{-1}$, with the blocks G_z, G_r of the chain lift of ι, which do not depend on the presentation. After the element of GL_2 that brings $-t, -s$ to s, t, we have $a_3 = -\gamma s$ and $a_8 = -\eta s - \beta t$, whose coefficients satisfy the conditions $\beta\gamma \neq 0$, $\eta = 0$, and $\xi \neq 0$. Since ι acts on A by constant changes of rows and columns, ι commutes with GL_2. Thus, possibly replacing (d_1, d_2) with the image of (d_1, d_2) under ι, which we use only for the chart ZN, we may assume that $\eta = 0$ or $\xi \neq 0$.

By Construction-Proposition 10.5(1), an element of $\mathcal{G}_{\mathrm{sh}}$ brings the s-coefficient of a_0 and the t-coefficient of a_{11} to zero, and it fixes a_1, a_{10}, a_3, a_8. From now on, write

$$b = es + kt, \qquad d = hs + ft.$$

The second identity of (10.1) at $(0,1)$ and the third identity at $(1,0)$ imply that

$$a_0 = -\frac{\eta^2}{\gamma} t, \qquad a_{11} = -\frac{\xi^2}{\beta} s.$$

Substituting into the second and third identities, we obtain

$$a_7 = -\beta^2 s - \left(2\beta\eta - \frac{\eta^2\xi}{\gamma}\right) t, \qquad a_2 = -\left(2\xi\gamma - \frac{\xi^2\eta}{\beta}\right) s - \gamma^2 t.$$

Thus the first identity at $(0,1)$ and the fourth identity at $(1,0)$ imply that

$$\eta(\eta k + 2\gamma^3) = 0, \qquad \xi(\xi h + 2\beta^3) = 0. \tag{10.2}$$

Consider an element of $\mathcal{G}_{\mathrm{tor}}$, followed by the diagonal element of GL_2 that brings a_1, a_{10} back to s, t. The composition preserves the vanishing of the s-coefficient of a_0 and of the t-coefficient of a_{11}. By Construction-Proposition 10.5(2), it multiplies β, γ, η, ξ by

$$\frac{\nu\alpha_1^2}{\alpha_2}, \quad \frac{\alpha_2^2}{\nu\alpha_1}, \quad \frac{\alpha_2^3}{\nu^2\alpha_1^2}, \quad \frac{\nu^2\alpha_1^3}{\alpha_2^2},$$

respectively. The exponent matrices of (β, γ, η) and (β, γ, ξ) on $\mathcal{G}_3$, and of (β, γ) on $\mathcal{G}_2$, satisfy

$$\det \begin{pmatrix} 1 & 2 & -1 \\ -1 & -1 & 2 \\ -2 & -2 & 3 \end{pmatrix} = -1, \qquad \det \begin{pmatrix} 1 & 2 & -1 \\ -1 & -1 & 2 \\ 2 & 3 & -2 \end{pmatrix} = 1, \qquad \det \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix} = 1.$$

Thus the multipliers of (β, γ, η) take any value in $(\mathbb{C}^*)^3$ on $\mathcal{G}_3$, and so do the multipliers of (β, γ, ξ), and the multipliers of (β, γ) take any value in $(\mathbb{C}^*)^2$ on $\mathcal{G}_2$. We distinguish three possibilities.

- If $\eta \neq 0$ and $\xi \neq 0$, then an element of $\mathcal{G}_3$ brings β, γ, η to 1, and $\xi =: p \neq 0$. By (10.2), $k = -2$ and $h = -2/p$. Then $a_0 = -t$ and $a_{11} = -p^2 s$, which is the chart NN.
- If $\eta = 0$ and $\xi \neq 0$, then an element of $\mathcal{G}_3$ brings β, γ, ξ to 1. By (10.2), $h = -2$. Then $a_0 = 0$ and $a_{11} = -s$, which is the chart ZN.
- If $\eta = \xi = 0$, then an element of $\mathcal{G}_2$ brings β, γ to 1. Then $a_0 = a_{11} = 0$, which is the chart ZZ.

Case 2. Otherwise, $a_{10} = \kappa a_1$ with $\kappa \neq 0$, since a_1, a_{10} are non-zero, and a_3, a_8 are non-zero at the zero of a_1 by Lemma 10.4(2)(3). An element of GL_2 brings a_1, a_3 to $a_1 = s$ and $a_3 = t$. Write

$$a_{10} = \kappa s, \qquad a_8 = \xi s + \gamma t,$$

where $\gamma = a_8(0,1) \neq 0$. By Construction-Proposition 10.5(1), an element of $\mathcal{G}_{\mathrm{sh}}$ brings the s-coefficients of a_0 and a_{11} to zero, and it fixes a_1, a_{10}, a_3, a_8. From now on, write

$$b = es + \mu t, \qquad d = fs + \psi t.$$

The second and third identities of (10.1) at $(0,1)$ imply that

$$a_0 = -\frac{1}{\gamma}t, \qquad a_{11} = -\gamma^2 t.$$

Substituting into the second and third identities, we obtain

$$a_7 = \frac{\xi}{\gamma}t, \qquad a_2 = -\frac{\xi^2}{\kappa}s - \frac{2\xi\gamma}{\kappa}t.$$

The first and fourth identities at $(0,1)$ imply that

$$\mu = -\frac{4\xi\gamma^2}{\kappa}, \qquad \psi = \frac{2\xi}{\gamma^2}.$$

Then the first and fourth identities determine

$$a_4 = \left(\frac{e}{\gamma} + \frac{2\xi^2}{\kappa}\right)t, \qquad a_6 = \frac{1}{\kappa}\left(\gamma^2 f - \frac{2\xi^2}{\gamma}\right)t.$$

Consider an element of $\mathcal{G}_{\mathrm{tor}}$, followed by the diagonal element of GL_2 that brings a_1, a_3 back to s, t. The composition preserves the vanishing of the s-coefficients of a_0 and a_{11}. By Construction-Proposition 10.5(2), it multiplies γ, κ by

$$\frac{\nu\alpha_1}{\alpha_2}, \quad \frac{\nu^3\alpha_1^4}{\alpha_2^4},$$

respectively. On $\mathcal{G}_2$, the exponent matrix satisfies

$$\det\begin{pmatrix} 1 & 1 \\ 3 & 4 \end{pmatrix} = 1.$$

Thus an element of $\mathcal{G}_2$ brings γ, κ to 1. Let $p := \xi$. Then $a_{10} = s$, $a_8 = ps + t$, $b = es - 4pt$, $d = fs + 2pt$, and $a_0 = a_{11} = -t$, which is the chart PN.

In each case, the elements used are elements of GL_2, of $\mathcal{G}_{\mathrm{sh}}$, and of $\mathcal{G}_3$ for NN and ZN, or of $\mathcal{G}_2$ for ZZ and PN. Their composition is an element of J_3 or of J_2, respectively. □

The following coefficient matrix is used for the types $(2,2)$, $(3,1)$, and (4). We use it to bound the normalized lower space over the charts of Table 12.

Construction-Proposition 10.7. Let $\mathbf{p}$ be one of the partitions $(2,2)$, $(3,1)$, (4), and let A be a first-stage matrix of type $\mathbf{p}$ as in Construction-Proposition 8.6. Let $x_0, \dots, x_{83}$ be the 84 scalar coefficients of Construction-Proposition 8.6, with $o_i = sx_{2i} + tx_{2i+1}$ and $c_j = s^2x_{24+3j} + stx_{25+3j} + t^2x_{26+3j}$, where the twelve slots are taken in lexicographic order and o_i is the coefficient at the i-th slot. This is the order of the lists of Tables 46 and 50 and of the matrix O of Table 34. We number the 96 coefficients of $y_0, \dots, y_3$ in the entries of $B_1N^t + CB_3^t - AO$, the negative of the matrix of the first equation of Construction-Proposition 8.6: the number $24i + 4j + l$ refers to the coefficient of y_l in the entry (i,j), with i, j, l starting at 0. The 96×104 coefficient matrix of the 96 coefficients in the entries of N and C has an 84×84 minor of non-zero constant determinant, listed for each type in Appendix A, and the 84 coefficients on the rows of the minor determine N and twelve entries of C in terms of the o_i and the c_j. By substitution, the other twelve coefficients, in increasing order of their numbers, are binary quadratic forms, linear in $o_0, \dots, o_{11}$. The entries of $AN - CP^t$ in row-major order, followed by the entries of UN^t in row-major order, after removing the entries that are identically zero as polynomials in the first-stage forms, the o_i, and the c_j, are m binary cubic forms, linear in the o_i and the c_j, where $m = 56, 56, 54$ for $\mathbf{p} = (2,2), (3,1), (4)$, respectively. We denote by $T_{\mathbf{p}}$ the $(36 + 4m) \times 84$ coefficient matrix of the twelve binary quadratic forms and the m binary cubic forms in $x_0, \dots, x_{83}$, so that $\mathcal{U}_A = \operatorname{Ker} T_{\mathbf{p}}$, with rows and columns numbered from 0 as follows.

(1) Column r is the coefficient of x_r.
(2) Row $3r + l$, where $0 \le r \le 11$ and $0 \le l \le 2$, is the coefficient of $s^{2-l}t^l$ in the r-th of the twelve binary quadratic forms.

(3) Row $36+4r+l$, where $0 \le r < m$ and $0 \le l \le 3$, is the coefficient of $s^{3-l}t^l$ in the r-th of the m binary cubic forms.

Proof. Solving the minor for N and twelve entries of C and expanding the three equations of Construction-Proposition 8.6 with the fixed blocks of type $\mathbf{p}$ is a direct calculation with the blocks of Tables 11, 14, and 16, which shows the determinants of the minors, the degrees and the linearity of the statement, that the twelve forms do not involve the c_j, and the values of m. The minor determines N and twelve entries of C, so the three equations of Construction-Proposition 8.6 are equivalent to the vanishing of the twelve forms and of the m forms. For $\mathbf{p}=(2,2)$, the twelve forms are the rows of the matrix H of Table 34 applied to $(o_0,\dots,o_{11})$, and their numbers are $0, 12, 16, 24, 36, 40, 54, 62, 70, 78, 86, 94$. □

Lemma 10.8. *Let Q be the parameter space of one of the charts NN, ZN, ZZ, PN of Table 12, and for $q \in Q$, let A_q be the first-stage matrix of the chart at q. Let $\mathcal{I}_Q \subset Q \times \mathbb{C}^{84}$ be the set of pairs (q,x) with x in the normalized lower space $\mathcal{U}_{A_q}$ of Construction-Proposition 8.6. Then $\dim \mathcal{I}_Q \le 8,8,9,9$ for the charts NN, ZN, ZZ, PN, respectively.*

Proof. For $q \in Q$, let $T_{(2,2)}(q)$ be the value at q of the matrix $T_{(2,2)}$ of Construction-Proposition 10.7, with 260 rows and 84 columns and with entries in the coordinate ring of Q. By Construction-Proposition 10.7,

$$\dim \mathcal{U}_{A_q} = 84 - \operatorname{rank} T_{(2,2)}(q).$$

Appendix A lists, for each chart, a pivot table of $T_{(2,2)}$ on the coordinate ring of Q, in the sense of Construction-Proposition 6.7. Each pivot table is a finite list of rational row operations with explicit pivots, the equality of the current entry with β at any triple (r,c,β) is a finite exact computation over the coordinate ring of Q, and the only divisions are by units of the chart. Table 13 lists the numbers of triples, and the last column of Table 13 follows from Construction-Proposition 6.7.

TABLE 13. The pivot tables of $T_{(2,2)}$ on the charts NN, ZN, ZZ, and PN.

chart	coordinate ring of Q	$\dim Q$	number of triples	upper bound of $\dim \mathcal{U}_{A_q}$
NN	$\mathbb{C}[p,p^{-1},e,f]$	3	77	7
ZN	$\mathbb{C}[k,e,f]$	3	79	5
ZZ	$\mathbb{C}[k,h,e,f]$	4	79	5
PN	$\mathbb{C}[p,e,f]$	3	77	7

For a chart, let T^{res} be the last matrix of the pivot table, with rows and columns numbered from 0 as in Appendix A, and let $T^{\mathrm{res}}(q)$ be the value of T^{res} at $q \in Q$. We call the entries of T^{res} *residual entries*. Since the pivots are units, any triple lowers the rank by exactly one at any point of Q, so $\operatorname{rank} T_{(2,2)}(q)$ is the number of triples plus $\operatorname{rank} T^{\mathrm{res}}(q)$.

We first consider the chart NN. The residual entry in row 45 and column 82 is $-e-2p$, and the residual entry in row 49 and column 82 is $p-1$. The 2×2 minor of T^{res} in rows $45, 49$ and columns $60, 82$, with entries in Table 35, is a regular function on Q with value 16 at $(p,e,f)=(2,0,0)$. Since Q is irreducible and $\dim Q = 3$, the set of the points $q \in Q$ with $\operatorname{rank} T^{\mathrm{res}}(q) \le 1$ is contained in a proper closed subset of Q, which is of dimension at most 2. The set of the points $q \in Q$ with $\operatorname{rank} T^{\mathrm{res}}(q) = 0$ is contained in the closed subset $\{p=1, e=-2\}$ of dimension 1. Over the three sets of q with $\operatorname{rank} T^{\mathrm{res}}(q) \ge 2$, $\operatorname{rank} T^{\mathrm{res}}(q) = 1$, and $\operatorname{rank} T^{\mathrm{res}}(q) = 0$, we have $\dim \mathcal{U}_{A_q} \le 5, 6, 7$, respectively. Since the fibers of the projection $\mathcal{I}_Q \to Q$ are the linear spaces $\mathcal{U}_{A_q}$, [Har77, Exercise II.3.22] implies

$$\dim \mathcal{I}_Q \le \max\{3+5,\ 2+6,\ 1+7\} = 8. \tag{10.3}$$

By Table 13, $\dim \mathcal{I}_Q \le 3+5=8$ for the chart ZN and $\dim \mathcal{I}_Q \le 4+5=9$ for the chart ZZ.

Finally, we consider the chart PN. The residual entry in row 41 and column 79 is $4p$. Hence $\dim \mathcal{U}_{A_q} \leq 6$ for $p \neq 0$, and $\dim \mathcal{U}_{A_q} \leq 7$ for $p = 0$. Since $\{p = 0\} \subset Q$ is of dimension 2, and since the fibers of $\mathcal{I}_Q \to Q$ are linear spaces, [Har77, Exercise II.3.22] implies

$$\dim \mathcal{I}_Q \leq \max\{3 + 6,\ 2 + 7\} = 9.$$

□

Proposition 10.9. *We have* $\dim \overline{\mathcal{M}^\circ_{(2,2)}} \leq 7$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we introduce the set of normalized lower blocks of non-exceptional presentations, and we show that J preserves it.

Let $\mathcal{L}_{(2,2)}$ be the affine space of Construction-Proposition 8.6, with the fixed blocks of Table 11. Let $\mathcal{L}^\circ \subset \mathcal{L}_{(2,2)}$ be the set of the points $g * \ell$, where $g \in J$ and $\ell \in \mathcal{L}_{(2,2)}$ is the lower blocks of a non-exceptional presentation of a surface as in Set-up 8.1 with $\mathbf{p} = (2, 2)$. By Construction-Proposition 8.4 and Lemma 4.2, a point of $\mathcal{L}^\circ$ is the lower blocks of a presentation with the fixed blocks of such a surface. We write a point of $\mathcal{L}^\circ$ as (A, x) with $x \in \mathcal{U}_A$, since the blocks of $d_1' d_2 = 0$ give the three equations of Construction-Proposition 8.6.

Since $(g, \ell) \to g * \ell$ is an algebraic action of J on $\mathcal{L}_{(2,2)}$ (Construction-Proposition 8.6), any $g \in J$ acts by an automorphism of $\mathcal{L}_{(2,2)}$. It preserves $\mathcal{L}^\circ$, so it preserves the closure $\overline{\mathcal{L}^\circ}$ of $\mathcal{L}^\circ$ in $\mathcal{L}_{(2,2)}$. Let $g \in J$ with component γ in GL_2. On the generators z_j and on the relations of degree 6, the chain map of Construction-Proposition 8.4 is constant. The image of a presentation is written in the shape of Set-up 4.1(7) by a change $z_j \to z_j + f$ with $f \in (S_0)_4$, which only alters the first row of d_1, and by a column change restoring $r_1, \dots, r_6$, which is constant. Thus g sends $A(u)$ to $MA(u\gamma)M'$, where u is a row vector and M, M' are constant invertible matrices, and the action of J preserves the condition $A \neq 0$.

Lemma 2.9 needs a constructible subset and no constructibility of $\mathcal{L}^\circ$ is proved, so we use $\overline{\mathcal{L}^\circ}$. Lemma 8.7 needs $A \neq 0$, so we add the condition $A \neq 0$. Let $\mathcal{W} \subset \Xi_{\mathcal{L}_{(2,2)}}$ be the subset of pairs $(\ell, [X])$ with $\ell \in \overline{\mathcal{L}^\circ}$ and with block $A \neq 0$, where $\Xi_{\mathcal{L}_{(2,2)}}$ is the set of Lemma 4.5(2) for $\mathcal{L} = \mathcal{L}_{(2,2)}$ and the quadrics $r_1, \dots, r_6$ of Construction-Proposition 10.2. By Lemma 4.5(2), $\mathcal{W}$ is constructible, and for any $\ell \in \mathcal{L}_{(2,2)}$, there exists at most one point $[X]$ with $(\ell, [X]) \in \mathcal{W}$.

Step 2. In this step we prove that $\dim \overline{\mathcal{L}^\circ} \leq 17$.

Let Q be the parameter space of one of the charts NN, ZN, ZZ, PN of Table 12, and for $q \in Q$, let A_q be the first-stage matrix of the chart at q. Let $\mathcal{I}_Q^{\mathrm{geo}}$ be the set of pairs (q, x) with $q \in Q$ and x a geometric point of $\mathcal{U}_{A_q}$. The set $\mathcal{I}_Q^{\mathrm{geo}}$ is contained in the set of the pairs (q, x) with $x \in \mathcal{U}_{A_q}$, which is closed in $Q \times \mathbb{C}^{84}$ since $\mathcal{U}_{A_q}$ is the kernel of the value at q of $T_{(2,2)}$. Thus, by Lemma 10.8,

$$\dim \overline{\mathcal{I}_Q^{\mathrm{geo}}} \leq 8, 8, 9, 9 \tag{10.4}$$

for the charts NN, ZN, ZZ, PN, respectively.

Let $\ell = (A, x)$ be a point of $\mathcal{L}^\circ$. Up to the gauge of Lemma 4.2, which keeps the first stage, ℓ is the lower blocks of a presentation (d_1, d_2) that is the image under an element of J of a non-exceptional presentation. By Construction-Proposition 10.6, there exist a chart and an element g of J_3 for the charts NN and ZN, and of J_2 for the charts ZZ and PN, such that g sends the first stage of (d_1, d_2) to a point q of the chart. For the chart ZN only, we may have to replace (d_1, d_2) with the image of (d_1, d_2) under ι. By Construction-Proposition 8.4 and Construction-Proposition 10.5(3), the presentation (d_1, d_2) moves with g and ι to a presentation with isomorphic canonical model. By Lemma 4.2, the gauge of the unique Γ of Construction-Proposition 8.6 does not change the first stage or the canonical model, and it brings the lower blocks to a point (A_q, x') with x' a geometric point of $\mathcal{U}_{A_q}$. Let $\mathcal{L}^\circ_Q \subset \mathcal{L}^\circ$ be the set of the points ℓ for which such a g exists with the chart of Q.

The element ι does not belong to $\mathcal{G}^0$. Since ι lifts to a chain map given by matrices with constant entries and permutes the gauge classes (Construction-Proposition 8.4), and since Γ is polynomial in the blocks because the minor of Construction-Proposition 8.6 has constant determinant, the map sending ℓ to the point $\iota \star \ell \in \mathcal{L}_{(2,2)}$ in the gauge class of $\iota \cdot \ell$ is a morphism, with inverse $\ell \to \iota^{-1} \star \ell$. Moreover, $(q, x') \to (A_q, x')$ is polynomial in q and x'. By Construction-Proposition 8.6, $(g, \ell) \to g * \ell$ is an algebraic action. Let $Z_Q \subset \mathcal{L}_{(2,2)}$ be the image of

$$J_i \times \mathcal{I}_Q^{\mathrm{geo}} \to \mathcal{L}_{(2,2)}, \qquad (g, (q, x')) \to g^{-1} * (A_q, x'),$$

where $i = 3$ for the charts NN and ZN and $i = 2$ for the charts ZZ and PN. Hence, $\mathcal{L}_Q^\circ \subset Z_Q$ for the charts NN, ZZ, PN, and $\mathcal{L}_Q^\circ \subset Z_Q \cup (\iota^{-1} \star Z_Q)$ for the chart ZN. Since $\dim J_3 = 9$ and $\dim J_2 = 8$, since the map of the display is a morphism defined on $J_i \times \overline{\mathcal{I}_Q^{\mathrm{geo}}}$, and since $\ell \to \iota^{-1} \star \ell$ is an automorphism of $\mathcal{L}_{(2,2)}$, inequality (10.4) and [Har77, Exercise II.3.22] imply that $\dim \overline{\mathcal{L}_Q^\circ}$ is at most

$$9 + 8, \quad 9 + 8, \quad 8 + 9, \quad 8 + 9$$

for the charts NN, ZN, ZZ, PN, respectively. Since $\mathcal{L}^\circ$ is the union of the four sets $\mathcal{L}_Q^\circ$,

$$\dim \overline{\mathcal{L}^\circ} \leq 17. \tag{10.5}$$

Step 3. In this step we conclude the proof by Lemma 2.9.

Let $\mathcal{M}' \subset \mathcal{M}$ be the image of $\mathcal{W}$ under the second projection. By **Step 1**, the non-empty fibers of $\mathcal{W} \to \mathcal{L}_{(2,2)}$ are of dimension 0.

Let $(\ell, [X]) \in \mathcal{W}$, and let $g \in J$. By Construction-Proposition 8.4 and Lemma 4.2, $g * \ell$ is the lower blocks of a presentation with the fixed blocks of a numerical Godeaux surface with canonical model isomorphic to X. By **Step 1**, $g * \ell \in \overline{\mathcal{L}^\circ}$ and the block A of $g * \ell$ is non-zero. Thus $J * \ell$ is contained in the fiber of $\mathcal{W} \to \mathcal{M}$ over $[X]$. Since Γ is unique, $J * \ell$ is the orbit modulo the gauge of Lemma 8.7, whose hypothesis $A \neq 0$ holds by the definition of $\mathcal{W}$. Since $\mathbf{p} = (2, 2)$ has $n = 2$ parts, Lemma 8.7 implies that $\dim J * \ell \geq 12 - n = 10$.

The first projection of $\mathcal{W}$ is contained in $\overline{\mathcal{L}^\circ}$. Thus, by Lemma 2.9 with $V = \mathcal{L}_{(2,2)}$, $W = \mathcal{M}$, $G = \mathcal{W}$, and $e = 10$, and by (10.5),

$$\dim \overline{\mathcal{M}'} \leq \dim \overline{\mathcal{L}^\circ} - 10 \leq 17 - 10 = 7.$$

It remains to prove that $\mathcal{M}_{(2,2)}^\circ \subset \mathcal{M}'$. Let $[X]$ be a point of $\mathcal{M}_{(2,2)}^\circ$, with a non-exceptional presentation. By Construction-Proposition 8.6 and Lemma 4.2, the gauge of the unique Γ moves the lower blocks to a point of $\mathcal{L}_{(2,2)}$ without changing A or the canonical model. Since the presentation stays non-exceptional, the point belongs to $\mathcal{L}^\circ$, and its block A is non-zero since $a_1 \neq 0$ by Construction-Proposition 10.2, so $[X] \in \mathcal{M}'$. Therefore, $\dim \overline{\mathcal{M}_{(2,2)}^\circ} \leq 7$. □

10.3. **Exceptional presentations and the bound.** In this subsection, we treat the exceptional presentations and prove Theorem 10.1.

After the normalization of the first stage, only two parameters remain, and the linear system has a nine-dimensional solution space. We therefore use the multiplication of R: multiplying the six relations of degree 6 by z_0 and eliminating the unknown products yields four linear conditions on the coordinates of the lower blocks, which cut the nine-dimensional space down to a five-dimensional space, and thus the bound $2 + 5 = 7$ holds.

Construction-Proposition 10.10. Notation and conditions as in Construction-Proposition 10.2. The first stage of any exceptional presentation is obtained by the action of Construction-Proposition 8.4 from the chart

$$(a_0, \ldots, a_{11}) = (t, s, -(e+1)s - pt, s, 2(e+1)s + 2pt, 0, 2s + 2t, -s - t, s, 0, s, es + pt)$$

with $e \in \mathbb{C}$ and $p \in \mathbb{C}^*$. For (e, p), let A be the first-stage matrix of the chart, and let $r := (r_1, \ldots, r_6)$. Then the following hold.

(1) We have $\dim \mathcal{U}_A = 9$.
(2) In the notation of Construction-Proposition 10.7, the functions
$$(q_0, \dots, q_8) := (x_{46}, x_{62}, x_{74}, x_{77}, x_{78}, x_{79}, x_{81}, x_{82}, x_{83})$$
are coordinates on $\mathcal{U}_A$, and any other x_n is a linear combination of $q_0, \dots, q_8$ with coefficients in $\mathbb{Q}[e, p, p^{-1}]$.
(3) For any point of $\mathcal{U}_A$, with blocks O, N, the equations
$$\Phi B_3^t = rO, \quad \Theta B_4 = \Phi P^t - rN, \quad vB_3 = -\Theta P, \quad \Psi B_3^t = vO - \Theta N^t \tag{10.6}$$
have a solution (Φ, Θ, v, Ψ), where Φ is a row of 8 forms of bidegree $(1,2)$ in (s,t) and $(y_0, \dots, y_3)$, Θ is a row of 12 forms of bidegree $(2,1)$, v is a row of 6 binary cubics, and Ψ is a row of 8 binary quartics.
(4) For any point of $\mathcal{U}_A$, the solution of (10.6) is unique once the free coefficients of Φ and Θ listed in Appendix A are set to 0.
(5) If a point of $\mathcal{U}_A$ is geometric, then it occurs in a presentation whose first row of d_1 is $(r + v, \Theta, \Phi + \Psi)$, where (Φ, Θ, v, Ψ) is the solution of (4).
(6) Any geometric point of $\mathcal{U}_A$ satisfies
$$q_1 + q_8/p = q_3 - 2q_8/p = q_5 + 2q_8 + q_8/p = q_2 + 2q_8/p^2 = 0.$$
In particular, the geometric points of $\mathcal{U}_A$ are contained in a linear subspace of dimension 5.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we normalize the first stage and obtain the chart.

Let (d_1, d_2) be an exceptional presentation, and let $b := a_5 - a_9$ and $d := a_9$ as in Construction-Proposition 10.5. By Lemma 10.4(3), $a_{10} = \kappa a_1$ with $\kappa \in \mathbb{C}^*$. Let $u \in \mathbb{P}^1$ be the zero of a_1. By Lemma 10.4(2)(3), the forms a_3, a_8, b, d vanish at u, $a_0(u)a_{11}(u) \neq 0$, and $\operatorname{rank} A(u) = 2$. Using the factor GL_2 of J, we may assume that $s = a_1$, and we write
$$a_3 = \beta s, \quad a_8 = \xi s, \quad b = \epsilon s, \quad d = \theta s$$
with $\beta, \xi, \epsilon, \theta \in \mathbb{C}$.

We show that $\beta\xi \neq 0$. Suppose that $\beta = 0$. Then the third equation of Construction-Proposition 10.2 is $\kappa s a_2 + \xi^2 s^2 = 0$, and thus $a_2(u) = 0$. Since a_3, a_8, b vanish at u, the k-th column of A is a zero column at u for $k = 2, 3$. Thus the span of the second and third basis vectors of $\mathbb{C}^6$ is a two-dimensional subspace of $\operatorname{Ker} A(u)$, and the corresponding quadrics $r_2 = y_1 y_2$ and $r_3 = y_1 y_3$ have the common linear factor y_1. This contradicts Lemma 4.3(2). Suppose that $\xi = 0$. Then the second equation of Construction-Proposition 10.2 is $s a_7 + \beta^2 s^2 = 0$, and thus $a_7(u) = 0$. Since a_3, a_8, a_9 vanish at u, the k-th column of A is a zero column at u for $k = 3, 5$. Thus the span of the third and fifth basis vectors of $\mathbb{C}^6$ is a two-dimensional subspace of $\operatorname{Ker} A(u)$, and the corresponding quadrics $r_3 = y_1 y_3$ and $r_5 = (y_2 - y_0) y_3$ have the common linear factor y_3. This contradicts Lemma 4.3(2).

Let $g_1 \in \mathcal{G}_{\mathrm{sh}}$ be the element with $\delta_1 = -\theta/\beta$ and $\delta_2 = \epsilon/\xi$. By Construction-Proposition 10.5(1), g_1 fixes a_1, a_{10}, a_3, a_8, it sends b to $b - \delta_2 a_8 = 0$ and d to $d + \delta_1 a_3 = 0$, and it does not change the values of a_0, a_{11} at u. Possibly replacing (d_1, d_2) with the image under g_1, we may assume that $b = d = 0$. Since $a_0(u) \neq 0$, we may also assume that $t = a_0$, using again the factor GL_2 of J.

Let $g_2 \in \mathcal{G}_{\mathrm{tor}}$ be an element with parameters $\alpha_1, \alpha_2, \rho_1, \rho_2, \chi, \omega_1, \omega_2$. By Construction-Proposition 10.5(2), g_2 multiplies $a_1, a_{10}, a_3, a_8, a_0, a_{11}$ by non-zero scalars, and it preserves $b = d = 0$. After the diagonal change of s, t that restores $a_1 = s$ and $a_0 = t$, the scalars β, ξ, κ are multiplied by
$$m_1 = \frac{\omega_2 \rho_1 \alpha_1}{\omega_1 \rho_2}, \quad m_2 = \frac{\rho_1 \alpha_1^2}{\rho_2 \alpha_2}, \quad m_3 = \frac{\omega_2 \rho_1^2 \alpha_1^2}{\omega_1 \rho_2^2 \alpha_2^2},$$
respectively. For any non-zero m_1, m_2, m_3, we take
$$\alpha_1 = \frac{m_1^2}{m_2}, \quad \alpha_2 = \frac{m_2^2}{m_1 m_3}, \quad \frac{\rho_1}{\rho_2} = \frac{m_2^5}{m_1^5 m_3}, \quad \frac{\omega_2}{\omega_1} = \frac{m_1^4 m_3}{m_2^4}.$$

Then $(\omega_2/\omega_1)^2 = \rho_2\alpha_1/(\rho_1\alpha_2)$, which is compatible with $\omega_i^2 = \chi\rho_i/\alpha_i$, and $\chi, \rho_2 \in \mathbb{C}^*$ are arbitrary. Taking $(m_1, m_2, m_3) = (1/\beta, 1/\xi, 1/\kappa)$, we may assume that $\beta = \xi = \kappa = 1$.

Now $a_1 = a_{10} = a_3 = a_8 = s$, $a_5 = a_9 = 0$, and $a_0 = t$. We write $a_{11} = es + pt$, and $p \neq 0$ since $a_{11}(u) \neq 0$. The image under J is a presentation with the fixed blocks (Construction-Proposition 8.4), so that the four equations of Construction-Proposition 10.2 hold, and they become

$$s(t + a_7 + s) = 0, \quad s(a_2 + es + pt + s) = 0, \quad s(a_4 + 2a_2) = 0, \quad s(a_6 + 2a_7) = 0.$$

Thus $a_7 = -s - t$, $a_2 = -(e+1)s - pt$, $a_4 = 2(e+1)s + 2pt$, and $a_6 = 2s + 2t$, which is the chart.

Step 2. In this step we prove (1) and (2).

By Construction-Proposition 10.7, $\mathcal{U}_A = \operatorname{Ker} T_{(2,2)}$, where $T_{(2,2)}$ is a 260×84 matrix. Substituting the chart, the entries of $T_{(2,2)}$ belong to $\mathbb{Q}[e, p, p^{-1}]$. Table 39 is a pivot table of $T_{(2,2)}$ with 75 triples, in the sense of Construction-Proposition 6.7. We apply Gauss–Jordan elimination over $\mathbb{Q}[e, p, p^{-1}]$ in the order of the pivot table. All pivot values are rational constants times powers of p, so that the elimination is valid for any $(e, p) \in \mathbb{C} \times \mathbb{C}^*$, and the vanishing of the rows without a pivot is a finite set of identities in $\mathbb{Q}[e, p, p^{-1}]$. A direct calculation shows that all rows without a pivot become zero, and that the columns without a pivot are the columns $46, 62, 74, 77, 78, 79, 81, 82, 83$. Thus $\operatorname{rank} T_{(2,2)} = 75$ and $\dim \mathcal{U}_A = 84 - 75 = 9$. Moreover, the rows with a pivot express any other x_n as a linear combination of $q_0, \dots, q_8$ with coefficients in $\mathbb{Q}[e, p, p^{-1}]$.

Step 3. In this step we prove (3), (4), and (5).

The systems of (10.6) are linear in the 160 coefficients of Φ, the 144 coefficients of Θ, the 24 coefficients of v, and the 40 coefficients of Ψ, respectively. By Tables 40, 41, 42, and 43, a direct calculation shows that the ranks are $136, 132, 24, 40$, respectively, and that any augmented row without a pivot is zero, for any (e, p) and any point of $\mathcal{U}_A$. Thus (10.6) has a solution, and the solution is unique once the free coefficients listed under Tables 40 and 41 are set to 0. We denote by (Φ, Θ, v, Ψ) the solution with free coefficients equal to 0.

By the self-dual minimal S_1-resolution of Construction-Proposition 8.2 and flat base change to $\mathbb{C}[s, t]$, the kernel of the map $\Phi \to \Phi B_3^t$ is the span of the three rows of B_2 with coefficients in $(S_0)_5$, and the dimension of the kernel is $3 \cdot 8 = 24 = 160 - 136$. Again by the self-dual minimal S_1-resolution of Construction-Proposition 8.2 and flat base change to $\mathbb{C}[s, t]$, the kernel of the map $\Theta \to \Theta B_4$ is the span of the four rows of B_1 with coefficients in $(S_0)_4$, and the dimension of the kernel is $4 \cdot 3 = 12 = 144 - 132$. The maps $v \to vB_3$ and $\Psi \to \Psi B_3^t$ are injective, since the ranks of the last two systems are 24 and 40.

Assume that the point is geometric, and let (d_1, d_2) be a presentation with the fixed blocks and with the lower blocks of the point. By Set-up 4.1, the columns of d_1 have degrees $6, 7, 8$, and thus the first row of d_1 is $(r + v', \Theta', \Phi' + \Psi')$, where Φ' is a row of 8 forms of bidegree $(1, 2)$, Θ' is a row of 12 forms of bidegree $(2, 1)$, v' is a row of 6 binary cubics, and Ψ' is a row of 8 binary quartics. The first row of $d_1 d_2 = 0$ is

$$\begin{aligned}(r + v')O - \Theta' N^t - (\Phi' + \Psi')B_3^t &= 0,\\ (r + v')N + \Theta' B_4 - (\Phi' + \Psi')P^t &= 0,\\ (r + v')B_3 + \Theta' P &= 0.\end{aligned}$$

Since $rB_3 = 0$ by Construction-Proposition 8.2, splitting by the degree in $y_0, \dots, y_3$ implies that $(\Phi', \Theta', v', \Psi')$ solves (10.6).

By the description of the first kernel, $\Phi' - \Phi = \sum_{i=0}^{2} f_i (B_2)_i$, where $f_i \in (S_0)_5$ and $(B_2)_i$ is the i-th row of B_2, counted from 0. The change of generators $w_i \to w_i + f_i$ subtracts f_i times the row of w_i of d_1 from the first row of d_1, and it does not change the lower blocks. Possibly replacing the generators w_i with $w_i + f_i$, we may assume that $\Phi' = \Phi$. Since $\Phi' = \Phi$, both Θ' and Θ solve the second system of (10.6), and thus $\Theta' - \Theta = \sum_{i=0}^{3} g_i (B_1)_i$ with $g_i \in (S_0)_4$, where

$(B_1)_i$ is the i-th row of B_1, counted from 0. The change of generators $z_i \to z_i + g_i$ subtracts g_i times the row of z_i of d_1 from the first row of d_1. It does not change Φ', since the entries of $g_i C_i$ are binary quartics, where C_i is the i-th row of C, and it does not change the lower blocks. Possibly replacing the generators z_i with $z_i + g_i$, we may assume that $\Theta' = \Theta$. By the injectivity of the last two maps, $v' = v$ and $\Psi' = \Psi$. Thus the first row of d_1 is $(r + v, \Theta, \Phi + \Psi)$. The free coefficients set to 0 correspond to the $24 + 12$ changes of generators above.

Step 4. In this step we prove (6).

Let (d_1, d_2) be a presentation of a geometric point as in (5), with $d_1 : F_1 \to F_0$. The cokernel of d_1 is R. Multiplication by z_0 on R is S_0-linear, so that multiplying the relations of Set-up 4.1(7) by z_0 implies

$$(r_k + v_k)z_0 + \sum_{j=0}^{3} A_{jk} z_0 z_j = 0 \quad \text{in } R_{10} \qquad (1 \le k \le 6). \tag{10.7}$$

Let $\zeta_j \in (F_0)_8$ be a lift of $z_0 z_j$ for $0 \le j \le 3$. We also write z_0 for the basis vector of F_0 of the generator z_0. Then (10.7) implies

$$\Big((r_k + v_k)z_0 + \sum_{j=0}^{3} A_{jk}\zeta_j\Big)_{1\le k\le 6} \in \bigoplus_{k=1}^{6} \big(d_1(F_1) \cap (F_0)_{10}\big).$$

Table 45, the table of linear functionals, defines four linear functionals $\ell_1, \dots, \ell_4$ on $\bigoplus_{k=1}^{6}(F_0)_{10}$ through the reduction of $(F_0)_{10}$ modulo the relations of degree 10, which is computed by Table 44. A direct calculation over $\mathbb{Q}[e, p, p^{-1}, q_0, \dots, q_8]$ shows the following for $1 \le i \le 4$. The functional ℓ_i is zero on $\bigoplus_{k=1}^{6}(d_1(F_1) \cap (F_0)_{10})$. The functional ℓ_i is zero on $(\sum_{j=0}^{3} A_{jk}\zeta_j)_{1\le k\le 6}$ for any $\zeta_0, \dots, \zeta_3 \in (F_0)_8$ (by the pivot table of degree 8 in Table 44, any ζ_j is congruent modulo $d_1(F_1)_8$ to a combination H_j of the basis vectors of $(F_0)_8$ at the columns without pivot, and $A_{jk} d_1(F_1)_8 \subset d_1(F_1)_{10}$, on which ℓ_i is zero). Finally, the value of ℓ_i on $((r_k + v_k)z_0)_{1\le k\le 6}$ is L_i, where

$$L_1 = q_1 + \frac{q_8}{p}, \quad L_2 = q_3 - \frac{2q_8}{p}, \quad L_3 = q_5 + 2q_8 + \frac{q_8}{p}, \quad L_4 = q_2 + \frac{2q_8}{p^2}.$$

Thus $L_1 = L_2 = L_3 = L_4 = 0$ at any geometric point. The coefficients of q_1, q_3, q_5, q_2 in L_1, L_2, L_3, L_4 form the identity matrix, and thus the subspace of $\mathcal{U}_A$ defined by the vanishing of L_1, L_2, L_3, L_4 is linear of dimension $9 - 4 = 5$, with coordinates q_0, q_4, q_6, q_7, q_8. □

Proposition 10.11. *We have* $\dim \overline{\mathcal{M}^{\mathrm{exc}}_{(2,2)}} \le 7$.

Proof. Let $\mathcal{L}_{(2,2)}$ be the affine space of Construction-Proposition 8.6, with the fixed blocks of Table 11, and fix the quadrics $r_1, \dots, r_6$ of Construction-Proposition 10.2. Let $V \subset \mathcal{L}_{(2,2)}$ be the image of the map

$$\mathbb{C} \times \mathbb{C}^* \times \mathbb{C}^5 \to \mathcal{L}_{(2,2)}, \qquad (e, p, q_0, q_4, q_6, q_7, q_8) \to (d_1', d_2),$$

where (d_1', d_2) is determined by the chart and by the point of $\mathcal{U}_A$ with coordinates q_0, q_4, q_6, q_7, q_8 and

$$q_1 = -\frac{q_8}{p}, \quad q_3 = \frac{2q_8}{p}, \quad q_5 = -2q_8 - \frac{q_8}{p}, \quad q_2 = -\frac{2q_8}{p^2}.$$

The map is a morphism for the following reasons. The blocks P, U are the solutions of linear systems with constant coefficients in terms of A (Construction-Proposition 8.2). The coefficients $x_0, \dots, x_{83}$ are regular functions (Construction-Proposition 10.10(2)). The block N and the other twelve entries of C are determined by a minor of non-zero constant determinant (Construction-Proposition 10.7). Thus V is constructible [Har77, Exercise II.3.19], and $\dim V \le 2 + 5 = 7$.

Let $[X] \in \mathcal{M}^{\mathrm{exc}}_{(2,2)}$. By Definition 10.3, $[X]$ has an exceptional presentation. By Construction-Propositions 10.10 and 8.4, we may assume that the first stage is the chart for some (e, p), without changing the isomorphism class of the canonical model. By Lemma 4.2 and Construction-Proposition 8.6, a gauge moves O to the twelve slots, without changing A, the quadrics r_k, or

the canonical model. Then the lower blocks of the presentation are determined by A and a geometric point of $\mathcal{U}_A$. By Construction-Proposition 10.10(6), the lower blocks belong to V. Thus $\mathcal{M}^{\mathrm{exc}}_{(2,2)} \subset \mathcal{M}(V)$, for the set $\mathcal{M}(V)$ of Lemma 4.5(2) with $\mathcal{L} = \mathcal{L}_{(2,2)}$ and this V, and by Lemma 4.5(2),

$$\dim \overline{\mathcal{M}^{\mathrm{exc}}_{(2,2)}} \leq \dim \overline{\mathcal{M}(V)} \leq \dim V \leq 7.$$

□

Proof of Theorem 10.1. By Construction-Proposition 10.2 and Definition 10.3,

$$\mathcal{M}_{(2,2)} = \mathcal{M}^{\circ}_{(2,2)} \cup \mathcal{M}^{\mathrm{exc}}_{(2,2)}.$$

Thus Theorem 10.1 follows from Propositions 10.9 and 10.11. □

11. The collision type $(3,1)$

In this section, we prove the following theorem.

Theorem 11.1. *We have* $\dim \overline{\mathcal{M}_{(3,1)}} \leq 7$.

We first normalize the first stage, then bound the rank of a linear system on the normal forms, and prove Theorem 11.1 at the end of this section.

Construction-Proposition 11.2. Notation and conditions as in Set-up 8.1 with $\mathbf{p} = (3,1)$. Choose the basis (e, z, z^2, p) of $\mathcal{A} = \mathbb{C}[z]/(z^3) \times \mathbb{C}$, where e, p are the idempotents, and normalize $\lambda(z^2) = \lambda(p) = 1$ and $\lambda(e) = \lambda(z) = 0$. Then S has a presentation with $y_0, \dots, y_3$ restricting to $(e\tau^3, z\tau^3, z^2\tau^3, p\tau^3)$ and $z_0, \dots, z_3$ restricting to $(e\tau^4, z\tau^4, z^2\tau^4, p\tau^4)$,

$$(r_1, \dots, r_6) = (y_1^2 - y_0y_2, y_1y_2, y_2^2, y_0y_3, y_1y_3, y_2y_3),$$

$$A = \begin{pmatrix} t_7 - t_8 & t_9 & 0 & t_{10} & 0 & 0 \\ -t_3 & t_7 & t_9 & -t_4 & t_{10} & 0 \\ t_{10} - t_2 & t_6 & t_8 & t_0 & -t_4 & t_{10} \\ 0 & 0 & 0 & t_1 & -t_5 & t_{11} \end{pmatrix},$$

where $t_0, \dots, t_{11}$ are binary linear forms with $q_1 = q_2 = q_3 = q_4 = 0$ for

$$q_1 := t_0t_7 - t_0t_8 - t_1t_{11} + t_{10}^2 - t_{10}t_2 + t_3t_4 + t_5^2,$$
$$q_2 := t_0t_9 + t_{10}t_6 - t_{11}t_5 - t_4t_7,$$
$$q_3 := t_{10}t_8 + t_{11}^2 - t_4t_9,$$
$$q_4 := -t_3t_9 - 2t_6t_9 - t_7^2 + t_7t_8 - t_8^2.$$

The blocks $B_1, \dots, B_4$ are the fixed blocks of Table 14, and P, U are determined by A. Moreover, $t_9 \neq 0$, the forms t_8, t_9 are linearly independent, and t_9 defines the member of $|2K_X|$ that is singular at the point of length 3 of B.

Proof of Construction-Proposition 11.2. The Gram matrix of (e, z, z^2, p) for λ is

$$\Lambda = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

so the pairing of λ is non-degenerate. By Construction-Theorem 3.2(4), λ has the form $a \to \lambda_0(g^5 a)$ for a unit $g \in \mathcal{A}$, where λ_0 is the functional of Set-up 8.1(4). Thus, possibly replacing τ with $g\tau$, we may assume that the functional of Set-up 8.1(4) is λ.

We apply Construction-Proposition 8.2 to $(b_0, b_1, b_2, b_3) = (e, z, z^2, p)$. Since $1 = e + p$, the elements $1, z, z^2, p$ form a basis of $\mathcal{A}$, and $y_* = y_0 + y_3$. The linear map from the quadrics in $y_0, \dots, y_3$ to $\mathcal{A}$ sending y_iy_k to b_ib_k is surjective, so the kernel is of dimension $10 - 4 = 6$. Thus $r_1, \dots, r_6$, which are linearly independent elements of the kernel, form a basis of the kernel. The

TABLE 14. The table lists the fixed blocks $B_1, \dots, B_4$ and the matrices P, U determined by A, where the entries of P, U are linear in the t_j and those of $B_1, \dots, B_4$ are linear in the y_i.

$$B_1 = \begin{pmatrix} y_1 & 0 & 0 & 0 & y_2 & 0 & 0 & 0 & y_3 & 0 & 0 & 0 \\ -y_0-y_3 & y_1 & 0 & 0 & 0 & y_2 & 0 & 0 & 0 & y_3 & 0 & 0 \\ 0 & -y_0-y_3 & y_1 & 0 & -y_0-y_3 & 0 & y_2 & 0 & 0 & 0 & y_3 & 0 \\ 0 & 0 & 0 & y_1 & 0 & 0 & 0 & y_2 & 0 & 0 & 0 & -y_0 \end{pmatrix}$$

$$B_2 = \begin{pmatrix} 0 & 0 & y_2 & 0 & -y_0 & -y_1 & y_0 & 0 \\ y_3 & 0 & 0 & y_2 & -y_1 & -y_2 & 0 & y_0 \\ 0 & y_3 & 0 & 0 & 0 & 0 & -y_2 & y_1 \end{pmatrix}, \quad B_3 = \begin{pmatrix} y_2 & 0 & 0 & -y_3 & 0 & 0 & 0 & 0 \\ -y_1 & -y_2 & 0 & 0 & -y_3 & 0 & -y_3 & 0 \\ y_0 & y_1 & 0 & 0 & 0 & 0 & 0 & -y_3 \\ 0 & 0 & -y_1 & -y_2 & 0 & -y_2 & 0 & 0 \\ 0 & 0 & y_0 & y_1 & y_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & y_0 & y_1 & y_2 \end{pmatrix}$$

$$B_4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & y_3 & 0 & 0 & 0 & -y_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & y_3 & 0 & 0 & 0 & -y_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & y_3 & 0 & 0 & 0 & -y_2 & 0 & y_0+y_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -y_0 & 0 & 0 & 0 & -y_2 \\ 0 & 0 & -y_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1 & 0 \\ 0 & -y_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1 & -y_0-y_3 & 0 \\ -y_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1 & -y_0-y_3 & 0 & 0 \\ 0 & 0 & 0 & y_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1 \\ 0 & 0 & y_2 & 0 & 0 & 0 & -y_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & y_2 & 0 & 0 & 0 & -y_1 & y_0+y_3 & 0 & 0 & 0 & 0 & 0 \\ y_2 & 0 & -y_0-y_3 & 0 & -y_1 & y_0+y_3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & y_2 & 0 & 0 & 0 & -y_1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$P = \begin{pmatrix} t_9 & 0 & t_{10} & 0 & 0 & 0 & 0 & 0 \\ t_7 & -t_9 & -t_4 & -t_{10} & 0 & 0 & 0 & 0 \\ t_6 & -t_8 & t_0 & t_4 & 0 & 0 & -t_{10} & 0 \\ 0 & 0 & t_1 & t_5 & 0 & 0 & -t_{11} & 0 \\ -t_7+t_8 & t_9 & 0 & t_{10} & 0 & t_{10} & 0 & 0 \\ t_3 & t_7 & 0 & -t_4 & -t_{10} & -t_4 & 0 & 0 \\ -t_{10}+t_2 & t_6 & 0 & t_0 & t_4 & t_0 & 0 & -t_{10} \\ 0 & 0 & 0 & t_1 & t_5 & t_1 & 0 & -t_{11} \\ 0 & 0 & 0 & t_7-t_8 & t_9 & 0 & t_9 & 0 \\ t_9 & 0 & t_{10} & -t_3 & t_7 & 0 & t_7 & t_9 \\ t_8 & 0 & -t_4 & t_{10}-t_2 & t_6 & t_{10} & t_6 & t_8 \\ 0 & 0 & -t_5 & 0 & 0 & t_{11} & 0 & 0 \end{pmatrix}$$

$$U = \begin{pmatrix} t_4 & -t_{10} & 0 & t_5 & -t_{10} & 0 & 0 & -t_{11} & t_0 & -t_4 & t_{10} & t_1 \\ t_2 & t_3 & -t_7+t_8 & t_{11} & -t_6 & -t_7 & -t_9 & 0 & t_4 & -t_{10} & 0 & t_5 \\ t_6 & t_7 & t_9 & 0 & t_8 & t_9 & 0 & 0 & t_{10} & 0 & 0 & t_{11} \end{pmatrix}$$

blocks B_1, B_4 of Table 14 are the formulas of Construction-Proposition 8.2 with $y_* = y_0 + y_3$, with the matrix Λ above, and with M_j having k-th column the coordinates of $b_j b_k$. By direct multiplication, $(r_1, \dots, r_6)B_3 = 0$ and $B_3 B_2^t = 0$. Modulo y_*, the quadrics $r_1, \dots, r_6$ reduce to $y_1^2 + y_2 y_3$, $y_1 y_2$, y_2^2, $-y_3^2$, $y_1 y_3$, $y_2 y_3$, which form a basis of $(y_1, y_2, y_3)^2$. The minimal resolution of $\mathbb{C}[y_1, y_2, y_3]/(y_1, y_2, y_3)^2$ has Betti numbers $(1, 6, 8, 3)$ [Eis95, Theorem A2.10, Corollary A2.12], and modulo y_* the eight columns of B_3 and the three rows of B_2 are linearly independent, so their reductions are the linear first and second syzygies of $(y_1, y_2, y_3)^2$. Thus the hypotheses of Construction-Proposition 8.2 hold. Let $y_i \in R_3$ and $z_i \in R_4$ be elements restricting to $b_i \tau^3$ and $b_i \tau^4$ (Construction-Theorem 3.2(2)). By Lemma 8.3, S has a presentation with the generators y_i, z_i, with the quadrics $r_1, \dots, r_6$, and with the blocks $B_1, \dots, B_4$ of Table 14.

Since $d_1 d_2 = 0$ (Theorem 2.10), the blocks of $d_1' d_2 = 0$ imply

$$AB_3 + B_1 P = 0, \qquad UB_4 - B_2 P^t = 0, \qquad UP = 0. \tag{11.1}$$

Taking the coefficients of $y_0, \dots, y_3$, a direct calculation shows that the linear system on the entries of A, P, U given by $AB_3 + B_1P = 0$ and $UB_4 - B_2P^t = 0$ has solution space parametrized by $t_0, \dots, t_{11}$ as in A and Table 14: the vector of coefficients of s and the vector of coefficients of t are both solutions of one linear system, with 156 unknowns and of rank 144. A direct calculation also shows that

$$UP = \begin{pmatrix} 0 & 0 & 0 & q_1 & q_2 & 0 & q_2 & q_3 \\ 0 & q_4 & -q_1 & -q_2 & 0 & -q_2 & -q_3 & 0 \\ -q_4 & 0 & q_2 & 0 & 0 & q_3 & 0 & 0 \end{pmatrix}.$$

Thus $q_1 = q_2 = q_3 = q_4 = 0$ by (11.1). By Construction-Proposition 8.2, P, U are determined by A.

Let x be the point of length 3 of B, $\mathfrak{m}$ the maximal ideal of $\mathcal{O}_{X,x}$, and τ a local generator of $\mathcal{O}_X(K_X)$ at x restricting to the generator of $\mathcal{O}_B(K_X)$. Let $\bar{s} := s/\tau^2$, $\bar{t} := t/\tau^2$, $\bar{y}_i := y_i/\tau^3$, $\bar{z}_j := z_j/\tau^4$, and $\mathfrak{a} := (\bar{s}, \bar{t})$, the ideal of B at x. By Construction-Proposition 3.3(2), $\mathcal{O}_{X,x}$ is a regular local ring of dimension 2 and $\mathcal{O}_{X,x}/\mathfrak{a} = \mathbb{C}[z]/(z^3)$, which has embedding dimension 1, so $\mathfrak{a} \not\subset \mathfrak{m}^2$. Moreover, $\bar{y}_1$ restricts to z and $\bar{y}_2 - \bar{y}_1^2 \in \mathfrak{a}$. Let $\mu \in \mathfrak{a} \backslash \mathfrak{m}^2$. Then $\mathfrak{m} = (\mu, \bar{y}_1)$, and since $\mathcal{O}_{X,x}/\mathfrak{a}$ is of length 3, we have $\mathfrak{a} = (\mu, \bar{y}_1^3)$. Since $\mathfrak{a}$ is $\mathfrak{m}$-primary, it is not principal by Krull's height theorem [Mat89, Theorem 13.5], so the class of $\bar{y}_1^3$ in $\mathfrak{a}/\mathfrak{m}\mathfrak{a}$ is non-zero by Nakayama's lemma. Since $z_0, \dots, z_3$ lift $(e, z, z^2, p)\tau^4$ (Lemma 8.3), z_0 is a unit at x, and z_1, z_2, z_3 vanish at x.

The relation of r_2 is

$$y_1y_2 + t_9z_0 + t_7z_1 + t_6z_2 + v_2(s,t) = 0.$$

Divide by τ^6 and work modulo $\mathfrak{m}\mathfrak{a}$. We have $\bar{y}_1\bar{y}_2 \equiv \bar{y}_1^3$, the terms $t_7(\bar{s}, \bar{t})\bar{z}_1$ and $t_6(\bar{s}, \bar{t})\bar{z}_2$ belong to $\mathfrak{m}\mathfrak{a}$, and

$$v_2(\bar{s}, \bar{t}) \in \mathfrak{a}^3 \subset \mathfrak{m}\mathfrak{a}.$$

Thus

$$t_9(\bar{s}, \bar{t}) \equiv -\bar{z}_0^{-1}\bar{y}_1^3 \mod \mathfrak{m}\mathfrak{a}.$$

Therefore, $t_9 \neq 0$ and $t_9(\bar{s}, \bar{t}) \in \mathfrak{m}^2$. Since $\mathfrak{a} = (\bar{s}, \bar{t}) \not\subset \mathfrak{m}^2$, any binary linear form independent of t_9, evaluated at $(\bar{s}, \bar{t})$, is outside $\mathfrak{m}^2$. Hence t_9 defines the member of $|2K_X|$ that is singular at x.

It remains to show that t_8, t_9 are linearly independent. Suppose that t_8, t_9 are linearly dependent, and let $u \in \mathbb{P}^1$ be the zero of t_9. Then $t_8(u) = 0$. The values of q_4, q_3 at u imply $t_7(u) = t_{11}(u) = 0$. Dividing q_4 by t_9 and taking the value at u, we have $t_3(u) = -2t_6(u)$. The value of q_2 at u implies $t_{10}(u)t_6(u) = 0$.

If $t_{10}(u) \neq 0$, then $t_6(u) = t_3(u) = 0$, so the columns of r_2, r_3 in A are zero columns at u, and the columns of r_1, r_6 of $A(u)$ are proportional and supported in the third row. If $t_{10}(u) = 0$, then the first row of $A(u)$ vanishes and the columns of r_3, r_6 in A are zero columns at u. In both cases, $\operatorname{rank} A(u) \leq 3$, and the basis vectors of $\mathbb{C}^6$ at the columns of r_2, r_3, respectively r_3, r_6, span a two-dimensional subspace $W \subset \operatorname{Ker} A(u)$ such that the quadrics $\sum_k c_k r_k$ with $c \in W$ have the common factor y_2. This contradicts Lemma 4.3(2). □

The next result lists the first stages of type $(3,1)$ up to the action of J.

Construction-Proposition 11.3. Notation and conditions as in Construction-Proposition 11.2. In Table 15, ζ runs over the roots of $\zeta^2 - \zeta + 1 = 0$, η runs over the roots of $\eta^2 + (\zeta - 1)\eta - 1 = 0$, (α, ρ, κ) runs over $\mathbb{C}^3$, (h_2, k_2) runs over $\mathbb{C}^2$, and

$$\theta_0 := (1-\alpha)^2 + \rho^2, \quad \theta_1 := \rho(-(\zeta+1)\alpha - 2\zeta) + 2\alpha - 2, \quad \theta_2 := \rho(\zeta - 1) + \alpha + 1 + (\zeta - 1)(\alpha + 1)^2,$$

$$\ell := (\theta_0 - \kappa\alpha)s + (\theta_1 - \kappa - \theta_2\alpha)t.$$

After the action of J of Construction-Proposition 8.4, the first stage of S belongs to one of the families of Table 15, of dimensions at most 3.

TABLE 15. The families of normal forms of type $(3,1)$.

Family	Constants	Parameters	$(t_0,\dots,t_{11})$
(G)	ζ	(α,ρ,κ)	$(\alpha\rho s+(\rho-\zeta\alpha)t,\ \ell+\kappa s+\theta_2 t,\ \ell,\ 0,\ \alpha^2 s+(\alpha+1)t,$ $\rho s-\zeta(\alpha+1)t,\ 0,\ \zeta t,\ t,\ s,\ (1-\alpha)s-t,\ \alpha s+t)$
(V_0)	(ζ,η)	(h_2,k_2)	$((\eta+\zeta)t,\ 0,\ h_2 s+k_2 t,\ 0,\ s,\ \eta t,\ 0,\ \zeta t,\ t,\ s,\ 0,\ s)$
(V_1)	(ζ,η)	(h_2,k_2)	$(s+(\eta+\zeta)t,\ s+(\zeta-1+2\eta)t,\ h_2 s+k_2 t,\ 0,\ s,\ s+\eta t,$ $0,\ \zeta t,\ t,\ s,\ 0,\ s)$

Proof of Construction-Proposition 11.3. The proof is long, so we divide it into several steps.

Step 1. In this step we compute the fields which we use to normalize the first stage.

By Construction-Proposition 8.4, J acts on the presentations with the fixed blocks and preserves the isomorphism class of X. We denote by D,V,W the derivatives at $\epsilon=0$ of $(t_0,\dots,t_{11})$ under g_ϵ of Lemma 8.5 with $(\delta,a)=(z^2\frac{\partial}{\partial z},0)$, $(0,z)$, $(0,z^2)$, respectively, where z is taken on the factor of length 3.

A direct calculation of L_zA-AL_r in Lemma 8.5 shows that the derivatives of $(t_0,\dots,t_{11})$ are

$$D=(-\tfrac12 t_4,\ 0,\ \tfrac12 t_3+2t_6,\ \tfrac12(5t_7-t_8),\ \tfrac12 t_{10},\ t_{11},\ \tfrac12 t_7-t_8,\ -\tfrac32 t_9,\ \tfrac12 t_9,\ 0,\ 0,\ 0),$$

$$V=(\tfrac12 t_4,\ t_5,\ \tfrac12 t_3+t_6,\ \tfrac12(t_7+t_8),\ \tfrac12 t_{10},\ t_{11},\ \tfrac12 t_7-t_8,\ -\tfrac12 t_9,\ \tfrac12 t_9,\ 0,\ 0,\ 0),$$

$$W=(-\tfrac12 t_{10},\ -t_{11},\ -\tfrac12(t_7+t_8),\ -t_9,\ 0,\ 0,\ \tfrac12 t_9,\ 0,\ 0,\ 0,\ 0,\ 0).$$

Under g_ϵ, the tuple transforms polynomially in ϵ, and the first-stage identities are preserved (Lemma 8.5).

The components of D,V,W are linear in $t_0,\dots,t_{11}$, so D,V,W are linear operators on the tuple. They are nilpotent, because they send any t_j to a combination of forms t_k with $k>j$. The elements $(\phi,v,1,w)$ of $\mathcal{G}^0$ with $\phi(z)-z\in(z^2)$ and with $v-1,w-1$ nilpotent form a unipotent subgroup $\mathcal{G}^{\mathrm{u}}$ of $\mathcal{G}^0$, which acts linearly on the tuple. The matrices L_y and L_z are linear in (δ,a), and so is L_r, which is induced by L_y on the quadrics $r_1,\dots,r_6$. Thus for any constant combination of D,V,W, the exponential of ϵ times the combination is the action of the one-parameter subgroup of $\mathcal{G}^{\mathrm{u}}$ whose derivative at $\epsilon=0$ equals the derivative of g_ϵ for the corresponding (δ,a), since the map $g\to(A\to G_zAG_r^{-1})$ is a homomorphism of algebraic groups: by Construction-Proposition 8.4, it is the action on the block A of the algebraic action of J in Construction-Proposition 8.6(4), which the gauge does not change (Lemma 4.2).

Step 2. In this step we normalize t_3,t_6,t_7,t_8,t_9.

Since $t_9\neq0$, possibly replacing s,t by an element of GL_2, we may assume that $t_9=s$. Write $t_7=h_7s+k_7t$ and $t_8=h_8s+k_8t$. Since t_8,t_9 are linearly independent, $k_8\neq0$. The value of $q_4=0$ at $s=0$ is

$$k_7^2-k_7k_8+k_8^2=0,$$

so $k_7=\zeta k_8$ with $\zeta^2-\zeta+1=0$. The element $\exp(\epsilon_1D)\exp(\epsilon_2V)$ fixes t_9,k_7,k_8, and translates (h_7,h_8) by

$$(-\tfrac32\epsilon_1-\tfrac12\epsilon_2,\ \tfrac12\epsilon_1+\tfrac12\epsilon_2),$$

where

$$\det\begin{pmatrix}-\frac32 & -\frac12\\ \frac12 & \frac12\end{pmatrix}=-\frac12\neq0.$$

Possibly replacing the presentation with the image under such an element and scaling t, we may assume that

$$t_7=\zeta t,\quad t_8=t,\quad t_9=s.$$

Let $\Psi:=(\zeta+1)D-(\zeta+3)V$, which corresponds to

$$(\delta,a)=\big((\zeta+1)z^2\tfrac{\partial}{\partial z},\ -(\zeta+3)z\big).$$

Since $\delta(a) = -(\zeta+1)(\zeta+3)z^2 \neq 0$, the elements g_ϵ of (δ, a) do not form a one-parameter subgroup, so we use $\exp(\epsilon\Psi)$ of **Step 1** in place of g_ϵ. The values of Ψ on t_7, t_8, t_9 are $-\zeta s$, $-s$, 0, and

$$\Psi(t_6) = -t_7 + 2t_8 = (2-\zeta)t,$$

while $\Psi^2(t_6)$ is a multiple of s. Thus the composition of $\exp(\epsilon\Psi)$ with the shear $t \to t + \epsilon s$ preserves t_7, t_8, t_9, and translates the coefficient of t in t_6 by $(2-\zeta)\epsilon$, because the shear does not change the coefficients of t. Since $\zeta^2 - \zeta + 1 = 0$, we have $\zeta \neq 2$, so we may assume that the coefficient of t in t_6 is 0. Next, $\exp(\epsilon W)$ preserves t_7, t_8, t_9, and translates the coefficient of s in t_6 by $\frac{1}{2}\epsilon$, so we may assume that $t_6 = 0$. Then

$$q_4 = -t_3 s - (\zeta^2 - \zeta + 1)t^2 = -t_3 s,$$

hence $t_3 = 0$.

Step 3. In this step we solve $q_2 = q_3 = 0$ and we describe the remaining diagonal action.

Write $t_{11} = \alpha s + \beta t$. Then $q_3 = t_{10}t + t_{11}^2 - t_4 s = 0$ implies that there exists $\gamma \in \mathbb{C}$ with

$$t_{10} = \gamma s - \beta^2 t, \qquad t_4 = \alpha^2 s + (\gamma + 2\alpha\beta)t.$$

Write $t_5 = \rho s + \rho' t$. Then $q_2 = t_0 s - t_{11}t_5 - \zeta t_4 t = 0$ implies

$$\gamma = -\frac{\beta\rho'}{\zeta} - 2\alpha\beta, \qquad t_0 = \alpha\rho s + (\alpha\rho' + \beta\rho + \zeta\alpha^2)t. \tag{11.2}$$

Finally, $q_1 = 0$ becomes

$$t_1 t_{11} + t_2 t_{10} = (\zeta - 1)t t_0 + t_{10}^2 + t_5^2. \tag{11.3}$$

The map $(\phi, \chi, w) \to (\phi, w^2(\chi b_\phi)^{-1}, \chi, w)$ is an isomorphism of varieties from $\mathrm{Aut}(\mathcal{A}) \times \mathbb{C}^* \times \mathcal{A}^*$ onto $\mathcal{G}$. Here $\mathrm{Aut}(\mathcal{A}) = \mathrm{Aut}(\mathbb{C}[z]/(z^3))$ and $\mathcal{A}^*$ are connected, so $\mathcal{G} = \mathcal{G}^0$, and the elements of $\mathcal{G}$ below, whose components involve a square root $\mu_2^{1/2}$, belong to $\mathcal{G}^0$.

Let $\mu_1, \mu_2 \in \mathbb{C}^*$, and let $\mu_2^{1/2}$ be a square root of μ_2. Consider the element of $\mathcal{G}$ with $\phi(z) = \mu_1 z$, with $\chi = 1$, with $v = 1$ and $w = \mu_1^{-1}$ on the factor of length 3, and with $v = \mu_2$ and $w = \mu_2^{1/2}$ on the simple factor, for which $b_\phi = \mu_1^{-2}e + p$. It multiplies (y_0, y_1, y_2, y_3) by $(1, \mu_1, \mu_1^2, \mu_2)$ and (z_0, z_1, z_2, z_3) by $(\mu_1^{-1}, 1, \mu_1, \mu_2^{1/2})$. Thus it multiplies t_9, t_8, t_7 by $\mu_1^{-4}, \mu_1^{-3}, \mu_1^{-3}$, it multiplies t_{11}, t_5 by $\mu_1^{-2}\mu_2^{-1/2}, \mu_1^{-1}\mu_2^{-1/2}$, and it preserves $t_3 = t_6 = 0$. The diagonal change of s, t that restores $t_9 = s$, $t_8 = t$, $t_7 = \zeta t$ multiplies the coefficients of s by μ_1^4 and the coefficients of t by μ_1^3. Hence the combined action is

$$(\beta, \alpha, \rho', \rho) \to (\mu_1\mu_2^{-1/2}\beta,\ \mu_1^2\mu_2^{-1/2}\alpha,\ \mu_1^2\mu_2^{-1/2}\rho',\ \mu_1^3\mu_2^{-1/2}\rho). \tag{11.4}$$

All equations of this step hold for the new coefficients, since they were derived from $q_1 = q_2 = q_3 = 0$ and from the normalized forms t_3, t_6, t_7, t_8, t_9.

Step 4. In this step we prove the following statement for the presentations of this proof:

(11.5) if the columns of r_5 and r_6 in A are zero columns at $u \in \mathbb{P}^1$, then $t_9(u) = 0$.

Let $P_0 := [1 : 0 : 0 : 0]$. By Construction-Proposition 11.2, $y_0, \ldots, y_3$ restrict to $(e\tau^3, z\tau^3, z^2\tau^3, p\tau^3)$, so P_0 is the image of the point of length 3 of B, and t_9 defines the member singular at that point. The binary changes used in this proof after $t_9 = s$ (the shear, the scaling of t, and the diagonal changes that restore $t_9 = s, t_8 = t, t_7 = \zeta t$) multiply s by a non-zero constant, and D, V, W have derivative 0 on t_9, so t_9 still defines the member singular at the point of length 3. The change $y(a) \to y(v\phi(a))$ preserves each factor of $\mathcal{A}$, so P_0 is still the image of that point.

Assume that the columns of r_5 and r_6 in A are zero columns at u. Then the fifth and sixth basis vectors of $\mathbb{C}^6$ belong to $\mathrm{Ker}\, A(u)$, and H_u has codimension at most one in $\mathrm{Ker}\, A(u)$, so H_u contains a non-zero element c with $\sum_k c_k r_k = y_3(\xi_1 y_1 + \xi_2 y_2)$. The quadric $y_3(\xi_1 y_1 + \xi_2 y_2)$

is reducible and singular at P_0. By Lemma 4.3(3), C_u is singular at the point of length 3, so $t_9(u) = 0$, since t_9 defines the unique member singular at that point.

Step 5. In this step we consider Case (G): $\beta \neq 0$ and $\Delta \neq 0$, where $\Delta := -\rho'/\zeta - \alpha$.

By (11.4), β and Δ are multiplied by $\mu_1\mu_2^{-1/2}$ and $\mu_1^2\mu_2^{-1/2}$, so we may assume that $\beta = \Delta = 1$. Then $\rho' = -\zeta(\alpha+1)$, and $\gamma = 1-\alpha$ by (11.2), so

$$t_{10} = (1-\alpha)s - t, \quad t_{11} = \alpha s + t, \quad t_4 = \alpha^2 s + (\alpha+1)t,$$
$$t_5 = \rho s - \zeta(\alpha+1)t, \quad t_0 = \alpha\rho s + (\rho - \zeta\alpha)t.$$

Using $\zeta^2 = \zeta - 1$, a direct calculation shows that the right side of (11.3) is $\theta_0 s^2 + \theta_1 st + \theta_2 t^2$. Since $t_{10} + t_{11} = s$, the left side of (11.3) is $st_2 + (t_1 - t_2)(\alpha s + t)$. The coefficient of t^2 implies that $t_1 - t_2 = \kappa s + \theta_2 t$ for some $\kappa \in \mathbb{C}$, and the coefficients of s^2 and st imply that $t_2 = \ell$. Thus the first stage belongs to the family (G).

Step 6. In this step we exclude Case (Z): $\beta \neq 0$ and $\Delta = 0$.

By (11.4), we may assume that $\beta = 1$. Then $\rho' = -\zeta\alpha$ and $\gamma = -\alpha$. Let $\ell_0 := \alpha s + t$. Then $t_{11} = -t_{10} = \ell_0$, so the left side of (11.3) is $(t_1 - t_2)\ell_0$. Moreover, $t_5 = \rho s - \zeta\alpha t$ and $t_0 = \rho\ell_0$, so the value of the right side of (11.3) at $t = -\alpha s$ is $(\rho + \zeta\alpha^2)^2 s^2$. Hence $\rho = -\zeta\alpha^2$, and

$$t_4 = \alpha\ell_0, \quad t_5 = -\zeta\alpha\ell_0, \quad t_0 = -\zeta\alpha^2\ell_0.$$

Let u be the root of ℓ_0. The non-zero entries of the columns of r_5 and r_6 in A are among $\pm t_4, \pm t_5, t_{10}, t_{11}$, so the columns of r_5 and r_6 in A are zero columns at u, while $t_9(u) = s(u) \neq 0$. This contradicts (11.5).

Step 7. In this step we consider Case (V): $\beta = 0$.

By (11.2), $\gamma = 0$, so

$$t_{10} = 0, \quad t_{11} = \alpha s, \quad t_4 = \alpha^2 s, \quad t_0 = \alpha\rho s + (\alpha\rho' + \zeta\alpha^2)t.$$

Suppose that $\alpha = 0$. Then $t_0 = t_4 = t_{10} = t_{11} = 0$, and (11.3) becomes $t_5^2 = 0$, so $t_5 = 0$. Thus the columns of r_5 and r_6 in A vanish identically. For any u with $t_9(u) \neq 0$, this contradicts (11.5).

Therefore $\alpha \neq 0$, and by (11.4) we may assume that $\alpha = 1$. Then (11.3) becomes

$$t_1 s = (\zeta - 1)t(\rho s + (\rho' + \zeta)t) + (\rho s + \rho' t)^2.$$

Since $\zeta(\zeta - 1) = -1$, the coefficient of t^2 implies that $\rho'^2 + (\zeta - 1)\rho' - 1 = 0$, so $\rho' = \eta$ with $\eta^2 + (\zeta - 1)\eta - 1 = 0$. The other coefficients imply that $t_1 = \rho^2 s + \rho(\zeta - 1 + 2\eta)t$. By (11.4), with $\mu_1^2\mu_2^{-1/2} = 1$, the action multiplies ρ by μ_1 and preserves $\alpha = 1$ and $\rho' = \eta$, so we may assume that $\rho = i$ with $i \in \{0, 1\}$. Since $t_{10} = 0$, the form $t_2 = h_2 s + k_2 t$ is arbitrary in (11.3). Thus

$$(t_0, \ldots, t_{11}) = (is + (\eta + \zeta)t,\ i^2 s + i(\zeta - 1 + 2\eta)t,\ h_2 s + k_2 t,\ 0,\ s,\ is + \eta t,\ 0,\ \zeta t,\ t,\ s,\ 0,\ s),$$

which is the family (V_i).

Steps 5–7 exhaust the values of β and Δ. A direct calculation shows that the forms of Table 15 satisfy $q_1 = q_2 = q_3 = q_4 = 0$ for any values of the constants and the parameters. By Table 15, (G) consists of two copies of $\mathbb{C}^3$, and $(\mathrm{V}_0), (\mathrm{V}_1)$ consist of four copies of $\mathbb{C}^2$ each. □

The pivots in the proof of the next lemma are constants, independent of the parameters of the families, so the rank bound holds at any point of a family, with no stratification by zero columns or by the rank of A.

Lemma 11.4. *Let A be a first-stage matrix in a family of Table 15. Then the normalized lower space $\mathcal{U}_A$ of Construction-Proposition 8.6 satisfies* $\dim \mathcal{U}_A \leq 4$.

Proof. By Construction-Proposition 10.7 for $\mathbf{p} = (3, 1)$, we have $\mathcal{U}_A = \operatorname{Ker} T_{(3,1)}$, where $T_{(3,1)}$ has 260 rows and 84 columns. The twelve forms of Construction-Proposition 10.7 have the numbers $0, 4, 12, 24, 28, 36, 48, 52, 60, 87, 91, 95$, listed in Table 46. A direct calculation over the polynomial rings $\mathbb{C}[\alpha, \rho, \kappa]$ and $\mathbb{C}[h_2, k_2]$ shows that Tables 47, 48, and 49 are pivot tables of $T_{(3,1)}$ in the

sense of Construction-Proposition 6.7, for (G) with the additional row operation between the last two triples of Table 47, which does not change the rank.

We show that the pivots are non-zero. The constants of (G) are in $\mathbb{Q}(\zeta)$, and they are non-zero rational numbers or non-zero linear expressions in ζ with rational coefficients. Since $\zeta \notin \mathbb{Q}$, they do not vanish. For (V_0) and (V_1), the relation $\zeta = 1 - \eta + \eta^{-1}$ eliminates ζ, and $f(\eta) = 0$ with

$$f(x) = x^4 - x^3 - x^2 + x + 1.$$

Modulo 2, $f = x^4 + x^3 + x^2 + x + 1$ has no root and is not divisible by $x^2 + x + 1$, so f is irreducible over $\mathbb{Q}$. Each pivot of (V_i) is a non-zero rational polynomial of degree at most 3 in η, so it is non-zero at any root of f. The roots of f correspond to the four pairs (ζ, η), and Galois conjugation carries the tables to any of the four pairs.

The pivots are non-zero constants in distinct rows and columns. By Construction-Proposition 6.7, $\operatorname{rank} T_{(3,1)} \geq 80$ at any point of a family. Thus

$$\dim \mathcal{U}_A \leq 84 - 80 = 4.$$

□

Proof of Theorem 11.1. Let $\mathcal{L}_{(3,1)}$ be the affine space of Construction-Proposition 8.6, with the fixed blocks of Table 14. For a family $\mathcal{B}$ of Table 15, let $\mathcal{I}_{\mathcal{B}}$ be the set of pairs of a point of $\mathcal{B}$ and a point of $\mathcal{U}_A$, where A is the first-stage matrix of the point of $\mathcal{B}$. Then $\mathcal{I}_{\mathcal{B}}$ is defined in the product of $\mathcal{B}$ and the space of the 84 coefficients by the vanishing of $T_{(3,1)}$ on the coefficients, so $\mathcal{I}_{\mathcal{B}}$ is constructible. By Lemma 11.4 and [Har77, Exercise II.3.22(c)],

$$\dim \mathcal{I}_{\mathcal{B}} \leq \dim \mathcal{B} + 4.$$

The map from $\mathcal{I}_{\mathcal{B}}$ to $\mathcal{L}_{(3,1)}$ is polynomial, since P, U are determined by A, and N and twelve entries of C are determined by Construction-Proposition 8.6 through a minor of constant determinant. Let $\mathcal{Z} \subset \mathcal{L}_{(3,1)}$ be the union of the images of the sets $\mathcal{I}_{\mathcal{B}}$. By [Har77, Exercise II.3.19], $\mathcal{Z}$ is constructible. Since $\dim(\mathrm{G}) = 3$ and $\dim(\mathrm{V}_0) = \dim(\mathrm{V}_1) = 2$, we have $3 + 4 = 7$ and $2 + 4 = 6$, so

$$\dim \mathcal{Z} \leq 7.$$

Let $[X]$ be a point of $\mathcal{M}_{(3,1)}$. Then $[X]$ is represented by a torsion-free numerical Godeaux surface S whose bicanonical pencil has genus 4 and whose scheme B has partition $(3, 1)$. By Construction-Proposition 3.3, B is supported in the smooth locus of X and is curvilinear of length 4, so $\mathcal{A} \cong \mathbb{C}[z]/(z^3) \times \mathbb{C}$, and S is as in Set-up 8.1 with $\mathbf{p} = (3, 1)$. We show that $[X]$ is the moduli point of a point of $\mathcal{Z}$. Construction-Proposition 11.2 provides a presentation with the fixed blocks. By Construction-Propositions 8.4 and 11.3, an element of J moves the first stage of the presentation into a family of Table 15 and keeps the isomorphism class of X. By Lemma 4.2 and Construction-Proposition 8.6, a gauge moves O onto the twelve slots and keeps A and the isomorphism class of X. The relations $r_1, \dots, r_6$ are kept, because the lift of Construction-Proposition 8.4 is a chain map of the fixed resolution, so the reduction of d_1 modulo (s, t) does not change. Thus, by $d_1' d_2 = 0$, $B_1 B_4 = 0$, $B_2 B_3^t = 0$, and the uniqueness of P, U, the lower blocks of the new presentation are a point of $\mathcal{Z}$ whose moduli point is $[X]$.

Hence $\mathcal{M}_{(3,1)} \subset \mathcal{M}(\mathcal{Z})$, for the set $\mathcal{M}(\mathcal{Z})$ of Lemma 4.5(2) with the quadrics $r_1, \dots, r_6$ of Construction-Proposition 11.2, $\mathcal{L} = \mathcal{L}_{(3,1)}$, and $V = \mathcal{Z}$, and Lemma 4.5(2) implies that

$$\dim \overline{\mathcal{M}_{(3,1)}} \leq \dim \overline{\mathcal{M}(\mathcal{Z})} \leq \dim \mathcal{Z} \leq 7.$$

□

12. The collision type (4)

In this section, we prove the following theorem.

Theorem 12.1. *We have* $\dim \overline{\mathcal{M}_{(4)}} \leq 7$.

We first normalize the first stage, and then we bound the rank of a linear system on the normal forms.

Construction-Proposition 12.2. Notation and conditions as in Set-up 8.1 with $\mathbf{p} = (4)$. Fix an isomorphism $\mathcal{A} \cong \mathbb{C}[z]/(z^4)$, and normalize τ so that $\lambda(z^3) = 1$ and $\lambda(1) = \lambda(z) = \lambda(z^2) = 0$. Then S has a presentation with $y_0, \dots, y_3$ restricting to $(\tau^3, z\tau^3, z^2\tau^3, z^3\tau^3)$, with $z_0, \dots, z_3$ restricting to $(\tau^4, z\tau^4, z^2\tau^4, z^3\tau^4)$, and

$$(r_1, \dots, r_6) = (y_1^2 - y_0y_2, y_1y_2 - y_0y_3, y_1y_3, y_2^2, y_2y_3, y_3^2)$$

and

$$A = \begin{pmatrix} t_7 - t_8 & -t_{10} + t_9 & t_{11} & t_{11} & 0 & 0 \\ -t_3 + t_4 + t_6 & -t_5 - t_8 & t_9 & -t_{10} + 2t_9 & t_{11} & 0 \\ t_1 & -t_3 & t_7 & -t_5 & t_9 & t_{11} \\ t_0 & -t_2 & t_6 & -t_4 & t_8 & t_{10} \end{pmatrix},$$

where $t_0, \dots, t_{11}$ are binary linear forms with $t_{11} \neq 0$ and with $q_1 = q_2 = q_3 = q_4 = 0$ for

$$q_1 := -t_0t_{11} + t_1t_{10} - 2t_1t_9 + 2t_{10}t_2 - 2t_2t_9 + t_3t_5 + 2t_3t_8 + t_4t_5 + t_4t_7 - t_4t_8 + t_5t_6,$$

$$q_2 := -t_1t_{11} - t_{10}t_6 - t_{11}t_2 - t_4t_9 - t_5t_7 - 2t_7t_8 + t_8^2,$$

$$q_3 := -t_{10}t_7 + t_{10}t_8 + t_{11}t_3 - t_{11}t_4 + t_{11}t_6 + 2t_7t_9,$$

$$q_4 := t_{10}^2 - t_{10}t_9 + t_{11}t_5 + t_{11}t_7 + 2t_{11}t_8 + t_9^2.$$

The blocks $B_1, \dots, B_4$ are the fixed blocks of Table 16, and P, U are determined by A.

Proof of Construction-Proposition 12.2. By Construction-Theorem 3.2(4), there exists a unit $g \in \mathcal{A}$ such that replacing the generator τ by $g\tau$ replaces λ by the functional with $\lambda(z^3) = 1$ and $\lambda(1) = \lambda(z) = \lambda(z^2) = 0$. Possibly replacing τ with $g\tau$, we may assume that λ is the functional in the statement. All restrictions below use the trivialization of $\mathcal{O}_B(K_X)$ with generator τ.

We use Construction-Proposition 8.2 with $(b_0, b_1, b_2, b_3) = (1, z, z^2, z^3)$. Then $y_* = y_0$, and $\Lambda = (\lambda(z^{j+k}))_{j,k}$ has entries 1 for $j + k = 3$ and 0 otherwise. Thus the formulas of Construction-Proposition 8.2 for B_1, B_4 yield the matrices of Table 16. We have $r_k(1, z, z^2, z^3) = 0$ in $\mathcal{A}$. The linear map from the quadrics in $y_0, \dots, y_3$ to $\mathcal{A}$ sending y_iy_k to b_ib_k is surjective, so the kernel is of dimension $10 - 4 = 6$, and thus $r_1, \dots, r_6$, which are linearly independent elements of the kernel, form a basis of the kernel. Direct multiplication shows that $(r_1, \dots, r_6)B_3 = 0$ and $B_3B_2^t = 0$, and modulo y_0 the quadrics r_k are a basis of $(y_1, y_2, y_3)^2$ and the columns of B_3, B_2^t are the linear first and second syzygies of $(y_1, y_2, y_3)^2$ (and there are no others, since the reductions modulo y_0 of the 8 columns of B_3 and of the 3 rows of B_2 are linearly independent over $\mathbb{C}$ and the Betti numbers of $\mathbb{C}[y_1, y_2, y_3]/(y_1, y_2, y_3)^2$ are $(1, 6, 8, 3)$ [Eis95, Theorem A2.10, Corollary A2.12]), so the hypotheses of Construction-Proposition 8.2 hold.

By Construction-Theorem 3.2(2), there exist elements $y_j \in R_3$ and $z_j \in R_4$ restricting to $z^j\tau^3$ and $z^j\tau^4$ for $j \in \{0, 1, 2, 3\}$. By Lemma 8.3, S has a presentation with the generators y_j, z_j, with the quadrics $r_1, \dots, r_6$, and with the fixed blocks $B_1, \dots, B_4$ of Table 16.

Since $d_1d_2 = 0$ (Theorem 2.10), the blocks of $d_1'd_2 = 0$ imply

$$AB_3 + B_1P = 0, \qquad UB_4 - B_2P^t = 0, \qquad UP = 0.$$

We define binary linear forms $t_0, \dots, t_{11}$ by the last two rows of A:

$$\begin{pmatrix} A_{21} & A_{22} & A_{23} & A_{24} & A_{25} & A_{26} \\ A_{31} & A_{32} & A_{33} & A_{34} & A_{35} & A_{36} \end{pmatrix} = \begin{pmatrix} t_1 & -t_3 & t_7 & -t_5 & t_9 & t_{11} \\ t_0 & -t_2 & t_6 & -t_4 & t_8 & t_{10} \end{pmatrix}.$$

A direct calculation of the coefficients of $y_0, \dots, y_3$ in $AB_3 + B_1P = 0$ shows that A is the matrix in the statement. A direct calculation shows that the matrix A in the statement and the matrices P, U of Table 16 satisfy $AB_3 + B_1P = 0$ and $UB_4 - B_2P^t = 0$. By Construction-Proposition 8.2,

TABLE 16. The table lists the fixed blocks $B_1, \dots, B_4$ and the matrices P, U determined by A in Construction-Proposition 12.2.

$$B_1 = \begin{pmatrix} y_1 & 0 & 0 & 0 & y_2 & 0 & 0 & 0 & y_3 & 0 & 0 & 0 \\ -y_0 & y_1 & 0 & 0 & 0 & y_2 & 0 & 0 & 0 & y_3 & 0 & 0 \\ 0 & -y_0 & y_1 & 0 & -y_0 & 0 & y_2 & 0 & 0 & 0 & y_3 & 0 \\ 0 & 0 & -y_0 & y_1 & 0 & -y_0 & 0 & y_2 & -y_0 & 0 & 0 & y_3 \end{pmatrix}$$

$$B_2 = \begin{pmatrix} -y_3 & 0 & y_2 & 0 & -y_0 & -y_1 & y_0 & 0 \\ 0 & -y_3 & y_3 & y_2 & -y_1 & -y_2 & 0 & y_0 \\ 0 & 0 & 0 & y_3 & 0 & 0 & -y_2 & y_1 \end{pmatrix}$$

$$B_3 = \begin{pmatrix} y_2 & y_3 & y_3 & 0 & 0 & 0 & 0 & 0 \\ -y_1 & -y_2 & 0 & 0 & 0 & y_3 & 0 & 0 \\ -y_0 & -y_1 & -y_1 & -y_2 & 0 & -y_2 & -y_3 & 0 \\ y_0 & y_1 & 0 & 0 & -y_3 & 0 & 0 & 0 \\ 0 & 0 & y_0 & y_1 & y_2 & 0 & 0 & -y_3 \\ 0 & 0 & 0 & 0 & 0 & y_0 & y_1 & y_2 \end{pmatrix}$$

$$B_4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_3 & 0 & 0 & 0 & -y_2 \\ 0 & 0 & 0 & 0 & 0 & 0 & y_3 & 0 & 0 & 0 & -y_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & y_3 & 0 & 0 & 0 & -y_2 & 0 & y_0 \\ 0 & 0 & 0 & 0 & y_3 & 0 & 0 & -y_0 & -y_2 & 0 & y_0 & 0 \\ 0 & 0 & 0 & -y_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1 \\ 0 & 0 & -y_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1 & -y_0 \\ 0 & -y_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1 & -y_0 & 0 \\ -y_3 & 0 & 0 & y_0 & 0 & 0 & 0 & 0 & y_1 & -y_0 & 0 & 0 \\ 0 & 0 & 0 & y_2 & 0 & 0 & 0 & -y_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & y_2 & 0 & 0 & 0 & -y_1 & y_0 & 0 & 0 & 0 & 0 \\ 0 & y_2 & 0 & -y_0 & 0 & -y_1 & y_0 & 0 & 0 & 0 & 0 & 0 \\ y_2 & 0 & -y_0 & 0 & -y_1 & y_0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$P = \begin{pmatrix} -t_{10}+t_9 & 0 & t_{11} & 0 & 0 & 0 & 0 & 0 \\ -t_5-t_8 & t_{10}-t_9 & t_9 & -t_{11} & 0 & 0 & 0 & 0 \\ -t_3 & t_5+t_7 & t_7 & -t_9 & 0 & 0 & -t_{11} & 0 \\ -t_2 & t_4+t_6 & t_6 & -t_8 & 0 & 0 & -t_{10} & 0 \\ -t_7+t_8 & -t_{10}+t_9 & 0 & t_{11} & 0 & t_{11} & 0 & 0 \\ t_3-t_4-t_6 & -t_5-t_8 & 0 & t_9 & -t_{11} & t_9 & 0 & 0 \\ -t_1 & -t_3 & 0 & t_7 & -t_9 & t_7 & 0 & -t_{11} \\ -t_0 & -t_2 & 0 & t_6 & -t_8 & t_6 & 0 & -t_{10} \\ 0 & -t_7+t_8 & -t_7+t_8 & 0 & t_{11} & t_{10}-t_9 & t_{11} & 0 \\ 0 & t_3-t_4-t_6 & t_3-t_4-t_6 & 0 & -t_{10}+2t_9 & t_5+t_8 & t_9 & t_{11} \\ 0 & -t_1 & -t_1 & 0 & -t_5 & t_3 & t_7 & t_9 \\ 0 & -t_0 & -t_0 & 0 & -t_4 & t_2 & t_6 & t_8 \end{pmatrix}$$

$$U = \begin{pmatrix} t_0 & t_1 & -t_3+t_4+t_6 & t_7-t_8 & -t_2 & -t_3 & -t_5-t_8 & -t_{10}+t_9 & t_6 & t_7 & t_9 & t_{11} \\ t_2 & t_3 & t_5+t_8 & t_{10}-t_9 & t_4 & t_5 & t_{10}-2t_9 & -t_{11} & -t_8 & -t_9 & -t_{11} & 0 \\ t_6 & t_7 & t_9 & t_{11} & t_8 & t_9 & t_{11} & 0 & t_{10} & t_{11} & 0 & 0 \end{pmatrix}$$

the matrices P, U of the presentation are determined by A, and thus they are the matrices of Table 16. A direct calculation shows that

$$UP = \begin{pmatrix} 0 & q_1 & 0 & q_2 & q_3 & 0 & q_3 & q_4 \\ -q_1 & -q_2 & -q_2 & -q_3 & 0 & -q_3 & -q_4 & 0 \\ q_2 & 0 & q_3 & 0 & 0 & q_4 & 0 & 0 \end{pmatrix}.$$

Since $UP = 0$, we have $q_1 = q_2 = q_3 = q_4 = 0$.

It remains to show that $t_{11} \neq 0$. Let x be the support of B, a smooth point of X by Construction-Proposition 3.3(2). Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{X,x}$. We also denote by τ a local generator of $\mathcal{O}_X(K_X)$ at x extending the generator of $\mathcal{O}_B(K_X)$. We write

$$\bar{s} := s/\tau^2, \quad \bar{t} := t/\tau^2, \quad \bar{y}_j := y_j/\tau^3, \quad \bar{z}_j := z_j/\tau^4$$

in $\mathcal{O}_{X,x}$. The ideal $\mathfrak{a}$ of B at x is $(\bar{s}, \bar{t})$, and $\bar{y}_j$ restricts to z^j in $\mathcal{O}_{X,x}/\mathfrak{a} = \mathbb{C}[z]/(z^4)$.

Since $\dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2 = 2$ and $\dim_{\mathbb{C}} \mathfrak{m}/(\mathfrak{m}^2 + \mathfrak{a}) = 1$, we have $\mathfrak{a} \not\subset \mathfrak{m}^2$. Let μ be an element of $\mathfrak{a} \backslash \mathfrak{m}^2$. Since $\bar{y}_1$ restricts to z, $\mu, \bar{y}_1$ are regular parameters of $\mathcal{O}_{X,x}$. Since $\mathcal{O}_{X,x}/(\mu, \bar{y}_1^4)$ and $\mathcal{O}_{X,x}/\mathfrak{a}$ have

length 4 and $(\mu, \bar{y}_1^4) \subset \mathfrak{a}$, we have $\mathfrak{a} = (\mu, \bar{y}_1^4)$. Since $\dim \mathcal{O}_{X,x}/(\mu) = 1$ and $\dim \mathcal{O}_{X,x}/\mathfrak{a} = 0$, Nakayama's lemma implies that $\bar{y}_1^4 \notin \mathfrak{ma}$.

The relation of Set-up 4.1(7) for $r_3 = y_1y_3$, divided by τ^6, is

$$\bar{y}_1\bar{y}_3 + t_{11}(\bar{s},\bar{t})\bar{z}_0 + t_9(\bar{s},\bar{t})\bar{z}_1 + t_7(\bar{s},\bar{t})\bar{z}_2 + t_6(\bar{s},\bar{t})\bar{z}_3 + v_3(\bar{s},\bar{t}) = 0.$$

Since $\bar{z}_1, \bar{z}_2, \bar{z}_3 \in \mathfrak{m}$, $v_3(\bar{s},\bar{t}) \in \mathfrak{a}^3$, and $\bar{y}_3 - \bar{y}_1^3 \in \mathfrak{a}$, we obtain

$$\bar{y}_1^4 + t_{11}(\bar{s},\bar{t})\bar{z}_0 \in \mathfrak{ma}.$$

Since $\bar{y}_1^4 \notin \mathfrak{ma}$, we have $t_{11} \neq 0$. □

The next result reduces the first stage to three parameters.

Construction-Proposition 12.3. Notation and conditions as in Construction-Proposition 12.2. In Table 17, ζ runs over the roots of $\zeta^2-\zeta+1 = 0$, i runs over the roots of $i^2 = -1$, $\kappa := -2i/(1+3i)$, and (α, β, γ) runs over $\mathbb{C}^3$. After the action of J of Construction-Proposition 8.4, the first stage of S belongs to one of the families of Table 17, of dimension 3.

TABLE 17. The families of normal forms of type (4).

Family	Constants	Parameters	$(t_0, \dots, t_{11})$
(G)	ζ	(α, β, γ)	$(\zeta\beta t,\ -\beta s - 2\zeta\alpha t,\ \beta s + (2\zeta-1)\alpha t,$ $\alpha s - (\zeta+1)\gamma t,\ \alpha s - \zeta\gamma t,\ 0,\ \gamma t,$ $0,\ 0,\ t,\ \zeta t,\ s)$
(E)	i	(α, β, γ)	$(-(1+3i)\alpha t,\ -\beta s,\ \beta s,$ $\alpha s + (\kappa-1)\gamma t,\ \alpha s + \kappa\gamma t,\ -(1+2i)t,\ \gamma t,$ $t,\ it,\ 0,\ 0,\ s)$

Proof of Construction-Proposition 12.3. The proof is long, so we divide it into several steps.

Step 1. In this step we solve the first-stage identities and divide the proof into Cases (G), (E), and (Z), and we exclude (Z).

Since $t_{11} \neq 0$, the factor GL_2 of J moves t_{11} to s. We write $t_j = a_j s + b_j t$ for $0 \le j \le 10$. The identities $q_4 = q_3 = q_2 = q_1 = 0$ are equivalent to

$$\begin{aligned} t_5 &= -(t_{10}^2 - t_{10}t_9 + t_9^2)/s - t_7 - 2t_8, \\ t_3 &= (t_{10}t_7 - t_{10}t_8 - 2t_7t_9)/s + t_4 - t_6, \\ t_1 &= (-t_{10}t_6 - t_4t_9 - t_5t_7 - 2t_7t_8 + t_8^2)/s - t_2, \\ t_0 &= (t_1t_{10} - 2t_1t_9 + 2t_{10}t_2 - 2t_2t_9 + t_3t_5 + 2t_3t_8 + t_4t_5 + t_4t_7 - t_4t_8 + t_5t_6)/s. \end{aligned} \tag{12.1}$$

Any numerator of (12.1) is divisible by s, hence vanishes at $(s,t) = (0,1)$. For the first numerator, we obtain

$$b_{10}^2 - b_{10}b_9 + b_9^2 = 0.$$

Case (G). Suppose that $b_9 \neq 0$. Then $b_{10} = \zeta b_9$, where $\zeta^2 - \zeta + 1 = 0$. The second numerator of (12.1) at $(0,1)$ is $b_{10}b_7 - b_{10}b_8 - 2b_7b_9$. Thus

$$b_8 = (2\zeta - 1)b_7. \tag{12.2}$$

Case (E). Suppose that $b_9 = 0$ and $b_7 \neq 0$. Then $b_{10} = 0$, and the first line of (12.1) implies $b_5 = -b_7 - 2b_8$. The third numerator at $(0,1)$ is $-b_5b_7 - 2b_7b_8 + b_8^2 = b_7^2 + b_8^2$. Thus $b_8 = ib_7$, where $i^2 = -1$.

Case (Z). Suppose that $b_9 = b_7 = 0$. Then $b_{10} = 0$, and $b_8 = 0$ by the equation $b_7^2 + b_8^2 = 0$ of Case (E), which holds as soon as $b_9 = b_{10} = 0$. Let $u = [0:1]$. The first row of $A(u)$ is zero, and the last two columns of A are zero columns at u. Thus $\operatorname{rank} A(u) \le 3$, the span of the fifth and sixth basis vectors of $\mathbb{C}^6$ is a two-dimensional subspace of $\operatorname{Ker} A(u)$, and the corresponding

quadrics $r_5 = y_2y_3$ and $r_6 = y_3^2$ have the common linear factor y_3. Since S is torsion-free, this contradicts Lemma 4.3(2).

Since Case (Z) is excluded, the first stage is in Case (G) or in Case (E).

Step 2. In this step we write down the fields which we use to normalize the coefficients a_j, b_j.

For a nilpotent derivation δ of $\mathcal{A}$ and a nilpotent element a of $\mathcal{A}$, let $g_\epsilon \in \mathcal{G}^0$ be the element of Lemma 8.5. We denote by D_2, D_3, V_1, V_2, and V_3 the derivatives at $\epsilon = 0$ of the tuple $(t_0, \dots, t_{11})$ under g_ϵ for

$$(\delta, a) = \Big(z^2\frac{\partial}{\partial z}, 0\Big), \quad \Big(z^3\frac{\partial}{\partial z}, 0\Big), \quad (0, z), \quad (0, z^2), \quad (0, z^3),$$

respectively. A direct calculation of $L_zA - AL_r$ in Lemma 8.5 shows that

$$\begin{aligned}
D_2 &= (t_1 + 2t_2,\ 2t_3,\ t_3 - t_4 + 2t_6,\ -t_5 + 2t_7,\ t_5 + 4t_8,\ 4t_9,\\
&\qquad t_7 - t_8,\ -t_9,\ -2t_{10} + t_9,\ -2t_{11},\ t_{11},\ 0),\\
D_3 &= ((-t_3 + t_4 - 3t_6)/2,\ (-5t_7 + t_8)/2,\ (t_5 + 3t_8)/2,\ (-t_{10} + 3t_9)/2,\ t_{10}/2 - t_9,\\
&\qquad t_{11}/2,\ -t_{10} + t_9/2,\ -3t_{11}/2,\ t_{11}/2,\ 0,\ 0,\ 0),\\
V_1 &= (t_1/2 + t_2,\ (t_3 + t_4 + t_6)/2,\ t_3/2 - t_4,\ (-t_5 + t_8)/2,\ t_5/2 + 2t_8,\\
&\qquad t_{10}/2 + t_9,\ t_7/2 - t_8,\ -t_9/2,\ -t_{10} + t_9/2,\ -t_{11}/2,\ t_{11}/2,\ 0),\\
V_2 &= (-(t_3 + t_4 + 3t_6)/2,\ -t_5 - 3t_7/2 - t_8/2,\ (t_5 + t_8)/2,\ (t_{10} - t_9)/2,\\
&\qquad t_{10}/2 - t_9,\ -t_{11}/2,\ -t_{10} + t_9/2,\ -t_{11}/2,\ t_{11}/2,\ 0,\ 0,\ 0),\\
V_3 &= ((t_7 + t_8)/2,\ t_9,\ -(t_{10} + t_9)/2,\ -t_{11},\ -t_{11}/2,\ 0,\ t_{11}/2,\ 0,\ 0,\ 0,\ 0,\ 0).
\end{aligned} \tag{12.3}$$

The matrices L_y and L_z are linear in (δ, a), so the derivative for $(\delta, a) = ((1+i)z^3\partial/\partial z, -(1+3i)z^2)$ is $(1+i)D_3 - (1+3i)V_2$.

For D_2, D_3, where $a = 0$, for V_1, V_2, V_3, where $\delta = 0$, and for $(1+i)D_3 - (1+3i)V_2$, where $\delta(a) = 0$ since $z^4 = 0$, Lemma 8.5(2) implies that the elements g_ϵ form a one-parameter subgroup of $\mathcal{G}^0$ and that the formula for the derivative holds at any ϵ, with the values of $t_0, \dots, t_{11}$ at ϵ. The shear $t \to t + \epsilon s$ belongs to the factor GL_2 of J and commutes with $\mathcal{G}^0$. For $t_{11} = s$, the shear acts by $a_j \to a_j + \epsilon b_j$, $b_j \to b_j$. We call the derivative of a coefficient at $\epsilon = 0$ its *velocity*. Thus on the coefficients normalized below, the action is affine with a constant velocity matrix. By Lemma 8.5, the identities $q_1 = q_2 = q_3 = q_4 = 0$ are preserved, so (12.1) holds after any of the elements above.

Step 3. In this step we treat Case (G).

By (12.3), the fields D_2, V_1 and the shear preserve $t_{11} = s$, b_9, and b_{10}. On (a_9, a_{10}, b_7), their velocities are the columns of

$$\begin{pmatrix} -2 & -1/2 & b_9 \\ 1 & 1/2 & b_{10} \\ -b_9 & -b_9/2 & 0 \end{pmatrix},$$

whose determinant is $-b_9b_{10}/2 = -\zeta b_9^2/2 \neq 0$. Thus a composition of the flows of D_2 and V_1 and of the shear sets $a_9 = a_{10} = b_7 = 0$, and $b_8 = 0$ by (12.2).

The components of D_3, V_2 in t_9, t_{10}, t_{11} are zero, and their components in t_7, t_8 are constant multiples of $t_{11} = s$. So D_3, V_2 preserve the normalization above, and on (a_7, a_8) their velocity matrix is

$$\begin{pmatrix} -3/2 & -1/2 \\ 1/2 & 1/2 \end{pmatrix},$$

whose determinant is $-1/2$. Thus we may set $a_7 = a_8 = 0$. Now $t_7 = t_8 = 0$, $t_9 = b_9t$, and $t_{10} = \zeta b_9 t$. Since $t \to b_9 t$ belongs to the factor GL_2 of J, possibly replacing t with b_9t, we may assume that $t_9 = t$ and $t_{10} = \zeta t$. By (12.3), V_3 preserves $t_7, \dots, t_{11}$, and the velocity of a_6 is $1/2$. Thus we may set $a_6 = 0$.

Let $\alpha := a_4$, $\beta := a_2$, and $\gamma := b_6$. Then the first two lines of (12.1) imply $t_5 = 0$ and $t_3 = t_4 - t_6$. The third numerator is $-\zeta\gamma t^2 - t_4 t$, so $b_4 = -\zeta\gamma$ and $t_1 = -\alpha t - t_2$. The fourth

numerator is $(\zeta-2)t_1t+2(\zeta-1)t_2t$, whose value at $(0,1)$ is $(2-\zeta)\alpha+\zeta b_2$. Thus $b_2=(2\zeta-1)\alpha$ and $t_0=\zeta\beta t$. So $(t_0,\dots,t_{11})$ is the row (G) of Table 17.

Step 4. In this step we treat Case (E).

By (12.3), D_2, V_1 preserve $t_{11}=s$, and on (a_9,a_{10}) their velocity matrix is

$$\begin{pmatrix}-2 & -1/2\\ 1 & 1/2\end{pmatrix},$$

whose determinant is $-1/2$. Since $b_9=b_{10}=0$, the components of D_2, V_1 in t_7, t_8 are multiples of s, so D_2, V_1 preserve b_7 and b_8. Thus we may set $a_9=a_{10}=0$, that is, $t_9=t_{10}=0$. By (12.3), D_3, V_2 preserve $t_9, t_{10}, t_{11}, b_7, b_8$, and on (a_7,a_8) their velocity matrix is the 2×2 matrix of **Step 3**. Thus we may set $a_7=a_8=0$. Since $t\to b_7t$ belongs to the factor GL_2 of J, possibly replacing t with b_7t, we may assume that

$$t_7=t,\quad t_8=it,\quad t_9=t_{10}=0,\quad t_{11}=s.$$

Then $b_5=-1-2i$ by (12.1).

We compose the shear $t\to t+\epsilon s$ with the element g_ϵ of $(1+i)D_3-(1+3i)V_2$. The shear preserves any b_j, and the components of D_3, V_2 in t_5, t_7, t_8 are multiples of s, so the composition preserves b_5, b_7, b_8. The velocities of a_7 and a_8 are

$$1-\frac{3}{2}(1+i)+\frac{1}{2}(1+3i)=0,\qquad i+\frac{1}{2}(1+i)-\frac{1}{2}(1+3i)=0.$$

The velocity of b_2 is

$$\frac{1}{2}(1+i)(b_5+3b_8)-\frac{1}{2}(1+3i)(b_5+b_8)=-ib_5+b_8=-2+2i\neq 0.$$

Thus we may set $b_2=0$, preserving $t_7,\dots,t_{11}$. By (12.3), V_3 preserves t_2 and $t_7,\dots,t_{11}$, because $t_9=t_{10}=0$, and the velocity of a_6 is $1/2$. Thus we may set $a_6=0$.

Let $\alpha:=a_4$, $\beta:=a_2$, and $\gamma:=b_6$. Since $t_3=t_4-t_6$ by the second line of (12.1), the fourth numerator at $(0,1)$ is $-(1+3i)b_4-2i\gamma$, so $b_4=\kappa\gamma$. By (12.1), $t_5=-(1+2i)t$ and $t_3=t_4-t_6=\alpha s+(\kappa-1)\gamma t$. The third numerator is $-t_5t_7-2t_7t_8+t_8^2=0$, so $t_1=-t_2=-\beta s$. The fourth numerator is $t_3t_5+2t_3t_8+t_4t_5+t_4t_7-t_4t_8+t_5t_6=-(1+3i)\alpha st$, so $t_0=-(1+3i)\alpha t$. So $(t_0,\dots,t_{11})$ is the row (E) of Table 17.

Step 5. In this step we compute the dimension of the families.

A direct calculation shows that the forms of Table 17 satisfy $q_1=q_2=q_3=q_4=0$ for any $(\alpha,\beta,\gamma)\in\mathbb{C}^3$. Moreover, α and β are the coefficients of s in t_4 and t_2, and γ is the coefficient of t in t_6. Thus each of (G) and (E) consists of two copies of $\mathbb{C}^3$. □

We now bound the normalized lower space over the families.

Lemma 12.4. *For any first stage in a family of Construction-Proposition 12.3, the normalized lower space $\mathcal{U}_A$ of Construction-Proposition 8.6 satisfies* $\dim\mathcal{U}_A\le 4$.

Proof. By Construction-Proposition 10.7 for $\mathbf{p}=(4)$, we have $\mathcal{U}_A=\operatorname{Ker}T_{(4)}$, where $T_{(4)}$ has 252 rows and 84 columns. The slots, the minors of Construction-Propositions 8.6 and 10.7, and the positions of $c_0,\dots,c_{19}$ are in Table 50. The twelve forms of Construction-Proposition 10.7 have the numbers

$$0,\ 4,\ 12,\ 24,\ 28,\ 36,\ 48,\ 52,\ 60,\ 72,\ 76,\ 84,$$

also in Table 50. By Table 17, the entries of $T_{(4)}$ are polynomials in α,β,γ with coefficients in $\mathbb{Q}(\zeta)$ for (G) and in $\mathbb{Q}(i)$ for (E).

We apply Gauss–Jordan elimination to $T_{(4)}$ over $\mathbb{Q}(\zeta)[\alpha,\beta,\gamma]$ and over $\mathbb{Q}(i)[\alpha,\beta,\gamma]$ in exact arithmetic, in the order of Table 51 and of Table 52, respectively. At a triple, we divide the row by the non-zero constant of the triple, and we subtract multiples of the row from any other row in order to clear the column. We do not divide by an element involving a parameter. A direct

calculation shows that at each of the 80 triples, the entry of the current matrix is the constant of the triple.

A row or a column of a triple is not used again, and on the other rows and columns, the subtractions are the replacements of Construction-Proposition 6.7. Thus Tables 51 and 52 are pivot tables of $T_{(4)}$ in the sense of Construction-Proposition 6.7. Either root of $\zeta^2-\zeta+1=0$ and either root of $i^2=-1$ define embeddings of $\mathbb{Q}(\zeta)$ and of $\mathbb{Q}(i)$ in $\mathbb{C}$, so Tables 51 and 52 are pivot tables of $T_{(4)}$ over $\mathbb{C}[\alpha,\beta,\gamma]$ for the families (G) and (E). By Construction-Proposition 6.7, $\operatorname{rank} T_{(4)}\geq 80$ at any point (α,β,γ). Thus

$$\dim \mathcal{U}_A = 84-\operatorname{rank} T_{(4)}\leq 4.$$

□

Proof of Theorem 12.1. Let $r_1,\dots,r_6$ and $B_1,\dots,B_4$ be the quadrics and the blocks of Construction-Proposition 12.2, and let $\mathcal{L}_{(4)}$ be the affine space of Construction-Proposition 8.6. Let $\mathcal{V}\subset\mathcal{L}_{(4)}$ be the set of lower blocks such that A belongs to one of the families of Table 17, the blocks P,U are determined by A as in Table 16, and

$$AO-B_1N^t-CB_3^t=0,\quad AN-CP^t=0,\quad UN^t=0.$$

The set $\mathcal{V}$ is closed in $\mathcal{L}_{(4)}$, since A is affine linear and injective in (α,β,γ), the blocks P,U are polynomial in (α,β,γ), and the three equations are polynomial. By Construction-Proposition 8.6, the first of the three equations determines N and twelve entries of C in terms of $o_0,\dots,o_{11}$ and $c_0,\dots,c_{19}$, and the three equations are the equations defining $\mathcal{U}_A$. Let $\mathbb{C}^3$ be a copy in (G) or (E). The fiber over $(\alpha,\beta,\gamma)\in\mathbb{C}^3$ of the map sending an element of $\mathcal{V}$ with A in $\mathbb{C}^3$ to the parameters of A is isomorphic to $\mathcal{U}_A$. By Lemma 12.4 and [Har77, Exercise II.3.22(b)(c)], we obtain

$$\dim\mathcal{V}\leq 3+4=7.$$

We show that $\mathcal{M}_{(4)}\subset\mathcal{M}(\mathcal{V})$, for the set $\mathcal{M}(\mathcal{V})$ of Lemma 4.5(2) with the quadrics $r_1,\dots,r_6$, $\mathcal{L}=\mathcal{L}_{(4)}$, and $V=\mathcal{V}$. Let $[X]\in\mathcal{M}_{(4)}$, and let S be the surface with canonical model X. By Construction-Proposition 3.3 and the definition of $\mathcal{M}_{(4)}$, the surface S satisfies Set-up 8.1 with $\mathbf{p}=(4)$ and $\mathcal{A}\cong\mathbb{C}[z]/(z^4)$. By Construction-Proposition 12.2, S has a presentation with the fixed blocks. By Construction-Proposition 12.3, after the action of J, the first stage belongs to one of the families. By Construction-Proposition 8.4, the action of J preserves the fixed blocks, which include the quadrics $r_1,\dots,r_6$, and the isomorphism class of the canonical model. By Construction-Proposition 8.6, a gauge moves O to the twelve slots, and by Lemma 4.2, the gauge preserves A, $r_1,\dots,r_6$, and the canonical model. Let $\ell\in\mathcal{L}_{(4)}$ be the lower blocks of the presentation after the action of J and the gauge. Since $d_1'd_2=0$, and $B_1B_4=0$ and $B_2B_3^t=0$ for the blocks of Table 16, we have $\ell\in\mathcal{V}$. Thus $[X]\in\mathcal{M}(\mathcal{V})$.

By Lemma 4.5(2), we obtain

$$\dim\overline{\mathcal{M}_{(4)}}\leq\dim\overline{\mathcal{M}(\mathcal{V})}\leq\dim\mathcal{V}\leq 7.$$

□

13. Genus-two fibrations

The goal of this section is to prove Corollary 13.6: if the bicanonical pencil of a torsion-free numerical Godeaux surface S has genus 3, then the general fiber of $f:Y\to\mathbb{P}^1$ is nonhyperelliptic, which is one of the hypotheses of [CP06, Theorem 7.13] in Section 14. In [CP06, Introduction], the hyperelliptic case is left to a sequel. A hyperelliptic general fiber yields an automorphism of order two of S, hence a fibration of genus 2, for which [CP06, Theorem 4.13] provides a structure theorem, and Theorem 13.5 shows that such a surface has a bicanonical pencil of genus 4.

Construction-Proposition 13.1. Let $E:=\mathcal{O}(3)\oplus\mathcal{O}(4)^{\oplus 2}$, let $[s:t]$ be homogeneous coordinates on $\mathbb{P}^1$, and let $z:=t/s$. Let x_0,x_1 be the generators of the summands of $\mathcal{O}(1)^{\oplus 2}$, and let y_0 and y_1,y_2 be the generators of the summands $\mathcal{O}(3)$ and $\mathcal{O}(4)^{\oplus 2}$ of E. On $s\neq 0$,

the local generators are x_0, x_1 and y_0, y_1, y_2. On $t \neq 0$, the local generators are zx_0, zx_1 and z^3y_0, z^4y_1, z^4y_2. Let

$$\sigma : \operatorname{Sym}^2(\mathcal{O}(1)^{\oplus 2}) \to E$$

be a homomorphism with non-zero determinant and fiber rank at least two at any point (cf. [CP06, Definition 4.10]). Let $A := \sigma(x_0^2)$, $B := \sigma(x_0x_1)$, and $C := \sigma(x_1^2)$ be the columns of the matrix of σ, let (L_0, L_1, L_2) be the first row, with the L_i binary linear forms, and let $D := L_0L_2 - L_1^2$. Let

$$q_\sigma := AC - B^2 \in H^0(\mathbb{P}^1, \operatorname{Sym}^2 E(-4)),$$

let $p : \mathcal{C} \to \mathbb{P}^1$ be the conic bundle $\mathcal{C} := \operatorname{Proj}(\operatorname{Sym} E/(q_\sigma))$, and let $\mathcal{P} \subset \mathcal{C}$ be the set of *isolated branch points* [CP06, Definition 4.6]: over a zero of $\det \sigma$, the common zero of A, B, C.

Let

$$F_\sigma := \operatorname{coker}\big(q_\sigma : E(-10) \to \operatorname{Sym}^3 E(-14)\big) = p_*\mathcal{O}_\mathcal{C}(3)(-14),$$

and let

$$M_\sigma : H^1(\mathbb{P}^1, E(-10)) \to H^1(\mathbb{P}^1, \operatorname{Sym}^3 E(-14))$$

be the map induced by q_σ. The connecting homomorphism yields $H^0(\mathbb{P}^1, F_\sigma) = \operatorname{Ker} M_\sigma$.

We use the basis $z^{-1}, \dots, z^{-(d-1)}$ of $H^1(\mathbb{P}^1, \mathcal{O}(-d))$. Thus an element of $H^1(\mathbb{P}^1, E(-10))$ is $v = Uy_0 + Vy_1 + Wy_2$, with U a combination of $z^{-1}, \dots, z^{-6}$ and V, W combinations of $z^{-1}, \dots, z^{-5}$, and the component of $M_\sigma(v)$ at a cubic monomial with exponent m of y_0 consists of the coefficients of $z^{-1}, \dots, z^{-(1+m)}$ in $q_\sigma v$. We write $U = \sum_{i=1}^6 u_i z^{-i}$, $V = \sum_{i=1}^5 V_i z^{-i}$, $W = \sum_{i=1}^5 W_i z^{-i}$, $\mathbf{V}_i := (V_i, W_i)^t$, and $\mathbf{y} := (y_1, y_2)^t$, and a dot denotes the bilinear product $(a_1, a_2)^t \cdot (b_1, b_2)^t = a_1b_1 + a_2b_2$. For a non-zero $v \in \operatorname{Ker} M_\sigma$ with class $[v] \in \mathbb{P}(\operatorname{Ker} M_\sigma)$, let $\varphi_v \in H^0(\mathbb{P}^1, F_\sigma)$ be the section corresponding to v. We denote by F_{br} the local representative of φ_v, which on $s \neq 0$ is the non-negative Laurent part of $q_\sigma v$, and on $t \neq 0$, up to sign, is the negative Laurent part of $q_\sigma v$. We call the divisor $\Delta_{[v]}$ of φ_v on $\mathcal{C}$ the *branch divisor* of $[v]$.

We say that $(\sigma, [v])$ is *admissible* if (i) $\mathcal{C}$ has at most rational double points, (ii) $\Delta_{[v]}$ contains no point of $\mathcal{P}$, and (iii) the double cover of $\mathcal{C}$ branched along $\Delta_{[v]}$ has at most rational double points at the points of $\Delta_{[v]}$ [CP06, Definition 4.12]. Then the following hold.

(1) Let S be a numerical Godeaux surface, and let $f_2 : S \to \mathbb{P}^1$ be a fibration whose fibers are connected of genus 2. Then $(f_2)_*\omega_{S/\mathbb{P}^1} \cong \mathcal{O}(1)^{\oplus 2}$ and $(f_2)_*\omega_{S/\mathbb{P}^1}^{\otimes 2} \cong E$. For any choice of such isomorphisms, f_2 corresponds by [CP06, Theorem 4.13] to an admissible pair $(\sigma, [v])$, where σ is the multiplication. Any change of coordinates on $\mathbb{P}^1$ and automorphisms of $\mathcal{O}(1)^{\oplus 2}$ and E transform $(\sigma, [v])$ into an admissible pair corresponding to f_2.

(2) In the situation of (1), $f_2 = p \circ \psi \circ r$, where $r : S \to X'$ is the minimal resolution of the relative canonical model

$$X' := \operatorname{Proj} \bigoplus_{n \geq 0} (f_2)_*\omega_{S/\mathbb{P}^1}^{\otimes n}$$

of f_2, and $\psi : X' \to \mathcal{C}$ is a double cover with branch locus $\mathcal{P} \cup \Delta_{[v]}$ [CP06, Theorem 4.7]. Outside $\mathcal{P}$, the cover ψ has the local equation $\rho^2 = F_{\mathrm{br}}$ in a local generator of $\mathcal{O}_\mathcal{C}(1)$. Moreover, the pullback of $\mathcal{O}_\mathcal{C}(1)$ under $\psi \circ r$ is $\omega_{S/\mathbb{P}^1}^{\otimes 2}$, and the pullbacks of y_1, y_2 form a basis of $H^0(\mathbb{P}^1, E(-4)) = H^0(S, 2K_S)$.

(3) Let S be a torsion-free numerical Godeaux surface admitting an automorphism of order two. Then S has a fibration f_2 as in (1).

(4) A change of coordinates on $\mathbb{P}^1$ and automorphisms of $\mathcal{O}(1)^{\oplus 2}$ and E transform an admissible pair into an admissible pair. An automorphism of E multiplies U by a non-zero constant.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove the claims on F_σ, M_σ, and F_{br}.

Let $u \in \mathbb{P}^1$ be a point, and let $\Pi \subset \mathbb{C}^3$ be the set of the values $(A(w), B(w), C(w))$, where w runs over the dual of the fiber of E at u. Then $\dim \Pi$ is the fiber rank of σ at u, so $\dim \Pi \geq 2$.

A non-degenerate quadratic form on a three-dimensional space has no totally isotropic plane, so the form $\eta_0\eta_2 - \eta_1^2$ on $\mathbb{C}^3$ with coordinates η_0, η_1, η_2 does not vanish identically on Π. Thus

$$q_\sigma(w) = A(w)C(w) - B(w)^2 \neq 0$$

for some w, that is, q_σ is non-zero in any fiber, so $\mathcal{C}$ is a conic bundle. Multiplication by a non-zero polynomial is injective, so $q_\sigma : E(-10) \to \operatorname{Sym}^3 E(-14)$ is injective in any fiber, and F_σ is locally free of rank $10 - 3 = 7$. Let $\mathbb{P}(E) := \operatorname{Proj}(\operatorname{Sym} E)$, and let $\pi : \mathbb{P}(E) \to \mathbb{P}^1$ be the projection. Since q_σ is a non-zero section of $\mathcal{O}_{\mathbb{P}(E)}(2) \otimes \pi^*\mathcal{O}(-4)$ with zero scheme $\mathcal{C}$, we have the exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}(E)}(1) \otimes \pi^*\mathcal{O}(4) \to \mathcal{O}_{\mathbb{P}(E)}(3) \to \mathcal{O}_{\mathcal{C}}(3) \to 0.$$

Since $R^1\pi_*\mathcal{O}_{\mathbb{P}(E)}(1) = 0$ [Har77, Exercise III.8.4], pushing forward and tensoring with $\mathcal{O}(-14)$ yield $F_\sigma = p_*\mathcal{O}_{\mathcal{C}}(3)(-14)$ (cf. [CP06, Lemma 4.4(1)]).

We have

$$E(-10) = \mathcal{O}(-7) \oplus \mathcal{O}(-6)^{\oplus 2}, \qquad \operatorname{Sym}^3 E(-14) = \mathcal{O}(-5) \oplus \mathcal{O}(-4)^{\oplus 2} \oplus \mathcal{O}(-3)^{\oplus 3} \oplus \mathcal{O}(-2)^{\oplus 4},$$

and

$$H^0(\mathbb{P}^1, E(-10)) = H^0(\mathbb{P}^1, \operatorname{Sym}^3 E(-14)) = 0.$$

Thus the long exact sequence of cohomology yields $H^0(\mathbb{P}^1, F_\sigma) = \operatorname{Ker} M_\sigma$.

Let $d \geq 2$ be an integer. A section of $\mathcal{O}(-d)$ on the intersection of $s \neq 0$ and $t \neq 0$ is a Laurent polynomial in z times the local generator on $s \neq 0$. The local generator on $t \neq 0$ is z^{-d} times the local generator on $s \neq 0$, so the section is regular on $s \neq 0$ if and only if all exponents are non-negative, and regular on $t \neq 0$ if and only if all exponents are at most $-d$. Thus $z^{-1}, \dots, z^{-(d-1)}$ is a basis of $H^1(\mathbb{P}^1, \mathcal{O}(-d))$ (cf. [Har77, Theorem III.5.1]). A cubic monomial with exponent m of y_0 generates a summand $\mathcal{O}(-2-m)$ of $\operatorname{Sym}^3 E(-14)$, since

$$3m + 4(3-m) - 14 = -2 - m.$$

This implies the description of the components of $M_\sigma(v)$.

Let $v \in \operatorname{Ker} M_\sigma$ be a non-zero element, and write $q_\sigma v = F_+ + F_-$, with F_+ the non-negative Laurent part and F_- the negative Laurent part. Since $M_\sigma(v) = 0$, F_+ is regular on $s \neq 0$ and F_- is regular on $t \neq 0$. Since $F_+ + F_- = q_\sigma v$, the images of F_+ and $-F_-$ in F_σ coincide on the intersection of $s \neq 0$ and $t \neq 0$. Thus F_+ and $-F_-$ glue to the section φ_v of F_σ, by the definition of the connecting homomorphism.

Step 2. In this step we prove the second sentence of (1).

Let $\mathcal{V}_n := (f_2)_*\omega_{S/\mathbb{P}^1}^{\otimes n}$. Since S is minimal, f_2 is relatively minimal. By [CP06, Lemma 4.1(1), Definition 4.9, Theorem 4.13], there exists an exact sequence

$$0 \to \operatorname{Sym}^2 \mathcal{V}_1 \xrightarrow{\sigma} \mathcal{V}_2 \to \mathcal{O}_\tau \to 0$$

with $\tau \geq 0$ a divisor on $\mathbb{P}^1$, where σ is the multiplication, and

$$\chi(\mathcal{O}_S) = \deg \mathcal{V}_1 - 1, \qquad K_S^2 = 2\deg \mathcal{V}_1 + \deg \tau - 8.$$

By Lemma 2.2, $\chi(\mathcal{O}_S) = K_S^2 = 1$, so $\deg \mathcal{V}_1 = 2$ and $\deg \tau = 5$.

By [Har77, Exercise V.2.6], $\mathcal{V}_1 = \mathcal{O}(a_1) \oplus \mathcal{O}(a_2)$ with $a_1 + a_2 = 2$. Since $\omega_{S/\mathbb{P}^1} = \omega_S \otimes f_2^*\mathcal{O}(2)$, the projection formula yields

$$h^0(\mathbb{P}^1, \mathcal{V}_1(-2)) = h^0(S, K_S) = 0.$$

Thus $a_1, a_2 \leq 1$, and hence $a_1 = a_2 = 1$. The exact sequence implies $\deg \mathcal{V}_2 = 6 + 5 = 11$. By [Har77, Exercise V.2.6],

$$\mathcal{V}_2 = \mathcal{O}(a_1') \oplus \mathcal{O}(a_2') \oplus \mathcal{O}(a_3')$$

with $a_1' \leq a_2' \leq a_3'$. Since

$$\omega_S^{\otimes 2} = \omega_{S/\mathbb{P}^1}^{\otimes 2} \otimes f_2^*\mathcal{O}(-4), \tag{13.1}$$

the projection formula and Lemma 2.2 yield

$$\sum_{i=1}^{3} \max\{a_i' - 3, 0\} = h^0(\mathbb{P}^1, \mathcal{V}_2(-4)) = h^0(S, 2K_S) = 2.$$

Since $\sum_{i=1}^{3}(a_i' - 3) = 11 - 9 = 2$, we have $a_i' \geq 3$ for any i. Thus (a_1', a_2', a_3') is $(3,3,5)$ or $(3,4,4)$.

Suppose that $(a_1', a_2', a_3') = (3,3,5)$. The inclusion $\mathcal{O}(5) \to \mathcal{V}_2$ is non-zero, so by adjunction the homomorphism

$$f_2^*\mathcal{O}(5) \to \omega_{S/\mathbb{P}^1}^{\otimes 2}$$

is non-zero. By (13.1), we obtain a non-zero homomorphism $f_2^*\mathcal{O}(1) \to \omega_S^{\otimes 2}$. Let $Z \geq 0$ be the divisor of zeros of $f_2^*\mathcal{O}(1) \to \omega_S^{\otimes 2}$. Since

$$\mathcal{V}_2(-4) = \mathcal{O}(-1)^{\oplus 2} \oplus \mathcal{O}(1),$$

any element of $H^0(S, 2K_S)$ is the product of the defining section of Z and the pullback of a linear form in s, t. Thus the rational map defined by $|2K_S|$ is f_2, and the bicanonical pencil of S has genus 2. This contradicts [CP06, Theorem 5.1]. Thus $\mathcal{V}_2 \cong E$.

Step 3. In this step we conclude the proof of (1).

Fix isomorphisms $\mathcal{V}_1 \cong \mathcal{O}(1)^{\oplus 2}$ and $\mathcal{V}_2 \cong E$. By the exact sequence of **Step 2**, σ is injective, so $\det \sigma \neq 0$. For any $u \in \mathbb{P}^1$, we have $\dim(\mathcal{O}_\tau \otimes k(u)) \leq 1$, so the fiber rank of σ is at least two at any point. Let $\mathcal{V}_3^+$ be the invariant part of $\mathcal{V}_3$ under the involution of S over $\mathbb{P}^1$ induced by the hyperelliptic involution of a general fiber [CP06, Section 3, after Remark 3.2], which is a line bundle [CP06, Remark 3.3]. By [CP06, Proposition 4.8],

$$\mathcal{V}_3^+ \cong \det \mathcal{V}_1 \otimes \mathcal{O}(\tau) \cong \mathcal{O}(7).$$

Thus the last entry of the 5-tuple of f_2 [CP06, Definition 4.9], whose other entries are $\mathbb{P}^1$, $\mathcal{V}_1$, τ, and the isomorphism class of the exact sequence of **Step 2**, is a non-zero section, up to scalar, of $p_*\mathcal{O}_\mathcal{C}(3)(-14) = F_\sigma$. By **Step 1**, the last entry is φ_v for a non-zero $v \in \operatorname{Ker} M_\sigma$, and $[v]$ is uniquely determined. By the first statement of [CP06, Theorem 4.13], the 5-tuple of f_2 is admissible in the sense of [CP06, Definition 4.12]. The conditions of [CP06, Definition 4.12] are (i), (ii), and (iii), so $(\sigma, [v])$ is admissible.

A change of coordinates on $\mathbb{P}^1$ and automorphisms of $\mathcal{O}(1)^{\oplus 2}$ and E replace the isomorphisms $\mathcal{V}_1 \cong \mathcal{O}(1)^{\oplus 2}$ and $\mathcal{V}_2 \cong E$ by other isomorphisms, and the pair of f_2 for the new isomorphisms is the transform of $(\sigma, [v])$. Thus the transform is an admissible pair corresponding to f_2.

Step 4. In this step we prove (2).

The factorization and the branch locus are [CP06, Theorem 4.7]. By [CP06, Lemma 4.3, Proposition 4.8], the algebra $\bigoplus_{n\geq 0} \mathcal{V}_n$ is, as a graded module over the subalgebra $\mathcal{Y}$ generated by x_0, x_1 and y_0, y_1, y_2, the direct sum of $\mathcal{Y}$ and of the submodule generated by $\mathcal{V}_3^+$, and the even part of $\mathcal{Y}$ is the invariant part of the even part of the algebra. Let ρ be a local generator of $\mathcal{V}_3^+$, and let $x \in \mathcal{C}$ be a point outside $\mathcal{P}$. Since $AC = B^2$ on $\mathcal{C}$, one of A, C does not vanish at x. Since $A = x_0^2$ and $C = x_1^2$ in $\bigoplus_{n\geq 0} \mathcal{V}_n$, there exists $i \in \{0,1\}$ such that x_i^2 is a local generator of $\mathcal{O}_\mathcal{C}(1)$ at x, and x_i is a unit at the points of X' over x. Thus the anti-invariant part of $\psi_*\mathcal{O}_{X'}$ near x has rank one, with generator ρx_i^{-3}. By [CP06, Section 4, the paragraph before Definition 4.6], the multiplication $\mathcal{V}_3^+ \otimes \mathcal{V}_3^+ \to \mathcal{V}_6$ has image in the part of degree 6 of $\mathcal{Y}$, and by [CP06, Theorem 4.7, Definition 4.9] and **Step 3**, the divisor on $\mathcal{C}$ of the image of ρ^2 is $\Delta_{[v]}$, which yields $\rho^2 = F_{\mathrm{br}}$ after replacing ρ by a unit multiple. Dividing by $x_i^6 = (x_i^2)^3$, we obtain the local equation of ψ near x.

The morphism $\psi \circ r : S \to \mathcal{C}$ is induced by the inclusion of $\mathcal{Y}$ in the relative canonical algebra, and the part of even degree of $\mathcal{Y}$ is the quotient of $\operatorname{Sym} E$ defining $\mathcal{C}$ [CP06, Remark 4.5]. Thus the pullback of $\mathcal{O}_\mathcal{C}(1)$ under $\psi \circ r$ is $\omega_{S/\mathbb{P}^1}^{\otimes 2}$.

By (13.1) and the projection formula as in **Step 2**, $H^0(\mathbb{P}^1, E(-4)) = H^0(S, 2K_S)$. Since $E(-4) = \mathcal{O}(-1) \oplus \mathcal{O}^{\oplus 2}$, y_1, y_2 form a basis of $H^0(\mathbb{P}^1, E(-4))$, and so their pullbacks form a basis of $H^0(S, 2K_S)$.

Step 5. In this step we prove (3).

Let W' be the surface of [CCML07, Proposition 3.9] for the automorphism of order two. By [CCML07, Lemma 4.11], $-2 \le K_{W'}^2 \le 0$, and $K_{W'}^2 = 0$ if and only if W' is a minimal Enriques surface. If $K_{W'}^2 = 0$, then S is of Enriques type [CCML07, Section 5], and the torsion subgroup of $\mathrm{Pic}(S)$ is $\mathbb{Z}/4$ [CCML07, Proposition 5.3]. If $K_{W'}^2 = -1$, then S is of Du Val type [CCML07, Corollary 7.6], and the torsion subgroup of $\mathrm{Pic}(S)$ is $\mathbb{Z}/2$ or $\mathbb{Z}/4$ [CCML07, Classification Theorem]. Since S is torsion-free, $K_{W'}^2 = -2$. By [CCML07, Lemma 6.1, Remark 6.3], S has a base point free pencil of curves of genus 2, and a general member of the pencil is a double cover of a general member of the base point free pencil of rational curves on W'. A disconnected double cover of $\mathbb{P}^1$ is a disjoint union of two copies of $\mathbb{P}^1$, which contains no curve of genus 2, so a general member is connected. Thus the morphism $S \to \mathbb{P}^1$ defined by the pencil is a fibration f_2 as in (1) (Stein factorization, [Har77, Corollaries III.11.3 and III.11.5]).

Step 6. In this step we prove (4).

Let $(\sigma, [v])$ be an admissible pair, let $\alpha \in \mathrm{Aut}(\mathcal{O}(1)^{\oplus 2}) = GL_2(\mathbb{C})$ and $\beta \in \mathrm{Aut}(E)$, and let $\sigma' := \beta \circ \sigma \circ \mathrm{Sym}^2 \alpha$. The element

$$(x_0^2)(x_1^2) - (x_0x_1)^2 \in \mathrm{Sym}^2 \mathrm{Sym}^2(\mathcal{O}(1)^{\oplus 2})$$

is multiplied by $\det(\alpha)^2$ under $\mathrm{Sym}^2 \mathrm{Sym}^2 \alpha$, so

$$q_{\sigma'} = \det(\alpha)^2 \mathrm{Sym}^2 \beta(q_\sigma), \qquad M_{\sigma'} \circ \beta = \det(\alpha)^2 \mathrm{Sym}^3 \beta \circ M_\sigma.$$

Thus β carries $\mathcal{C}$, $\mathcal{P}$, $\mathrm{Ker}\, M_\sigma$, $\Delta_{[v]}$, and the double cover of $\mathcal{C}$ for σ isomorphically onto those for σ', with v replaced by $\beta(v)$ (the remaining constructions are natural). Pullback by a change of coordinates on $\mathbb{P}^1$ also carries $\mathcal{C}$, $\mathcal{P}$, $\mathrm{Ker}\, M_\sigma$, $\Delta_{[v]}$, and the double cover of $\mathcal{C}$ isomorphically onto those for the pulled-back homomorphism. Thus the transformed pair is admissible. Since $\mathrm{Hom}(\mathcal{O}(4), \mathcal{O}(3)) = 0$, β carries y_0 to cy_0 plus a combination of y_1, y_2, with $c \in \mathbb{C}^*$, and it carries y_1, y_2 to linear combinations of y_1, y_2. Thus the coefficient of y_0 in $\beta(v)$ is cU. □

We distinguish the cases where D has two distinct zeros or a double zero, since p maps the base scheme on $\mathcal{C}$ of the pencil spanned by y_1, y_2 isomorphically onto the zero scheme of D.

Lemma 13.2. *Notation and conditions as in Construction-Proposition 13.1, and let $(\sigma, [v])$ be an admissible pair. Then L_0, L_1, L_2 span $H^0(\mathbb{P}^1, \mathcal{O}(1))$. Moreover, $D \neq 0$, and after a change of coordinates on $\mathbb{P}^1$ and automorphisms of $\mathcal{O}(1)^{\oplus 2}$ and E, the first row is $(s, 0, t)$ if D has two distinct zeros, and $(s, t, 0)$ if D has a double zero.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show how a change of coordinates and automorphisms act on the first row.

Let $\alpha \in \mathrm{Aut}(\mathcal{O}(1)^{\oplus 2}) = GL_2(\mathbb{C})$ and $\beta \in \mathrm{Aut}(E)$. Since $\mathrm{Hom}(\mathcal{O}(4), \mathcal{O}(3)) = 0$, β is a non-zero scalar c on $\mathcal{O}(3)$, an element of $GL_2(\mathbb{C})$ on $\mathcal{O}(4)^{\oplus 2}$, and linear-form multiples of the first row added to the lower two rows. By Construction-Proposition 13.1(4), possibly replacing $(\sigma, [v])$ with the transformed pair, we may assume any normal form of the first row obtained by such changes.

The form D is multiplied by $\det(\alpha)^2 c^2$ under α and β, and a change of coordinates on $\mathbb{P}^1$ transforms D as a binary form. Thus such changes preserve the number of distinct zeros of D.

Let $\Gamma \subset \mathbb{P}^2$ be the smooth conic $\eta_0\eta_2 = \eta_1^2$ in the coordinates $[\eta_0 : \eta_1 : \eta_2]$. Then Γ consists of the points $[a^2 : ab : b^2]$, and a triple (η_0, η_1, η_2) is the linear functional on binary quadratic forms with values η_0, η_1, η_2 at x_0^2, x_0x_1, x_1^2. For $u \in \mathbb{P}^1$, the first row of σ at u is the composition of σ with the projection of E onto $\mathcal{O}(3)$, so the value $(L_0(u), L_1(u), L_2(u))$ is such a functional. The first row of $\beta \circ \sigma \circ \mathrm{Sym}^2 \alpha$ is c times the composition of the first row of σ with $\mathrm{Sym}^2 \alpha$.

Thus α acts on the first row at u through the action on $\mathbb{P}^2$ by composition with $\operatorname{Sym}^2 \alpha$. The automorphism α acts on $\mathbb{P}^2$ preserving Γ, and it acts on Γ through the transpose of α acting on $[a : b]$. Thus the action on Γ is doubly transitive. The tangent lines of Γ at $[1 : 0 : 0]$ and $[0 : 0 : 1]$ are $\eta_2 = 0$ and $\eta_0 = 0$.

Step 2. In this step we normalize a first row of rank one to $(s, 0, 0)$ or $(0, s, 0)$, and we exclude $(s, 0, 0)$.

Suppose that L_0, L_1, L_2 do not span $H^0(\mathbb{P}^1, \mathcal{O}(1))$. Since $\det \sigma \neq 0$, the first row is non-zero, so it is a non-zero linear form times a vector (c_0, c_1, c_2). After a change of coordinates on $\mathbb{P}^1$, the linear form is s. If $[c_0 : c_1 : c_2] \in \Gamma$, then we move it to $[1 : 0 : 0]$, and the first row is $(s, 0, 0)$. Otherwise, $[c_0 : c_1 : c_2]$ is the intersection point of the tangent lines of Γ at two distinct points. We move the points of tangency to $[1 : 0 : 0]$ and $[0 : 0 : 1]$, and the first row is $(0, s, 0)$.

In both cases, the first row vanishes at $[s : t] = [0 : 1]$. Since the fiber rank of σ at $[0 : 1]$ is at least two, the lower two rows at $[0 : 1]$ have rank two. Thus the common zero of A, B, C over $[0 : 1]$ is the point $x := [1 : 0 : 0]$ in the coordinates $[y_0 : y_1 : y_2]$, and $x \in \mathcal{P}$. By condition (ii), $\varphi_v(x) \neq 0$. The monomial y_0^3 generates a summand $\mathcal{O}(-5)$ of $\operatorname{Sym}^3 E(-14)$, so a coefficient $f(z)$ of y_0^3 on $s \neq 0$ is the coefficient $z^5 f(z)$ in the local generators on $t \neq 0$. Thus, up to sign, $\varphi_v(x)$ is the value at $z^{-1} = 0$ of z^5 times the coefficient of y_0^3 in the negative Laurent part of $q_\sigma v$.

If the first row is $(s, 0, 0)$, then q_σ has no y_0^2 term, since the coefficient of y_0^2 in q_σ is $D = 0$. Thus $q_\sigma v$ has no y_0^3 term and $\varphi_v(x) = 0$. This is a contradiction.

Step 3. In this step we exclude the first row $(0, s, 0)$.

Suppose that the first row is $(0, s, 0)$. Adding linear-form multiples of the first row to the lower two rows, we may assume that on $s \neq 0$,

$$A = \mathbf{a}(z) \cdot \mathbf{y}, \qquad B = y_0 + z^2(\mathbf{b}_2 \cdot \mathbf{y}), \qquad C = \mathbf{c}(z) \cdot \mathbf{y},$$

where $\mathbf{a}(z) = \mathbf{a}_0 + \mathbf{a}_1 z + \mathbf{a}_2 z^2$, $\mathbf{c}(z) = \mathbf{c}_0 + \mathbf{c}_1 z + \mathbf{c}_2 z^2$, and $\mathbf{a}_i, \mathbf{b}_2, \mathbf{c}_i \in \mathbb{C}^2$. Then

$$q_\sigma = -y_0^2 - 2z^2(\mathbf{b}_2 \cdot \mathbf{y})y_0 + (\mathbf{a}(z) \cdot \mathbf{y})(\mathbf{c}(z) \cdot \mathbf{y}) - z^4(\mathbf{b}_2 \cdot \mathbf{y})^2.$$

Write

$$v = U y_0 + \sum_{i=1}^{5} (\mathbf{V}_i \cdot \mathbf{y}) z^{-i}.$$

The component of $M_\sigma(v)$ at y_0^3 consists of the coefficients of $z^{-1}, \dots, z^{-4}$ in $-U$, so we have $U = u_5 z^{-5} + u_6 z^{-6}$. The components at $y_0^2 y_1$ and $y_0^2 y_2$ consist of the coefficients of z^{-1}, z^{-2}, z^{-3} in

$$-\sum_{i=1}^{5} \mathbf{V}_i z^{-i} - 2z^2 U \mathbf{b}_2,$$

so $\mathbf{V}_1 = \mathbf{V}_2 = 0$ and $\mathbf{V}_3 = -2u_5 \mathbf{b}_2$. In the components at $y_0 y_i y_j$, the coefficient of z^{-1} is

$$u_5(\mathbf{a}_2 \cdot \mathbf{y})(\mathbf{c}_2 \cdot \mathbf{y}) - u_5(\mathbf{b}_2 \cdot \mathbf{y})^2 - 2(\mathbf{b}_2 \cdot \mathbf{y})(\mathbf{V}_3 \cdot \mathbf{y}) = u_5\big((\mathbf{a}_2 \cdot \mathbf{y})(\mathbf{c}_2 \cdot \mathbf{y}) + 3(\mathbf{b}_2 \cdot \mathbf{y})^2\big),$$

so

$$u_5\big((\mathbf{a}_2 \cdot \mathbf{y})(\mathbf{c}_2 \cdot \mathbf{y}) + 3(\mathbf{b}_2 \cdot \mathbf{y})^2\big) = 0. \tag{13.2}$$

The coefficient of y_0^3 in the negative Laurent part of $q_\sigma v$ is $-U$, and

$$z^5(-U) = -u_5 - u_6 z^{-1}.$$

By **Step 2**, $\varphi_v(x) \neq 0$ implies $u_5 \neq 0$. Thus (13.2) yields

$$(\mathbf{a}_2 \cdot \mathbf{y})(\mathbf{c}_2 \cdot \mathbf{y}) = -3(\mathbf{b}_2 \cdot \mathbf{y})^2.$$

If $\mathbf{b}_2 \neq 0$, then unique factorization in $\mathbb{C}[y_1, y_2]$ implies that $\mathbf{a}_2$ and $\mathbf{c}_2$ are multiples of $\mathbf{b}_2$. If $\mathbf{b}_2 = 0$, then $\mathbf{a}_2 = 0$ or $\mathbf{c}_2 = 0$. In both cases,

$$\dim \operatorname{Span}(\mathbf{a}_2, \mathbf{b}_2, \mathbf{c}_2) \leq 1.$$

Since the first row vanishes at $[0:1]$ and the lower part of the columns at $[0:1]$ is $\mathbf{a}_2, \mathbf{b}_2, \mathbf{c}_2$, the fiber rank of σ at $[0:1]$ is at most one. This is a contradiction.

Thus L_0, L_1, L_2 span $H^0(\mathbb{P}^1, \mathcal{O}(1))$.

Step 4. In this step we prove the claims on D and the normal forms.

By **Step 3**, the map $u \to [L_0(u) : L_1(u) : L_2(u)]$ is an isomorphism from $\mathbb{P}^1$ onto a line $\ell \subset \mathbb{P}^2$. A smooth conic contains no line, so $\ell \not\subset \Gamma$, and $D \neq 0$. The zeros of D are the points of $\mathbb{P}^1$ over $\ell \cap \Gamma$.

If D has two distinct zeros, then $\ell \cap \Gamma$ consists of two distinct points, and by **Step 1** we may assume that they are $[1:0:0]$ and $[0:0:1]$. Then ℓ is the line $\eta_1 = 0$, so $L_1 = 0$, and after a change of coordinates on $\mathbb{P}^1$, the first row is $(s, 0, t)$. Substitution yields $D = st$.

If D has a double zero, then ℓ is tangent to Γ, and by **Step 1** we may assume that the point of tangency is $[1:0:0]$. Then ℓ is the line $\eta_2 = 0$, so $L_2 = 0$, and after a change of coordinates on $\mathbb{P}^1$, the first row is $(s, t, 0)$. Substitution yields $D = -t^2$. □

Lemma 13.3. *Notation and conditions as in Construction-Proposition 13.1, and let $(\sigma, [v])$ be an admissible pair. Assume that D has two distinct zeros. Then, after a change of coordinates on $\mathbb{P}^1$ and automorphisms of $\mathcal{O}(1)^{\oplus 2}$ and E,*

$$\sigma = \begin{pmatrix} s & 0 & t \\ ast & bs^2 + est + ht^2 & s^2 + cst \\ t^2 + dst & js^2 + kst + lt^2 & fst \end{pmatrix}$$

for complex numbers $a, b, c, d, e, f, h, j, k, l$. Moreover, $(jh)^2 = 1$, and U is a non-zero multiple of $z^{-1} - j^2 z^{-6}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce the lower rows of σ and solve the first equations of $M_\sigma(v) = 0$.

By Construction-Proposition 13.1(4) and Lemma 13.2, we may assume that the first row of σ is $(s, 0, t)$. Adding to a lower row of σ a linear form times the first row is an automorphism of E. The coefficient of s in the linear form controls the coefficient of s^2 in the first column, and the coefficient of t controls the coefficient of t^2 in the third column. Thus, on $s \neq 0$,

$$A = y_0 + ((\lambda z + \alpha z^2) \cdot \mathbf{y}), \quad B = (\beta + \gamma z + \delta z^2) \cdot \mathbf{y}, \quad C = z y_0 + ((\epsilon + \phi z) \cdot \mathbf{y}), \tag{13.3}$$

where the seven vectors $\lambda, \alpha, \beta, \gamma, \delta, \epsilon, \phi$ belong to $\mathbb{C}^2$. Then

$$q_\sigma = z y_0^2 + ((\epsilon + \phi z + \lambda z^2 + \alpha z^3) \cdot \mathbf{y}) y_0 + \sum_{i=0}^{4} \Psi_i z^i,$$

where

$$\begin{aligned}
\Psi_0 &= -(\beta \cdot \mathbf{y})^2, \\
\Psi_1 &= (\lambda \cdot \mathbf{y})(\epsilon \cdot \mathbf{y}) - 2(\beta \cdot \mathbf{y})(\gamma \cdot \mathbf{y}), \\
\Psi_2 &= (\lambda \cdot \mathbf{y})(\phi \cdot \mathbf{y}) + (\alpha \cdot \mathbf{y})(\epsilon \cdot \mathbf{y}) - (\gamma \cdot \mathbf{y})^2 - 2(\beta \cdot \mathbf{y})(\delta \cdot \mathbf{y}), \\
\Psi_3 &= (\alpha \cdot \mathbf{y})(\phi \cdot \mathbf{y}) - 2(\gamma \cdot \mathbf{y})(\delta \cdot \mathbf{y}), \\
\Psi_4 &= -(\delta \cdot \mathbf{y})^2.
\end{aligned}$$

By Construction-Proposition 13.1, $v = U y_0 + \sum_{i=1}^{5} (\mathbf{V}_i \cdot \mathbf{y}) z^{-i}$. The y_0^3 component of $q_\sigma v$ is zU, and the coefficients of $z^{-1}, \dots, z^{-4}$ in zU vanish. Thus $u_2 = u_3 = u_4 = u_5 = 0$. The coefficients of z^{-1}, z^{-2}, z^{-3} in the $y_0^2 \mathbf{y}$ components imply

$$\mathbf{V}_2 = -u_1 \epsilon, \qquad \mathbf{V}_3 = 0, \qquad \mathbf{V}_4 = -u_6 \alpha. \tag{13.4}$$

The coefficients of z^{-1} and z^{-2} in the part of $q_\sigma v$ that is linear in y_0 are

$$u_1 \Psi_0 + (\epsilon \cdot \mathbf{y})(\mathbf{V}_1 \cdot \mathbf{y}) + (\phi \cdot \mathbf{y})(\mathbf{V}_2 \cdot \mathbf{y}) + (\alpha \cdot \mathbf{y})(\mathbf{V}_4 \cdot \mathbf{y})$$

and

$$u_6\Psi_4 + (\epsilon\cdot\mathbf{y})(\mathbf{V}_2\cdot\mathbf{y}) + (\lambda\cdot\mathbf{y})(\mathbf{V}_4\cdot\mathbf{y}) + (\alpha\cdot\mathbf{y})(\mathbf{V}_5\cdot\mathbf{y}).$$

Both vanish. By (13.4), we have the identities of quadratic forms in $\mathbf{y}$

$$(\epsilon\cdot\mathbf{y})((\mathbf{V}_1 - u_1\phi)\cdot\mathbf{y}) = u_1(\beta\cdot\mathbf{y})^2 + u_6(\alpha\cdot\mathbf{y})^2, \tag{13.5}$$

$$(\alpha\cdot\mathbf{y})((\mathbf{V}_5 - u_6\lambda)\cdot\mathbf{y}) = u_1(\epsilon\cdot\mathbf{y})^2 + u_6(\delta\cdot\mathbf{y})^2. \tag{13.6}$$

Since F_{br} is the non-negative Laurent part of $q_\sigma v$ on $s\neq 0$, for $\xi\in\mathbb{C}^2$ the value of F_{br} at the point $[0:\xi_1:\xi_2]$ of the fiber $z=0$ is

$$\Psi_1(\xi)(\mathbf{V}_1\cdot\xi) + \Psi_2(\xi)(\mathbf{V}_2\cdot\xi) + \Psi_4(\xi)(\mathbf{V}_4\cdot\xi). \tag{13.7}$$

Step 2. In this step we prove that α and ϵ are linearly independent.

Suppose that $\xi\in\mathbb{C}^2$ is a non-zero vector with $\beta\cdot\xi = \epsilon\cdot\xi = 0$. At $z=0$ the forms A, B, C are y_0, $\beta\cdot\mathbf{y}$, and $\epsilon\cdot\mathbf{y}$, so $[0:\xi_1:\xi_2]$ is the isolated branch point of the fiber $z=0$. By condition (ii) of admissibility, F_{br} does not vanish at $[0:\xi_1:\xi_2]$. Moreover, $\Psi_1(\xi)=0$ and $\mathbf{V}_2\cdot\xi = 0$. Thus, for any non-zero ξ with $\beta\cdot\xi = \epsilon\cdot\xi = 0$, (13.4) and (13.7) imply

$$u_6(\delta\cdot\xi)^2(\alpha\cdot\xi)\neq 0. \tag{13.8}$$

Let the interchange be the operation that interchanges s,t and simultaneously x_0, x_1. By Construction-Proposition 13.1(4), the interchange transforms $(\sigma,[v])$ into an admissible pair. It preserves the first row $(s,0,t)$ and the shape (13.3). It interchanges α with ϵ, β with δ, and λ with ϕ, and it fixes γ. Up to a common sign of v, it interchanges u_1 with u_6, and $\mathbf{V}_i$ with $\mathbf{V}_{6-i}$ (in $H^1(\mathbb{P}^1,\mathcal{O}(-n))$, with $n=7$ for U and $n=6$ for V and W, the basis element z^{-i} is represented by $s^{i-n}t^{-i}$, which the interchange sends to $s^{-i}t^{i-n}$, the representative of $z^{-(n-i)}$, and the interchange of the open sets $s\neq 0$ and $t\neq 0$ of the cover changes the sign of the Čech cocycle). Thus (13.8) for the new pair reads: for any non-zero ξ with $\delta\cdot\xi = \alpha\cdot\xi = 0$,

$$u_1(\beta\cdot\xi)^2(\epsilon\cdot\xi)\neq 0. \tag{13.9}$$

Suppose that $\epsilon = 0$. Since the fiber rank of σ at $z=0$ is at least two, $\beta\neq 0$. Let ξ be a non-zero vector with $\beta\cdot\xi = 0$. Substituting $\mathbf{y}=\xi$ in (13.5) implies $u_6(\alpha\cdot\xi)^2 = 0$, which contradicts (13.8). Thus $\epsilon\neq 0$. Since $(\sigma,[v])$ is an arbitrary admissible pair of the shape (13.3), the interchange implies $\alpha\neq 0$.

Suppose that α and ϵ are linearly dependent, and let ξ be a non-zero vector with $\alpha\cdot\xi = \epsilon\cdot\xi = 0$. If $\beta\cdot\xi = 0$, then (13.8) fails. Thus $\beta\cdot\xi\neq 0$. If $\delta\cdot\xi = 0$, then (13.9) fails, since $\epsilon\cdot\xi = 0$. Thus $\delta\cdot\xi\neq 0$. Substituting $\mathbf{y}=\xi$ in (13.5) and (13.6) implies $u_1(\beta\cdot\xi)^2 = 0$ and $u_6(\delta\cdot\xi)^2 = 0$, so $u_1 = u_6 = 0$. Since $\epsilon\neq 0$ and $\alpha\neq 0$, (13.5) and (13.6) now imply $\mathbf{V}_1 = \mathbf{V}_5 = 0$. Together with (13.4) and $U=0$, we obtain $v=0$, a contradiction.

Step 3. In this step we obtain the normal form and eliminate the coefficients of v.

An automorphism of the summand $\mathcal{O}(4)^{\oplus 2}$ of E sends ϵ to $(1,0)^t$ and α to $(0,1)^t$. Five vectors remain free, and we write $\lambda = (a,d)^t$, $\beta = (b,j)^t$, $\gamma = (e,k)^t$, $\delta = (h,l)^t$, and $\phi = (c,f)^t$. Then σ is the matrix of the statement.

Let $g_1 := b + ez + hz^2$ and $g_2 := j + kz + lz^2$. Then

$$q_\sigma = zy_0^2 + p_1y_0y_1 + p_2y_0y_2 + T_{11}y_1^2 + T_{12}y_1y_2 + T_{22}y_2^2,$$

where $p_1 = 1 + cz + az^2$, $p_2 = fz + dz^2 + z^3$, and

$$\begin{aligned} T_{11} &= az(1+cz) - g_1^2,\\ T_{12} &= afz^2 + (dz+z^2)(1+cz) - 2g_1g_2,\\ T_{22} &= (dz+z^2)fz - g_2^2. \end{aligned}$$

The constant terms of T_{11}, T_{12}, T_{22} are $-b^2, -2bj, -j^2$, and the coefficients of z^4 are $-h^2, -2hl, -l^2$. By **Step 1**, we have $V_2 = -u_1$, $V_3 = V_4 = 0$, $W_2 = W_3 = 0$, and $W_4 = -u_6$. The coefficients of z^{-1} and z^{-2} in $T_{11}U + p_1V$, in $T_{12}U + p_1W + p_2V$, and in $T_{22}U + p_2W$ vanish, and they imply

$$\begin{aligned} V_1 &= (b^2 + c)u_1, & u_1 &= -h^2u_6, \\ W_1 &= (2bj + f)u_1, & V_5 &= (a + 2hl)u_6, \\ u_6 &= -j^2u_1, & W_5 &= (l^2 + d)u_6. \end{aligned}$$

If $u_1 = 0$, then $u_6 = 0$, and all the coefficients of v vanish. Since $v \neq 0$, we have $u_1 \neq 0$. Moreover, $u_1 = -h^2u_6 = (jh)^2u_1$, hence $(jh)^2 = 1$. Finally, $U = u_1z^{-1} + u_6z^{-6} = u_1(z^{-1} - j^2z^{-6})$. □

The torsion-free hypothesis enters in the next proposition: if $\det\sigma$ vanishes at the double zero of D, then the section y_1 of an $\mathcal{O}(4)$ summand of E has an even divisor on S, which yields non-trivial 2-torsion by Lemma 2.3.

Proposition 13.4. *Notation and conditions as in Construction-Proposition 13.1(1), and let $(\sigma, [v])$ be an admissible pair corresponding to f_2. Assume that S is torsion-free and that D has a double zero. Then $\det\sigma$ does not vanish at the zero of D. Moreover, if the first row of σ is $(s, t, 0)$, then $U = Pz^{-1} + Qz^{-2}$ with $(P, Q) \neq (0, 0)$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we reduce the lower rows of σ and solve the first equations of $M_\sigma(v) = 0$.

By Lemma 13.2 and Construction-Proposition 13.1(1), we may assume that the first row of σ is $(s, t, 0)$. Then $D = -t^2$, and the zero of D is $z = 0$. The coefficient of y_0^2 in q_σ is D for any lower rows of σ, so the y_0^3 component of $q_\sigma v$ is $-z^2U$. The coefficients of $z^{-1}, \dots, z^{-4}$ in $-z^2U$ vanish, hence $U = Pz^{-1} + Qz^{-2}$ for complex numbers P, Q. By Construction-Proposition 13.1(1) and (4), an automorphism of E transforms $(\sigma, [v])$ into an admissible pair corresponding to f_2, and it multiplies U by a non-zero constant. Thus we may change the lower rows of σ by automorphisms of E in the proof of both conclusions.

Adding to a lower row a linear form times the first row removes the coefficients of s^2 and st in the first column. Thus, on $s \neq 0$,

$$A = y_0 + z^2(\alpha_2 \cdot \mathbf{y}), \quad B = zy_0 + ((\beta_0 + \beta_1 z + \beta_2 z^2) \cdot \mathbf{y}), \quad C = (\gamma_0 + \gamma_1 z + \gamma_2 z^2) \cdot \mathbf{y},$$

where $\alpha_2, \beta_i, \gamma_i$ belong to $\mathbb{C}^2$. Let

$$\mathbf{r}_0 := \gamma_0, \quad \mathbf{r}_1 := \gamma_1 - 2\beta_0, \quad \mathbf{r}_2 := \gamma_2 - 2\beta_1, \quad \mathbf{r}_3 := -2\beta_2.$$

Then

$$q_\sigma = -z^2y_0^2 + \Big(\sum_{i=0}^{3}(\mathbf{r}_i \cdot \mathbf{y})z^i\Big)y_0 + \sum_{i=0}^{4}\Psi_i z^i,$$

where $\Psi_0 = -(\beta_0 \cdot \mathbf{y})^2$, $\Psi_1 = -2(\beta_0 \cdot \mathbf{y})(\beta_1 \cdot \mathbf{y})$, and

$$\Psi_2 = (\alpha_2 \cdot \mathbf{y})(\gamma_0 \cdot \mathbf{y}) - (\beta_1 \cdot \mathbf{y})^2 - 2(\beta_0 \cdot \mathbf{y})(\beta_2 \cdot \mathbf{y}).$$

By Construction-Proposition 13.1, $v = Uy_0 + \sum_{i=1}^{5}(\mathbf{V}_i \cdot \mathbf{y})z^{-i}$. The coefficients of z^{-1}, z^{-2}, z^{-3} in the $y_0^2\mathbf{y}$ components of $q_\sigma v$ imply

$$\mathbf{V}_3 = P\mathbf{r}_0 + Q\mathbf{r}_1, \qquad \mathbf{V}_4 = Q\mathbf{r}_0, \qquad \mathbf{V}_5 = 0. \tag{13.10}$$

The coefficients of z^{-2} and z^{-1} in the part of $q_\sigma v$ that is linear in y_0 vanish, that is,

$$Q\Psi_0 + (\mathbf{r}_0 \cdot \mathbf{y})(\mathbf{V}_2 \cdot \mathbf{y}) + (\mathbf{r}_1 \cdot \mathbf{y})(\mathbf{V}_3 \cdot \mathbf{y}) + (\mathbf{r}_2 \cdot \mathbf{y})(\mathbf{V}_4 \cdot \mathbf{y}) = 0, \tag{13.11}$$

$$\begin{aligned} &P\Psi_0 + Q\Psi_1 + (\mathbf{r}_0 \cdot \mathbf{y})(\mathbf{V}_1 \cdot \mathbf{y}) + (\mathbf{r}_1 \cdot \mathbf{y})(\mathbf{V}_2 \cdot \mathbf{y}) \\ &\quad + (\mathbf{r}_2 \cdot \mathbf{y})(\mathbf{V}_3 \cdot \mathbf{y}) + (\mathbf{r}_3 \cdot \mathbf{y})(\mathbf{V}_4 \cdot \mathbf{y}) = 0. \end{aligned} \tag{13.12}$$

At $z = 0$ the forms A, B, C are y_0, $\beta_0 \cdot \mathbf{y}$, and $\gamma_0 \cdot \mathbf{y}$. Thus $\det\sigma$ vanishes at $z = 0$ if and only if β_0 and γ_0 are linearly dependent. From now until the end of **Step 4**, suppose that $\det\sigma$ vanishes at $z = 0$.

Step 2. In this step we prove that $\gamma_0 \neq 0$.

Suppose that $\gamma_0 = 0$. Since the fiber rank of σ at $z = 0$ is at least two, $\beta_0 \neq 0$. Let $\xi \in \mathbb{C}^2$ be a non-zero vector with $\beta_0 \cdot \xi = 0$. Then $[0 : \xi_1 : \xi_2]$ is the isolated branch point of the fiber $z = 0$, and F_{br} does not vanish at $[0 : \xi_1 : \xi_2]$ by condition (ii) of admissibility. By (13.10), we have $\mathbf{V}_4 = 0$ and $\mathbf{V}_3 \cdot \xi = Q(\gamma_1 \cdot \xi)$. Substituting $\mathbf{y} = \xi$ in (13.11) implies $Q(\gamma_1 \cdot \xi)^2 = 0$, so $\mathbf{V}_3 \cdot \xi = 0$. Substituting $\mathbf{y} = \xi$ in (13.12) now implies $(\gamma_1 \cdot \xi)(\mathbf{V}_2 \cdot \xi) = 0$. Since F_{br} is the non-negative Laurent part of $q_\sigma v$, the value of F_{br} at $[0 : \xi_1 : \xi_2]$ is

$$\sum_{i=1}^{4} \Psi_i(\xi)(\mathbf{V}_i \cdot \xi) = -(\beta_1 \cdot \xi)^2(\mathbf{V}_2 \cdot \xi).$$

If $\gamma_1 \cdot \xi \neq 0$, then $\mathbf{V}_2 \cdot \xi = 0$ and F_{br} vanishes at $[0 : \xi_1 : \xi_2]$, a contradiction. Thus $\gamma_1 = f\beta_0$ for a complex number f, and

$$q_\sigma \equiv -(\beta_0 \cdot \mathbf{y})^2 + z(\beta_0 \cdot \mathbf{y})\big((f-2)y_0 - 2\beta_1 \cdot \mathbf{y}\big) \pmod{z^2}.$$

Therefore q_σ belongs to the square of the ideal $(z, \beta_0 \cdot \mathbf{y})$, and $\mathcal{C}$ is singular along the line $z = 0$, $\beta_0 \cdot \mathbf{y} = 0$. This contradicts condition (i) of admissibility, since rational double points are isolated.

Step 3. In this step we obtain the normal form of σ and prove that $m_1 = 0$, $Q \neq 0$, $k \neq 0$, and $m_2 \neq 2k$.

By **Step 2**, an automorphism of the summand $\mathcal{O}(4)^{\oplus 2}$ of E sends γ_0 to $(1, 0)^t$ and β_0 to $(b, 0)^t$ for a complex number b. Thus, on $s \neq 0$,

$$\sigma = \begin{pmatrix} 1 & z & 0 \\ az^2 & g_1 & 1 + n_1 z + n_2 z^2 \\ dz^2 & g_2 & m_1 z + m_2 z^2 \end{pmatrix}, \qquad g_1 := b + ez + hz^2, \quad g_2 := kz + lz^2.$$

Here $\mathbf{r}_0 = (1, 0)^t$, $\mathbf{r}_1 = (n_1 - 2b, m_1)^t$, $\mathbf{r}_2 = (n_2 - 2e, m_2 - 2k)^t$, and $\mathbf{r}_3 = (-2h, -2l)^t$. By (13.10), we have $\mathbf{V}_3 = (P + (n_1 - 2b)Q, m_1 Q)^t$ and $\mathbf{V}_4 = (Q, 0)^t$. The forms Ψ_0 and Ψ_1 have no y_2^2 term. The coefficients of y_2^2 and $y_1 y_2$ in (13.11) and the coefficient of y_2^2 in (13.12) are

$$\begin{aligned} &Qm_1^2 = 0, \\ &W_2 + m_1 P + 2(n_1 - 2b)m_1 Q + (m_2 - 2k)Q = 0, \\ &m_1 W_2 + (m_2 - 2k)m_1 Q = 0. \end{aligned}$$

Eliminating W_2, we obtain $Qm_1^2 = 0$ and $m_1^2(P + 2(n_1 - 2b)Q) = 0$. If $m_1 \neq 0$, then $P = Q = 0$, and then (13.11) and (13.12) read $y_1(\mathbf{V}_2 \cdot \mathbf{y}) = 0$ and $y_1(\mathbf{V}_1 \cdot \mathbf{y}) = 0$, so $v = 0$, a contradiction. Thus $m_1 = 0$ and $W_2 = -(m_2 - 2k)Q$. Moreover, $W_3 = W_4 = W_5 = 0$.

Now

$$q_\sigma = -z^2 y_0^2 + p_1 y_0 y_1 + p_2 y_0 y_2 + T_{11} y_1^2 + T_{12} y_1 y_2 + T_{22} y_2^2,$$

where $p_1 = 1 + (n_1 - 2b)z + (n_2 - 2e)z^2 - 2hz^3$, $p_2 = z^2(m_2 - 2k - 2lz)$, and

$$\begin{aligned} T_{11} &= az^2(1 + n_1 z + n_2 z^2) - g_1^2, \\ T_{12} &= z^2\big(am_2 z^2 + d(1 + n_1 z + n_2 z^2)\big) - 2g_1 g_2, \\ T_{22} &= z^2\big(dm_2 z^2 - (k + lz)^2\big). \end{aligned}$$

At $z = 0$ the forms A, B, C are y_0, by_1, and y_1, so $[0 : 0 : 1]$ is the isolated branch point of the fiber $z = 0$. The value of F_{br} at $[0 : 0 : 1]$ is the constant term of $T_{22}W$, namely

$$-k^2 W_2 = Qk^2(m_2 - 2k).$$

By condition (ii) of admissibility, $Q \neq 0$, $k \neq 0$, and $m_2 \neq 2k$. We normalize $Q = 1$, so that

$$V = V_1 z^{-1} + V_2 z^{-2} + (P + n_1 - 2b)z^{-3} + z^{-4}, \qquad W = W_1 z^{-1} + (2k - m_2)z^{-2}.$$

Step 4. In this step we prove that y_1 has an even divisor on S, which contradicts the torsion-free hypothesis.

Setting $y_1 = 0$, we obtain $q_\sigma|_{y_1=0} = z^2\Theta$, where

$$\Theta = -y_0^2 + (m_2 - 2k - 2lz)y_0y_2 + \big(dm_2z^2 - (k+lz)^2\big)y_2^2. \tag{13.13}$$

Since U and W have poles of order at most two and z^2 divides the coefficients $-z^2$, p_2, and T_{22} of y_0^2, y_0y_2, and y_2^2 in q_σ, all negative Laurent terms of $q_\sigma v$ are divisible by y_1. Thus

$$q_\sigma v = F_{\mathrm{br}} - y_1N$$

for a quadratic form N in y_0, y_1, y_2, whose coefficients are polynomials in z^{-1} without constant term and with poles of order at most four. In particular, $F_{\mathrm{br}} = y_1N$ on $\mathcal{C}$ over $z \neq 0, \infty$. We prove that

$$N|_{y_1=0} = \frac{k^2y_2^2}{z^2}. \tag{13.14}$$

The coefficients of y_0^2 and y_0y_2 in $N|_{y_1=0}$ vanish because the $y_0^2y_1$ and $y_0y_1y_2$ components of $q_\sigma v$ have no negative Laurent part. The $y_0^2y_1$ component is $-z^2V + p_1U$ and the $y_0y_1y_2$ component is $p_1W + p_2V + T_{12}U$. Both have poles of order at most two, since U and W have poles of order at most two, V has poles of order at most four, and z^2 divides p_2. The coefficients of z^{-1} and z^{-2} in both components vanish by $M_\sigma(v) = 0$. The coefficient of y_2^2 in $N|_{y_1=0}$ is minus the negative Laurent part of the $y_1y_2^2$ component $T_{12}W + T_{22}V$ of $q_\sigma v$. Since T_{12} is divisible by z, T_{22} is divisible by z^2, W has poles of order at most two, and V has poles of order at most four, $T_{12}W + T_{22}V$ has poles of order at most two. The z^{-2} coefficient of $T_{12}W + T_{22}V$ is $-k^2$, the z^2 coefficient of T_{22} times the z^{-4} coefficient 1 of V. The z^{-1} coefficient of $T_{12}W + T_{22}V$ vanishes since $M_\sigma(v) = 0$.

By Construction-Proposition 13.1(2), y_1 is a non-zero section of $2K_S$, pulled back from $\mathcal{C}$ under $\psi \circ r$. Let ℓ be the line $y_1 = 0$ in the fiber $z = 0$. Let $x \in \mathcal{C}$ be a point with $y_1 = 0$. We show that near any point of S over x, the local function given by y_1 is a square times a unit. The equalities between y_1, ρ, N, and F_{br} below are read after dividing by a power of a local generator of $\mathcal{O}_{\mathcal{C}}(1)$ and pulling back by r, and ρ is scaled so that $\rho^2 = \varphi_v$. We distinguish three cases.

Case 1. x is not contained in the fiber $z = 0$. Let $y_0^\infty := z^3y_0$ and $y_i^\infty := z^4y_i$ for $i = 1, 2$ be the generators on $t \neq 0$, and let

$$N^\infty := z^{10}N(z, z^{-3}y_0^\infty, z^{-4}y_1^\infty, z^{-4}y_2^\infty).$$

On $t \neq 0$, up to sign, the local representative of φ_v is $F_{\mathrm{br}} = y_1^\infty N^\infty$, and (13.14) reads $N^\infty|_{y_1^\infty=0} = k^2(y_2^\infty)^2$. The coefficient of y_0^2 in q_σ is $-z^2$, and the coefficient of $(y_0^\infty)^2$ in the equation of $\mathcal{C}$ on $t \neq 0$ is -1. Thus $y_2 \neq 0$ at x if x is over $z \neq 0, \infty$, and $y_2^\infty \neq 0$ at x if x is over $[s : t] = [0 : 1]$. By (13.14), N or N^∞, respectively, is a unit at x. Since F_{br} vanishes at x, condition (ii) of admissibility implies $x \notin \mathcal{P}$. By Construction-Proposition 13.1(2), the cover ψ has the local equation $\rho^2 = F_{\mathrm{br}}$ near x. Thus $y_1 = \rho^2N^{-1}$ if x is over $z \neq 0, \infty$, and, up to sign, $y_1^\infty = \rho^2(N^\infty)^{-1}$ if x is over $[s : t] = [0 : 1]$.

Case 2. $x \in \ell$ and $x \neq [0 : 0 : 1]$. Then $y_0 \neq 0$ at x, and we write $q_\sigma = z^2\Theta + y_1G$ with $G|_{z=y_1=0} = y_0$. Thus G is a unit at x, $\mathcal{C}$ is smooth at x, and on $\mathcal{C}$ near x we have $y_1 = z^2w$ with $w := -\Theta/G$. If $w(x) \neq 0$, then $y_1 = z^2w$ is a square times a unit. Suppose that $w(x) = 0$. Write $N = N|_{y_1=0} + y_1N'$, where the coefficients of N' have poles of order at most four. By (13.14),

$$z^2N = k^2y_2^2 + z^4wN',$$

which is regular near x and equal to $k^2y_2^2$ at x. By (13.13), the equalities $\Theta(x) = 0$ and $y_0(x) \neq 0$ imply $y_2(x) \neq 0$. Thus z^2N is a unit at x. Moreover, $F_{\mathrm{br}} = y_1N = w(z^2N)$ on $z \neq 0$ near x. Since F_{br} and $w(z^2N)$ are regular near x and z is a non-zero-divisor in the local ring of $\mathcal{C}$ at x, we have $F_{\mathrm{br}} = w(z^2N)$ near x. Since $[0 : 0 : 1]$ is the only point of $\mathcal{P}$ in the fiber $z = 0$, Construction-Proposition 13.1(2) implies the local equation $\rho^2 = w(z^2N)$. Thus $y_1 = z^2w = (z\rho)^2(z^2N)^{-1}$.

Case 3. $x = [0:0:1]$. We first prove that X' is smooth over x. Let

$$\mathcal{R} := \bigoplus_{i\geq 0} (f_2)_* \omega_{S/\mathbb{P}^1}^{\otimes i},$$

so that $X' = \operatorname{Proj} \mathcal{R}$. Let $\mathcal{V}_3^+$ be the invariant part of $(f_2)_*\omega_{S/\mathbb{P}^1}^{\otimes 3}$ under the involution of S over $\mathbb{P}^1$ induced by the hyperelliptic involution of a general fiber [CP06, Section 3, after Remark 3.2], let $\mathcal{Y} \subset \mathcal{R}$ be the subalgebra generated by $(f_2)_*\omega_{S/\mathbb{P}^1}$ and $(f_2)_*\omega_{S/\mathbb{P}^1}^{\otimes 2}$ [CP06, Definition 4.2], and let $\mathcal{Y}_i$ be the part of degree i of $\mathcal{Y}$. Since σ is the multiplication (Construction-Proposition 13.1(1)) and $\det \sigma$ is a non-zero section of $\mathcal{O}(11-6) = \mathcal{O}(5)$, [CP06, Definition 3.1, Lemma 4.1(1), Proposition 4.8] yield

$$\mathcal{V}_3^+ \cong \det \mathcal{O}(1)^{\oplus 2} \otimes \mathcal{O}(5) = \mathcal{O}(7).$$

We fix such an isomorphism. In the exact sequence [CP06, Lemma 4.4(1)] for $\mathcal{Y}_6$, the first map is the multiplication by q_σ on $(\det \mathcal{O}(1)^{\oplus 2})^{\otimes 2} \otimes E = E(4)$, so

$$\mathcal{Y}_6 = \operatorname{coker}\big(q_\sigma : E(4) \to \operatorname{Sym}^3 E\big) = F_\sigma(14).$$

By [CP06, Section 4, the paragraph before Definition 4.6], $\mathcal{V}_3^+ \cdot \mathcal{V}_3^+ \subset \mathcal{Y}_6$, so the multiplication induces

$$\delta : (\mathcal{V}_3^+)^{\otimes 2} \to \mathcal{Y}_6, \qquad \delta \in H^0\big(\mathbb{P}^1, \mathcal{Y}_6 \otimes (\mathcal{V}_3^+)^{-2}\big) \cong H^0(\mathbb{P}^1, F_\sigma).$$

By [CP06, Theorem 4.7, Definition 4.9], $[\delta]$ is the last entry of the 5-tuple of f_2, so $[\delta] = [\varphi_v]$ by Construction-Proposition 13.1(1), that is, $\delta = \lambda\varphi_v$ for some $\lambda \in \mathbb{C}^*$. We take for ρ the generator of $\mathcal{V}_3^+$ on $s \neq 0$ corresponding to $\lambda^{-1/2}$ times the local generator of $\mathcal{O}(7)$ there, for a square root $\lambda^{1/2}$ of λ. Then

$$\rho^2 = \delta(\rho \otimes \rho) = F_{\mathrm{br}} \quad \text{on } s \neq 0.$$

By [CP06, Lemma 4.3], on $s \neq 0$ we have $\mathcal{R} = \mathcal{Y} \oplus \eta\mathcal{Y}$ for an element η of degree 3, and $\mathcal{Y}_3$ is the anti-invariant part of $(f_2)_*\omega_{S/\mathbb{P}^1}^{\otimes 3}$. Thus $\eta \in \mathbb{C}^*\rho + \mathcal{Y}_3$, so $\mathcal{R} = \mathcal{Y} \oplus \rho\mathcal{Y}$, and $\mathcal{R}$ is generated over $\mathbb{C}[z]$ by $x_0, x_1, y_0, y_1, y_2, \rho$ on $s \neq 0$.

Near x, y_2 is a unit since $x = [0:0:1]$, and ρ is a unit since $\rho^2 = F_{\mathrm{br}}$ and $F_{\mathrm{br}}(x) \neq 0$ by condition (ii) of admissibility ($x \in \mathcal{P}$). We localize at the element ρ/y_2 of degree one and set $\rho = y_2$. Since $\mathcal{P}$ is contained in the branch locus $\mathcal{P} \cup \Delta_{[v]}$ of ψ (Construction-Proposition 13.1(2)), we have $\psi^{-1}(x) = \{x'\}$ for a point $x' \in X'$. Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{X',x'}$. Since A, B, C vanish at x, we have $x_0, x_1 \in \mathfrak{m}$. The multiplication relations are

$$\begin{aligned} x_0^2 &= y_0 + az^2y_1 + dz^2y_2, \\ x_0x_1 &= z\kappa + g_1y_1, \qquad \kappa := y_0 + (k+lz)y_2, \\ x_1^2 &= ny_1 + m_2z^2y_2, \qquad n := 1 + n_1z + n_2z^2. \end{aligned}$$

Here $n(x) = 1$ and $\kappa(x) = ky_2(x) \neq 0$, so n and κ are units. Eliminating y_1 from the middle and the third relations, we obtain

$$z\big(\kappa - g_1n^{-1}m_2zy_2\big) = x_1\big(x_0 - g_1n^{-1}x_1\big),$$

and $\kappa' := \kappa - g_1n^{-1}m_2zy_2$ is a unit. Thus $z = x_1\tau$ with $\tau := \kappa'^{-1}(x_0 - g_1n^{-1}x_1) \in (x_0, x_1)$. The third and first relations now imply $y_1, y_0 \in (x_0, x_1)$. Let $c_0 := k^2(m_2 - 2k) \neq 0$. The relation $\rho^2 = F_{\mathrm{br}}$ reads $y_2^2 \equiv c_0y_2^3$ modulo (x_0, x_1), and y_2 is a unit, so $y_2 - c_0^{-1} \in (x_0, x_1)$. Thus $\mathfrak{m} = (x_0, x_1)$. Since X' is a surface, $\mathcal{O}_{X',x'}$ is regular of dimension two, and r is an isomorphism over x. Finally, the third relation and $z = x_1\tau$ imply

$$y_1 = x_1^2n^{-1}\big(1 - m_2\tau^2y_2\big),$$

and $1 - m_2\tau^2y_2$ is a unit because $\tau \in \mathfrak{m}$.

By Cases 1, 2, and 3, any prime component of the divisor of y_1 on S has even multiplicity. Thus the divisor of y_1 on S is $2H$ for a divisor $H \geq 0$, and $2H \sim 2K_S$. By Lemma 2.3(2),

$\mathcal{O}_S(H - K_S)$ is a non-trivial 2-torsion line bundle, which contradicts the assumption that S is torsion-free. Thus $\det \sigma$ does not vanish at $z = 0$.

Step 5. In this step we prove that $(P, Q) \neq (0, 0)$.

By **Step 4**, β_0 and γ_0 are linearly independent, so $\mathbf{r}_0 = \gamma_0 \neq 0$. Suppose that $P = Q = 0$. Then $U = 0$, and $\mathbf{V}_3 = \mathbf{V}_4 = \mathbf{V}_5 = 0$ by (13.10). Thus (13.11) reads $(\gamma_0 \cdot \mathbf{y})(\mathbf{V}_2 \cdot \mathbf{y}) = 0$, so $\mathbf{V}_2 = 0$, and then (13.12) reads $(\gamma_0 \cdot \mathbf{y})(\mathbf{V}_1 \cdot \mathbf{y}) = 0$, so $\mathbf{V}_1 = 0$. Hence $v = 0$, a contradiction. Thus $(P, Q) \neq (0, 0)$. □

For a fibration f_2 as in Construction-Proposition 13.1(1), the bicanonical pencil is spanned by y_1, y_2, the generators of the summand $\mathcal{O}(4)^{\oplus 2}$ of E, and the intersection of $\mathcal{C}$ with the common zero section of y_1, y_2 is the zero scheme of D. Thus the base scheme of the pencil can be read off from the first row of σ and the Čech representative v. We prove that the base scheme consists of smooth points at which a general member is smooth, which implies that the genus is 4.

Theorem 13.5. *Let S be a torsion-free numerical Godeaux surface with a fibration $f_2 : S \to \mathbb{P}^1$ whose fibers are connected of genus 2. Then the bicanonical pencil of S has genus 4.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we describe the base scheme of the bicanonical pencil on X' in terms of D and the double cover ψ.

By Construction-Proposition 13.1(1), f_2 corresponds to an admissible pair $(\sigma, [v])$. By Construction-Proposition 13.1(2), $f_2 = p \circ \psi \circ r$, and the pullbacks of y_1, y_2 form a basis of the space $H^0(\mathbb{P}^1, E(-4)) = H^0(S, 2K_S)$. Let $\mathbb{P}(E) := \operatorname{Proj}(\operatorname{Sym} E)$ and

$$\Sigma := (y_1 = y_2 = 0) \subset \mathbb{P}(E).$$

By Construction-Proposition 13.1(1), a change of coordinates on $\mathbb{P}^1$ and automorphisms of $\mathcal{O}(1)^{\oplus 2}$ and E transform $(\sigma, [v])$ into an admissible pair corresponding to f_2, so we may use the coordinates of Lemmas 13.2 and 13.3 in **Steps 2 and 3**.

Among the monomials of $q_\sigma = AC - B^2$, only y_0^2 has no factor y_1 or y_2, and the coefficient of y_0^2 is $L_0 L_2 - L_1^2 = D$. Thus

$$q_\sigma|_\Sigma = D y_0^2.$$

Let $\mathcal{Z} := \mathcal{C} \cap \Sigma$. By Lemma 13.2, $D \neq 0$, so $\mathcal{Z}$ is finite and p maps $\mathcal{Z}$ isomorphically onto the zero scheme of D. The base scheme of the sections y_1, y_2 on X' is $\psi^{-1}(\mathcal{Z})$.

Assume that $\det \sigma \neq 0$ at a zero of D. Then A, B, C are linearly independent in the fiber of E over that zero, so the conic of $\mathcal{C}$ over that zero is smooth. Thus the point of $\mathcal{Z}$ over that zero is not contained in $\mathcal{P}$, and p is smooth at that point.

Over a point of $\mathcal{C}$ outside $\mathcal{P}$, we write the cover as $\rho^2 = F_{\mathrm{br}}$ (Construction-Proposition 13.1(2)). Since $v = U y_0 + V y_1 + W y_2$, any monomial of $q_\sigma v$ other than y_0^3 has a factor y_1 or y_2, and the coefficient of y_0^3 in $q_\sigma v$ is DU. Thus the restriction of F_{br} to Σ is determined by the Laurent parts of DU.

Step 2. In this step we assume that D has two distinct zeros, and we prove that $\psi^{-1}(\mathcal{Z})$ consists of four distinct reduced smooth points of X'.

We use the coordinates and the complex numbers j, h of Lemma 13.3. Then the first row of σ is $(s, 0, t)$, so $D = st$, and

$$\det \sigma = -js^5 \text{ at } t = 0, \qquad \det \sigma = -ht^5 \text{ at } s = 0.$$

Since $(jh)^2 = 1$, we have $j \neq 0$ and $h \neq 0$. Thus $\det \sigma \neq 0$ at both zeros of D, and by **Step 1** the conics of $\mathcal{C}$ over both zeros of D are smooth and $\mathcal{Z} \cap \mathcal{P} = \emptyset$. Let $\nu \neq 0$ be the scalar with $U = \nu(z^{-1} - j^2 z^{-6})$ (Lemma 13.3). On $s \neq 0$ we have $D = z$, so

$$DU = \nu - \nu j^2 z^{-5}.$$

On $s \neq 0$, F_{br} is the non-negative Laurent part of $q_\sigma v$, so the value of F_{br} at the point of $\mathcal{Z}$ over $z = 0$ is ν. On $t \neq 0$, up to sign, F_{br} is the negative Laurent part of $q_\sigma v$, and the coefficient

of y_0^3 in the negative Laurent part is $-\nu j^2 z^{-5}$. The monomial y_0^3 belongs to the summand $\mathcal{O}(-5)$ of $\mathrm{Sym}^3 E(-14)$, and the local generator of $\mathcal{O}(-5)$ on $t \neq 0$ is $z^{-5} y_0^3$. Thus the value of F_{br} at the point of $\mathcal{Z}$ over $s = 0$ is $-\nu j^2$, up to sign. Since $\nu \neq 0$ and $j \neq 0$, both values are non-zero, so ψ is étale over $\mathcal{Z}$.

Since D has simple zeros, $\mathcal{Z}$ consists of two distinct reduced points, and $\mathcal{C}$ is smooth along $\mathcal{Z}$ by **Step 1**. Thus $\psi^{-1}(\mathcal{Z})$ consists of four distinct reduced smooth points of X'. Moreover, y_1, y_2 generate the maximal ideal of the local ring of X' at any point of $\psi^{-1}(\mathcal{Z})$.

Step 3. In this step we assume that D has a double zero, and we prove that X' is smooth along $\psi^{-1}(\mathcal{Z})$ and that a linear combination of y_1, y_2 has non-zero differential at any point of $\psi^{-1}(\mathcal{Z})$.

We use the coordinates of Lemma 13.2. Then the first row of σ is $(s, t, 0)$, so $D = -t^2$. By Proposition 13.4, $\det \sigma \neq 0$ at $z = 0$, and

$$U = Pz^{-1} + Qz^{-2}, \qquad (P, Q) \neq (0, 0).$$

Let $x = [1 : 0 : 0]$ be the point of Σ over $z = 0$, so that $\mathcal{Z}$ is supported at x. By **Step 1**, $x \notin \mathcal{P}$, the conic of $\mathcal{C}$ over $z = 0$ is smooth, $\mathcal{C}$ is smooth at x, and $dz \neq 0$ at x.

On the chart $y_0 = 1$, the functions z, y_1, y_2 generate the maximal ideal of the local ring of $\mathcal{C}$ at x. Thus a linear combination w of y_1, y_2 has $dw \neq 0$ at x, and dz, dw are independent at x. The ideal (y_1, y_2) of the local ring of $\mathcal{C}$ at x has quotient $\mathbb{C}[z]/(z^2)$ by **Step 1**. Since z generates the maximal ideal of the local ring of $\mathcal{C}$ at x modulo w, the ideal (y_1, y_2) equals (w, z^2).

By **Step 1**, F_{br} modulo (y_1, y_2) is the non-negative Laurent part of DU, and

$$DU = -z^2(Pz^{-1} + Qz^{-2}) = -Pz - Q.$$

If $Q \neq 0$, then $F_{\mathrm{br}}(x) = -Q \neq 0$, so ψ is étale over x, and X' is smooth at both points over x. If $Q = 0$, then $P \neq 0$. Since $F_{\mathrm{br}} + Pz \in (w, z^2)$, we have

$$\frac{\partial F_{\mathrm{br}}}{\partial z}(x) = -P \neq 0$$

in the local coordinates z, w of $\mathcal{C}$ at x. Thus F_{br}, w are local coordinates of $\mathcal{C}$ at x, and the cover $\rho^2 = F_{\mathrm{br}}$ is smooth over x with local coordinates ρ, w. In both cases w has non-zero differential at any point of X' over x.

Step 4. In this step we prove that $|2K_S|$ has no fixed divisor.

By **Steps 2 and 3**, the base scheme of the sections y_1, y_2 on X' is a finite scheme supported at smooth points of X'. Thus at any singular point of X' some linear combination y of y_1, y_2 is non-zero, and $r^* y$ is non-zero along the fiber of r over that point. Since r is an isomorphism over the smooth locus of X', r creates no base point and no fixed curve. Thus $Z = 0$ in the notation of Theorem 2.4, so $|M| = |2K_S|$, and r is an isomorphism near the base points of $|2K_S|$.

Step 5. In this step we compute the genus.

If D has two distinct zeros, then y_1, y_2 generate the maximal ideal at any base point of $|2K_S|$ by **Step 2**. If D has a double zero, then w of **Step 3** has non-zero differential at any base point of $|2K_S|$. Thus at any base point of $|2K_S|$ some linear combination of y_1, y_2 has non-zero differential, so the linear combinations of y_1, y_2 with zero differential at a fixed base point form a proper linear subspace of $H^0(S, 2K_S)$. Since the base scheme is finite, a general member G of $|2K_S|$ is smooth at any base point.

By Theorem 2.4 with $Z = 0$, G is reduced and irreducible. Let $\beta : Y \to S$ and $f : Y \to \mathbb{P}^1$ be as in Construction-Proposition 3.1, so that the genus g of the bicanonical pencil is the genus of a general fiber of f. By Construction-Proposition 3.1, the strict transform of G on Y is a smooth connected curve, and it is a general fiber of f. Since β is an isomorphism away from the base points and G is smooth at the base points, G is smooth and isomorphic to the strict transform. By adjunction,

$$2g - 2 = 2K_S \cdot 3K_S = 6,$$

so $g = 4$. □

A hyperelliptic general fiber of a pencil of genus 3 would imply that S has an involution, so Theorem 13.5 excludes it.

Corollary 13.6. *Let S be a torsion-free numerical Godeaux surface whose bicanonical pencil has genus 3. Then the general fiber of the morphism $f : Y \to \mathbb{P}^1$ of Construction-Proposition 3.1 is nonhyperelliptic.*

Proof. Suppose that a general fiber of f is hyperelliptic. The relative canonical map

$$Y \dashrightarrow \mathbb{P}(f_*\omega_{Y/\mathbb{P}^1})$$

restricts to the canonical map on a general fiber, so the relative canonical map has degree 2 on a general fiber. Thus Y has a birational involution ι over $\mathbb{P}^1$. By [CP06, Section 3, after Remark 3.2], ι acts biregularly on the minimal model S (Y is of general type, hence not birationally ruled). The automorphism of S induced by ι is not the identity because ι is not the identity on a general fiber. Thus S admits an automorphism of order two, and by Construction-Proposition 13.1(3), S has a fibration f_2 as in Construction-Proposition 13.1(1). By Theorem 13.5, the bicanonical pencil of S has genus 4, which contradicts the assumption that the genus is 3. □

14. Pencils of genus three

The goal of this section is to prove Theorem 14.19. Unlike in Construction-Proposition 3.3(2), if the bicanonical pencil of S has genus 3, then the scheme-theoretic base locus of $|2K_X|$ is not a curvilinear scheme in the smooth locus of X, and the local computations of Sections 9–12 do not apply, so we use the relative canonical algebra of the morphism $f : Y \to \mathbb{P}^1$ of Construction-Proposition 3.1 [CP06, Lemma 7.2, Proposition 7.9].

14.1. **Profiles.** In this subsection, we describe the resolution of the bicanonical pencil of genus 3, the relative canonical algebra of f, the fibers of the relative canonical model of f, and the fixed divisor Z in Profile II.

Construction-Proposition 14.1. Notation and conditions as in Construction-Proposition 3.1. Assume that S is torsion-free and that the bicanonical pencil of S has genus 3. Then Y is relatively minimal over $\mathbb{P}^1$, and any fiber of f is numerically 2-connected: for any $u \in \mathbb{P}^1$ and any expression $f^*(u) = D_1 + D_2$ with non-zero divisors $D_1, D_2 \geq 0$, we have $D_1 \cdot D_2 \geq 2$. Moreover, exactly one of the following holds, where F is the fiber class, and E_i is the total transform on Y of the exceptional curve of the i-th blow-up of β:

(1) (Profile I) $Z = 0$, β is one blow-up, $m_1 = 2$, the curve $E := E_1$ is a (-1)-bisection, and $F \sim 2K_Y - 4E$.
(2) (Profile II) $K_S \cdot Z = 0$, $Z^2 = -2$, β consists of two blow-ups, $m_1 = m_2 = 1$, and $2K_Y \sim F + \beta^*Z + 3(E_1 + E_2)$.

In Profile II, for distinct centers, E_1, E_2 are (-1)-sections, and for infinitely near centers, the final exceptional curve E_2 is a (-1)-section and $E_1 - E_2$ is a vertical (-2)-curve. In both profiles, $K_Y \sim \beta^*K_S + \sum_i E_i$ and $F \sim \beta^*M - \sum_i m_iE_i$. Moreover, $K_Y^2 = 0$ in Profile I and $K_Y^2 = -1$ in Profile II.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we determine the multiplicities m_i, the divisor Z, and the irreducible exceptional curves of β. By [Har77, Proposition V.3.2, Proposition V.3.3, Proposition V.3.6], we have $E_i \cdot E_j = -\delta_{ij}$ and $\beta^*D \cdot E_i = 0$ for any divisor D on S, and

$$K_Y \sim \beta^*K_S + \sum_i E_i, \qquad F \sim \beta^*M - \sum_i m_iE_i.$$

Thus $K_Y^2 = 1 - k$, where k is the number of blow-ups of β. By Construction-Proposition 3.1(2) with $g = 3$, we have

$$\sum_i m_i = 2 \quad \text{and} \quad \sum_i m_i^2 = M^2.$$

Since the m_i are positive integers, either $k = 1$, $m_1 = 2$, and $M^2 = 4$, which is Profile I, or $k = 2$, $m_1 = m_2 = 1$, and $M^2 = 2$, which is Profile II. By Theorem 2.4, we have $Z = 0$ in Profile I, and $K_S \cdot Z = 0$ and $Z^2 = -2$ in Profile II. Since k differs, exactly one of Profile I and Profile II holds. Substituting $2K_S \sim M + Z$, we obtain

$$2K_Y \sim F + \beta^* Z + \sum_i (m_i + 2)E_i.$$

Thus $F \sim 2K_Y - 4E$ in Profile I and $2K_Y \sim F + \beta^* Z + 3(E_1 + E_2)$ in Profile II. Moreover, $K_Y^2 = 0$ in Profile I and $K_Y^2 = -1$ in Profile II.

In Profile I, E is an irreducible smooth rational curve, $E^2 = -1$, and $F \cdot E = 2$. In Profile II with distinct centers, E_1 and E_2 are disjoint irreducible smooth rational curves, $E_i^2 = -1$, and $F \cdot E_i = 1$. In Profile II with infinitely near centers, the irreducible exceptional curves of β are $E_1 - E_2$ and E_2, and we have

$$(E_1 - E_2)^2 = -2, \quad E_2^2 = -1, \quad F \cdot (E_1 - E_2) = 0, \quad F \cdot E_2 = 1.$$

Thus the only irreducible vertical curve contracted by β is $E_1 - E_2$, in the case of infinitely near centers, and

$$K_Y \cdot (E_1 - E_2) = (E_1 + E_2) \cdot (E_1 - E_2) = 0.$$

Step 2. In this step we prove that $K_Y \cdot C \geq 0$ for any irreducible vertical curve C on Y, and that Y is relatively minimal over $\mathbb{P}^1$. If β contracts C, then $K_Y \cdot C = 0$ by **Step 1**. Otherwise, C is not a component of any E_i, and since K_S is nef, we have

$$K_Y \cdot C = K_S \cdot \beta_* C + \sum_i E_i \cdot C \geq 0.$$

Since a (-1)-curve C satisfies $K_Y \cdot C = -1$, there exists no vertical (-1)-curve on Y. Moreover, $K_Y \cdot D \geq 0$ for any vertical divisor $D \geq 0$ on Y.

Step 3. In this step we prove that f has no multiple fiber. Let $u \in \mathbb{P}^1$, and suppose that $f^*(u) = nG$ for an integer $n \geq 2$ and a divisor $G \geq 0$.

In Profile II, we have $n(G \cdot E_2) = F \cdot E_2 = 1$, which is not possible.

In Profile I, we have $n(G \cdot E) = F \cdot E = 2$, so $n = 2$. Then $2G \sim F \sim 2K_Y - 4E$, so

$$2(K_Y - 2E - G) \sim 0.$$

By [Har77, Proposition V.3.2], we have $\operatorname{Pic}(Y) = \beta^* \operatorname{Pic}(S) \oplus \mathbb{Z}E$. Since S is torsion-free, $\operatorname{Pic}(Y)$ has no non-trivial torsion. Thus $K_Y \sim G + 2E \geq 0$, which contradicts $h^0(Y, K_Y) = p_g(S) = 0$.

Thus, if we write $f^*(u) = \sum_j n_j C_j$ with distinct prime divisors C_j, then $\gcd_j(n_j) = 1$.

Step 4. In this step we reduce the numerical 2-connectedness to the exclusion of $D_1 \cdot D_2 = 1$. Let $u \in \mathbb{P}^1$, and let $f^*(u) = D_1 + D_2$ with non-zero divisors $D_1, D_2 \geq 0$. Since $F \cdot D_1 = F \cdot D_2 = 0$, we have

$$D_1 \cdot D_2 = -D_1^2 = -D_2^2.$$

Since the general fiber of f is connected (Construction-Proposition 3.1), any fiber of f is connected [Har77, Corollary III.11.5]. By Zariski's lemma [BHPV04, Chapter III, Lemma 8.2], we have $D_1^2 \leq 0$, and if $D_1^2 = 0$, then $D_1 = qf^*(u)$ for a rational number q. Thus $D_1 \cdot D_2 \geq 0$. If $D_1 \cdot D_2 = 0$, then $0 < q < 1$ and $qn_j \in \mathbb{Z}$ for any j, where n_j are the multiplicities of $f^*(u)$. Since $\gcd_j(n_j) = 1$ by **Step 3**, we have $q \in \mathbb{Z}$, which is not possible. Thus $D_1 \cdot D_2 \geq 1$.

Suppose that $D_1 \cdot D_2 = 1$. Then $D_1^2 = D_2^2 = -1$. By the Riemann–Roch theorem [Har77, Theorem V.1.6], $D^2 + K_Y \cdot D$ is even for any divisor D on Y. Thus $K_Y \cdot D_1$ and $K_Y \cdot D_2$ are odd. By **Step 2**, they are non-negative. Moreover, by Construction-Proposition 3.1(2),

$$K_Y \cdot D_1 + K_Y \cdot D_2 = K_Y \cdot F = 4.$$

Possibly interchanging D_1 and D_2, we may assume that $K_Y \cdot D_1 = 1$ and $K_Y \cdot D_2 = 3$.

Step 5. In this step we exclude $D_1 \cdot D_2 = 1$ in Profile I. Since $F \sim 2K_Y - 4E$, we have

$$0 = F \cdot D_1 = 2K_Y \cdot D_1 - 4E \cdot D_1.$$

Thus $K_Y \cdot D_1 = 2E \cdot D_1$ is even, which contradicts $K_Y \cdot D_1 = 1$.

Step 6. In this step we exclude $D_1 \cdot D_2 = 1$ in Profile II. Let $G_1 := \beta_* D_1$, $G_2 := \beta_* D_2$, and $r_i := E_i \cdot D_1$. Then $G_1, G_2 \geq 0$, $G_1 + G_2 \sim M$, and $E_i \cdot D_2 = F \cdot E_i - r_i = 1 - r_i$. By [Har77, Proposition V.3.2], we have

$$D_1 = \beta^* G_1 - r_1 E_1 - r_2 E_2, \qquad D_2 = \beta^* G_2 - (1 - r_1) E_1 - (1 - r_2) E_2.$$

Thus

$$1 = K_Y \cdot D_1 = K_S \cdot G_1 + r_1 + r_2.$$

Since K_S is nef, $K_S \cdot G_1 \geq 0$. Moreover, $r_1 + r_2 = (E_1 + E_2) \cdot D_1 \geq 0$: if a component C of D_1 is not contracted by β, then $E_i \cdot C \geq 0$, and if C is contracted by β, then $C = E_1 - E_2$ by **Step 1** and $(E_1 + E_2) \cdot C = 0$.

Suppose that $K_S \cdot G_1 = 0$. If $G_1 \neq 0$, then $G_1 \cdot A > 0$ for an ample divisor A on S, so G_1 is not numerically trivial. By Lemma 2.2(4), $G_1^2 < 0$ and G_1^2 is even, so $G_1^2 \leq -2$. Thus

$$D_1^2 = G_1^2 - r_1^2 - r_2^2 \leq -2,$$

which contradicts $D_1^2 = -1$. If $G_1 = 0$ (for distinct centers, no vertical curve is contracted by β, so $G_1 \neq 0$), then by **Step 1**, D_1 is a multiple of $E_1 - E_2$, so $K_Y \cdot D_1 = 0$, which contradicts $K_Y \cdot D_1 = 1$. Thus $K_S \cdot G_1 = 1$ and $r_1 + r_2 = 0$.

By Theorem 2.4, we have $K_S \cdot G_2 = K_S \cdot M - K_S \cdot G_1 = 1$. By $D_2^2 = -1$ and $r_1 + r_2 = 0$, we have

$$G_2^2 = -1 + (1 - r_1)^2 + (1 - r_2)^2 = 1 + r_1^2 + r_2^2 \geq 1.$$

By Lemma 2.2(4), we have $G_2^2 \leq (K_S \cdot G_2)^2 = 1$. Thus $r_1 = r_2 = 0$ and $G_2^2 = 1$, so

$$(G_2 - K_S) \cdot K_S = 0 \quad \text{and} \quad (G_2 - K_S)^2 = 0.$$

By Lemma 2.2(4), $G_2 - K_S$ is numerically trivial. By Lemma 2.3(1) and since S is torsion-free, we have $G_2 \sim K_S$. Since $G_2 \geq 0$, we have $h^0(S, K_S) \geq 1$, which contradicts $p_g(S) = 0$.

By **Steps 4, 5, and 6**, we have $D_1 \cdot D_2 \geq 2$. □

In Profile I, a general member of $|2K_S|$ has multiplicity 2 at the center of β, so any member has multiplicity at least 2 there, since this is a linear condition on the members of the pencil, and in Profile II, any member of $|2K_S|$ contains Z, so the scheme-theoretic base locus of $|2K_X|$ contains the singular points of X to which the components of Z are contracted.

The relation of degree four of the relative canonical algebra does not lift to the symmetric square of the part of degree two of the algebra, and Theorem 14.12, Lemma 14.15, and Theorem 14.19 depend on the non-zero connecting class of the relation.

Construction-Proposition 14.2. Notation and conditions as in Construction-Proposition 14.1, and let

$$V_n := f_* \omega_{Y/\mathbb{P}^1}^{\otimes n} = f_* \mathcal{O}_Y(nK_Y + 2nF).$$

Then $V_1 = \mathcal{O}(1)^{\oplus 3}$, and

$$V_2 = \begin{cases} \mathcal{O}(2) \oplus \mathcal{O}(3)^{\oplus 4} \oplus \mathcal{O}(5) & \text{in Profile I,} \\ \mathcal{O}(2)^{\oplus 2} \oplus \mathcal{O}(3)^{\oplus 3} \oplus \mathcal{O}(5) & \text{in Profile II.} \end{cases}$$

The multiplication $\sigma_2 : \operatorname{Sym}^2 V_1 \to V_2$ has cokernel $\mathcal{O}_\tau$ for a divisor $\tau \geq 0$ on $\mathbb{P}^1$. Let ζ be the section of $V_2(-5)$, unique up to a non-zero scalar. We also regard ζ as a section of $\mathcal{O}_Y(2K_Y - F)$ through $H^0(\mathbb{P}^1, V_2(-5)) = H^0(Y, \mathcal{O}_Y(2K_Y - F))$, with divisor $\beta^* Z + \sum_i (m_i + 2)E_i$.

Let

$$c_\sigma : \operatorname{Sym}^2(\wedge^2 V_1) \to \operatorname{Sym}^2 V_2$$

be the *minor map* defined by

$$c_\sigma((g_1 \wedge g_2)(g_3 \wedge g_4)) = \sigma_2(g_1 g_3)\sigma_2(g_2 g_4) - \sigma_2(g_1 g_4)\sigma_2(g_2 g_3),$$

and let $N := 10$ in Profile I and $N := 9$ in Profile II. Let $U := V_2/\mathcal{O}(5)\zeta$, let $\bar{\sigma}_2 : \operatorname{Sym}^2 V_1 \to U$ be the composition of σ_2 with the projection, and let

$$\bar{c}_\sigma : \operatorname{Sym}^2(\wedge^2 V_1) \to \operatorname{Sym}^2 U$$

be the composition of c_σ with the projection. Then c_σ is injective with locally free cokernel, and there exists an exact sequence

$$0 \to \mathcal{O}(N) \xrightarrow{\iota} \operatorname{coker}(c_\sigma) \to V_4 \to 0.$$

We call ι the *relation of degree four*. Regarded as a section of $\operatorname{coker}(c_\sigma)(-N)$, ι is not the image of a section of $\operatorname{Sym}^2 V_2(-N)$. Thus the image of ι under the connecting homomorphism of

$$0 \to \operatorname{Sym}^2(\wedge^2 V_1)(-N) \to \operatorname{Sym}^2 V_2(-N) \to \operatorname{coker}(c_\sigma)(-N) \to 0$$

is a non-zero element of $\operatorname{Ker} H^1(c_\sigma(-N))$, which we call the *connecting class*.

Proof. By Corollary 13.6, the general fiber of f is nonhyperelliptic, and by Construction-Proposition 14.1, Y is relatively minimal with numerically 2-connected fibers, so [CP06, Lemma 7.2, Lemma 7.3, Lemma 7.7, Proposition 7.9] hold for f. The proof is long, so we divide it into several steps.

Step 1. In this step we prove that $h^0(Y, 2K_Y - F) = 1$. Let $D := \beta^* Z + \sum_i (m_i + 2)E_i$. By Construction-Proposition 14.1, we have $D \sim 2K_Y - F$ and $D \geq 0$. Since K_S is nef, K_X is ample, and $K_S \cdot Z = 0$, the morphism π contracts Z. Thus $\pi \circ \beta : Y \to X$ contracts $\operatorname{Supp} D$. Let h be a rational function on Y with $\operatorname{div}(h) + D \geq 0$. Then h induces a regular function on the complement of a finite set in the normal surface X, so h induces a regular function on X [Har77, Proposition II.6.3A], and h is constant [Har77, Theorem I.3.4]. Thus $h^0(Y, \mathcal{O}_Y(D)) = 1$, and the section of $\mathcal{O}_Y(2K_Y - F)$, unique up to a non-zero scalar, has divisor D.

Step 2. In this step we determine the splitting types of V_1 and V_2. By [CP06, Remark 2.10, Remark 2.11], we have

$$\operatorname{rank} V_1 = 3, \quad \deg V_1 = 3, \quad \operatorname{rank} V_2 = 6, \quad \deg V_2 = K_Y^2 + 19.$$

Since $h^0(\mathbb{P}^1, V_1(-2)) = h^0(Y, K_Y) = p_g(S) = 0$, any line bundle in a splitting of V_1 [Har77, Exercise V.2.6] has degree at most 1. Thus $V_1 = \mathcal{O}(1)^{\oplus 3}$.

By [CP06, Lemma 7.2(1)], we have $\operatorname{coker}(\sigma_2) \cong \mathcal{O}_\tau$ for a divisor $\tau \geq 0$ on $\mathbb{P}^1$. Thus $\sigma_2 : \mathcal{O}(2)^{\oplus 6} \to V_2$ is injective and generically an isomorphism. Write $V_2 = \bigoplus_{j=1}^6 \mathcal{O}(b_j)$. If $b_j < 2$ for some j, then the projection of σ_2 to $\mathcal{O}(b_j)$ is zero since $\operatorname{Hom}(\mathcal{O}(2), \mathcal{O}(b_j)) = 0$, which is not possible. Thus $b_j \geq 2$ for any j. By Lemma 2.2(2) and **Step 1**, we have

$$\sum_j \max\{b_j - 3, 0\} = h^0(\mathbb{P}^1, V_2(-4)) = h^0(Y, 2K_Y) = h^0(S, 2K_S) = 2$$

and

$$\sum_j \max\{b_j - 4, 0\} = h^0(\mathbb{P}^1, V_2(-5)) = h^0(Y, 2K_Y - F) = 1.$$

Thus exactly one b_j equals 5, and the other five b_j belong to $\{2, 3\}$. By $\sum_j b_j = K_Y^2 + 19$ and Construction-Proposition 14.1, four of the five are 3 in Profile I, and three of the five are 3 in

Profile II. Moreover, the section ζ of $V_2(-5)$, unique up to a non-zero scalar, has divisor D on Y by **Step 1**.

Step 3. In this step we compute $\deg \tau$ and obtain the exact sequence of the statement. By the exact sequence

$$0 \to \operatorname{Sym}^2 V_1 \to V_2 \to \mathcal{O}_\tau \to 0,$$

we have

$$\deg \tau = \deg V_2 - \deg \operatorname{Sym}^2 V_1 = (K_Y^2 + 19) - 12 = K_Y^2 + 7.$$

The formula of c_σ is [CP06, Definition 7.6], and c_σ is injective with locally free cokernel by [CP06, Lemma 7.7]. By [CP06, Definition 7.8, Proposition 7.9], the composition of the natural morphisms

$$\operatorname{Sym}^2 V_2 \to \operatorname{coker}(c_\sigma) \to V_4$$

is the multiplication of sections, and $\operatorname{coker}(c_\sigma) \to V_4$ is surjective. By [CP06, Proposition 7.9(3)], the kernel of $\operatorname{coker}(c_\sigma) \to V_4$ is $\det V_1 \otimes \mathcal{O}(\tau)$. Since

$$\deg V_1 + \deg \tau = K_Y^2 + 10 = N,$$

we obtain the exact sequence of the statement.

Step 4. In this step we prove that ι does not lift. By **Step 2**, the degree of any line bundle in the splitting of $\operatorname{Sym}^2 V_2$ induced by the splitting of V_2 is at most 8, except for the line bundle $\mathcal{O}(10)$ generated by ζ^2. Since $N \geq 9$, we have

$$H^0(\mathbb{P}^1, \operatorname{Sym}^2 V_2(-N)) = H^0(\mathbb{P}^1, \mathcal{O}(10-N))\zeta^2.$$

Suppose that ι is the image of $a\zeta^2$ for some $a \in H^0(\mathbb{P}^1, \mathcal{O}(10-N))$. Since $\iota \neq 0$, we have $a \neq 0$. By **Step 3**, the image of $a\zeta^2$ in $V_4(-N)$ is the section $(f^*a)\zeta^2$ of $\mathcal{O}_Y(4K_Y + (8-N)F)$, which is non-zero since Y is integral. However, the image of ι in $V_4(-N)$ is zero. Thus ι is not the image of a section of $\operatorname{Sym}^2 V_2(-N)$.

Since c_σ is injective, we have the short exact sequence of the statement. By exactness of the long cohomology sequence, the image of ι under the connecting homomorphism is a non-zero element of $\operatorname{Ker} H^1(c_\sigma(-N))$. □

Construction-Proposition 14.3 collects the description of the fibers of f used in the rest of this section.

Construction-Proposition 14.3. Notation and conditions as in Construction-Proposition 14.2. Let $H := K_Y + F$, let $X' := \operatorname{Proj} \bigoplus_{n \geq 0} V_n$ be the relative canonical model of f, let $r : Y \to X'$ be the natural morphism, and let x_0, x_1, x_2 be a basis of $H^0(Y, \mathcal{O}_Y(H)) = H^0(\mathbb{P}^1, V_1(-1))$. For $u \in \mathbb{P}^1$, let $F_u := f^*(u)$, and let X'_u be the fiber of $X' \to \mathbb{P}^1$ over u. Then the following hold.

(1) The following hold.
  (i) The sections x_0, x_1, x_2 of $\mathcal{O}_Y(H)$ have no common zero.
  (ii) In Profile II, we have $H \sim \beta^*(3K_S - Z)$.
(2) The surface X' has only rational double points, and r is the minimal resolution of X'.
(3) For any $u \in \mathbb{P}^1$, the following hold.
  (i) For any integer $n \geq 1$, the fiber of V_n at u is
  $$H^0(F_u, \omega_{F_u}^{\otimes n}) = H^0(X'_u, \omega_{X'_u}^{\otimes n}).$$
  (ii) The fiber of σ_2 at u is the multiplication of sections.
  (iii) The canonical ring of X'_u is the direct sum of the fibers of the V_n at u for $n \geq 0$.
  (iv) The restrictions of x_0, x_1, x_2 form a basis of $H^0(X'_u, \omega_{X'_u})$.
(4) If $u \notin \operatorname{Supp} \tau$, then the morphism $X'_u \to \mathbb{P}^2$ defined by x_0, x_1, x_2 embeds X'_u as a plane quartic.
(5) If $u \in \operatorname{Supp} \tau$, then the following hold.
  (i) The kernel of the fiber of σ_2 at u is spanned by a non-zero quadratic form q_u.

(ii) The canonical ring of X'_u is $\mathbb{C}[x_0, x_1, x_2, v]/(q_u, v^2 - G_u)$, with $\deg x_i = 1$ and $\deg v = 2$, where $G_u \in \mathbb{C}[x_0, x_1, x_2]$ is a form of degree 4.
(iii) The morphism $X'_u \to Q_u := (q_u = 0) \subset \mathbb{P}^2$ defined by x_0, x_1, x_2 is a finite flat double cover with algebra $\mathcal{O}_{Q_u} \oplus \mathcal{O}_{Q_u}(-2)$, where the conic Q_u is possibly reducible or non-reduced.
(iv) $\omega_{X'_u}$ is the pullback of $\mathcal{O}_{Q_u}(1)$.
(v) The image of the fiber of σ_2 at u consists of the pullbacks of the sections of $\mathcal{O}_{Q_u}(2)$.

(6) In the case of (5), a smooth point of X'_u maps to a smooth point of Q_u.

Proof. We prove (1). Let $u \in \mathbb{P}^1$. By Construction-Proposition 14.1, F_u is numerically 2-connected, so the canonical map of F_u is a morphism [CP06, Lemma 2.5]. Since $f_*\mathcal{O}_Y = \mathcal{O}_{\mathbb{P}^1}$, relative duality [CP06, Remark 2.10] implies that $R^1 f_* \omega_{Y/\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}$, so the map from the fiber of V_1 at u to $H^0(F_u, \omega_{F_u})$ is an isomorphism [Har77, Theorem III.12.11]. Since $\omega_{F_u} = \mathcal{O}_Y(H)|_{F_u}$ and $V_1(-1) \cong \mathcal{O}^{\oplus 3}$ is generated by x_0, x_1, x_2, the restrictions of x_0, x_1, x_2 to F_u span $H^0(F_u, \omega_{F_u})$, so they have no common zero on F_u. In Profile II, by Construction-Proposition 14.1 and $M \sim 2K_S - Z$, we have

$$H = K_Y + F \sim \beta^* K_S + E_1 + E_2 + \beta^* M - E_1 - E_2 \sim \beta^*(3K_S - Z).$$

(2) is [CP06, Remark 7.1].

We prove (3). Let $f' : X' \to \mathbb{P}^1$ be the induced morphism. By [CP06, Remark 2.10], we have $R^1 f_* \omega_{Y/\mathbb{P}^1}^{\otimes n} = 0$ for any integer $n \geq 2$. Thus, for any integer $n \geq 1$, the fiber of V_n at u is $H^0(F_u, \omega_{F_u}^{\otimes n})$ [Har77, Theorem III.12.11], where the case $n = 1$ was shown above. By (2), we have $\omega_{Y/\mathbb{P}^1} = r^* \omega_{X'/\mathbb{P}^1}$, $r_*\mathcal{O}_Y = \mathcal{O}_{X'}$, and $R^1 r_* \mathcal{O}_Y = 0$. Thus, for any integer $n \geq 1$,

$$f'_* \omega_{X'/\mathbb{P}^1}^{\otimes n} = V_n \quad \text{and} \quad R^1 f'_* \omega_{X'/\mathbb{P}^1}^{\otimes n} = R^1 f_* \omega_{Y/\mathbb{P}^1}^{\otimes n}.$$

Since $R^1 f_* \omega_{Y/\mathbb{P}^1} \cong \mathcal{O}$ and $R^1 f_* \omega_{Y/\mathbb{P}^1}^{\otimes n} = 0$ for $n \geq 2$, the fiber of V_n at u is $H^0(X'_u, \omega_{X'_u}^{\otimes n})$ for any integer $n \geq 1$ (X' is Gorenstein by (2)) [Har77, Theorem III.12.11]. Since restriction to a fiber is compatible with multiplication of sections, the fiber of σ_2 at u is the multiplication of sections. Moreover, $V_0 = \mathcal{O}$, and $R^1 f'_* \mathcal{O}_{X'} = R^1 f_* \mathcal{O}_Y$ is dual to V_1 by relative duality, so it is locally free, and $h^0(X'_u, \mathcal{O}_{X'_u}) = 1$ [Har77, Theorem III.12.11]. Thus the canonical ring of X'_u is the direct sum of the fibers of the V_n at u for $n \geq 0$, and the restrictions of x_0, x_1, x_2 form a basis of $H^0(X'_u, \omega_{X'_u})$.

We prove (4) and (5). Let $u \in \mathbb{P}^1$. By [CP06, Lemma 2.7], one of the following holds.

(a) $\omega_{X'_u}$ embeds X'_u as a plane quartic.
(b) The morphism $\varphi_u : X'_u \to \mathbb{P}^2$ defined by x_0, x_1, x_2 is a finite double cover of a plane conic, and X'_u is a complete intersection of type $(2, 4)$ in $\mathbb{P}(1, 1, 1, 2)$.

Assume (b), let v be the coordinate of degree 2 of $\mathbb{P}(1, 1, 1, 2)$, and write the relation of degree 2 as $q_u + \lambda v$, where $\lambda \in \mathbb{C}$ and q_u is a quadratic form in x_0, x_1, x_2. Since φ_u is not an embedding, $\lambda = 0$, and $q_u \neq 0$. Write the relation of degree 4 as $cv^2 + \ell v + G$, where $c \in \mathbb{C}$ and ℓ, G are forms in x_0, x_1, x_2 of degrees $2, 4$. If $c = 0$, then X'_u contains the point $x_0 = x_1 = x_2 = 0$ of $\mathbb{P}(1, 1, 1, 2)$, which is not possible since φ_u is a morphism. Thus $c \neq 0$, and completing the square, we may write the relation of degree 4 as $v^2 - G_u$, with $\deg G_u = 4$. Let

$$\mathcal{A}_u := \mathbb{C}[x_0, x_1, x_2, v]/(q_u, v^2 - G_u).$$

Since $\mathcal{A}_u$ is a complete intersection ring of dimension 2, we have $(\mathcal{A}_u)_n = H^0(X'_u, \mathcal{O}_{X'_u}(n))$ for any integer n. By adjunction (X'_u avoids the vertex, the only singular point of $\mathbb{P}(1, 1, 1, 2)$, since $c \neq 0$), $\omega_{X'_u} = \mathcal{O}_{X'_u}(-5 + 6) = \mathcal{O}_{X'_u}(1)$. Thus $\mathcal{A}_u$ is the canonical ring of X'_u, and $\omega_{X'_u}$ is the pullback of $\mathcal{O}_{Q_u}(1)$. Moreover, $\mathcal{A}_u$ is free over $\mathbb{C}[x_0, x_1, x_2]/(q_u)$ with basis $1, v$, so $X'_u \to Q_u$ is a finite flat double cover with algebra $\mathcal{O}_{Q_u} \oplus \mathcal{O}_{Q_u}(-2)$. By (3), the image of the fiber of σ_2 at u is spanned by the products $x_i x_j$, which are the pullbacks of the sections of $\mathcal{O}_{Q_u}(2)$, and the kernel of the fiber of σ_2 at u is spanned by q_u.

In case (a), the canonical ring of X'_u is the quotient of $\mathbb{C}[x_0, x_1, x_2]$ by a quartic form, so the fiber of σ_2 at u is bijective by (3). Since $\operatorname{coker}(\sigma_2) \cong \mathcal{O}_\tau$ and $\operatorname{Sym}^2 V_1$ and V_2 have rank 6, the fiber of σ_2 at u is not bijective if and only if $u \in \operatorname{Supp}\tau$. Thus case (b) holds exactly when $u \in \operatorname{Supp}\tau$. By (3), the restrictions of x_0, x_1, x_2 form a basis of $H^0(X'_u, \omega_{X'_u})$, so the morphism defined by x_0, x_1, x_2 is the morphism defined by $\omega_{X'_u}$. This implies (4) and (5).

We prove (6). Let P be a singular point of Q_u, and let Λ be the local ring of Q_u at P with maximal ideal $\mathfrak{m}_\Lambda$. Since Q_u is a plane curve, $\dim_{\mathbb{C}} \mathfrak{m}_\Lambda/\mathfrak{m}_\Lambda^2 = 2$. Let $g \in \Lambda$ be the image of G_u under a trivialization of $\mathcal{O}_{Q_u}(2)$ near P. Then the algebra of (5) at P is $\Lambda' := \Lambda[v]/(v^2 - g) = \Lambda \oplus \Lambda v$.

If g is a unit, then $\Lambda \to \Lambda'$ is étale, and since Λ is not regular, the local rings of Λ' are not regular. If $g \in \mathfrak{m}_\Lambda$, then Λ' is a local ring with maximal ideal $\mathfrak{n} := \mathfrak{m}_\Lambda \oplus \Lambda v$, and

$$\mathfrak{n}^2 = (\mathfrak{m}_\Lambda^2 + g\Lambda) \oplus \mathfrak{m}_\Lambda v.$$

Thus

$$\dim_{\mathbb{C}} \mathfrak{n}/\mathfrak{n}^2 = \dim_{\mathbb{C}} \mathfrak{m}_\Lambda/(\mathfrak{m}_\Lambda^2 + g\Lambda) + 1 \geq 2,$$

so Λ' is not regular, since $\dim \Lambda' = 1$. Thus any point of X'_u over P is a singular point of X'_u, which implies (6). □

Lemma 14.4. *Notation and conditions as in Construction-Proposition 14.1. Assume that Profile II holds. Then the following hold.*

(1) Any component of Z is a smooth rational (-2)-curve, the intersection matrix of the components of Z is negative definite, and the support of Z is connected.
(2) Any $W \in |2K_S - 2Z|$ is numerically 1-connected, that is, $W_1 \cdot W_2 \geq 1$ for any expression $W = W_1 + W_2$ with non-zero divisors $W_1, W_2 \geq 0$, and $h^0(\mathcal{O}_W) = 1$.
(3) If $|2K_S - 2Z| \neq \emptyset$, then $h^0(S, 3K_S - 2Z) = 0$.

Proof. We first prove (1). Since K_S is nef and $K_S \cdot Z = 0$, we have $K_S \cdot G = 0$ for any component G of Z. By Lemma 2.2(4), $D^2 \leq (K_S \cdot D)^2$ and $D^2 + K_S \cdot D$ is even for any divisor D on S, and a divisor D with $D^2 = (K_S \cdot D)^2$ is numerically equivalent to $(K_S \cdot D)K_S$. Since $K_S^2 = 1$, the Hodge index theorem [Har77, Theorem V.1.9, Remark V.1.9.1] also implies that $D^2 < 0$ for any non-zero $\mathbb{R}$-divisor $D \geq 0$ with $K_S \cdot D = 0$, and by adjunction [Har77, Proposition V.1.5, Exercise V.1.3], $2p_a(D) - 2 = D^2 + K_S \cdot D$ for any divisor $D \geq 0$. Thus $G^2 = 2p_a(G) - 2 < 0$ and $p_a(G) \geq 0$, so $G^2 = -2$ and $p_a(G) = 0$, and $G \cong \mathbb{P}^1$ [Har77, Exercise IV.1.8(b)].

Let D be a non-zero real linear combination of the components of Z, and write $D = D_+ - D_-$ with $\mathbb{R}$-divisors $D_+, D_- \geq 0$ without common components. Then

$$D^2 = D_+^2 + D_-^2 - 2D_+ \cdot D_- < 0.$$

Thus the intersection matrix of the components of Z is negative definite. Suppose that the support of Z is not connected. Then $Z = Z_1 + Z_2$ with non-zero divisors $Z_1, Z_2 \geq 0$ with disjoint supports. Since $K_S \cdot Z_i = 0$, we have $Z_i^2 < 0$ and Z_i^2 is even, so $Z_i^2 \leq -2$. Thus $Z^2 = Z_1^2 + Z_2^2 \leq -4$, which contradicts $Z^2 = -2$.

Next we prove (2). Since $|M|$ has no fixed component, M is nef. Let $W \in |2K_S - 2Z|$. Then

$$K_S \cdot W = 2, \qquad W^2 = 4K_S^2 + 4Z^2 = -4. \tag{14.1}$$

Let $W = W_1 + W_2$ with non-zero divisors $W_1, W_2 \geq 0$. Since K_S is nef, $K_S \cdot W_1$ and $K_S \cdot W_2$ are non-negative integers with $K_S \cdot W_1 + K_S \cdot W_2 = 2$. Suppose that $K_S \cdot W_1 = 0$. Then $W_1^2 < 0$ and $-Z \cdot W_1 = M \cdot W_1 \geq 0$, so

$$W_1 \cdot W_2 = W \cdot W_1 - W_1^2 = -2Z \cdot W_1 - W_1^2 > 0.$$

The case $K_S \cdot W_2 = 0$ follows by exchanging W_1 and W_2.

It remains to consider the case $K_S \cdot W_1 = K_S \cdot W_2 = 1$. Then $W_1^2 \leq 1$. If $W_1^2 = 1$, then $(W_1 - K_S)^2 = 0$ and $K_S \cdot (W_1 - K_S) = 0$, so $W_1 - K_S$ is numerically trivial. Since S is torsion-free, Lemma 2.3(1) implies that $W_1 \sim K_S$, which contradicts $p_g(S) = 0$. Since $W_1^2 + K_S \cdot W_1$ is even,

we obtain that W_1^2 is odd and $W_1^2 \le -1$. Exchanging W_1 and W_2, we obtain that W_2^2 is odd and $W_2^2 \le -1$. Since $W \sim 2(K_S - Z)$, $W \cdot W_1 = W_1^2 + W_1 \cdot W_2$ is even, so $W_1 \cdot W_2$ is odd. Suppose that $W_1 \cdot W_2 \le 0$. Then $W_1 \cdot W_2 \le -1$. Since $-4 = W^2 = W_1^2 + W_2^2 + 2W_1 \cdot W_2$, we have

$$W_1^2 = W_2^2 = W_1 \cdot W_2 = -1.$$

Thus $(W_1 - W_2)^2 = 0$ and $K_S \cdot (W_1 - W_2) = 0$, so $W_1 - W_2$ is numerically trivial, and $W_1 \sim W_2$ by Lemma 2.3(1). Thus $2W_1 \sim W \sim 2(K_S - Z)$, and $W_1 \sim K_S - Z$ by Lemma 2.3(1). Hence $K_S \sim W_1 + Z \ge 0$, which contradicts $p_g(S) = 0$. Thus $W_1 \cdot W_2 \ge 1$, and W is numerically 1-connected. By [KML08, Lemma 1.2, Remark 1.3(b)], $h^0(\mathcal{O}_W) = 1$.

Finally we prove (3). Let $W \in |2K_S - 2Z|$. By (14.1), $2p_a(W) - 2 = W^2 + K_S \cdot W = -2$, so $p_a(W) = 0$. By (2), $h^1(\mathcal{O}_W) = 0$, so $h^0(\omega_W) = 0$ [Har77, Corollary III.7.7]. By [Har77, Theorem III.7.11] applied to W and S in a projective space, together with the ideal sheaf sequence of W in S, there exists an exact sequence

$$0 \to \mathcal{O}_S(K_S) \to \mathcal{O}_S(K_S + W) \to \omega_W \to 0.$$

Since $p_g(S) = 0$, we obtain $h^0(S, 3K_S - 2Z) = h^0(S, K_S + W) = 0$. □

14.2. **Distinct centers.** In this subsection, we prove that in Profile II the centers of β are infinitely near (Proposition 14.7). Construction-Proposition 14.5 collects the restrictions of V_2 to E_1, E_2 and the quadratic form q that we use to prove Lemma 14.6 and Proposition 14.7.

Construction-Proposition 14.5. Notation and conditions as in Construction-Proposition 14.3. Assume that Profile II holds and that the centers of β are two distinct points. Then the morphism $h : Y \to \mathbb{P}^2$ defined by x_0, x_1, x_2 maps E_1, E_2 to two distinct points P_1, P_2. We take the basis x_0, x_1, x_2 with $P_1 = [1 : 0 : 0]$ and $P_2 = [0 : 1 : 0]$. Then the following hold.

(1) The morphism r is an isomorphism near $E_1 \cup E_2$.
(2) After a change of basis of the summand $\mathcal{O}(2)^{\oplus 2}$ of V_2, the composition of σ_2 with the projection $V_2 \to \mathcal{O}(2)^{\oplus 2}$ sends x_0^2, x_1^2 to the generators of $\mathcal{O}(2)^{\oplus 2}$, and sends $x_0x_1, x_0x_2, x_1x_2, x_2^2$ to zero.
(3) If $\bar{\sigma}_2$ is surjective, then $\operatorname{Ker} \bar{\sigma}_2 \cong \mathcal{O}(-1)$, the inclusion $\mathcal{O}(-1) \to \operatorname{Sym}^2 V_1$ is a quadratic form $q = Ax_0x_1 + B_1x_0x_2 + B_2x_1x_2 + Dx_2^2$ with A, B_1, B_2, D a basis of $H^0(\mathcal{O}(3))$, and $q = g\zeta$ on Y, where $g \in H^0(\mathcal{O}(6))$ has zero divisor τ.
(4) The summand $\mathcal{O}(3)^{\oplus 3} \oplus \mathcal{O}(5)$ of V_2 restricts to zero on E_1 and on E_2.

Proof. By Construction-Proposition 14.1 and Construction-Proposition 14.3(1),

$$K_Y \sim \beta^* K_S + E_1 + E_2, \quad F \sim \beta^* M - E_1 - E_2, \quad H = K_Y + F \sim \beta^*(3K_S - Z).$$

Since E_1, E_2 are disjoint (-1)-sections of f (Construction-Proposition 14.1), we have

$$E_i \cdot F = 1, \quad K_Y \cdot E_i = -1, \quad H \cdot E_i = 0.$$

By Construction-Proposition 14.3(1), x_0, x_1, x_2 have no common zero, so h is a morphism. Since $H \cdot E_i = 0$, the sections x_0, x_1, x_2 restrict to constants on E_i, so h maps E_i to a point P_i. For a general point u, the curve F_u is smooth and connected (Construction-Proposition 3.1(1)), $u \notin \operatorname{Supp} \tau$, and r induces an isomorphism $F_u \to X'_u$, because r contracts only finitely many curves and they are contained in fibers of f. Thus $h|_{F_u}$ is the canonical embedding of F_u as a plane quartic (Construction-Proposition 14.3(4)). Since $E_1 \cap E_2 = \emptyset$ and $E_i \cdot F_u = 1$, the morphism $h|_{F_u}$ separates the point of $E_1 \cap F_u$ and the point of $E_2 \cap F_u$. Thus $P_1 \ne P_2$.

We prove (1). Let Γ be a curve on Y contracted by r. Then Γ is contained in a fiber of f, and $K_Y \cdot \Gamma = 0$ because r is the minimal resolution of a surface with only rational double points (Construction-Proposition 14.3(2)). The curves contracted by β are E_1, E_2, which are sections of f, so $\beta_*\Gamma$ is a curve. Since S is minimal, K_S is nef, so

$$0 = K_Y \cdot \Gamma = K_S \cdot \beta_*\Gamma + E_1 \cdot \Gamma + E_2 \cdot \Gamma$$

is a sum of non-negative numbers. Thus $E_1 \cdot \Gamma = E_2 \cdot \Gamma = 0$, and Γ is disjoint from $E_1 \cup E_2$. Since r is an isomorphism outside the curves it contracts, this proves (1).

We prove (2) and (4). Let

$$\rho_i : V_2 \to f_*(\omega_{Y/\mathbb{P}^1}^{\otimes 2}|_{E_i})$$

be the restriction. Since $f|_{E_i}$ is an isomorphism and $(2K_Y + 4F) \cdot E_i = 2$, we have

$$f_*(\omega_{Y/\mathbb{P}^1}^{\otimes 2}|_{E_i}) \cong \mathcal{O}(2).$$

Since $\mathrm{Hom}(\mathcal{O}(k), \mathcal{O}(2)) = 0$ for any $k \geq 3$, the map ρ_i vanishes on $\mathcal{O}(3)^{\oplus 3} \oplus \mathcal{O}(5)$. Thus the map

$$\rho := (\rho_1, \rho_2) : V_2 \to \mathcal{O}(2)^{\oplus 2}$$

is the composition of the projection $V_2 \to \mathcal{O}(2)^{\oplus 2}$ with a constant 2×2 matrix. The sections x_1, x_2 vanish on E_1, the sections x_0, x_2 vanish on E_2, and $x_0|_{E_1}, x_1|_{E_2}$ are non-zero constants. Thus, after scaling the generators of $\mathcal{O}(2)^{\oplus 2}$, the composition $\rho \circ \sigma_2$ sends x_0^2, x_1^2 to the generators of $\mathcal{O}(2)^{\oplus 2}$, and sends $x_0x_1, x_0x_2, x_1x_2, x_2^2$ to zero. In particular, the constant matrix is invertible. After the change of basis of the summand $\mathcal{O}(2)^{\oplus 2}$ given by the constant matrix, the projection $V_2 \to \mathcal{O}(2)^{\oplus 2}$ coincides with ρ. This proves (2) and (4).

We prove (3). Assume that $\bar{\sigma}_2$ is surjective. Since

$$\mathrm{Sym}^2 V_1 = \mathcal{O}(2)^{\oplus 6}, \quad U = \mathcal{O}(2)^{\oplus 2} \oplus \mathcal{O}(3)^{\oplus 3},$$

and $12 - 13 = -1$, we have an exact sequence

$$0 \to \mathcal{O}(-1) \to \mathcal{O}(2)^{\oplus 6} \to \mathcal{O}(2)^{\oplus 2} \oplus \mathcal{O}(3)^{\oplus 3} \to 0. \tag{14.2}$$

Let q be the inclusion in (14.2), as in (3). The coefficients of q belong to $H^0(\mathcal{O}(3))$. By (2), the projection of $\bar{\sigma}_2(q) = 0$ to $\mathcal{O}(2)^{\oplus 2}$ has the coefficients of x_0^2 and x_1^2 in q as coordinates. Thus q has no term in x_0^2 or x_1^2, and

$$q = Ax_0x_1 + B_1x_0x_2 + B_2x_1x_2 + Dx_2^2.$$

Twisting (14.2) by $\mathcal{O}(-2)$ and dualizing, we obtain the exact sequence

$$0 \to \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-1)^{\oplus 3} \to \mathcal{O}^{\oplus 6} \to \mathcal{O}(3) \to 0,$$

where $\mathcal{O}^{\oplus 6} \to \mathcal{O}(3)$ is given by the six coefficients of q. Since $h^0(\mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-1)^{\oplus 3}) = 2$ and $h^0(\mathcal{O}(3)) = 4 = 6 - 2$, the six coefficients of q span $H^0(\mathcal{O}(3))$. Thus A, B_1, B_2, D is a basis of $H^0(\mathcal{O}(3))$.

Since $\bar{\sigma}_2 \circ q = 0$, the composition $\sigma_2 \circ q$ factors through the summand $\mathcal{O}(5)$ of V_2. Thus there exists $g \in H^0(\mathcal{O}(6))$ such that $\sigma_2 \circ q$ is g times the inclusion $\mathcal{O}(5) \to V_2$. Since $\bar{\sigma}_2$ is surjective, the snake lemma for the maps g, σ_2, and the identity of U implies that the cokernel of $g : \mathcal{O}(-1) \to \mathcal{O}(5)$ is isomorphic to $\mathrm{coker}(\sigma_2) \cong \mathcal{O}_\tau$ (Construction-Proposition 14.2). Thus the zero divisor of g is τ.

Since the inclusion $\mathcal{O}(5) \to V_2$ corresponds to the section ζ of $\mathcal{O}_Y(2K_Y - F)$ (Construction-Proposition 14.2), we have $q = g\zeta$ as sections of $\mathcal{O}_Y(2H + 3F) = \mathcal{O}_Y(2K_Y + 5F)$. □

We first show that the centers belong to Z. A center outside Z implies that on any fiber over a point of $\mathrm{Supp}\,\tau$ the value of ζ is outside the image of σ_2, so $\bar{\sigma}_2$ is surjective and we obtain the quadratic form q of Construction-Proposition 14.5(3), whose coefficient A of x_0x_1 is a non-zero section of $\mathcal{O}(3)$ and has three zeros counted with multiplicity. Since $(\mathrm{div}(x_2) - E_1 - E_2) \cdot E_i = 1$, and any zero of A is simple and corresponds to a point of intersection of $\mathrm{div}(x_2) - E_1 - E_2$ with E_1 or E_2, the form A has at most two zeros, a contradiction.

Lemma 14.6. *Notation and conditions as in Construction-Proposition 14.1. Assume that we are in Profile II and that the centers of β are two distinct points p_1, p_2. Then $p_1, p_2 \in Z$, the divisor Z has multiplicity one at each p_i, and a general member of $|M|$ intersects Z only at p_1, p_2, with intersection multiplicity one at each p_i.*

Proof. The proof is long, so we divide it into several steps. We use the notation of Construction-Proposition 14.5. Since $m_1 = m_2 = 1$, Construction-Proposition 14.2 implies that

$$\operatorname{div}(\zeta) = \beta^* Z + 3E_1 + 3E_2.$$

Let $u \in \mathbb{P}^1$. Since $E_i \cdot F_u = 1$, the set $E_i \cap F_u$ is one point e_i, the divisor F_u is reduced and smooth at e_i, and E_i intersects F_u transversely at e_i. By Construction-Proposition 14.5(1), r is an isomorphism near e_i, so we also regard e_i as a smooth point of X'_u. The value of ζ at u is $\zeta|_{F_u}$, up to a non-zero scalar. For $u \in \operatorname{Supp}\tau$, let $\pi_u : X'_u \to Q_u$ be the finite flat double cover of the plane conic Q_u [CP06, Lemma 2.7] defined by x_0, x_1, x_2 (Construction-Proposition 14.3(5)(iii)).

Step 1. In this step we suppose that one of p_1, p_2 is outside Z, and we prove that $\bar{\sigma}_2$ is surjective.

Suppose that at least one of p_1, p_2 does not belong to Z. Possibly interchanging p_1 and p_2, we may assume that $p_2 \notin Z$. We keep the assumption $p_2 \notin Z$ until the end of **Step 5**. Then $\beta^* Z$ is disjoint from E_2, so $\operatorname{div}(\zeta) = 3E_2$ near E_2. Thus, for any $u \in \mathbb{P}^1$,

$$\operatorname{mult}_{e_2}(\zeta|_{F_u}) = 3. \tag{14.3}$$

The cokernel of the map

$$\mathcal{O}(5) \to V_2 \to \operatorname{coker}(\sigma_2) \cong \mathcal{O}_\tau$$

is isomorphic to $\operatorname{coker}(\bar{\sigma}_2)$, so it suffices to prove that $\mathcal{O}(5) \to \mathcal{O}_\tau$ is surjective. Suppose that $\mathcal{O}(5) \to \mathcal{O}_\tau$ is not surjective. Since $\mathcal{O}_\tau$ is a cyclic module, Nakayama's lemma [AM69, Corollary 2.7] implies that there exists $u \in \operatorname{Supp}\tau$ such that the value of ζ at u belongs to the image of the fiber of σ_2 at u. By Construction-Proposition 14.3(3) and (5)(v), the section $\zeta|_{X'_u}$ is the pullback by π_u of a section of $\mathcal{O}_{Q_u}(2)$. By Construction-Proposition 14.3(6), the point $\pi_u(e_2) = P_2$ is a smooth point of Q_u.

If π_u is ramified at e_2, then the order of zero of $\zeta|_{X'_u}$ at e_2 is even, which contradicts (14.3). Thus π_u is unramified at e_2. Since π_u is finite and flat of degree 2, there exists a second point $e'_2 \neq e_2$ of X'_u over P_2, and π_u is étale at e'_2. Thus e'_2 is a smooth point of X'_u, and $\zeta|_{X'_u}$ has a zero of order exactly three at e'_2. Since $\pi_u(e_1) = P_1 \neq P_2$, we have $e'_2 \neq e_1$.

A rational double point has embedding dimension three [KM98, Theorem 4.22], and X'_u is cut out by one equation on X', so a rational double point of X' on X'_u is a singular point of X'_u. Thus X' is smooth at e'_2, and r is an isomorphism near $r^{-1}(e'_2)$ (Construction-Proposition 14.3(2)). We also regard e'_2 as a point of Y. Then F_u is reduced and smooth at e'_2. Since $E_i \cap F_u = \{e_i\}$, we have $e'_2 \notin E_1 \cup E_2$, so $\operatorname{div}(\zeta) = \beta^* Z$ near e'_2. A component of $\beta^* Z$ through e'_2 contained in a fiber of f would coincide with F_u near e'_2, and $\zeta|_{F_u}$ would vanish identically near e'_2, which is not the case. Let $Z' \leq \beta^* Z$ be the sum, with multiplicities, of the components of $\beta^* Z$ not contained in fibers of f. Then the local intersection number at e'_2 satisfies $(F_u \cdot Z')_{e'_2} = 3$. On the other hand, since $K_S \cdot Z = 0$ and $Z^2 = -2$,

$$F \cdot Z' = F \cdot \beta^* Z = M \cdot Z = (2K_S - Z) \cdot Z = 2.$$

Since $(F_u \cdot Z')_y \geq 0$ for any $y \in F_u \cap Z'$, we obtain $3 = (F_u \cdot Z')_{e'_2} \leq F \cdot Z' = 2$, a contradiction. Thus $\mathcal{O}(5) \to \mathcal{O}_\tau$ is surjective, and $\bar{\sigma}_2$ is surjective.

By Construction-Proposition 14.5(3), we obtain the quadratic form q with

$$Ax_0x_1 + B_1x_0x_2 + B_2x_1x_2 + Dx_2^2 = q = g\zeta, \tag{14.4}$$

where A, B_1, B_2, D is a basis of $H^0(\mathcal{O}(3))$, and the zero divisor of g is τ. In particular, A, B_1, B_2, D have no common zero. If $A(u) = 0$, then the value of q at u is the non-zero quadratic form

$$q(u) = x_2\lambda_u, \quad \text{where } \lambda_u := B_1(u)x_0 + B_2(u)x_1 + D(u)x_2. \tag{14.5}$$

We let $\ell := (x_2 = 0) \subset \mathbb{P}^2$. Then $P_1, P_2 \in \ell$.

Step 2. In this step we prove that any zero of A is outside $\operatorname{Supp}\tau$.

Suppose that $A(u) = 0$ and $u \in \operatorname{Supp}\tau$. Then $g(u) = 0$, so (14.4) implies that $q(u)$ belongs to the kernel of the fiber of σ_2 at u. Since $q(u) \neq 0$, we may take $q_u = q(u) = x_2\lambda_u$ (Construction-Proposition 14.3(5)(i)). Since e_1, e_2 are smooth points of X'_u, the points P_1, P_2 are smooth points

of Q_u (Construction-Proposition 14.3(6)). If $B_1(u) = B_2(u) = 0$, then Q_u is a double line, which has no smooth point, a contradiction. If $B_1(u) = 0 \neq B_2(u)$, then P_1 is the intersection point of the distinct lines ℓ and $(\lambda_u = 0)$, which is a singular point of Q_u, a contradiction. If $B_2(u) = 0 \neq B_1(u)$, then P_2 is the intersection point of the distinct lines ℓ and $(\lambda_u = 0)$, which is a singular point of Q_u, a contradiction. Thus $B_1(u)B_2(u) \neq 0$. In particular, $\ell \neq (\lambda_u = 0)$, and $P_1, P_2 \notin (\lambda_u = 0)$.

Let T be the sum, with the multiplicities of F_u, of the irreducible components of F_u that r does not contract and on which x_2 vanishes identically. Since $F_u = r^*X'_u$, for a component G of F_u that r does not contract, the multiplicity of G in F_u equals the multiplicity of $r(G)$ in X'_u. Since π_u is finite and flat of degree 2, and Q_u consists of the distinct lines ℓ and $(\lambda_u = 0)$, the pushforward by π_u of the cycle of X'_u is $2\ell + 2(\lambda_u = 0)$. Thus

$$h_*T = 2\ell.$$

We show that $E_i \cdot T = 1$. Since F_u is smooth and reduced at e_i, and Q_u coincides with ℓ near P_i, we have $T = F_u$ near e_i. Since $T \leq F_u$ and $E_i \cap F_u = \{e_i\}$, we have $E_i \cap T = \{e_i\}$, and

$$E_i \cdot T = E_i \cdot F_u = 1.$$

Since T is supported in a fiber of f, we have $F \cdot T = 0$, so $K_Y \cdot T = H \cdot T$. The divisor H is the pullback of a line under h, so the projection formula implies that $H \cdot T = \mathcal{O}_{\mathbb{P}^2}(1) \cdot 2\ell = 2$. Thus

$$K_S \cdot \beta_*T = K_Y \cdot T - (E_1 + E_2) \cdot T = 2 - 2 = 0.$$

The components of T are vertical and E_1, E_2 are sections, so $\beta_*T \geq 0$ is a non-zero divisor. Since $e_2 \in T \cap E_2$, we have $p_2 \in \operatorname{Supp} \beta_*T$. Let Γ be a component of β_*T through p_2. Since K_S is nef, $K_S \cdot \Gamma = 0$. Since $p_2 \notin Z$, the curve Γ is not a component of Z, so

$$M \cdot \Gamma = (2K_S - Z) \cdot \Gamma = -Z \cdot \Gamma \leq 0.$$

On the other hand, a general member of $|M|$ does not contain Γ and passes through $p_2 \in \Gamma$, so $M \cdot \Gamma > 0$, a contradiction. Thus $u \notin \operatorname{Supp} \tau$.

Step 3. In this step we subtract E_1, E_2 from the divisor of x_2, and we compute the intersection numbers of the remaining divisor with E_1, E_2.

Since $P_1, P_2 \in \ell$, the section x_2 vanishes on E_1, E_2. We show that E_2 has multiplicity one in $\operatorname{div}(x_2)$. Near E_2, the section x_1 has no zero, and we have the regular functions $\xi := x_0/x_1$ and $\eta := x_2/x_1$. Let u be a general point of $\mathbb{P}^1$. Then $A(u)g(u) \neq 0$, and $h|_{F_u}$ is the canonical embedding of the smooth nonhyperelliptic curve F_u (by Corollary 13.6 and Construction-Proposition 14.3(3)(4), since r contracts only finitely many curves and they are contained in fibers of f, so that r maps F_u isomorphically onto X'_u), so $\xi|_{F_u}, \eta|_{F_u}$ generate the maximal ideal of F_u at e_2. If E_2 had multiplicity at least two in $\operatorname{div}(x_2)$, then $\eta|_{F_u}$ would have a zero of order at least two at e_2, and $\xi|_{F_u}$ would be a parameter at e_2. Dividing (14.4) by x_1^2, we obtain on F_u

$$A(u)\xi + B_1(u)\xi\eta + B_2(u)\eta + D(u)\eta^2 = g(u)\frac{\zeta}{x_1^2}.$$

The left side has a zero of order one at e_2, and the right side has a zero of order three at e_2, a contradiction. Interchanging x_0 and x_1 and using $\operatorname{div}(\zeta) \geq 3E_1$, the argument for E_2 shows that E_1 has multiplicity one in $\operatorname{div}(x_2)$.

Let

$$R := \operatorname{div}(x_2) - E_1 - E_2.$$

Then $R \geq 0$ is a divisor containing neither E_1 nor E_2, and $R \sim H - E_1 - E_2$. Since $H \cdot E_i = 0$, $E_i^2 = -1$, and $E_1 \cdot E_2 = 0$, we have

$$R \cdot E_1 = R \cdot E_2 = 1. \tag{14.6}$$

Step 4. In this step we prove that any zero of A is of one of two types, and that the type determines a section E_i with center outside Z.

Fix $u \in \mathbb{P}^1$ with $A(u) = 0$. By **Step 2**, $g(u) \neq 0$ and $u \notin \operatorname{Supp}\tau$. By Construction-Proposition 14.3(4), h embeds X'_u as a plane quartic, and X'_u is smooth at P_1, P_2. By (14.4) and (14.5), on F_u we have

$$x_2\lambda_u = g(u)\zeta.$$

Suppose that ℓ is a component of X'_u. Since X'_u is smooth at P_2 and $P_2 \in \ell$, the curve X'_u coincides with ℓ near P_2. Thus $x_2\lambda_u$ vanishes identically on F_u near e_2, which contradicts (14.3). Thus ℓ is not a component of X'_u.

The quartic form of X'_u restricted to $\ell \cong \mathbb{P}^1$ is a non-zero binary form of degree 4, the intersection multiplicity of ℓ with X'_u at a point is the order of zero of the restriction at that point, and the orders of zero sum to 4. Let I_i be the intersection multiplicity of ℓ with X'_u at P_i. Then $I_i \geq 1$, $I_1 + I_2 \leq 4$, and I_i is the order of zero of $x_2|_{F_u}$ at e_i.

Suppose that $B_1(u)B_2(u) \neq 0$. Then λ_u has no zero at P_1, P_2. Thus I_i is the order of zero of $\zeta|_{F_u}$ at e_i, which is at least 3. This contradicts $I_1 + I_2 \leq 4$.

Suppose that $B_1(u) = B_2(u) = 0$. Then $D(u) \neq 0$, and $q(u) = D(u)x_2^2$. Thus the order of zero of $q(u)$ at e_2 is $2I_2$, which contradicts (14.3).

Thus exactly one of the following holds:

(i) $B_1(u) = 0$ and $B_2(u) \neq 0$.
(ii) $B_2(u) = 0$ and $B_1(u) \neq 0$.

In type (i), λ_u has no zero at P_2, so (14.3) implies that $I_2 = 3$. In type (ii), λ_u has no zero at P_1, so I_1 is the order of zero of $\zeta|_{F_u}$ at e_1. If $p_1 \in Z$, then $\beta^*Z \geq E_1$, so $\operatorname{div}(\zeta) \geq 4E_1$ near e_1, and $I_1 \geq 4$. Then $I_1 + I_2 \geq 5$, a contradiction. Thus, in type (ii), we have $p_1 \notin Z$, the divisor β^*Z is disjoint from E_1, and $I_1 = 3$.

Step 5. In this step we prove that any zero of A is simple, that the point of E_2 over a zero of type (i) is the only point of $R \cap E_2$, and that the point of E_1 over a zero of type (ii) is the only point of $R \cap E_1$, and we obtain a contradiction.

Let u be a zero of A of type (i), and let w be a local coordinate of $\mathbb{P}^1$ centered at u. Recall that $\xi = x_0/x_1$ and $\eta = x_2/x_1$ are regular near e_2. Since X'_u is a plane curve smooth at P_2 and $I_2 = 3 \geq 2$, the function $\xi|_{F_u}$ is a parameter at e_2. Since $F_u = (w = 0)$ is reduced and smooth at e_2, the pair (ξ, w) is a regular system of parameters of Y at e_2. Since ξ vanishes on E_2 and $\xi|_{F_u}$ has a zero of order one at e_2, we have $E_2 = (\xi = 0)$ near e_2. Thus

$$\eta = \xi\psi(\xi, w)$$

for a regular function ψ, and $R = (\psi = 0)$ near e_2. Trivializing $\mathcal{O}(3)$ and $\mathcal{O}(6)$ near u, we regard A, B_1, B_2, D, g as functions of w. Dividing (14.4) by x_1^2, we obtain

$$A(w)\xi + B_1(w)\xi^2\psi + B_2(w)\xi\psi + D(w)\xi^2\psi^2 = g\frac{\zeta}{x_1^2}.$$

Since $g(u) \neq 0$ and $\operatorname{div}(\zeta) = 3E_2$ near e_2, the right side is ξ^3 times a unit. Dividing by ξ and setting $\xi = 0$, we obtain

$$A(w) + B_2(w)\psi(0, w) = 0.$$

Since $B_2(u) \neq 0$, the function $B_2(w)$ is a unit near $w = 0$, so

$$\operatorname{mult}_u A = \operatorname{mult}_0(\psi(0, w)),$$

and $\operatorname{mult}_0(\psi(0, w))$ is the local intersection number of R and E_2 at e_2. Moreover, $\operatorname{mult}_u A > 0$ because $A(u) = 0$, and the local intersection number is at most $R \cdot E_2 = 1$ because R does not contain E_2 (**Step 3**). Thus $\operatorname{mult}_u A = 1$, and e_2 is the only point of $R \cap E_2$.

The argument for type (ii) is obtained by interchanging the indices: x_0 and x_1 are interchanged, $B_1(u) \neq 0$, and E_1 is disjoint from β^*Z by **Step 4**. Thus, for a zero u of A of type (ii), we have $\operatorname{mult}_u A = 1$, and the point of E_1 over u is the only point of $R \cap E_1$.

Thus A has at most one zero of type (i), at most one zero of type (ii), and all zeros of A are simple. Hence the zero divisor of A has degree at most 2. Since A is a non-zero section of $\mathcal{O}(3)$, the zero divisor of A has degree 3, a contradiction. Thus $p_1, p_2 \in Z$.

Step 6. In this step we prove the statements on multiplicities.

Let M' be a general member of $|M|$. Since $m_1 = m_2 = 1$, M' is smooth at p_1, p_2. Since $|M|$ has no fixed component, M' contains no component of Z. A local equation of Z at p_i belongs to the $\mathrm{mult}_{p_i} Z$-th power of the maximal ideal and M' is smooth at p_i, so the local intersection number $(M' \cdot Z)_{p_i}$ of M' and Z at p_i is at least $\mathrm{mult}_{p_i} Z$. Since $p_1, p_2 \in Z$, we obtain

$$2 = M \cdot Z \geq (M' \cdot Z)_{p_1} + (M' \cdot Z)_{p_2} \geq \mathrm{mult}_{p_1} Z + \mathrm{mult}_{p_2} Z \geq 2.$$

Thus $\mathrm{mult}_{p_1} Z = \mathrm{mult}_{p_2} Z = 1$, $(M' \cdot Z)_{p_1} = (M' \cdot Z)_{p_2} = 1$, and M' intersects Z only at p_1, p_2. □

Proposition 14.7. *Notation and conditions as in Construction-Proposition 14.1. Assume that Profile II holds. Then the centers of β are infinitely near.*

Proof. The proof is long, so we divide it into several steps.

Suppose that the centers of β are two distinct points p_1, p_2. We use the notation of Construction-Proposition 14.5. By Construction-Proposition 14.1, $F \sim \beta^* M - E_1 - E_2$, $K_Y \sim \beta^* K_S + E_1 + E_2$, and E_1, E_2 are (-1)-sections of f. Since $|M|$ has no fixed component, M is nef. Let $\ell := (x_2 = 0) \subset \mathbb{P}^2$, the line through P_1, P_2. For $u \in \mathbb{P}^1$, let $e_i(u)$ be the point of E_i over u. Since $E_i \cdot F_u = 1$, exactly one component of F_u passes through $e_i(u)$, with coefficient one in F_u, and F_u is smooth at $e_i(u)$ and transverse to E_i. By Construction-Proposition 14.5(1), r is an isomorphism near $E_1 \cup E_2$, so we also regard $e_i(u)$ as a smooth point of X'_u.

Step 1. In this step we show that Z is a reduced chain of smooth rational curves, and that the strict transform of Z is contained in one fiber of f.

By Lemma 14.6, exactly one component of Z passes through p_i, with coefficient one in Z. For a component G of Z, let z_G be the coefficient of G in Z, and let n_G be the number of the points p_1, p_2 on G. By Lemma 14.6, a general member of $|M|$ intersects Z only at p_1, p_2, with intersection multiplicity one at each p_i. Since $K_S \cdot G = 0$ and $M \sim 2K_S - Z$, we have

$$-Z \cdot G = M \cdot G = n_G, \quad \text{that is,} \quad \sum_{G'' \neq G} z_{G''}(G'' \cdot G) = 2z_G - n_G. \tag{14.7}$$

By Lemma 14.4(1), $G \cdot G'' \leq 1$ for any distinct components G, G'' of Z, because a matrix with diagonal -2 and off-diagonal entry at least 2 is not negative definite.

Suppose that p_1, p_2 are on one component G. Then $z_G = 1$ and $n_G = 2$, so the sum in (14.7) is zero, and $Z = G$ since the support of Z is connected. Suppose that p_1, p_2 are on different components, and let G_1 be the component through p_1. Then $z_{G_1} = n_{G_1} = 1$, so by (14.7) exactly one other component G_2 intersects G_1, with $z_{G_2} = 1$ and $G_1 \cdot G_2 = 1$. Let $k \geq 2$, and assume that we have distinct components $G_1, \ldots, G_k$ with coefficient one in Z, with $G_j \cdot G_{j+1} = 1$ for $1 \leq j < k$, and for any $j < k$ the components intersecting G_j are exactly G_{j-1}, G_{j+1}, where G_0 is omitted. Since $p_1 \in G_1$, we have $n_{G_k} \leq 1$. If $n_{G_k} = 1$, then $p_2 \in G_k$, the sum in (14.7) for G_k is 1, so G_{k-1} is the only component intersecting G_k, and we stop. If $n_{G_k} = 0$, then the sum is 2, so besides G_{k-1} exactly one further component G_{k+1} intersects G_k, with $z_{G_{k+1}} = 1$ and $G_k \cdot G_{k+1} = 1$, and $G_{k+1} \notin \{G_1, \ldots, G_{k-1}\}$ by the assumption on G_j with $j < k$. Since Z has finitely many components, we stop at some $G_k \ni p_2$. Since the support of Z is connected, $G_1, \ldots, G_k$ are all the components of Z. Thus Z is a reduced chain of smooth rational curves with p_1, p_2 on the end components and away from the nodes.

Let $Z' := \beta^* Z - E_1 - E_2$ be the strict transform of Z. For a component G of Z, the strict transform $\beta^* G - \sum_{p_i \in G} E_i$ of G satisfies

$$F \cdot \Big(\beta^* G - \sum_{p_i \in G} E_i\Big) = M \cdot G - n_G = 0.$$

Since p_1, p_2 are not nodes of Z, Z' is connected and $E_i \cdot Z' = 1$. Since Z' is connected and reduced, there exists a point $u_0 \in \mathbb{P}^1$ with $Z' \leq F_{u_0}$. Since E_i is not a component of $F_{u_0} - Z'$ and $E_i \cdot F_{u_0} = E_i \cdot Z' = 1$, the divisor $F_{u_0} - Z' \geq 0$ is disjoint from $E_1 \cup E_2$. Since a divisor on Y disjoint from $E_1 \cup E_2$ is a pullback of a divisor on S [Har77, Proposition V.3.2], $F_{u_0} - Z' = \beta^* W$ for a divisor $W \geq 0$ on S, and $\beta^* W \sim \beta^*(M - Z)$. Hence

$$W \in |2K_S - 2Z|. \tag{14.8}$$

By Construction-Proposition 14.2, regarded as a section of $\mathcal{O}_Y(2K_Y - F)$, ζ has the divisor

$$\beta^* Z + 3E_1 + 3E_2 = Z' + 4E_1 + 4E_2. \tag{14.9}$$

Step 2. In this step we show that h maps Z' onto a reduced plane conic which does not contain ℓ, and we define the linear forms a_i, b_i on $\mathbb{P}^1$.

By Construction-Proposition 14.3(1),

$$H - Z' \sim \beta^*(3K_S - 2Z) + E_1 + E_2.$$

Since $E_1 + E_2$ is β-exceptional, $\beta_* \mathcal{O}_Y(E_1 + E_2) = \mathcal{O}_S$. Thus Lemma 14.4(3) and (14.8) imply that

$$h^0(Y, H - Z') = h^0(S, 3K_S - 2Z) = 0.$$

Hence the restriction $H^0(Y, \mathcal{O}_Y(H)) \to H^0(Z', H|_{Z'})$ is injective.

By (14.7), for any component G of Z, we have

$$H \cdot \Big(\beta^* G - \sum_{p_i \in G} E_i\Big) = (3K_S - Z) \cdot G = n_G.$$

Thus H has multidegree (2) or $(1, 0, \ldots, 0, 1)$ on the connected reduced nodal chain Z' of copies of $\mathbb{P}^1$. Let Σ be a connected reduced nodal chain of copies of $\mathbb{P}^1$, and let L be a line bundle on Σ of non-negative degree on any component of Σ. Then $h^0(\Sigma, L) = \deg L + 1$ and $h^1(\Sigma, L) = 0$. If $\Sigma \cong \mathbb{P}^1$, this is clear. Otherwise, let Σ_1 be an end component of Σ with node v, and let Σ' be the closure of $\Sigma - \Sigma_1$. We use the exact sequence

$$0 \to L|_{\Sigma_1}(-v) \to L \to L|_{\Sigma'} \to 0,$$

with $h^0(\Sigma_1, L|_{\Sigma_1}(-v)) = \deg L|_{\Sigma_1}$ and $h^1(\Sigma_1, L|_{\Sigma_1}(-v)) = 0$, and we induct on the number of components. Since $\deg H|_{Z'} = 2$, we obtain $h^0(Z', H) = 3$ and $h^1(Z', H) = 0$. Since $h^0(Y, H) = 3$, the restriction above is an isomorphism, and $h|_{Z'}$ is the morphism defined by the complete linear system of $H|_{Z'}$.

Let $\Gamma := h(Z')$ with the reduced structure. If Z' is irreducible, then $h|_{Z'}$ embeds Z' as a smooth conic Γ. Otherwise, $H^0(Z', H)$ is the fiber product of the spaces $H^0(\mathcal{O}(1))$ on the end components over the values at the nodes, so h maps the end components of Z' isomorphically onto distinct lines, and contracts the other components of Z' to the intersection point of the lines. In both cases Γ is a reduced plane conic, $h|_{Z'}$ is a local isomorphism at $e_i(u_0)$, and $P_1 \neq P_2$ are smooth points of Γ, on different lines if Γ is reducible. Thus ℓ is not a component of Γ, and

$$(\ell \cdot \Gamma)_{P_1} = (\ell \cdot \Gamma)_{P_2} = 1. \tag{14.10}$$

We identify E_i with $\mathbb{P}^1$ through f. We trivialize $H|_{E_1}$ by $x_0|_{E_1}$ and $H|_{E_2}$ by $x_1|_{E_2}$, and we fix isomorphisms $N^*_{E_i/Y} \cong \mathcal{O}(1)$. The sections x_1, x_2 vanish on E_1, and x_0, x_2 vanish on E_2. Restricting $x_1, x_2 \in H^0(Y, \mathcal{O}_Y(H - E_1))$ to E_1, and $x_0, x_2 \in H^0(Y, \mathcal{O}_Y(H - E_2))$ to E_2, we obtain linear forms

$$a_1, b_1 \in H^0(\mathcal{O}(1)), \qquad a_2, b_2 \in H^0(\mathcal{O}(1)), \tag{14.11}$$

where a_1, b_1 are the restrictions of x_1, x_2 to E_1, and a_2, b_2 are the restrictions of x_0, x_2 to E_2. Since F_u is transverse to E_i at $e_i(u)$, $N^*_{E_i/Y}$ at $e_i(u)$ is the cotangent space of F_u at $e_i(u)$. Thus, up to a non-zero scalar, $(a_1(u), b_1(u))$ is the derivative of $(x_1/x_0, x_2/x_0)$ at $e_1(u)$ along F_u, and $(a_2(u), b_2(u))$ is the derivative of $(x_0/x_1, x_2/x_1)$ at $e_2(u)$ along F_u. Near $e_i(u_0)$, F_{u_0} coincides

with Z', so (14.10) implies that $b_1(u_0) \neq 0$ and $b_2(u_0) \neq 0$. In particular, b_1 and b_2 are non-zero linear forms.

Step 3. In this step we show that $\bar{\sigma}_2$ is surjective.

Let

$$y_1, y_2, y_3 \in H^0(\mathbb{P}^1, V_2(-3)) = H^0(Y, \mathcal{O}_Y(2H - F))$$

be generators of the summand $\mathcal{O}(3)^{\oplus 3}$ of V_2. By Construction-Proposition 14.5(2) and (4), for each of the four products $x_i x_j \in \{x_0x_1, x_0x_2, x_1x_2, x_2^2\}$, we have

$$x_i x_j = \sum_{k=1}^{3} \Lambda_{k,ij} y_k + \psi_{ij}\zeta, \tag{14.12}$$

where $\Lambda_{k,ij}$ are linear forms and ψ_{ij} are cubic forms on $\mathbb{P}^1$, and the sections y_k, ζ vanish on E_1 and on E_2. Let Λ be the 3×4 matrix $(\Lambda_{k,ij})$, with columns in the order $x_0x_1, x_0x_2, x_1x_2, x_2^2$, and let ψ be the row (ψ_{ij}). In the order $x_0^2, x_1^2, x_0x_1, x_0x_2, x_1x_2, x_2^2$ of the columns and $\mathcal{O}(2)^{\oplus 2}, y_1, y_2, y_3, \zeta$ of the rows, we have

$$\sigma_2 = \begin{pmatrix} I_2 & 0 \\ * & \Lambda \\ * & \psi \end{pmatrix}$$

by Construction-Proposition 14.5(2), where I_2 is the identity matrix of size two.

We trivialize $(2H)|_{E_1}$ by $x_0^2|_{E_1}$ and $(2H)|_{E_2}$ by $x_1^2|_{E_2}$. Then

$$\mathcal{O}_Y(2H - F - E_i)|_{E_i} \cong \mathcal{O}(-1) \otimes N^*_{E_i/Y} \cong \mathcal{O},$$

so the restriction of $y_k \in H^0(Y, \mathcal{O}_Y(2H - F - E_i))$ to E_i is a constant J_{ik}. Let $J := (J_{ik})$, a constant 2×3 matrix. We restrict (14.12) to $\mathcal{O}_Y(2H - E_i)|_{E_i} \cong \mathcal{O}(1)$. By (14.9), ζ contributes zero. By (14.11) and the product rule, we obtain

$$J\Lambda = \begin{pmatrix} a_1 & b_1 & 0 & 0 \\ a_2 & 0 & b_2 & 0 \end{pmatrix}. \tag{14.13}$$

Since $b_1 b_2 \neq 0$, the right-hand side has rank two at a general point of $\mathbb{P}^1$. Thus $\operatorname{rank} J = 2$.

For $u \in \mathbb{P}^1$, let $\zeta_u \in H^0(X'_u, \omega_{X'_u}^{\otimes 2})$ be the value of ζ at u (Construction-Proposition 14.3(3)). Since $\mathcal{O}(5)$ is a summand of V_2, $\zeta_u \neq 0$. Since $Z' \leq F_{u_0}$, $r^*\zeta_u = \zeta|_{F_u}$, and r is an isomorphism near E_1 and E_2, (14.9) implies that for $u \neq u_0$,

$$\text{the zero scheme of } \zeta_u \text{ on } X'_u \text{ is } 4e_1(u) + 4e_2(u). \tag{14.14}$$

Since $\operatorname{coker}(\sigma_2) \cong \mathcal{O}_\tau$, the map $\bar{\sigma}_2$ is surjective if and only if the map $\mathcal{O}(5) \to \mathcal{O}_\tau$ is surjective. By Nakayama's lemma, it is enough to show that for any $u \in \operatorname{Supp} \tau$, the section ζ_u is not in the image of the fiber of σ_2 at u.

Suppose that $u \in \operatorname{Supp} \tau$ and that ζ_u is in the image of the fiber of σ_2 at u. By Construction-Proposition 14.3(5)(iii) and (5)(v), the morphism $\pi : X'_u \to Q_u$ defined by x_0, x_1, x_2 is a finite flat double cover of a plane conic with algebra $\mathcal{O}_{Q_u} \oplus \mathcal{O}_{Q_u}(-2)$, and $\zeta_u = \pi^* s$ for a section s of $\mathcal{O}_{Q_u}(2)$. We have $\pi(e_i(u)) = P_i$, and P_i is a smooth point of Q_u by Construction-Proposition 14.3(6).

First suppose that $u \neq u_0$. Then $\zeta_u(e_1(u)) = 0$, so $s(P_1) = 0$, and ζ_u vanishes on $\pi^{-1}(P_1)$. By (14.14), since $P_1 \neq P_2$, the fiber $\pi^{-1}(P_1)$ is supported at $e_1(u)$. Since π is flat of degree two, $\pi^{-1}(P_1)$ has length two. Since X'_u is smooth at $e_1(u)$ and Q_u is smooth at P_1, any regular function on Q_u near P_1 which vanishes at P_1 pulls back to a function with a zero of order at least two at $e_1(u)$. Thus $a_1(u) = b_1(u) = 0$. Replacing $P_1, e_1(u)$ by $P_2, e_2(u)$, we obtain $a_2(u) = b_2(u) = 0$. By (14.13), $J\Lambda(u) = 0$. Since $\operatorname{rank} J = 2$, we have $\operatorname{rank} \Lambda(u) \leq 1$. By the form of σ_2, the fiber of σ_2 at u has rank at most

$$2 + \operatorname{rank} \Lambda(u) + \operatorname{rank} \psi(u) \leq 2 + 1 + 1 = 4.$$

However, since $\operatorname{coker}(\sigma_2) \cong \mathcal{O}_\tau$, the rank of the fiber of σ_2 at $u \in \operatorname{Supp} \tau$ is five, a contradiction.

Now suppose that $u = u_0$. Since $h|_{F_{u_0}} = \pi \circ r|_{F_{u_0}}$, we have $\Gamma \subset Q_{u_0}$, and hence $Q_{u_0} = \Gamma$ because both are plane conics. By (14.9), ζ vanishes on Z', so s vanishes at all points of Γ. Since Γ is reduced, $s = 0$, so $\zeta_{u_0} = 0$, a contradiction.

Step 4. In this step we derive the identities between the coefficients A, B_1, B_2 of the quadratic form q of Construction-Proposition 14.5(3) and the linear forms a_i, b_i of **Step 2**, and we show that $A(u_0) \neq 0$.

By **Step 3** and Construction-Proposition 14.5(3), the kernel of $\bar{\sigma}_2$ is $\mathcal{O}(-1)$, and the inclusion $\mathcal{O}(-1) \to \operatorname{Sym}^2 V_1$ is the quadratic form

$$q = Ax_0x_1 + B_1x_0x_2 + B_2x_1x_2 + Dx_2^2, \qquad q = g\zeta \text{ on } Y, \tag{14.15}$$

where A, B_1, B_2, D is a basis of $H^0(\mathcal{O}(3))$ and $g \in H^0(\mathcal{O}(6))$ has zero divisor τ. In particular, $q(u)$ is a non-zero quadratic form for any $u \in \mathbb{P}^1$. We restrict $q \in H^0(Y, \mathcal{O}_Y(2H + 3F - E_i))$ to E_i, in the trivializations of **Step 3**. By (14.9), the right-hand side $g\zeta$ contributes zero, so (14.11) implies that

$$Aa_1 + B_1b_1 = 0, \qquad Aa_2 + B_2b_2 = 0. \tag{14.16}$$

Recall that $\Gamma = h(Z')$ is the reduced plane conic of **Step 2**. By (14.9) and (14.15), q vanishes on Z', so $q(u_0)$ vanishes on Γ. Since $q(u_0) \neq 0$, we have $\Gamma = (q(u_0) = 0)$. If $A(u_0) = 0$, then x_2 divides $q(u_0)$, so ℓ is a component of Γ, which contradicts **Step 2**. Thus

$$A(u_0) \neq 0. \tag{14.17}$$

Step 5. In this step we show that $b_1(u) = b_2(u) = 0$ at any zero u of A.

Let u be a zero of A. Then

$$q(u) = x_2\big(B_1(u)x_0 + B_2(u)x_1 + D(u)x_2\big).$$

We first show that $u \notin \operatorname{Supp}\tau$. Suppose that $u \in \operatorname{Supp}\tau$. Since $g(u) = 0$, the non-zero form $q(u)$ spans the kernel of the fiber of σ_2 at u, so $Q_u = (q(u) = 0)$ and $Q_u = \ell + \ell'$ with $\ell' := (B_1(u)x_0 + B_2(u)x_1 + D(u)x_2 = 0)$. Since $e_i(u)$ is a smooth point of X'_u, Construction-Proposition 14.3(6) implies that P_i is a smooth point of Q_u. Since $P_1, P_2 \in \ell$, we obtain $\ell \neq \ell'$ and $P_1, P_2 \notin \ell'$.

Let $\pi : X'_u \to Q_u$ be the double cover of Construction-Proposition 14.3(5)(iii), and let T be the sum of the components of F_u mapped onto ℓ by h, with the coefficients they have in F_u. Let C be a component of T, and let c_C be the coefficient of C in F_u. Since $h|_{F_u} = \pi \circ r|_{F_u}$, the morphism r does not contract C. Since $F_u = r^*X'_u$, the number c_C equals the multiplicity of $r(C)$ in X'_u. Since H is the pullback of a line under h, we have

$$H \cdot T = \sum_C c_C \deg(h|_C : C \to \ell).$$

Since ℓ is generically reduced in Q_u, the right-hand side is the length of the fiber of π over the generic point of ℓ. Since π is finite and flat of degree two, we obtain

$$H \cdot T = 2.$$

Since $F \cdot T = 0$, we have $K_Y \cdot T = 2$. The component of F_u through $e_i(u)$ has coefficient one and is not contracted by r, because r is an isomorphism near E_i. Thus h maps the component of F_u through $e_i(u)$ onto a component of Q_u containing P_i, that is, onto ℓ. Hence $E_i \cdot T = E_i \cdot F_u = 1$. Thus

$$K_S \cdot \beta_*T = K_Y \cdot T - E_1 \cdot T - E_2 \cdot T = 0.$$

Let G' be the image under β of the component of F_u through $e_1(u)$. Then G' is an irreducible curve through p_1. Since G' is a component of β_*T, $K_S \cdot \beta_*T = 0$, and K_S is nef, we have $K_S \cdot G' = 0$. If G' is not a component of Z, then $Z \cdot G' > 0$ because $p_1 \in Z$, so $M \cdot G' = -Z \cdot G' < 0$, which contradicts the nefness of M. Thus G' is a component of Z, and the strict transform of G' is contained in $Z' \leq F_{u_0}$ and in F_u. Hence $u = u_0$, which contradicts (14.17). Thus $u \notin \operatorname{Supp}\tau$.

By Construction-Proposition 14.3(4), X'_u is a plane quartic embedded by x_0, x_1, x_2. By (14.17), $u \neq u_0$. Since $g(u) \neq 0$, (14.15) and (14.14) imply that the zero scheme of $q(u)$ on X'_u is $4e_1(u) + 4e_2(u)$. In particular, neither factor of $q(u)$ vanishes on a component of X'_u, and $\ell \cap X'_u \subset \{P_1, P_2\}$. Thus

$$(\ell \cdot X'_u)_{P_1} + (\ell \cdot X'_u)_{P_2} = 4, \qquad (\ell \cdot X'_u)_{P_i} \geq 1. \tag{14.18}$$

If $B_1(u) \neq 0$, then the second factor of $q(u)$ does not vanish at P_1, so $(\ell \cdot X'_u)_{P_1} = 4$, which contradicts (14.18). Thus $B_1(u) = 0$. Replacing P_1, B_1 by P_2, B_2, we obtain $B_2(u) = 0$. Since A, B_1, B_2, D have no common zero, $D(u) \neq 0$, and $q(u) = D(u)x_2^2$. Thus $(\ell \cdot X'_u)_{P_i} = 2$ for $i = 1, 2$. Hence x_2/x_0 has a zero of order two at $e_1(u)$ along X'_u, and x_2/x_1 has a zero of order two at $e_2(u)$ along X'_u. Thus $b_1(u) = b_2(u) = 0$.

Step 6. In this step we conclude the proof.

The form A is a non-zero cubic form, so A has a zero. Since b_1, b_2 are non-zero linear forms, **Step 5** implies that A, b_1, b_2 have exactly one common zero u_*, and that A has no other zero. Let w be a non-zero linear form vanishing at u_*. Then

$$A = cw^3, \qquad b_1 = d_1 w, \qquad b_2 = d_2 w$$

with non-zero constants c, d_1, d_2. By (14.16),

$$B_1 = -\frac{c}{d_1} w^2 a_1, \qquad B_2 = -\frac{c}{d_2} w^2 a_2.$$

Thus $A, B_1, B_2 \in w^2 H^0(\mathcal{O}(1))$, and $\dim w^2 H^0(\mathcal{O}(1)) = 2$. Hence $\dim \operatorname{Span}(A, B_1, B_2, D) \leq 3$, which contradicts that A, B_1, B_2, D is a basis of $H^0(\mathcal{O}(3))$. □

14.3. **Profile I.** In this subsection, we prove Theorem 14.12. In Profile I, the curve E is a (-1)-bisection, and comparing multiplication with restriction to E fixes the first two rows of σ_2. We will use the first two rows, the normal coefficients along E, and the connecting class to exclude Profile I.

Construction-Proposition 14.8. Notation and conditions as in Construction-Proposition 14.3. Assume that Profile I holds. Then there exist homogeneous coordinates $[s : t]$ on $\mathbb{P}^1$, coordinates $[u : v]$ on E with $f|_E([u : v]) = [u^2 : v^2]$, a basis x_0, x_1, x_2 of $H^0(Y, \mathcal{O}_Y(H))$, and sections w of $V_2(-2)$, $z_0, \ldots, z_3$ of $V_2(-3)$, and ζ of $V_2(-5)$ generating

$$V_2 = \mathcal{O}(2) \oplus \mathcal{O}(3)^{\oplus 4} \oplus \mathcal{O}(5),$$

for which the following hold.

(1) The sections $x_0, x_1, x_2, w, z_0, z_1, z_2, z_3, \zeta$ restrict to $u, v, 0, uv, 1, 0, 0, 0, 0$ on E.
(2) With rows indexed by $w, z_0, z_1, z_2, z_3, \zeta$ and columns indexed by $x_0^2, x_0x_1, x_1^2, x_0x_2, x_1x_2, x_2^2$, the rows of the matrix of σ_2 are $(0, 1, 0, 0, 0, 0)$, $(s, 0, t, 0, 0, 0)$, three rows of linear forms, and a row of cubic forms.
(3) The last four rows define a map
$$\Psi : \mathcal{K} \to \mathcal{O}(3)^{\oplus 3} \oplus \mathcal{O}(5),$$
where $\mathcal{K}$ is the kernel of the first two rows, and Ψ has fiber rank at least three at any point.
(4) We have $\mathcal{K} = \mathcal{O}(1) \oplus \mathcal{O}(2)^{\oplus 3}$, with generators $-tx_0^2 + sx_1^2$, x_0x_2, x_1x_2, and x_2^2.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the coordinates and compute the restriction to E of the generators of V_1 and V_2.

By Construction-Proposition 14.1, E is a smooth rational curve with $E^2 = K_Y \cdot E = -1$, and $F \sim 2K_Y - 4E$. Thus

$$F \cdot E = 2(K_Y \cdot E) - 4(E^2) = 2, \qquad H \cdot E = K_Y \cdot E + F \cdot E = 1.$$

In particular, $\deg(f|_E) = 2$. By the Hurwitz formula [Har77, Corollary IV.2.4], the ramification divisor of $f|_E$ has degree two. A ramification point of a morphism of degree two has index two, and distinct ramification points have distinct images, because a fiber of $f|_E$ has degree two. Thus $f|_E$ has exactly two ramification points, and their images are distinct. We take the ramification points and their images to 0 and ∞, and we obtain homogeneous coordinates $[u : v]$ on E and $[s : t]$ on $\mathbb{P}^1$ with $f|_E([u : v]) = [u^2 : v^2]$. We identify $\mathcal{O}_E(H) = \mathcal{O}_E(1)$, and we identify $\mathcal{O}_E(F) = \mathcal{O}_E(2)$ so that s, t restrict to u^2, v^2.

By Construction-Proposition 14.3(1), the elements of a basis of $H^0(Y, \mathcal{O}_Y(H))$ have no common zero. Thus their restrictions to E are sections of $\mathcal{O}_E(1)$ without common zero, and their restrictions to E span $H^0(E, \mathcal{O}_E(1))$, since a one-dimensional span would have a common zero. After a constant change of basis, we obtain x_0, x_1, x_2 restricting to $u, v, 0$ on E.

We have $V_2(-2) = f_*\mathcal{O}_Y(2H)$, $V_2(-3) = f_*\mathcal{O}_Y(2H - F)$, and $V_2(-5) = f_*\mathcal{O}_Y(2H - 3F)$. Let w', $z'_0, \dots, z'_3$, and ζ be generators of the summands of

$$V_2 = \mathcal{O}(2) \oplus \mathcal{O}(3)^{\oplus 4} \oplus \mathcal{O}(5)$$

(Construction-Proposition 14.2). Since

$$2H \cdot E = 2, \qquad (2H - F) \cdot E = 0, \qquad (2H - 3F) \cdot E = -4,$$

the section w' restricts to a binary quadratic form $a(u, v)$, the section z'_k restricts to a constant c_k for $k = 0, \dots, 3$, and ζ restricts to 0. For any $i \le j$, write

$$x_i x_j = q_{ij} w' + \sum_{k=0}^{3} L_{ijk}(s, t) z'_k + G_{ij}(s, t)\zeta,$$

where q_{ij} is a constant, L_{ijk} is a linear form, and G_{ij} is a cubic form. Restricting to E, we obtain

$$(x_i x_j)|_E = q_{ij} a(u, v) + \sum_{k=0}^{3} c_k L_{ijk}(u^2, v^2). \tag{14.19}$$

If $c_0 = c_1 = c_2 = c_3 = 0$, then all the restrictions in (14.19) are multiples of a, whereas x_0^2, x_0x_1, x_1^2 restrict to the linearly independent forms u^2, uv, v^2. Thus some c_k is non-zero, and after a constant change of $z'_0, \dots, z'_3$, we may assume that $(c_0, c_1, c_2, c_3) = (1, 0, 0, 0)$. Now (14.19) for x_0x_1 reads

$$uv = q_{01} a(u, v) + L_{010}(u^2, v^2),$$

so the coefficient of uv in a is non-zero. For a linear form λ, the section $w' + \lambda(s, t)z'_0$ of $V_2(-2)$ restricts to $a(u, v) + \lambda(u^2, v^2)$. Thus there exist a non-zero constant γ and a linear form λ such that

$$w := \gamma(w' + \lambda(s, t)z'_0)$$

restricts to uv on E. Let $z_k := z'_k$ for $k = 0, \dots, 3$. Since γ is non-zero, this is a triangular change of generators. Then $w, z_0, z_1, z_2, z_3, \zeta$ restrict to $uv, 1, 0, 0, 0, 0$ on E. This proves (1).

Step 2. In this step we compute the first two rows of σ_2.

Let q be the row of coefficients of w in σ_2, which has constant entries, and let $L(s, t)$ be the row of coefficients of z_0, which has linear entries. Since z_1, z_2, z_3, ζ restrict to 0 on E, restriction to E implies the identity of rows

$$(u^2, uv, v^2, 0, 0, 0) = uvq + L(u^2, v^2).$$

The entries of $L(u^2, v^2)$ belong to the span of u^2 and v^2. Comparing the coefficients of u^2, uv, v^2, we obtain

$$q = (0, 1, 0, 0, 0, 0), \qquad L(s, t) = (s, 0, t, 0, 0, 0).$$

The next three rows are maps $\mathcal{O}(2)^{\oplus 6} \to \mathcal{O}(3)$, and the last row is a map $\mathcal{O}(2)^{\oplus 6} \to \mathcal{O}(5)$, so their entries are linear forms and cubic forms, respectively. This proves (2).

Step 3. In this step we compute $\mathcal{K}$ and Ψ.

By **Step 2**, the first two rows define a map

$$\mathcal{O}(2)^{\oplus 6} \to \mathcal{O}(2) \oplus \mathcal{O}(3),$$

and the map is surjective on any fiber, since the first row is non-zero on x_0x_1 and s, t have no common zero. A local section

$$\alpha x_0^2 + \alpha' x_0x_1 + \alpha'' x_1^2 + \alpha_0 x_0x_2 + \alpha_1 x_1x_2 + \alpha_2 x_2^2$$

belongs to $\mathcal{K}$ if and only if $\alpha' = 0$ and $s\alpha + t\alpha'' = 0$. The kernel of $(s, t) : \mathcal{O}(2)^{\oplus 2} \to \mathcal{O}(3)$ is $\mathcal{O}(1)$, with inclusion $(-t, s)$. Thus $\mathcal{K} = \mathcal{O}(1) \oplus \mathcal{O}(2)^{\oplus 3}$, with generators $-tx_0^2 + sx_1^2$, x_0x_2, x_1x_2, and x_2^2.

The snake lemma, applied to the map from the exact sequence

$$0 \to \mathcal{K} \to \mathcal{O}(2)^{\oplus 6} \to \mathcal{O}(2) \oplus \mathcal{O}(3) \to 0$$

to the exact sequence

$$0 \to \mathcal{O}(3)^{\oplus 3} \oplus \mathcal{O}(5) \to V_2 \to \mathcal{O}(2) \oplus \mathcal{O}(3) \to 0$$

defined by Ψ, σ_2, and the identity, implies that

$$\operatorname{coker}(\Psi) \cong \operatorname{coker}(\sigma_2).$$

By Construction-Proposition 14.2, $\operatorname{coker}(\sigma_2) \cong \mathcal{O}_\tau$. For any $p \in \mathbb{P}^1$, the cokernel of the fiber of Ψ at p is the fiber $\mathcal{O}_\tau \otimes \mathcal{O}_p$ of $\mathcal{O}_\tau$ at p, and $\dim(\mathcal{O}_\tau \otimes \mathcal{O}_p) \leq 1$. Thus Ψ has fiber rank at least three at p. This proves (3) and (4). □

We now study x_2 and the sections vanishing on $2E$ near E, and the fibers of f over the branch values of $f|_E$.

Construction-Proposition 14.9. Notation and conditions as in Construction-Proposition 14.8. Fix an isomorphism $\mathcal{O}_E(-E) \cong \mathcal{O}_E(1)$, and identify $\mathcal{O}_E(H) = \mathcal{O}_E(1)$ through $x_0, x_1 \to u, v$ and $\mathcal{O}_E(F) = \mathcal{O}_E(2)$ through $s, t \to u^2, v^2$. For a section θ of a line bundle $\mathcal{L}$ on Y vanishing on jE, we call the restriction to E of the section of $\mathcal{L}(-jE)$ defined by θ the *j-th normal coefficient* of θ along E. Thus normal coefficients of sections of $\mathcal{O}_Y(aH + bF)$ are binary forms on E, and the normal coefficient of a product is the product of the normal coefficients. Then the following hold.

(1) We have $H^1(C, \mathcal{O}_C(2E)) = H^0(C, \omega_C(-2E)) = 0$ for any fiber C of f.
(2) We have
$$f_*\omega_{Y/\mathbb{P}^1}^{\otimes 2}(-2E) = \mathcal{O}(3) \oplus \mathcal{O}(5),$$
with generators z and ζ, where z is a section of $\mathcal{O}_Y(2H - F)$ vanishing on $2E$.
(3) The section x_2 of $\mathcal{O}_Y(H)$ satisfies $x_2^2 = lz + g_3\zeta$, with l a non-zero binary linear form and g_3 a binary cubic form.
(4) There exist linear forms R, T on E with distinct zeros, which are the ramification points of $f|_E$, for which the following hold.
  (i) Up to non-zero scalars, the first normal coefficient of x_2 along E is $RT = uv$.
  (ii) Up to a non-zero scalar, the second normal coefficient of z along E is T^2.
  (iii) The form l vanishes at the image of the zero of R.
(5) Any fiber C of f over a branch value of $f|_E$ is singular at the point of $C \cap E$.

Proof. By Construction-Proposition 14.1, we have $F \sim 2K_Y - 4E$, $F \cdot E = 2$, $E^2 = K_Y \cdot E = -1$, and any fiber of f is numerically 2-connected. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1) and (2).

Let $\mathcal{N} := \mathcal{O}_Y(K_Y - 2E)$, so $\mathcal{N}^{\otimes 2} = \mathcal{O}_Y(F)$. For any fiber C of f, by adjunction [Har77, Theorem III.7.11], we have $\omega_C = \mathcal{O}_C(K_Y + C) = \mathcal{O}_C(K_Y)$, so $\mathcal{N}|_C = \omega_C(-2E)$.

We prove that the restriction of $\mathcal{N}$ to the generic fiber of f is non-trivial. Suppose that the restriction is trivial. Then there exist a positive integer n and a section of $\mathcal{N}(nF)$ with vertical divisor $D \geq 0$, and

$$2D \sim (2n+1)F.$$

For $p \in \mathbb{P}^1$, let D_p be the part of D supported on $f^{-1}(p)$. For any curve $\Gamma \subset f^{-1}(p)$, we have $D_p \cdot \Gamma = D \cdot \Gamma = 0$, so $D_p^2 = 0$. By Zariski's lemma [BHPV04, Chapter III, Lemma 8.2], $D_p = r_p f^*(p)$ for a rational number r_p. If $f^*(p) = n_0 D_0$ with $n_0 \geq 2$ and $D_0 > 0$, then $f^*(p)^2 = 0$ implies $D_0^2 = 0$, so $D_0 \cdot (n_0 - 1)D_0 = 0 < 2$, contrary to the numerical 2-connectedness in Construction-Proposition 14.1. Thus the coefficients of $f^*(p)$ have greatest common divisor one, and r_p is an integer. Therefore $D \sim kF$ for an integer k, and intersecting $2D \sim (2n+1)F$ with E implies

$$2k(F \cdot E) = (2n+1)(F \cdot E),$$

that is, $2k = 2n+1$, which is not possible.

The restriction of $\mathcal{N}$ to the generic fiber of f is thus a non-trivial line bundle of degree zero ($K_Y \cdot F = 4 = 2E \cdot F$), and h^0 of the restriction is zero (a non-zero section of a line bundle of degree zero on an integral curve has no zero). By semicontinuity [Har77, Theorem III.12.8], $h^0(C, \mathcal{N}|_C) = 0$ for a general fiber C. By duality [Har77, Corollary III.7.7], for any fiber C we have

$$h^1(C, \mathcal{O}_C(2E)) = h^0(C, \omega_C(-2E)) = h^0(C, \mathcal{N}|_C).$$

Thus $R^1 f_* \mathcal{O}_Y(2E)$ is a torsion sheaf [Har77, Theorem III.12.11]. Let c be the length of $R^1 f_* \mathcal{O}_Y(2E)$, and let $G := f_* \mathcal{O}_Y(2E)$. Since $2E \cdot F = 4$ and any fiber has arithmetic genus three, the general fiber C satisfies $h^0(C, \mathcal{O}_C(2E)) = 2$, so $\operatorname{rank} G = 2$. By Riemann–Roch on Y [Har77, Theorem V.1.6],

$$\chi(\mathcal{O}_Y(2E)) = \chi(\mathcal{O}_Y) + \frac{1}{2}(2E \cdot (2E - K_Y)) = 1 + \frac{1}{2}(-4+2) = 0,$$

so by the Leray spectral sequence and Riemann–Roch on $\mathbb{P}^1$,

$$0 = \chi(G) - c = \deg G + 2 - c,$$

that is, $\deg G = c - 2$. Since $K_Y \sim \beta^* K_S + E$ and $F \sim 2K_Y - 4E$, we have $F + 2E \sim 2\beta^* K_S$, so

$$h^0(\mathbb{P}^1, G(1)) = h^0(Y, \mathcal{O}_Y(F + 2E)) = h^0(S, 2K_S) = 2.$$

Moreover,

$$\chi(G(1)) = \deg G + 4 = c + 2 \leq h^0(\mathbb{P}^1, G(1)) = 2,$$

so $c = 0$. Thus $R^1 f_* \mathcal{O}_Y(2E) = 0$ and $\deg G = -2$. By cohomology and base change [Har77, Theorem III.12.11], for any fiber C we have $H^1(C, \mathcal{O}_C(2E)) = 0$, and the fiber of G at $f(C)$ is $H^0(C, \mathcal{O}_C(2E))$. By duality, $H^0(C, \omega_C(-2E)) = 0$. This proves (1).

Since E is exceptional for β, we have $h^0(\mathbb{P}^1, G) = h^0(Y, \mathcal{O}_Y(2E)) = 1$. Since G is a direct sum of line bundles [Har77, Exercise V.2.6], $\operatorname{rank} G = 2$, and $\deg G = -2$, we obtain

$$G = \mathcal{O} \oplus \mathcal{O}(-2).$$

The section of $\mathcal{O}_Y(2E)$ with divisor $2E$ spans the summand $\mathcal{O}$. Since $H = K_Y + F$ and $F \sim 2K_Y - 4E$, we have

$$\omega_{Y/\mathbb{P}^1}^{\otimes 2} = \mathcal{O}_Y(2H + 2F) = \mathcal{O}_Y(5F + 4E),$$

so

$$f_* \omega_{Y/\mathbb{P}^1}^{\otimes 2}(-2E) = G(5) = \mathcal{O}(3) \oplus \mathcal{O}(5).$$

The generator of the summand $\mathcal{O}(5)$ is a section of $\mathcal{O}_Y(2H - 3F) = \mathcal{O}_Y(4E)$ with divisor $4E$. By Construction-Proposition 14.2, ζ has divisor $\beta^* Z + (m_1 + 2)E = 4E$, so ζ generates $\mathcal{O}(5)$. Let z be a generator of the summand $\mathcal{O}(3)$, which is a section of $\mathcal{O}_Y(2H - F)$ vanishing on $2E$. This proves (2).

Step 2. In this step we prove that, for a divisor A of degree two on a numerically 2-connected curve C of arithmetic genus three, under the conditions below, the complete linear system of $\mathcal{O}_C(2A)$ has no base point in $\operatorname{Supp} A$.

Let W be a smooth surface and let $C \geq 0$ be a numerically 2-connected divisor on W of arithmetic genus three. Let $A \geq 0$ be a Cartier divisor on C of degree two with $\operatorname{Supp} A$ contained in the smooth locus of C, and let $\eta := \omega_C(-2A)$. Assume that $\eta^{\otimes 2} = \mathcal{O}_C$ and $H^0(C, \eta) = 0$. We also prove that $h^0(C, \mathcal{O}_C(2A)) = 2$.

By Riemann–Roch and duality on C [BHPV04, Chapter II, Theorems 3.1 and 6.1], for any line bundle $\mathcal{L}$ on C we have

$$h^0(C, \mathcal{L}) - h^0(C, \omega_C \otimes \mathcal{L}^{-1}) = \deg \mathcal{L} - 2. \tag{14.20}$$

For $\mathcal{L} = \mathcal{O}_C(2A)$, (14.20) and $H^0(C, \eta) = 0$ imply that $h^0(C, \mathcal{O}_C(2A)) = 2$.

Suppose that $p \in \operatorname{Supp} A$ is a base point. Then $h^0(C, \mathcal{O}_C(2A - p)) = 2$, and (14.20) implies that $\eta(p)$ has a non-zero section ξ. We prove that ξ is a nonzerodivisor. Let $\mathcal{L}_0 := \omega_C \otimes \eta^{-1}(-p)$, and view ξ as a map $\mathcal{L}_0 \to \omega_C$. By [CFHR99, Lemma 2.4], ξ factors through a Cohen–Macaulay subcurve B of C as

$$\mathcal{L}_0 \to \mathcal{L}_0|_B \hookrightarrow \omega_B \subset \omega_C,$$

where the map $\mathcal{L}_0|_B \to \omega_B$ is generically onto, and it is injective since $\mathcal{L}_0$ is invertible and B is defined by $\operatorname{Ann}(\xi)$. Since B is a pure subcurve of a divisor on W, B is a divisor with $0 < B \leq C$. Suppose that $B \neq C$. Since η is torsion, $\deg_B \eta = 0$. The injection of line bundles on B (the cokernel has finite length, since the map is generically onto, and $\omega_B = \mathcal{O}_B(K_W + B)$ by adjunction) implies that

$$(K_W + C) \cdot B - \deg_B(p) \leq (K_W + B) \cdot B = (K_W + C) \cdot B - (C - B) \cdot B.$$

Since p is a smooth point of C, we have $\deg_B(p) \leq 1$, so $(C - B) \cdot B \leq 1$, contrary to numerical 2-connectedness. Thus $B = C$, and ξ is a nonzerodivisor.

The zero divisor of ξ is a Cartier divisor of degree $\deg \eta(p) = 1$. The local equation of the zero divisor generates the maximal ideal at the point of the zero divisor, so the zero divisor of ξ is a single smooth point q of C. Thus

$$\eta \cong \mathcal{O}_C(q - p), \qquad 2q \sim 2p, \qquad q \neq p,$$

where $2q \sim 2p$ follows from $\eta^{\otimes 2} = \mathcal{O}_C$, and $q \neq p$ follows from $H^0(C, \eta) = 0$. Since

$$\eta(q) = \mathcal{O}_C(2q - p) = \mathcal{O}_C(p),$$

we have $h^0(C, \eta(q)) \geq 1$. For $\mathcal{L} = \mathcal{O}_C(2A - q)$, (14.20) implies that $h^0(C, \mathcal{O}_C(2A - q)) \geq 2$, so q is a base point of the complete linear system of $\mathcal{O}_C(2A)$. The section of $\mathcal{O}_C(2A)$ with zero divisor $2A$ vanishes at q, so $q \in \operatorname{Supp} A$. If $A = 2P$ for a point P, then $p = P = q$, contrary to $q \neq p$. Thus p and q are distinct points of $\operatorname{Supp} A$, and since $\deg A = 2$, we have $A = p + q$. Therefore

$$\mathcal{O}_C(2A) = \mathcal{O}_C(2p + 2q) = \mathcal{O}_C(4p).$$

Since $2q \sim 2p$, the divisors $2p$ and $2q$ are the zero divisors of sections ξ_0, ξ_1 of $\mathcal{O}_C(2p)$. Let C_0 be the irreducible component of C through p, which is reduced at p since p is a smooth point of C. Since ξ_1 is a nonzerodivisor and $q \neq p$, the ratio ξ_0/ξ_1 on C_0 has a zero of order two at p, so the ratio is not constant on C_0. Thus $\xi_0^2, \xi_0\xi_1, \xi_1^2$ are linearly independent sections of $\mathcal{O}_C(4p)$, contrary to $h^0(C, \mathcal{O}_C(2A)) = 2$.

Step 3. In this step we prove (3) and (4).

Recall that x_2 restricts to 0 on E. Since $V_1 = \mathcal{O}(1)^{\oplus 3}$, we have

$$H^0(Y, \mathcal{O}_Y(H - F)) = H^0(\mathbb{P}^1, V_1(-2)) = 0,$$

so x_2 does not vanish on any fiber C of f. Thus $x_2|_C$ is a non-zero section of $\mathcal{O}_C(H) = \omega_C$. If x_2 vanished on $2E$, then for a general fiber C, $x_2|_C$ would be a non-zero section of $\omega_C(-2E)$, contrary to (1). Let ν_{x_2} be the first normal coefficient of x_2 along E. Then ν_{x_2} is a non-zero section of $\mathcal{O}_E(H - E) = \mathcal{O}_E(2)$.

We prove that there does not exist $p \in \mathbb{P}^1$ with $(f|_E)^{-1}(p) \leq \operatorname{div}(\nu_{x_2})$. Suppose that such a point p exists, and let $C := f^*(p)$. Let $e = 0$ be a local equation of E, let $\gamma = 0$ be a local equation of C, and write $x_2 = ea$ in local trivializations. The containment implies that $a \in (e, \gamma)$ near $E \cap C$, so $x_2 \in (e^2, \gamma)$ near $E \cap C$. Thus $x_2|_C$ is a non-zero section of $\omega_C(-2E)$ (E is horizontal, so e is a nonzerodivisor on $\mathcal{O}_C$), contrary to (1).

The section x_2^2 of $\mathcal{O}_Y(2H)$ vanishes on $2E$. By (2),

$$H^0(Y, \mathcal{O}_Y(2H - 2E)) = H^0(\mathbb{P}^1, \mathcal{O}(1))z \oplus H^0(\mathbb{P}^1, \mathcal{O}(3))\zeta,$$

so there exist a binary linear form l and a binary cubic form g_3 with $x_2^2 = lz + g_3\zeta$. Suppose that $l = 0$. Then for a general fiber C, the restriction of $x_2^2 = g_3\zeta$ to C and $\operatorname{div}(\zeta) = 4E$ imply that $\operatorname{div}(x_2|_C) = 2E|_C$, so $x_2|_C$ is a non-zero section of $\omega_C(-2E)$, contrary to (1). Thus $l \neq 0$. This proves (3).

The second normal coefficient ν_z of z along E is a section of $\mathcal{O}_E(2H - F - 2E) = \mathcal{O}_E(2)$. Since ζ vanishes on $4E$, taking second normal coefficients in $x_2^2 = lz + g_3\zeta$ implies the identity of binary quartics

$$\nu_{x_2}(u, v)^2 = l(u^2, v^2)\nu_z(u, v). \tag{14.21}$$

Suppose that the zero of l is not a branch value of $f|_E$. Then $l(u^2, v^2)$ has two distinct zeros, and by (14.21) both zeros are zeros of ν_{x_2}. Thus ν_{x_2} is a non-zero scalar multiple of $l(u^2, v^2)$, and $\operatorname{div}(\nu_{x_2})$ is a fiber of $f|_E$, which we have excluded. Therefore the zero of l is a branch value of $f|_E$, and up to a non-zero scalar, $l(u^2, v^2) = R(u, v)^2$, where R is the linear form vanishing at the ramification point over the zero of l. Now (14.21) implies that, up to non-zero scalars,

$$\nu_{x_2} = RT, \qquad \nu_z = T^2$$

for a linear form T. The zeros of R and T are distinct, since otherwise ν_{x_2} would be a scalar multiple of R^2, and $\operatorname{div}(\nu_{x_2})$ would be a fiber of $f|_E$.

Let P be the zero of T, and let $C := f^*(f(P))$. Near C, the sections ζ and z define generators of $G = f_*\mathcal{O}_Y(2E)$, up to nowhere vanishing functions on the base. By **Step 1**, the restrictions to C of the corresponding sections of $\mathcal{O}_Y(2E)$ form a basis of $H^0(C, \mathcal{O}_C(2E))$. The section defined by ζ has divisor $2E$, and the section defined by z restricts to $\nu_z = T^2$ on E. Thus P is a base point of the complete linear system of $\mathcal{O}_C(2E)$.

We apply **Step 2** to C, $A := E|_C$, and $\eta := \omega_C(-2E) = \mathcal{N}|_C$. The fiber C is numerically 2-connected of arithmetic genus three, $\deg A = F \cdot E = 2$, $\eta^{\otimes 2} = \mathcal{O}_C(F) = \mathcal{O}_C$, and $H^0(C, \eta) = 0$ by (1). If P is not a ramification point of $f|_E$, then $E \cap C$ consists of two distinct points at which $\gamma|_E$ has a simple zero, where $\gamma = 0$ is a local equation of C. Thus $d\gamma \neq 0$ at both points, both points are smooth points of C, and **Step 2** implies that P is not a base point, a contradiction. If P is a ramification point of $f|_E$ and C is smooth at P, then $A = 2P$, and **Step 2** again implies that P is not a base point, a contradiction. Thus P is a ramification point of $f|_E$, and C is singular at P.

Since $f|_E$ has exactly two ramification points and the zeros of R and T are distinct ramification points, the zeros of R and T are the ramification points of $f|_E$. Thus $RT = uv$ up to a non-zero scalar. By the definition of R, the form l vanishes at the image of the zero of R. This proves (4).

Step 4. In this step we prove (5).

By **Step 3**, the branch values of $f|_E$ are $f(P)$ and $f(P_0)$, where P_0 is the zero of R, and $f^*(f(P))$ is singular at the point P of $f^*(f(P)) \cap E$. Let $C := f^*(f(P_0))$, so $C \cap E = \{P_0\}$. Since $l(f(P_0)) = 0$ and l has a simple zero, in a local trivialization near P_0 the function $\gamma := l \circ f$ is a local equation of C. Let $e = 0$ be a local equation of E, and write

$$x_2 = ea, \qquad z = e^2 b, \qquad \zeta = e^4\epsilon,$$

with ϵ a unit. Cancelling e^2 in $x_2^2 = lz + g_3\zeta$, we obtain

$$a^2 = \gamma b + (g_3 \circ f)e^2\epsilon$$

near P_0. We have $a(P_0) = \nu_{x_2}(P_0) = 0$, $e(P_0) = 0$, and $\gamma(P_0) = 0$. Moreover, $b(P_0) \neq 0$, since $b|_E = \nu_z = T^2$ does not vanish at P_0. Differentiating at P_0, we obtain $b(P_0)d\gamma(P_0) = 0$, so $d\gamma(P_0) = 0$. Thus C is singular at P_0. This proves (5). □

We refine the normal form of Construction-Proposition 14.8 along E, and then we prove that Profile I does not occur.

Construction-Proposition 14.10. Notation and conditions as in Construction-Proposition 14.9. Then there exist coordinates and generators as in Construction-Proposition 14.8 for which, with rows indexed by $w, z_0, z_1, z_2, z_3, \zeta$ and columns indexed by $x_0^2, x_0x_1, x_1^2, x_0x_2, x_1x_2, x_2^2$, the matrix of σ_2 is

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ s & 0 & t & 0 & 0 & 0 \\ -bt & 0 & 0 & 0 & t & 0 \\ 0 & 0 & as & s & 0 & 0 \\ -c_1s - c_2t & 0 & c_0s & pt & qs & s \\ -d_1s^3 - d_2s^2t - d_3st^2 - d_4t^3 & 0 & d_0s^3 & et^3 & hs^3 & mt^3 \end{pmatrix} \tag{14.22}$$

for complex numbers $a, b, c_0, c_1, c_2, p, q, d_0, \dots, d_4, e, h, m$. Thus, with rows indexed by z_1, z_2, z_3, ζ and columns indexed by the generators $-tx_0^2 + sx_1^2, x_0x_2, x_1x_2, x_2^2$ of $\mathcal{K}$, the matrix of Ψ is

$$\begin{pmatrix} bt^2 & 0 & t & 0 \\ as^2 & s & 0 & 0 \\ c_0s^2 + c_1st + c_2t^2 & pt & qs & s \\ d_0s^4 + d_1s^3t + d_2s^2t^2 + d_3st^3 + d_4t^4 & et^3 & hs^3 & mt^3 \end{pmatrix}. \tag{14.23}$$

Proof. The proof is long, so we divide it into several steps. Let $V_2^{(j)} := f_*\mathcal{O}_Y(2H + 2F - jE)$ for $j = 1, 2$.

Step 1. In this step we replace z_1, z_2, z_3 by generators with first normal coefficients $u, v, 0$ along E.

By Construction-Proposition 14.8(1), the sections w, z_0 restrict to the generators $uv, 1$ of $(f|_E)_*\mathcal{O}_E(6)$ (by the projection formula, since $\mathcal{O}_E(6) = (f|_E)^*\mathcal{O}(3)$ and $(f|_E)_*\mathcal{O}_E = \mathcal{O} \oplus \mathcal{O}(-1)$), and the sections z_1, z_2, z_3, ζ restrict to 0. Thus z_1, z_2, z_3, ζ generate $V_2^{(1)}$, and $V_2^{(1)} = \mathcal{O}(3)^{\oplus 3} \oplus \mathcal{O}(5)$. Since $(2H + 2F - E) \cdot E = 7$, taking the first normal coefficient along E defines a map

$$\pi : V_2^{(1)} \to (f|_E)_*\mathcal{O}_E(7) = \mathcal{O}(3)^{\oplus 2}$$

with kernel $V_2^{(2)}$, where the summands $\mathcal{O}(3)$ are generated by u and v (by the projection formula and $(f|_E)_*\mathcal{O}_E(1) = \mathcal{O}^{\oplus 2}$).

Since $F \sim 2K_Y - 4E$ (Construction-Proposition 14.1), we have

$$\mathcal{O}_Y(2H + 2F - 2E) \cong \mathcal{O}_Y(5F + 2E).$$

By Construction-Proposition 14.9(1), $H^1(C, \mathcal{O}_C(2E)) = 0$ for any fiber C of f. Thus $R^1f_*\mathcal{O}_Y(2H + 2F - 2E) = 0$ [Har77, Theorem III.12.11], and π is surjective by the long exact sequence of f_* applied to

$$0 \to \mathcal{O}_Y(2H + 2F - 2E) \to \mathcal{O}_Y(2H + 2F - E) \to \mathcal{O}_E(7) \to 0.$$

Since ζ vanishes on $4E$, we have $\pi(\zeta) = 0$. Thus the constant 2×3 matrix of π on z_1, z_2, z_3 is of rank 2, and there exist constant combinations of z_1, z_2, z_3 with first normal coefficients u and v, which we take as the new z_1, z_2.

Let the new z_3 be the generator z of Construction-Proposition 14.9(2). Since z is a section of $V_2(-3)$ vanishing on E, z is a constant combination of the old z_1, z_2, z_3 plus a quadratic form times ζ. Since z, ζ generate $V_2^{(2)}$, the new z_1, z_2, z_3, ζ generate $V_2^{(1)}$. Thus w, z_0, the new z_1, z_2, z_3, and ζ generate V_2. The first two rows of σ_2 are the coefficients of w and z_0, so they stay as they are, and Construction-Proposition 14.8(1) and (2) hold for $w, z_0, z_1, z_2, z_3, \zeta$.

Step 2. In this step we reduce to the case when $l = s$.

Possibly scaling x_2, we may assume that the first normal coefficient of x_2 along E is uv (Construction-Proposition 14.9(4)). By Construction-Proposition 14.9(3),

$$x_2^2 = lz_3 + g_3\zeta,$$

where l is a non-zero linear form and g_3 is a binary cubic form. By Construction-Proposition 14.9(4), l vanishes at one of the branch values $[0:1]$ and $[1:0]$ of $f|_E$. If l vanishes at $[1:0]$, we exchange s, t, also u, v, also x_0, x_1, and also z_1, z_2, all at once. The relation $f|_E([u:v]) = [u^2 : v^2]$, the restrictions $u, v, 0$ of x_0, x_1, x_2, the first normal coefficients $u, v, 0$ of z_1, z_2, z_3, and the first two rows $(0,1,0,0,0,0)$ and $(s,0,t,0,0,0)$ of σ_2 stay as they are. Thus l vanishes at $[0:1]$. Possibly scaling z_3, we may assume that $l = s$. Now the linear forms of Construction-Proposition 14.9(4) are u and v up to non-zero scalars, with u the form whose zero maps to the zero of l. Since $s|_E = u^2$ and the second normal coefficient of x_2^2 is u^2v^2, the second normal coefficient of z_3 is v^2.

Step 3. In this step we compute the first normal coefficient of $\rho := -tx_0^2 + sx_1^2$, a section of $\mathcal{O}_Y(2H + F)$.

Since $s|_E = u^2$, $t|_E = v^2$, $x_0|_E = u$, and $x_1|_E = v$, we have $\rho|_E = -v^2u^2 + u^2v^2 = 0$. Since $(2H + F - E) \cdot E = 5$, the first normal coefficient ν_ρ of ρ is an element of $H^0(E, \mathcal{O}_E(5))$.

We prove that ν_ρ is divisible by uv. Let $y_0 \in E$ be the zero of u, let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{Y,y_0}$, and let ξ be a local equation of E at y_0. Since $f(y_0) = [0:1]$, Construction-Proposition 14.9(5) implies that the fiber $f^*([0:1])$ is singular at y_0. Thus the local equation s/t of $f^*([0:1])$ satisfies $s/t \in \mathfrak{m}^2$, that is, the differential of s/t vanishes at y_0. Since $x_0(y_0) = 0$, in a local trivialization we have $x_0^2 \in \mathfrak{m}^2$, hence $\rho \in \mathfrak{m}^2$. Write $\rho = \xi\rho'$, where $\rho'(y_0)$ is a non-zero multiple of $\nu_\rho(y_0)$. Since E is smooth, $\xi \in \mathfrak{m}\backslash\mathfrak{m}^2$, and since $\mathcal{O}_{Y,y_0}$ is regular, $\xi\rho' \in \mathfrak{m}^2$ implies that $\rho' \in \mathfrak{m}$. Thus $\nu_\rho(y_0) = 0$. Exchanging the roles of s, x_0 and t, x_1, we obtain that ν_ρ vanishes at the zero of v. Thus uv divides ν_ρ.

Let $\alpha_0, \alpha_1 \in \mathbb{C}$. Replacing x_0 by $x_0 + \alpha_0 x_2$ and x_1 by $x_1 + \alpha_1 x_2$ replaces ρ by

$$\rho + 2x_2(-\alpha_0 t x_0 + \alpha_1 s x_1) + (-\alpha_0^2 t + \alpha_1^2 s)x_2^2.$$

Since x_2 vanishes on E, the restrictions $u, v, 0$ stay as they are. Thus the first two rows of σ_2, which are determined by the restrictions of $x_i x_j$ to E, stay as they are. Since x_2^2 vanishes on $2E$, ν_ρ is replaced by

$$\nu_\rho + uv(-2\alpha_0 uv^2 + 2\alpha_1 u^2 v).$$

Choosing α_0, α_1 to remove the monomials u^2v and uv^2 of the cubic form $\nu_\rho/(uv)$, we may assume that

$$\nu_\rho = au^4v + buv^4 \tag{14.24}$$

for complex numbers a, b.

Step 4. In this step we reduce the columns of $x_0x_1, x_0x_2, x_1x_2, x_2^2$ of σ_2 to those of (14.22).

A combination $\phi_1 z_1 + \phi_2 z_2 + \phi_3 z_3 + \phi_4 \zeta$ with binary forms ϕ_i has first normal coefficient $\phi_1(u^2, v^2)u + \phi_2(u^2, v^2)v$. The sections $x_0x_2, x_1x_2, x_2^2, \rho$ vanish on E, and their first normal coefficients are $u^2 \cdot v$, $v^2 \cdot u$, 0, and $bv^4 \cdot u + au^4 \cdot v$ by (14.24). Since the terms $\phi_1(u^2, v^2)u$ and $\phi_2(u^2, v^2)v$ are separated by the parity of the exponent of u, the pairs of coefficients of z_1, z_2 in $x_0x_2, x_1x_2, x_2^2, \rho$ are $(0, s)$, $(t, 0)$, $(0, 0)$, and (bt^2, as^2).

We now change the generators, in the following order. Each change is an invertible triangular change of the split bundle V_2.

(1) Replace w by x_0x_1. Since w appears only in the column of x_0x_1, only the column of x_0x_1 changes, and it becomes $(1, 0, 0, 0, 0, 0)^t$. Also $x_0x_1|_E = uv$.
(2) The coefficients of z_3 in x_0x_2 and x_1x_2 are linear forms, which we write as $\gamma_2 s + pt$ and $qs + \gamma_1 t$ with $\gamma_1, \gamma_2, p, q \in \mathbb{C}$. Replace z_2 by $z_2 + \gamma_2 z_3$ and z_1 by $z_1 + \gamma_1 z_3$. Then the

coefficients of z_3 in x_0x_2 and x_1x_2 become pt and qs. Since z_3 vanishes on $2E$, the first normal coefficients of z_1, z_2 stay as they are.

(3) Write $g_3 = sg + mt^3$ with a quadratic form g and $m \in \mathbb{C}$, and replace z_3 by $z_3 + g\zeta$. Then $x_2^2 = sz_3 + mt^3\zeta$. Since ζ vanishes on $4E$, the second normal coefficient of z_3 stays as it is.

(4) Write the coefficient of ζ in x_0x_2 as $sg_2 + et^3$ and the coefficient of ζ in x_1x_2 as $tg_1 + hs^3$, with quadratic forms g_1, g_2 and $e, h \in \mathbb{C}$. Replace z_2 by $z_2 + g_2\zeta$ and z_1 by $z_1 + g_1\zeta$. Then the coefficients of ζ in x_0x_2 and x_1x_2 become et^3 and hs^3. Since z_1, z_2 do not appear in x_2^2, the coefficient mt^3 stays as it is.

Step 5. In this step we reduce the columns of x_0^2, x_1^2, and we conclude the proof.

Let (ϕ, ϕ') be the entries of one of the rows of z_1, z_2, z_3, ζ in the columns of x_0^2, x_1^2. Replacing z_0 by z_0 plus a constant multiple of z_1, z_2, or z_3, or plus a quadratic form times ζ, replaces (ϕ, ϕ') by $(\phi - \psi s, \phi' - \psi t)$ with an arbitrary form ψ of the corresponding degree. It does not change any other column, and it does not change the coefficient $-t\phi + s\phi'$ of ρ. The replacement of z_0 is also an invertible triangular change of the split bundle V_2. Since s, t are coprime, the solutions of the relations $-t\phi + s\phi' = bt^2$ in the row of z_1 and $-t\phi + s\phi' = as^2$ in the row of z_2 are $(-bt, 0) + c(s, t)$ and $(0, as) + c(s, t)$ for a constant c, which depends on the row. Thus the replacements of z_0 with $\psi = c$ reduce the rows of z_1, z_2 to $(-bt, 0)$ and $(0, as)$. In the rows of z_3 and ζ, we reduce ϕ' modulo t, and we obtain $(-c_1s - c_2t, c_0s)$ and $(-d_1s^3 - d_2s^2t - d_3st^2 - d_4t^3, d_0s^3)$ for complex numbers $c_0, c_1, c_2, d_0, \dots, d_4$.

We obtain (14.22). Since w, z_0 still restrict to $uv, 1$ on E and the sections z_1, z_2, z_3, ζ still vanish on E, Construction-Proposition 14.8(1) and (2) hold for the new generators. The formula of $\mathcal{K}$ stays as it is. The rank of Ψ at a point is the rank of the fiber of σ_2 at the point minus 2, since the first two rows are surjective on any fiber, with kernel the fiber of $\mathcal{K}$, so Construction-Proposition 14.8(3) also holds. Finally, the column of ρ in the matrix of Ψ is $-t$ times the column of x_0^2 plus s times the column of x_1^2, which implies (14.23). □

Lemma 14.11. *Notation and conditions as in Construction-Proposition 14.10. Assume that $p \neq 0$. Then $a = 0$ and $c_0 = q^2$.*

Proof. By Construction-Proposition 14.2, the connecting class is non-zero, and it is an element of $\operatorname{Ker} H^1(c_\sigma(-10))$. Since $\bar{c}_\sigma$ is the composition of c_σ with the projection, we have $\operatorname{Ker} H^1(c_\sigma(-10)) \subset \operatorname{Ker} H^1(\bar{c}_\sigma(-10))$, so $\operatorname{Ker} H^1(\bar{c}_\sigma(-10)) \neq 0$. We prove that if $a \neq 0$ or $c_0 \neq q^2$, then $\operatorname{Ker} H^1(\bar{c}_\sigma(-10)) = 0$.

Let Q_{ij} be the column of x_ix_j in (14.22) after dropping ζ, regarded as a combination of w, z_0, z_1, z_2, z_3:

$$\begin{aligned} &Q_{00} = sz_0 - btz_1 - (c_1s + c_2t)z_3, \quad Q_{01} = w, \qquad\qquad Q_{11} = tz_0 + asz_2 + c_0sz_3, \\ &Q_{02} = sz_2 + ptz_3, \qquad\qquad Q_{12} = tz_1 + qsz_3, \quad Q_{22} = sz_3. \end{aligned}$$

By the formula of c_σ (Construction-Proposition 14.2, [CP06, Definition 7.6]), on the products $(x_0 \wedge x_1)^2$, $(x_0 \wedge x_1)(x_0 \wedge x_2)$, $(x_0 \wedge x_1)(x_1 \wedge x_2)$, $(x_0 \wedge x_2)^2$, $(x_0 \wedge x_2)(x_1 \wedge x_2)$, and $(x_1 \wedge x_2)^2$, the six columns of $\bar{c}_\sigma(-10)$ are, in order,

$$\begin{aligned} &Q_{00}Q_{11} - Q_{01}^2, \quad Q_{00}Q_{12} - Q_{01}Q_{02}, \quad Q_{01}Q_{12} - Q_{02}Q_{11}, \\ &Q_{00}Q_{22} - Q_{02}^2, \quad Q_{01}Q_{22} - Q_{02}Q_{12}, \quad Q_{11}Q_{22} - Q_{12}^2. \end{aligned} \tag{14.25}$$

The domain of $\bar{c}_\sigma(-10)$ is

$$\operatorname{Sym}^2(\wedge^2 V_1)(-10) = \mathcal{O}(-6)^{\oplus 6}.$$

Let $\lambda \in H^1(\mathcal{O}(-6))^{\oplus 6}$ be an element of $\operatorname{Ker} H^1(\bar{c}_\sigma(-10))$, write $\lambda = (\lambda_1, \dots, \lambda_6)$, and let $\lambda_i = (\lambda_{i,0}, \dots, \lambda_{i,4})$, where $\lambda_{i,j}$ is the coordinate dual to $s^{4-j}t^j$ under the perfect pairing [Har77, Theorem III.5.1(d)]. Multiplication by a binary form $P = \sum_{r=0}^{d} P_r s^{d-r} t^r$ is the transpose of

multiplication by P on H^0, so

$$(P\lambda_i)_k = \sum_{r=0}^{d} P_r \lambda_{i,k+r} \quad \text{for } 0 \le k \le 4-d. \tag{14.26}$$

Since the image of λ vanishes, the coefficient of any monomial of degree 2 in $w, z_0, \dots, z_3$ vanishes. A direct calculation from (14.25) shows the coefficients in Table 18.

TABLE 18. The coefficients in the image of λ under $H^1(\bar{c}_\sigma(-10))$ of the monomials other than $wz_0, z_0^2, z_0z_1, z_0z_2$, where the terms in λ_1 are omitted in all rows but the first.

monomial	coefficient of the monomial in the image of λ
w^2	$-\lambda_1$
wz_1	$t\lambda_3$
wz_2	$-s\lambda_2$
wz_3	$-pt\lambda_2 + qs\lambda_3 + s\lambda_5$
z_0z_3	$qs^2\lambda_2 - pt^2\lambda_3 + s^2\lambda_4 + st\lambda_6$
z_1^2	$-bt^2\lambda_2 - t^2\lambda_6$
z_2^2	$-as^2\lambda_3 - s^2\lambda_4$
z_1z_2	$-st\lambda_5$
z_1z_3	$-((bq+c_1)st + c_2t^2)\lambda_2 - bst\lambda_4 - pt^2\lambda_5 - 2qst\lambda_6$
z_2z_3	$-(c_0s^2 + apst)\lambda_3 - 2pst\lambda_4 - qs^2\lambda_5 + as^2\lambda_6$
z_3^2	$-(qc_1s^2 + qc_2st)\lambda_2 - c_0pst\lambda_3 - (c_1s^2 + c_2st + p^2t^2)\lambda_4$ $-pqst\lambda_5 + (c_0 - q^2)s^2\lambda_6$

We apply (14.26) to the rows of Table 18. The coefficient of w^2 implies that $\lambda_1 = 0$. The coefficients of wz_1 and wz_2 imply that $\lambda_{3,1} = \cdots = \lambda_{3,4} = 0$ and $\lambda_{2,0} = \cdots = \lambda_{2,3} = 0$. Let $\mu_1 := \lambda_{2,4}$ and $\mu_2 := \lambda_{3,0}$. The coefficient of wz_3 is $-p\lambda_{2,k+1} + q\lambda_{3,k} + \lambda_{5,k}$ for $0 \le k \le 3$, thus

$$(\lambda_{5,0}, \lambda_{5,1}, \lambda_{5,2}, \lambda_{5,3}) = (-q\mu_2, 0, 0, p\mu_1).$$

The coefficient of z_0z_3 is

$$q\lambda_{2,k} - p\lambda_{3,k+2} + \lambda_{4,k} + \lambda_{6,k+1} \quad \text{for } 0 \le k \le 2,$$

thus $\lambda_{4,k} = -\lambda_{6,k+1}$ for $k = 0, 1, 2$. The coefficient of z_1^2 implies that $\lambda_{6,k+2} = -b\lambda_{2,k+2}$ for $k = 0, 1, 2$, thus $\lambda_{6,2} = \lambda_{6,3} = 0$ and $\lambda_{6,4} = -b\mu_1$. The coefficient of z_2^2 implies that $\lambda_{4,k} = -a\lambda_{3,k}$ for $k = 0, 1, 2$, thus $\lambda_{4,0} = -a\mu_2$ and $\lambda_{4,1} = \lambda_{4,2} = 0$. Therefore, there exist complex numbers $\mu_3 := \lambda_{4,3}$, $\mu_4 := \lambda_{4,4}$, $\mu_5 := \lambda_{5,4}$, and $\mu_6 := \lambda_{6,0}$ with

$$\begin{aligned}
\lambda_1 &= (0, 0, 0, 0, 0), \\
\lambda_2 &= (0, 0, 0, 0, \mu_1), \\
\lambda_3 &= (\mu_2, 0, 0, 0, 0), \\
\lambda_4 &= (-a\mu_2, 0, 0, \mu_3, \mu_4), \\
\lambda_5 &= (-q\mu_2, 0, 0, p\mu_1, \mu_5), \\
\lambda_6 &= (\mu_6, a\mu_2, 0, 0, -b\mu_1).
\end{aligned} \tag{14.27}$$

Substituting (14.27) into the last four rows of Table 18 and using (14.26), we obtain the following equations, where k is the index of (14.26):

$$\begin{aligned}
&\text{the coefficient of } z_1z_2,\ k=2: && -p\mu_1=0,\\
&\text{the coefficient of } z_1z_3,\ k=2: && -c_2\mu_1-b\mu_3-p\mu_5=0,\\
&\text{the coefficient of } z_2z_3,\ k=0: && (q^2-c_0)\mu_2+a\mu_6=0,\\
&\text{the coefficient of } z_2z_3,\ k=1: && a^2\mu_2=0,\\
&\text{the coefficient of } z_2z_3,\ k=2: && -2p\mu_3=0,\\
&\text{the coefficient of } z_3^2,\ k=0: && ac_1\mu_2+(c_0-q^2)\mu_6=0,\\
&\text{the coefficient of } z_3^2,\ k=2: && -p^2q\mu_1-c_2\mu_3-p^2\mu_4=0.
\end{aligned} \tag{14.28}$$

Since $p\neq 0$, (14.28) implies that $\mu_1=\mu_3=\mu_5=\mu_4=0$. If $a\neq 0$, then $a^2\mu_2=0$ implies that $\mu_2=0$, and $a\mu_6=0$ implies that $\mu_6=0$. If $a=0$ and $c_0\neq q^2$, then the third and the sixth equations of (14.28) imply that $\mu_2=\mu_6=0$. In both cases, (14.27) implies that $\lambda=0$. Since $\operatorname{Ker} H^1(\bar{c}_\sigma(-10))\neq 0$, we have $a=0$ and $c_0=q^2$. □

Theorem 14.12. *Let S be a torsion-free numerical Godeaux surface whose bicanonical pencil has genus 3. Then Profile II of Construction-Proposition 14.1 holds.*

Proof. By Construction-Proposition 14.1, exactly one of Profile I and Profile II holds. Suppose that Profile I holds, and take the coordinates and generators of Construction-Proposition 14.10.

Let C be any fiber of f. Then $\mathcal{O}_C(H)\cong\omega_C$ and $\mathcal{O}_C(2K_Y-F)\cong\omega_C^{\otimes 2}$ by adjunction [Har77, Theorem III.7.11] and $\mathcal{O}_C(F)=\mathcal{O}_C$, so x_0,x_1,x_2 restrict to sections of ω_C and ζ restricts to a section of $\omega_C^{\otimes 2}$. We prove that multiplication by ζ,

$$H^0(C,\omega_C^{\otimes 2})\xrightarrow{\cdot\zeta}H^0(C,\omega_C^{\otimes 4}), \tag{14.29}$$

is injective. By Construction-Proposition 14.2, the divisor of ζ is $4E$, so ζ is a unit on $C\backslash E$. Since $E\cdot F=2$, the curve E is horizontal, so E and C have no common component. Since Y is smooth, at any point of $C\cap E$ a local equation of E is a nonzerodivisor on $\mathcal{O}_C$, also when C is non-reduced [Mat89, Theorem 17.4(iii)]. Thus the fourth power of a local equation of E is a nonzerodivisor on $\mathcal{O}_C$, and (14.29) is injective.

By Construction-Proposition 14.3(3), the fiber of V_2 at a point of $\mathbb{P}^1$ is $H^0(C,\omega_C^{\otimes 2})$ for the fiber C of f over the point, and the fiber of σ_2 is the multiplication of sections. Since w,z_0,z_1,z_2,z_3,ζ generate V_2, the sections w,z_0,z_1,z_2,z_3,ζ restrict to a basis of $H^0(C,\omega_C^{\otimes 2})$.

Step 1. In this step we prove that $p\neq 0$.

Suppose that $p=0$. By (14.23), at $[s:t]=[0:1]$ the matrix of Ψ has the rows

$$(b,0,1,0),\quad (0,0,0,0),\quad (c_2,0,0,0),\quad (d_4,e,0,m).$$

Since Ψ has fiber rank at least three (Construction-Proposition 14.8(3)), we have $c_2\neq 0$ and $(e,m)\neq(0,0)$.

Let C be the fiber of f over $[0:1]$. By (14.22), in $H^0(C,\omega_C^{\otimes 2})$ we have

$$x_0^2=-bz_1-c_2z_3-d_4\zeta,\quad x_0x_2=e\zeta,\quad x_2^2=m\zeta.$$

The relation $(x_0x_2)^2=x_0^2x_2^2$ in $H^0(C,\omega_C^{\otimes 4})$ implies that

$$\zeta\big(mbz_1+mc_2z_3+(md_4+e^2)\zeta\big)=0.$$

By (14.29), $mbz_1+mc_2z_3+(md_4+e^2)\zeta=0$. Since z_1,z_3,ζ are linearly independent in the fiber of V_2 at $[0:1]$, we have $mc_2=0$ and $md_4+e^2=0$. Since $c_2\neq 0$, we have $m=0$, and then $e=0$. This contradicts $(e,m)\neq(0,0)$. Thus

$$p\neq 0. \tag{14.30}$$

Step 2. In this step we determine a and c_0.

By (14.30) and Lemma 14.11,

$$a = 0, \quad c_0 = q^2. \tag{14.31}$$

Step 3. In this step we obtain a contradiction at the fiber of f over $[1:0]$.

Let C be the fiber of f over $[1:0]$. By (14.22) and (14.31), in $H^0(C, \omega_C^{\otimes 2})$ we have

$$x_1^2 = q^2 z_3 + d_0\zeta, \quad x_1x_2 = qz_3 + h\zeta, \quad x_2^2 = z_3.$$

Thus

$$0 = x_1^2x_2^2 - (x_1x_2)^2 = \zeta\big((d_0 - 2qh)z_3 - h^2\zeta\big)$$

in $H^0(C, \omega_C^{\otimes 4})$. By (14.29), $(d_0 - 2qh)z_3 - h^2\zeta = 0$. Since z_3, ζ are linearly independent in the fiber of V_2 at $[1:0]$, we have $h^2 = 0$ and $d_0 - 2qh = 0$. Thus $h = d_0 = 0$. By (14.23) and (14.31), at $[s:t] = [1:0]$ the matrix of Ψ has the rows

$$(0,0,0,0), \quad (0,1,0,0), \quad (q^2,0,q,1), \quad (0,0,0,0),$$

so the rank of Ψ at $[1:0]$ is at most 2. This contradicts Construction-Proposition 14.8(3). Thus Profile I does not hold, and Profile II holds. □

14.4. **Infinitely near centers.** In this subsection, we write the multiplication of Profile II with infinitely near centers in a normal form. Since Profile I does not occur (Theorem 14.12) and the centers are not distinct (Proposition 14.7), only this case remains in genus 3. For infinitely near centers, the equations of the connecting class leave seven parameters, which yields the bound of Theorem 14.19.

Construction-Proposition 14.13. Notation and conditions as in Construction-Proposition 14.3. Assume that Profile II holds and that the centers of β are infinitely near. Let E be the final exceptional (-1)-section, let C be the strict transform of the first exceptional curve, and let $b_C := f(C)$. Then the following hold.

(1) For any $u \in \mathbb{P}^1$ with $u \neq b_C$, the curve X'_u is smooth at the point of $r(E)$ over u.

(2) There exist splittings

$$V_1 = \bigoplus_{i=0}^{2} \mathcal{O}(1)x_i, \quad V_2 = \mathcal{O}(2)y_0 \oplus \mathcal{O}(2)y_1 \oplus \bigoplus_{i=2}^{4} \mathcal{O}(3)y_i \oplus \mathcal{O}(5)\zeta,$$

such that the following hold.

(a) Multiplication is

$$x_0^2 = y_0, \quad x_0x_1 = y_1,$$
$$x_1^2 = wy_2 + ay_3 + cy_4 + f_3\zeta, \quad x_0x_2 = vy_2 + by_3 + dy_4 + g_3\zeta,$$
$$x_1x_2 = vy_3 + ly_4 + h_3\zeta, \quad x_2^2 = vy_4 + j_3\zeta,$$

where $w, v, a, b, c, d, l \in H^0(\mathcal{O}(1))$ and $f_3, g_3, h_3, j_3 \in H^0(\mathcal{O}(3))$.

(b) The forms w, v are non-zero, and the zero of w is b_C.

(c) The sections x_1, x_2 of $\mathcal{O}_Y(H)$ vanish along $C + E$ and $C + 2E$ respectively, and under isomorphisms

$$\mathcal{O}_Y(H - E)|_E \cong \mathcal{O}_Y(H - C - 2E)|_E \cong (f|_E)^*\mathcal{O}(1),$$

the forms w and v are the restrictions to E of x_1/E and $x_2/(C + 2E)$.

(d) The section x_0 of $\mathcal{O}_Y(H)$ has no zero on E.

Proof. The proof is long, so we divide it into several steps. For a section s of a line bundle L on Y vanishing along a divisor $D \geq 0$, we write

$$(s/D)|_E \in H^0(E, L(-D)|_E), \quad (s/D)|_C \in H^0(C, L(-D)|_C)$$

for the restrictions of the section s/D of $L(-D)$.

Step 1. In this step we compute the classes on Y and prove (1).

Since $E_1 = C+E$, $E_2 = E$, and $m_1 = m_2 = 1$, Construction-Proposition 14.1 and Construction-Proposition 14.3(1) imply that

$$K_Y \sim \beta^* K_S + C + 2E, \quad F \sim \beta^* M - C - 2E, \quad H \sim \beta^*(3K_S - Z),$$

and that

$$C^2 = -2, \quad E^2 = -1, \quad C \cdot E = 1, \quad F \cdot C = 0, \quad F \cdot E = 1, \quad H \cdot C = H \cdot E = 0.$$

The morphism $f|_E : E \to \mathbb{P}^1$ is an isomorphism. For $u \in \mathbb{P}^1$, let e_u be the point of E over u.

Let $G \neq C$ be an irreducible curve contracted by r. Since r is the minimal resolution of rational double points (Construction-Proposition 14.3(2)), we have $K_Y \cdot G = 0$. The curve G is vertical, so $G \neq E$. Thus $\beta_* G$ is a curve, and

$$0 = K_Y \cdot G = \beta^* K_S \cdot G + C \cdot G + 2(E \cdot G),$$

where $\beta^* K_S \cdot G \geq 0$, $C \cdot G \geq 0$, and $E \cdot G \geq 0$. Therefore $E \cdot G = 0$, and $G \cap E = \emptyset$.

Let $u \in \mathbb{P}^1$ with $u \neq b_C$. Then $e_u \notin C$, so no curve contracted by r contains e_u, and r is an isomorphism on a neighborhood of e_u by Zariski's main theorem [Har77, Corollary III.11.4]. Since $f|_E$ is an isomorphism, f is smooth at e_u. Thus F_u is smooth at e_u, and X'_u is smooth at $r(e_u)$. This proves (1).

Step 2. In this step we fix the line bundles on E and choose x_0, x_1, x_2.

Let $\mathcal{O}_E(d) := (f|_E)^* \mathcal{O}(d)$, and read sections of $\mathcal{O}_E(d)$ as binary forms of degree d on $\mathbb{P}^1$. We have $\mathcal{O}_Y(F)|_E = \mathcal{O}_E(1)$. Since

$$H \cdot E = 0, \quad (H - C - E) \cdot E = 0, \quad (2H - F - C - 2E) \cdot E = 0,$$

we fix isomorphisms of $\mathcal{O}_Y(H)|_E$, $\mathcal{O}_Y(H - C - E)|_E$, and $\mathcal{O}_Y(2H - F - C - 2E)|_E$ with $\mathcal{O}_E$. Tensor products of the fixed isomorphisms and of $\mathcal{O}_Y(F)|_E = \mathcal{O}_E(1)$ define an isomorphism $\mathcal{O}_Y(L)|_E \cong \mathcal{O}_E(L \cdot E)$ for any integral combination L of H, F, C, E, since

$$E = H + (H - C - E) - F - (2H - F - C - 2E), \quad C = F + (2H - F - C - 2E) - 2(H - C - E).$$

Thus, for sections s_1, s_2 vanishing along integral combinations $D'_1, D'_2 \geq 0$ of C, E, we have the equality of binary forms

$$(s_1 s_2/(D'_1 + D'_2))|_E = (s_1/D'_1)|_E \cdot (s_2/D'_2)|_E. \tag{14.32}$$

By Construction-Proposition 14.3(1), the sections of $\mathcal{O}_Y(H)$ have no common zero, so the restriction

$$H^0(Y, \mathcal{O}_Y(H)) \to H^0(E, \mathcal{O}_Y(H)|_E) = \mathbb{C}$$

is surjective. Choose a basis x_0, x_1, x_2 of $H^0(Y, \mathcal{O}_Y(H))$ with $x_0|_E = 1$ and $x_1|_E = x_2|_E = 0$. Since $H \cdot C = 0$ and $C \cong \mathbb{P}^1$, sections of $\mathcal{O}_Y(H)|_C$ are constant. Thus x_0 has no zero on $C \cup E$, and x_1, x_2 vanish on C.

Let $u \in \mathbb{P}^1$ be general. Then $u \neq b_C$, the curve F_u is smooth and nonhyperelliptic of genus 3 (Corollary 13.6), and $\omega_{F_u} = \mathcal{O}_Y(H)|_{F_u}$ by adjunction. Since

$$H^0(Y, \mathcal{O}_Y(H - F)) = H^0(Y, \mathcal{O}_Y(K_Y)) = 0,$$

the restrictions of x_0, x_1, x_2 form a basis of $H^0(F_u, \omega_{F_u})$. Thus x_0, x_1, x_2 embed F_u as a smooth plane quartic [Har77, Example IV.5.2.1]. Since $F \cdot E = 1$ and $e_u \notin C$, a section s of $\mathcal{O}_Y(H)$ vanishing along iE has $\mathrm{mult}_{e_u}(s|_{F_u}) \geq i$, with equality if and only if $(s/iE)|_E$ does not vanish at u.

Consider the linear map from $\mathbb{C}x_1 + \mathbb{C}x_2$ to $H^0(\mathcal{O}_E(1))$ sending s to $(s/E)|_E$, where $(H - E) \cdot E = 1$. Since s/E vanishes on C, the image consists of multiples of one form with zero b_C. The image is non-zero, as otherwise $\mathrm{mult}_{e_u}(x_1|_{F_u}) \geq 2$ and $\mathrm{mult}_{e_u}(x_2|_{F_u}) \geq 2$, which contradicts that x_0, x_1, x_2 embed F_u. Thus we may choose x_1, x_2 with $(x_1/E)|_E \neq 0$ and $(x_2/E)|_E = 0$.

Therefore x_1 vanishes along $C+E$, and x_2 vanishes along $D_0 := C+2E$. The constant $(x_1/(C+E))|_E \in H^0(\mathcal{O}_E)$ is non-zero, because $(x_1/E)|_E \neq 0$, and we rescale x_1 so that $(x_1/(C+E))|_E = 1$. Let

$$w := (x_1/E)|_E \in H^0(\mathcal{O}_E(1)), \quad v := (x_2/D_0)|_E \in H^0(\mathcal{O}_E((H-D_0)\cdot E)) = H^0(\mathcal{O}_E(1)).$$

Then $w \neq 0$, and the zero of w is b_C. By (14.32), we have

$$\begin{aligned} (x_1^2/D_0)|_E = w, \quad (x_0x_2/D_0)|_E = v, \\ (x_1x_2/(2C+3E))|_E = v, \quad (x_2^2/(2C+4E))|_E = v^2. \end{aligned} \tag{14.33}$$

Step 3. In this step we prove that $v \neq 0$.

Suppose that $v = 0$. Then x_2 vanishes along $C+3E$. Let $u \in \mathbb{P}^1$ be general, and let $k := \operatorname{mult}_{e_u}(x_2|_{F_u})$. Then $k \geq 3$, that is, e_u is a flex of the plane quartic F_u, and $k \leq \deg \omega_{F_u} = 4$. By **Step 2**, we have $\operatorname{mult}_{e_u}(x_0|_{F_u}) = 0$ and $\operatorname{mult}_{e_u}(x_1|_{F_u}) = 1$. Thus the multiplicities at e_u of $x_0^2, x_0x_1, x_1^2, x_0x_2, x_1x_2, x_2^2$ on F_u are

$$0,1,2,3,4,6 \quad \text{if } k=3, \qquad 0,1,2,4,5,8 \quad \text{if } k=4.$$

The smooth plane quartic F_u is not contained in a conic, so the map

$$\operatorname{Sym}^2 H^0(F_u, \omega_{F_u}) \to H^0(F_u, \omega_{F_u}^{\otimes 2})$$

is injective, and the Riemann–Roch theorem implies that $h^0(F_u, \omega_{F_u}^{\otimes 2}) = 8+1-3 = 6$. Thus the six products form a basis of $H^0(F_u, \omega_{F_u}^{\otimes 2})$. Since their multiplicities are pairwise distinct, a non-zero element of $H^0(F_u, \omega_{F_u}^{\otimes 2})$ with multiplicity at least 6 at e_u is a non-zero multiple of $x_2^2|_{F_u}$.

Regard ζ as a section of $\mathcal{O}_Y(2K_Y - F) = \mathcal{O}_Y(2H-3F)$. Since $\sum_i (m_i+2)E_i = 3(C+E)+3E$, Construction-Proposition 14.2 implies that the divisor of ζ is

$$\Delta := \beta^* Z + 3C + 6E.$$

By Construction-Proposition 14.3(3), the fiber of V_2 at u is $H^0(F_u, \omega_{F_u}^{\otimes 2})$, and $\zeta|_{F_u}$ is non-zero with multiplicity at least 6 at e_u. Thus $x_2^2|_{F_u}$ is a multiple of $\zeta|_{F_u}$ for general u. Since

$$f_*\mathcal{O}_Y(2H) = V_2(-2) = \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(1)^{\oplus 3} \oplus \mathcal{O}(3)\zeta,$$

the components of x_2^2 in the first two summands vanish identically, and

$$x_2^2 = p\zeta \tag{14.34}$$

for a non-zero form $p \in H^0(\mathcal{O}(3))$.

Let u be a zero of p. Then the fiber of σ_2 at u sends x_2^2 to zero. Since $\operatorname{Sym}^2 V_1$ and V_2 have rank 6, the cokernel of σ_2 is non-zero at u, so $u \in \operatorname{Supp}\tau$. By Construction-Proposition 14.3(5)(i), q_u is a non-zero multiple of x_2^2. Thus the conic Q_u of Construction-Proposition 14.3(5)(iii) is a double line, which has no smooth point, and X'_u has no smooth point by Construction-Proposition 14.3(6). By (1), we have $u = b_C$. Therefore the divisor of p is $3b_C$.

Comparing the divisors of x_2^2 and $p\zeta$ in (14.34) as sections of $\mathcal{O}_Y(2H)$, we obtain

$$2\operatorname{div}(x_2) = \Delta + 3F_{b_C}.$$

Let $D := \beta_* \operatorname{div}(x_2)$ and $M_0 := \beta_* F_{b_C} \in |M|$. Pushing forward by β, we obtain

$$2D = Z + 3M_0.$$

Let $T := D - M_0$. Then $2T = Z + M_0 \geq 0$, so $T \geq 0$, and $2T \sim 2K_S$. By Lemma 2.3(2), $\mathcal{O}_S(T - K_S)$ is a non-trivial 2-torsion line bundle. This contradicts that S is torsion-free. Thus $v \neq 0$.

Step 4. In this step we construct y_0, y_1.

We have the exact sequence

$$0 \to \mathcal{O}_{D_0} \to \mathcal{O}_C \oplus \mathcal{O}_{2E} \to \mathcal{O}_{C\cap 2E} \to 0.$$

We have

$$f_*\mathcal{O}_{2E} = \mathcal{O} \oplus f_*\mathcal{O}_E(-E) = \mathcal{O} \oplus \mathcal{O}(1),$$

where the summand $\mathcal{O}$ is the image of $\mathcal{O}_{\mathbb{P}^1}$, and $f_*\mathcal{O}_C$ is the skyscraper sheaf $\mathbb{C}$ at b_C. Since $C \cdot E = F \cdot E = 1$, a local parameter of $\mathbb{P}^1$ at b_C pulls back to a local equation of C at the point $C \cap E$. Thus a section t of $f_*\mathcal{O}_{2E}$ restricts to a constant on $C \cap 2E$ if and only if the component of t in $\mathcal{O}(1)$ vanishes at b_C. Applying f_* to the exact sequence, we see that the kernel of

$$f_*\mathcal{O}_C \oplus f_*\mathcal{O}_{2E} \to f_*\mathcal{O}_{C\cap 2E}$$

consists of the triples whose component in $\mathcal{O}(1)$ vanishes at b_C and whose components in $\mathcal{O}$ and $f_*\mathcal{O}_C$ take equal values at b_C, so

$$f_*\mathcal{O}_{D_0} \cong \mathcal{O} \oplus \mathcal{O}(1)(-b_C) \cong \mathcal{O}^{\oplus 2}.$$

By **Step 2**, x_0^2 has no zero on $C \cup E$, so x_0^2 induces an isomorphism $f_*(\mathcal{O}_Y(2H)|_{D_0}) \cong f_*\mathcal{O}_{D_0}$. Let

$$\rho : V_2(-2) = f_*\mathcal{O}_Y(2H) \to f_*\mathcal{O}_{D_0}$$

be the restriction. Then $\rho(x_0^2) = 1$, and $\rho(x_0x_1)$ is the form w in $\mathcal{O}(1)(-b_C)$, because $x_0|_E = 1$ and $(x_1/E)|_E = w$. Since the zero of w is b_C, the sections $\rho(x_0^2), \rho(x_0x_1)$ form a basis of $f_*\mathcal{O}_{D_0}$. The summands of positive degree of $V_2(-2)$ restrict to zero, since $\operatorname{Hom}(\mathcal{O}(k), \mathcal{O}) = 0$ for $k > 0$. Thus $\operatorname{Ker}\rho$ is the sum of the summands of positive degree. Let $y_0 := x_0^2$ and $y_1 := x_0x_1$. Then

$$V_2(-2) = \mathcal{O}y_0 \oplus \mathcal{O}y_1 \oplus \operatorname{Ker}\rho, \quad \operatorname{Ker}\rho \cong \mathcal{O}(1)^{\oplus 3} \oplus \mathcal{O}(3)\zeta.$$

The divisors of $x_1^2, x_0x_2, x_1x_2, x_2^2$ contain D_0 by **Step 2**, so $x_1^2, x_0x_2, x_1x_2, x_2^2 \in H^0(\mathbb{P}^1, \operatorname{Ker}\rho)$.

Step 5. In this step we construct y_2, y_3, y_4.

Let $W_2 := H^0(Y, \mathcal{O}_Y(2H - F))$. By **Step 4** and the projection formula [Har77, Exercise II.5.1(d)], we have

$$f_*(\mathcal{O}_Y(2H - F)|_{D_0}) \cong \mathcal{O}(-1)^{\oplus 2}.$$

Since $\mathcal{O}(-1)^{\oplus 2}$ has no non-zero section, any section in W_2 vanishes along D_0, and $W_2 = H^0(\mathbb{P}^1, \operatorname{Ker}\rho(-1))$. Choose $y_2', y_3', y_4' \in W_2$ generating the summands $\mathcal{O}(1)^{\oplus 3}$ of $\operatorname{Ker}\rho$, so that

$$W_2 = \mathbb{C}y_2' \oplus \mathbb{C}y_3' \oplus \mathbb{C}y_4' \oplus H^0(\mathcal{O}(2))\zeta.$$

Since $\Delta \geq 3C + 6E$, we have $(\zeta/D')|_E = 0$ for any divisor $D' \leq 3C + 5E$. Thus, for any section $s = \sum_i \ell_i y_i' + g\zeta$ of $\operatorname{Ker}\rho$, with $\ell_i \in H^0(\mathcal{O}(1))$ and $g \in H^0(\mathcal{O}(3))$, and for any divisor $D' \leq 3C + 5E$ along which each y_i' with $\ell_i \neq 0$ vanishes, we have

$$(s/D')|_E = \sum_i \ell_i \cdot (y_i'/D')|_E. \tag{14.35}$$

Let $\lambda_0 : W_2 \to \mathbb{C}$ be the linear map $\lambda_0(y) := (y/D_0)|_E$, where $(2H - F - C - 2E) \cdot E = 0$. By (14.33) and (14.35), we have $\sum_i \ell_i\lambda_0(y_i') = w \neq 0$ for the coefficients ℓ_i of x_1^2. Thus λ_0 is non-zero on $\mathbb{C}y_2' + \mathbb{C}y_3' + \mathbb{C}y_4'$. Possibly replacing y_2', y_3', y_4' with another basis of $\mathbb{C}y_2' + \mathbb{C}y_3' + \mathbb{C}y_4'$, we may assume that $\lambda_0(y_2') = 1$ and $\lambda_0(y_3') = \lambda_0(y_4') = 0$, and we let $y_2 := y_2'$. By (14.33) and (14.35), the coefficient of y_2 is w in x_1^2 and v in x_0x_2. The divisors of x_1x_2 and x_2^2 contain $D_0 + E$, so the coefficient of y_2 is zero in x_1x_2 and in x_2^2.

Let $y \in W_2$ with $\lambda_0(y) = 0$. Then y vanishes along $C + 3E$. Since $(2H - F - C - 3E) \cdot C = -1$, we have $(y/(C + 3E))|_C = 0$, so y vanishes along $D_1 := 2C + 3E$. Let $\lambda_1(y) := (y/D_1)|_E \in \mathbb{C}$, where $(2H - F - D_1) \cdot E = 0$.

By (14.33) and (14.35), the coefficients ℓ_3, ℓ_4 of y_3', y_4' in x_1x_2 satisfy $\ell_3\lambda_1(y_3') + \ell_4\lambda_1(y_4') = v$. Since $v \neq 0$ by **Step 3**, λ_1 is non-zero on $\mathbb{C}y_3' + \mathbb{C}y_4'$. Choose a basis y_3, y_4 of $\mathbb{C}y_3' + \mathbb{C}y_4'$ with $\lambda_1(y_3) = 1$ and $\lambda_1(y_4) = 0$. Then the coefficient of y_3 in x_1x_2 is v. The divisor of x_2^2 contains $D_1 + E$, so the coefficient of y_3 in x_2^2 is zero. Thus

$$x_2^2 = my_4 + j_3\zeta$$

for some $m \in H^0(\mathcal{O}(1))$ and $j_3 \in H^0(\mathcal{O}(3))$.

Since $\lambda_1(y_4) = 0$, the section y_4 vanishes along $D_2 := 2C + 4E$. Let

$$n := (y_4/D_2)|_E \in H^0(\mathcal{O}_E(1)),$$

where $(2H - F - D_2) \cdot E = 1$. By (14.33) and (14.35), we have

$$v^2 = mn.$$

Since $v \neq 0$, the forms m, n are non-zero multiples of v. Possibly replacing y_4 with a non-zero multiple, we may assume that $m = v$.

Now x_0, x_1, x_2 is a basis of $H^0(\mathbb{P}^1, V_1(-1))$, and $V_1(-1) = \mathcal{O}^{\oplus 3}$. The sections y_0, y_1 of $V_2(-2)$, the sections y_2, y_3, y_4 of $V_2(-3)$, and the section ζ of $V_2(-5)$ define the splitting of V_2 by **Step 4** and the choice of y_2', y_3', y_4'. Let a, c, f_3 be the remaining coefficients of x_1^2, let b, d, g_3 be the remaining coefficients of x_0x_2, and let l, h_3 be the remaining coefficients of x_1x_2. Then

$$x_1^2 = wy_2 + ay_3 + cy_4 + f_3\zeta, \quad x_0x_2 = vy_2 + by_3 + dy_4 + g_3\zeta,$$

$$x_1x_2 = vy_3 + ly_4 + h_3\zeta, \quad x_2^2 = vy_4 + j_3\zeta,$$

with $a, b, c, d, l \in H^0(\mathcal{O}(1))$ and $f_3, g_3, h_3, j_3 \in H^0(\mathcal{O}(3))$. Together with $x_0^2 = y_0$, $x_0x_1 = y_1$, **Step 2**, and **Step 3**, this proves (2). □

Consider the forms w and v of Construction-Proposition 14.13: the zero of w is b_C, and for $u \neq b_C$ and the point e_u of E over u, $v(u) = 0$ if and only if $\mathrm{mult}_{e_u}(x_2|_{F_u}) > 2$ (Construction-Proposition 14.13(2)(c)), so Lemma 14.15 says that $\mathrm{mult}_{e_u}(x_2|_{F_u}) = 2$ for any $u \neq b_C$. That w and v are proportional is the first step in the normalization of σ_2. The next lemma is a computation with principal parts on $\mathbb{P}^1$, and it is used only in the proof of Lemma 14.15.

Lemma 14.14. *Let $[s : t]$ be homogeneous coordinates on $\mathbb{P}^1$, and let $a, b, c, d, l \in H^0(\mathcal{O}(1))$ with b and l non-zero at the point $[0 : 1]$. Let $U := \mathcal{O}(2)^{\oplus 2} \oplus \mathcal{O}(3)^{\oplus 3}$ with generators $y_0, \dots, y_4$, and let*

$$\bar\sigma_{11} := ty_2 + ay_3 + cy_4, \quad \bar\sigma_{02} := sy_2 + by_3 + dy_4, \quad \bar\sigma_{12} := sy_3 + ly_4, \quad \bar\sigma_{22} := sy_4.$$

Then the homomorphism $\mathcal{O}(4)^{\oplus 6} \to \mathrm{Sym}^2 U$ given by

$$\bar N_1 := y_0\bar\sigma_{11} - y_1^2, \quad \bar N_2 := y_0\bar\sigma_{12} - y_1\bar\sigma_{02}, \quad \bar N_3 := y_1\bar\sigma_{12} - \bar\sigma_{11}\bar\sigma_{02},$$

$$\bar N_4 := y_0\bar\sigma_{22} - \bar\sigma_{02}^2, \quad \bar N_5 := y_1\bar\sigma_{22} - \bar\sigma_{02}\bar\sigma_{12}, \quad \bar N_6 := \bar\sigma_{11}\bar\sigma_{22} - \bar\sigma_{12}^2$$

induces an injective map

$$H^1(\mathbb{P}^1, \mathcal{O}(-5)^{\oplus 6}) \to H^1(\mathbb{P}^1, \mathrm{Sym}^2 U(-9)).$$

Proof. For any integer $n \geq 2$, the Čech complex of the cover of $\mathbb{P}^1$ by $s \neq 0$ and $t \neq 0$ computes $H^1(\mathbb{P}^1, \mathcal{O}(-n))$ [Har77, Theorem III.4.5]. With $t = 1$, the sections of $\mathcal{O}(-n)$ over $s \neq 0$, over $t \neq 0$, and over $st \neq 0$ are spanned by the s^j with $j \leq -n$, with $j \geq 0$, and with $j \in \mathbb{Z}$, respectively. Thus a Laurent polynomial in s represents a class of $H^1(\mathbb{P}^1, \mathcal{O}(-n))$, and the class is zero if and only if the coefficients of $s^{-1}, \dots, s^{-(n-1)}$ vanish. A homomorphism between line bundles acts on Laurent polynomials by multiplication by a polynomial in s.

We have

$$\mathrm{Sym}^2 U(-9) = \mathcal{O}(-5)^{\oplus 3} \oplus \mathcal{O}(-4)^{\oplus 6} \oplus \mathcal{O}(-3)^{\oplus 6},$$

where the first summand corresponds to the monomials in y_0, y_1, the second summand to the products of one of y_0, y_1 and one of y_2, y_3, y_4, and the third summand to the monomials in y_2, y_3, y_4. Thus a coefficient in the first, second, and third summand represents zero if and only if the coefficients of $s^{-1}, \dots, s^{-k}$ vanish, with $k = 4, 3, 2$, respectively.

Let $\kappa_1, \dots, \kappa_6$ be linear combinations of $s^{-1}, s^{-2}, s^{-3}, s^{-4}$ representing a class in the kernel, so that $\sum_{i=1}^6 \kappa_i \bar N_i$ represents zero. For a monomial $\mathfrak{m}$, we denote by $[\mathfrak{m}]$ the coefficient of $\mathfrak{m}$ in $\sum_i \kappa_i \bar N_i$. The coefficients of eleven monomials are listed in Table 19. Let $b(0)$, $d(0)$, and $l(0)$ be the values at $s = 0$ of b, d, and l, and write $l = l(0) + l_1 s$. By assumption, $b(0) \neq 0$ and $l(0) \neq 0$.

TABLE 19. The coefficients of eleven monomials in $\sum_i \kappa_i \bar{N}_i$, with $t = 1$.

$\mathfrak{m}$	$[\mathfrak{m}]$
y_1^2	$-\kappa_1$
y_0y_3	$a\kappa_1 + s\kappa_2$
y_0y_4	$c\kappa_1 + l\kappa_2 + s\kappa_4$
y_1y_3	$-b\kappa_2 + s\kappa_3$
y_1y_4	$-d\kappa_2 + l\kappa_3 + s\kappa_5$
y_2^2	$-s\kappa_3 - s^2\kappa_4$
y_2y_3	$-(b + as)\kappa_3 - 2sb\kappa_4 - s^2\kappa_5$
y_2y_4	$-(d + cs)\kappa_3 - 2sd\kappa_4 - sl\kappa_5 + s\kappa_6$
y_3^2	$-ab\kappa_3 - b^2\kappa_4 - bs\kappa_5 - s^2\kappa_6$
y_3y_4	$-(ad + bc)\kappa_3 - 2bd\kappa_4 - (bl + ds)\kappa_5 + s(a - 2l)\kappa_6$
y_4^2	$-cd\kappa_3 - d^2\kappa_4 - dl\kappa_5 + (cs - l^2)\kappa_6$

The coefficient of y_1^2 implies that $\kappa_1 = 0$. The coefficient of y_0y_3 is then $s\kappa_2$, so $\kappa_2 = \gamma_1 s^{-1}$ for some $\gamma_1 \in \mathbb{C}$. The coefficients of y_0y_4, y_1y_3, and y_1y_4 imply, one after another, that there exist $\gamma_2, \gamma_3, \gamma_4 \in \mathbb{C}$ with

$$\kappa_4 = \gamma_2 s^{-1} - l(0)\gamma_1 s^{-2}, \quad \kappa_3 = \gamma_3 s^{-1} + b(0)\gamma_1 s^{-2},$$
$$\kappa_5 = \gamma_4 s^{-1} + \big(d(0)\gamma_1 - l(0)\gamma_3 - l_1 b(0)\gamma_1\big)s^{-2} - l(0)b(0)\gamma_1 s^{-3}.$$

The coefficient of s^{-1} in $-s\kappa_3 - s^2\kappa_4$ is $-b(0)\gamma_1$, so $\gamma_1 = 0$. Thus

$$\kappa_2 = 0, \quad \kappa_3 = \gamma_3 s^{-1}, \quad \kappa_4 = \gamma_2 s^{-1}, \quad \kappa_5 = \gamma_4 s^{-1} - l(0)\gamma_3 s^{-2}.$$

The coefficient of s^{-1} in the row of y_2y_3 is $-b(0)\gamma_3$, so $\gamma_3 = 0$. Thus $\kappa_3 = 0$ and $\kappa_5 = \gamma_4 s^{-1}$. In the row of y_2y_4, all terms except $s\kappa_6$ are now polynomials in s, so $\kappa_6 = \gamma_5 s^{-1} + \gamma_6 s^{-4}$ for some $\gamma_5, \gamma_6 \in \mathbb{C}$. In the row of y_3^2, the coefficient of s^{-2} is $-\gamma_6$ and the coefficient of s^{-1} is $-b(0)^2\gamma_2$, so $\gamma_6 = \gamma_2 = 0$. In the row of y_3y_4, the coefficient of s^{-1} is now $-b(0)l(0)\gamma_4$, so $\gamma_4 = 0$. In the row of y_4^2, the coefficient of s^{-1} is now $-l(0)^2\gamma_5$, so $\gamma_5 = 0$. Therefore, $\kappa_1 = \cdots = \kappa_6 = 0$. □

Lemma 14.15. *Notation and conditions as in Construction-Proposition 14.13. Then the forms w and v are proportional.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we compute the divisor of ζ and describe Z when the strict transform of Z on Y intersects E.

Let Z' be the strict transform of Z on Y. Let ν_0 be the multiplicity of Z at the first center of β, and let ν_1 be the multiplicity at the second center of the strict transform of Z on the first blow-up. Since $m_1 = m_2 = 1$, $E_1 = C + E$, and $E_2 = E$, [Har77, Proposition V.3.6] and Construction-Proposition 14.2 imply that

$$\beta^*Z = Z' + \nu_0 C + (\nu_0 + \nu_1)E, \quad \operatorname{div}(\zeta) = Z' + (3 + \nu_0)C + (6 + \nu_0 + \nu_1)E. \tag{14.36}$$

We have $C^2 = -2$, $C \cdot E = 1$, $E^2 = -1$, $F \cdot C = 0$, $F \cdot E = 1$, and $M \cdot Z = -Z^2 = 2$. Thus

$$F \cdot Z' = 2 - \nu_0 - \nu_1, \quad Z' \cdot C = \nu_0 - \nu_1, \quad Z' \cdot E = \nu_1.$$

Since F is nef and C is not a component of Z', we have $\nu_0 + \nu_1 \le 2$ and $\nu_1 \le \nu_0$.

Assume that $Z' \cdot E > 0$. Then $\nu_0 = \nu_1 = 1$. By Lemma 14.4(1), any component of Z is a smooth rational (-2)-curve, and the support of Z is connected. Since $\nu_0 = 1$, exactly one component T of Z contains the first center, the coefficient of T in Z is one, and T is smooth at the first center. Since $\nu_1 = 1$, the strict transform of T contains the second center. Let T' be the

strict transform of T on Y. Then $\beta^*T = T' + C + 2E$, so $F \cdot T' = M \cdot T - 2$. Since F is nef, $M \cdot T \geq 2$.

For any component G of Z, we have $K_S \cdot G = 0$, so $-Z \cdot G = M \cdot G \geq 0$ because M is nef. In the identity

$$2 = -Z^2 = \sum_G (\mathrm{mult}_G Z)(-Z \cdot G),$$

all summands are non-negative, and the summand of T is $M \cdot T \geq 2$. Thus $Z \cdot T = -2$. Since $T^2 = -2$, we have $(Z - T) \cdot T = 0$, so no other component of Z intersects T. Since the support of Z is connected, $Z = T$.

Therefore, $Z' = T'$ is a smooth rational curve with $F \cdot Z' = 0$ and $Z' \cdot E = 1$, and Z' is contained in a fiber F_{b_Z}. We have $F \cdot E = 1$ and $C \cdot E = Z' \cdot E = 1$, so $F_{b_Z} \neq F_{b_C}$: otherwise C and Z' are two distinct components of one fiber and $F \cdot E \geq 2$. Moreover, the coefficient of Z' in F_{b_Z} is one, and $(F_{b_Z} - Z') \cdot E = 0$. Since F_{b_Z} and C are disjoint, $F_{b_Z} - Z' = \beta^*W$ for $W := \beta_*(F_{b_Z} - Z') \geq 0$, because the intersection numbers of $F_{b_Z} - Z' - \beta^*W$ with C, E vanish and the intersection matrix of C, E is negative definite. Pushing forward by β, we obtain $W \sim M - Z \sim 2K_S - 2Z$. Thus $|2K_S - 2Z| \neq \emptyset$, and $h^0(S, 3K_S - 2Z) = 0$ by Lemma 14.4(3).

By Construction-Proposition 14.3(1) and (14.36),

$$H - Z' \sim \beta^*(3K_S - 2Z) + C + 2E.$$

Since $C + 2E \geq 0$ is contracted by β, we have $h^0(Y, H - Z') = h^0(S, 3K_S - 2Z) = 0$. We have $H \cdot Z' = (3K_S - Z) \cdot Z = 2$ and $Z' \cong \mathbb{P}^1$, so the restriction

$$H^0(Y, \mathcal{O}_Y(H)) \to H^0(Z', \mathcal{O}_{Z'}(H))$$

is an injective map between spaces of dimension 3, hence an isomorphism. Thus x_0, x_1, x_2 map Z' isomorphically onto a smooth conic $\Gamma \subset \mathbb{P}^2$.

Step 2. In this step we prove that $\bar{\sigma}_2$ is surjective at any point $u \neq b_C$.

Let $u \in \mathbb{P}^1$ with $u \neq b_C$, let $V_2(u)$ be the fiber of V_2 at u, and let $\zeta_u \in V_2(u)$ be the value of ζ at u. Then $\zeta_u \neq 0$, since $\mathcal{O}(5)\zeta$ is a direct summand of V_2 (Construction-Proposition 14.13(2)).

Suppose that $\bar{\sigma}_2$ is not surjective at u. Since $\mathrm{coker}(\sigma_2) \cong \mathcal{O}_\tau$ (Construction-Proposition 14.2), we have $u \in \mathrm{Supp}\,\tau$, the image of the fiber of σ_2 at u is a hyperplane of $V_2(u)$, and ζ_u belongs to that hyperplane. Let $\pi : X'_u \to Q_u$ be the double cover of Construction-Proposition 14.3(5)(iii). By Construction-Proposition 14.3(3) and (5)(v), $\zeta_u = \pi^* g$ for a section g of $\mathcal{O}_{Q_u}(2)$.

Let e be the point of $r(E)$ over u. Then X'_u is smooth at e (Construction-Proposition 14.13(1)). Since X' is integral and dominates $\mathbb{P}^1$, X' is flat over $\mathbb{P}^1$ [Har77, Proposition III.9.7], so X' is regular at e by [Mat89, Theorem 23.7(ii)]. Since r is the minimal resolution of a surface with rational double points (Construction-Proposition 14.3(2), [CP06, Remark 7.1]), r is an isomorphism over a neighborhood of e. We regard e as a point of F_u. Since $F \cdot E = 1$, a local equation of E restricts to a local parameter of F_u at e.

We show that the differential of π at e is non-zero. By Construction-Proposition 14.13(2), x_0 has no zero on E, x_1 vanishes on E, and w, whose zero is b_C, is the restriction to E of x_1/E. Since $u \neq b_C$, the restriction $x_1|_{F_u}$ vanishes to order one at e. Thus the map on F_u defined by x_0, x_1, x_2 has non-zero differential at e. Near e, the map on F_u defined by x_0, x_1, x_2 is π.

By Construction-Proposition 14.3(6), $\pi(e)$ is a smooth point of Q_u. Since the differential of π at e is non-zero, the fiber of π over $\pi(e)$ has length one at e. Since π is finite and flat of degree two, the fiber of π over $\pi(e)$ consists of e and another point e', and π is étale at e and e'. Thus X'_u is smooth at e'. The argument at e implies that r is an isomorphism over a neighborhood of e', and we regard e' as a point of F_u. Since $F \cdot E = 1$ and $u \neq b_C$, we have $e' \notin E \cup C$.

Suppose first that no vertical component of $\mathrm{div}(\zeta)$ contains e. By (14.36), ζ_u is non-zero near e and vanishes to order at least 6 at e. Since π is étale at e and e' and $\zeta_u = \pi^* g$, ζ_u is non-zero near e' and

$$6 \leq \mathrm{mult}_e \,\mathrm{div}(\zeta_u) = \mathrm{mult}_{\pi(e)} \,\mathrm{div}(g) = \mathrm{mult}_{e'} \,\mathrm{div}(\zeta_u).$$

Since ζ_u is non-zero near e', no vertical component of $\operatorname{div}(\zeta)$ contains e'. Since $e' \notin E \cup C$, near e', $\operatorname{div}(\zeta)$ is a sum of horizontal components of Z'. Since $F \cdot G = 0$ for any vertical component G of Z', and $(F_u \cdot G)_y \geq 0$ for any horizontal component G of Z' and any $y \in F_u \cap G$ [Har77, Proposition V.1.4], we have

$$\operatorname{mult}_{e'} \operatorname{div}(\zeta_u) \leq F \cdot Z' = 2 - \nu_0 - \nu_1 \leq 2,$$

a contradiction.

Suppose now that a vertical component of $\operatorname{div}(\zeta)$ contains e. Since E is horizontal and $e \notin C$, that component is a component of Z' by (14.36), so $Z' \cdot E > 0$. By **Step 1**, Z' is a smooth rational curve in F_{b_Z}, so $u = b_Z$, and x_0, x_1, x_2 map Z' isomorphically onto the smooth conic Γ. Thus $\Gamma \subset Q_u$. Since Γ and Q_u are plane conics, $Q_u = \Gamma$. Since $\operatorname{div}(\zeta) \geq Z'$, ζ_u vanishes on Z'. Hence g vanishes on the reduced conic $\Gamma = Q_u$, so $g = 0$ and $\zeta_u = 0$, a contradiction.

Step 3. In this step we conclude the proof.

Suppose that w and v are not proportional. Use the homogeneous coordinates $s := v$ and $t := w$ on $\mathbb{P}^1$. Since b_C is the zero of w, the point $[0:1]$ is different from b_C. We write y_i also for the image of y_i in $U = \mathcal{O}(2)^{\oplus 2} \oplus \mathcal{O}(3)^{\oplus 3}$. By Construction-Proposition 14.13(2), $\bar{\sigma}_2$ maps

$$x_0^2,\ x_0x_1,\ x_1^2,\ x_0x_2,\ x_1x_2,\ x_2^2 \quad \text{to} \quad y_0,\ y_1,\ \bar{\sigma}_{11},\ \bar{\sigma}_{02},\ \bar{\sigma}_{12},\ \bar{\sigma}_{22},$$

respectively, where

$$\bar{\sigma}_{11} = ty_2 + ay_3 + cy_4, \quad \bar{\sigma}_{02} = sy_2 + by_3 + dy_4, \quad \bar{\sigma}_{12} = sy_3 + ly_4, \quad \bar{\sigma}_{22} = sy_4.$$

At $[0:1]$, we have $\bar{\sigma}_{22} = 0$, so the image of $\bar{\sigma}_2$ is spanned by y_0, y_1, $y_2 + ay_3 + cy_4$, $by_3 + dy_4$, and ly_4. By **Step 2**, $\bar{\sigma}_2$ is surjective at $[0:1]$, so b and l are non-zero at $[0:1]$.

We have $\operatorname{Sym}^2(\wedge^2 V_1) = \mathcal{O}(4)^{\oplus 6}$ with generators

$$(x_0 \wedge x_1)^2,\ (x_0 \wedge x_1)(x_0 \wedge x_2),\ (x_0 \wedge x_1)(x_1 \wedge x_2),\ (x_0 \wedge x_2)^2,\ (x_0 \wedge x_2)(x_1 \wedge x_2),\ (x_1 \wedge x_2)^2.$$

The map $\bar{c}_\sigma$ sends the i-th generator above to $\bar{N}_i$ of Lemma 14.14. By Lemma 14.14, $H^1(\bar{c}_\sigma(-9))$ is injective, so $H^1(c_\sigma(-9))$ is injective. This contradicts Construction-Proposition 14.2, since the connecting class is a non-zero element of $\operatorname{Ker} H^1(c_\sigma(-9))$. □

The next lemma is used in **Step 2** of the proof of Theorem 14.19.

Lemma 14.16. *Notation and conditions as in Construction-Proposition 14.13. Let $u \in \mathbb{P}^1$, and let*

$$\mu_u : H^0(F_u, \omega_{F_u}^{\otimes 2}) \to H^0(F_u, \omega_{F_u}^{\otimes 4})$$

be the map given by multiplication by $\zeta|_{F_u}$, after trivializing the fiber of $\mathcal{O}(-5)$ at u. Then $\dim \operatorname{Ker} \mu_u \leq 1$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we attach to a non-zero element of $\operatorname{Ker} \mu_u$ a divisor $0 \leq B \leq F_u$ with $2K_Y \cdot B + B^2 \geq 0$.

Since $m_1 = m_2 = 1$, $E_1 = C + E$, and $E_2 = E$, Construction-Proposition 14.2 implies that $\Delta := \beta^* Z + 3C + 6E$ is the divisor of ζ. Let η be a non-zero element of $\operatorname{Ker} \mu_u$. Since $\mathcal{O}_{F_u}(F_u)$ is trivial, $\omega_{F_u}^{\otimes 2} = \mathcal{O}_{F_u}(2K_Y)$. Write $F_u = \sum_i n_i G_i$, where the G_i are the prime components of F_u. Choose local trivializations. At the generic point of G_i, the local ring of Y is a discrete valuation ring, and η is an element modulo the n_i-th power of a uniformizer. Let $d_i \in \{0, \dots, n_i\}$ be the minimum of n_i and the order along G_i of a lift of η, which does not depend on the lift. Let $D := \sum_i d_i G_i$ and $B := F_u - D$.

Since the local rings of Y are unique factorization domains, a local lift of η is divisible by a local equation of D. Thus η is the image of a section $\tilde{\eta}$ under the injection on global sections induced by

$$0 \to \mathcal{O}_B(2K_Y - D) \to \mathcal{O}_{F_u}(2K_Y) \to \mathcal{O}_D(2K_Y) \to 0.$$

Since $\eta \neq 0$, we have $B \neq 0$. Since $\mathcal{O}_B(F_u)$ is trivial, we regard $\tilde{\eta}$ as a section of $\mathcal{O}_B(2K_Y + B)$.

By the definition of d_i, $\tilde{\eta}$ is a unit at the generic point of any component of B. Since B is a divisor on a smooth surface, B is Cohen–Macaulay and has no embedded points [Har77, Theorem II.8.21A(a)(c)(d)], so $\tilde{\eta}$ is a nonzerodivisor on B. Thus the zero scheme of $\tilde{\eta}$ is zero-dimensional, and the length of the zero scheme of $\tilde{\eta}$ is the degree of $\mathcal{O}_B(2K_Y + B)$ [Har77, Exercise V.1.1]. Therefore,

$$2K_Y \cdot B + B^2 \geq 0, \tag{14.37}$$

and if equality holds in (14.37), then $\tilde{\eta}$ has no zero.

Since $\zeta|_{F_u}\eta = 0$, at the generic point of G_i we have $d_i + \mathrm{mult}_{G_i} \Delta \geq n_i$. Thus $B \leq \Delta$.

Step 2. In this step we prove that $B = \beta^* Z - C - 2E$ and conclude the proof.

Let $A := \beta_* B$. Since $0 \leq B \leq \Delta$, we have $0 \leq A \leq Z$. Suppose that $A = 0$. Then B is contracted by β and is contained in a fiber of f. Since E is a section and C is a vertical (-2)-curve (Construction-Proposition 14.1), $B = nC$ for a positive integer n, and

$$2K_Y \cdot B + B^2 = -2n^2 < 0,$$

which contradicts (14.37). Thus $A \neq 0$.

Write

$$B = \beta^* A - e_1 C - e_2 E$$

with integers e_1, e_2. Since K_S is nef, $K_S \cdot Z = 0$, and $0 \leq A \leq Z$, we have $K_S \cdot A = 0$. Since $A \geq 0$ and $A \neq 0$, A has positive intersection with an ample divisor on S, so A is not numerically trivial, and Lemma 2.2(4) implies that $A^2 < 0$ and that A^2 is even, and hence $A^2 \leq -2$.

Let $k := K_Y \cdot B$. Since $m_1 = m_2 = 1$, $E_1 = C + E$, and $E_2 = E$, Construction-Proposition 14.1 implies that $K_Y \sim \beta^* K_S + C + 2E$ and $F \sim \beta^* M - C - 2E$. Moreover, $C^2 = -2$, $C \cdot E = 1$, $E^2 = -1$, and $\beta^* D' \cdot C = \beta^* D' \cdot E = 0$ for any divisor D' on S, so $K_Y \cdot C = F \cdot C = 0$, $K_Y \cdot E = -1$, and $F \cdot E = 1$. Since $F \cdot B = 0$ and $M \sim 2K_S - Z$, we obtain

$$k = e_2 = M \cdot A = -Z \cdot A, \quad B^2 = A^2 - e_1^2 - (e_2 - e_1)^2.$$

Since M is nef and $0 \leq A \leq Z$, we have $0 \leq k \leq M \cdot Z = -Z^2 = 2$. If $k = 0$, then $2k + B^2 \leq A^2 \leq -2$. If $k = 1$, then $e_1^2 + (1 - e_1)^2 \geq 1$ and $2k + B^2 \leq 2 - 2 - 1 = -1$. Both cases contradict (14.37). Thus $k = 2$, so $e_1^2 + (2 - e_1)^2 \geq 2$ and

$$0 \leq 2k + B^2 = 4 + A^2 - e_1^2 - (2 - e_1)^2 \leq 4 - 2 - 2 = 0.$$

Therefore,

$$A^2 = -2, \quad e_1 = 1, \quad e_2 = 2, \quad B^2 = -4, \quad K_Y \cdot B = 2.$$

Since $Z \cdot A = -2$, we have

$$(Z - A)^2 = Z^2 - 2(Z \cdot A) + A^2 = -2 + 4 - 2 = 0.$$

We also have $K_S \cdot (Z - A) = 0$ and $Z - A \geq 0$. By Lemma 2.2(4), $Z - A$ is numerically trivial, so $Z - A = 0$, since a non-zero divisor $D' \geq 0$ on S has positive intersection with an ample divisor. Thus

$$B = B_0 := \beta^* Z - C - 2E.$$

Moreover, equality holds in (14.37), so $\tilde{\eta}$ has no zero. We have proved that any non-zero element of $\operatorname{Ker} \mu_u$ is the image of a nowhere vanishing section of $\mathcal{O}_{B_0}(2K_Y + B_0)$ under the injection of **Step 1** for $B = B_0$. Let y be a closed point of B_0. Then the evaluation map at y on the subspace of $H^0(B_0, \mathcal{O}_{B_0}(2K_Y + B_0))$ corresponding to $\operatorname{Ker} \mu_u$ is injective, and the fiber of $\mathcal{O}_{B_0}(2K_Y + B_0)$ at y is one-dimensional. Thus $\dim \operatorname{Ker} \mu_u \leq 1$. □

14.5. **The bound.** In this subsection, we prove Theorem 14.19. We first solve the coefficient equations of the connecting class.

Lemma 14.17. *Let $a_0, b_0, c_0, d_0, l_0 \in \mathbb{C}$ with $b_0 \neq 0$. Let $s, y_0, \dots, y_4$ be variables, where y_0, y_1 have degree 2 and y_2, y_3, y_4 have degree 3. Let*

$$\bar{\sigma}_{11} := sy_2 + a_0y_3 + c_0y_4, \quad \bar{\sigma}_{02} := sy_2 + b_0y_3 + d_0y_4, \quad \bar{\sigma}_{12} := sy_3 + l_0y_4, \quad \bar{\sigma}_{22} := sy_4,$$
$$\bar{N}_1 := y_0\bar{\sigma}_{11} - y_1^2, \quad \bar{N}_2 := y_0\bar{\sigma}_{12} - y_1\bar{\sigma}_{02}, \quad \bar{N}_3 := y_1\bar{\sigma}_{12} - \bar{\sigma}_{11}\bar{\sigma}_{02},$$
$$\bar{N}_4 := y_0\bar{\sigma}_{22} - \bar{\sigma}_{02}^2, \quad \bar{N}_5 := y_1\bar{\sigma}_{22} - \bar{\sigma}_{02}\bar{\sigma}_{12}, \quad \bar{N}_6 := \bar{\sigma}_{11}\bar{\sigma}_{22} - \bar{\sigma}_{12}^2,$$

and let $\kappa_1, \dots, \kappa_6$ be combinations of $s^{-1}, \dots, s^{-4}$. Assume that for any monomial $\mathfrak{m}$ of degree e in $y_0, \dots, y_4$, the coefficient of $\mathfrak{m}$ in $\sum_i \kappa_i\bar{N}_i$ has no term s^{-j} with $1 \leq j \leq 8-e$. Then the following hold.

(1) There exist constants p, r, q, k with $\kappa_1 = 0$, $\kappa_2 = p/s$, $\kappa_3 = r/s + b_0p/s^2$, $\kappa_4 = q/s - l_0p/s^2$, and $\kappa_5 = k/s + (d_0p - l_0r)/s^2 - l_0b_0p/s^3$.

(2) If $p \neq 0$, then $a_0 + b_0 = 3l_0$ and $b_0^2 - b_0l_0 + l_0^2 = 0$.

(3) If $b_0 = p = 1$ and $c_0 = 0$, then, with $\lambda := l_0$, we have $r = (2\lambda - 1)d_0/3$, $q = 2(\lambda - 2)d_0/3$, $k = (2 - 7\lambda)d_0^2/3$, and

$$\kappa_6 = d_0^3/s + rd_0/s^2 - q/s^3 - \lambda^2/s^4.$$

Proof. For a monomial $\mathfrak{m}$, we denote by $[\mathfrak{m}]$ the coefficient of $\mathfrak{m}$ in $\sum_i \kappa_i\bar{N}_i$. We write $\kappa_6 = \sum_{i=1}^4 \kappa_{6,i}s^{-i}$.

We prove (1). Since $[y_1^2] = -\kappa_1$ and y_1^2 is of degree 4, we have $\kappa_1 = 0$. The coefficients of the monomials of degree 5 are

$$[y_0y_2] = 0, \quad [y_1y_2] = -s\kappa_2, \quad [y_0y_3] = s\kappa_2, \quad [y_1y_3] = s\kappa_3 - b_0\kappa_2,$$
$$[y_0y_4] = s\kappa_4 + l_0\kappa_2, \quad [y_1y_4] = s\kappa_5 + l_0\kappa_3 - d_0\kappa_2,$$

and none of them has a term s^{-1}, s^{-2}, or s^{-3}. From $[y_0y_3]$, we have $\kappa_2 = p/s$ for a constant p. Then $[y_1y_3] = s\kappa_3 - b_0p/s$ implies $\kappa_3 = r/s + b_0p/s^2$ for a constant r, and $[y_0y_4] = s\kappa_4 + l_0p/s$ implies $\kappa_4 = q/s - l_0p/s^2$ for a constant q. Finally,

$$[y_1y_4] = s\kappa_5 + \frac{l_0r - d_0p}{s} + \frac{l_0b_0p}{s^2},$$

which implies the formula for κ_5 with a constant k.

We prove (2). Table 20 lists the coefficients of the monomials of degree 6, which have no terms s^{-1} and s^{-2} by assumption. By (1), the coefficient of s^{-1} in $[y_2y_3]$ is $b_0p(3l_0 - a_0 - b_0)$. The coefficient of s^{-2} in $[y_2y_4]$ is $\kappa_{6,4} + l_0^2b_0p$, so $\kappa_{6,4} = -l_0^2b_0p$. The coefficient of s^{-2} in $[y_3^2]$ is $-a_0b_0^2p + 2b_0^2l_0p - \kappa_{6,4}$, which is equal to $b_0p(-a_0b_0 + 2b_0l_0 + l_0^2)$. Since $b_0p \neq 0$, we have $a_0 + b_0 = 3l_0$ and $-a_0b_0 + 2b_0l_0 + l_0^2 = 0$. Eliminating a_0, we obtain $b_0^2 - b_0l_0 + l_0^2 = 0$.

TABLE 20. The coefficients of the monomials of degree 6 in $\sum_i \kappa_i\bar{N}_i$.

$\mathfrak{m}$	$[\mathfrak{m}]$
y_2^2	$-s^2(\kappa_3 + \kappa_4)$
y_2y_3	$-(a_0 + b_0)s\kappa_3 - 2b_0s\kappa_4 - s^2\kappa_5$
y_2y_4	$-(c_0 + d_0)s\kappa_3 - 2d_0s\kappa_4 - l_0s\kappa_5 + s^2\kappa_6$
y_3^2	$-a_0b_0\kappa_3 - b_0^2\kappa_4 - b_0s\kappa_5 - s^2\kappa_6$
y_3y_4	$-(a_0d_0 + b_0c_0)\kappa_3 - 2b_0d_0\kappa_4 - (b_0l_0 + d_0s)\kappa_5 + (a_0 - 2l_0)s\kappa_6$
y_4^2	$-c_0d_0\kappa_3 - d_0^2\kappa_4 - d_0l_0\kappa_5 + (c_0s - l_0^2)\kappa_6$

We prove (3). Assume that $b_0 = p = 1$ and $c_0 = 0$. By (2), we have $a_0 = 3\lambda - 1$ and $\lambda^2 = \lambda - 1$, so $a_0 - 2\lambda = \lambda^2$, $\lambda^{-1} = 1 - \lambda$, and $(2\lambda - 1)^2 = -3$. Substituting (1) into Table 20 and taking the

coefficients of s^{-1} and s^{-2}, we obtain, among others, the equations of Table 21, in the order in which we use them.

TABLE 21. Vanishing of the terms s^{-1}, s^{-2} in the coefficients of the monomials of degree 6, after substituting (1) with $b_0 = p = 1$ and $c_0 = 0$.

$\mathfrak{m}$	term	equation
y_2y_4	s^{-2}	$\kappa_{6,4} = -\lambda^2$
y_2y_4	s^{-1}	$\kappa_{6,3} = (1-\lambda)d_0 - \lambda^2 r$
y_3^2	s^{-1}	$q = -\lambda r - 2d_0 + \lambda d_0$
y_3y_4	s^{-2}	$(2\lambda - 1)r + d_0 = 0$
y_4^2	s^{-2}	$\kappa_{6,2} = rd_0$
y_3y_4	s^{-1}	$\lambda k = -\lambda r d_0 - 2d_0 q - d_0^2$
y_4^2	s^{-1}	$\lambda^2\kappa_{6,1} = -d_0^2 q - \lambda d_0 k$

Since $(2\lambda - 1)^2 = -3$, the fourth equation implies $r = (2\lambda - 1)d_0/3$. The third equation then implies

$$q = \frac{d_0}{3}\big(-2\lambda^2 + \lambda - 6 + 3\lambda\big) = \frac{2(\lambda - 2)d_0}{3}.$$

The sixth equation implies

$$k = (1-\lambda)\cdot\frac{(7-5\lambda)d_0^2}{3} = \frac{(2-7\lambda)d_0^2}{3}.$$

Moreover, the second equation implies $\kappa_{6,3} = (4 - 2\lambda)d_0/3 = -q$, and the last equation implies $\lambda^2\kappa_{6,1} = (\lambda - 1)d_0^3$, so $\kappa_{6,1} = d_0^3$. □

Lemma 14.18. *Let $\lambda, d_0 \in \mathbb{C}$ with $\lambda^2 - \lambda + 1 = 0$. Let $s, y_0, \dots, y_4$ be variables, where y_0, y_1 have degree 2 and y_2, y_3, y_4 have degree 3. Let*

$$\bar\sigma_{11} := sy_2 + (3\lambda - 1)y_3, \quad \bar\sigma_{02} := sy_2 + y_3 + d_0y_4, \quad \bar\sigma_{12} := sy_3 + \lambda y_4, \quad \bar\sigma_{22} := sy_4,$$

$$\bar N_1 := y_0\bar\sigma_{11} - y_1^2, \quad \bar N_2 := y_0\bar\sigma_{12} - y_1\bar\sigma_{02}, \quad \bar N_3 := y_1\bar\sigma_{12} - \bar\sigma_{11}\bar\sigma_{02},$$
$$\bar N_4 := y_0\bar\sigma_{22} - \bar\sigma_{02}^2, \quad \bar N_5 := y_1\bar\sigma_{22} - \bar\sigma_{02}\bar\sigma_{12}, \quad \bar N_6 := \bar\sigma_{11}\bar\sigma_{22} - \bar\sigma_{12}^2.$$

Let

$$r := \frac{(2\lambda - 1)d_0}{3}, \quad q := \frac{2(\lambda - 2)d_0}{3}, \quad k := \frac{(2 - 7\lambda)d_0^2}{3},$$

and let

$$\kappa_1 := 0, \quad \kappa_2 := \frac{1}{s}, \quad \kappa_3 := \frac{r}{s} + \frac{1}{s^2}, \quad \kappa_4 := \frac{q}{s} - \frac{\lambda}{s^2},$$
$$\kappa_5 := \frac{k}{s} + \frac{d_0 - \lambda r}{s^2} - \frac{\lambda}{s^3}, \quad \kappa_6 := \frac{d_0^3}{s} + \frac{rd_0}{s^2} - \frac{q}{s^3} - \frac{\lambda^2}{s^4}.$$

Then for any monomial $\mathfrak{m}$ of degree e in $y_0, \dots, y_4$, the coefficient of $\mathfrak{m}$ in $\sum_i \kappa_i\bar N_i$ has no term s^{-j} with $1 \le j \le 8 - e$.

Proof. For a monomial $\mathfrak{m}$, we denote by $[\mathfrak{m}]$ the coefficient of $\mathfrak{m}$ in $\sum_i \kappa_i\bar N_i$. The polynomials $\bar N_i$ are combinations of the quadratic monomials in $y_0, \dots, y_4$, which are of degree 4, 5, or 6. The sum of the terms of degree 4 in $\sum_i \kappa_i\bar N_i$ is $-\kappa_1 y_1^2 = 0$, and the coefficients of the monomials of degree 5 are

$$[y_0y_2] = s\kappa_1 = 0, \quad [y_0y_3] = s\kappa_2 + (3\lambda - 1)\kappa_1 = 1, \quad [y_0y_4] = s\kappa_4 + \lambda\kappa_2 = q,$$
$$[y_1y_2] = -s\kappa_2 = -1, \quad [y_1y_3] = s\kappa_3 - \kappa_2 = r, \quad [y_1y_4] = s\kappa_5 + \lambda\kappa_3 - d_0\kappa_2 = k,$$

which have no terms s^{-1}, s^{-2}, s^{-3}. The coefficients of the monomials of degree 6 are

$$[y_2^2] = -s^2(\kappa_3+\kappa_4), \quad [y_2y_3] = -3\lambda s\kappa_3 - 2s\kappa_4 - s^2\kappa_5,$$
$$[y_2y_4] = -d_0s\kappa_3 - 2d_0s\kappa_4 - \lambda s\kappa_5 + s^2\kappa_6, \quad [y_3^2] = -(3\lambda-1)\kappa_3 - \kappa_4 - s\kappa_5 - s^2\kappa_6,$$
$$[y_3y_4] = -(3\lambda-1)d_0\kappa_3 - 2d_0\kappa_4 - (\lambda+d_0s)\kappa_5 + (\lambda-1)s\kappa_6, \quad [y_4^2] = -d_0^2\kappa_4 - \lambda d_0\kappa_5 - \lambda^2\kappa_6,$$

and by the formulas for $\kappa_3, \dots, \kappa_6$, their coefficients of s^{-2} and s^{-1} are

$\mathfrak{m}$	coefficient of s^{-2} in $[\mathfrak{m}]$	coefficient of s^{-1} in $[\mathfrak{m}]$
y_2^2	0	0
y_2y_3	0	0
y_2y_4	0	$(\lambda-1)d_0 + \lambda^2 r - q$
y_3^2	$\lambda^2-\lambda+1$	$-(2\lambda-1)r - d_0$
y_3y_4	$(1-\lambda)d_0 + \lambda^2 r - (\lambda-1)q$	$-\lambda d_0 r - 2d_0 q - \lambda k - d_0^2$
y_4^2	0	$-d_0^2 q - \lambda d_0 k - \lambda^2 d_0^3$

Since $\lambda^2 = \lambda - 1$, the coefficient of s^{-2} in $[y_3^2]$ is 0, $(2\lambda-1)^2 = 4(\lambda^2-\lambda+1) - 3 = -3$, and the formulas for r, q, k imply

$$(2\lambda-1)r = -d_0, \quad \lambda r = \frac{(\lambda-2)d_0}{3}, \quad \lambda^2 r = -\frac{(\lambda+1)d_0}{3},$$
$$(\lambda-1)q = -\frac{2(2\lambda-1)d_0}{3}, \quad \lambda k = \frac{(7-5\lambda)d_0^2}{3}.$$

Hence, by $q = 2(\lambda-2)d_0/3$ and $\lambda^2 = \lambda - 1$,

$$(\lambda-1)d_0 + \lambda^2 r - q = \frac{d_0}{3}\big(3(\lambda-1) - (\lambda+1) - 2(\lambda-2)\big) = 0, \quad -(2\lambda-1)r - d_0 = d_0 - d_0 = 0,$$
$$(1-\lambda)d_0 + \lambda^2 r - (\lambda-1)q = \frac{d_0}{3}\big(3(1-\lambda) - (\lambda+1) + 2(2\lambda-1)\big) = 0,$$
$$-\lambda d_0 r - 2d_0 q - \lambda k - d_0^2 = \frac{d_0^2}{3}\big(-(\lambda-2) - 4(\lambda-2) - (7-5\lambda) - 3\big) = 0,$$
$$-d_0^2 q - \lambda d_0 k - \lambda^2 d_0^3 = -\frac{d_0^3}{3}\big(2(\lambda-2) + (7-5\lambda) + 3(\lambda-1)\big) = 0,$$

so $[\mathfrak{m}]$ has no terms s^{-1}, s^{-2} for any monomial $\mathfrak{m}$ of degree 6. □

The proof of Theorem 14.19 reduces the multiplication and the relation of degree four to seven constants, for each root of $\lambda^2 - \lambda + 1 = 0$.

Theorem 14.19. *We have* $\dim \overline{\mathcal{L}_3} \le 7$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we bring the multiplication of a surface in $\mathcal{L}_3$ to the form (14.38), where a_0, b_0, c_0, d_0, l_0 are constants.

Let S be a torsion-free numerical Godeaux surface whose bicanonical pencil has genus 3. By Theorem 14.12, Profile II holds, and by Proposition 14.7, the centers of β are infinitely near. Thus Construction-Proposition 14.13 holds for S, and the exact sequence of Construction-Proposition 14.2 reads

$$0 \to \mathcal{O}(9) \xrightarrow{\iota} \operatorname{coker}(c_\sigma) \to V_4 \to 0.$$

We write $\sigma_{ij} := \sigma_2(x_ix_j)$, so that

$$\sigma_{00} = y_0, \quad \sigma_{01} = y_1, \quad \sigma_{11} = wy_2 + ay_3 + cy_4 + f_3\zeta,$$
$$\sigma_{02} = vy_2 + by_3 + dy_4 + g_3\zeta, \quad \sigma_{12} = vy_3 + ly_4 + h_3\zeta, \quad \sigma_{22} = vy_4 + j_3\zeta.$$

By Lemma 14.15, w and v are proportional. Rescaling $x_2 \to \delta x_2$ and $y_4 \to \delta y_4$, where δ is the constant with $\delta v = w$, we may assume that $v = w$ by a direct calculation. We choose coordinates $[s:t]$ on $\mathbb{P}^1$ with $v = w = s$, so the zero of s is b_C.

We remove the multiples of s from a, b, c, d, l. First, the replacement $y_3 \to y_3 + \epsilon y_4$ for a constant ϵ removes the multiple of s from l. Next, replacing y_2 with y_2 plus constant multiples of y_3 and y_4 removes the multiples of s from a and c. Finally, for constants η_1, η_2, the replacement of x_0 by $x_0 + \eta_1 x_1 + \eta_2 x_2$ changes σ_{02} by $\eta_1\sigma_{12} + \eta_2\sigma_{22}$. We choose η_1 to remove the multiple of s from the coefficient of y_3 in σ_{02}, and then η_2 to remove the multiple of s from the coefficient of y_4. We define y_0 and y_1 again by $y_0 = x_0^2$ and $y_1 = x_0x_1$. The new elements y_0, y_1 are lifts of a basis of the quotient of $V_2(-2)$ by the summand $\mathcal{O}(1)^{\oplus 3} \oplus \mathcal{O}(3)$, so $y_0, \dots, y_4, \zeta$ still split V_2 into summands of degrees $2, 2, 3, 3, 3, 5$. The sections x_1, x_2 do not change, so they still vanish on E.

Thus we have

$$\begin{gathered} \sigma_{00} = y_0, \quad \sigma_{01} = y_1, \quad \sigma_{11} = sy_2 + a_0ty_3 + c_0ty_4 + f_3\zeta, \\ \sigma_{02} = sy_2 + b_0ty_3 + d_0ty_4 + g_3\zeta, \quad \sigma_{12} = sy_3 + l_0ty_4 + h_3\zeta, \quad \sigma_{22} = sy_4 + j_3\zeta, \end{gathered} \tag{14.38}$$

where a_0, b_0, c_0, d_0, l_0 are constants. From now on, the letters a, b, c, d, l are not used. We write $f_3 = \sum_{i=0}^3 f_{3,i}s^it^{3-i}$, $g_3 = \sum_{i=0}^3 g_{3,i}s^it^{3-i}$, $h_3 = \sum_{i=0}^3 h_{3,i}s^it^{3-i}$, and $j_3 = \sum_{i=0}^3 j_{3,i}s^it^{3-i}$. Let $N_1, \dots, N_6$ be the images under c_σ of the elements $(x_0 \wedge x_1)^2$, $(x_0 \wedge x_1)(x_0 \wedge x_2)$, $(x_0 \wedge x_1)(x_1 \wedge x_2)$, $(x_0 \wedge x_2)^2$, $(x_0 \wedge x_2)(x_1 \wedge x_2)$, $(x_1 \wedge x_2)^2$ of the basis of $\operatorname{Sym}^2(\wedge^2 V_1)$, that is,

$$\begin{gathered} N_1 = \sigma_{00}\sigma_{11} - \sigma_{01}^2, \quad N_2 = \sigma_{00}\sigma_{12} - \sigma_{01}\sigma_{02}, \quad N_3 = \sigma_{01}\sigma_{12} - \sigma_{11}\sigma_{02}, \\ N_4 = \sigma_{00}\sigma_{22} - \sigma_{02}^2, \quad N_5 = \sigma_{01}\sigma_{22} - \sigma_{02}\sigma_{12}, \quad N_6 = \sigma_{11}\sigma_{22} - \sigma_{12}^2. \end{gathered} \tag{14.39}$$

On the chart $t = 1$, the projections of σ_{ij} and N_i to $\operatorname{Sym}^\bullet U$ are the polynomials $\bar\sigma_{ij}$ and $\bar N_i$ of Lemma 14.17.

Step 2. In this step we prove that $b_0l_0 \neq 0$.

We prove that for any $u \in \mathbb{P}^1$,

$$\dim \operatorname{Ker} \bar c_\sigma(u) \leq 1. \tag{14.40}$$

Let $\theta \in \operatorname{Ker} \bar c_\sigma(u)$ be a non-zero vector. Since c_σ is injective with locally free cokernel (Construction-Proposition 14.2), we have $c_\sigma(u)(\theta) \neq 0$. The kernel of the projection on the fiber at u is $\zeta \cdot V_2(u)$, so $c_\sigma(u)(\theta) = \zeta \cdot \theta'$ for a unique non-zero $\theta' \in V_2(u)$, and $\theta \to \theta'$ is linear and injective. The multiplication $\operatorname{Sym}^2 V_2 \to V_4$ vanishes on the image of c_σ, because $(\xi_1\xi_3)(\xi_2\xi_4) = (\xi_1\xi_4)(\xi_2\xi_3)$ for local sections $\xi_1, \dots, \xi_4$ of V_1. Thus $\zeta\theta' = 0$ in $V_4(u)$. By Construction-Proposition 14.3(3), θ' belongs to the kernel of the map μ_u of Lemma 14.16, so (14.40) follows from Lemma 14.16.

We consider the fiber at b_C, where $s = 0$ and $t = 1$. By (14.38), there we have $\bar\sigma_{02} = b_0y_3 + d_0y_4$, $\bar\sigma_{12} = l_0y_4$, and $\bar\sigma_{22} = 0$. If $l_0 = 0$, then $\bar\sigma_{12} = \bar\sigma_{22} = 0$, so $\bar N_5 = \bar N_6 = 0$ at b_C, which implies $\dim \operatorname{Ker} \bar c_\sigma(b_C) \geq 2$, a contradiction to (14.40). If $b_0 = 0$, then at b_C,

$$\bar N_4 = -d_0^2y_4^2, \quad \bar N_5 = -d_0l_0y_4^2, \quad \bar N_6 = -l_0^2y_4^2.$$

The polynomials $\bar N_4, \bar N_5, \bar N_6$ span a space of dimension at most one, so $\dim \operatorname{Ker} \bar c_\sigma(b_C) \geq 2$, again a contradiction to (14.40). Thus $b_0l_0 \neq 0$.

Step 3. In this step we fix the charts and the representatives of H^1, and prove that $b_0 = 1$, $l_0 = \lambda$, and $a_0 = 3\lambda - 1$ after a change of t, where $\lambda^2 - \lambda + 1 = 0$.

We use the charts $t \neq 0$ and $s \neq 0$ of $\mathbb{P}^1$, and write sections on the chart $t \neq 0$ by setting $t = 1$. A monomial $\mathfrak{m}$ of degree e in $y_0, \dots, y_4, \zeta$, where ζ has degree 5, spans a summand $\mathcal{O}(e-9)\mathfrak{m}$ of $\operatorname{Sym}^2 V_2(-9)$. On $\mathcal{O}(e-9)\mathfrak{m}$, a Laurent monomial $s^n\mathfrak{m}$ is regular on the chart $t \neq 0$ if and only if $n \geq 0$, and regular on the chart $s \neq 0$ if and only if $n \leq e - 9$. Thus s^{-j}, $1 \leq j \leq 8 - e$, represent a basis of $H^1(\mathbb{P}^1, \mathcal{O}(e-9))$, computed by the Čech complex of the cover by the charts $t \neq 0$ and $s \neq 0$ [Har77, Theorem III.4.5]. For a Laurent polynomial φ in s, we denote by $[\varphi]_{\geq 0}$ and $[\varphi]_{<0}$ the sums of the terms with non-negative and negative exponents of s.

Since

$$\operatorname{Sym}^2(\wedge^2 V_1)(-9) = \mathcal{O}(-5)^{\oplus 6},$$

the connecting class has six components. Let $\kappa_1, \dots, \kappa_6$ be their representatives, each a combination of $s^{-1}, \dots, s^{-4}$, with respect to the basis of **Step 1**. The connecting class belongs to $\operatorname{Ker} H^1(c_\sigma(-9))$, so for any monomial $\mathfrak{m}$ of degree e,

$$\text{the coefficient of } \mathfrak{m} \text{ in } \sum_i \kappa_i N_i \text{ has no term } s^{-j} \text{ with } 1 \le j \le 8-e. \tag{14.41}$$

There exist lifts Φ_t and Φ_s of ι to $\operatorname{Sym}^2 V_2(-9)$ on the charts $t \neq 0$ and $s \neq 0$ with $\Phi_t - \Phi_s = \sum_i \kappa_i N_i$, by the Čech description of the connecting map on the cover by the charts $t \neq 0$ and $s \neq 0$, where the terms s^n with $n \ge 0$ or $n \le -5$ of a component are absorbed into Φ_t or Φ_s. The sections Φ_t, Φ_s are local equations of the relation of degree four of the relative canonical algebra. The global sections of $\operatorname{Sym}^2 V_2(-9)$ are $(\alpha_1 t + \alpha_2 s)\zeta^2$, $\alpha_1, \alpha_2 \in \mathbb{C}$, because ζ^2 spans $\mathcal{O}(1)\zeta^2$ and all other summands are of negative degree. Thus, by the description of the regular Laurent monomials above,

$$\Phi_t = \Big[\sum_i \kappa_i N_i\Big]_{\ge 0} + (\alpha_1 t + \alpha_2 s)\zeta^2, \quad \Phi_s = -\Big[\sum_i \kappa_i N_i\Big]_{<0} + (\alpha_1 t + \alpha_2 s)\zeta^2 \tag{14.42}$$

for constants α_1, α_2. Moreover, since $H^0(\mathbb{P}^1, \mathcal{O}(-5)) = 0$, the kernel of

$$H^0(\operatorname{Sym}^2 V_2(-9)) \to H^0(\operatorname{coker}(c_\sigma)(-9))$$

is zero, so α_1, α_2 are determined by ι.

For monomials in $y_0, \dots, y_4$, (14.41) is the assumption of Lemma 14.17 for the polynomials $\bar{N}_i$. Since $b_0 \neq 0$, Lemma 14.17(1) implies the formulas for $\kappa_1, \dots, \kappa_5$ with constants p, r, q, k. The coefficient of $y_0\zeta$ in $\sum_i \kappa_i N_i$ is $\kappa_2 h_3 + \kappa_4 j_3$. Since $y_0\zeta$ is of degree 7, (14.41) implies that the coefficient of s^{-1} is zero, that is,

$$p h_{3,0} + q j_{3,0} - l_0 p j_{3,1} = 0. \tag{14.43}$$

We prove that $l_0 p j_{3,0} \neq 0$. Let $u \in \mathbb{P}^1$ be a general point, so that $u \notin \operatorname{Supp} \tau$, $t(u) = 1$, and $s(u) \neq 0$. Then $\sigma_2(u)$ is an isomorphism, and

$$\operatorname{coker}(c_\sigma)(u) = \operatorname{Sym}^4 V_1(u).$$

Let Ψ_u be the image of $\Phi_t(u)$ in $\operatorname{Sym}^4 V_1(u)$. Since $\iota(u) \neq 0$ (V_4 is locally free, so ι is injective on fibers) and $\iota(u)$ goes to zero in $V_4(u)$, the equation of the plane quartic X'_u of Construction-Proposition 14.3(4) is $\Psi_u = 0$. By Construction-Proposition 14.13(2), the image of E in X'_u is the point $[1:0:0]$. Let $\mathfrak{n}$ be the ideal of $[1:0:0]$ in the local ring of the plane, with $x_0 = 1$. Under $\sigma_2(u)$, we have $y_0 = 1$, and any one of $y_1, y_2, y_3, y_4, \zeta$ is a combination of x_1, x_2, and $x_i x_j$, $i, j \ge 1$, so it belongs to $\mathfrak{n}$. Moreover, $\sigma_{12}, \sigma_{22} \in \mathfrak{n}^2$. By (14.39), (14.42), and $\kappa_1 = 0$, the terms of $\Phi_t(u)$ which contain y_0 are y_0 times

$$L := p y_3 + q y_4 + \big[\kappa_2 h_3 + \kappa_4 j_3\big]_{\ge 0}\zeta,$$

and all other terms belong to $\mathfrak{n}^2$. Therefore $\Psi_u \equiv L$ modulo $\mathfrak{n}^2$. On the other hand,

$$\kappa_2\sigma_{12} + \kappa_4\sigma_{22} = p y_3 + q y_4 + (\kappa_2 h_3 + \kappa_4 j_3)\zeta \in \mathfrak{n}^2.$$

Thus, by (14.43),

$$\Psi_u \equiv -\big[\kappa_2 h_3 + \kappa_4 j_3\big]_{<0}\zeta = \frac{l_0 p j_{3,0}}{s^2}\zeta \quad \text{modulo } \mathfrak{n}^2. \tag{14.44}$$

If $l_0 p j_{3,0} = 0$, then $\Psi_u \in \mathfrak{n}^2$, so X'_u is singular at the image of E, a contradiction to Construction-Proposition 14.13(1). Thus $p \neq 0$ and $j_{3,0} \neq 0$.

By Lemma 14.17(2), we have

$$a_0 + b_0 = 3l_0, \quad b_0^2 - b_0 l_0 + l_0^2 = 0. \tag{14.45}$$

Replacing t with $b_0 t$, which divides a_0, b_0, c_0, d_0, l_0 by b_0 and multiplies p and $j_{3,0}$ by non-zero constants, we may assume that $b_0 = 1$, $l_0 = \lambda$, and $a_0 = 3\lambda - 1$, where $\lambda^2 - \lambda + 1 = 0$.

Step 4. In this step we remove c_0 and solve for $\kappa_1, \dots, \kappa_6$.

For a constant ν, consider the replacements

$$x_0 \to x_0 + 2\nu x_1 + \nu^2 x_2, \quad x_1 \to x_1 + \nu x_2, \quad y_3 \to y_3 + \nu y_4, \quad y_2 \to y_2 + 2\nu y_3 + \nu^2 y_4,$$

where x_2 does not change and y_0, y_1 are defined again by $y_0 = x_0^2$ and $y_1 = x_0 x_1$. Then σ_{11} and σ_{02} change by $2\nu\sigma_{12} + \nu^2\sigma_{22}$, and σ_{12} changes by $\nu\sigma_{22}$. Expanding in the new y_2, y_3, y_4, we obtain (14.38) again, where a_0, b_0, l_0 do not change, while c_0 and d_0 become $c_0 + (2\lambda - a_0)\nu = c_0 + (1-\lambda)\nu$ and $d_0 + (2\lambda - b_0)\nu = d_0 + (2\lambda - 1)\nu$. The sections x_1, x_2 still vanish on E. We replace the basis of **Step 1** with the new one, and the argument of **Step 3** implies $p \neq 0$ and $j_{3,0} \neq 0$ for the new form. Since $\lambda \neq 1$, the choice $\nu = -c_0/(1-\lambda)$ fixes $c_0 = 0$, and we denote the new constant again by d_0.

The map ι is defined up to a non-zero scalar, so we may assume that $p = 1$. By Lemma 14.17(1) and (3),

$$\begin{gathered} \kappa_1 = 0, \quad \kappa_2 = \frac{1}{s}, \quad \kappa_3 = \frac{r}{s} + \frac{1}{s^2}, \quad \kappa_4 = \frac{q}{s} - \frac{\lambda}{s^2}, \\ \kappa_5 = \frac{k}{s} + \frac{d_0 - \lambda r}{s^2} - \frac{\lambda}{s^3}, \quad \kappa_6 = \frac{d_0^3}{s} + \frac{r d_0}{s^2} - \frac{q}{s^3} - \frac{\lambda^2}{s^4}, \end{gathered} \tag{14.46}$$

where

$$r = \frac{(2\lambda - 1)d_0}{3}, \quad q = \frac{2(\lambda - 2)d_0}{3}, \quad k = \frac{(2 - 7\lambda)d_0^2}{3}.$$

The monomials which contain ζ and for which the range $1 \le j \le 8 - e$ of (14.41) is not empty are $y_0\zeta$ and $y_1\zeta$, of degree 7, since $y_2\zeta, y_3\zeta, y_4\zeta$ are of degree 8 and ζ^2 is of degree 10. The coefficient of $y_1\zeta$ is $-\kappa_2 g_3 + \kappa_3 h_3 + \kappa_5 j_3$. Taking the coefficient of s^{-1} for $y_0\zeta$ and $y_1\zeta$, we obtain

$$\begin{gathered} h_{3,0} + q j_{3,0} - \lambda j_{3,1} = 0, \\ -g_{3,0} + r h_{3,0} + h_{3,1} + k j_{3,0} + (d_0 - \lambda r) j_{3,1} - \lambda j_{3,2} = 0. \end{gathered} \tag{14.47}$$

Step 5. In this step we remove nine cubic coefficients and the scale of ζ, and describe the relation of degree four by two further constants.

Since $j_{3,0} \neq 0$, rescaling ζ, we may assume that $j_{3,0} = 1$. For binary quadratic forms ρ_2, ρ_3, ρ_4 in s, t, the replacements $y_i \to y_i + \rho_i\zeta$, $i = 2, 3, 4$, are automorphisms of V_2. They do not change the constants of (14.38), and they do not change $\kappa_1, \dots, \kappa_6$. First, the choice $\rho_4 = (j_3 - t^3)/s$ implies $j_3 = t^3$. Next, for the current form h_3, the choice $\rho_3 = (h_3 - h_{3,0}t^3)/s$ implies that h_3 is a multiple of t^3, and it does not change j_3. Finally, for the current form f_3, the choice $\rho_2 = (f_3 - f_{3,0}t^3)/s$ implies $f_3 = f_{3,0}t^3$, and it does not change h_3 and j_3. By the first equation of (14.47), $h_3 = -qt^3$. By the second equation of (14.47),

$$g_{3,0} = k - rq = \frac{(2 - 5\lambda)d_0^2}{3},$$

because $(2\lambda - 1)(\lambda - 2) = -3\lambda$. Therefore

$$\begin{gathered} \sigma_{00} = y_0, \quad \sigma_{01} = y_1, \quad \sigma_{11} = sy_2 + (3\lambda - 1)ty_3 + f_{3,0}t^3\zeta, \\ \sigma_{02} = sy_2 + ty_3 + d_0 t y_4 + g_3\zeta, \quad \sigma_{12} = sy_3 + \lambda t y_4 - qt^3\zeta, \quad \sigma_{22} = sy_4 + t^3\zeta, \end{gathered} \tag{14.48}$$

where $g_3 = \sum_{i=0}^{3} g_{3,i} s^i t^{3-i}$ with $g_{3,0} = (2 - 5\lambda)d_0^2/3$, and $q = 2(\lambda - 2)d_0/3$. The relation of degree four is given by (14.42) with $\kappa_1, \dots, \kappa_6$ of (14.46). Thus (σ_2, ι) is determined by λ and the seven constants

$$(d_0, f_{3,0}, g_{3,1}, g_{3,2}, g_{3,3}, \alpha_1, \alpha_2) \in \mathbb{C}^7.$$

Conversely, let λ be a complex number with $\lambda^2 - \lambda + 1 = 0$, and let $d_0, f_{3,0}, g_{3,1}, g_{3,2}, g_{3,3}$ be arbitrary constants. We prove that the polynomials N_i of (14.39) for (14.48) and the κ_i of (14.46) satisfy (14.41) for any monomial $\mathfrak{m}$. We set $t = 1$. For the monomials in $y_0, \dots, y_4$, (14.41) holds by Lemma 14.18, since the projections of the N_i to $\mathrm{Sym}^\bullet U$ are the $\bar{N}_i$ of Lemma 14.18 (**Step 1**), and the κ_i of (14.46) are those of Lemma 14.18. Among the monomials which contain

ζ, it suffices to consider $y_0\zeta$ and $y_1\zeta$, of degree 7 (**Step 4**). By (14.39), (14.48), and (14.46), the coefficients of $y_0\zeta$ and $y_1\zeta$ in $\sum_i \kappa_i N_i$ are

$$f_{3,0}\kappa_1 - q\kappa_2 + \kappa_4 = -\frac{\lambda}{s^2} \quad \text{and} \quad -g_3\kappa_2 - q\kappa_3 + \kappa_5,$$

and the coefficient of s^{-1} in $-g_3\kappa_2 - q\kappa_3 + \kappa_5$ is $k - rq - g_{3,0} = 0$, since $k = (2-7\lambda)d_0^2/3$, $g_{3,0} = (2-5\lambda)d_0^2/3$, and $rq = -2\lambda d_0^2/3$ by $(2\lambda-1)(\lambda-2) = -3\lambda$. Thus, for any seven constants, the formulas (14.42) define sections Φ_t and Φ_s of $\operatorname{Sym}^2 V_2(-9)$ on the charts $t \neq 0$ and $s \neq 0$ with $\Phi_t - \Phi_s = \sum_i \kappa_i N_i$.

Step 6. In this step we pass from the seven constants to the moduli space.

Fix λ with $\lambda^2 - \lambda + 1 = 0$, and let $T_\lambda := \mathbb{A}^7$ with coordinates $(d_0, f_{3,0}, g_{3,1}, g_{3,2}, g_{3,3}, \alpha_1, \alpha_2)$. Let

$$V_2 := \mathcal{O}(2)^{\oplus 2} \oplus \mathcal{O}(3)^{\oplus 3} \oplus \mathcal{O}(5)$$

on $\mathbb{P}^1$, with symbols $y_0, \dots, y_4, \zeta$, and let $P(V_2) := \operatorname{Proj} \operatorname{Sym} V_2$. Let $\mathcal{X}_\lambda \subset P(V_2) \times T_\lambda$ be the closed subscheme defined by $N_1, \dots, N_6$ and by Φ_t and Φ_s on the charts $t \neq 0$ and $s \neq 0$ of $\mathbb{P}^1$. It is well defined because $\Phi_t - \Phi_s$ is a combination of the minors. For $\mathbf{t} \in T_\lambda$, let $\mathcal{X}_{\lambda,\mathbf{t}}$ be the fiber of $\mathcal{X}_\lambda$ over $\mathbf{t}$. We denote by $T_\lambda^{\mathrm{geo}} \subset T_\lambda$ the set of $\mathbf{t}$ for which $\mathcal{X}_{\lambda,\mathbf{t}}$ is a connected normal surface with only rational double points and the minimal model of the minimal resolution of $\mathcal{X}_{\lambda,\mathbf{t}}$ is a numerical Godeaux surface. A point $\mathbf{t} \in T_\lambda^{\mathrm{geo}}$ determines $\mathcal{X}_{\lambda,\mathbf{t}}$ by equations, and so the minimal resolution, the minimal model $S_{\mathbf{t}}$, and the canonical model $X_{\mathbf{t}}$.

We prove that any surface of $\mathcal{L}_3$ occurs. Let S be as in **Step 1**, with $f : Y \to \mathbb{P}^1$, and let $\mathbf{t} \in T_\lambda$ be the point of **Step 5** for S, for one of the values of λ. By [CP06, Definition 7.8], the bundle attached to σ_2 is $\operatorname{coker}(c_\sigma)$. By [CP06, Definition 7.10], a genus three 5-tuple consists of $\mathbb{P}^1$, V_1, τ, the class of

$$0 \to \operatorname{Sym}^2 V_1 \to V_2 \to \mathcal{O}_\tau \to 0,$$

and a line $\mathcal{O}(9)$ in $\operatorname{coker}(c_\sigma)$. Let $W_\sigma \subset P(V_2)$ be the image of the rational map $P(V_1) \dashrightarrow P(V_2)$ induced by σ_2. By [CP06, Definition 7.11(2)], the relative canonical model of a 5-tuple is the divisor on W_σ cut by the line. By Corollary 13.6, the general fiber of f is nonhyperelliptic, and by Construction-Proposition 14.1, Y is relatively minimal over $\mathbb{P}^1$ and any fiber of f is numerically 2-connected. Thus [CP06, Theorem 7.13] implies that X' is the surface defined by the 5-tuple in which τ and the class are those determined by σ_2 and the line is $\iota(\mathcal{O}(9))$.

We prove that the zero scheme of $N_1, \dots, N_6$ is W_σ. Over $u \notin \operatorname{Supp}\tau$, the fiber $\sigma_2(u)$ is an isomorphism and $N_1, \dots, N_6$ are the 2×2 minors of the symmetric matrix $(\sigma_{ij}(u))$, which vanish on the Veronese plane by $x_i x_j = \sigma_{ij}$ and generate the ideal of the Veronese plane, so the zero scheme is the Veronese plane. Over $u \in \operatorname{Supp}\tau$, the matrix $(\sigma_{ij}(u))$ satisfies one linear relation, given by the quadratic form q_u of Construction-Proposition 14.3(5)(i) at u, and the zero scheme is the cone over the image of $\{q_u = 0\}$ in the second Veronese embedding. Over any u, the Hilbert polynomial is $(2n+1)(n+1)$ (the homogeneous coordinate ring $\bigoplus_{i \geq 0} H^0(\mathbb{P}^2, \mathcal{O}(2i))$ of the Veronese plane is a domain, so the non-zero linear relation is a nonzerodivisor on it, the hyperplane section has Hilbert function $4i+1$, and the cone over the hyperplane section has Hilbert polynomial $\sum_{i=0}^{n}(4i+1) = (2n+1)(n+1)$), so the zero scheme is flat over $\mathbb{P}^1$ [Har77, Theorem III.9.9], and it is the closure of the part over $\mathbb{P}^1 \setminus \operatorname{Supp}\tau$ [Har77, Proposition III.9.7]. Thus $\mathcal{X}_{\lambda,\mathbf{t}} = X'$. By Construction-Proposition 14.3(2), X' has only rational double points and Y is the minimal resolution of X'. Since $\beta : Y \to S$ is a composition of blow-ups, the minimal model of Y is S. Thus $\mathbf{t} \in T_\lambda^{\mathrm{geo}}$, $S_{\mathbf{t}} \cong S$, and $X_{\mathbf{t}} \cong X$.

We prove that T_λ^{geo} is constructible. By [Mat89, Theorem 24.1] and noetherian induction on T_λ, there exists a finite stratification $T_\lambda = \bigsqcup_i T_i$ into integral locally closed subschemes such that

$$\mathcal{X}_i := \mathcal{X}_\lambda \times_{T_\lambda} T_i \to T_i$$

is flat. Let $T_i^{\mathrm{rdp}} \subset T_i$ be the subset of points whose fibers are normal with only rational double points. The subset T_i^{rdp} is open in T_i, for the following reason: rational singularities are

open in a flat family [Elk78, Théorème 4], a flat deformation of a hypersurface singularity is a hypersurface singularity (the dimension of the Zariski tangent space of the fiber at a point is upper semicontinuous on $\mathcal{X}_i$ [Cat13, p. 184]), a rational singularity with Cartier canonical divisor is canonical [KM98, Corollary 5.24], so Du Val [KM98, Theorem 4.20], and $\mathcal{X}_i \to T_i$ is proper. Thus, if T_i^{rdp} is non-empty, then it is irreducible, so connected in the analytic topology, and any two points of T_i^{rdp} are joined by a chain of holomorphic discs in T_i^{rdp}, since any two points of T_i^{rdp} belong to an irreducible curve $\Gamma \subset T_i^{\mathrm{rdp}}$, and finitely many coordinate discs join any two points of the normalization of Γ (cf. [Cat13, p. 186]).

For $\mathbf{t} \in T_i^{\mathrm{rdp}}$ and $m \geq 0$, the number $h^0(\mathcal{X}_{\lambda,\mathbf{t}}, \omega^{\otimes m})$ is the m-th plurigenus of the minimal resolution, because the resolution is crepant. Near any point of a holomorphic disc in T_i^{rdp}, after shrinking the disc and passing to a finite cover, the family of fibers has a simultaneous resolution [Cat13, Theorem 3.6] (we use that theorem at each singular point of the fiber over that point, which is a rational double point, and the local resolutions glue with the family, since they are isomorphisms elsewhere), and plurigenera are invariant in a smooth family by Iitaka's theorem (the N.B. after [BHPV04, Chapter VI, Theorem 8.1]). Thus $h^0(\mathcal{X}_{\lambda,\mathbf{t}}, \omega^{\otimes m})$ is locally constant on the disc, so it is constant on the disc, and so on T_i^{rdp} for any $m \geq 0$. A smooth projective surface has a numerical Godeaux surface as minimal model if and only if the plurigenera are $P_0 = 1$, $P_1 = 0$, and $P_m = 1 + m(m-1)/2$ for $m \geq 2$, since quadratic growth implies general type, and Riemann–Roch on the minimal model implies $K^2 = \chi = 1$, $p_g = 0$, and $h^1(\mathcal{O}) = 0$. Since $h^0(\mathcal{X}_{\lambda,\mathbf{t}}, \mathcal{O})$ is the number of connected components of the normal surface $\mathcal{X}_{\lambda,\mathbf{t}}$, the condition $P_0 = 1$ includes connectedness. Therefore $T_\lambda^{\mathrm{geo}} \cap T_i$ is empty or equal to T_i^{rdp}, and T_λ^{geo} is constructible.

We construct the family of canonical models. Let $T_i^{\mathrm{geo}} := T_\lambda^{\mathrm{geo}} \cap T_i$ be non-empty, and let $\pi : \mathcal{X}_i|_{T_i^{\mathrm{geo}}} \to T_i^{\mathrm{geo}}$ be the projection. The fibers of π are Gorenstein, so ω_π is a line bundle compatible with base change, by relative duality. Since

$$h^0(\mathcal{X}_{\lambda,\mathbf{t}}, \omega^{\otimes m}) = h^0(S_\mathbf{t}, mK_{S_\mathbf{t}})$$

is constant and T_i^{geo} is integral, [Har77, Corollary III.12.9] implies that $\pi_*\omega_\pi^{\otimes m}$ is locally free and compatible with base change for any $m \geq 0$. By Theorem 2.10(1), the fiber algebras are generated in degrees at most 5, so $\mathcal{R} := \bigoplus_{m\geq 0} \pi_*\omega_\pi^{\otimes m}$ is generated in degrees at most 5, by Nakayama's lemma in each degree, since the graded pieces of $\mathcal{R}$ are locally free and compatible with base change. Then $\operatorname{Proj}\mathcal{R} \to T_i^{\mathrm{geo}}$ is a flat family with fibers the canonical models

$$X_\mathbf{t} = \operatorname{Proj} \bigoplus_{m\geq 0} H^0(S_\mathbf{t}, mK_{S_\mathbf{t}}).$$

By Lemma 4.5(1), $\operatorname{Proj}\mathcal{R}$ induces a morphism $\gamma_i : T_i^{\mathrm{geo}} \to \mathcal{M}$ with $\gamma_i(\mathbf{t}) = [X_\mathbf{t}]$.

We conclude the proof. By Theorem 2.7(1), $\mathcal{M}$ is a quasi-projective scheme. The stratum T_i^{geo} is a reduced scheme of finite type with $\dim T_i^{\mathrm{geo}} \leq 7$, so $\dim \overline{\gamma_i(T_i^{\mathrm{geo}})} \leq 7$ by [Har77, Exercise II.3.22(c)]. Since any surface of $\mathcal{L}_3$ occurs, $\mathcal{L}_3$ is contained in the union of the images $\gamma_i(T_i^{\mathrm{geo}})$ over the finitely many strata and both values of λ. Thus $\dim \overline{\mathcal{L}_3} \leq 7$. □

15. Proof of the main theorem

In this section, we prove Theorem 1.5.

Lemma 15.1. *The function sending $[X] \in \mathcal{M}$ to the isomorphism class of $\pi_1(S)$, where S is the minimal resolution of X, is locally constant on the analytic space underlying $\mathcal{M}$, and so are the function sending $[X]$ to the isomorphism class of the torsion subgroup of $\operatorname{Pic}(S)$ and the function sending $[X]$ to the deformation class of S. In particular, each of the functions is constant on each irreducible component of $\mathcal{M}$, irreducible components with different values are disjoint, and for any value the corresponding locus is open and closed in $\mathcal{M}$ in the Zariski topology.*

Proof. The proof is long, so we divide it into several steps.

Fix a point $[X] \in \mathcal{M}$ and let S be the minimal resolution of X. Let $\psi : \mathrm{Def}(S) \to \mathrm{Def}(X)$ be the morphism of Theorem 2.7, and let $f : \mathrm{Def}(S) \to (\mathcal{M}, [X])$ be the composition of ψ with the quotient by $\mathrm{Aut}(X)$. If K_S is ample, then $S = X$ and ψ is the identity. By Theorem 2.7(2)(3), f is a finite surjective morphism of germs. Since a finite morphism of germs has a finite representative, we take a finite surjective representative $f : \mathrm{Def}(S) \to W$, where W is an open neighborhood of $[X]$ in $\mathcal{M}$ and $f^{-1}([X]) = \{0\}$. Let $\Omega \subset \mathrm{Def}(S)$ be a small connected open neighborhood of 0, and we denote by S_a the fiber over $a \in \Omega$ of the Kuranishi family of S.

Step 1. In this step we show that the fibers over Ω of the Kuranishi family of S are deformation equivalent to S.

The points a and 0 are joined by a chain of irreducible curves in the reduction of Ω [Cat13, p. 186]. The pull-back of the Kuranishi family to the normalization of any such curve is a proper holomorphic submersion with connected fibers over a smooth connected curve, and finitely many coordinate discs join any two points of the normalization. Thus S_a is deformation equivalent to S in the sense of [Cat13, Corollary 1.6]. By Ehresmann's theorem [Voi02, Theorem 9.3], S_a and S are diffeomorphic. In particular,

$$\pi_1(S_a) \cong \pi_1(S) \quad \text{and} \quad H^2(S_a, \mathbb{Z}) \cong H^2(S, \mathbb{Z}).$$

Step 2. In this step we show that S_a is a numerical Godeaux surface whose point in $\mathcal{M}$ is $f(a)$, and that the torsion subgroups of $\mathrm{Pic}(S_a)$ and $\mathrm{Pic}(S)$ are isomorphic.

By **Step 1** and [BHPV04, Chapter VI, Theorem 8.1], S_a is a minimal surface of general type. In any family over a disc of **Step 1**, the canonical bundles of the fibers are the restrictions of the relative canonical bundle, so $K_{S_a}^2 = K_S^2 = 1$. By upper semicontinuity on the reduction of Ω [BHPV04, Chapter I, Theorem 8.5(ii)], possibly shrinking Ω, we may assume that

$$h^1(S_a, \mathcal{O}_{S_a}) = h^2(S_a, \mathcal{O}_{S_a}) = 0.$$

Thus $p_g(S_a) = q(S_a) = 0$ and $\chi(\mathcal{O}_{S_a}) = 1$.

For $a \in \Omega$ with $t = \psi(a)$, the surface S_a is the minimal resolution of the fiber X_t of the Kuranishi family of X [Cat13, Theorem 3.6] and [BW74], see [Cat13, Theorem 3.8], and $f(a) = [X_t]$ [Cat13, Remark 3.7]. Since $f(a)$ is a point of $\mathcal{M}$, the space X_t is projective, and so is the minimal resolution S_a. Thus S_a is a numerical Godeaux surface with canonical model X_t. By **Step 1** and Lemma 2.3(1) for S and for S_a, we have

$$\mathrm{Pic}(S_a) \cong H^2(S_a, \mathbb{Z}) \cong H^2(S, \mathbb{Z}) \cong \mathrm{Pic}(S).$$

Thus the torsion subgroups of $\mathrm{Pic}(S_a)$ and $\mathrm{Pic}(S)$ are isomorphic. Therefore, for each of the three functions, the value at $f(a)$ is equal to the value at $[X]$.

Step 3. In this step we prove that the three functions are constant on a neighborhood of $[X]$ in $\mathcal{M}$.

Since f is finite and $f^{-1}([X]) = \{0\}$, the set $f(\mathrm{Def}(S)\backslash\Omega)$ is closed in W and does not contain $[X]$. Thus

$$W' := W\backslash f(\mathrm{Def}(S)\backslash\Omega)$$

is an open neighborhood of $[X]$. Since f is surjective, any point of W' is $f(a)$ for some $a \in \mathrm{Def}(S)$, and then $a \in \Omega$. By **Step 2**, the three functions are constant on W'. Thus the three functions are locally constant on the analytic space underlying $\mathcal{M}$.

Step 4. In this step we conclude the proof.

An irreducible variety is connected in the analytic topology [Sha13, Chapter 7, Section 2]. By **Step 3**, each of the three functions is constant on each irreducible component of $\mathcal{M}$. If irreducible components with different values had a common point x, then the corresponding function would not be constant on any neighborhood of x. Thus irreducible components with different values are disjoint. By Theorem 2.7(1), $\mathcal{M}$ is quasi-projective, so $\mathcal{M}$ has finitely many irreducible components. Thus, for any value, the corresponding locus and the complement of the

locus are finite unions of irreducible components of $\mathcal{M}$. Therefore, the locus is open and closed in $\mathcal{M}$ in the Zariski topology. □

Construction-Proposition 15.2. Let

$$\mathcal{E} := \overline{\mathcal{L}_3} \cup \overline{\mathcal{M}_{\mathrm{ev}}} \cup \overline{\mathcal{M}_{\mathrm{ne}} \backslash \mathcal{B}_{\mathrm{SS}}} \cup \overline{\mathcal{M}_{(2,1,1)}} \cup \overline{\mathcal{M}_{(2,2)}} \cup \overline{\mathcal{M}_{(3,1)}} \cup \overline{\mathcal{M}_{(4)}}.$$

Then $\dim \mathcal{E} \leq 7$ and

$$\mathcal{M}_{\mathrm{tf}} \backslash \mathcal{B}_{\mathrm{SS}} \subset \mathcal{E}.$$

Proof. By Definition 2.6, the closures in the definition of $\mathcal{E}$ are Zariski closures in $\mathcal{M}$, so $\mathcal{E}$ is a finite union of Zariski closed subsets of $\mathcal{M}$, and $\mathcal{E}$ is Zariski closed. By Theorems 14.19, 6.9, 7.10, 9.1, 10.1, 11.1, and 12.1, the dimension of each of the seven closures is at most 7. Thus $\dim \mathcal{E} \leq 7$.

Let $[X] \in \mathcal{M}_{\mathrm{tf}} \backslash \mathcal{B}_{\mathrm{SS}}$, let S be the minimal resolution of X, and let g be the genus of the bicanonical pencil of S. By Construction-Proposition 3.1(3), $g \in \{3, 4\}$. If $g = 3$, then $[X] \in \mathcal{L}_3$. If $g = 4$, then Construction-Proposition 3.3(3) implies that $[X] \in \mathcal{M}_{\mathbf{p}}$, where $\mathbf{p}$ is one of the partitions $(1,1,1,1)$, $(2,1,1)$, $(2,2)$, $(3,1)$, and (4). If $\mathbf{p} \neq (1,1,1,1)$, then $[X] \in \overline{\mathcal{M}_{\mathbf{p}}} \subset \mathcal{E}$.

Assume that $\mathbf{p} = (1,1,1,1)$. By Construction-Proposition 5.1, S has a standard pencil matrix A. If A has an event, then $[X] \in \mathcal{M}_{\mathrm{ev}}$. If A is event-free, then $[X] \in \mathcal{M}_{\mathrm{ne}}$, and so $[X] \in \mathcal{M}_{\mathrm{ne}} \backslash \mathcal{B}_{\mathrm{SS}}$. Thus $[X] \in \mathcal{E}$ in any case. □

Theorem 15.3. *We have $\mathcal{M}_{\mathrm{tf}} = \mathcal{B}_{\mathrm{SS}}$. In particular, $\mathcal{M}_{\mathrm{tf}}$ is non-empty, irreducible, of dimension 8, and a connected component of $\mathcal{M}$. Any torsion-free numerical Godeaux surface is deformation equivalent to the Barlow surface, and in particular simply connected.*

Proof. By Lemma 15.1, $\mathcal{M}_{\mathrm{tf}}$ is open and closed in $\mathcal{M}$ in the Zariski topology.

Let S be a fiber of $\mathcal{S}$. By Theorem 1.4, $\pi_1(S)$ is trivial, so $H^2(S, \mathbb{Z})$ is torsion-free by the universal coefficient theorem. By Lemma 2.3(1), $\mathrm{Pic}(S)$ is torsion-free. Thus the point in $\mathcal{M}$ of the canonical model of S is contained in $\mathcal{M}_{\mathrm{tf}}$. Since $\mathcal{M}_{\mathrm{tf}}$ is Zariski closed, Definition 2.6 implies that

$$\mathcal{B}_{\mathrm{SS}} \subset \mathcal{M}_{\mathrm{tf}}.$$

With $\mathcal{D}$ of Construction-Proposition 7.7, we have a morphism $\mathcal{D} \to \mathcal{M}$, and $\mathcal{B}_{\mathrm{SS}}$ is the Zariski closure of the image of the morphism. By [SS20, Corollary 5.2], $\mathcal{D}$ is irreducible of dimension 11, and the family over $\mathcal{D}$ modulo an action of $(\mathbb{C}^*)^3$ is $\mathcal{S}$, whose base is of dimension 8. Thus the points of an orbit of the action define one fiber of $\mathcal{S}$, so the morphism is constant on the orbit, and a general orbit is of dimension 3. Therefore, $\mathcal{B}_{\mathrm{SS}}$ is non-empty and irreducible, and $\dim \mathcal{B}_{\mathrm{SS}} \leq 11 - 3 = 8$, since the closure of the image of an irreducible scheme is irreducible, and, for a general point x of $\mathcal{D}$, the fiber of the morphism through x is of dimension at most $11 - \dim \mathcal{B}_{\mathrm{SS}}$ at x [Har77, Exercise II.3.22(c)] and contains the orbit of x, which is of dimension 3.

The set $\mathcal{M}_{\mathrm{tf}} \backslash \mathcal{B}_{\mathrm{SS}}$ is analytically open in $\mathcal{M}$, since it is Zariski open. By Construction-Proposition 15.2, $\mathcal{M}_{\mathrm{tf}} \backslash \mathcal{B}_{\mathrm{SS}}$ is contained in the Zariski closed subset $\mathcal{E}$ of $\mathcal{M}$ and $\dim \mathcal{E} \leq 7$. By Lemma 2.8, $\mathcal{M}_{\mathrm{tf}} \backslash \mathcal{B}_{\mathrm{SS}}$ is empty. Thus

$$\mathcal{M}_{\mathrm{tf}} = \mathcal{B}_{\mathrm{SS}}.$$

In particular, $\mathcal{M}_{\mathrm{tf}}$ is non-empty and irreducible, and $\dim \mathcal{M}_{\mathrm{tf}} \leq 8$. Since $\mathcal{M}_{\mathrm{tf}}$ is non-empty and Zariski open, hence analytically open, in $\mathcal{M}$, Lemma 2.8 implies that $\dim \mathcal{M}_{\mathrm{tf}} \geq 8$. Thus $\dim \mathcal{M}_{\mathrm{tf}} = 8$. Since $\mathcal{M}_{\mathrm{tf}}$ is irreducible, open, and closed in $\mathcal{M}$, $\mathcal{M}_{\mathrm{tf}}$ is a connected component and an irreducible component of $\mathcal{M}$ (an irreducible space is connected, and an irreducible component of $\mathcal{M}$ that intersects an open and closed subset is contained in it).

Let S be a torsion-free numerical Godeaux surface with canonical model X. Then $[X] \in \mathcal{M}_{\mathrm{tf}}$. Since $\mathcal{M}_{\mathrm{tf}}$ is an irreducible component of $\mathcal{M}$, Lemma 15.1 implies that the deformation class of the minimal resolution is constant on $\mathcal{M}_{\mathrm{tf}}$. A general member of the family $\mathcal{S}$ is deformation equivalent to the Barlow surface by Theorem 1.4, and its canonical model defines a point of $\mathcal{B}_{\mathrm{SS}} = \mathcal{M}_{\mathrm{tf}}$. Thus S is deformation equivalent to the Barlow surface. The Barlow surface is

simply connected [Bar85, Theorem 2.5 and the subsequent Corollary], and the fundamental group is a deformation invariant by Ehresmann's theorem [Voi02, Theorem 9.3]. Thus S is simply connected. □

We classify the points of $\mathcal{M}$ by the torsion subgroup of the Picard group of the minimal resolution, whose possible values are known, and then we determine the fundamental group.

Definition-Proposition 15.4. For $m \in \{1, \dots, 5\}$, let $\mathcal{T}_m \subset \mathcal{M}$ be the locus of points $[X]$ such that the torsion subgroup of $\mathrm{Pic}(S)$, where S is the minimal resolution of X, is isomorphic to $\mathbb{Z}/m$, so that $\mathcal{T}_1 = \mathcal{M}_{\mathrm{tf}}$. Then $\mathcal{M} = \bigsqcup_{m=1}^{5} \mathcal{T}_m$, each $\mathcal{T}_m$ is non-empty and irreducible, and $\mathcal{T}_m = \mathcal{M}_m$ for any m.

Proof. **Step 1.** In this step we prove that $\mathcal{M} = \bigsqcup_{m=1}^{5} \mathcal{T}_m$.

Let S be a numerical Godeaux surface. The torsion subgroup of $\mathrm{Pic}(S)$ is cyclic of order at most 5 [SS20, Lemma 1.6]. Thus any point of $\mathcal{M}$ belongs to $\mathcal{T}_m$ for exactly one $m \in \{1, \dots, 5\}$.

Step 2. In this step we prove that each $\mathcal{T}_m$ is non-empty and irreducible.

By Theorem 15.3, $\mathcal{T}_1 = \mathcal{M}_{\mathrm{tf}}$ is non-empty and irreducible. By Theorem 1.3, $\mathcal{T}_2$ is irreducible and $\dim \mathcal{T}_2 = 8$, so $\mathcal{T}_2$ is non-empty.

Let $m \in \{3, 4, 5\}$. By Theorem 1.2, $\mathcal{T}_m$ is non-empty and irreducible. More precisely, by Lemma 4.5(1), there exists a morphism to $\mathcal{M}$ with image $\mathcal{T}_m$ from an irreducible reduced scheme of finite type carrying a flat family of canonical models (the family is the quotient by the free action of $\mathbb{Z}/m$ of the family of covers Y over the open locus where Y has the general Hilbert function, only rational double points and a free action of $\mathbb{Z}/m$, flat since $\mathcal{O}_X$ is a direct summand of the push-forward of $\mathcal{O}_Y$, and the image of the morphism is $\mathcal{T}_m$ by the theorems cited below and [SS20, Lemma 1.6], since by [Rei78, Proposition 0.4], the canonical rings of the covers of the points of $\mathcal{T}_m$, which include the covers over a general parameter, have equal Hilbert functions). Here the general value of the Hilbert function is an open condition (the locus of the general value is stable under generization by [Har77, Exercise II.5.8(a)], and for any point of the locus, the locus contains a non-empty open subset of the closure of the point by generic freeness [Mat89, Theorem 24.1], so the locus is open by [Mat89, Theorem 24.2]), over which the family of covers is flat since the graded pieces are locally free [Har77, Exercise II.5.8(c)]. Rational double points and the freeness of the group action are open conditions in a flat proper family, for rational double points by [Gro66, Théorème 12.2.4(iv)], [Elk78, Théorème 4], and [Cat13, p. 184] after a relative Veronese embedding over an affine cover of the base, and for the freeness since the fixed loci of the non-trivial elements of $\mathbb{Z}/m$ are closed.

For $m = 5$, the scheme is a Zariski open subset of the projective space of the quintics spanned by the monomials (1) of [Rei78, Section 1], see [Rei78, Theorem 1.1] and the paragraph after it.

For $m = 4$, the scheme is a Zariski open subset of the affine space of the pairs (q_0, q_2) of [Rei78, Section 2], see [Rei78, Theorem 2.2] and [CU18, Section 4.1].

For $m = 3$, the scheme is a Zariski open subset of the affine space whose coordinates are the nine free parameters of the relations of [Rei78, Section 3], see [Rei78, Theorem 3.9] and [CU18, Proposition 2.2].

Step 3. In this step we prove that $\mathcal{T}_m = \mathcal{M}_m$ for any m.

By Theorem 15.3, any numerical Godeaux surface S with canonical model X such that $[X] \in \mathcal{T}_1$ is simply connected, so $\mathcal{T}_1 \subset \mathcal{M}_1$. By Theorem 1.3, any numerical Godeaux surface S with canonical model X such that $[X] \in \mathcal{T}_2$ satisfies $\pi_1(S) \cong \mathbb{Z}/2$, so $\mathcal{T}_2 \subset \mathcal{M}_2$.

Let $m \in \{3, 4, 5\}$. By Lemma 15.1, $\mathcal{T}_m$ is open and closed in $\mathcal{M}$. Since $\mathcal{T}_m$ is irreducible, $\mathcal{T}_m$ is an irreducible component of $\mathcal{M}$ (an irreducible component of $\mathcal{M}$ that intersects an open and closed subset is contained in it), and $\pi_1(S)$ is constant on $\mathcal{T}_m$. For $m = 3$, we have $\pi_1(S) \cong \mathbb{Z}/3$ for the surfaces of [CU18, Theorem 1.1]. For $m = 5, 4$, we have $\pi_1(S) \cong \mathbb{Z}/m$ for the surfaces of [Rei78, Theorems 1.1 and 2.2], respectively, by [CU18, Introduction] and [DR26, Introduction]. Thus $\pi_1(S) \cong \mathbb{Z}/m$ for any point $[X] \in \mathcal{T}_m$, so $\mathcal{T}_m \subset \mathcal{M}_m$.

Hence $\mathcal{T}_m \subset \mathcal{M}_m$ for any $m \in \{1,\dots,5\}$. Let $[X] \in \mathcal{M}_m$. By **Step 1**, there exists $m' \in \{1,\dots,5\}$ such that $[X] \in \mathcal{T}_{m'}$, so $[X] \in \mathcal{M}_{m'}$ and

$$\mathbb{Z}/m \cong \pi_1(S) \cong \mathbb{Z}/m'.$$

Thus $m' = m$ and $[X] \in \mathcal{T}_m$. Therefore, $\mathcal{T}_m = \mathcal{M}_m$. □

Proof of Theorem 1.5. By Definition-Proposition 15.4, $\mathcal{M} = \bigsqcup_{m=1}^{5} \mathcal{M}_m$, each $\mathcal{M}_m$ is non-empty and irreducible, and $\mathcal{M}_m$ is the locus where the torsion subgroup of $\mathrm{Pic}(S)$ is isomorphic to $\mathbb{Z}/m$.

By Lemma 15.1, $\mathcal{M}_m$ is open and closed in $\mathcal{M}$. Since $\mathcal{M}_m$ is irreducible, $\mathcal{M}_m$ is a connected component and an irreducible component of $\mathcal{M}$ (an irreducible space is connected, and an irreducible component of $\mathcal{M}$ that intersects an open and closed subset is contained in it).

We have $\dim \mathcal{M}_1 = 8$ by Theorem 15.3, and $\dim \mathcal{M}_2 = 8$ by Theorem 1.3. For $m \in \{3,4,5\}$, we have $\dim \mathcal{M}_m = 8$ by [DR26, Introduction].

Let S and S' be numerical Godeaux surfaces with canonical models X and X' such that $[X] \in \mathcal{M}_m$ and $[X'] \in \mathcal{M}_{m'}$, so that $|\pi_1(S)| = m$ and $|\pi_1(S')| = m'$. If $m = m'$, then S and S' are deformation equivalent, since by Lemma 15.1 the deformation class is constant on the irreducible component $\mathcal{M}_m$. If $m \neq m'$, then $\pi_1(S) \ncong \pi_1(S')$, so S and S' are not deformation equivalent, since the fundamental group is a deformation invariant by Ehresmann's theorem [Voi02, Theorem 9.3].

In particular, there exist exactly five deformation classes of numerical Godeaux surfaces, and two numerical Godeaux surfaces are deformation equivalent if and only if their fundamental groups have equal order. □

Appendix A. Rank computations

Subsection A.1 contains the tables for the families of Section 6, and Subsections A.2, A.3, A.4, A.5, and A.6 contain the tables for the types $(2,1,1)$, $(2,2)$, $(2,2)$ with exceptional presentations, $(3,1)$, and (4), respectively. In the appendix, the rows and columns of matrices and the slots are numbered from 0, while the index k of the relations r_k and of the entries A_{jk} runs through $1,\dots,6$, as in Set-up 4.1. The row i of O and of B_3 corresponds to r_{i+1}, that is, $(AO)_{jm} = \sum_{i=0}^{5} A_{j,i+1} O_{im}$.

A.1. **Pencil matrices with events.** Each table of this subsection is a pivot table, over the coordinate ring of a family of Construction-Proposition 6.8, of the matrix T of Construction-Proposition 6.7, which also fixes the row and column order of T and the meaning of a triple (r,c,β). The blocks $B_1,\dots,B_4$ are those of Table 1, and the slots of C and O are those of Construction-Proposition 5.3. To check a table, one verifies that, taking the triples in the listed order, the entry of the current matrix in row r and column c equals β, and that β is a unit of the ring. Such a check is a finite exact computation over the coordinate ring of the family, which a computer algebra system can carry out from the definition of T.

TABLE 22. A pivot table of T with 78 pivots over the coordinate ring of the triangle family of Construction-Proposition 6.3, the polynomial ring in g, h localized at gh.

$(12, 60, 1)$, $(36, 64, 1)$, $(40, 65, 1)$, $(125, 74, 1)$, $(129, 75, 1)$, $(178, 69, 1)$, $(206, 72, 1)$, $(210, 73, 1)$, $(279, 80, 1)$, $(287, 83, g)$, $(308, 18, 1)$, $(315, 71, g)$, $(395, 8, 1)$, $(522, 28, 1)$, $(523, 29, 1)$, $(16, 61, 1)$, $(73, 66, gh)$, $(170, 68, 1)$, $(174, 76, h)$, $(267, 78, 1)$, $(271, 79, 1)$, $(283, 81, 1)$, $(316, 0, 1)$, $(320, 33, 1)$, $(328, 3, 1)$, $(331, 14, h)$, $(332, 6, 1)$, $(337, 62, 1)$, $(338, 63, 1)$, $(347, 11, 1)$, $(360, 24, h)$, $(371, 44, 1)$, $(399, 32, 1)$, $(404, 21, 1)$, $(369, 42, 1)$, $(423, 38, g)$, $(451, 77, h)$, $(484, 70, 1)$, $(485, 1, 1)$, $(486, 2, 1)$, $(493, 4, 1)$, $(494, 5, 1)$, $(496, 51, 1)$, $(521, 27, 1)$, $(524, 12, 1)$, $(532, 39, -h)$, $(456, 54, h)$, $(551, 59, g)$, $(546, 53, 1)$, $(304, 15, 1)$, $(309, 19, 1)$, $(310, 20, 1)$, $(317, 13, h)$, $(319, 7, g)$, $(335, 56, g)$, $(370, 43, 1)$, $(373, 25, h)$, $(380, 36, g)$, $(394, 41, h/g)$, $(498, 58, h)$, $(422, 40, h)$, $(427, 23, h)$, $(488, 48, 1)$, $(536, 30, 1)$, $(381, 52, g)$, $(434, 67, g)$, $(549, 55, 1)$, $(334, 10, g)$, $(538, 50, 2)$, $(490, 47, 1)$, $(307, 26, g)$, $(450, 37, 2g)$, $(306, 17, 1)$, $(318, 35, g)$, $(421, 57, 2gh)$, $(460, 45, 2)$, $(305, 16, 1)$, $(333, 9, g)$

TABLE 23. A pivot table of T with 78 pivots over the coordinate ring of the path family of Construction-Proposition 6.3, the polynomial ring in g, h localized at gh.

$(36, 70, g)$, $(40, 71, g)$, $(121, 66, 1)$, $(129, 75, 1)$, $(206, 72, 1)$, $(214, 77, 1)$, $(267, 78, h)$, $(271, 79, h)$, $(287, 81, h)$, $(302, 61, g)$, $(335, 8, h)$, $(494, 7, g)$, $(520, 18, 1)$, $(523, 29, 1)$, $(97, 62, 1)$, $(101, 63, 1)$, $(125, 67, 1)$, $(170, 68, 1)$, $(174, 69, 1)$, $(210, 73, 1)$, $(279, 80, h)$, $(300, 60, g)$, $(308, 21, g)$, $(311, 23, g)$, $(319, 14, 1)$, $(323, 41, g)$, $(371, 44, 1)$, $(444, 51, 1)$, $(447, 50, 1)$, $(423, 53, 1)$, $(456, 54, 1)$, $(492, 6, g)$, $(493, 64, 1)$, $(503, 56, h)$, $(514, 17, 1)$, $(521, 19, 1)$, $(544, 82, 1)$, $(320, 39, g)$, $(283, 65, h)$, $(483, 83, 1)$, $(545, 52, h)$, $(546, 2, g)$, $(307, 11, 1)$, $(321, 40, g)$, $(331, 5, h)$, $(360, 24, 1)$, $(368, 33, 1)$, $(427, 35, h)$, $(445, 48, 1)$, $(446, 49, 1)$, $(505, 27, 1)$, $(507, 74, 1)$, $(524, 36, 1)$, $(550, 55, h)$, $(309, 3, 1)$, $(318, 13, 1)$, $(322, 9, 1)$, $(330, 4, h)$, $(363, 26, 1)$, $(482, 1, -g)$, $(490, 28, 1)$, $(383, 20, h)$, $(306, 10, 1)$, $(333, 76, g/h)$, $(370, 43, 1)$, $(509, 31, 1)$, $(310, 22, g)$, $(341, 12, 1)$, $(496, 25, 1)$, $(382, 47, 2)$, $(549, 57, 1/h)$, $(426, 34, h)$, $(525, 42, 1)$, $(381, 46, 2)$, $(425, 32, 2/h)$, $(316, 0, 1)$, $(488, 15, 2)$, $(369, 30, 1)$

TABLE 24. A pivot table of T with 79 pivots over the coordinate ring of the family of Construction-Proposition 6.4, the polynomial ring in p, q, w localized at $pqw(p-1)(q-1)$.

$(36, 70, w)$, $(40, 71, w)$, $(121, 66, 1)$, $(129, 75, 1)$, $(206, 72, 1)$, $(214, 77, 1)$, $(520, 18, 1)$, $(523, 29, 1)$, $(97, 62, 1)$, $(101, 63, 1)$, $(125, 67, 1)$, $(170, 68, 1)$, $(174, 69, 1)$, $(210, 73, 1)$, $(255, 78, -1)$, $(263, 81, -1)$, $(279, 82, 1)$, $(283, 83, 1)$, $(300, 60, w)$, $(308, 21, w)$, $(371, 44, 1)$, $(395, 11, 1)$, $(420, 36, 1)$, $(492, 6, w)$, $(494, 8, w)$, $(514, 17, 1)$, $(516, 51, 1)$, $(521, 19, 1)$, $(259, 79, -1)$, $(301, 61, w)$, $(320, 39, w)$, $(368, 33, 1)$, $(375, 26, 1)$, $(423, 41, 1)$, $(493, 7, w)$, $(495, 64, p^3)$, $(335, 76, pw)$, $(500, 24, 1)$, $(505, 74, 1)$, $(513, 16, 1)$, $(543, 56, -1)$, $(548, 57, 1)$, $(550, 59, 1)$, $(311, 23, pw)$, $(321, 40, w)$, $(328, 48, w)$, $(332, 9, 1)$, $(340, 0, 1)$, $(438, 14, 1)$, $(511, 32, -1)$, $(519, 35, 1)$, $(522, 20, 1)$, $(525, 25, 1)$, $(540, 65, -1/p)$, $(517, 52, 1)$, $(541, 3, 2/w)$, $(316, 30, w)$, $(380, 27, 1)$, $(542, 53, -1)$, $(526, 43, 1)$, $(549, 58, 1)$, $(309, 22, w)$, $(322, 12, p)$, $(310, 4, p)$, $(534, 47, 1)$, $(333, 10, 1)$, $(370, 38, 1)$, $(421, 37, 1)$, $(481, 1, -w)$, $(482, 2, -w)$, $(398, 31, 1)$, $(422, 28, 1)$, $(425, 42, 1)$, $(462, 50, 2)$, $(529, 15, -1)$, $(329, 49, w)$, $(381, 46, 2)$, $(499, 55, 2pq/((p-1)(q-1)))$, $(518, 13, 2/(pw))$

TABLE 25. A pivot table of T with 79 pivots over the coordinate ring of family (I) of Table 3, the polynomial ring in p, q, w localized at $pqw(1-pq)(1-w)$.

$(36, 64, 1)$, $(44, 71, 1)$, $(97, 62, 1)$, $(170, 68, p)$, $(495, 8, 1)$, $(12, 60, 1)$, $(16, 61, 1)$, $(40, 65, 1)$, $(101, 63, 1)$, $(105, 75, -q)$, $(121, 66, 1)$, $(125, 67, 1)$, $(174, 69, p)$, $(178, 77, -1)$, $(206, 72, p)$, $(210, 73, p)$, $(255, 78, -w)$, $(263, 81, -1)$, $(279, 82, w)$, $(283, 83, w)$, $(323, 41, 1)$, $(372, 18, w)$, $(492, 3, 1)$, $(494, 5, 1)$, $(512, 15, 1)$, $(514, 17, 1)$, $(515, 29, -q)$, $(540, 51, -w)$, $(259, 79, -w)$, $(311, 23, 1)$, $(316, 0, p)$, $(320, 33, 1)$, $(375, 26, 1)$, $(420, 36, w)$, $(459, 56, 1)$, $(486, 2, 1)$, $(493, 4, 1)$, $(513, 16, 1)$, $(522, 20, 1)$, $(541, 52, -w)$, $(542, 80, 1)$, $(308, 70, p^2)$, $(319, 14, 1)$, $(380, 27, w)$, $(427, 44, 1)$, $(532, 45, w)$, $(508, 30, w/(pq-1))$, $(488, 74, w^2)$, $(547, 59, -1)$, $(548, 57, w)$, $(549, 58, w)$, $(329, 12, 1)$, $(332, 6, 2p)$, $(485, 1, 1)$, $(309, 19, 1)$, $(363, 53, q)$, $(422, 38, w)$, $(491, 11, q/(w-1))$, $(373, 24, 1)$, $(340, 21, pw)$, $(421, 37, w)$, $(435, 47, 1)$, $(489, 31, -p)$, $(490, 28, q)$, $(538, 32, p)$, $(537, 42, -1)$, $(533, 46, w)$, $(509, 39, -1)$, $(322, 35, 1)$, $(362, 25, 1)$, $(530, 7, -p)$, $(543, 43, q/p)$, $(499, 50, 2)$, $(321, 34, 2)$, $(539, 40, q/w)$, $(310, 22, 1)$, $(318, 13, 1)$, $(426, 49, 2q/(w-1))$, $(374, 54, 2q/w)$

TABLE 26. A pivot table of T with 79 pivots over the coordinate ring of family (II) of Table 3, the polynomial ring in p, q, w localized at $pqw(1-pqw)(1-pq)(1-w)(1-q)(1-pw)$.

$(36, 64, 1)$, $(44, 71, 1)$, $(97, 62, 1)$, $(267, 78, q)$, $(389, 68, w)$, $(391, 73, 1)$, $(455, 83, p)$, $(495, 8, 1)$, $(544, 51, q)$, $(12, 60, 1)$, $(16, 61, 1)$, $(40, 65, 1)$, $(101, 63, 1)$, $(105, 75, -p)$, $(121, 66, 1)$, $(125, 67, 1)$, $(206, 76, w)$, $(210, 77, w)$, $(271, 79, q)$, $(279, 80, q)$, $(283, 81, q)$, $(323, 41, 1)$, $(328, 3, q)$, $(390, 69, w)$, $(486, 2, 1)$, $(512, 15, 1)$, $(515, 29, -p)$, $(383, 59, p)$, $(522, 20, 1)$, $(308, 18, 1)$, $(311, 23, 1)$, $(320, 33, 1)$, $(363, 26, 1)$, $(456, 48, w)$, $(459, 56, 1)$, $(484, 0, 1)$, $(493, 4, 1)$, $(494, 5, 1)$, $(513, 16, 1)$, $(514, 17, 1)$, $(541, 54, -1)$, $(545, 52, q)$, $(546, 53, q)$, $(380, 21, q)$, $(427, 44, 1)$, $(485, 1, 1)$, $(491, 72, 1/w)$, $(307, 11, p)$, $(500, 12, 1)$, $(521, 19, 1)$, $(549, 55, q)$, $(319, 14, 1)$, $(321, 34, 1)$, $(322, 32, w)$, $(368, 9, w)$, $(501, 13, 1)$, $(508, 30, -w)$, $(421, 39, 1)$, $(422, 40, 1)$, $(516, 36, -w)$, $(531, 35, -1)$, $(374, 25, 1)$, $(537, 49, q)$, $(543, 38, p(pw-1)/(pqw-1))$, $(331, 50, 2)$, $(310, 22, 1)$, $(304, 45, w(pq-1)/(pqw-1))$, $(538, 43, -1)$, $(434, 7, q)$, $(489, 31, -q(w-1)/(pqw-1))$, $(309, 27, -2(pw-1)/(pqw-1))$, $(492, 70, q^2(w-1)^2/(pqw-1)^2)$, $(505, 24, -pqw(w-1)/(pw-1))$, $(317, 6, -2pq(w-1)/(pqw-1))$, $(497, 42, -1/p)$, $(498, 10, 2)$, $(425, 46, -2q(w-1)/(pqw-1))$, $(318, 47, qw(pq-1)/(pqw-1))$, $(306, 58, -2p)$

TABLE 27. A pivot table of T with 78 pivots over the coordinate ring of family (III) of Table 3, the polynomial ring in p, q localized at $pq(p-1)(q+1)$.

$(36, 64, 1)$, $(44, 71, 1)$, $(97, 62, 1)$, $(267, 78, 1)$, $(287, 83, q)$, $(366, 75, 1)$, $(389, 68, 1)$, $(391, 73, 1)$, $(495, 8, 1)$, $(544, 51, 1)$, $(12, 60, 1)$, $(16, 61, 1)$, $(40, 65, 1)$, $(81, 63, q)$, $(121, 66, 1)$, $(125, 67, 1)$, $(206, 76, 1)$, $(210, 77, 1)$, $(271, 79, 1)$, $(279, 80, 1)$, $(283, 81, 1)$, $(323, 41, 1)$, $(328, 3, 1)$, $(372, 18, p)$, $(390, 69, 1)$, $(407, 44, q)$, $(486, 2, 1)$, $(512, 15, 1)$, $(515, 29, -1)$, $(383, 59, 1)$, $(522, 20, 1)$, $(311, 11, q)$, $(316, 36, p)$, $(320, 33, 1)$, $(363, 26, 1)$, $(423, 38, q)$, $(484, 0, 1)$, $(420, 9, p)$, $(493, 4, 1)$, $(494, 5, 1)$, $(496, 74, 1-p)$, $(360, 30, p-1)$, $(507, 17, q)$, $(513, 16, 1)$, $(539, 72, q+1)$, $(541, 52, -p)$, $(546, 53, 1)$, $(548, 54, 1)$, $(550, 56, 1)$, $(304, 24, p)$, $(380, 27, p)$, $(439, 14, q)$, $(331, 50, 1)$, $(343, 23, q+1)$, $(382, 28, q)$, $(447, 32, 2q)$, $(482, 7, p)$, $(502, 40, q+1)$, $(508, 6, p)$, $(549, 55, 1)$, $(422, 37, q)$, $(332, 45, 2)$, $(329, 48, 2)$, $(370, 35, 1)$, $(530, 1, 1)$, $(334, 10, q)$, $(361, 21, 1)$, $(368, 42, 1-p)$, $(426, 43, q+1)$, $(497, 19, 1)$, $(506, 57, 1)$, $(362, 25, 1)$, $(374, 49, q)$, $(421, 39, 1)$, $(489, 31, p)$, $(501, 13, 2)$, $(369, 34, 1)$, $(461, 47, 2/q)$

TABLE 28. A pivot table of T with 78 pivots over the coordinate ring of family (IV) of Table 3, the polynomial ring in p, q localized at $pq(1+p^2q)(1+pq^2)$.

$(36,64,1)$, $(44,71,1)$, $(16,61,1)$, $(311,23,1)$, $(319,32,1)$, $(495,8,1)$, $(12,60,1)$, $(40,65,1)$, $(97,62,1)$, $(121,66,1)$, $(125,67,1)$, $(166,69,1/p)$, $(170,68,p)$, $(174,77,-1)$, $(206,72,p)$, $(210,73,p)$, $(267,78,p)$, $(279,80,p)$, $(323,41,1)$, $(338,75,1/p)$, $(101,63,1)$, $(371,29,1/p)$, $(423,50,1/p)$, $(435,47,1)$, $(486,2,1)$, $(492,3,1)$, $(494,5,1)$, $(514,17,1)$, $(522,20,1)$, $(259,79,-q)$, $(283,81,p)$, $(484,0,1)$, $(485,1,1)$, $(493,4,1)$, $(499,11,1)$, $(512,15,1)$, $(520,18,1)$, $(521,19,1)$, $(544,51,p)$, $(546,53,p)$, $(548,54,p)$, $(320,33,1)$, $(370,35,1)$, $(513,16,1)$, $(527,44,-1/p)$, $(328,12,1)$, $(374,26,1)$, $(488,48,p)$, $(536,30,p)$, $(496,57,1)$, $(498,10,1)$, $(503,76,-1/p^2)$, $(510,31,1/p)$, $(519,38,1/p)$, $(462,59,1)$, $(530,28,2/p)$, $(541,52,-q)$, $(308,70,p^2)$, $(318,14,1)$, $(332,6,p)$, $(404,21,p)$, $(500,24,1)$, $(524,39,p)$, $(525,13,1)$, $(549,55,p)$, $(501,40,-p)$, $(420,36,2q)$, $(460,45,2p)$, $(369,34,1)$, $(361,22,1)$, $(433,25,1)$, $(497,58,1)$, $(333,46,1)$, $(345,37,2p)$, $(393,7,2p)$, $(330,49,1)$, $(502,43,2/p)$, $(538,56,(p^2q+1)/(pq^2+1))$

TABLE 29. A pivot table of T with 77 pivots over the coordinate ring of family (V) of Table 3, the polynomial ring in w localized at w.

$(36,64,1)$, $(44,71,1)$, $(16,61,1)$, $(214,77,1)$, $(275,79,w)$, $(287,81,w)$, $(311,23,1)$, $(319,32,1)$, $(335,8,w)$, $(338,67,w)$, $(367,75,w)$, $(391,69,1)$, $(407,29,1)$, $(447,53,w)$, $(459,50,1)$, $(12,60,1)$, $(40,65,1)$, $(97,62,1)$, $(101,63,1)$, $(121,66,1)$, $(210,76,1)$, $(255,78,1)$, $(271,80,w)$, $(307,11,w)$, $(323,41,1)$, $(331,5,w)$, $(363,17,1)$, $(383,59,w)$, $(389,72,1)$, $(390,68,1)$, $(427,35,w)$, $(453,82,w)$, $(486,2,1)$, $(492,3,1)$, $(494,7,1)$, $(334,56,w)$, $(507,74,w)$, $(514,38,w)$, $(406,83,w)$, $(531,20,w)$, $(318,14,1)$, $(328,12,1)$, $(374,26,1)$, $(551,44,1)$, $(484,0,1)$, $(485,1,1)$, $(493,4,1)$, $(502,52,w)$, $(512,15,1)$, $(520,18,1)$, $(521,19,1)$, $(540,51,1)$, $(310,22,1)$, $(320,6,1)$, $(316,9,1)$, $(322,28,1)$, $(360,24,1)$, $(526,31,3)$, $(544,54,-1)$, $(550,43,2/3)$, $(308,36,1)$, $(332,33,1)$, $(488,30,1)$, $(362,16,1)$, $(422,40,1)$, $(306,10,w)$, $(426,34,w)$, $(530,49,2)$, $(536,42,-1)$, $(524,57,2)$, $(420,39,1)$, $(424,45,2)$, $(304,48,w)$, $(309,55,1)$, $(317,27,2)$, $(329,13,2)$, $(321,37,1)$

TABLE 30. A pivot table of T with 78 pivots over the coordinate ring of family I of Table 4.

$(36,64,1)$, $(44,71,1)$, $(105,75,-w)$, $(495,8,1)$, $(515,29,-w)$, $(4,60,-1)$, $(8,61,-1)$, $(40,65,1)$, $(81,67,-w)$, $(97,62,1)$, $(101,63,1)$, $(77,66,-w)$, $(170,68,1)$, $(206,72,1)$, $(255,78,-1)$, $(279,82,1)$, $(492,3,1)$, $(494,5,1)$, $(514,17,1)$, $(174,69,1)$, $(210,73,1)$, $(259,79,-1)$, $(271,80,-qw)$, $(323,41,1)$, $(375,53,-w)$, $(481,0,-1)$, $(483,2,-1)$, $(493,4,1)$, $(507,20,-w)$, $(512,15,1)$, $(520,18,1)$, $(527,83,-w^2)$, $(214,77,-w/((1-q)(w+v)))$, $(383,59,w)$, $(511,23,-w)$, $(287,81,-w(p-q)/(p(v(p-q)+1)))$, $(540,51,-1)$, $(548,57,1)$, $(319,32,1)$, $(320,6,-1)$, $(506,19,-w)$, $(513,16,1)$, $(541,52,-1)$, $(545,54,-qw)$, $(308,70,1)$, $(322,35,1)$, $(331,47,-1)$, $(368,21,-1)$, $(488,30,-1)$, $(499,11,1)$, $(501,13,1)$, $(525,40,1)$, $(503,56,-w(p-q)/(p(v(p-q)+1)))$, $(551,44,-w/((1-q)(w+v)))$, $(321,7,-1)$, $(496,36,-1)$, $(500,12,1)$, $(524,39,1)$, $(536,74,1)$, $(369,22,-1)$, $(489,27,w)$, $(537,31,1)$, $(332,24,1)$, $(334,50,w/((1-q)(w+v)))$, $(502,26,(p-q)/(p(v(p-q)+1)))$, $(422,38,1)$, $(538,28,w)$, $(309,33,2)$, $(310,34,2)$, $(482,1,-1)$, $(508,45,-1)$, $(509,46,-1)$, $(306,58,-w)$, $(382,25,2w)$, $(362,37,-w)$, $(542,14,-2p(v(p-q)+1)(pv(p-q)+q)/(w(p-q)^2))$, $(307,10,w^3(pv(p-q)+q)^2/(p^2(1-q)^2(w+v)^2(v(p-q)+1)^2))$, $(526,43,-w/((1-q)(w+v)))$

TABLE 31. A pivot table of T with 80 pivots over the coordinate ring of family II of Table 4.

$(0, 64, q(1-q)/(p(p-q)))$, $(8, 61, -1)$, $(16, 71, q(1-q)/(p(p-q)))$, $(105, 75, -w)$,
$(178, 77, -q(pw(p-q)-q+1)/(p(p-q)))$, $(275, 79, -w)$, $(259, 81, q)$, $(335, 8, -w)$, $(515, 29, -w)$,
$(4, 60, -1)$, $(12, 65, q(1-q)/(p(p-q)))$, $(73, 62, -q(p-q)/((1-q)(pw(p-q)-q)))$,
$(81, 63, q(pw(p-q)-q+1)/((1-q)(pw(p-q)-q)))$,
$(101, 67, w(1-q)(pw(p-q)-q)/(q(pw(p-q)-q+1)))$, $(121, 66, 1)$,
$(158, 68, -q(p-q)/((1-q)(pw(p-q)-q)))$, $(166, 69, pw(p-q)/((1-q)(pw(p-q)-q)))$,
$(174, 73, q(1-q)(pw(p-q)-q)(pw(p-q)-q+1)/(p^2w(p-q)^2))$, $(206, 72, 1)$, $(255, 78, -1)$, $(267, 82, q)$,
$(271, 80, -qw)$, $(311, 11, q(pw(p-q)-q+1)/((1-q)(pw(p-q)-q)))$,
$(319, 14, q(pw(p-q)-q+1)/(p(p-q)))$, $(323, 23, -pw(p-q)/(q(pw(p-q)-q+1)))$,
$(371, 44, -q(pw(p-q)-q+1)/((1-q)(pw(p-q)-q)))$,
$(375, 50, q(pw(p-q)-q+1)/((1-q)(pw(p-q)-q)))$, $(423, 53, pw^2(p-q)/(q(pw(p-q)-q+1)))$,
$(480, 3, q(1-q)/(p(p-q)))$, $(483, 2, -1)$, $(491, 56, -q(1-q)(pw(p-q)-q)(pw(p-q)-q+1)/(p^2(p-q)^2))$,
$(494, 5, 1)$, $(514, 17, 1)$, $(539, 76, q(pw(p-q)-q+1)/(p(p-q)))$,
$(307, 32, pw(p-q)/(q(pw(p-q)-q+1)))$, $(331, 7, w)$, $(383, 26, w)$, $(427, 35, -w)$, $(481, 0, -1)$,
$(484, 4, q(1-q)/(p(p-q)))$, $(504, 15, -q(p-q)/((1-q)(pw(p-q)-q)))$, $(507, 20, -w)$, $(520, 18, 1)$,
$(540, 51, -1)$, $(541, 52, -1)$, $(544, 57, q)$, $(546, 55, -qw)$, $(502, 41, -2(1-q)(pw(p-q)-q)/(pw(p-q)))$,
$(526, 43, -pw(p-q)/((1-q)(pw(p-q)-q)))$, $(320, 6, -1)$, $(363, 28, -pw^2(p-q)/((1-q)(pw(p-q)-q)))$,
$(422, 38, 1)$, $(482, 1, -1)$, $(505, 16, -q(p-q)/((1-q)(pw(p-q)-q)))$, $(545, 54, -qw)$,
$(308, 9, -q(p-q)/((1-q)(pw(p-q)-q)))$, $(322, 10, pw(p-q)/((1-q)(pw(p-q)-q)))$, $(368, 21, -1)$,
$(374, 49, 2q(pw(p-q)-q+1)/((1-q)(pw(p-q)-q)))$, $(382, 22, -w)$, $(501, 13, 1)$, $(525, 40, 1)$,
$(528, 45, q)$, $(537, 27, w)$, $(328, 12, -q)$, $(426, 34, -w)$, $(536, 48, -2q(1-q)/(p(p-q)))$, $(488, 30, -1)$,
$(521, 19, 1)$, $(489, 31, -1)$, $(309, 33, 2)$, $(508, 42, -2q/(p^2w))$, $(304, 24, -q(1-q)/(p(p-q)))$,
$(524, 58, -2q(1-q)/(pw(p-q)))$, $(306, 25, w)$, $(305, 47, 2q(1-q)/(pw(p-q)))$,
$(316, 36, -q(1-q)/(p(p-q)))$, $(330, 46, -1)$, $(362, 37, -w)$, $(321, 39, 1)$

TABLE 32. A pivot table of T with 78 pivots over the coordinate ring of family III of Table 4.

$(36, 64, 1)$, $(44, 71, 1)$, $(105, 75, -w)$, $(214, 77, -w/p)$, $(287, 81, -w/q)$, $(495, 8, 1)$, $(515, 29, -w)$, $(4, 60, -1)$,
$(8, 61, -1)$, $(40, 65, 1)$, $(73, 62, -q/p)$, $(81, 67, -w)$, $(101, 63, 1)$, $(121, 66, 1)$, $(158, 68, -q/p)$, $(166, 69, w/p)$,
$(206, 72, 1)$, $(210, 73, 1)$, $(255, 78, -1)$, $(259, 79, -1)$, $(267, 82, q)$, $(271, 80, -w)$, $(323, 11, w/p)$, $(492, 3, 1)$,
$(494, 5, 1)$, $(503, 56, -w/q)$, $(514, 17, 1)$, $(527, 44, -w/p)$, $(551, 76, w/p)$, $(311, 23, 1)$, $(375, 53, -w)$,
$(423, 50, w/p)$, $(481, 0, -1)$, $(483, 2, -1)$, $(493, 4, 1)$, $(504, 15, -q/p)$, $(507, 20, -w)$, $(511, 32, w/p)$,
$(520, 18, 1)$, $(540, 51, -1)$, $(544, 57, q)$, $(550, 55, -w/q)$, $(307, 14, -w)$, $(320, 6, -1)$, $(383, 26, w/q)$,
$(502, 41, -2p/w)$, $(526, 43, -w/p)$, $(331, 7, w/q)$, $(363, 28, -w^2/p)$, $(370, 35, 1)$, $(505, 16, -q/p)$,
$(541, 52, -1)$, $(545, 54, -w)$, $(308, 9, -q/p)$, $(322, 10, w/p)$, $(368, 21, -1)$, $(422, 38, 1)$, $(501, 13, 1)$,
$(525, 40, 1)$, $(528, 45, q)$, $(328, 12, -q)$, $(360, 30, -q/p)$, $(382, 22, -w/q)$, $(426, 34, -w/q)$, $(496, 27, -q/p)$,
$(372, 24, -q)$, $(420, 36, 1)$, $(524, 39, 1)$, $(498, 25, 1)$, $(310, 42, 2q)$, $(306, 70, w^2/q^2)$, $(318, 31, 1)$, $(329, 48, 2)$,
$(321, 37, 1)$, $(373, 19, 1)$, $(309, 33, 2)$, $(305, 1, -1)$, $(330, 46, -1)$

Lemma A.1. *Let T_{I} be the matrix left after the replacements of Construction-Proposition 6.7 for the 78 triples of Table 30, starting from T on family I. Then*

$$\begin{aligned}(T_{\mathrm{I}})_{328,49} &= -2q^2(pw+pv-v)(pv-qv+1)/((p-q)(q-1)(w+v)),\\ (T_{\mathrm{I}})_{374,49} &= -2(qw+qv-v)/((q-1)(w+v)),\\ (T_{\mathrm{I}})_{328,49}(T_{\mathrm{I}})_{374,9} - (T_{\mathrm{I}})_{328,9}(T_{\mathrm{I}})_{374,49} &= -2qv^2H_0/(p^2(q-1)(w+v)^3),\end{aligned} \tag{A.1}$$

where

$$\begin{aligned}H_0 := {}& p^3w^2+2p^3wv+p^3v^2-2p^2qw^2-4p^2qwv-2p^2qv^2\\ &+p^2w^2-p^2v^2+pq^2w^2+2pq^2wv+pq^2v^2-pqw^2\\ &+pqwv+3pqv^2-pwv-pv^2-q^2wv-2q^2v^2\\ &+qwv-qw+qv^2.\end{aligned}$$

Let T_{III} be the matrix left after the replacements of Construction-Proposition 6.7 for the 78 triples of Table 32, starting from T on family III, and let $G := (p-w)(w-1)$. Then

$$(T_{\mathrm{III}})_{405,74} = w(p-w)(w-1)/p^2 = wG/p^2. \tag{A.2}$$

Proof. This is a direct calculation with the replacements of Construction-Proposition 6.7. □

A.2. **Type** $(2,1,1)$**.** The block B_3 is the block of Table 10, and $y_0,\dots,y_3$ are those of Construction-Proposition 9.2.

The slot (i,j,k) is the coefficient of y_k in O_{ij}, with $0 \le i < j \le 5$ and $0 \le k \le 3$, and the slots are in lexicographic order. Let M be the 48×48 constant matrix of the minor of Construction-Proposition 8.6: the rows of M are the 48 slots outside the twelve slots of Table 33, the columns of M are the entries Γ_{ij} in row-major order, and the entry of M at the slot (a,b,k) and the column (i,j) is the coefficient of $\Gamma_{ij}y_k$ in the (a,b) entry of $B_3\Gamma^t - \Gamma B_3^t$. Table 33 lists the twelve slots, the pivots, and the row exchanges.

The a-th term of the list of pivots is the pivot in column a. A pair (a,b) means that rows a and b are exchanged before the elimination in column a. Then any row below row a is replaced by that row minus the ratio of the entry of that row in column a to the pivot, times row a. The list has 13 pivots -1, 4 pivots -2, and 39 exchanges, so

$$\det M = (-1)^{39}(-1)^{13}(-2)^4 = 16.$$

Construction-Proposition 8.6 refers to M for the type $(2,1,1)$.

TABLE 33. The twelve slots, the pivots, and the row exchanges of the matrix M of type $(2,1,1)$.

The twelve slots: $(0,1,0)$, $(0,2,0)$, $(0,3,2)$, $(0,5,2)$, $(1,2,0)$, $(1,3,0)$, $(1,4,3)$, $(1,5,3)$, $(2,3,0)$, $(2,4,0)$, $(3,5,2)$, $(4,5,3)$

The pivots: 1, 1, 1, 1, −1, −1, −1, −1, 1, 1, 1, 1, −1, −1, −1, −1, 1, 1, 1, 1, 1, 1, −1, −1, −1, 1, 1, −1, 1, 1, 1, 1, −1, 1, −2, 1, −2, 1, 1, 1, 1, 1, −2, −2, 1, 1, 1, 1

The row exchanges: $(0,1)$, $(1,13)$, $(2,6)$, $(3,9)$, $(6,8)$, $(7,11)$, $(9,25)$, $(10,13)$, $(11,22)$, $(12,17)$, $(13,18)$, $(14,21)$, $(15,24)$, $(16,18)$, $(17,19)$, $(18,25)$, $(19,25)$, $(20,28)$, $(21,31)$, $(22,30)$, $(23,32)$, $(24,28)$, $(25,31)$, $(26,30)$, $(27,34)$, $(28,29)$, $(29,35)$, $(30,40)$, $(31,43)$, $(32,40)$, $(33,43)$, $(34,38)$, $(35,40)$, $(36,39)$, $(37,43)$, $(38,41)$, $(39,46)$, $(43,45)$, $(44,46)$

A.3. **Type** $(2,2)$**.** We consider the equations

$$AO - B_1N^t - CB_3^t = 0, \quad AN - CP^t = 0, \quad UN^t = 0,$$

with A of Construction-Proposition 10.2 and with B_1, B_3, P, U of Table 11.

The slot (i,j,l) is the coefficient of y_l in O_{ij}, with $0 \le i < j \le 5$ and $0 \le l \le 3$. We number the 60 slots in lexicographic order, and the entries of Γ in row-major order. The 48 rows of the 60×48 constant matrix of the map $\Gamma \to B_3\Gamma^t - \Gamma B_3^t$ with the numbers listed in Table 34 form a

square minor of determinant 16. Setting the 48 slots of the minor to zero implies the normalized shape of Construction-Proposition 8.6, which is the matrix O of Table 34, where $o_0, \dots, o_{11}$ are binary linear forms.

We use the numbers $24i + 4j + l$ of Construction-Proposition 10.7 for the 96 coefficients of $y_0, \dots, y_3$ in the entries of $B_1N^t + CB_3^t - AO$, the negative of the left-hand side of the first equation, and we number the 104 entries of N and C from 0, with the entries of N in row-major order followed by the entries of C in row-major order. The 84×84 coefficient minor on the rows and columns listed in Table 34, formed with the blocks B_1 and B_3 of Table 11, has determinant 1. Solving the square system of the minor implies the formulas of Table 34, which lists all 72 positions of N and all 32 positions of C, row by row and entry by entry: 84 positions, namely N and twelve entries of C, are solved, and the other twenty entries of C are the binary quadratic forms $c_0, \dots, c_{19}$. The positions of $c_0, \dots, c_{19}$ in C are numbered from 0 to 31 in row-major order. The other twelve coefficients of the entries of $B_1N^t + CB_3^t - AO$ have the numbers $0, 12, 16, 24, 36, 40, 54, 62, 70, 78, 86, 94$, outside the minor. They are the entries of Ho, where H is the matrix of Table 34 and $o = (o_0, \dots, o_{11})^t$, so the rows of H are the twelve forms of Construction-Proposition 10.7. The matrix $AN - CP^t$ has 48 entries, the matrix UN^t has 18 entries, and 56 of the entries of $AN - CP^t$ and UN^t are not identically zero as polynomials in the a_i, o_i, c_j.

Let $T_{(2,2)}$ be the 260×84 matrix of Construction-Proposition 10.7. Each of Tables 35, 36, 37, and 38 begins with the list $(a_0, \dots, a_{11})$ of a chart of Table 12, with parameters among p, e, f, k, h, and the matrix of the chart is $T_{(2,2)}$ for the list $(a_0, \dots, a_{11})$, the coefficients in s, t being taken after the substitution. The coefficient rings are $\mathbb{Q}[p, p^{-1}, e, f]$ for NN, $\mathbb{Q}[k, e, f]$ for ZN, $\mathbb{Q}[k, h, e, f]$ for ZZ, and $\mathbb{Q}[p, e, f]$ for PN, and extension of scalars to $\mathbb{C}$ gives the coordinate rings of Table 13.

A triple (r, c, β) of the four tables means the following. After the row operations of the triples listed before (r, c, β), the entry in row r and column c equals β. Divide row r by β, and from any other row, including the rows of earlier triples, subtract the current entry of that row in column c times the new row r. Rows and columns are not renumbered. In each of the four tables, the rows of the triples are pairwise distinct, and so are the columns. Over the coefficient ring of each chart, the row operations follow the triples in the listed order, the only divisions are by units of that ring, and each equality of an entry with β is a finite exact computation. On the rows and columns of no triple up to and including (r, c, β), the row operations of (r, c, β) change the entries by the formula of Construction-Proposition 6.7, and the changes in the rows of earlier triples do not enter the formula. Thus each list of triples is a pivot table of the matrix of the chart in the sense of Construction-Proposition 6.7, with the values β of the list, and it has $77, 79, 79, 77$ triples for NN, ZN, ZZ, PN, respectively. The four tables also list the columns without pivot. The rows and the columns of the last matrix of the pivot table of a chart are the rows without pivot and the columns without pivot. Table 35 lists the 2×2 minor of the last matrix of the pivot table of NN in rows $45, 49$ and columns $60, 82$, and Table 38 lists the entry of the last matrix of the pivot table of PN in row 41 and column 79. Lemma 10.8 uses the four tables.

A.4. **Type $(2, 2)$, exceptional presentations.** In this subsection, we list the pivot tables and the linear functionals used for exceptional presentations in Subsection 10.3.

The quadrics $r_1, \dots, r_6$ are those of Construction-Proposition 10.2, the blocks $B_1, \dots, B_4, P, U$ are those of Table 11 (Section 10), and the blocks O, N, C are those of Table 34 (Subsection A.3), while the numbering $x_0, \dots, x_{83}$ of the 84 coefficients and the 260×84 matrix $T_{(2,2)}$ are those of Construction-Proposition 10.7. Tuples are exponent tuples in the order $(s, t, y_0, y_1, y_2, y_3)$.

All tables of this subsection come from Gauss–Jordan elimination over $\mathbb{Q}[e, p, p^{-1}]$ in the order of the tables, with each pivot value a rational constant times a power of p. Where an entry depends on a point of $\mathcal{U}_A$, the variables $q_0, \dots, q_8$ are adjoined to the ring.

TABLE 34. The rows of the 48×48 minor and the matrices O, N, and C of type $(2, 2)$.

The rows of the 48×48 minor: 0, 1, 2, 3, 4, 5, 6, 7, 10, 11, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 27, 28, 29, 30, 31, 32, 33, 35, 36, 37, 39, 40, 41, 42, 43, 44, 45, 47, 50, 51, 52, 53, 55, 56, 57, 59
O, row 0: 0, 0, 0, $o_0y_0 + o_1y_1$, $o_2y_0 + o_3y_1$, 0
O, row 1: 0, 0, 0, o_4y_2, 0, o_5y_2
O, row 2: 0, 0, 0, o_6y_2, 0, o_7y_2
O, row 3: $-o_0y_0 - o_1y_1$, $-o_4y_2$, $-o_6y_2$, 0, $o_8y_0 + o_9y_1$, $o_{10}y_2$
O, row 4: $-o_2y_0 - o_3y_1$, 0, 0, $-o_8y_0 - o_9y_1$, 0, $o_{11}y_2$
O, row 5: 0, $-o_5y_2$, $-o_7y_2$, $-o_{10}y_2$, $-o_{11}y_2$, 0
N, row 0: $a_5o_1 + a_8o_3 - a_9o_1$, $-a_4o_1 - a_7o_3$, $-a_5o_1 - a_8o_3$, $-a_6o_1 - a_9o_3$, $a_5o_0 + a_8o_2 + c_5$, $-c_0$, $-c_5$, $-c_{12}$, $a_6o_0 + a_9o_2 + c_{12} + c_6$, $-c_1$, $-c_6$, $-c_{13}$
N, row 1: $-a_5o_0 - a_8o_2 - c_5$, c_0, c_5, c_{12}, $a_5o_4 - a_9o_4$, $-a_4o_4 - a_5o_0 - a_8o_2 - c_5$, $-a_5o_4 + a_9o_4 - c_7$, $a_{10}o_7 + a_9o_6 - c_{14}$, $-a_{10}o_7 - a_9o_6 + c_8$, $-a_6o_0 - a_9o_2 - c_{12} + c_2 - c_6$, $-c_8$, $-c_{15}$
N, row 2: $-a_6o_0 - a_9o_2 - c_{12} - c_6$, c_1, c_6, c_{13}, $a_{10}o_7 + a_5o_6 - c_8$, $-a_4o_6 - c_2$, $-a_{10}o_7 - a_5o_6 - c_9$, $-a_{11}o_7 - a_6o_6 - c_{16}$, $a_{11}o_7 + a_6o_6 + c_{10} - c_{15}$, $-c_3$, $-c_{10}$, $-c_{17}$
N, row 3: $-a_1o_1 + a_8o_9$, $a_0o_1 + a_5o_0 - a_7o_9 + a_8o_2 + c_5$, $a_1o_1 - a_8o_9 + c_7$, $-a_9o_9 + c_{14}$, 0, $a_5o_4 + a_8o_6 - a_9o_4$, $-a_1o_0 + a_8o_8$, $-a_1o_2 + a_{10}o_{11} - a_5o_8 + a_9o_8$, $a_1o_2 - a_{10}o_{11} + a_5o_8 + c_{11}$, $-c_4$, $-c_{11}$, $-c_{18}$
N, row 4: $-a_1o_3 - a_5o_9 + a_9o_9$, $a_0o_3 + a_4o_9 + a_6o_0 + a_9o_2 + c_{12} + c_6$, $a_1o_3 + a_5o_9 + c_8 + c_9$, $a_6o_9 + c_{15} + c_{16}$, $-a_1o_2 + a_{10}o_{11} - a_5o_8 - c_{11}$, c_4, $-a_{10}o_{11} + c_{11}$, $-a_{11}o_{11} + c_{18}$, 0, $a_{11}o_7 + a_6o_6 + c_{10} - c_{15}$, $a_{11}o_{11} - a_6o_8 - c_{18}$, $-c_{19}$
N, row 5: $-a_{11}o_7 - a_6o_6 - c_{10} + c_{15}$, c_3, c_{10}, c_{17}, $-a_5o_{10} - a_8o_{11} + a_9o_{10}$, $-a_{11}o_7 + a_4o_{10} + a_5o_5 - a_6o_6 + a_7o_{11} + a_8o_7 - a_9o_5 - c_{10} + c_{15}$, $-a_{11}o_{11} + a_5o_{10} + a_6o_8 + a_8o_{11} - a_9o_{10} + c_{18}$, c_{19}, 0, 0, 0, 0
C, row 0: $-a_5o_0 - a_8o_2 - c_5$, $-a_6o_0 - a_9o_2 - c_{12} - c_6$, 0, $a_{10}o_7 + a_9o_6 - c_8$, $-a_{10}o_7 - a_9o_6 + c_8$, $-a_{11}o_7 - a_6o_6 - c_{10} + c_{15}$, $-a_1o_2 + a_{10}o_{11} - a_5o_8 - c_{11}$, 0
C, row 1: c_0, c_1, $a_5o_0 + a_8o_2 + c_5$, $a_6o_0 + a_9o_2 + c_{12} - c_2 + c_6$, c_2, c_3, c_4, $-a_{11}o_7 - a_6o_6 - c_{10} + c_{15}$
C, row 2: c_5, c_6, c_7, c_8, c_9, c_{10}, c_{11}, $-a_{11}o_{11} + a_6o_8 + c_{18}$
C, row 3: c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_{17}, c_{18}, c_{19}

TABLE 34. The rows and the columns of the 84×84 minor, the positions of $c_0, \ldots, c_{19}$ in C, and the matrix H of type $(2, 2)$ (continued).

The rows of the 84×84 minor: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 17, 18, 19, 20, 21, 22, 23, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 37, 38, 39, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 55, 56, 57, 58, 59, 60, 61, 63, 64, 65, 66, 67, 68, 69, 71, 72, 73, 74, 75, 76, 77, 79, 80, 81, 82, 83, 84, 85, 87, 88, 89, 90, 91, 92, 93, 95
The columns of the 84×84 minor: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 82, 83, 87, 95
The positions of $c_0, \ldots, c_{19}$ in C: 8, 9, 12, 13, 14, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 27, 28, 29, 30, 31
H, row 0: $-a_5 + a_9$, 0, $-a_8$, 0, 0, 0, 0, 0, 0, 0, 0, 0
H, row 1: a_1, 0, 0, 0, 0, 0, 0, 0, $-a_8$, 0, 0, 0
H, row 2: 0, 0, a_1, 0, 0, 0, 0, 0, $a_5 - a_9$, 0, 0, 0
H, row 3: a_4, $-a_5 + a_9$, a_7, $-a_8$, 0, 0, 0, 0, 0, 0, 0, 0
H, row 4: $-a_0$, a_1, 0, 0, 0, 0, 0, 0, a_7, $-a_8$, 0, 0
H, row 5: 0, 0, $-a_0$, a_1, 0, 0, 0, 0, $-a_4$, $a_5 - a_9$, 0, 0
H, row 6: 0, 0, 0, 0, a_9, a_{10}, 0, 0, 0, 0, 0, 0
H, row 7: $-a_1$, 0, 0, 0, $-a_3$, 0, 0, 0, a_8, 0, a_{10}, 0
H, row 8: 0, 0, 0, 0, 0, $-a_3$, 0, 0, 0, 0, $-a_9$, 0
H, row 9: 0, 0, 0, 0, a_6, a_{11}, a_9, a_{10}, 0, 0, 0, 0
H, row 10: 0, 0, $-a_1$, 0, $-a_2$, 0, $-a_3$, 0, $-a_5 + a_9$, 0, a_{11}, a_{10}
H, row 11: 0, 0, 0, 0, 0, $-a_2$, 0, $-a_3$, 0, 0, $-a_6$, $-a_9$

TABLE 35. The list $(a_0, \ldots, a_{11})$, the triples, the columns without pivot, and the 2×2 minor of the chart NN.

The list $(a_0, \ldots, a_{11})$: $-t$, s, $p^2s - 2ps - t$, $s + t$, $et - 2p^2s - 2p^2t + 4ps + 4pt + 2t$, $es + ft - 2t - 2s/p$, $fp^2s - 2p^2s + 4ps - 2pt + 2s + 4t$, $pt - s - 2t$, $ps + t$, $ft - 2s/p$, t, $-p^2s$
The triples: $(5, 17, -1)$, $(18, 8, -2/p)$, $(0, 4, -p)$, $(2, 1, 2)$, $(3, 0, 1)$, $(4, 5, 1/2)$, $(14, 16, 1)$, $(19, 9, -2/p)$, $(20, 11, 1)$, $(22, 10, -p/2)$, $(23, 20, -1)$, $(27, 12, -2/p)$, $(12, 2, 1)$, $(32, 13, -1)$, $(203, 19, -1)$, $(29, 15, 1)$, $(31, 22, 1)$, $(28, 14, 1)$, $(240, 21, -2p^3)$, $(9, 6, -p)$, $(13, 3, 1)$, $(10, 7, -p)$, $(200, 18, -1)$, $(251, 45, p^{-4})$, $(59, 50, 1)$, $(191, 41, -2)$, $(243, 71, 1)$, $(227, 38, -1)$, $(249, 46, -1)$, $(236, 24, -p)$, $(188, 42, p)$, $(71, 56, -1)$, $(211, 47, 4)$, $(235, 77, -1)$, $(135, 53, 1)$, $(255, 68, -2)$, $(43, 44, 1/2)$, $(47, 74, -1)$, $(119, 62, 2)$, $(95, 32, 1)$, $(107, 35, 1)$, $(250, 78, -1)$, $(212, 27, -p)$, $(242, 70, 1)$, $(241, 69, 1)$, $(40, 39, 1)$, $(58, 49, 1)$, $(208, 33, -1)$, $(232, 75, -1)$, $(234, 76, -1)$, $(202, 48, -p)$, $(76, 58, -4/p)$, $(79, 26, -1)$, $(201, 40, -1)$, $(83, 29, -1)$, $(224, 36, -p)$, $(252, 57, -1)$, $(192, 51, -p)$, $(48, 66, 2)$, $(57, 61, 1)$, $(216, 30, p)$, $(60, 54, -2p)$, $(41, 52, -p)$, $(42, 55, 1)$, $(44, 72, -p)$, $(189, 43, p)$, $(194, 65, -1)$, $(226, 37, -1)$, $(193, 25, -1)$, $(77, 67, 1)$, $(134, 81, -1)$, $(78, 34, -1)$, $(68, 63, 1)$, $(238, 64, 1)$, $(46, 73, -1)$, $(245, 31, -p)$, $(197, 28, -1)$
The columns without pivot: 23, 59, 60, 79, 80, 82, 83
The rows 45, 49 in the columns 60, 82: row 45:
$-efp^2/4 + ep^3/2 - ep^2/2 - ep + e/2 - fp^3/2 + p^4 - p^3 - 2p^2 + p$, $-e - 2p$, row 49:
$3fp^3/4 - fp^2/4 - p^4/2 + p^3 + 3p^2/2 - 3p/2 + 1/2$, $p - 1$
At $(p, e, f) = (2, 0, 0)$: row 45: 2, -4, row 49: $7/2$, 1, of determinant 16

TABLE 36. The list $(a_0, \ldots, a_{11})$, the triples, and the columns without pivot of the chart ZN.

The list $(a_0, \ldots, a_{11})$: 0, s, $-2s - t$, s, $4s + 2t$, $es + ft + kt - 2s$, $fs + 2s$, $-s$, $s + t$, $ft - 2s$, t, $-s$
The triples: $(5, 17, -1)$, $(14, 19, -1)$, $(18, 8, -2)$, $(23, 21, 1)$, $(0, 4, -1)$, $(2, 5, -1)$, $(3, 0, 1)$, $(4, 1, 1)$, $(13, 3, 1)$, $(19, 9, -2)$, $(20, 11, 1)$, $(22, 10, -1/2)$, $(27, 12, -2)$, $(29, 15, 1)$, $(32, 20, 1)$, $(203, 23, -1)$, $(59, 80, -1)$, $(135, 47, -1)$, $(251, 13, 1)$, $(255, 38, 1)$, $(227, 83, 2)$, $(51, 50, 1)$, $(31, 22, 1)$, $(235, 71, 1)$, $(28, 14, 1)$, $(63, 56, -2)$, $(43, 53, -1)$, $(250, 79, 1)$, $(210, 35, -1)$, $(11, 7, -1)$, $(12, 2, 1)$, $(9, 6, -1)$, $(240, 16, -2)$, $(202, 41, -1)$, $(241, 45, 2)$, $(242, 70, 1)$, $(70, 49, 1)$, $(234, 77, -1)$, $(249, 78, 1)$, $(209, 34, -1)$, $(208, 33, -1)$, $(232, 75, -1)$, $(50, 26, 1)$, $(226, 37, -1)$, $(194, 65, -2)$, $(239, 46, 1)$, $(134, 62, -1)$, $(233, 76, -1)$, $(47, 58, -1)$, $(201, 40, -1)$, $(191, 44, 1)$, $(107, 32, 1)$, $(196, 27, -1)$, $(192, 24, -1)$, $(48, 36, -1)$, $(200, 18, -1)$, $(224, 39, 2)$, $(40, 51, -1)$, $(252, 48, -1)$, $(42, 52, -1)$, $(193, 25, -1)$, $(56, 57, -2)$, $(76, 66, 2)$, $(188, 42, 1)$, $(216, 30, 1)$, $(104, 54, 2)$, $(41, 55, -1)$, $(230, 74, -4)$, $(62, 29, 1)$, $(57, 61, 1)$, $(77, 67, 1)$, $(225, 81, -1)$, $(68, 63, 1)$, $(44, 72, -1)$, $(61, 28, 1)$, $(197, 73, -1)$, $(238, 64, 1)$, $(81, 43, 2)$, $(93, 31, 1)$
The columns without pivot: 59, 60, 68, 69, 82

TABLE 37. The list $(a_0, \ldots, a_{11})$, the triples, and the columns without pivot of the chart ZZ.

The list $(a_0, \ldots, a_{11})$: 0, s, $-t$, s, $2t$, $es + ft + hs + kt$, $2s$, $-s$, t, $ft + hs$, t, 0
The triples: $(3, 0, 1)$, $(5, 17, -1)$, $(14, 19, -1)$, $(21, 8, -1)$, $(23, 21, 1)$, $(248, 45, -1)$, $(1, 4, -1)$, $(2, 5, -1)$, $(4, 1, 1)$, $(30, 12, -1)$, $(12, 2, 1)$, $(10, 6, -1)$, $(13, 3, 1)$, $(19, 10, 1)$, $(20, 11, 1)$, $(22, 9, -1)$, $(29, 15, 1)$, $(31, 13, -1)$, $(32, 20, 1)$, $(40, 39, 1)$, $(192, 24, -1)$, $(200, 18, -1)$, $(203, 23, -1)$, $(59, 80, -1)$, $(135, 47, -1)$, $(251, 22, 1)$, $(255, 38, 1)$, $(227, 83, 2)$, $(51, 50, 1)$, $(11, 7, -1)$, $(28, 14, 1)$, $(235, 71, 1)$, $(250, 79, 1)$, $(210, 35, -1)$, $(63, 56, -2)$, $(43, 53, -1)$, $(48, 57, -1)$, $(224, 66, -2)$, $(201, 40, -1)$, $(237, 63, 1)$, $(60, 27, 1)$, $(202, 41, -1)$, $(208, 33, -1)$, $(228, 69, -1)$, $(249, 16, 1)$, $(50, 26, 1)$, $(70, 49, 1)$, $(193, 25, -1)$, $(209, 34, -1)$, $(226, 37, -1)$, $(239, 46, -3)$, $(234, 70, 1)$, $(242, 77, 1)$, $(194, 65, -2)$, $(56, 48, 1)$, $(216, 60, -2)$, $(76, 51, -1)$, $(44, 42, 1)$, $(104, 81, 2)$, $(92, 30, 1)$, $(241, 72, -1)$, $(41, 54, 1)$, $(58, 62, 1)$, $(47, 58, -1)$, $(49, 36, -1)$, $(225, 74, -2)$, $(215, 29, -1)$, $(77, 52, -1)$, $(191, 44, 1)$, $(107, 32, 1)$, $(42, 55, 1)$, $(57, 61, 1)$, $(45, 43, 1)$, $(69, 64, 1)$, $(105, 75, -1)$, $(233, 76, -1)$, $(81, 28, -1)$, $(46, 73, -1)$, $(106, 31, 1)$
The columns without pivot: 59, 67, 68, 78, 82

TABLE 38. The list $(a_0, \dots, a_{11})$, the triples, the columns without pivot, and the entry in the row 41 and the column 79 of the chart PN.

The list $(a_0, \dots, a_{11})$: $-t$, s, $-p^2s - 2pt$, t, $t(e + 2p^2)$, $-2pt + s(e + f)$, $t(f - 2p^2)$, pt, $ps + t$, $fs + 2pt$, s, $-t$
The triples: $(5, 17, -1)$, $(23, 9, -1)$, $(2, 5, -1)$, $(3, 0, 1)$, $(21, 20, 1)$, $(4, 1, 1)$, $(1, 4, -1)$, $(12, 2, 1)$, $(14, 16, 1)$, $(15, 6, 1)$, $(18, 10, 1)$, $(19, 11, 1)$, $(22, 8, -1)$, $(27, 14, 1)$, $(30, 22, 1)$, $(32, 13, -1)$, $(203, 19, -1)$, $(240, 69, 1)$, $(13, 3, 1)$, $(31, 12, -1)$, $(200, 39, -1)$, $(208, 33, -1)$, $(236, 63, 1)$, $(251, 21, -1)$, $(195, 41, 1)$, $(252, 81, 1)$, $(11, 7, -1)$, $(43, 50, -2)$, $(51, 38, -1)$, $(227, 68, -2)$, $(250, 46, -1)$, $(40, 54, 1)$, $(28, 15, 1)$, $(248, 78, 1)$, $(242, 71, 1)$, $(59, 23, -1)$, $(192, 24, -1)$, $(202, 18, 1)$, $(60, 27, 1)$, $(204, 42, -1)$, $(191, 44, 1)$, $(63, 56, -2)$, $(71, 83, 1)$, $(219, 32, 1)$, $(103, 62, -2)$, $(47, 74, -1)$, $(241, 70, 1)$, $(223, 35, -1)$, $(107, 77, -1)$, $(79, 26, -1)$, $(249, 45, -1)$, $(56, 60, 1)$, $(83, 29, -1)$, $(201, 40, -1)$, $(42, 49, -1)$, $(226, 37, -1)$, $(244, 75, 1)$, $(50, 52, -1)$, $(194, 67, -2)$, $(58, 57, -1)$, $(210, 55, 1)$, $(234, 76, -1)$, $(193, 25, -1)$, $(190, 43, 1)$, $(218, 31, 1)$, $(225, 36, -1)$, $(49, 48, 2)$, $(57, 61, 1)$, $(238, 65, 1)$, $(253, 51, -1)$, $(133, 66, -2)$, $(77, 34, -1)$, $(94, 73, -1)$, $(61, 28, 1)$, $(82, 64, 1)$, $(45, 72, -1)$, $(217, 30, 1)$
The columns without pivot: 47, 53, 58, 59, 79, 80, 82
The entry in the row 41 and the column 79: $4p$

A triple is (row, column, value), as in the pivot tables of Construction-Proposition 6.7. The row operations of a triple are those of the triples of Subsection A.3, so a pivot row is divided by its value and the pivot column is cleared in all other rows. Table 39 is the pivot table of $T_{(2,2)}$ after substitution of the chart of Construction-Proposition 10.10. The columns without pivot are $46, 62, 74, 77, 78, 79, 81, 82, 83$, assigned to $q_0, \dots, q_8$, respectively. All rows without pivot reduce to zero, and the final pivot rows express each pivot coefficient as a linear form in $q_0, \dots, q_8$.

TABLE 39. The pivot table of $T_{(2,2)}$ on the chart of exceptional presentations.

$(0, 4, -1)$, $(1, 5, -1)$, $(11, 1, 2p)$, $(4, 17, -1)$, $(18, 10, 1)$, $(19, 11, 1)$, $(29, 9, 2)$, $(22, 21, 1)$, $(3, 0, 1)$, $(10, 7, -1)$, $(21, 8, -1)$, $(27, 14, 1)$, $(28, 15, 1)$, $(51, 41, -1)$, $(71, 50, p)$, $(195, 47, p)$, $(202, 3, -2)$, $(239, 59, -1)$, $(242, 20, -2p)$, $(9, 6, -1)$, $(63, 44, -1)$, $(79, 53, -1)$, $(119, 56, -p)$, $(240, 69, 1)$, $(241, 70, 1)$, $(12, 2, 1)$, $(13, 16, -2)$, $(31, 13, -1)$, $(235, 80, -2)$, $(42, 23, 2)$, $(30, 12, -1)$, $(200, 18, -1)$, $(56, 48, 1)$, $(107, 32, p)$, $(194, 26, -1)$, $(224, 22, -1)$, $(226, 38, -1)$, $(50, 40, -2)$, $(234, 19, 2)$, $(201, 36, -2)$, $(70, 49, p)$, $(218, 68, 2/p)$, $(189, 43, 1)$, $(62, 29, 1)$, $(225, 37, -1)$, $(134, 39, -p^2)$, $(246, 71, -2p)$, $(188, 42, 1)$, $(210, 35, -1)$, $(232, 75, -1)$, $(233, 76, -1)$, $(60, 27, 1)$, $(212, 54, -2)$, $(100, 30, -1)$, $(244, 60, 2)$, $(216, 45, 2)$, $(44, 58, -1)$, $(61, 28, 1)$, $(213, 55, -2)$, $(40, 51, -1)$, $(132, 66, -1)$, $(208, 33, -1)$, $(46, 72, -2p)$, $(45, 73, -1)$, $(117, 61, 2)$, $(57, 57, 2)$, $(133, 52, 1)$, $(253, 67, -2)$, $(41, 65, -1)$, $(48, 24, 1)$, $(93, 31, 1)$, $(77, 25, -1)$, $(69, 64, 1)$, $(49, 34, -1)$, $(68, 63, 1)$
Columns without pivot: 46, 62, 74, 77, 78, 79, 81, 82, 83

We call the four equations of (10.6), in the order written there, the *first*, *second*, *third*, and *fourth equation*, with unknowns Φ, Θ, v, and Ψ, respectively. The equations are solved in order. For each equation, we take the coefficients of all monomials of the bidegree of the equation in (s, t) and $(y_0, \dots, y_3)$, namely $(1, 3)$, $(2, 2)$, $(3, 1)$, and $(4, 1)$, respectively, zero rows being allowed. A row of the linear system is labeled $(j; m)$, where j is the entry of the equation, numbered from 0, and m is the exponent tuple of the monomial, and the augmented last column is minus the right-hand side. The unknowns are numbered by ℓ, starting at 0, in the lexicographic order of the indices

Φ :	(i, l, a, b), $0 \le i < 8$, $0 \le l \le 1$, $0 \le a \le b < 4$,	coefficient of $s^{1-l}t^l y_a y_b$ in entry i,
Θ :	(i, l, a), $0 \le i < 12$, $0 \le l \le 2$, $0 \le a < 4$,	coefficient of $s^{2-l}t^l y_a$ in entry i,
v :	(i, l), $0 \le i < 6$, $0 \le l \le 3$,	coefficient of $s^{3-l}t^l$ in v_{i+1},
Ψ :	(i, l), $0 \le i < 8$, $0 \le l \le 4$,	coefficient of $s^{4-l}t^l$ in entry i.

An entry $((j; m), \ell, \beta)$ of the tables consists of a row label, an unknown index, and the pivot value β. The free unknowns, whose numbers are listed in Tables 40 and 41, are the free coefficients of Construction-Proposition 10.10(4), and they are set to zero, and each pivot unknown is minus

the final augmented entry of the pivot row. All rows without pivot of the augmented matrix are zero.

TABLE 40. The pivot table of the first equation of (10.6), with unknown Φ.

$((0;0,1,0,0,0,3),39,1)$, $((0;0,1,0,0,3,0),17,1)$, $((0;0,1,0,1,0,2),36,1)$, $((0;0,1,0,2,0,1),34,1)$, $((0;0,1,0,2,1,0),14,1)$, $((0;0,1,1,0,0,2),33,1)$, $((0;0,1,1,0,2,0),12,1)$, $((0;0,1,2,0,0,1),30,1)$, $((0;0,1,2,0,1,0),10,1)$, $((0;1,0,0,0,0,3),29,1)$, $((0;1,0,0,0,3,0),7,1)$, $((0;1,0,0,1,0,2),26,1)$, $((0;1,0,0,2,0,1),24,1)$, $((0;1,0,0,2,1,0),4,1)$, $((0;1,0,1,0,0,2),23,1)$, $((0;1,0,2,0,0,1),20,1)$, $((1;0,1,0,0,0,3),79,1)$, $((1;0,1,0,0,3,0),57,1)$, $((1;0,1,0,1,2,0),55,1)$, $((1;0,1,1,0,2,0),52,1)$, $((1;0,1,1,1,1,0),51,1)$, $((1;0,1,2,0,1,0),50,1)$, $((1;1,0,0,0,0,3),69,1)$, $((1;1,0,0,1,2,0),45,1)$, $((1;1,0,3,0,0,0),40,-1)$, $((1;1,0,2,0,1,0),42,-1)$, $((2;0,1,0,0,0,3),119,1)$, $((2;0,1,0,1,0,2),116,1)$, $((2;0,1,0,2,0,1),114,1)$, $((2;1,0,0,0,0,3),109,1)$, $((2;1,0,0,1,0,2),106,1)$, $((2;1,0,0,2,0,1),104,1)$, $((3;0,1,0,0,0,3),139,1)$, $((3;0,1,0,0,1,2),138,1)$, $((3;0,1,0,0,2,1),137,1)$, $((3;0,1,1,0,0,2),133,1)$, $((3;0,1,1,0,1,1),132,1)$, $((3;0,1,2,0,0,1),130,1)$, $((3;1,0,0,0,0,3),129,1)$, $((3;1,0,0,0,2,1),127,1)$, $((3;1,0,1,0,0,2),123,1)$, $((4;0,1,0,0,0,3),159,1)$, $((4;0,1,0,0,2,1),157,1)$, $((4;0,1,1,0,0,2),153,1)$, $((4;0,1,1,0,1,1),152,1)$, $((4;0,1,2,0,0,1),150,1)$, $((4;1,0,0,0,0,3),149,1)$, $((4;1,0,1,0,0,2),143,1)$, $((4;1,0,1,0,1,1),142,1)$, $((4;1,0,2,0,0,1),140,1)$, $((0;0,1,0,0,1,2),19,1)$, $((0;0,1,0,0,2,1),18,1)$, $((0;0,1,0,1,2,0),15,1)$, $((0;0,1,1,0,1,1),13,1)$, $((0;0,1,1,1,0,1),31,1)$, $((0;0,1,1,1,1,0),11,1)$, $((0;1,0,0,0,1,2),9,1)$, $((0;1,0,0,0,2,1),8,1)$, $((0;1,0,0,1,2,0),5,1)$, $((0;1,0,1,0,1,1),3,1)$, $((0;1,0,1,0,2,0),2,1)$, $((0;1,0,1,1,0,1),21,1)$, $((0;1,0,1,1,1,0),1,1)$, $((0;1,0,2,0,1,0),0,1)$, $((1;0,1,0,0,1,2),59,1)$, $((1;0,1,0,0,2,1),58,1)$, $((1;0,1,0,1,0,2),38,1)$, $((1;0,1,0,2,0,1),16,-1)$, $((1;0,1,0,2,1,0),54,1)$, $((1;0,1,1,0,0,2),73,1)$, $((1;0,1,2,0,0,1),53,-1)$, $((1;1,0,0,0,1,2),49,1)$, $((1;1,0,0,0,2,1),48,1)$, $((1;1,0,0,0,3,0),47,1)$, $((1;1,0,0,1,0,2),28,1)$, $((1;1,0,0,2,0,1),6,-1)$, $((1;1,0,0,2,1,0),44,1)$, $((1;1,0,1,0,0,2),63,1)$, $((1;1,0,1,1,1,0),41,1)$, $((1;1,0,2,0,0,1),43,-1)$, $((2;0,1,0,0,2,1),98,1)$, $((2;0,1,0,0,3,0),97,1)$, $((2;0,1,0,1,2,0),37,-1)$, $((2;0,1,0,2,1,0),35,-1)$, $((2;0,1,1,0,0,2),99,-1)$, $((2;0,1,1,0,2,0),77,-1)$, $((2;0,1,2,1,0,0),71,-1)$, $((2;0,1,3,0,0,0),70,-1)$, $((2;1,0,0,0,2,1),88,1)$, $((2;1,0,0,0,3,0),87,1)$, $((2;1,0,0,1,2,0),27,-1)$, $((2;1,0,0,2,1,0),25,-1)$, $((2;1,0,1,0,0,2),89,-1)$, $((2;1,0,1,0,2,0),67,-1)$, $((2;1,0,2,1,0,0),61,-1)$, $((2;1,0,3,0,0,0),60,-1)$, $((3;0,1,0,1,0,2),78,1)$, $((3;0,1,0,2,0,1),56,-1)$, $((3;1,0,0,0,1,2),128,1)$, $((3;1,0,0,1,0,2),68,1)$, $((3;1,0,0,2,0,1),46,-1)$, $((3;1,0,1,0,1,1),122,1)$, $((3;1,0,2,0,0,1),120,1)$, $((4;0,1,0,0,1,2),158,1)$, $((4;0,1,0,1,0,2),113,-1)$, $((4;0,1,0,1,1,1),117,1)$, $((4;1,0,0,0,1,2),148,1)$, $((4;1,0,0,0,2,1),147,1)$, $((4;1,0,0,1,0,2),103,-1)$, $((4;1,0,0,1,1,1),107,1)$, $((5;0,1,0,2,1,0),115,-1)$, $((5;0,1,1,2,0,0),111,-1)$, $((5;0,1,2,1,0,0),110,-1)$, $((5;1,0,0,2,1,0),105,-1)$, $((5;1,0,1,2,0,0),101,-1)$, $((5;1,0,2,1,0,0),100,-1)$, $((0;0,1,0,1,1,1),74,1)$, $((0;1,0,0,1,1,1),64,1)$, $((1;0,1,0,1,1,1),75,1)$, $((1;0,1,1,0,1,1),72,1)$, $((1;0,1,1,1,0,1),32,1)$, $((1;1,0,0,1,1,1),65,1)$, $((1;1,0,1,0,1,1),62,1)$, $((1;1,0,1,1,0,1),22,1)$, $((2;0,1,0,0,1,2),118,1)$, $((2;0,1,0,1,1,1),76,1)$, $((2;0,1,2,0,0,1),93,-1)$, $((2;0,1,1,0,1,1),112,1)$, $((2;1,0,0,0,1,2),108,1)$, $((2;1,0,0,1,1,1),66,1)$, $((2;1,0,2,0,0,1),83,-1)$, $((2;1,0,1,0,1,1),102,1)$, $((3;0,1,0,1,1,1),92,1)$, $((3;0,1,1,1,0,1),90,1)$, $((3;1,0,0,1,1,1),82,1)$, $((3;1,0,1,1,0,1),80,1)$

Free unknowns: 81, 84, 85, 86, 91, 94, 95, 96, 121, 124, 125, 126, 131, 134, 135, 136, 141, 144, 145, 146, 151, 154, 155, 156

For Tables 44 and 45, let d_1 be the matrix with first row $(r+v, \Theta, \Phi+\Psi)$, where (Φ, Θ, v, Ψ) is the solution of Construction-Proposition 10.10(4), given by the tables above, and with lower rows d_1' as in Set-up 4.1.

Monomials of a given weight in $s, t, y_0, \ldots, y_3$, of weights $(2,2,3,3,3,3)$, are listed in lexicographically increasing order of the exponent tuples. Let g_i be the degree of generator i of F_0, with $i = 0, \ldots, 7$ for $1, z_0, \ldots, z_3, w_0, w_1, w_2$, so that $(g_0, \ldots, g_7) = (0,4,4,4,4,5,5,5)$. Let δ_a be the degree of column a of d_1, with $a = 0, \ldots, 25$, so that $\delta_a = 6$ for $0 \le a \le 5$, $\delta_a = 7$ for $6 \le a \le 17$, and $\delta_a = 8$ for $18 \le a \le 25$.

The *relation matrix in degree* n is the matrix whose rows are the coefficient vectors of the products of a monomial of weight $n - \delta_a$ and column a of d_1, in the basis of $(F_0)_n$. The columns of the relation matrix are numbered by the basis of $(F_0)_n$, ordered by i first and then by the monomials of weight $n - g_i$. The rows are ordered by a first and then by the monomials of weight

TABLE 41. The pivot table of the second equation of (10.6), with unknown Θ.

$((0;0,2,0,0,2,0),118,1)$, $((0;0,2,1,0,0,1),68,-1)$, $((0;0,2,1,0,1,0),116,1)$, $((2;0,2,1,1,0,0),141,-1)$, $((2;0,2,2,0,0,0),140,-1)$, $((2;1,1,2,0,0,0),136,-1)$, $((4;0,2,0,0,0,2),23,1)$, $((7;0,2,1,0,1,0),46,-1)$, $((7;0,2,2,0,0,0),44,-1)$, $((0;0,2,0,0,0,2),71,-1)$, $((0;0,2,0,0,1,1),70,-1)$, $((0;0,2,0,1,0,1),69,-1)$, $((0;0,2,0,1,1,0),117,1)$, $((0;1,1,0,0,0,2),67,-1)$, $((0;1,1,0,0,2,0),114,1)$, $((0;1,1,0,1,0,1),65,-1)$, $((0;1,1,0,1,1,0),113,1)$, $((0;1,1,1,0,0,1),64,-1)$, $((0;1,1,1,0,1,0),112,1)$, $((0;2,0,0,0,0,2),63,-1)$, $((0;2,0,0,0,2,0),110,1)$, $((0;2,0,0,1,0,1),61,-1)$, $((0;2,0,0,1,1,0),109,1)$, $((0;2,0,1,0,0,1),60,-1)$, $((0;2,0,1,0,1,0),108,1)$, $((1;0,2,0,0,2,0),106,1)$, $((1;0,2,1,0,1,0),104,1)$, $((2;0,2,0,0,0,2),95,-1)$, $((2;0,2,0,0,2,0),142,1)$, $((2;0,2,0,1,0,1),93,-1)$, $((2;1,1,0,0,0,2),91,-1)$, $((2;1,1,0,0,2,0),138,1)$, $((2;1,1,0,1,0,1),89,-1)$, $((2;1,1,0,1,1,0),137,1)$, $((2;2,0,0,0,0,2),87,-1)$, $((2;2,0,0,0,2,0),134,1)$, $((2;2,0,0,1,0,1),85,-1)$, $((2;2,0,0,1,1,0),133,1)$, $((2;2,0,1,0,1,0),132,1)$, $((3;0,2,0,0,0,2),83,-1)$, $((1;0,2,0,0,0,2),59,-1)$, $((3;0,2,0,0,2,0),130,1)$, $((3;0,2,0,1,0,1),81,-1)$, $((1;0,2,0,1,0,1),57,-1)$, $((3;0,2,0,1,1,0),129,1)$, $((1;0,2,0,1,1,0),105,1)$, $((3;0,2,1,0,1,0),94,1)$, $((3;0,2,2,0,0,0),92,1)$, $((3;1,1,0,0,0,2),79,-1)$, $((1;1,1,0,0,0,2),55,-1)$, $((3;1,1,0,0,2,0),126,1)$, $((1;1,1,0,0,2,0),102,1)$, $((3;1,1,0,1,0,1),77,-1)$, $((1;1,1,0,1,0,1),53,-1)$, $((3;1,1,0,1,1,0),125,1)$, $((1;1,1,0,1,1,0),101,1)$, $((3;2,0,0,0,0,2),75,-1)$, $((1;2,0,0,0,0,2),51,-1)$, $((3;2,0,0,0,2,0),122,1)$, $((1;2,0,0,0,2,0),98,1)$, $((3;2,0,0,1,0,1),73,-1)$, $((1;2,0,0,1,0,1),49,-1)$, $((3;2,0,0,1,1,0),121,1)$, $((1;2,0,0,1,1,0),97,1)$, $((4;0,2,0,0,1,1),22,1)$, $((4;0,2,1,0,0,1),20,1)$, $((4;1,1,0,0,0,2),19,1)$, $((4;1,1,0,0,1,1),18,1)$, $((4;1,1,1,0,0,1),16,1)$, $((4;2,0,0,0,0,2),15,1)$, $((4;2,0,0,0,1,1),14,1)$, $((4;2,0,1,0,0,1),12,1)$, $((6;0,2,0,0,0,2),47,1)$, $((6;1,1,0,0,0,2),43,1)$, $((6;1,1,0,0,1,1),42,1)$, $((6;1,1,1,0,0,1),40,1)$, $((6;2,0,0,0,0,2),39,1)$, $((6;2,0,0,0,1,1),38,1)$, $((6;2,0,1,0,0,1),36,1)$, $((7;0,2,0,0,0,2),35,1)$, $((5;0,2,0,0,0,2),11,1)$, $((7;0,2,0,0,1,1),34,1)$, $((5;0,2,0,0,1,1),10,1)$, $((7;0,2,1,0,0,1),32,1)$, $((7;0,2,1,1,0,0),45,-1)$, $((7;1,1,0,0,0,2),31,1)$, $((5;1,1,0,0,0,2),7,1)$, $((7;1,1,0,0,1,1),30,1)$, $((5;1,1,0,0,1,1),6,1)$, $((7;1,1,1,0,0,1),28,1)$, $((7;2,0,0,0,0,2),27,1)$, $((5;2,0,0,0,0,2),3,1)$, $((7;2,0,0,0,1,1),26,1)$, $((5;2,0,0,0,1,1),2,1)$, $((7;2,0,1,0,0,1),24,1)$, $((10;1,1,1,1,0,0),41,1)$, $((10;2,0,1,1,0,0),37,1)$, $((11;0,2,1,1,0,0),33,1)$, $((11;1,1,1,1,0,0),29,1)$, $((11;2,0,1,1,0,0),25,1)$, $((0;1,1,0,0,1,1),66,-1)$, $((0;2,0,0,0,1,1),62,-1)$, $((1;0,2,1,0,0,1),56,-1)$, $((1;1,1,1,0,0,1),52,-1)$, $((1;1,1,1,0,1,0),100,1)$, $((1;2,0,1,0,0,1),48,-1)$, $((1;2,0,1,0,1,0),96,1)$, $((2;0,2,0,0,1,1),128,1)$, $((2;1,1,0,0,1,1),90,-1)$, $((2;1,1,1,0,0,1),88,-1)$, $((2;2,0,0,0,1,1),86,-1)$, $((2;2,0,1,0,0,1),84,-1)$, $((3;0,2,0,0,1,1),82,-1)$, $((1;0,2,0,0,1,1),58,-1)$, $((3;0,2,1,0,0,1),80,-1)$, $((3;1,1,0,0,1,1),78,-1)$, $((1;1,1,0,0,1,1),54,-1)$, $((3;1,1,1,0,0,1),76,-1)$, $((3;1,1,1,0,1,0),124,1)$, $((3;2,0,0,0,1,1),74,-1)$, $((1;2,0,0,0,1,1),50,-1)$, $((3;2,0,1,0,0,1),72,-1)$, $((3;2,0,1,0,1,0),120,1)$, $((4;0,2,0,1,0,1),21,1)$, $((4;1,1,0,1,0,1),17,1)$, $((4;2,0,0,1,0,1),13,1)$, $((5;0,2,0,1,0,1),9,1)$, $((5;0,2,1,0,0,1),8,1)$, $((5;1,1,0,1,0,1),5,1)$, $((5;1,1,1,0,0,1),4,1)$, $((5;2,0,0,1,0,1),1,1)$, $((5;2,0,1,0,0,1),0,1)$

Free unknowns: 99, 103, 107, 111, 115, 119, 123, 127, 131, 135, 139, 143

TABLE 42. The pivot table of the third equation of (10.6), with unknown v.

$((0;0,3,0,0,1,0),3,1)$, $((0;0,3,0,1,0,0),7,-1)$, $((0;1,2,0,0,1,0),2,1)$, $((0;1,2,0,1,0,0),6,-1)$, $((0;2,1,0,0,1,0),1,1)$, $((0;2,1,0,1,0,0),5,-1)$, $((0;3,0,0,0,1,0),0,1)$, $((0;3,0,0,1,0,0),4,-1)$, $((1;0,3,0,1,0,0),11,-1)$, $((1;1,2,0,1,0,0),10,-1)$, $((1;2,1,0,1,0,0),9,-1)$, $((1;3,0,0,1,0,0),8,-1)$, $((2;0,3,0,1,0,0),15,-1)$, $((2;1,2,0,1,0,0),14,-1)$, $((2;2,1,0,1,0,0),13,-1)$, $((2;3,0,0,1,0,0),12,-1)$, $((3;0,3,0,1,0,0),19,-1)$, $((3;1,2,0,1,0,0),18,-1)$, $((3;2,1,0,1,0,0),17,-1)$, $((3;3,0,0,1,0,0),16,-1)$, $((5;0,3,0,1,0,0),23,-1)$, $((5;1,2,0,1,0,0),22,-1)$, $((5;2,1,0,1,0,0),21,-1)$, $((5;3,0,0,1,0,0),20,-1)$

$n-\delta_a$. Thus the relation matrix has 49 columns for $n=8$ and 116 columns for $n=10$. In Table 44, all rows without pivot reduce to zero.

For $n=8,10$, let π_n be the map from $(F_0)_n$ to the vector space with basis the columns without pivot in degree n, in the increasing order listed in Table 44, sending the basis vector of a column without pivot to itself, and the basis vector of a pivot column to minus the entries in the columns without pivot of the final row of that pivot. Thus π_n is zero on the relations of degree n, and there exist 29 columns without pivot for $n=8$ and 46 for $n=10$, for all values of e, p, and $q_0,\dots,q_8$.

TABLE 43. The pivot table of the fourth equation of (10.6), with unknown Ψ.

$((3;0,4,0,0,0,1),34,1)$, $((0;0,4,0,0,0,1),9,1)$, $((0;0,4,0,0,1,0),4,1)$, $((0;1,3,0,0,0,1),8,1)$, $((0;1,3,0,0,1,0),3,1)$, $((0;2,2,0,0,0,1),7,1)$, $((0;2,2,0,0,1,0),2,1)$, $((0;3,1,0,0,0,1),6,1)$, $((0;3,1,0,0,1,0),1,1)$, $((0;4,0,0,0,0,1),5,1)$, $((0;4,0,0,0,1,0),0,1)$, $((1;0,4,0,0,0,1),19,1)$, $((1;0,4,0,0,1,0),14,1)$, $((1;1,3,0,0,0,1),18,1)$, $((1;1,3,0,0,1,0),13,1)$, $((1;2,2,0,0,0,1),17,1)$, $((1;2,2,0,0,1,0),12,1)$, $((1;3,1,0,0,0,1),16,1)$, $((1;3,1,0,0,1,0),11,1)$, $((1;4,0,0,0,0,1),15,1)$, $((1;4,0,0,0,1,0),10,1)$, $((2;0,4,0,0,0,1),29,1)$, $((2;0,4,0,0,1,0),24,1)$, $((2;1,3,0,0,0,1),28,1)$, $((2;1,3,0,0,1,0),23,1)$, $((2;2,2,0,0,0,1),27,1)$, $((2;2,2,0,0,1,0),22,1)$, $((2;3,1,0,0,0,1),26,1)$, $((2;3,1,0,0,1,0),21,1)$, $((2;4,0,0,0,0,1),25,1)$, $((2;4,0,0,0,1,0),20,1)$, $((3;1,3,0,0,0,1),33,1)$, $((3;2,2,0,0,0,1),32,1)$, $((3;3,1,0,0,0,1),31,1)$, $((3;4,0,0,0,0,1),30,1)$, $((4;0,4,0,0,0,1),39,1)$, $((4;1,3,0,0,0,1),38,1)$, $((4;2,2,0,0,0,1),37,1)$, $((4;3,1,0,0,0,1),36,1)$, $((4;4,0,0,0,0,1),35,1)$

Let $\mu_0,\dots,\mu_{28}$ be the basis vectors of $(F_0)_8$ at the columns without pivot in degree 8, in increasing order of the columns, let $\eta_0,\dots,\eta_{115}$ be formal variables, and let

$$H_j := \sum_{n=0}^{28} \eta_{29j+n}\mu_n \qquad (0 \le j \le 3).$$

For $1 \le k \le 6$ and $0 \le l < 46$, we define

$$E_{46(k-1)+l} := \text{coordinate } l \text{ of } \pi_{10}\Big((r_k + v_k)z_0 + \sum_{j=0}^{3} A_{jk}H_j\Big),$$

where v_k is the binary cubic of Set-up 4.1. Thus $E_0,\dots,E_{275}$ are polynomials of degree at most 1 in $\eta_0,\dots,\eta_{115}$, with coefficients in $\mathbb{Q}[e,p,p^{-1},q_0,\dots,q_8]$. The rows of Table 45 define the linear functionals of **Step 4** of the proof of Construction-Proposition 10.10: the term E_ρ with $\rho = 46(k-1)+l$ stands for coordinate l of π_{10} on the k-th summand of $\bigoplus_{k=1}^{6}(F_0)_{10}$. A row of the table of linear functionals, Table 45, is a list of pairs (ρ,ν), meaning $\sum_{(\rho,\nu)} \nu E_\rho$.

TABLE 44. The pivot tables of the relation matrices in degrees 8 and 10.

Degree 8:

$(0,5,1)$, $(1,16,1)$, $(4,3,1)$, $(5,14,1)$, $(2,4,1)$, $(3,15,1)$, $(10,0,1)$, $(11,11,1)$, $(8,1,1)$, $(9,12,1)$, $(6,2,1)$, $(7,13,1)$, $(18,43,1)$, $(19,47,1)$, $(13,38,1)$, $(17,46,-1)$, $(14,41,1)$, $(12,37,-1)$, $(15,39,-1)$, $(16,42,-1)$

Columns without pivot in degree 8: 6, 7, 8, 9, 10, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 40, 44, 45, 48

Degree 10:

$(0,5,1)$, $(1,16,1)$, $(2,27,1)$, $(6,3,1)$, $(7,14,1)$, $(8,25,1)$, $(3,4,1)$, $(4,15,1)$, $(5,26,1)$, $(65,84,1)$, $(15,0,1)$, $(16,11,1)$, $(17,22,1)$, $(57,56,1)$, $(12,1,1)$, $(13,12,1)$, $(14,23,1)$, $(9,2,1)$, $(10,13,1)$, $(11,24,1)$, $(33,86,1)$, $(41,57,1)$, $(25,58,1)$, $(61,70,1)$, $(49,85,1)$, $(53,42,1)$, $(29,72,1)$, $(55,51,1)$, $(45,71,1)$, $(21,44,1)$, $(54,50,1)$, $(38,105,-1)$, $(78,102,1)$, $(79,106,1)$, $(64,81,1)$, $(80,110,1)$, $(81,114,1)$, $(37,43,1)$, $(68,93,1)$, $(69,97,1)$, $(76,109,-1)$, $(77,113,-1)$, $(22,53,1)$, $(30,104,-1)$, $(70,100,1)$, $(56,96,1)$, $(66,92,-1)$, $(62,78,1)$, $(24,55,1)$, $(72,94,-1)$, $(74,101,-1)$, $(73,98,-1)$, $(32,83,1)$, $(23,54,1)$, $(39,52,1)$, $(26,67,1)$, $(46,79,1)$, $(31,82,1)$, $(58,64,1)$, $(18,39,1)$, $(27,68,1)$, $(28,69,1)$, $(50,36,1)$, $(47,80,1)$, $(20,41,1)$, $(19,40,1)$, $(42,65,1)$, $(34,37,1)$, $(43,66,1)$, $(35,38,1)$

Columns without pivot in degree 10: 6, 7, 8, 9, 10, 17, 18, 19, 20, 21, 28, 29, 30, 31, 32, 33, 34, 35, 45, 46, 47, 48, 49, 59, 60, 61, 62, 63, 73, 74, 75, 76, 77, 87, 88, 89, 90, 91, 95, 99, 103, 107, 108, 111, 112, 115

Define the coefficients x_n by Table 39, (Φ,Θ,v,Ψ) by Tables 40, 41, 42, and 43, and E_ρ by the formula for $E_{46(k-1)+l}$, with π_{10} given by Table 44. A direct calculation shows that, for $i = 1,\dots,4$, the coefficient of any η_m in $\sum_{(\rho,\nu)} \nu E_\rho$, the sum over the i-th row of Table 45, is 0 and the remaining term is L_i. This is the calculation needed in **Step 4**, since a lift ζ_j of $z_0 z_j$ differs from H_j, with η_{29j+n} the coordinate n of $\pi_8(\zeta_j)$, by a relation of degree 8, and s and t times a relation of degree 8 is a relation of degree 10.

TABLE 45. The table of linear functionals for exceptional presentations.

$\sum_{(\rho,\nu)} \nu E_\rho = L_1$, with $L_1 = q_1 + q_8/p$:
$(47, 1/(2p) + 1/(2p^2))$, $(231, 1/(2p) + 1/(2p^2))$, $(97, 1)$, $(0, -1)$, $(46, 1/p)$
$\sum_{(\rho,\nu)} \nu E_\rho = L_2$, with $L_2 = q_3 - 2q_8/p$:
$(97, -2)$, $(0, 2)$, $(46, -2/p)$, $(47, 1 - 1/p - 1/p^2)$, $(231, 1 - 1/p - 1/p^2)$
$\sum_{(\rho,\nu)} \nu E_\rho = L_3$, with $L_3 = q_5 + 2q_8 + q_8/p$:
$(1, -1)$, $(185, -1)$, $(47, 1 + 1/p + 1/(2p^2))$, $(231, 1 + 1/p + 1/(2p^2))$, $(97, 2p + 1)$, $(0, -2p - 1)$, $(46, 2 + 1/p)$
$\sum_{(\rho,\nu)} \nu E_\rho = L_4$, with $L_4 = q_2 + 2q_8/p^2$:
$(80, -1/(2p))$, $(264, -1/(2p))$, $(47, -1 - 3/(2p) + 3/(2p^2) + p^{-3})$, $(231, -1 - 3/(2p) + 3/(2p^2) + p^{-3})$, $(97, 1 + 2/p)$, $(0, -1 - 2/p)$, $(46, 1/p + 2/p^2)$, $(1, -p/2)$, $(185, -p/2)$

A.5. **Type** $(3, 1)$**.** Table 46 lists the data of Construction-Propositions 8.6 and 10.7 for the fixed blocks $B_1, \dots, B_4$ of Table 14. Tables 47, 48, and 49 are pivot tables, in the sense of Construction-Proposition 6.7, of the matrix $T_{(3,1)}$ of Construction-Proposition 10.7, which has 260 rows and 84 columns, for the families (G), (V_0), and (V_1) of Construction-Proposition 11.3 and Table 15, with the additional row operation in the caption of Table 47, and a triple is (row, column, pivot). Lemma 11.4 uses Tables 46, 47, 48, and 49.

TABLE 46. The slots, the minors, the twelve forms, and the entries $c_0, \dots, c_{19}$ of type $(3, 1)$.

The twelve slots, where (i, j, k) with $i < j$ is the coefficient of y_k in O_{ij}, are: $(0, 1, 0)$, $(0, 2, 0)$, $(0, 3, 0)$, $(0, 4, 0)$, $(1, 2, 0)$, $(1, 3, 0)$, $(1, 4, 0)$, $(2, 3, 0)$, $(2, 4, 0)$, $(3, 4, 3)$, $(3, 5, 3)$, $(4, 5, 3)$. The form o_r is the coefficient at the r-th listed slot, counted from 0.

The rows of the 60×48 matrix of the map $\Gamma \to B_3\Gamma^t - \Gamma B_3^t$ are the sixty slots in lexicographic order, numbered from 0, and its columns are the entries of Γ in row-major order. The 48×48 minor of Construction-Proposition 8.6 has determinant -16, and its rows are the slots other than the twelve slots above, with the numbers: 1, 2, 3, 5, 6, 7, 9, 10, 11, 13, 14, 15, 16, 17, 18, 19, 21, 22, 23, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 37, 38, 39, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 52, 53, 54, 56, 57, 58.

The 84×84 minor of Construction-Proposition 10.7 has determinant -1. With the entries of N and of C numbered from 0 in row-major order, the columns of the minor, where the number $n \le 71$ is the entry of N numbered n and the number $72 + n$ is the entry of C numbered n, are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 81, 99, 100, 103.

The rows of the minor are the coefficients of Construction-Proposition 10.7 with the numbers: 1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 49, 50, 51, 53, 54, 55, 56, 57, 58, 59, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 88, 89, 90, 92, 93, 94.

The twelve forms of Construction-Proposition 10.7 have the numbers: 0, 4, 12, 24, 28, 36, 48, 52, 60, 87, 91, 95, in increasing order, which are the complement in $0, \dots, 95$ of the numbers of the rows of the minor.

The entries of C numbered 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 29, 30 are $c_0, \dots, c_{19}$, respectively.

A.6. **Type** (4)**.** Table 50 lists the data of Construction-Propositions 8.6 and 10.7 for the fixed blocks $B_1, \dots, B_4$ of Table 16. Tables 51 and 52 are pivot tables, in the sense of Construction-Proposition 6.7, of the matrix $T_{(4)}$ of Construction-Proposition 10.7, which has 252 rows and 84 columns, for the families (G) and (E) of Table 17, and the third entry of any triple in them is a non-zero constant.

Each pivot table ends with the columns of $T_{(4)}$ without a pivot. Lemma 12.4 uses Tables 50, 51, and 52.

TABLE 47. A pivot table of $T_{(3,1)}$ for the family (G) of Table 15, with ζ either root of $\zeta^2-\zeta+1=0$, where after the first 79 triples, row 27 of the current matrix is replaced by row 27 minus row 81 of the current matrix, which does not change the rank, and then the last triple is taken.

$(2,5,-1)$, $(6,10,-1)$, $(0,0,1)$, $(1,1,1)$, $(5,4,1-\zeta)$, $(11,7,-1)$, $(15,12,-1)$, $(24,16,-1)$, $(26,13,-\zeta)$, $(16,6,1-\zeta)$, $(29,21,1)$, $(32,23,1)$, $(251,15,-1)$, $(9,2,1)$, $(10,3,1)$, $(25,14,\zeta-1)$, $(28,20,1)$, $(31,22,1)$, $(203,80,-1)$, $(207,83,-1)$, $(248,30,1)$, $(18,8,1)$, $(19,9,1)$, $(47,38,-1)$, $(60,33,1)$, $(215,26,\zeta)$, $(236,24,-1)$, $(238,17,-\zeta)$, $(252,39,2)$, $(259,41,1-\zeta)$, $(43,28,1)$, $(67,56,\zeta-1)$, $(107,44,-1)$, $(202,79,-1)$, $(206,82,-1)$, $(228,57,1)$, $(231,35,\zeta)$, $(237,25,-1)$, $(243,32,-\zeta)$, $(63,19,-1)$, $(242,27,\zeta-1)$, $(55,47,1-\zeta)$, $(59,62,\zeta-1)$, $(235,59,\zeta-1)$, $(46,37,-1)$, $(91,50,\zeta)$, $(95,65,-\zeta)$, $(135,68,1)$, $(224,54,-1)$, $(216,48,-1)$, $(42,31,-\zeta)$, $(214,52,-2)$, $(62,18,-1)$, $(64,63,1)$, $(150,61,-1)$, $(190,55,-1)$, $(201,78,-1)$, $(205,81,-1)$, $(256,42,-1)$, $(40,45,1)$, $(41,36,1)$, $(213,46,2)$, $(212,71,-2)$, $(44,69,1)$, $(45,70,1)$, $(56,72,1)$, $(104,66,1)$, $(189,60,1)$, $(188,51,1)$, $(225,75,-1)$, $(105,67,1)$, $(229,58,1)$, $(65,64,1)$, $(253,34,-1)$, $(61,40,2)$, $(53,49,1)$, $(54,74,1-2\zeta)$, $(57,73,1)$, $(90,43,-1)$, $(27,76,2-2\zeta)$

The four columns without a pivot are 11, 29, 53, 77.

TABLE 48. A pivot table of $T_{(3,1)}$ for the family (V_0) of Table 15, with (ζ,η) any of the four pairs of roots of $\zeta^2-\zeta+1=0$ and $\eta^2+(\zeta-1)\eta-1=0$, and with the pivots written as polynomials in η by the relation $\zeta=1-\eta+\eta^{-1}$.

$(0,0,1)$, $(1,1,1)$, $(6,10,-1)$, $(8,5,\eta^3-\eta^2+1)$, $(10,3,1)$, $(27,20,1)$, $(29,19,-\eta)$, $(30,22,1)$, $(31,23,1)$, $(120,63,1)$, $(196,42,-1)$, $(92,78,-1)$, $(203,77,\eta)$, $(205,82,-1)$, $(206,83,-1)$, $(236,24,-1)$, $(7,4,\eta^3-\eta^2+1)$, $(9,2,1)$, $(15,12,-1)$, $(17,7,\eta^3-\eta^2+1)$, $(18,6,-1)$, $(19,9,1)$, $(24,16,-1)$, $(26,13,\eta^3-\eta^2)$, $(219,47,1)$, $(247,15,-\eta^3+\eta)$, $(16,8,\eta^3-\eta^2+1)$, $(47,29,\eta)$, $(42,25,-1)$, $(28,18,-\eta)$, $(43,11,-\eta^3+\eta^2+\eta)$, $(44,36,-1)$, $(83,32,-\eta)$, $(95,35,-\eta)$, $(123,56,-\eta)$, $(148,60,-1)$, $(188,30,-1)$, $(201,75,\eta)$, $(202,76,\eta)$, $(204,21,-\eta^3+\eta^2+\eta-1)$, $(244,48,-1)$, $(52,51,1)$, $(259,41,\eta^3-\eta^2+1)$, $(67,62,\eta^3-\eta^2+1)$, $(25,14,-\eta^3+\eta^2-1)$, $(40,27,1)$, $(212,45,2)$, $(64,57,1)$, $(45,37,-1)$, $(189,31,-1)$, $(82,70,-\eta^3+\eta^2-\eta)$, $(58,52,\eta)$, $(226,74,-2\eta^3+4\eta-2)$, $(53,49,1)$, $(56,39,-1)$, $(197,43,-1)$, $(60,33,1)$, $(88,54,-1)$, $(104,66,1)$, $(121,64,1)$, $(133,67,1)$, $(135,59,-\eta)$, $(149,61,-1)$, $(220,72,-2)$, $(41,17,3\eta^3-2\eta^2-\eta)$, $(213,28,9\eta^3/49-12\eta^2/49-5\eta/49-22/49)$, $(190,46,\eta^3-\eta^2-\eta)$, $(46,38,-1)$, $(59,53,\eta)$, $(227,71,-4\eta^3+8\eta^2-2\eta-6)$, $(54,50,1)$, $(65,58,1)$, $(106,68,1)$, $(57,40,-1)$, $(198,44,-1)$, $(221,69,-2\eta^3+2\eta^2+2\eta-2)$, $(61,34,1)$, $(66,55,-\eta^3+\eta^2-1)$, $(89,65,\eta^3-\eta^2)$, $(107,73,2\eta^2-2\eta)$

The four columns without a pivot are 26, 79, 80, 81.

TABLE 49. A pivot table of $T_{(3,1)}$ for the family (V_1) of Table 15, with (ζ,η) any of the four pairs of roots of $\zeta^2-\zeta+1=0$ and $\eta^2+(\zeta-1)\eta-1=0$, and with the pivots written as polynomials in η by the relation $\zeta=1-\eta+\eta^{-1}$.

$(0,0,1)$, $(1,1,1)$, $(6,10,-1)$, $(8,5,\eta^3-\eta^2+1)$, $(10,3,1)$, $(29,19,-\eta)$, $(203,77,\eta)$, $(7,4,\eta^3-\eta^2+1)$, $(9,2,1)$, $(15,12,-1)$, $(17,7,\eta^3-\eta^2+1)$, $(19,9,1)$, $(24,16,-1)$, $(26,13,\eta^3-\eta^2)$, $(16,6,\eta^3-\eta^2+1)$, $(27,18,-1)$, $(28,20,-\eta)$, $(30,21,\eta^3-\eta^2-\eta+1)$, $(31,22,\eta^3-\eta^2-2\eta+1)$, $(43,11,-\eta^3+\eta^2+\eta)$, $(200,75,1)$, $(202,76,\eta)$, $(206,80,\eta^3-2\eta^2+3)$, $(219,26,2\eta^3-3\eta^2+\eta+2)$, $(247,15,-\eta^3+\eta)$, $(18,8,1)$, $(47,29,\eta)$, $(42,25,-1)$, $(236,24,-1)$, $(259,41,\eta^3-\eta^2+1)$, $(25,14,-\eta^3+\eta^2-1)$, $(60,33,1)$, $(67,56,-\eta^3+\eta^2-1)$, $(83,32,-\eta)$, $(201,78,\eta)$, $(205,79,\eta^3-2\eta^2+3)$, $(212,27,-1)$, $(40,17,\eta^3-\eta^2)$, $(44,36,-1)$, $(104,23,3\eta^3/13-\eta^2/13-8\eta/13+2/13)$, $(188,30,-1)$, $(231,35,-\eta^3+\eta^2)$, $(248,45,-2)$, $(52,48,1)$, $(88,39,2)$, $(91,38,\eta^3-\eta^2-\eta)$, $(227,50,\eta-1)$, $(95,53,-2\eta^2+2\eta+1)$, $(254,31,\eta^3-\eta^2+1)$, $(226,37,\eta^3-\eta^2)$, $(62,52,2)$, $(90,49,-\eta^3+\eta^2)$, $(249,51,-\eta^3/2+\eta^2/2-1/2)$, $(41,28,1)$, $(213,46,2)$, $(45,62,-\eta^3+3\eta^2-1)$, $(53,70,-131\eta^3/247+412\eta^2/247-123\eta/247-18/247)$, $(150,74,558\eta^3/721-870\eta^2/721+68\eta/721+582/721)$, $(46,71,-\eta^2+\eta+1)$, $(64,57,1)$, $(135,59,-\eta)$, $(105,42,-1)$, $(107,44,-1)$, $(235,65,-2\eta^3+2\eta^2-2)$, $(94,34,-\eta)$, $(122,55,-\eta)$, $(149,61,-1)$, $(89,69,\eta^3-\eta^2-1)$, $(225,73,4\eta^3/3-10\eta^2/3+2\eta+2/3)$, $(57,40,-1)$, $(148,60,-1)$, $(224,54,-1)$, $(56,72,1)$, $(93,43,1)$, $(92,67,1)$, $(106,58,-\eta^3+\eta^2)$, $(65,68,-\eta^3+\eta^2-1)$, $(120,63,1)$, $(121,64,1)$, $(132,66,1)$

The four columns without a pivot are 47, 81, 82, 83.

TABLE 50. The slots, the minors, the twelve forms, and the entries $c_0, \dots, c_{19}$ of type (4).

The twelve slots, where (m, l, k) with $m < l$ is the coefficient of y_k in the entry of O in row m and column l, are: $(0,1,0)$, $(0,2,0)$, $(0,3,0)$, $(0,4,0)$, $(1,2,0)$, $(1,2,1)$, $(1,3,0)$, $(1,4,0)$, $(2,3,0)$, $(2,4,0)$, $(3,4,0)$, $(3,5,0)$. The form o_r is the coefficient at the r-th listed slot, counted from 0.

The rows of the 60×48 matrix of the map $\Gamma \to B_3\Gamma^t - \Gamma B_3^t$ are the sixty slots in lexicographic order, numbered from 0, and its columns are the entries of Γ in row-major order. The 48×48 minor of Construction-Proposition 8.6 has determinant 32, and its rows are the slots other than the twelve slots above, with the numbers: 1, 2, 3, 5, 6, 7, 9, 10, 11, 13, 14, 15, 16, 17, 18, 19, 22, 23, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 37, 38, 39, 41, 42, 43, 44, 45, 46, 47, 49, 50, 51, 53, 54, 55, 56, 57, 58, 59.

The 84×84 minor of Construction-Proposition 10.7 has determinant -1. With the entries of N and of C numbered from 0 in row-major order, the columns of the minor, where the number $n \le 71$ is the entry of N numbered n and the number $72 + n$ is the entry of C numbered n, are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 81, 83, 84, 87.

The rows of the minor are the coefficients of Construction-Proposition 10.7 with the numbers: 1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 49, 50, 51, 53, 54, 55, 56, 57, 58, 59, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 73, 74, 75, 77, 78, 79, 80, 81, 82, 83, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95.

The twelve forms of Construction-Proposition 10.7 have the numbers: 0, 4, 12, 24, 28, 36, 48, 52, 60, 72, 76, 84, in increasing order, which are the complement in $0, \dots, 95$ of the numbers of the rows of the minor.

The entries of C numbered 8, 10, 13, 14, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31 are $c_0, \dots, c_{19}$, respectively.

TABLE 51. A pivot table of $T_{(4)}$ for the family (G) of Table 17, with ζ either root of $\zeta^2 - \zeta + 1 = 0$.

$(2,1,1-\zeta)$, $(9,6,1)$, $(0,2,1)$, $(3,8,1)$, $(4,9,1)$, $(10,4,1-\zeta)$, $(11,3,1)$, $(1,0,1-\zeta)$, $(18,16,1)$, $(20,7,1)$, $(21,20,1)$, $(30,22,2)$, $(248,57,-1)$, $(12,10,1)$, $(19,12,1-\zeta)$, $(22,21,1)$, $(27,18,1)$, $(29,11,\zeta-1)$, $(231,26,\zeta-1)$, $(13,15,1)$, $(40,30,-1)$, $(52,33,-1)$, $(67,35,1)$, $(207,29,1)$, $(251,77,1)$, $(28,14,1-\zeta)$, $(230,25,\zeta-1)$, $(31,23,2)$, $(43,19,1)$, $(56,69,1)$, $(116,78,-1)$, $(128,81,-1)$, $(180,63,-1)$, $(204,39,1)$, $(228,27,-1)$, $(232,42,-2)$, $(44,66,1)$, $(208,51,1)$, $(47,32,-\zeta)$, $(239,62,-2)$, $(55,41,2-\zeta)$, $(183,44,-1)$, $(203,83,\zeta-1)$, $(227,59,2-\zeta)$, $(42,28,-1)$, $(206,37,2-2\zeta)$, $(229,24,\zeta-1)$, $(64,48,1)$, $(68,72,1)$, $(80,75,1)$, $(185,76,-1)$, $(195,50,1)$, $(103,56,2)$, $(95,71,1)$, $(205,36,1-\zeta)$, $(236,45,2)$, $(65,49,1)$, $(45,60,1-\zeta)$, $(59,53,-1)$, $(66,34,1)$, $(58,52,-1)$, $(82,64,-1)$, $(181,67,-\zeta-1)$, $(53,46,1)$, $(83,65,-1)$, $(215,47,\zeta-2)$, $(41,31,-1)$, $(233,40,2)$, $(62,68,-2\zeta)$, $(46,61,1-\zeta)$, $(182,43,-1)$, $(212,54,-1)$, $(57,70,1)$, $(93,73,-1)$, $(69,79,1)$, $(101,58,1)$, $(105,82,1)$, $(213,55,-1)$, $(94,74,-1)$, $(70,80,1)$

The four columns without a pivot are 5, 13, 17, 38.

TABLE 52. A pivot table of $T_{(4)}$ for the family (E) of Table 17, with i either root of $i^2 = -1$.

$(5, 1, -1+i)$, $(9, 6, 1)$, $(14, 3, 1-i)$, $(1, 5, 1)$, $(0, 2, 1)$, $(3, 8, 1)$, $(10, 0, 1+i)$, $(18, 16, 1)$, $(21, 20, 1)$, $(29, 7, i)$, $(4, 9, -i)$, $(30, 22, 2)$, $(43, 13, 2i)$, $(248, 57, -1)$, $(12, 10, 1)$, $(19, 4, 1+i)$, $(27, 18, 1)$, $(40, 30, -1)$, $(47, 15, -1+i)$, $(55, 38, -1+i)$, $(239, 17, 1+2i)$, $(63, 11, 2)$, $(13, 29, 1/2 - i/2)$, $(22, 12, -1+i)$, $(28, 19, 1)$, $(52, 33, -1)$, $(228, 27, -1)$, $(31, 14, -1+i)$, $(56, 69, 1)$, $(59, 21, -1)$, $(219, 41, 1+i)$, $(83, 44, -i)$, $(67, 32, 1)$, $(42, 25, -2i)$, $(45, 67, 1)$, $(64, 48, 1)$, $(80, 75, 1)$, $(92, 72, -1)$, $(68, 78, 1)$, $(128, 81, -1)$, $(180, 63, -1)$, $(204, 39, 1)$, $(229, 23, 1)$, $(46, 28, -2-i)$, $(232, 42, -2)$, $(44, 66, 1)$, $(208, 51, 1)$, $(119, 47, i)$, $(181, 36, -i)$, $(182, 37, -i)$, $(54, 35, -1)$, $(71, 53, i)$, $(103, 65, 4+4i)$, $(238, 68, -2/5 - 4i/5)$, $(107, 56, i)$, $(227, 77, -2-2i)$, $(131, 50, i)$, $(203, 59, -1-i)$, $(236, 45, 2)$, $(41, 24, -2i)$, $(57, 60, -1+i)$, $(58, 40, i)$, $(218, 61, -2-2i)$, $(82, 43, -i)$, $(66, 31, 1)$, $(69, 73, 1)$, $(143, 74, 1+2i)$, $(209, 52, 1)$, $(212, 54, -1)$, $(129, 82, -1)$, $(221, 58, -1)$, $(53, 34, -1)$, $(94, 46, -2i)$, $(70, 64, 2i)$, $(237, 80, -2/5 + 4i/5)$, $(81, 70, -1+i)$, $(213, 55, -1)$, $(65, 49, 1)$, $(93, 79, 1)$, $(106, 76, -1-i)$

The four columns without a pivot are 26, 62, 71, 83.

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